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BPSC CONCEPT WALLAH

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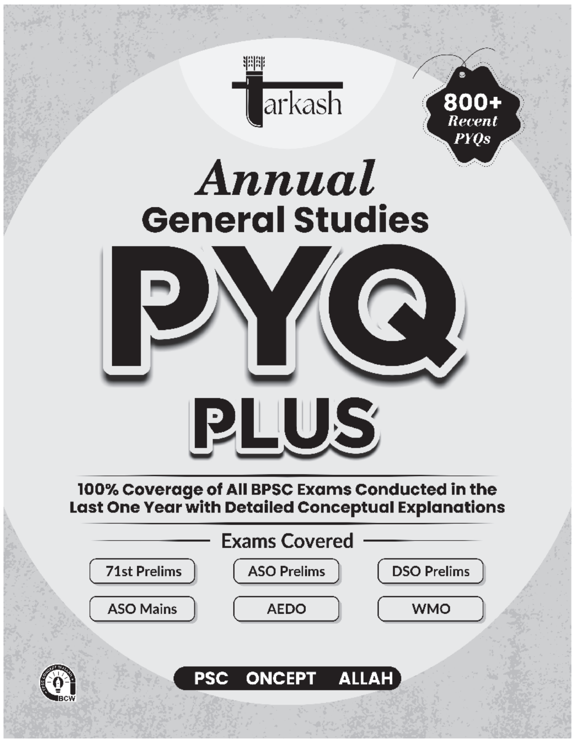

This book is a supplementary volume to Tarkash Prelims PYQ, already available in the market. It is designed to complement the existing book and provide aspirants with additional exam-oriented focus with shifting pattern and broad coverage.

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PREFACE

The BPSC examinations remain the most prestigious yet challenging hurdles for civil services aspirants in Bihar. In recent times, the Commission has radically transformed its examination pattern, shifting toward deeper conceptual clarity, analytical evaluation, and multi-statement questions that demand a much higher level of precision. Success in this rapidly changing environment requires more than just hard work; it demands a surgical approach to the evolving trends and a deep familiarity with the modern mindset of the paper setters.

At BPSC Concept Wallah, our journey has always been defined by the success of our students. We take immense pride in the fact that our methodologies and resources have consistently helped candidates clear the BPSC Combined Competitive Examination (CCE), placing them in key administrative roles across the state. This legacy of excellence is perhaps best embodied by our flagship publication, Tarkash, which has become a household name among Bihar’s civil services aspirants for its precision and depth.

Building on the trust and success of Tarkash, we have designed this Annual Compilation to be a definitive, all-in-one companion for your BPSC preparation. As the BPSC continues to change its trends, navigating the vast syllabus without a map can be daunting. This book is specifically crafted to address this challenge, serving as an invaluable asset for aspirants attempting to master the new pattern.

We have structured this resource to follow a "Concept-to-Confidence" framework: by meticulously compiling and providing detailed, multidimensional explanations for every single question asked by BPSC over the last one year, this book bridges the gap between reading an old concept and understanding its new application. Decoding these recent questions will give you an unparalleled edge in predicting the trajectory of upcoming exams.

Our mission is to democratize quality education and simplify the path to public service. This book is not just a retrospective collection of past questions; it is a forward-looking strategic asset built on the feedback of successful candidates and the expertise of seasoned mentors who understand the changing pulse of Bihar’s competitive landscape.

Team BPSC Concept Wallah

ACKNOWLEDGEMENT

This Annual Compilation is the culmination of months of meticulous tracking and rigorous analysis of shifting exam trends by the academic team at BPSC Concept Wallah. We extend our sincere gratitude to the dedicated subject matter experts who painstakingly decoded every question asked by the Commission over the last year, ensuring the utmost accuracy and depth of the explanations.

Most importantly, we thank you—the aspirant. Your resilience in adapting to BPSC's evolving patterns inspires us to keep refining our methods. We are confident that if you engage with this book as rigorously as our students did with Tarkash, your dream of joining the Bihar administration will soon be a reality.

Prepare with purpose. Achieve with pride.

Team BPSC Concept Wallah

Polity

1

1. Consider the following subjects belonging to different lists of the Constitution of India. Which of the following subjects belong to the concurrent list? *[71st BPSC]*
1. Protection of wildlife
2. Income on Agriculture
3. Tax on Electricity consumption or sale
4. Price control

Out of these:
A. 2 and 3 are correct
B. 1 and 4 are correct
C. Only 4 is correct
D. Only 2 is correct

Ans. B

Exp: Seventh Schedule in the Constitution of India deals with the specified separation of powers and functions between the Union & States on different subjects. It contains three lists; i.e.

i. Union List - Presently, it consists of 98 entries in which the center has the exclusive powers to make law. It consists of subjects like defense, banking, currency, foreign affairs, inter-state trade and commerce, census, etc.

ii. State List - Presently, it consists of 59 entries in which a state in "normal circumstances" has the power to make laws. It consists of subjects like public order, police, public health and sanitation, local government, etc.

iii. Concurrent List - Presently, it consists of 52 entries in which both the Parliament and the state legislature have the power to make laws. It consists of subjects like criminal law and procedure, civil procedure, marriage and divorce, population control and family planning, electricity, etc.

- **Protection of Wildlife:** Originally in the State List, it was moved to the Concurrent List (Entry 17B) by the **42nd Amendment in 1976**.
- **Price Control:** This is found in Entry 34 of the Concurrent List.
- **Agriculture Income & Electricity Tax:** These are generally State List subjects (Entries 46 and 53, respectively).

BCW Bits:
- According to Article 254 (1), if any provision of state law is repugnant to a provision in a law made by the Parliament, which the Parliament is competent to enact, or with any existing law regarding any matter in the Concurrent List, then the Parliamentary law would prevail over the State law.

2. Which of the following Directive Principles do not follow Gandhian Principles? *[71st BPSC]*
1. Organization of Village Panchayats
2. Common Civil Code
3. Promotion of cottage Industry in rural areas
4. Right to work

Out of these:
A. Only 2 B. 1 and 2 C. 2 and 4 D. 1 and 3

Ans. C

Exp: Directive Principles of the State Policy:
- Directive Principles of the State Policy are the instrument of instruction to the state.
- Articles 36-51, Part-IV of the Indian Constitution deals with Directive Principles of the State Policy (DPSP).
- DPSPs are positive obligations on the state
- The provisions contained in this part cannot be enforced by any court, but these principles are fundamental in the governance of the country and it shall be the duty of the State to apply these principles in making laws.
- They are borrowed from the Constitution of Ireland.
- The Directive Principles are classified based on ideological sources and objectives. i.e., Socialist, Gandhian, Liberal-Intellectual:
  - **Gandhian Principles-** These reflect the program of reconstruction outlined by Mahatma Gandhi during the national movement. It includes Art. 40 (Panchayats), Art. 43 (Cottage Industries), Art. 47 (prohibition of intoxicants).etc
  - **Liberal-Intellectual Principles -** These principles reflect the ideology of liberalism and emphasize a modern, scientific state. It includes Art. 44 (Uniform Civil Code).
  - **Socialist Principles-**These articles aim to provide social and economic justice and lay the foundation for a welfare state. It includes Art.39 (promoting justice and minimizing inequalities), Art 39-A (free legal aid). Art 41 (Right to work) etc.

3. The power to recruit the Secretarial staff of the House of the People is vested in: *[71st BPSC]*
A. Union Public Service Commission
B. Prime Minister's Office (PMO)
C. Staff Selection Commission
D. President, after consultation with the Speaker of the House of the People

Ans. D

Exp: Under **Article 98** of the Indian Constitution, each House of Parliament has its own separate secretarial staff. Parliament has the power to enact laws governing recruitment and terms of service; in the absence of such legislation, the President may establish these regulations. But before making such appointments or regulations, the President must confer with the Speaker of the Lok Sabha or the Chairman of the Rajya Sabha in order to preserve the Legislature's independence from the Executive.

4. Attorney General of India- ***[71st BPSC]***

A. He gives advice to the President on legal matters

B. Is not entitled of audience except the Supreme Court

C. Cannot participate in the proceedings in the House of Parliament

D. Is a whole-time counsel for the government

Ans. A

Exp: **Article 76** defines the Attorney General (AG) as the highest law officer in the country. Their primary duty is to advise the Government of India (via the President) on legal issues.

- The Attorney General (AG) of India has the right to take part in the proceedings of the Parliament of India even though he is not a member of any house. He is the central chief legal advisor and primarily a lawyer in the supreme court of India.
- AG is appointed by the president under article 76 (1) and he is a part of the union executive.
- He is the **highest law officer** in the country. He must be a person qualified to be appointed as a judge of the Supreme Court.
- The term of his office is not fixed by the constitution and hence he holds office at the pleasure of the president.
- According to Article 165 of the Indian constitution, the Governor of each State shall appoint a person who is qualified to be appointed a Judge of a High Court to be Advocate-General for the State.

5. Consider the following statements about the Central Government: ***[71st BPSC]***

1. Ministries and Departments are formed by the Prime Minister with consultation of the Cabinet Secretary.
2. Every Minister is allocated Ministry by the President on the advice of the Prime Minister.

Out of these:

A. Only 1 is correct B. Only 2 is correct

C. 1 and 2 are correct D. Neither 1 nor 2 is correct

Ans. B

Exp: **Statement 1 is false:** The creation of ministries is an executive decision governed by the "Government of India (Allocation of Business) Rules." While the PM decides, it isn't a constitutional requirement to consult the Cabinet Secretary for the formation of the ministry itself.

- **Statement 2 is true:** Under Article 75, the President appoints ministers on the advice of the Prime Minister. The PM effectively chooses the team, and the President formally signs off.

6. What is true about the proceedings of the State Legislature? ***[71st BPSC]***

A. Advocate General has the right to vote.

B. Business in the legislature shall be transacted only in official language of the state or in Hindi or in English.

C. It can discuss the conduct of any judge of the High Court also.

D. Validity of proceedings shall be called in question on the ground of any alleged irregularity.

Ans. B

Exp: **Article 210** states that the State Legislature may conduct business in Hindi, English, or the State's official language or languages. However, the Speaker or Chairman may allow a member to address the House in their mother tongue if they are unable to communicate effectively in any of these.

- **Article 120 (1)** of the Indian Constitution pertains to the language used in the Parliament. Parliamentary proceedings must take place in either Hindi or English. However, the **Chairman of the Rajya Sabha** or the **Speaker of the Lok Sabha** may **allow** a **member,** unable to effectively communicate in Hindi or English, to **speak in their mother tongue.**

7. Which one of the following subjects fall within the jurisdiction of both High Court and the Supreme Court? ***[71st BPSC]***

A. Dispute among States

B. Dispute between Centre and State

C. Protection of fundamental rights.

D. Protection from the violation of the Constitution

Ans. C

Exp: The **protection of fundamental rights falls within the jurisdiction of both the High Court and the Supreme Court.** To effectively protect these rights, both courts possess the power to review and invalidate any government action or law that violates them. The concept of "Judicial Review" is not explicitly mentioned in the Constitution; this shared power is derived from several articles:

- **Article 13:** Declares that all laws that are inconsistent with or in derogation of the **Fundamental Rights** shall be void.
  - **Supreme Court:** Under **Article 32**, the Supreme Court can be approached for the enforcement of Fundamental Rights. It can also review the constitutionality of any law under Articles 131–136.
  - **High Courts:** Under **Article 226**, High Courts have the power to issue writs for the enforcement of Fundamental Rights and for "any other purpose." Their power of judicial review is actually broader in

scope than that of the Supreme Court because of the phrase "any other purpose."

8. Who is empowered to review financial position of the Panchayats? ***[71st BPSC]***

A. Chairman of Block Committee
B. Chief Minister
C. Chairman of District Council
D. Finance Commission constituted by the Governor

Ans. D

Exp: Under **Article 243-I**, the Governor of a state must constitute a **State Finance Commission** every five years. Its specific job is to review the financial health of Panchayats and recommend how state tax revenues should be shared with local bodies.

- **Financial powers:** A Finance Commission in each State is established to determine the principles based on which adequate financial resources would be ensured for panchayats and municipalities (Article 243I). This makes the decentralization of Financial powers possible.

9. Who among the following addresses the first session after each General Election to the House of People? ***[BPSC ASO, Pre]***

A. President
B. Speaker of the Lok Sabha
C. Prime Minister
D. Parliamentary Affairs Minister

Ans. A

Exp: Under **Article 87** of the Indian Constitution, the President is required to deliver a special address at the commencement of:

- The first session after each **General Election** to the Lok Sabha (House of the People).
- The first session of **each year** (usually the Budget Session).
- **Nature of the Speech:** The address is drafted by the Government (the Executive) which outlines the government's policies, programs, and achievements from the previous year, as well as the plan for the upcoming year.
- **Joint Sitting:** This address takes place in a **joint sitting** of both the Lok Sabha and the Rajya Sabha.
- **Motion of Thanks:** After the address, both Houses discuss the speech through a "Motion of Thanks." If this motion is not passed in the Lok Sabha, it is seen as a defeat for the government and can lead to its resignation.

10. If any question as to the disqualification of member of state legislature arises, the question shall be referred for the decision of: ***[BPSC ASO, Pre]***

A. Governor B. Election Commission
C. Supreme Court D. President

Ans. A

Exp: **Article 192** of the Indian Constitution states that the Governor has the authority to decide whether to disqualify a member of the State Legislative Assembly for reasons other than defection, which are addressed in the Tenth Schedule.

11. Which one of the following pairs is not matched? ***[BPSC ASO, Pre]***

A. Powers of Parliament to legislate on a State subject – Article 248
B. Freedom of trade and commerce – Article 301
C. Administration of Scheduled and Tribal areas – Article 244
D. Consolidated Fund of India – Article 266

Ans. A

Exp: **Article 248 (Residuary Powers):** It gives Parliament the exclusive power to make any law with respect to any matter not enumerated in the Concurrent List or State List.

- The power of Parliament to legislate on a State subject in the national interest is covered under **Article 249.**
- **Freedom of trade and commerce** – Article 301 of Part XIII guarantees that trade, commerce, and intercourse throughout the territory of India shall be free.
- **Administration of Scheduled and Tribal areas** – Article 244 in Part X of the Constitution deals with the administration of Scheduled Areas and Tribal Areas.
- **Consolidated Fund of India** – Article 266 of the Constitution provides for the establishment of the Consolidated Funds of India and of the States.

12. Consider the following statement:

When a bill has been passed by the legislative assembly or by both houses of legislature of the state (In case of legislative council) then the Governor may ***[BPSC ASO, Pre]***

1. Assent to the bill.
2. Return for reconsideration.
3. Withhold his assent.
4. Return the money bill.

A. Only 4 is correct B. 1 and 3 are correct
C. 1, 2, and 3 are correct D. Only 2 is correct

Ans. C

Exp: Under **Article 200** of the Indian Constitution, when a bill passed by the State Legislature is presented to the Governor, they have four primary options.

- Give assent to the bill
- Withhold assent on the bill
- Return a non-Money Bill for reconsideration. If the House passes the bill again (with or without amendments) and sends it back, the Governor **must** give assent.
- Reserve it for the President.

13. Who has the power to transfer a Judge from one High Court to another? ***[BPSC ASO, Pre]***

A. Chief Justice of High Court
B. Parliament
C. Prime Minister's Office
D. President

Ans. D

Exp: According to **Article 222**, the President may, after consultation with the Chief Justice of India, transfer a judge from one High Court to any other High Court. This "consultation" currently involves the Collegium of the Supreme Court.

14. Appellate Jurisdiction of Supreme Court in regard to criminal matters is: ***[BPSC ASO, Pre]***

A. Not defined in the Constitution
B. Limited
C. Related to political matters only
D. Unlimited

Ans. B

Exp: The Supreme Court's criminal appellate jurisdiction (Article 134) is "limited" because it only hears appeals in specific cases:

- if a High Court has reversed an acquittal to a death sentence, withdrawn a case to itself and sentenced an accused to death, or certified that the case is fit for appeal to the Supreme Court.

15. Which of the following Committees is *not* related to the Panchayati Raj System? ***[BPSC ASO, Pre]***

A. Ashok Mehta Committee
B. G. V. K. Rao Committee
C. Swaran Singh Committee
D. L. M. Singhvi Committee

Ans. C

Exp: The Swaran Singh Committee (1976) was formed to make recommendations about Fundamental Duties, which were later added by the 42nd Amendment. The other three committees were specifically formed to study and provide recommendations for the development of local self-government in India.

16. Consider the following statements regarding Municipalities: ***[BPSC ASO, Pre]***

1. An election to constitute a municipality shall be completed before the expiration of a period of six months from the date of its dissolution.
2. Election to constitute a municipality should be before three months of its expiration.
3. A municipality constituted upon dissolution of a municipality before expiration of its term shall continue only for a remainder period.
4. If a municipality is dissolved before expiration of its term then the new municipality will be constituted for five years.

A. 2 and 4 are correct B. Only 2 is correct
C. 1 and 3 are correct D. Only 4 is correct

Ans. C

Exp: The provisions regarding the duration and elections of Municipalities are governed by **Article 243U** of the Constitution of India.

- According to **Article 243U(3)(b),** if a municipality is dissolved, the election to reconstitute it must be completed within a period of **six months** from the date of dissolution.
- The Constitution **(Article 243U(3)(a))** states that the election to constitute a municipality must be completed **before the expiry of its five-year duration**. It does not specify a "three-month" window.
- If a municipality is dissolved mid-term, the newly elected body does not get a fresh 5-year term. It serves only for the **remainder of the period** for which the dissolved municipality would have continued.

17. Which article of the Indian Constitution enables the government to enact separate laws for the governance and administration of tribal areas? ***[BPSC ASO, Mains]***

A. Article 244 B. Article 243
C. Article 245 D. Article 249

Ans. A

Exp: **Article 244** of the Indian Constitution is the primary provision that enables the government to enact separate systems of governance and administration for "Scheduled Areas" and "Tribal Areas." This article facilitates the application of the **Fifth Schedule** for the administration of scheduled areas and tribes in states other than Assam, Meghalaya, Tripura, and Mizoram, and the **Sixth Schedule** specifically for the tribal areas within those four northeastern states.

- The intent behind this article is to provide a degree of autonomy and specialized legal frameworks to protect the social, cultural, and economic interests of indigenous communities, often bypassing or modifying general state and central laws to suit their unique customs.

BCW Bits:

- **Article 243:** This article (and its subsequent parts 243A to 243O) pertains to the **Panchayati Raj System**, defining the constitution, powers, and authority of local self-government in rural areas.
- **Article 245:** This governs the **extent of laws** made by Parliament and by the Legislatures of States, establishing the territorial jurisdiction for federal and state legislation.
- **Article 249:** This grants **Parliament the power to legislate** with respect to a matter in the State List if it is deemed necessary or expedient in the national interest, provided the Rajya Sabha passes a resolution to that effect by a two-thirds majority.

18. Which among the following are the examples of asymmetry in Indian federalism? ***[BPSC ASO, Mains]***

(1) Representation of different states in Rajya Sabha.
(2) Provisions with regard to administration of tribal areas under Fifth and Sixth Schedule.
(3) Composition of NITI Aayog.
(4) Provisions for Union Territories as federal units.

Select the correct answer using the codes given below:

A. Only (1) and (2)
B. All of the above
C. Only (2) and (3)
D. Only (1), (2) and (4)

Ans. D

Exp: Federalism is a system in which power is divided between a central authority and constituent units, which, in India, is termed "quasi-federal" by scholar **K.C. Wheare** due to its strong unitary bias. Unlike the **USA**, which is a "coming together" federation with **symmetric federalism** (equal powers for all states), India features **asymmetric federalism** where certain regions have special provisions under the Fifth and Sixth Schedules.

Statements (1), (2), and (4) are established examples of **asymmetry in Indian federalism**, whereas statement (3) represents a mechanism of cooperative, uniform federalism. Asymmetric federalism occurs when different constituent units of a federation are granted unequal powers or status. This is evident in:

- **Representation in Rajya Sabha (1):** Unlike federal systems in which all states have equal seats (as in the US Senate), India allocates seats based on population, giving states varying levels of influence in the upper house.
- **Special Provisions for Tribal Areas (2):** The **Fifth and Sixth Schedules** create unique administrative structures for specific regions to protect tribal interests, deviating from the standard state governance model.
- **Union Territories as Federal Units (4):** Union Territories do not have the same constitutional status or autonomy as States; some have legislatures while others are governed directly by the Centre, creating a multi-tiered federal structure.

Statement (3) is excluded because **NITI Aayog** is designed to foster "Cooperative Federalism" by providing a common platform for all States and Union Territories to interact with the Centre on equal terms on policy and development matters.

19. Which among the following is not a part of Fundamental Duties enlisted under Part IVA of Constitution? ***[BPSC ASO, Mains]***

A. To develop the scientific temper
B. To keep our surroundings clean
C. To safeguard public property
D. To preserve rich heritage of our composite culture

Ans. B

Exp: Fundamental Duties, absent in the original Constitution, were incorporated into **Part IVA** under **Article 51A** via the **42nd Amendment Act (1976)** based on the **Swaran Singh Committee** recommendations. Inspired by the **USSR** constitution, they originally numbered ten but increased to eleven after the **86th Amendment (2002)**, serving as non-justiciable moral obligations for citizens to uphold national unity.

- "To keep our surroundings clean" is not explicitly listed as a Fundamental Duty under **Article 51A (Part IVA)** of the Constitution. While cleanliness is a vital civic responsibility and part of initiatives like the Swachh Bharat Mission, it is not one of the eleven specific duties codified in the constitutional text.
- The other options are fundamental duties as follows:

A. To develop the scientific temper: Article 51A(h) mandates citizens to develop a scientific temper, humanism, and the spirit of inquiry and reform.

C. To safeguard public property: Article 51A(i) requires citizens to protect public property and to abjure violence.

D. To preserve the rich heritage of our composite culture: Article 51A(f) emphasizes the duty to value and preserve the rich heritage of the nation's composite culture.

20. Consider the following statements about Habeas Corpus: ***[BPSC ASO, Mains]***

(1) The writ of habeas corpus is mentioned in both Articles 32 and 226.
(2) The primary objective of the writ is to secure the release of a person found to be detained illegally.

Which of the above statements is/are correct?

A. Both (1) and (2)
B. Only (1)
C. Only (2)
D. None of the above

Ans. A

Exp: Writs are formal written orders issued by the **Supreme Court** under **Article 32** and **High Courts** under **Article 226** to enforce Fundamental Rights. **Borrowed from British "Prerogative Writs, "** these formal orders—**Habeas Corpus, Mandamus, Prohibition, Certiorari, and Quo-Warranto**—empower the judiciary to restrain administrative overreach and safeguard individual liberty.

- **Habeas Corpus**, specifically, acts as the primary constitutional shield against arbitrary detention, mandating the legal justification for any person's loss of freedom.
- Both statements regarding the writ of **Habeas Corpus** are fundamental truths of the Indian legal system.

Statement 1: The power to issue this writ is enshrined in both **Article 32** and **Article 226**, creating a dual jurisdiction that ensures citizens have accessible legal remedies at both the national and state levels.

Statement 2: Literally translating "*to have the body,* " the writ commands a detaining authority to produce the individual before the court. If the court finds the confinement lacks legal justification, it orders the person's immediate release.

21. Who among the following was not a member of the Drafting Committee of the Constitution? ***[BPSC ASO, Mains]***

A. N. Madhava Rau B. S. M. Saadulla
C. Damodar Swarup Seth D. N. G. Ayyangar

Ans. C

Exp: The idea of a Constituent Assembly was first proposed by **M.N. Roy** in 1934 and officially accepted by the British government through the **August Offer** of 1940. It was eventually formed in November 1946 under the **Cabinet Mission Plan**, with members elected indirectly by provincial assemblies and nominated by princely states.

Damodar Swarup Seth was not a member of the Drafting Committee. While he was a prominent member of the Constituent Assembly, he was known for his socialist views and for proposing various amendments during the debates, but he did not serve on the seven-member drafting team.

The Drafting Committee, appointed on August 29, 1947, was chaired by **Dr. B.R. Ambedkar**. The other members included:

- **N. Madhava Rau** (who replaced B.L. Mitter after his resignation due to ill health).
- **S.M. Saadulla**.
- **N.G. Ayyangar**.
- **Alladi Krishnaswami Ayyar**.
- **K.M. Munshi**.
- **T.T. Krishnamachari** (who replaced D.P. Khaitan after his death in 1948).

22. Which among the following is true about the Central Vigilance Commission? ***[BPSC ASO, Mains]***

(1) It was formed on the recommendation of Santhanam Committee.
(2) It got a statutory status in 1964.
(3) It constitutes of a Chief Vigilance Commissioner and three other commissioners.

Select the correct answer using the codes given below:

A. Only (1) B. (1) and (2)
C. (1) and (3) D. None of the above

Ans. A

Exp: The **Central Vigilance Commission (CVC)** was established in 1964, based on the **Santhanam Committee's** recommendations, to combat corruption and eventually received statutory status under the **CVC Act, 2003.** It functions as an independent apex body consisting of a **Central Vigilance Commissioner** and not more than **two** Vigilance Commissioners, appointed by the President of India.

- **Statement (1) is correct:** The CVC was indeed created in February 1964 following the recommendations of the Committee on Prevention of Corruption, headed by **K. Santhanam**.
- **Statement (2) is incorrect:** While it was *formed* in 1964, it was initially an executive body. It only received **statutory status** much later, in **September 2003**, following a Supreme Court directive in the *Vineet Narain* case.
- **Statement (3) is incorrect:** The Commission is a multi-member body consisting of a Central Vigilance Commissioner (Chairperson) and **not more than two** Vigilance Commissioners. The statement claiming "three other commissioners" is inaccurate.

23. Which among the following is true about Money Bill? ***[BPSC ASO, Mains]***

(1) Any bill that provides for imposition or abolition of any 'tax' by any local authority, automatically qualifies it to be a Money Bill.
(2) If the Council of States makes some recommendations on the Money Bill, the House of People is bound to accept those recommendations.
(3) If the Money Bill is not returned to the House of the People within fourteen days by the Council of States, it is deemed to be passed.

Select the correct answer using the codes given below:

A. Only (1) B. (1) and (3)
C. (2) and (3) D. Only (3)

Ans. D

Exp: A **Money Bill**, as defined under **Article 110**, deals exclusively with matters such as taxation, public expenditure, and government borrowings. It can be introduced in the **Lok Sabha** only on the President's recommendation, **thereby** granting the lower house ultimate authority over the nation's financial decisions and limiting the **Rajya Sabha's** role to a purely advisory capacity.

Only statement (3) aligns with the special procedure for Money Bills outlined in **Articles 109 and 110**:

- **The 14-Day Rule (Statement 3):** This is a key constitutional mandate. Once the Lok Sabha passes a Money Bill, the Rajya Sabha has only **14 days** to return it with or without recommendations. If it fails to do so, the bill is automatically "deemed passed" by both Houses in the form originally approved by the Lok Sabha.
- **Exclusion of Local Taxes (Statement 1):** According to **Article 110(2)**, a bill is *not* considered a Money Bill if it merely provides for the imposition, abolition, or regulation of any tax by a **local authority** for local purposes. This distinguishes national fiscal policy from local administrative levies.
- **Non-Binding Recommendations (Statement 2):** The Lok Sabha holds supreme financial power. While the

Rajya Sabha may suggest amendments, the Lok Sabha is **not bound** to accept them. It can choose to pass the bill while rejecting all recommendations, and the bill is still considered passed by both Houses.

24. In which year did UNESCO define the scope of Political Science? ***[BPSC ASO, Mains]***

A. 1946 B. 1948 C. 1950 D. 1952

Ans. B

Exp: Political Science underwent a significant global transformation post-World War II as international bodies sought to standardize its academic boundaries. This effort culminated in a specialized convention aimed at distinguishing political study from general history or law, thereby establishing it as a rigorous, independent social science.

- In **September 1948**, UNESCO organized a historic **conference of experts** in **Paris** to deliberate on the **subject matter and methodology of Political Science**. This convention was fundamental in providing the discipline with a universal academic identity through the following classifications:
- **Categorization of Scope:** The experts divided the field into **four distinct pillars**:
  1. **Political Theory:** Including the history of political ideas.
  2. **Political Institutions:** Covering constitutions, national, regional, and local governments.
  3. **Parties, Groups, and Public Opinion:** Focusing on political dynamics and citizen participation.
  4. **International Relations:** Addressing international politics and organizations.
- By defining these boundaries, UNESCO helped transition the study of politics from a branch of philosophy or legal studies into a **systematic social science** recognized by universities worldwide.
- This framework ensured that the study of governance and power remained consistent across different nations, facilitating better academic exchange and research.

BCW Bits:

- **1946:** The UNESCO was officially established after the signing of its constitution.
- **1950:** The **International Political Science Association (IPSA)**—which resulted from the 1948 conference—began its full-scale operations.
- **1952:** A period marked by the further expansion of UNESCO's social science programs into developing nations.

25. "Political Science is concerned with the state and instruments of Government". Who said this? ***[BPSC ASO, Mains]***

A. Dimock B. Robert Dahl
C. David Easton D. Dante Germino

Ans. A

Exp: Political Science definitions range from studying formal government structures to analyzing informal power dynamics. This evolution reflects the transition from "Traditionalism, " which focuses on the state's legal architecture, to "Behaviouralist, " which examines the human and systemic influences behind political decisions.

Marshall Edward Dimock defined Political Science through an institutional lens, asserting it is fundamentally "concerned with the state and the instruments of Government." His definition is characterized by:

- **Institutional Framework:** He viewed the discipline as the study of the **State** as a sovereign entity and its **administrative machinery**.
- **Operational Focus:** By "instruments, " Dimock emphasized the **Executive, Legislative, and Judicial branches**, focusing on how these departments execute public policy.
- **Legal-Formal Approach:** This perspective prioritizes **statutory laws and organizational structures** over individual social behavior or abstract theory.

BCW Bits:

- **Robert Dahl:** Defined politics as the study of **Power and Influence**, moving focus from formal "instruments" to how individuals actually exercise authority.
- **David Easton:** Proposed the **Systems Theory**, defining politics as the **"authoritative allocation of values"** for an entire society.
- **Dante Germino:** Advocated for a **Philosophical Approach**, viewing Political Science as a moral inquiry into the "Open Society" and the human condition.

26. Consider the following statements about the Preamble of the Indian Constitution: ***[BPSC ASO, Mains]***

(1) The Preamble is a part of the Constitution as affirmed by the Supreme Court in the Kesavananda Bharati Case (1973).
(2) The Preamble can be amended under Article 368, but amendments cannot alter the basic structure of the Constitution.
(3) The Preamble is justiciable and its provisions are enforceable in Courts of Law.

Which of the statements given above is/are correct?

A. Only (1) and (3) B. (1), (2) and (3)
C. Only (1) D. Only (1) and (2)

Ans. D

Exp: The **Preamble**, based on the "Objectives Resolution" drafted by **Jawaharlal Nehru**, serves as the introduction to the Constitution, outlining its ideology and objectives. It was adopted by the Constituent Assembly on **January**

22, 1947, and has been amended only once by the **42nd Amendment Act (1976)**, which added the words "Socialist, " "Secular, " and "Integrity."

Statements (1) and (2) accurately reflect the legal status of the Preamble, while statement (3) is incorrect.

- **Status as Part of the Constitution (Statement 1):** While the Supreme Court initially ruled the Preamble was not part of the Constitution in the *Berubari Union* case (1960), it famously reversed this position in the **Kesavananda Bharati Case (1973)**, declaring it an integral part of the constitutional framework.
- **Power of Amendment (Statement 2):** Under **Article 368**, the Parliament can amend the Preamble, provided the "Basic Structure" remains intact. This was demonstrated in 1976 when the Preamble was expanded to include new socialist and secular ideals.
- **Non-Justiciability (Statement 3):** The Preamble is **non-justiciable**, meaning its provisions cannot be enforced in a court of law to override specific constitutional articles. It acts as a "key to the minds of the makers" during legal interpretation but does not grant independent substantive powers to the government or citizens.

27. Which among the following is true about the Public Accounts Committee? ***[BPSC ASO, Mains]***

(1) It was first set up in 1921.
(2) It cannot have more than 22 members.
(3) The functions of the Committee are enshrined in Rule 308(1) of the Rules of Procedure and Conduct of Business in the Lok Sabha.

Select the correct answer using the codes given below:

A. Only (2) and (3) B. Only (1) and (2)
C. All of the above D. None of the above

Ans. C

Exp: The **Public Accounts Committee (PAC)** is a powerful Indian parliamentary standing committee that audits government revenue and expenditure. Established in 1921 under the Government of India Act of 1919, its primary role is to examine audit reports submitted by the Comptroller and Auditor General (CAG) to ensure executive financial accountability.

Key Facts About the Public Accounts Committee (PAC):

- **Composition and Strength:** The committee comprises up to **22 members**—15 elected from the Lok Sabha (Lower House) and 7 from the Rajya Sabha (Upper House).
- **Election and Term:** Members are elected annually from amongst themselves by the parliamentarians using the principle of proportional representation by means of a single transferable vote. The term of office of the members is **one year**.
- **Restrictions:** A **Minister cannot be elected** to the committee. If an existing member is appointed as a minister, they cease to be a member from the date of such appointment.
- **The Chairman:** The Chairman of the Committee is appointed by the Speaker of the Lok Sabha from amongst its members. By an established convention since 1967, the Chairman is selected from the **Opposition**.
- **Mandate and Scope:** The functions and powers of the PAC are explicitly detailed and enshrined under **Rule 308(1)** of the Rules of Procedure and Conduct of Business in Lok Sabha. Its main function is to examine the appropriation accounts showing the appropriation of sums granted by the House for the expenditure of the Government of India, the annual finance accounts, and other such accounts laid before the House.

28. A citizen of Canada residing in India cannot claim the right to: ***[BPSC DSO, Pre]***

A. Equality before the law
B. Freedom of speech and expression
C. Freedom of religion
D. Protection against arrest and detention in certain cases

Ans. B

Exp: Fundamental Rights only for Citizens of India are:

- **Article 15 –** Prohibition of discrimination on grounds of religion, race, caste, sex, or place of birth.
- **Article 16 –** Equality of opportunity in matters of public employment.
- **Article 19 –** Protection of six rights related to freedom – (a) of speech and expression; (b) to assemble peaceably and without arms; (c) to form associations or unions; (d) to move freely throughout the territory of India; (e) to reside and settle in any part of the territory of India; and (f) to practice any profession, or to carry on any occupation, trade or business.
- **Article 29 –** Protection of language, script, and culture of minorities.
- **Article 30 –** Right of minorities to establish and administer educational institutions.

Note: Articles 15, 16, 19, 29, and 30 are only available to the citizens of India and not to the aliens.

29. Which of the following Articles was not inserted by the 42nd Constitutional Amendment Act of 1976? ***[BPSC DSO, Pre]***

A. Article 38 B. Article 39
C. Article 39A D. Article 43A

Ans. A

Exp: The Directive Principles along with the Fundamental Rights contain the philosophy of the Constitution and is the soul of the Constitution. Granville Austin has described the Directive Principles and Fundamental Rights as the **'Conscience of the Constitution'.**

- Dr. B R Ambedkar described these principles as **'novel features'** of the Indian Constitution. He also called the Directive Principles of the State Policy as **'charter of social and economic democracy of India'.**

- The idea of the Directive Principles is borrowed from the **Irish Constitution** of 1937, which had copied it from the **Spanish Constitution.**
- **42nd Amendment Act of 1976**
  - To secure opportunities for the healthy development of children (Article 39).
  - To promote equal justice and to provide free legal aid to the poor (Article 39 A).
  - To take steps to secure the participation of workers in the management of industries (Article 43 A).
  - To protect and improve the environment and to safeguard forests and wildlife (Article 48 A)
- **44th Amendment Act of 1978**
  - State to minimize inequalities in income, status, facilities, and opportunities (Article 38)
  - **86th Amendment Act of 2002**
  - Changed the subject matter of Article 45
  - **97th Amendment Act of 2011**
  - State to promote voluntary formation, autonomous functioning, democratic control, and professional management of cooperative societies (Article 43B).

30. From which source did the Indian Constitution borrow the provision related to the election of Rajya Sabha members? ***[BPSC DSO, Pre]***

A. British Constitution
B. Canadian Constitution
C. South African Constitution
D. Government of India Act, 1935

Ans. C

Exp: While the nomination of members to the Rajya Sabha is borrowed from Ireland, the election procedure of its members is borrowed from the South African Constitution. We also borrowed the procedure for a constitutional amendment from them.

31. Who among the following was elected unopposed to the position of Vice President of India? ***[BPSC DSO, Pre]***

A. Varahagiri Venkata Giri
B. Kocheril Raman Narayanan
C. Mohammad Hidayatullah
D. Mohammad Hamid Ansari

Ans. C

Exp: **Justice Mohammad Hidayatullah** (the 6th Vice President of India) was elected **unopposed in 1979.**
He was a consensus candidate supported by all political parties, which is a rare feat in Indian politics. He also holds the unique distinction of having served as the Chief Justice of India, the Vice President of India, and the Acting President of India.

- **V.V. Giri:** He served as the 3rd Vice President, who won as an **independent candidate** against the official Congress nominee in 1969.
- **K.R. Narayanan:** He served as the 9th Vice President and later became the first person to become the president of India from deemed to be Scheduled Caste.
- **Mohammad Hamid Ansari:** He served **two consecutive terms** as Vice President, the only person to do so besides Dr. S. Radhakrishnan.

32. The removal of the Speaker of the Lok Sabha requires: ***[BPSC DSO, Pre]***

A. Simple majority B. Effective majority
C. Absolute majority D. Special majority

Ans. B

Exp: Under **Article 94**, the Speaker is removed by a resolution passed by a majority of all the then members of the Lok Sabha. This is known as an "**Effective Majority**" (i.e, Total strength – vacancies).

33. Which of the following statements are correct? ***[BPSC DSO, Pre]***

1. Calling Attention Motion is an Indian innovation.
2. Calling Attention Motion is not mentioned in the Rules of Procedure.

A. Only (1) B. Only (2)
C. Both (1) and (2) D. None of the above

Ans. A

Exp: Calling attention motion is an Indian innovation in parliamentary practice since 1954. It is mentioned in the Rules of Procedure, unlike the zero hour, which is also an Indian innovation but not mentioned in the Rules of Procedure.

34. Which of the following statements are correct regarding the Estimates Committee of the Parliament? ***[BPSC DSO, Pre]***

1. The Estimates Committee was constituted on the recommendation of the John Mathai Committee.
2. The Rajya Sabha has no representation in this committee.

A. Only (1) B. Only (2)
C. Both (1) and (2) D. None of the above

Ans. C

Exp: The first Estimates Committee in the post-independence era was set up in 1950. It was created on the **recommendation of John Mathai**, who was the Finance Minister at that time. The origin of the committee goes back to the Standing Financial Committee of 1921.

- The Estimates Committee is a committee of the Lok Sabha only. It has **30 members**, and all of them are elected from the Lok Sabha. The **Rajya Sabha has no members** in this committee.
- They are elected every year by the Lok Sabha using the system of proportional representation by means of a single transferable vote. The members are elected for a term of

one year. A Minister cannot be elected as a member of this committee.

- The **Speaker appoints the Chairman** of the committee from among its members. The Chairman always belongs to the ruling party. The committee examines the budget estimates and suggests ways to bring economies in public expenditure. Therefore, it is also called a **'Continuous Economy Committee'.**

35. The Constitution of India confers the power of judicial review on: ***[BPSC DSO, Pre]***

A. Supreme Court only
B. Both the Supreme Court and the High Courts
C. All the courts
D. None of the above

Ans. **B**

Exp: "Judicial Review" is not explicitly mentioned in the Constitution of India. The power is derived from several articles:

- **Article 13:** Declares that all laws that are inconsistent with or in derogation of the **Fundamental Rights** shall be void.
- **Supreme Court:** Under **Article 32**, the Supreme Court can be approached for the enforcement of Fundamental Rights. It can also review the constitutionality of any law under Articles 131–136.
- **High Courts:** Under **Article 226**, High Courts have the power to issue writs for the enforcement of Fundamental Rights and for "any other purpose." Their power of judicial review is actually broader in scope than that of the Supreme Court because of the phrase "any other purpose."

36. The power to establish a common High Court for two or more states is mentioned under which Article of the Constitution of India? ***[BPSC DSO, Pre]***

A. Article 229 B. Article 230
C. Article 231 D. Article 232

Ans. **C**

Exp: **Article 231** gives Parliament the power to establish a common High Court for two or more States or for two or more States and a Union Territory, e.g., the Punjab and Haryana High Court.

- **Article 230:** Extension of jurisdiction of High Courts to **Union Territories**.
- **Article 229:** Deals with the officers and servants and the expenses of High Courts (administrative powers of the Chief Justice).
- **Article 214:** Mandates that there shall be a High Court for **each State**. (Note: Article 231 acts as an exception to this general rule).

37. Which of the following is the correct chronological sequence of committees related to local government? ***[BPSC DSO, Pre]***

1. Balwant Mehta Committee
2. GVK Rao Committee
3. Ashok Mehta Committee
4. Thungon Committee

A. (2), (4), (1), (3) B. (3), (1), (4), (2)
C. (4), (2), (3), (1) D. (1), (3), (2), (4)

Ans. **D**

Exp: Panchayati Raj was inserted through the 73rd Constitutional Amendment Act, 1992.

- This Act added Part IX to the Constitution and added the 11th Schedule which contains 29 functional items of the panchayat.

Panchayati Raj

1. Gram Panchayat at village level
2. Panchayat Samiti at the block level (Executive body)
3. Zila parishad at the District level (Advisory, Supervisory, and Coordinating body).

Panchayati Raj Committees	
Balwant Rai Mehta Committee	1957-Recommended a Three-tier Panchayati Raj System
Ashok Mehta Committee	1978-Recommended a Two-Tier system.
Hanumantha Rao Committee	1984-On District Planning
G.V.K. Rao Committee	1985-Recommended regular elections to the PRIs.
L. M. Singhvi Committee	1986-Recommended constitutional recognition to the Panchayati Raj Institution.
Thungon Committee	1988-Recommended financial autonomy of the Panchayati Raj Institutions.

38. Which among the following is true about the Panchayats Extension to the Scheduled Areas (PESA) Act? ***[BPSC AEDO]***

1. It was enacted in 1996.
2. It accords statutory status to the Gram Sabhas in Scheduled V areas.
3. Section 4(i) of the Act mandates that the Gram Sabha or Panchayat should consent before any land acquisition for development projects.

Select the correct answer using the code given below:

A. Only 1 B. 1 and 2
C. All of the above D. 2 and 3

Ans. C

Exp:

- The Provisions of the **Panchayats (Extension to the Scheduled Areas)** Act (PESA) was enacted by the Parliament of India on **December 24, 1996**, to extend the Panchayati Raj framework (Part IX of the Constitution) to tribal areas
- PESA directly accords statutory status to the Gram Sabhas in **Scheduled V areas**, making them the nucleus of decentralized democracy and community-driven governance.
- Under Section 4(i) of the PESA Act, it is mandatory that the Gram Sabha or the Panchayats at the appropriate level must be consulted (often acting as a mandatory consent mechanism) before acquiring land in Scheduled Areas for development projects, as well as before resettling or rehabilitating affected persons.

BCW Bits:

- PESA is often called a **"bottom-up democracy law"** because power flows from the village level upward.
- It applies only to Fifth Schedule Areas, not to Sixth Schedule Areas.
- It covers states like **Jharkhand, Chhattisgarh, Odisha, Madhya Pradesh, Maharashtra, Gujarat, Rajasthan, Andhra Pradesh, Telangana, Himachal Pradesh.**
- The Gram Sabha under PESA can control Minor forest produce, Local resources, and Social customs and traditions.
- PESA is all about: **"Local tribal people decide for themselves".**

39. Match the List – I with List – II. ***[BPSC AEDO]***

List – I (Article of Constitution)		List – II (Fundamental Duties)	
a.	Article 51A (c)	1.	Educate one's children/ward between 6 and 14 years
b.	Article 51A (f)	2.	To uphold the sovereignty of India
c.	Article 51A (k)	3.	To value and preserve the rich heritage of our composite culture
d.	Article 51A (i)	4.	To safeguard public property

Select the correct answer using the code given below:

	a	b	c	d
A.	2	4	3	1
B.	4	3	2	1
C.	1	4	2	3
D.	2	3	1	4

Ans. D

Exp: Fundamental Duties are constitutional obligations of citizens mentioned under Article 51A of the Constitution. They were added by the 42nd Constitutional Amendment Act, 1976.

- **Article 51A(c): -** Uphold sovereignty, unity and integrity of India;
- **Article 51A (f):-** Value and preserve the rich heritage of our composite culture;
- **Article 51A (k):-** Provide education to children (6–14 years), Added later by **86th Amendment (2002);**
- **Article 51A (i):-** Safeguard public property and avoid violence

BCW Bits:

- The Constitution of India, as adopted in 1950, did not contain any Fundamental Duties.
- Then Initial 10 Duties were officially incorporated into the Constitution (Part IV-A, Article 51A) in 1976 via the 42nd Constitutional Amendment Act, based on the recommendations of the Swaran Singh Committee.
- The 11th Duty was added in 2002 by the 86th Constitutional Amendment Act.
- Fundamental Duties are non-justiciable (not enforceable by courts directly)
- The inclusion of Fundamental Duties is inspired by the Soviet Constitution.

40. Consider the following sentences about writ jurisdiction in India. ***[BPSC AEDO]***

1. The writ jurisdiction is a form of original jurisdiction and not appellate jurisdiction.
2. It can be issued either by High Courts or the Supreme Court.

Which of the above statements is/are correct?

A. Both 1 and 2 B. Only 1
C. None of the above D. Only 2

Ans. A

Exp:

- The writ jurisdiction is fundamentally a form of original jurisdiction. This means an aggrieved person can approach the higher judiciary directly at the first instance to seek a remedy for the violation of their rights, rather than having to arrive via a lower court appeal.
- Constitutional writs can be issued by both the Supreme Court of India and the various High Courts. The Supreme Court exercises this power under Article 32 for enforcing Fundamental Rights only, while the High Courts exercise it under Article 226 for enforcing both Fundamental Rights and other legal rights. Thus, High Courts have wider writ jurisdiction than the Supreme Court.

BCW Bits:

- There are 5 types of writs**: Habeas Corpus, Mandamus, Certiorari, Prohibition, Quo Warranto.**

- Article 32 is called the "heart and soul" of the Constitution (Dr B.R. Ambedkar)
- **Writ jurisdiction** is a key part of judicial review.

41. What is the minimum age required to become the Vice President of India? ***[BPSC AEDO]***

A. 25 years B. 30 years
C. 40 years D. 35 years

Ans. D

Exp: According to **Article 66** of the Indian Constitution, a candidate must meet the following criteria to be eligible for the office of Vice President:

- Be a citizen of India.
- Have completed the age of **35 years**.
- Be qualified for election as a member of the Rajya Sabha.
- Not hold any office of profit under the Government of India, any State Government, or any local or other authority.

Election of Vice President:

- The Vice President is elected by **both Houses of Parliament** (Lok Sabha + Rajya Sabha)**.** The State legislatures do not participate.
- Election is done by proportional representation with single transferable vote; The oath is administered by the President**.**

Duties of Vice President: The Vice President is the second-highest constitutional office in India.

- **Chairman of Rajya Sabha-** conducts proceedings, maintains order,
- **Acts as President of India** - when President is absent or President resigns or dies (Ex- **V. V. Giri** → after the death of President Zakir Hussain; **B. D. Jatti** → after the resignation of President Fakhruddin Ali Ahmed)

Important Constitutional Provisions:

- Article 63 - There shall be a Vice President;
- Article 64 –Vice President is ex-officio Chairman of Rajya Sabha;
- Article 65-Acts as President when needed;
- Article 66 -Election and qualifications;
- Article 67 -Term (5 years);
- Article 68 - Vacancy and election timing.

BCW Bits:

- Unlike the impeachment process for the President of India, the removal of the Vice President of India requires a parliamentary resolution passed by an **effective majority** in the Rajya Sabha and agreed to by a **simple majority** in the Lok Sabha, following a mandatory 14-day advance notice.
- Under Article 100(1), the Chairman **cannot vote in the first instance** but casts a deciding vote to break a deadlock. So, Vice President casts the vote only in case of a tie.
- The office is modeled on the US Vice **President**, but roles differ.

42. Which among the following are valid conditions that can give recognition to any political party as a National Party in India? ***[BPSC AEDO]***

1. The party should have polled at least six per cent votes in four or more states in Lok Sabha or State Assembly elections and secured at least four seats in Lok Sabha from one or more states.
2. The party has won two per cent of total seats in the Lok Sabha from at least three states.
3. In the last General Elections to the Lok Sabha, the party should have won at least one seat for every 25 seats allotted in any one state.
4. The party is recognised as a state party in four or more states.

Select the correct answer using the code given below:

A. Only 1 and 2 B. 1, 2 and 3
C. 1, 3 and 4 D. 1, 2 and 4

Ans. D

Exp: According to the Election Commission of India (ECI), a registered political party is recognized as a National Party if it meets any of the following criteria:

- Recognized as a State Party in at least four states.
- Secures at least 6% of valid votes across four or more states in a general election (Lok Sabha or Assembly) and wins at least 4 Lok Sabha seats.
- Wins at least 2% of total Lok Sabha seats (11 out of 543) with candidates elected from at least three different states.

  National Party status provides benefits like reserved symbols, free airtime on public broadcasters, and the ability to designate 40 star campaigners, whose travel costs are not included in candidate expense limits.
- However, winning at least **one seat for every 25 seats** in any state is a criteria for recognition as a State Party, not a National Party.

43. Which Article of the Constitution provides that no person accused of any offence can be compelled to be a witness against himself? ***[BPSC AEDO]***

A. Article 21 B. Article 20
C. Article 23 D. Article 22.

Ans. B

Exp: **Article 20** [Specifically Article 20(3)] of the Indian Constitution establishes the **Right against Self-Incrimination**. It explicitly states that *"No person accused of any offence shall be compelled to be a witness against himself."* This fundamental safeguard guarantees that an accused individual cannot be forced to give oral or documentary testimony that could establish their own guilt.

Article 20 – Protection in criminal cases

- 20(1): No ex post facto punishment
- 20(2): No double punishment (double jeopardy)
- 20(3): No self-incrimination (this is what the question asks) (Focus: fair treatment of accused persons)

BCW Bits:

- Article 20 is available to both citizens and non-citizens.
- Self-incrimination applies only to "accused persons", not witnesses.
- It protects against compelled testimony, not voluntary statements.
- It does not bar collection of physical evidence (like fingerprints, DNA)

44. Which Article of the Constitution provides that all proceedings in the Supreme Court and in every High Court shall be in English language? ***[BPSC AEDO]***

A. Article 345 B. Article 350
C. Article 348 D. None of the above.

Ans. C

Exp: **Article 348(1)(a)** of the Constitution of India explicitly mandates that all proceedings in the Supreme Court and in every High Court shall be in the English language until Parliament provides otherwise.

- **Article 348(1)(b)** also states that the authoritative texts of all Bills, Acts, Ordinances, and orders must be in English.
- **Article 348(2)** provides an exception, allowing the Governor of a State to authorize the use of Hindi or any other official state language in High Court proceedings, but with the prior consent of the President.

BCW Bits:

Article 345: Empowers the Legislature of a State to adopt any language(s) in use in that state or Hindi as its official language(s).

Article 350: Provides every citizen the legal right to submit a formal complaint or request to any government authority using any official language used in their state or across the country. This ensures that language barriers do not prevent anyone from seeking help or justice.

45. Which among the following are included in Directive Principles of State Policy? ***[BPSC AEDO]***

1. Equal pay for equal work for men and women
2. Right to work
3. Public assistance in case of unemployment

Select the correct answer using the code given below:

A. 1 and 2 B. 2 and 3
C. All of the above D. None of the above

Ans. C

Exp: The Directive Principles of State Policy (DPSPs), enshrined in **Part IV (Articles 36–51)** of the Indian Constitution, are guidelines for the government to create a just, equitable welfare state. Though **non-justiciable** (not enforceable by courts), they serve as the **moral compass** for framing laws and socio-economic policies. DPSPs are traditionally categorized into three core philosophies:

- **Socialist Principles:** Aim to secure social and economic justice, equitable wealth distribution, and fair working conditions (Articles 38, 39, 39A, 41, 42, 43, 43A).
- **Gandhian Principles:** Reflect the constructive program of Gandhi, focusing on village panchayats, cottage industries, and prohibition (Articles 40, 43, 46, 47, 48).
- **Liberal-Intellectual Principles:** Focus on modern and progressive ideologies, including a Uniform Civil Code, environmental protection, and separation of the judiciary (Articles 44, 48, 48A, 50, 51).
- **Equal pay for equal work for men and women:** This is explicitly included under **Article 39(d)** of the Directive Principles of State Policy (DPSP) in Part IV of the Indian Constitution. It directs the state to prevent wage discrimination based on gender.
- **Right to work:** This is explicitly listed under **Article 41.** It states that the State shall make effective provisions for securing the right to work within the limits of its economic capacity.
- **Public assistance in case of unemployment:** This is also explicitly listed under **Article 41.** The article specifically mandates the State to provide public assistance in cases of unemployment, old age, sickness, and disablement.

BCW Bits:

- **Dr. B.R. Ambedkar** described DPSPs as the **"novel features"** of the Indian Constitution.
- **Granville Austin** described the combination of Fundamental Rights and DPSPs as the **"Conscience of the Constitution".**
- **Sir B.N. Rau** viewed DPSPs as **"Instruments of Instructions"** for the legislature and executive.

46. Consider the following statements about Panchayats in India: ***[BPSC AEDO]***

1. States with a population less than twenty lakh have an option not to constitute Panchayats at the intermediate level.
2. Any person who has attained the age of twenty one years is eligible to be chosen as a member of Panchayat.
3. The provisions regarding Panchayats are not applicable for Union Territories.

Select the correct answer using the code given below:

A. Only 1 B. 1 and 2
C. 2 and 3 D. All of the above

Ans. B

Exp: The **73rd Constitutional Amendment Act of 1992,** added Part IX (Articles 243 to 243-O) and the Eleventh

Schedule to the Constitution. The Eleventh Schedule outlines 29 subjects transferred to Panchayats, including agriculture, rural housing, and drinking water.

- According to **Article 243B** of the Indian Constitution, while a mandatory three-tier Panchayati Raj system is established across the country, states with a total population not exceeding twenty lakhs (2 million) are given the explicit option not to constitute Panchayats at the intermediate (block) level.
- **Article 243F** clarifies the age qualifications for contesting elections. While the minimum age to contest elections for the State Legislature is 25 years, a person cannot be disqualified if they have attained the age of **twenty-one years for the purposes of Panchayat elections**. Therefore, anyone aged 21 or older is fully eligible to be chosen as a member.
- The provisions regarding Panchayats (Part IX) are applicable to Union Territories. Under **Article 243L**, the President of India may direct that the provisions of Part IX apply to any Union Territory subject to specified exceptions or modifications. The State Election Commission for UTs manages these local body structures.

BCW Bits:

- Balwant Rai Mehta Committee recommended the three-tier Panchayati Raj system
- Nagaur (Rajasthan) was the first to implement Panchayati Raj on October 2, 1959.
- One-third seats are reserved for women in Panchayats
- Panchayat elections are conducted by State Election Commission

47. What is the correct chronological sequence of the below? Choose the right answers from the codes: ***[BPSC AEDO]***

1. Joint Parliamentary Committee on Reform of Electoral Law
2. Goswami Committee on Electoral Reforms
3. Tarkunde Committee on Electoral Reforms
4. Election Commission Campaign against Criminalization of Politics

Choose the correct answer from the options given below:

A. 4 1 3 2　　B. 4 1 2 3
C. 1 4 2 3　　D. 1 3 2 4

Ans. D

Exp:

- **Joint Parliamentary Committee on Reform of Electoral Law:** Constituted and submitted reports during 1971–1972 to examine election laws and suggest comprehensive administrative changes.
- **Tarkunde Committee on Electoral Reforms:** Formed in 1974 by Jayaprakash Narayan, this committee submitted its landmark report in 1975, famously recommending the voting age be lowered to 18. It is considered one of the earliest major non-government efforts for electoral reforms. The committee recommended measures like lowering the voting age, state funding of elections, and strengthening democratic participation.
- **Goswami Committee on Electoral Reforms:** Appointed in 1990 under Dinesh Goswami, to review the electoral system, this committee provided significant structural recommendations regarding election expenses, EVMs, and the misuse of official machinery.
- **Election Commission Campaign against Criminalization of Politics:** While continuously advocated, the Election Commission of India intensified its major proactive campaign in 1997. This effort was heavily bolstered by the 2002 Supreme Court judgment mandating criminal record disclosures for candidates.

BCW Bits:

- Article 324 deals with the Election Commission of India.
- Sukumar Sen was the first Chief Election Commissioner of India.
- T.N. Seshan became famous for strict enforcement of election rules.
- Voting age in India was reduced from 21 to 18 years by the 61st Constitutional Amendment.

48. In the question given below, there are two statements marked as Assertion (A) and Reason (R). Mark your answer as per the codes provided: ***[BPSC AEDO]***

Assertion (A): The Constitution has described India as a 'Union of States', and not a Federation.

Reason (R): Indian polity is a result of agreement between States to form an indestructible Union.

Which of the options given below is/are correct?

A. (A) is false, while (R) is true
B. Both (A) and (R) are true but (R) is not the correct explanation for (A)
C. (A) is true, while (R) is false
D. Both (A) and (R) are true and (R) is the correct explanation for (A)

Ans. C

Exp: Article 1 of the Indian Constitution describes India as a **"Union of States."** The Constitution deliberately avoided using the term "Federation" directly.

- The **reason** behind this wording was explained by B. R. Ambedkar in the Constituent Assembly. India is called a "Union" because the Indian federation was **not formed by an agreement** among independent states voluntarily coming together like the USA. Instead, the states derive their authority from the Constitution itself. Thus, Indian polity is NOT the result of an agreement between states.
- In countries like the United States, independent states agreed to form a federation. But in India, the Constitution created the Union, and states do not have the right to separate from it. This is why India is often described as "**an indestructible Union of destructible States.**"

- The Parliament has the power to alter state boundaries, create new states, rename states, or reorganize territories under Article 3. This shows the stronger position of the Union government compared to classical federations.

BCW Bits:

- The Seventh Schedule divides powers into Union, State, and Concurrent Lists.
- Residuary powers in India belong to the Union government.
- Emergency provisions strengthen the unitary character of the Indian Constitution.

49. Which of the following statements is/are incorrect? ***[BPSC AEDO]***

1. The Constitution does not contain any specific procedure for the selection and appointment of the Chief Minister.
2. Upon death of an incumbent Chief Minister, if the ruling party appoints a new leader, then the Governor has to appoint the new leader as Chief Minister.

Select the correct option using the codes given below:

A. Neither 1 nor 2 B. 2 only
C. Both 1 and 2 D. 1 only

Ans. A

Exp: The Constitution of India does not lay down any specific or detailed selection procedure for the Chief Minister. **Article 164** simply states that the Chief Minister shall be appointed by the Governor. In practice, the Governor appoints the leader of the majority party or coalition in the Legislative Assembly as Chief Minister.

- According to established constitutional conventions in India, upon the sudden death of an incumbent Chief Minister, if the ruling party immediately chooses and elects a new leader, the Governor has no personal discretion and is bound to appoint that newly elected leader as the Chief Minister. The Governor only exercises personal discretion if the ruling party fails to decide on a clear successor or splits.

BCW Bits:

- Article 163 deals with the Council of Ministers aiding and advising the Governor.
- A person who is not a member of the legislature can become Chief Minister or Minister, for six months. This provision is is governed by **Article 164(4)** of the Indian Constitution for state ministers/Chief Ministers, and **Article 75(5)** for Union Ministers.
- Collective responsibility of the Council of Ministers is towards the Legislative Assembly.
- Hung assembly means no party gets clear majority.
- Floor test is used to determine majority support in the Assembly.
- Sarkaria Commission gave recommendations on Centre-State relations.
- Parliamentary government is also known as Cabinet government.

50. Match List-I with List-II and select the correct answer using the codes given below the lists. ***[BPSC AEDO]***

List-I		List-II	
a.	Speaker	1.	Fixes the time limit for discussion of demands
b.	Finance Minister	2.	Examines the financial operation of the executive
c.	President	3.	Makes mention in Parliament about the annual financial statement
d.	C & AG	4.	Participates in the formulation of Five Year Plan

Codes:

	a	b	c	d
A.	4	1	3	2
B.	4	1	2	3
C.	1	4	2	3
D.	1	4	3	2

Ans. D

Exp:

- The **Speaker** of the Lok Sabha controls proceedings of the House and fixes the time limit for discussion of demands for grants during budget discussions. This is an important parliamentary function connected with financial business.
- The **Finance Minister** of India participates in economic planning and policy formulation. Earlier, during the Planning Commission era, the Finance Minister played a major role in formulation of Five Year Plans along with the Planning Commission and Union government.
- The **President of India** causes the Annual Financial Statement (Budget) to be laid before Parliament under Article 112. In constitutional language, the budget is presented "on behalf of the President, " though practically it is introduced by the Finance Minister.
- The **Comptroller and Auditor General of India (CAG)** examines government accounts and audits the financial operations of the executive. This makes the CAG one of the most important authorities for ensuring financial accountability in India.

BCW Bits:

- The Speaker decides whether a bill is a Money Bill.
- Money Bills can only be introduced in Lok Sabha, and the Demands for Grants are discussed only in Lok Sabha.
- The CAG is appointed by the President of India, and is called the "guardian of the public purse."

- The Union Budget is presented annually in Parliament.

51. Which of the following is/are probable cause(s) of the ineffectiveness of the PRIs in India? ***[BPSC AEDO]***

1. The transfer of various governance functions—like the provision of education, health, sanitation and water—was not mandated by the 73rd Amendment Act.
2. Lack of finances for PRIs.
3. Despite there being mandatory provisions for reservations for women and SC/STs in the PRIs, there is hardly any representation of these vulnerable groups.

Select the correct option using the codes given below:

A. 1, 2 and 3 B. 1 and 2 only
C. 2 and 3 only D. 1 only

Ans. B

Exp:

- The 73rd Constitutional Amendment Act gave constitutional status to Panchayati Raj Institutions, but it did not make compulsory transfer of all governance functions to PRIs. The Eleventh Schedule contains 29 subjects related to local governance, such as education, health, sanitation, agriculture, and water management, but actual transfer of powers and functions **depends largely on state governments.** This is one of the major reasons for the weak functioning of PRIs in many states. In several cases, PRIs exist constitutionally but do not receive sufficient administrative authority or control over officials and departments.
- **Lack of adequate finances** is another major problem faced by PRIs. Many Panchayats depend heavily on grants from state governments and do not have strong independent revenue sources. Because of financial weakness, they often struggle to implement development schemes effectively.
- The Constitution has made mandatory provisions for reservation of seats for women and SC/ST communities in Panchayati Raj Institutions, and these groups now have significant representation numerically in local bodies. In fact, women's participation in Panchayats has increased substantially after the 73rd Amendment.

  However, the real issue is not lack of representation, but sometimes lack of effective empowerment or proxy representation in certain areas. Therefore, saying there is "hardly any representation" is factually incorrect.

BCW Bits:

- Reservation for women in PRIs is at least one-third, though many states increased it to 50%.
- Panchayati Raj Institutions are considered instruments of grassroots democracy.
- Urban local bodies are dealt with under the 74th Constitutional Amendment Act.

52. Consider the following bills in State Legislature: ***[BPSC AEDO]***

1. A bill passed by the assembly but pending in the council.
2. A bill passed by the council but pending in the assembly.
3. A bill passed by the state legislature but pending the assent of the governor.
4. A bill passed by the state legislature but returned by the president for reconsideration.

Which of the above bills lapse when the State Legislative Assembly is dissolved?

A. 3 and 4 only B. 1 and 2 only
C. 2 and 3 only D. 1 only

Ans. B

Exp: **Article 196** of the Indian Constitution outlines the procedural rules for introducing and passing bills in State Legislatures. It mandates that a bill must be passed by both houses in states with bicameral legislatures (or the single assembly in others) and also dictates when pending bills lapse upon assembly dissolution.

Effect of Dissolution (The Lapsing Rules):

- If a bill is pending in the Legislative Assembly, or was passed by the Assembly but is pending in the Legislative Council, it lapses when the Assembly is dissolved.
- However, if a bill originated in the Legislative Council and was never sent to or considered by the Assembly, it does not lapse when the Assembly is dissolved.

Therefore,

- **Statement 1**: A bill passed by the Legislative Assembly but pending in the Legislative Council lapses upon the dissolution of the assembly. This happens because the bill originated or was handled by the dissolved house, and the permanent house (Council) cannot pass it independently once the originating Assembly ceases to exist.
- **Statement 2**: A bill passed by the Council but pending in the Assembly lapses. According to Article 196, any bill that is currently pending consideration within the Legislative Assembly at the time of its dissolution automatically lapses, regardless of where it originated.
- **Statement 3:** A bill passed by the state legislature but pending the assent of the Governor (or reserved for the President) does not lapse. The closing of the assembly does not affect a bill that has already concluded its passage through the house(s).
- **Statement 4:** A bill passed by the state legislature but returned by the President for reconsideration does not lapse. It can be reconsidered and passed again by the newly elected successor Assembly.

BCW Bits:

- **Article 200** governs the powers of the State Governor when a state bill is presented to them for assent. The

Governor can grant assent, withhold assent, or return a bill (other than a Money Bill) to the state legislature for reconsideration. It also allows the Governor to reserve a state bill for the President's consideration. If the state legislature passes a returned bill again, the Governor *must* give assent.

- **Article 201** comes into play only if the Governor decides to reserve the bill for the consideration of the President. It applies to the bills specifically reserved by the Governor under Article 200. The President can grant assent, withhold assent, or direct the Governor to return the bill to the state legislature for reconsideration. If a returned non-Money Bill is passed again by the legislature, it is presented to the President again, but the President is **not bound** to give assent. However, Money Bills **cannot** be returned to the state legislature for reconsideration.

53. Which of the following committees are helped by the CAG in his work of exercising legislative control over the executive? ***[BPSC AEDO]***

1. Public Accounts Committee
2. Privileges Committee
3. Estimates Committee
4. Committee on Public Undertakings

Select the correct answer using the codes given below:

A. 1, 3 and 4 B. 1, 2 and 4
C. 1, 2 and 3 D. 1 and 4

Ans. D

Exp: The Comptroller and Auditor General (CAG) explicitly acts as a "***guide, friend, and philosopher***" to this financial committee. It helps Parliament in exercising financial control over the executive by assisting mainly the Public Accounts Committee (PAC), and the Committee on Public Undertakings (COPU).

- The **Public Accounts Committee** examines the audit reports submitted by the CAG. It checks whether public money granted by Parliament has been spent properly and legally by the government. The CAG assists the PAC by explaining technical nuances while scrutinizing the annual Audit Report on Appropriation Accounts and the Audit Report on Finance Accounts
- Similarly, the **Committee on Public Undertakings** examines the functioning and audit reports of public sector undertakings (PSUs). The CAG directly assists this committee by preparing and submitting the Audit Report on Public Undertakings, which evaluates the financial transparency and operational efficiency of commercial state enterprises..
- **Privileges Committee :** This non-financial standing committee evaluates breaches of parliamentary discipline, conduct, and contempt issues. It does not oversee executive financial spending, and the CAG has no operational role in its workflow.
- **Estimates Committee** : Known as the "***continuous economy committee***, " its main job is to evaluate pre-budget estimates and suggest policy alternatives to improve financial administration. It acts independently of the audit work completed by the CAG.

BCW Bits:

- **Public Accounts Committee** was introduced in 1921 under the Government of India Act of 1919 (Montford Reforms). It consists of **22 Members** of Parliament (15 elected by the Lok Sabha and 7 by the Rajya Sabha.). Members serve a one-year term. No Minister can be elected as a member of the PAC. By a convention established in 1967, the Chairperson is always chosen from the Opposition party to ensure impartial oversight.
- The **Estimates Committee** in India traces its origins to the Standing Finance Committee, which was first established in 1921 under the provisions of the Government of India Act, 1919. The present Estimates Committee was formally constituted in April 1950, on the recommendation of the then-Finance Minister, **John Mathai**. It consists of **30 members**, all are drawn exclusively from the Lok Sabha. No Minister can be elected to serve on this committee. Members serve a term of one year. Chairperson appointed traditionally from the ruling party.

54. Consider the following statements regarding the Citizenship (Amendment) Bill, 2019: ***[BPSC AEDO]***

1. The Act allows the specified class of illegal migrants to apply for citizenship by naturalization after five years of residency in India.
2. The provisions of the Bill are applicable across the country.
3. The cutoff date for the immigration is on or before December 31, 2014.

Which of the above statements is/are correct?

A. 1 and 3 only B. 2 only
C. 2 and 3 only D. 1 only

Ans. A

Exp: The Citizenship (Amendment) Act, 2019 (CAA) reduces the residency requirement for obtaining Indian citizenship by naturalization from 11 years to **5 years** for specific undocumented communities (Hindus, Sikhs, Buddhists, Jains, Parsis, and Christians) coming from **Afghanistan, Bangladesh,** and **Pakistan**.

- The designated cutoff date for entry into India to be eligible for citizenship under this framework is on or before **December 31, 2014.**
- The provisions of the Act are **not applicable across the entire country**. It explicitly exempts tribal areas of Assam, Meghalaya, Mizoram, and Tripura under the Sixth Schedule of the Indian Constitution, as well as regions regulated under the Inner Line Permit (ILP) system (such as Arunachal Pradesh, Mizoram, Manipur, and Nagaland). This exemption was provided mainly for

protecting indigenous culture and demographic balance in parts of Northeast India.

- The Citizenship Amendment Act does not grant automatic citizenship. It only relaxes eligibility conditions for certain categories of migrants.

BCW Bits:

- Citizenship provisions are contained in Articles 5 to 11 of the Constitution.
- Parliament has the power to regulate citizenship under Article 11.
- The Citizenship Act was originally enacted in 1955.
- Assam Accord was signed in 1985.
- NRC was first prepared in Assam.

55. Which of the following statements is/are correct regarding the 98th Constitutional Amendment Act? ***[BPSC AEDO]***

1. 98th Constitutional Amendment Act inserted Article 371-J, which provided for special provision for the Hyderabad-Karnataka region.
2. The special provisions aim to establish an institutional mechanism for an equitable allocation of funds to meet the development needs over the region.
3. Article 371-J provided for reservation of seats in educational and vocational training institutions in the region for students who belong to the region.

Select the correct answer using the codes given below:

A. 1 and 3 only B. 2 and 3 only
C. 1, 2 and 3 D. 1 and 2 only

Ans. C

Exp: The **98th Constitutional Amendment Act, 2012**, inserted **Article 371-J** into the Constitution of India. This article grants special provisions to the **Hyderabad-Karnataka region** (now known as ***Kalyana-Karnataka***) to address its socio-economic backwardness.

- The amendment empowers the Governor of Karnataka to establish an institutional framework (**a separate Development Board**) to ensure an equitable allocation of funds tailored specifically for the development needs of this backward region.
- Article 371-J explicitly allows for the **reservation of seats in educational and vocational training institutions** within the region for students who belong to that locality by birth or domicile. It also provides local reservation benefits in state government jobs.

BCW Bits:

- **Articles 371 to 371-J** of the Indian Constitution (**Part XXI**) grant "special, transitional, and temporary provisions" to specific states. These provisions aim to preserve local culture, protect indigenous land rights, and promote equitable regional development and self-governance across 12 different states.

56. In which order is the following placed in the Constitution of India? ***[BPSC AEDO]***

(i) Right to freedom of religion
(ii) Change of state name
(iii) Imposition of President's rule in a state
(iv) Mention about the Indian Administrative Services

Options:

A. (iv) (ii) (iii) (i) B. (ii) (i) (iv) (iii)
C. (iii) (iv) (ii) (i) D. (i) (iii) (iv) (ii)

Ans. B

Exp:

- The provision related to **change of state name** comes under **Article 3** of the Constitution. It empowers Parliament to form new states, alter boundaries, increase or decrease the area of states, and change state names. For example, the renaming of Orissa to Odisha or Uttaranchal to Uttarakhand was done through this constitutional mechanism.
- The **Right to Freedom of Religion** is covered under **Articles 25 to 28** in Part III of the Constitution dealing with Fundamental Rights. These Articles guarantee religious freedom to individuals and religious groups.
- The mention regarding **Indian Administrative Services (IAS)** comes under **Article 312**, which deals with **All India Services**. It empowers Parliament to create new All India Services if the Rajya Sabha passes a resolution supported by not less than two-thirds of members present and voting. The IAS and IPS are examples of All India Services.
- The provision regarding **President's Rule in a state** comes under **Article 356.** It allows the President to impose President's Rule if the constitutional machinery in a state fails. This provision is based on the report of the Governor or other information available to the President. During President's Rule, the state government may be dismissed and Parliament takes over legislative functions of the state.

BCW Bits:

- The landmark **S. R. Bommai** case judgment imposed important judicial limitations on arbitrary use of President's Rule.
- All India Services currently include **IAS, IPS,** and **IFoS (Indian Forest Service)**.
- Emergency provisions are borrowed mainly from the **German Constitution (Weimar Constitution)**.
- Article 356 is related to **State Emergency**, while Article 352 deals with National Emergency.
- Fundamental Rights are enforceable through **Article 32**, called the "Heart and Soul" of the Constitution by B. R. Ambedkar.
- The Indian Constitution originally had **395 Articles and 8 Schedules**.

57. The 93rd Amendment of the Constitution relates to the following: ***[BPSC AEDO]***

A. Panchayati Raj Institutions
B. Reservation for women in legislative assemblies
C. Reservation for Other Backward Classes in educational institutions
D. Fundamental Duties

Ans. C

Exp: The 93rd Amendment (2005) inserted **Article 15(5)** into the Constitution, which states that nothing in Article 15 or Article 19(1)(g) shall prevent the State from making special provisions for the advancement of socially and educationally backward classes or for SCs and STs regarding their admission to educational institutions. This includes private educational institutions, whether aided or unaided by the State, except minority educational institutions protected under Article 30(1).

- Article 15 originally prohibits discrimination by the State on grounds of religion, race, caste, sex, or place of birth. However, over time, constitutional amendments and judicial interpretations allowed the State to make special provisions for disadvantaged groups in order to achieve substantive equality.
- The amendment provided constitutional backing for laws enabling reservation for OBCs in centrally funded educational institutions. This eventually led to the **Central Educational Institutions (Reservation in Admission) Act, 2006**, which introduced 27% reservation for OBCs in many central educational institutions.
- The Supreme Court later examined the validity of the 93rd Amendment in the famous **Ashoka Kumar Thakur v. Union of India** case and upheld the constitutional validity of OBC reservations in central educational institutions, though the creamy layer principle was emphasized for OBCs.
- Reservation policies in India are based on the idea of substantive equality and social justice.

BCW Bits:

- 42nd Amendment – related to Fundamental Duties (Swarn Singh Committee)
- 44th Amendment - Emergency reforms
- 52nd Amendment - Anti-defection
- 61st Amendment - Voting age 18 years
- 73rd / 74th Amendment - Local self-government
- 86th Amendment - Right to Education
- 93rd Amendment - Reservation in educational institutions.
- 106th Amendment - Nari Shakti Vandan Adhiniyam.

58. Consider the following provisions given in the Indian Constitution: ***[BPSC AEDO]***

1. Article 280 – Election Commission of India
2. Article 315 – Union Public Service Commission
3. Article 324 – Finance Commission

Which of the above statements is not correct?

A. Only one B. Only one and three
C. All three D. None of the above

Ans. B

Exp:

- **Article 280** actually deals with the **Finance Commission,** a constitutional body appointed by the President of India to recommend distribution of financial resources between the Centre and the States. It plays a key role in Indian fiscal federalism. The Commission recommends Distribution of tax revenues between Union and States, Grants-in-aid to states, Measures to improve state finances. The Finance Commission is constituted every five years or earlier if needed.
- **Article 315** provides for the establishment of the Union Public Service Commission (UPSC), State Public Service Commissions (SPSC), Joint State Public Service Commissions.
- **Article 324** vests the superintendence, direction, and control of elections in the Election Commission. It conducts Elections to Lok Sabha

BCW Bits:

- The first Finance Commission was constituted in **1951** under K. C. Neogy.
- The Chief Election Commissioner enjoys protections similar to a Supreme Court judge regarding removal.
- UPSC provisions are contained in **Part XIV** of the Constitution.
- Articles 315–323 deal with Public Service Commissions in the Constitution.

59. During which Prime Minister's tenure the National Commission to Review the Constitution was formed? ***[BPSC AEDO]***

A. Morarji Desai B. V.P. Singh
C. Atal Bihari Vajpayee D. Manmohan Singh

Ans. C

Exp:

- The National Commission to Review the Working of the Constitution was formed during the tenure of Prime Minister **Atal Bihari Vajpayee**. The 11-member commission, established on **February 22, 2000**, was headed by former Chief Justice of India Justice **M.N. Venkatachaliah**.
- The Government of India established this commission to examine how the Constitution had functioned over the decades and to suggest possible reforms for improving governance, efficiency, accountability, and democratic functioning. It was not formed to rewrite or replace the Constitution. Its objective was to review the "working" of

the Constitution and recommend suitable changes while preserving its basic structure.

- It submitted its report in 2002 and made a large number of recommendations. Some important recommendations included Strengthening anti-defection laws, Electoral reforms, Judicial accountability, Decentralization of governance, Review of emergency provisions, Improvement in Centre-State relations, Criminalization of politics.

BCW Bits:

- Morarji Desai served as Prime Minister from 1977 to 1979 after the Emergency period. His government is mainly remembered for the **44th Constitutional Amendment (1978)**, which reversed several Emergency-era provisions introduced by the 42nd Amendment.
- V. P. Singh implemented the **Mandal Commission recommendations** regarding OBC reservation in government jobs.
- The 42nd Amendment (1976) is often called the **"Mini Constitution."**
- Anti-defection provisions were added through the **52nd Constitutional Amendment (1985)**.
- The Sarkaria Commission and Punchhi Commission dealt with **Centre-State relations**.

60. Match List I with List II and give answer from the code given below: ***[BPSC AEDO]***

List I		List II	
a.	Estimates Committee	(i)	Examination of Appropriated Accounts
b.	Public Accounts Committee	(ii)	Suggestions for Economy in Expenditure
c.	Subordinate Legislation	(iii)	Examination of rules made by executive departments under laws passed by the legislature
d.	Public Sector Undertakings Committee	(iv)	Review of the performance of public sector undertakings

Code:

	a	b	c	d
A.	(i)	(ii)	(iii)	(iv)
B.	(ii)	(i)	(iii)	(iv)
C.	(iii)	(iv)	(i)	(ii)
D.	(iv)	(i)	(ii)	(iii)

Ans. B

Exp:

- The **Estimates Committee** is a key parliamentary standing committee in India that examines the budget estimates presented to the Parliament. Its primary purpose is to scrutinize government spending, **suggest economies in public expenditure**, and ensure the efficient utilization of public funds. This committee is often called a "continuous economy committee" because it studies how government departments can reduce wasteful expenditure and improve administrative efficiency.
- The **Public Accounts Committee (PAC)** examines whether the money granted by Parliament has been spent properly and legally. It scrutinizes Appropriation Accounts, Audit Reports of the Comptroller and Auditor General (CAG). PAC is extremely important because it acts as a watchdog over public expenditure.
- The **Committee on Subordinate Legislation** examines rules, regulations, bylaws, and orders made by the executive under powers delegated by Parliament through legislation. Modern legislatures often pass broad laws and allow the executive to frame detailed rules later. This is called delegated legislation or subordinate legislation. This committee ensures that the executive does not misuse delegated powers beyond the authority granted by Parliament.
- The **Committee on Public Sector Undertakings (PSUs)** examines reports and accounts of government-owned enterprises and evaluates their efficiency, autonomy, and performance.

BCW Bits:

- Rajya Sabha has no representation in the Estimates Committee.
- Committee on Public Undertakings was created in 1964.
- Financial committees are generally classified into Estimates Committee, Public Accounts Committee, Committee on Public Undertakings.

61. **Assertion (A):** According to Article 312 of the Indian Constitution, the Rajya Sabha can create new All India Services in the national interest.

Reason (R): The Rajya Sabha can define national interest in a comprehensive manner. ***[BPSC AEDO]***

Options:

A. Both (A) and (R) are true and (R) is the correct explanation of (A)

B. Both (A) and (R) are true, but (R) is not the correct explanation of (A)

C. (A) is true but (R) is false

D. (A) is false but (R) is true

Ans. C

Exp:

- Article 312 provides that if the Rajya Sabha declares by a resolution supported by **not less than two-thirds of the members present and voting** that it is necessary or expedient in the national interest to create one or more All

India Services, then Parliament may create such services by law.

- This provision reflects the federal nature of India. Since All India Services function both under the Union and the States, the Constitution gives a special role to the Rajya Sabha, which represents the States in the Indian federal system.
- The Constitution nowhere says that the Rajya Sabha has any special constitutional authority to "define national interest in a comprehensive manner." The Rajya Sabha only has the constitutional power to pass a resolution stating that creation of a new All India Service is necessary in national interest.

BCW Bits:

- **Article 249** → Rajya Sabha can authorize Parliament to legislate on State List subjects in national interest.
- **Article 312** → Rajya Sabha can initiate creation of new All India Services.
- **Article 315** deals with Public Service Commissions.
- Central Services work only under the Union Government, while All India Services serve both the Union and State Governments.

62. In which article the provision on reservation of Scheduled Castes and Scheduled Tribes to Panchayat has been given? ***[BPSC AEDO]***

A. 243A B. 243B C. 243C D. 243D

Ans. **D**

Exp:

- The 73rd Constitutional Amendment (1992) inserted **Part IX** into the Constitution, covering Articles **243 to 243O**. Its main objective was to strengthen local self-government in rural India by giving constitutional recognition to Panchayats. This amendment is considered a major step toward democratic decentralization and grassroots governance.
- Article 243D specifically deals with **reservation in Panchayats**. It provides reservation of seats for Scheduled Castes (SCs), Scheduled Tribes (STs), Women, The reservation for SCs and STs must be proportional to their population in the Panchayat area. Additionally, not less than one-third of total seats, including reserved seats, must be reserved for women. Many states later increased women's reservation to 50%.

BCW Bits:

- Article 243A deals with the **Gram Sabha**. It provides powers and functions to the Gram Sabha at the village level as determined by state legislatures. Gram Sabha is considered the foundation of direct democracy in rural governance.
- Article 243B provides for the **constitution of Panchayats** at village, intermediate, and district levels. It establishes the three-tier Panchayati Raj structure in states with population above a specified limit.
- Article 243C deals with the **composition of Panchayats**. It empowers state legislatures to determine Number of seats, Territorial constituencies, Representation structure.

Note:

- The 73rd Constitutional Amendment Act came into force on **April 24, 1993.**
- The Eleventh Schedule contains **29 subjects** related to Panchayats.
- The 74th Constitutional Amendment relates to **Urban Local Bodies**.
- Balwant Rai Mehta Committee recommended the three-tier Panchayati Raj system in 1957.
- Nagaur (Rajasthan) was the first to implement Panchayati Raj on **October 2, 1959**.
- State Election Commission conducts Panchayat elections.
- State Finance Commission reviews Panchayat finances every five years.

63. In January 2015, who among the following was appointed as the first C.E.O. of NITI Aayog of India? ***[BPSC AEDO]***

A. Sindhushree Kullar B. B.V.R. Subrahmanyam
C. Cyrus Mistry D. S. S. Mudra

Ans. **A**

Exp: Sindhushree Khullar, a retired IAS officer, became the first CEO of NITI Aayog in January 2015. She had earlier served in important administrative positions including the Ministry of Urban Development.

- NITI Aayog was created by the Government of India on **1 January 2015**, replacing the old Planning Commission.
- The full form of NITI is: **National Institution for Transforming India**.
- The creation of NITI Aayog marked a major shift in India's development planning approach. Unlike the Planning Commission, which followed a centralized top-down model, NITI Aayog was designed as a policy think tank promoting Cooperative federalism, Competitive federalism, **Bottom-up planning**, Innovation-driven governance.
- The Prime Minister of India serves as the Chairperson of NITI Aayog. The organization includes the Governing Council consisting of Chief Ministers and Lt. Governors, Vice-Chairperson, Full-time members, Part-time members, CEO.
- The old Planning Commission, established in 1950 under Jawaharlal Nehru, was associated with centralized economic planning and Five-Year Plans. NITI Aayog was therefore introduced to Encourage state participation, Improve policy coordination, Promote innovation
- Unlike the Planning Commission, NITI Aayog does not allocate funds to states. Its role is mainly advisory, strategic, and policy-oriented.

BCW Bits:
- Arvind Panagariya was the first Vice-Chairperson of NITI Aayog.
- NITI Aayog does not prepare **Five-Year Plans**.
- The Planning Commission was neither a constitutional nor a statutory body.
- The Aspirational Districts Programme is associated with NITI Aayog.
- The Planning Commission was inspired partly by the Soviet-style planning model.

64. In India, the enactment of the budget in the Parliament goes through the following five stages. ***[BPSC WMO]***
1. General discussion
2. Voting on demands
3. Introduction
4. Finance bill
5. Appropriation bill

Arrange the above in a sequential order.

A. 3, 1, 2, 5, 4 B. 1, 2, 3, 4, 5
C. 3, 2, 1, 4, 5 D. 3, 1, 2, 4, 5

Ans. A

Exp: The Constitution of India does not use the term "budget." Instead, **Article 112** defines it as the **'Annual Financial Statement'**. It is a comprehensive estimate of the Government of India's projected receipts and expenditures for the upcoming financial year. The President is constitutionally obligated to ensure this statement is laid before both Houses of Parliament.

Six Stages of Enactment
1. **Presentation of the Budget:** The finance minister introduces the budget in the Lok Sabha, presenting the government's fiscal policy, tax proposals, and spending plans.
2. **General Discussion:** Both Houses hold a broad debate on the budget. At this stage, the House does not vote on specific demands; instead, members discuss the overall economic policy and the state of the economy.
3. **Scrutiny by Departmental Committees:** After the general discussion, Parliament is adjourned. During this time, the various Departmental Standing Committees examine the detailed demands for grants of individual ministries, facilitating an in-depth and non-partisan review.
4. **Voting on Demands for Grants:** The Lok Sabha then discusses and votes on the specific demands for grants presented by each ministry. This is where the power of the purse is exercised; members can move "cut motions" to reduce the amount of a demand.
5. **Passing of the Appropriation Bill:** Once the demands are voted upon, the Appropriation Bill is introduced. This bill is critical because, according to the Constitution, no money can be withdrawn from the **Consolidated Fund of India** except under an appropriation made by law.
6. **Passing of the Finance Bill:** The final stage involves the Finance Bill, which legalizes the government's taxation proposals. Once passed by Parliament and assented to by the President, the budget enactment process is complete, providing the legal authority to collect revenue and spend funds.

65. Consider the following statements regarding the use of the Electronic Voting Machines in India. ***[BPSC WMO]***
1. EVMs were first used in the Assembly Constituency of Kerala in 1982.
2. EVMs do not run where there is no electricity.
3. The EVMs have been devised and designed by the Indian Public Sector Undertakings.
4. VVPATs with EVMs were used for the first time in Nagaland.

Which of the above statements is/are incorrect?

A. Only 2 B. Only 1, 2 and 4
C. Only 4 D. Only 3 and 4

Ans. A

Exp: **Statement 1 is correct:** Electronic Voting Machines (EVMs) were first used on an experimental basis in the Paravur Assembly constituency of Kerala in May 1982.
- **Statement 2 is incorrect:** EVMs are stand-alone machines that operate on battery power (an ordinary 7.5-volt alkaline power pack). They do not require an external electricity connection and can function effectively in areas without electricity.
- **Statement 3 is correct:** The EVMs were designed and developed by the Technical Experts Committee of the Election Commission of India in collaboration with two Public Sector Undertakings (PSUs): **Bharat Electronics Ltd. (BEL)**, Bangalore, and **Electronics Corporation of India Ltd. (ECIL)**, Hyderabad.
- **Statement 4 is correct:** Voter Verifiable Paper Audit Trail (VVPAT) systems, when attached to EVMs, were used for the first time in an election in the 51-Noksen (ST) Assembly Constituency of Nagaland in September 2013.

66. Which of the following is/are correct about the scope of judicial review? ***[BPSC WMO]***
1. Authority derived from a statute of the Parliament may be challenged on the ground that the Act of Parliament itself is unconstitutional.
2. Delegated legislation can be challenged on the ground of legality.
3. Sanctity of the Fundamental Rights is violated by the administration.

Select the correct answer using the codes given below.
Code:
A. Only 1 and 2 B. 1, 2 and 3
C. Only 1 D. Only 2 and 3

Ans. B

Exp: **Judicial Review** is the power of courts to invalidate legislative and executive actions that contravene the Constitution. While not explicitly defined in the Constitution, it is a judicially evolved doctrine derived from the judiciary's role as the constitutional interpreter.

The scope of judicial review in India is comprehensive, covering all three mentioned grounds:

1. **Parliamentary Acts:** Under **Article 13(2)**, the judiciary can strike down any law passed by Parliament that violates **Fundamental Rights** or exceeds its legislative competence. The Basic Structure doctrine (e.g., **Kesavananda Bharati v. State of Kerala, 1973**) further empowers courts to review even constitutional amendments.
2. **Delegated Legislation:** Rules and regulations framed by the executive under statutory authority are subject to the doctrine of **ultra vires**. If such legislation conflicts with the parent Act or the Constitution, it is rendered void. E.g. **Supreme Court Advocates-on-Record Association v. Union of India (2015)** (Striking down the NJAC Act).
3. **Administrative Actions:** Under **Articles 32 and 226**, the judiciary reviews administrative decisions to ensure they do not violate fundamental rights, intervening whenever actions are found to be arbitrary, mala fide, or unreasonable, as illustrated by the evolution from the restrictive view in **Kharak Singh v. State of U.P. (1962)**—which laid the groundwork for privacy—to the landmark recognition of the right to privacy as a fundamental right in **Justice K.S. Puttaswamy v. Union of India (2017),** alongside established precedents like **S.R. Bommai v. Union of India (1994)** (Reviewing the Governor's power to recommend President's Rule) regarding the scope of judicial review over executive power.

67. Given below are two statements, one is labelled as Assertion (A) and the other is labelled as Reason (R). ***[BPSC WMO]***

Assertion (A): The President can dismiss the Prime Minister at any time.

Reason (R): The Prime Minister holds office during the pleasure of the President.

Select the correct answer from the code given below.
Code:
A. Both (A) and (R) are true, but (R) is not the correct explanation ofA.
B. (A) is false, but (R) is true
C. Both (A) and (R) are true, and (R) is the correct explanation ofA.
D. (A) is true, but (R) is false

Ans. B

Exp: The office of the President of India is governed by several key provisions: **Article 52** establishes the office, **Article 53** vests executive power, and **Article 75** mandates that the Prime Minister and Council of Ministers are appointed by and hold office during the "pleasure of the President."

- **Reason (R) is true:** As per **Article 75**, the formal tenure of the Prime Minister is indeed tied to the "pleasure of the President, " serving as a constitutional mechanism to define the executive's appointment.
- **Assertion (A) is false:** The phrase "pleasure of the President" is not an absolute power. In India's parliamentary democracy, the President acts on the aid and advice of the Council of Ministers (**Article 74**). The President cannot dismiss a Prime Minister who commands a majority in the Lok Sabha. The power to remove the executive is effectively reserved for the House of the People, ensuring the government remains accountable to the legislature rather than the personal whim of the President.

68. Match List - I with List - II and select the correct answer using the codes given below the lists. ***[BPSC WMO]***

List - I		List - II	
a.	Habeas Corpus	1.	Warrant or authority
b.	Mandamus	2.	Certified or to be made certain
c.	Certiorari	3.	We command
d.	Quo Warranto	4.	Produce the body of a person

Code:

	a	b	c	d
A.	1	2	3	4
B.	2	3	1	4
C.	4	3	2	1
D.	3	4	2	1

Ans. C

Exp: The writs issued by the courts (under Article 32 for the Supreme Court and Article 226 for High Courts) serve specific legal purposes:

- **Habeas Corpus (a-4):** Literally means "to produce the body of." It is issued to ensure a detained person is brought before the court to determine the legality of their detention.
- **Mandamus (b-3):** Translates to "We command." It is issued by a court to a public official, directing them to perform their mandatory official duties.

- **Certiorari (c-2):** Translates to "to be certified" or "to be made certain." It is issued by a higher court to a lower court or tribunal to quash an order or transfer a case, acting as both a preventive and curative measure.
- **Quo Warranto (d-1):** Translates to "by what warrant or authority." It is used to legally challenge an individual's claim or title to a public office.

BCW Bits:

- **Prohibition** is a judicial writ issued by a higher court to prevent a lower court or quasi-judicial body from acting beyond its legal jurisdiction. It is strictly **preventive**, acting as a check to halt proceedings before they conclude, and applies exclusively to judicial and quasi-judicial authorities.

69. The 'Transferred Subjects' under Dyarchy in the provinces were to be looked after by the Governor and his Council of Ministers. Which of the following was/were not included in them? ***[BPSC WMO]***

i. Education ii. Local self-government
iii. Labour iv. Land Revenue

Choose the answer from the code given below:
Code:
A. (iii) and (iv) B. Only (iv)
C. (i) and (ii) D. (ii) and (iv)

Ans. A

Exp: Under the **Government of India Act, 1919** (Montagu-Chelmsford Reforms), the system of **Dyarchy** was introduced in the provinces, which divided provincial subjects into two categories:

- **Transferred Subjects:** Administered by the Governor with the aid of Indian Ministers responsible to the provincial Legislative Council. These included:
  - **Education** (i)
  - **Local self-government** (ii)
  - Public health, agriculture, and public works.
- **Reserved Subjects:** Administered by the Governor and his Executive Council, who were not responsible to the legislature. These included "vital" and financial areas such as:
  - **Land Revenue** (iv)
  - **Labour** (iii) (often categorized under industrial disputes or welfare, which were reserved)
  - Police, justice, and finance.
- Since **Labour (iii)** and **Land Revenue (iv)** were classified as Reserved Subjects, they were not included in the Transferred Subjects. Therefore, the correct option is **(A)**.

70. Under the Indian Constitutional Scheme, how the independence of Comptroller and Auditor General (CAG) is secured? ***[BPSC WMO]***

1. He is barred from holding any government office after completion of his tenure.
2. His salary, except that of the staff is charged on the Public Account of India.

Select the correct answer using the code given below:
Code:
A. Only 2 B. Neither 1 nor 2
C. Only 1 D. Both 1 and 2

Ans. C

Exp: The Comptroller and Auditor General (CAG) of India is the constitutional head of the Indian Audit and Accounts Department and is famously described as the "guardian of the public purse." Appointed by the President under **Article 148**, the CAG is tasked with ensuring the financial accountability of the executive to the Parliament.

- To secure the independence of this vital office, the Constitution provides several safeguards: the CAG is ineligible for further office under the Government of India or any State government after completing their tenure, which prevents the officeholder from being swayed by the prospect of future political or administrative appointments (Statement 1 is correct).
- Regarding financial independence, while the CAG's office functions with autonomy, their salary, allowances, and pensions are charged upon the **Consolidated Fund of India (CFI)**—not the Public Account—which ensures that these expenses are not subject to the annual vote of Parliament, thereby insulating the CAG from executive or legislative pressure. (Statement 2 is incorrect).
- Under **Article 149**, the Parliament has further defined the CAG's authority through the *CAG's (Duties, Powers and Conditions of Service) Act, 1971*, reinforcing that the CAG's service conditions cannot be varied to their disadvantage after appointment, mirroring the protections granted to Supreme Court judges to ensure a non-partisan and fearless audit of public accounts.

71. Given below are two statements, one is labeled as Assertion (A) and other labeled as Reason (R). ***[BPSC WMO]***

Assertion (A): Implementing development programs is the only task of Gram Sabha.

Reason (R): In some States, Gram Sabha forms committees like Construction and Development committees.

In the light of the above statements choose the most appropriate answer from the options given below.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A)
B. (A) is false, but (R) is true
C. Both (A) and (R) are true, and (R) is the correct explanation of (A)
D. (A) is true, but (R) is false

Ans. B

Exp: The **Panchayati Raj Institution (PRI)** framework is constitutionally anchored by the **11th Schedule**, which lists 29 functional areas for panchayats. It was formally institutionalized by the **73rd Constitutional Amendment Act (1992)**, which added **Part IX** and **Article 243-243O** to the Constitution. Furthermore, committees like the **Balwant Rai Mehta Committee (1957)** and **Ashok Mehta Committee (1977)** were pivotal in recommending the democratic decentralization and three-tier structure that defines grassroots governance in India today.

- **Assertion(A) is false:** While development implementation is a core function, it is not the only task. Under **Article 243A**, the Gram Sabha acts as a deliberative, supervisory, and auditing body. Its mandate includes social audits, beneficiary identification, and ensuring fiscal accountability, making it the primary legislative organ at the village level.
- **Reason (R) is true:** To enhance specialized oversight, various State Panchayati Raj Acts empower the Gram Sabha to form standing committees. For example, the **Bihar Panchayat Raj Act, 2006**, explicitly mandates the formation of various standing committees (Sthayi Samitis)—such as those for **Planning and Coordination**, **Social Justice**, and **Infrastructure Development**—which empower the Gram Sabha to manage local affairs beyond mere development implementation.

72. Which statement correctly describes a feature of the Montagu-Chelmsford reforms under the Government of India Act, 1919? ***[BPSC WMO]***

A. Provincial subjects were divided into 'reserved subjects' under the Governor and 'transferred subjects' handled by ministers responsible to the legislatures.

B. The central legislature gained complete control over the Viceroy and his Executive Council.

C. All members of the provincial legislatures were to be nominated directly by the Governor.

D. The right to vote was granted to every adult citizen of British India, irrespective of property or education.

Ans. A

Exp: The **Government of India Act, 1919** (Montagu-Chelmsford Reforms) introduced **Dyarchy** in the provinces. This system divided provincial subjects into two lists:

- **Reserved Subjects:** Administered by the Governor and his Executive Council (not responsible to the legislature).
- **Transferred Subjects:** Administered by the Governor with the aid of ministers responsible to the provincial Legislative Council.

73. The famous song 'Vande Mataram' was written in which of the following books? ***[BPSC WMO]***

A. Naivedya
B. Vidrohi
C. Devi-Chaudhurani
D. Anandamath

Ans. D

Exp: The famous song 'Vande Mataram' was written by Bankim Chandra Chatterjee and is featured in his 1882 Bengali novel, **Anandamath**. The novel is set against the backdrop of the Sannyasi Rebellion in the late 18th century and became a significant source of inspiration for the Indian independence movement.

74. Which Article of the Indian Constitution deals with Panchayati Raj Institutions? ***[BPSC WMO]***

A. Article 243
B. Article 300
C. Article 40
D. Article 51

Ans. A

Exp: **Article 243:** This article, along with the subsequent clauses (**Articles 243A to 243O**), was added by the **73rd Constitutional Amendment Act, 1992.** It gave PRIs constitutional recognition and established them as the third tier of governance in India and provides the constitutional framework for the Panchayati Raj Institutions (PRIs) in India, covering their organization, elections, powers, and responsibilities.

BCW Bits:

- **Article 40:** While this Article (part of the Directive Principles of State Policy) directs the State to organize village panchayats, it was a guiding principle for future legislation rather than the source of the modern constitutional status held by PRIs.
- **Article 300:** Deals with suits and proceedings by or against the Government of India or the States.
- **Article 51:** Deals with the promotion of international peace and security.

Physics

2

1. We apply a force of 200 Newtons on a wooden box and push it on the floor at constant velocity. The marginal friction force will be: *[71st BPSC]*
   A. 100 Newtons B. 200 Newtons
   C. 300 Newtons D. 400 Newtons

Ans. B

Exp: When a wooden box is pushed on the floor at a constant velocity, the net force acting on it is zero, meaning the applied force is exactly balanced by the kinetic friction force. Since the applied force is 200 N, the marginal friction force will also be 200 Newtons (opposite to the direction of motion).

BCW Bits:

- When an object moves at a constant velocity, the opposing frictional force is equal to the applied force, as the net force is zero.
- $F_{applied} = F_{friction}$ (when acceleration is zero).
- This is an application of Newton's First Law (Law of Inertia), a body moving at constant velocity has zero net force acting on it. Note that the friction acting here is **kinetic friction** (not static friction), since the box is already in motion; static friction only acts before the object begins to move.

2. A truck starts from rest down a hill with a constant acceleration. It achieves 400 meters in 20 seconds. If the weight of the truck is 7 tons then what will be the force acting on it? *[71st BPSC]*
   A. 11000 Newtons B. 12000 Newtons
   C. 13000 Newtons D. 14000 Newtons

Ans. D

Exp: Force (F) = Mass (m) x Acceleration (a) .... (i)

By using the kinematic equation,

Given, M = 7 tons (= 7000 Kg)

S = 400 meters

u = 0 (initial speed is zero because the truck starts from rest)

t = 20 seconds

Now using, $s = ut + (1/2)\,at^2$

$\Rightarrow 400 = (0 \times 20) + (1/2).a.20^2$

$a = 2\ m/s^2$

Using the value of (M) and (a) in eqn (i).

Force (F) = 7000 (Kg) x 2 (m/s^2)

= 14000 $Kg.m/s^2$ = 14000 Newtons

Therefore, the force acting on the truck is 14000 N.

BCW Bits:

- The force calculated here (F = ma) represents the **net force** acting on the truck, the resultant of all forces including gravity component along the slope, friction, and air resistance. This question is a classic two-step application: first find acceleration using kinematics (equations of motion), then apply Newton's Second Law; a pattern frequently tested in BPSC.

3. In which medium, the speed of sound is maximum? *[71st BPSC]*
   A. Steel B. Water
   C. Air D. Hydrogen

Ans. A

Exp: Sound travels fastest in solids, slower in liquids, and slowest in gases because particles are most tightly packed in solids, allowing vibrations (sound waves) to transmit more quickly. Among the given options, as Steel is a solid, while water is a liquid, and air and hydrogen are gases, the speed of sound is maximum in Steel.

Approximate speed values for reference:

- Steel: ~5100–5960 m/s
- Water: ~1480 m/s
- Air: ~343 m/s
- Hydrogen: ~1270 m/s

BCW Bits:

- Among gases, hydrogen has a much higher speed of sound (~1270 m/s) than air (~343 m/s) due to its very low molecular mass, making it a common exam distractor. However, it still falls far short of steel (~5960 m/s), as the defining factor for maximum sound speed is the high elasticity and rigid particle arrangement of solids, not molecular mass alone.

4. The potential energy of a freely falling body continuously decreases. *[71st BPSC]*
   A. The principle of conservation of energy is violated.
   B. The principle of conservation of energy is not violated.
   C. Gravitational force is violated.
   D. Gravitational force is not violated.

Ans. B

Exp:

- When a body falls freely under gravity, its **height decreases**, so its **gravitational potential energy (PE = mgh)** continuously decreases. At the same time, its **speed increases**, so its **kinetic energy (KE = ½mv²)** increases.

- The **loss in potential energy** is exactly equal to the **gain in kinetic energy**, so the **total mechanical energy (PE + KE)** remains constant throughout the fall.
- Thus, even though potential energy decreases continuously, **total energy is conserved**, and the **principle of conservation of energy is not violated**.

BCW Bits:

- At every instant during free fall: **Loss in PE = Gain in KE**, so total mechanical energy (PE + KE) remains constant throughout the motion.
- At the highest point, energy is entirely potential (KE = 0), and just before hitting the ground, it is entirely kinetic (PE = 0), a perfect illustration of energy transformation without violation of conservation.

5. A pair of Oxen exerts a force of 140 Newton while ploughing the field. The field ploughed is 15 meter long. The work done in ploughing the length of the field is: ***[71st BPSC]***

A. 1900 Joule B. 2000 Joule
C. 2100 Joule D. 2200 Joule

Ans. C

Exp: 1 Joule is the amount of work done when a force of one Newton displaces a body by one meter in the direction of the force applied.
Therefore, Work done = Force x Displacement
Given, Force = 140 N, displacement = 15 meter
So, Work done = 140 N x 15 m
= 2100 Joule
Hence, the work done in ploughing the field is 2100 Joule.

BCW Bits:

- The general formula for work is $\mathbf{W = F \times d \times \cos\theta}$, where θ is the angle between the force and displacement. Here, since the oxen pull horizontally along the direction of motion ($\theta = 0°$, $\cos 0° = 1$), the formula simplifies to $W = F \times d$.

6. The focal distance of a plane mirror is ***[71st BPSC]***

A. +1 cm B. -1 cm
C. 2 cm D. Infinity

Ans. D

Exp: The focal length (focal distance) is defined as the distance where parallel incident rays converge or appear to converge. For a plane mirror, because the reflected rays are parallel, they only meet at an infinite distance. Therefore, the focal distance of a plane mirror is at infinity.

BCW Bits:

- A plane mirror is a flat reflective surface that does not converge or diverge light rays in the same way curved mirrors do. Thus, a plane mirror can be considered a part of a sphere with an infinite radius of curvature.

7. Energy consumed in a home is 250 units, then the total energy in Joule will be: [71st BPSC]

A. 9×10^5 B. 8×10^6
C. 9×10^8 D. 10^5

Ans. C

Exp: The electricity consumption is usually expressed in Units, such that 1 unit of energy is universally defined as 1 **kilowatt-hour** (kWh).
Since, 1 kWh = 3.6×10^6 J
So, for 250 units, the total energy is calculated in Joule as,
Energy Consumed = 250 KWh
= $250 \times 3.6 \times 10^6$ J
= 900×10^6 J
= 9×10^8 J

BCW Bits:

- The conversion 1 kWh = 3.6×10^6 J comes from: 1 kW = 1000 W = 1000 J/s, and 1 hour = 3600 seconds, so 1 kWh = $1000 \times 3600 = 3.6 \times 10^6$ J.
- On your electricity bill, each "unit" represents the energy consumed by a 1 kilowatt appliance running continuously for 1 hour, it is a unit of **energy**, not power.

8. Which term in the following does not denote the electric power in electric circuit? ***[71st BPSC]***

A. I^2R B. IR^2
C. VI D. V^2/R

Ans. B

Exp: Electric power (P) in an electric circuit is defined as the rate at which electrical energy is consumed or converted. It is measured in Watts (W) or Joules per second (J/s). The fundamental formula of Power is, **P = VI**, which can also be expressed using resistance (R) as
$\mathbf{P = I^2R}$ or $\mathbf{P = V^2/R}$.

9. In short circuit, the value of electric current in a time circuit: ***[71st BPSC]***

A. Very low B. Do not change
C. Increases very high D. Continuously changing

Ans. C

Exp: During a short circuit, the current in the circuit increases heavily (or rises abruptly). This sudden, dangerous spike occurs because a fault, such as damaged insulation or direct contact between live and neutral wires, creates an unintended, very low-resistance path, allowing for massive current flow. In a short circuit, the resistance becomes close to zero so that the current increases to a very high value, as per Ohm's law.

BCW Bits:

- By Ohm's Law, I = V/R, when resistance (R) approaches zero in a short circuit, current (I) rises to dangerously high values even at normal voltage.

- This excessive current generates intense heat ($P = I^2R$), which can melt wires, damage appliances, or cause fires, which is precisely why **fuses and circuit breakers** are installed as protective devices to break the circuit when current exceeds safe limits.

10. What is the device to protect equipments from the electric shock? ***[71st BPSC]***

A. Fuse B. Generator
C. Motor D. Current controller

Ans. A

Exp: A **fuse** is an essential electrical safety device designed to protect appliances and circuits from damage caused by excessive current. It operates on the principle of the heating effect of electric current, where the heat generated in a conductor is proportional to the square of the current, the resistance, and the time of flow (expressed by the relationship $\mathbf{H = I^2Rt}$).

A fuse wire is engineered with a low melting point and high resistivity, commonly made from an alloy of tin and lead. When an abnormally high current flows through the circuit due to overloading or a short circuit, the wire rapidly heats up, reaches its melting point, and breaks. This action creates an open circuit, instantly interrupting the electrical flow and shielding connected equipment from severe damage or fire hazards.

- In modern electrical layouts, **Miniature Circuit Breakers** (MCBs) are preferred over traditional fuses because they automatically switch off (trip) during faults and can be easily reset without the hassle of replacing a wire.
- A **generator** converts mechanical energy into electrical energy to produce power, a **motor** converts electrical energy into mechanical energy to drive machinery, and a **current controller** simply regulates current flow to a steady level rather than cutting off the circuit during a dangerous power spike.

BCW Bits:

- It is worth noting that a fuse protects **equipment** from damage due to excess current, it does not protect humans from electric shock directly.
- For personal shock protection, **ELCB (Earth Leakage Circuit Breaker)** or **RCD (Residual Current Device)** are used, which detect leakage current and trip the circuit within milliseconds.

11. Pressure has the same dimension as: ***[BPSC ASO, Pre]***

A. Force per unit volume B. Energy per unit volume
C. Force D. Energy

Ans. B

Exp: Pressure (P) is defined as force per unit area. (P = F/A)

- The dimension for force is $[MLT^{-2}]$ and dimension for area is $[L^2]$.
- Thus, the dimension for pressure is, $[P] = [ML^{-1}T^{-2}]$ ....(i)
- The dimension of energy [E] is same as work (i.e – force x displacement), which is equal to, $[E] = [MLT^{-2}] \times [L] = [ML^2T^{-2}]$.
- The dimension of volume $[V] = [L^3]$.
- The dimension for energy per unit volume is, $= [ML^{-1}T^{-2}]$. ....(ii)
- Since, both the dimensions are identical, Pressure has the same dimension as that of energy per unit volume.

12. If the focal length of the objective lens is increased ***[BPSC ASO, Pre]***

A. Magnifying power of microscope and telescope both will decrease
B. Magnifying power of the microscope will increase but that of telescope will decrease.
C. Magnifying power of microscope will decrease but that of telescope will increase.
D. Magnifying power of microscope and telescope both will increase.

Ans. C

Exp: The magnifying power of a compound microscope is inversely proportional to the focal length of the objective lens (f_o). It is given by the formula, $M = \frac{L}{f_o} \times \frac{D}{f_e}$

- Where, L is the tube length, D is the least distance of distinct vision, (f_o) is the focal length of objective lens and (f_e) is the focal length of the eyepiece. Thus, in case of microscope, increasing (f_o) causes the magnifying power to decrease.
- The magnifying power of a telescope is directly proportional to the focal length of the objective lens (F_o), and is represented by the formula, $M = \frac{f_o}{f_e}$
- Where, (F_o) is the focal length of the objective lens of telescope and (F_e) is the focal length of the eyepiece. Thus, in case of telescope, increasing (F_o) causes the magnifying power to increase.

BCW Bits:

- The contrasting behaviour arises from the fundamental difference in design purpose: in a microscope, a **shorter** objective focal length brings the lens closer to the object, producing a highly magnified real image, so increasing f_o reduces magnification.
- In a telescope, a **longer** objective focal length collects more detail from distant objects and increases the angular magnification ratio (F_o/F_e), so increasing F_o increases magnifying power.

13. Which is the correct ascending order of the electromagnetic spectrum in terms of increasing frequency? ***[BPSC ASO, Pre]***

A. Radio < visible < infrared < uv < gamma
B. Infrared < radio < visible < uv < gamma
C. Infrared <visible < radio < uv < gamma
D. Radio < infrared < visible < uv < gamma

Ans. D

Exp: The electromagnetic spectrum arranged in order of increasing frequency (and decreasing wavelength/ increasing energy) is, **Radio waves < Microwaves < Infrared < Visible light < Ultraviolet < X-rays < Gamma rays.**

The electromagnetic spectrum covers an immense, continuous range of frequencies, spanning from below (1 Hz) to over (10^{25} Hz). It includes low frequency Radio waves (<3 x 10^{11} Hz) to high frequency Gamma rays (> 10^{24} Hz).

Short wavelength ⟶ Long wavelength
High frequency ⟵ Low frequency

Gamma Waves	X Rays	UV Waves	Infrared Waves	Micro-Waves	Radio Waves

Visible Light

BCW Bits:

- A useful mnemonic for the full EM spectrum in increasing frequency order: **"Raging Martians Invaded Venus Using X-ray Guns"**- Radio, Microwaves, Infrared, Visible, Ultraviolet, X-rays, Gamma rays.
- Note that frequency (f) and wavelength (λ) are inversely related by c = fλ, so the highest frequency (Gamma) has the shortest wavelength and the highest energy (E = hf).

14. Light is an electromagnetic wave, is the principle of **[BPSC ASO, Pre]**

A. Michael Faraday B. Isaac Newton
C. Oersted D. James Clark Maxwell

Ans. D

Exp: **Electromagnetic waves:** It is also called as EM waves. It is produced due to an accelerated electric charge and travels in space due to the periodic oscillation of electric as well as magnetic fields. These waves do not need a medium to travel; hence they can travel in vacuum, air or through any state of matter. The speed of the electromagnetic waves is equal to the speed of light i.e. 3 × 10^8 m/s. Light is an electromagnetic wave.

- **James Clerk Maxwell** (a Scottish scientist) **explained Electromagnetic waves** through his equations named Maxwell's equations theoretically. **His theories were later demonstrated experimentally by Heinrich Hertz** (a German physicist) while producing and receiving radio waves. He proved that the velocity of radio waves was the same as that of light.

BCW Bits:

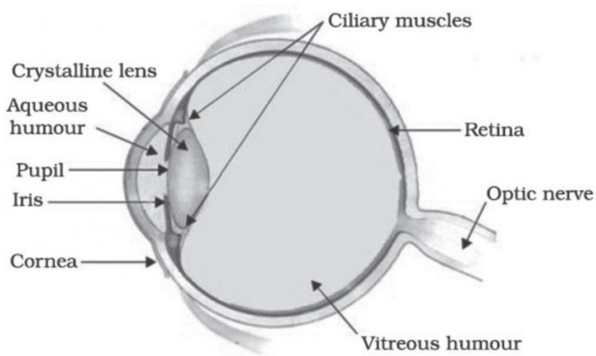
- **Michael Faraday discovered electromagnetic induction,** proving that a changing magnetic field induces an electric current, which laid the experimental groundwork for Maxwell's mathematical synthesis.
- **Isaac Newton advocated for the corpuscular theory of light**, proposing that light consists of a stream of tiny particles rather than waves.
- **Hans Christian Oersted discovered electromagnetism** by observing that an electric current flowing through a wire deflects a nearby compass needle, proving for the first time that electricity produces a magnetic field.

15. The focal length of lens of the eye is under the control of? **[BPSC ASO, Pre]**

A. Ciliary body B. Iris
C. Cornea D. Pupil

Ans. A

Exp: The focal length of the eye's lens is controlled by the **ciliary muscles**, which adjust the lens's curvature (thickness) to focus images of nearby or distant objects onto the retina, a process called **accommodation**. When muscles contract, the lens thickens (shorter focal length for near vision) and when they relax, the lens flattens (longer focal length for far vision).



16. Which of the following materials is ferromagnetic? **[BPSC ASO, Pre]**

A. Wood B. Nickel C. Manganese D. Gold

Ans. B

Exp: **Ferromagnetic substances** are those that are strongly magnetized when placed in an external magnetic field. **Examples:** Iron, Cobalt and Nickel.

BCW Bits:

- Magnetic materials are classified into three types: **Ferromagnetic** (strongly attracted-Iron, Cobalt, Nickel), **Paramagnetic** (weakly attracted- Aluminium, Manganese, Platinum), and **Diamagnetic** (weakly repelled- Gold, Silver, Wood, Copper).

- Among the options, Manganese is a common trap: it is paramagnetic, not ferromagnetic, despite being a metal. Only Iron, Cobalt, and Nickel are the classic ferromagnetic elements.

17. How much more heat is produced if the current is doubled? ***[BPSC ASO, Pre]***

A. Four times the original amount
B. Twice the original amount
C. Five times the original amount
D. Thrice the original amount

Ans. **A**

Exp: According to **Joule's law of heating**, the heat produced (H) in a conductor is directly proportional to the square of the current (I^2), resistance (R), and time (t). Thus, the Heat produced in a resistor is given by, $H_1 = I^2R.t$,

When current is doubled, heat produced is,

- $H_2 = (2I)^2$ R .t = 4 (I^2 R.t)
- The question asks **"how much more heat is produced"**- and it is visible that $H_2 = 4H_1$, meaning total heat becomes 4 times the original amount.

18. Given below are two statements, one is labelled as Assertion (A) and the other is labelled as Reason (R). ***[BPSC ASO, Mains]***

Assertion (A): A myopic person cannot see distant objects clearly.

Reason (R): The image of distant objects forms behind the retina.

A. Both (A) and (R) are true and (R) is the correct explanation of (A)
B. Both (A) and (R) are true, but (R) is not the correct explanation of (A)
C. (A) is true, but (R) is false
D. (A) is false, but (R) is true

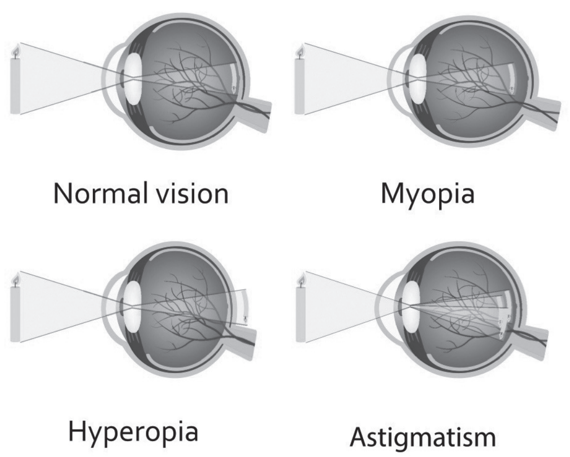
Ans. **C**

Exp: Vision defects arise when the eye's natural focusing mechanism fails to align light directly on the retina. These refractive errors are primarily caused by irregularities in the shape of the eyeball or the flexibility of the lens, requiring corrective optics to restore clarity.

- **Myopia** occurs when the eye is too long or the cornea is too curved, causing light to focus **in front of the retina** instead of on it. This results in clear vision for nearby objects but blurred vision for distant ones, making Assertion (A) correct.
- However, Reason (R) is false because it incorrectly locates the focal point behind the retina. To correct this, a **concave lens** is used to spread the light rays outward, pushing the focal point further back so it lands precisely on the retina.

Other common refractive errors:

Defect	Description	Image Formation	Correction
Hypermetropia (Far-sightedness)	Near objects appear blurry; distant objects are clear.	Behind the retina (due to a short eyeball or flat lens).	Convex (converging) lens
Presbyopia	Age-related loss of near-focusing ability due to a stiffening lens.	Behind the retina (similar to hypermetropia).	Bifocal lenses
Astigmatism	Blurred vision at all distances due to irregular corneal curvature.	Multiple focal points instead of one.	Cylindrical lenses
Cataract	Gradual blurring/loss of vision due to the lens becoming opaque.	Light is blocked/scattered by the cloudy lens.	Surgical replacement (IOL)



BCW Bits:

The power of the corrective lens directly indicates the defect: **negative power (concave/diverging)** corrects myopia, while **positive power (convex/converging)** corrects hypermetropia.

Note that cataract differs from the other three, it is not a refractive error but a structural opacity of the lens, corrected by surgery rather than spectacles.

19. If the mass of the Earth were doubled and its radius halved, what would be the weight of a body of mass "m" on its surface? (Consider "g" as the acceleration due to gravity on the unchanged Earth.) ***[BPSC ASO, Mains]***

A. 0.5mg B. 4mg
C. 8mg D. mg

Ans. C

Exp: Weight is the gravitational force acting on an object, defined by the formula **Weight = mass × gravity**.

- In physics, the acceleration due to gravity is strongly influenced by a planet's physical characteristics, with changes in size often having a more dramatic effect than changes in mass.

Mathematical Derivation:

- The weight of a body is given by **W = mg**, where the acceleration due to gravity is defined by the formula: **g = (G * M) / R^2**
- When the mass is doubled (**M' = 2M**) and the radius is halved (**R' = R/2**), the new acceleration due to gravity (**g'**) is calculated as:

$$g' = \frac{G \cdot 2M}{\left(\frac{R}{2}\right)^2}$$

$$g' = \frac{G \cdot 2M}{\frac{R^2}{4}}$$

$$g' = \frac{G \cdot 2M \times 4}{R^2}$$

$$g' = 8 \times \frac{G \cdot M}{R^2}$$

- Since **(G * M) / R^2** is the original gravity (**g**), we find that **g' = 8g**. Therefore, the new weight of the body on the modified Earth is: **New Weight = m * 8g = 8mg**

20. Two identical-sized balls, one of Lead (mass = 12 kg) and another of aluminium (mass = 3 kg), both initially at rest are dropped from a height of 50 m simultaneously. When they are 10 m above the ground, which of the following quantities remains the same? Neglect air resistance. ***[BPSC ASO, Mains]***

A. Speed and acceleration
B. Kinetic energy and speed
C. Potential energy and momentum
D. Acceleration and kinetic energy

Ans. A

Exp:

- In the absence of air resistance, all objects undergo free fall under gravity regardless of their mass.
- Acceleration → Both experience $g \approx 9.8\, m/s^2$ (mass-independent).
- Height fallen = 50 – 10 = 40m.
- Using $v^2 = 2gs$,
- $v^2 = 2 \times 10 \times 40 = 800$
- $v = \sqrt{800} \approx 28.3\, m/s$
- Thus, both balls have the **same speed** and **same acceleration**.
- However, kinetic energy (1/2 mv^2), momentum (*mv*), and potential energy (*mgh*) depend on mass, so they are different for the two balls.

21. Two long solenoids, P and Q each have the same total number of turns and carry the same current. Solenoid P is twice as long as solenoid Q. Find the ratio of the magnetic field strengths inside them ($B_P : B_Q$). ***[BPSC ASO, Mains]***

A. 1 : 1
B. 1 : 2
C. 2 : 1
D. 4 : 1

Ans. B

Exp:

- The magnetic field inside a long solenoid is:
- $B = \mu_0 nI = \mu_0 \frac{N}{L} I$
- where is the total number of turns and is the length of the solenoid.

Given:
Same total turns (*N*)
Same current (I)
$L_p = 2L_Q$
Therefore,

$$n_P = \frac{N}{2L_Q}, n_Q = \frac{N}{L_Q} \frac{B_P}{B_Q} = \frac{n_P}{n_Q} = \frac{1}{2}$$

Hence,
$B_P : B_Q = 1:2$

- Since solenoid P is longer, its turns are more spread out, resulting in a smaller turns-per-unit-length and therefore a weaker magnetic field.
- 22.If the same current passes through a copper wire, and an iron wire of equal length and cross-sectional area, then which statement is true? ***[BPSC ASO, Mains]***

A. The heat produced is the same in both copper wire and iron wire
B. The heat produced is more in copper wire than in iron wire
C. The heat produced is more in iron wire than in copper wire
D. Cannot be compared without potential difference

Ans. C

Exp: Metals differ in their atomic structure, which determines how much opposition they offer to current flow. Iron offers greater resistance to the flow of electrons than copper.

- According to Joule's law of heating:
  - $H = 1^2 Rt$
- Since current (I) and time (t) are the same for both wires, $H \propto R$
- The resistance of a wire is:

$$R = \frac{\rho L}{A}$$

- Since the length (L) and cross-sectional area (A) are equal for both wires,
$$R \propto \rho$$
- Resistivity values:
  - Copper: $\rho \approx 1.7 \times 10^{-8}\ \Omega \cdot m$
  - Iron: $\rho \approx 10 \times 10^{-8}\ \Omega \cdot m$

  Thus,
  $\rho_{iron} > \rho_{copper} R_{iron} > R_{copper} H_{iron} > H_{copper}$
- Therefore, more heat is produced in the **iron wire** than in the **copper wire** for the same current.

23. Arrange the speeds of sound in the following media in ascending order at room temperature: Air, Water, Iron.
[BPSC ASO, Mains]

A. Iron < Water < Air
B. Air < Water < Iron
C. Water < Iron < Air
D. Air < Iron < Water

Ans. B

Exp: Sound is a mechanical wave and requires a material medium for propagation. Its speed depends mainly on the elasticity and density of the medium.

- In **air (gas)**, molecules are far apart, so sound travels slowest.
- In **water (liquid)**, molecules are closer together, so sound travels faster than in air.
- In **iron (solid)**, particles are tightly packed and the medium is highly elastic, allowing vibrations to be transmitted most rapidly.
- Therefore, the speed of sound increases in the order:
  - Air < Water < Iron

24. Given below are two statements, one is labelled as Assertion (A) and the other is labelled as Reason (R).
[BPSC ASO, Mains]

Assertion (A) : Kerosene has a higher refractive index (1.44) than water (1.33).

Reason (R) : Refractive index of a medium depends on its mass density—the greater the mass density, the greater the speed of light through it.

A. Both (A) and (R) are true, and (R) is the correct explanation of (A)
B. Both (A) and (R) are true, but (R) is not the correct explanation of (A)
C. (A) is true, but (R) is false
D. (A) is false, but (R) is true

Ans. C

Exp:

- **Assertion (A) is True:** Kerosene has a higher absolute refractive index (1.44) than water (1.33). Therefore, kerosene is **optically denser** than water.
- **Reason (R) is False:** The reason statement contains two major scientific errors:
  - **Optical Density Mass Density:** A medium's refractive index depends on its *optical density*, which is completely independent of its *mass density*. For example, kerosene is optically denser than water, but it has a lower mass density (which is why kerosene floats on water).
  - **The Speed of Light Relationship:** The speed of light in a medium (v) is **inversely proportional** to its refractive index (n), as shown by the formula: $n = \frac{c}{v}$

  (where c is the speed of light in a vacuum).
- This means that as the refractive index increases, the speed of light **decreases**. The reason incorrectly claims that light travels faster through denser media.

BCW Bits:

- Light travels slower in kerosene than in water because kerosene is optically denser, despite being physically lighter (lower mass density).

25. A fuse wire made of tin is designed to melt when the current exceeds 10 A. What change should be made to the fuse wire (of the same material) to safely carry a current up to 30 A to an electrical appliance?
[BPSC ASO, Mains]

A. Increase the cross-sectional area (thickness) of the fuse wire and connect it in series to the appliance
B. Increase the cross-sectional area (thickness) of the fuse wire and connect it in parallel to the appliance
C. Decrease the cross-sectional area (thickness) of the fuse wire and connect it in series to the appliance
D. Decrease the cross-sectional area (thickness) of the fuse wire and connect it in parallel to the appliance

Ans. A

Exp: An electric fuse is a critical safety device that protects electrical circuits from overcurrent. It functions as a "weak link" in the circuit, sacrificial by nature, to prevent fires or damage to expensive appliances. Its operation is governed by the thermal properties of conductors and the laws of thermodynamics.

The functionality of a fuse is explained through the following scientific principles:

- **Joule's Law of Heating:** The heat (**H**) generated in a wire is determined by the formula **H = I² * R * t**. This shows that heat increases with the square of the current (**I**). To prevent a wire from melting at a higher current (30 A instead of 10 A), we must significantly reduce its resistance (**R**).
- **Law of Resistance:** The resistance of a conductor is determined by the formula **R = (ρ * L) / A**. Since the material (resistivity **ρ**) and length (**L**) remain the same, the only way to decrease resistance is to increase the

cross-sectional area (A). A thicker wire has more path for electrons, reducing collisions and heat buildup.

- **Circuit Theory (Series Connection):** For a fuse to fulfill its purpose, it must be connected in **series**. In a series circuit, the same current flows through every component. If the fuse melts due to a surge, the path is physically broken, and the current flow to the appliance stops immediately. If connected in parallel, the current would simply bypass the blown fuse and continue to damage the appliance.

Key Safety Facts

- **Low Melting Point:** Fuse wires are typically made of alloys (such as tin-lead) with a low melting point so they can react quickly to excess heat.
- **Current Rating:** Every fuse has a specific "blow current." By increasing the thickness, you effectively increase this rating, allowing the electrical appliance to draw more power safely.
- **Inverse Relationship:** Resistance and thickness are inversely related. As **Thickness (Area)** goes **UP**, the **Resistance** goes **DOWN**, and the **Current Capacity** goes **UP**.

26. Match the concepts in List-I with their correct examples or explanations given in List-II. ***[BPSC ASO, Mains]***

	List-I (Concept)		List-II (Example/ Explanation)
a.	Balanced Forces	1.	Recoil of a gun when a bullet is fired
b.	First Law of Motion	2.	A wooden block pulled equally from both sides remains at rest
c.	Second Law of Motion	3.	Passenger falling backwards when the bus starts moving from rest
d.	Third Law of Motion	4.	A fielder pulls his hands gradually while catching a fast-moving ball

Codes:

A. a-2, b-3, c-4, d-1 B. a-2, b-4, c-3, d-1
C. a-1, b-3, c-4, d-2 D. a-1, b-4, c-3, d-2

Ans. A

Exp: The principles of classical mechanics define how physical systems achieve equilibrium or undergo acceleration in response to external stimuli.

- These interactions are categorized by the following physical laws and mathematical frameworks:

Laws of Motion and Force Analysis:

1. **Balanced Forces (Net Force = 0):** When multiple forces act upon a body such that their vector sum is zero, the object maintains its state of translational equilibrium. This is demonstrated by a **wooden block pulled equally from both sides**; the opposing force vectors cancel, resulting in zero net acceleration.
2. **Newton's First Law: The Law of Inertia (If F = 0, then v = constant):** This law states that an object will remain at rest or move at a constant velocity unless acted upon by a net external force. A **passenger falling backwards when a bus starts moving** is a direct result of inertia: while the bus and the passenger's lower body move forward, the passenger's upper body tends to remain at rest.
3. **Newton's Second Law: Law of Momentum ($F = \Delta p / \Delta t$):** Force is mathematically defined as the rate of change of linear momentum over time. A **fielder pulling their hands gradually while catching a fast-moving ball** utilizes this principle by increasing the time interval (**t**) of the impact. According to the formula, as the duration of momentum change increases, the impact force (**F**) exerted on the hands decreases, preventing injury.
4. **Newton's Third Law: Action and Reaction ($F\_action = -F\_reaction$):** For every force exerted by one object on another, there is a force of equal magnitude and opposite direction exerted back on the first object. The **recoil of a gun** occurs because the force required to accelerate the bullet forward (action) generates an instantaneous and equivalent force in the opposite direction (reaction) against the firearm and the shooter.

27. When sunlight enters a raindrop and a primary rainbow is formed, which sequence of optical phenomena correctly describes the path of light? ***[BPSC ASO, Mains]***

A. Dispersion → Refraction → Total Internal Reflection → Refraction
B. Refraction → Internal Reflection → Refraction → Dispersion
C. Dispersion → Refraction → Internal Reflection → Refraction
D. Refraction → Dispersion → Internal Reflection → Refraction

Ans. D

Exp: Light is an electromagnetic wave that travels in a straight line in a uniform medium. However, when light encounters a boundary between two media, such as air and water, its velocity changes due to differences in **optical density**. This interaction triggers a series of events that can redistribute energy or separate light into its constituent wavelengths. A primary rainbow is a meteorological phenomenon caused by these specific interactions—**refraction**, **dispersion**, and **reflection**—within spherical raindrops acting as tiny prisms.

- The formation of a rainbow involves several distinct physical processes that occur in a precise chronological order:
- **Refraction:** This occurs when light changes speed and direction as it passes from one medium to another (e.g., from air into a water drop). It is governed by **Snell's Law**, which states that the ratio of the sines of the angles of incidence and refraction equals the ratio of the refractive indices ($\mathbf{n_1 \sin \theta_1 = n_2 \sin \theta_2}$).
- **Dispersion:** Because the refractive index of water varies slightly for different wavelengths (colors), violet light bends more than red light upon entering the drop. This separates white sunlight into a continuous spectrum of colors.
- **Internal Reflection:** Once the dispersed light reaches the back surface of the raindrop, it strikes the water-air interface from the inside. Instead of exiting, a portion of the light reflects back into the droplet. In a primary rainbow, this occurs exactly **once**.
- **Total Internal Reflection (TIR):** While often confused with internal reflection, TIR only occurs when light travels from a denser to a rarer medium at an angle greater than the **critical angle**. In most primary rainbows, the reflection at the back of the drop is standard internal reflection rather than 100% TIR.

BCW Bits:

- **Scattering:** The redirection of light in various directions due to particles in the atmosphere. Rayleigh scattering (shorter wavelengths like blue scatter more) is why the sky appears blue.
- **Diffraction:** The slight bending of light waves as they pass around the edge of an object or through a narrow opening.
- **Interference:** The phenomenon where two light waves superimpose to form a resultant wave of greater, lower, or the same amplitude. This is often seen in the iridescent colours of soap bubbles.
- **Polarization:** The process of restricting the vibrations of light waves to a single plane, often used in sunglasses to reduce glare from reflective surfaces.

28. What will happen if a running fridge door is opened in a room? ***[BPSC DSO, Pre]***

A. Room temperature will rise
B. Room temperature will fall
C. Room temperature will not be affected
D. Room temperature first will go up and later on will go down

Ans. A

Exp: The average **room temperature** in the room **will rise.** The reason is that with the door open, the temperature will start to rise inside the refrigerator. The thermostat will kick in and try to cool it back down. This means the motor is running, which means **heat is being added to the room.**

BCW Bits:

- This is a direct application of the **Second Law of Thermodynamics**, a refrigerator is a heat pump, not a cooling device for the room. It absorbs heat from inside (Q_cold) and expels a larger amount of heat at the condenser coils (Q_hot = Q_cold + W, where W is the compressor work). With the door open, the refrigerator simply pumps room heat back into the room plus adds extra heat from the compressor motor, resulting in a net temperature rise.

29. In the winter, we get warmer by using two blankets instead of one, why? ***[BPSC DSO, Pre]***

A. Two blankets compress the air between the body and blanket
B. Two blankets are thicker, so generate more heat
C. Two blankets enclose air which does not allow the cold to penetrate
D. None of the above

Ans. C

Exp: Two thin blankets are warmer than one thick one because they trap a layer of air between them, reducing the heat transfer through conduction and convection, and this acts as an excellent insulator, preventing body heat from escaping to the cold surroundings. Since air is a poor conductor of heat, this extra, still air layer provides superior insulation compared to a single thick blanket.

30. Longitudinal waves cannot travel through? ***[BPSC DSO, Pre]***

A. Gas B. Liquid C. Solid D. Vacuum

Ans. D

Exp: **Longitudinal waves** are waves in which the particles of the medium oscillate along the direction of wave propagation. Longitudinal waves cannot travel through a vacuum. As mechanical waves, they require a material medium—such as solids, liquids, or gases—to propagate. E.g. Sound waves, ultrasound waves, etc.

31. Which specific waves are used in SONAR? ***[BPSC DSO, Pre]***

A. Audible Sound Waves
B. Radio Waves
C. Infrasonic Waves
D. Ultrasonic Waves

Ans. D

Exp: **SONAR** is a human-made technology used in **submarines and ships** to detect underwater objects. In SONAR, the emitter emits **ultrasonic waves of high frequency** (not audible to human ears). These high-frequency sound waves get reflected as echoes upon hitting an obstacle and are sensed by the receiver. This phenomenon is also known as echolocation as it helps in the calculation of the distance at which an obstacle is present.

BCW Bits:

- SONAR is primarily a **naval technology** used in submarines and ships to detect underwater objects, measure ocean depth, and locate enemy vessels, it emits ultrasonic pulses (>20,000 Hz) and calculates distance using the formula $Distance = \frac{Speed \times Time}{2}$
- Bats, dolphins, and whales use the same principle biologically (**echolocation**).

32. What is the function of a microphone? It converts ***[BPSC DSO, Pre]***

A. Electromagnetic waves to sound signals
B. Sound signals to electromagnetic waves
C. Sound signals to Electric signals
D. Electric signals to sound signals

Ans. **C**

Exp: A **microphone changes sound waves into electrical signals**. When we speak, our sound waves hit a tiny moving part called a diaphragm inside the microphone. This movement creates a matching electrical current that copies our voice so it can be recorded or sent to a speaker.

- Converting **electrical signals back into physical sound waves** is the exact opposite process, which is performed by a **loudspeaker or headphones.** Working with **electromagnetic waves involves wireless radio broadcasting**; turning **electrical currents into wireless radio waves requires a transmitter antenna**, while catching those **wireless waves and translating them back into audio is the function of a complete radio receiver system**.

33. Which rays can be used to read very old written documents? ***[BPSC DSO, Pre]***

A. X Rays B. γ Rays C. β Rays D. IR Rays

Ans. **D**

Exp: Old written materials, such as manuscripts that have become faded, charred, or obscured over time, can be read using Infrared (IR) rays. Infrared imaging and photography assists in reading manuscripts by enhancing the contrast between the ink and the background material.

BCW Bits:

- While ultraviolet (UV) light is also commonly used for fluorescence imaging of palimpsests, infrared spectrum imaging is a standard non-destructive technique for recovering text from cultural heritage objects that are otherwise illegible.
- IR rays are particularly effective because many inks absorb or reflect infrared light differently than the underlying parchment or paper, allowing hidden or faded text to become visible to specialized sensors.
- Although X-ray fluorescence (XRF) is used for high-value obscured texts, IR-rays remain a primary and more accessible method for general manuscript recovery and legibility enhancement.

34. Which force is responsible for separation of cream from the churned milk? ***[BPSC DSO, Pre]***

A. Centripetal force B. Centrifugal force
C. Gravitational force D. Cohesive force

Ans. **B**

Exp: The separation of cream from milk relies on centrifugal force, an outward pseudo-force experienced by objects in a rotating frame of reference. When milk is spun rapidly in a churner, the components rotate around a central axis at high speed.

- This inertia pushes the denser skimmed milk outward toward the container walls, displacing the lighter, less dense cream particles inward toward the center, where they collect to be skimmed off.
  - **Centripetal force** is the actual inward force that pulls an object toward the center to maintain its circular path, rather than pushing components outward.
  - **Gravitational force** is the natural attraction between masses; while it can cause natural sedimentation over time, it is too weak to quickly separate cream without rapid rotation.
  - **Cohesive force** is the intermolecular attraction holding similar molecules together, which works to resist separation rather than driving it.

Uses of centrifugal force:

- In diagnostic laboratories for blood and urine tests.
- In the washing machine, to squeeze out water from wet clothes.
- In dairies and home to separate butter from cream.

35. Who is considered as the father of 'Artificial Intelligence'? ***[BPSC DSO, Pre]***

A. Geoffrey Hinton B. Sam Altman
C. John McCarthy D. Yann Le Cun

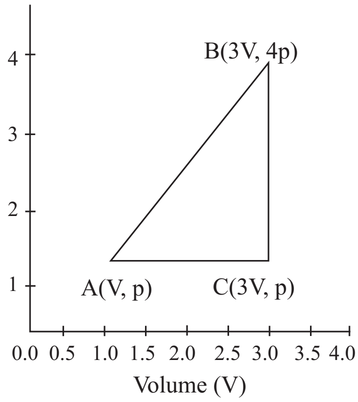
Ans. **C**

Exp: John McCarthy is generally recognized as the "father of AI" for coining the term in 1956 and organizing the Dartmouth Conference, which established the field. He was an American computer scientist who developed the Lisp programming language and created the concept of "time-sharing".

- **Geoffrey** Everest **Hinton** is widely regarded as the

father of **deep learning**, who won the **Turing Award (2018) along with Yann LeCun** and **Yoshua Bengio** for breakthroughs in deep learning and the **Nobel Prize in Physics (2024)** for foundational discoveries enabling machine learning with artificial neural networks. **Sam Altman is** the **CEO of OpenAI**, responsible for the commercial scaling of generative AI tools like **ChatGPT**.

36. An ideal gas is taken around a closed cycle ABCA as shown in the p–V diagram below. The coordinates of the points are: A(V, p); B(3V, 4p); C(3V, p). What is the work done during one complete cycle? ***[BPSC AEDO]***



A. pV B. 2pV
C. 6pV D. 3pV

Ans. D

Exp:

- In a p–V diagram, the work done in one complete cycle = area enclosed by the cycle , So instead of doing complicated formulas, just: Find the area of the shape formed by A, B, C.
- **Step 1: Understand the shape,** Points are: A → (V, p), B → (3V, 4p), C → (3V, p) form a triangle on the p–V graph.
- **Step 2: Calculate base and height,** Base (along volume axis): = 3V − V = **2V,** Height (along pressure axis): = 4p – p = **3p**
- **Step 3: Area of triangle,** Work done = Area enclosed = **½ × base × height** = ½ × (2V) × (3p) = **3pV**

BCW Bits:

- Clockwise cycle- work done by gas (positive)
- Anticlockwise- work done on gas (negative)
- Here, the path ABCA is clockwise — hence work is done by the gas (positive work).
- The SI unit of work done is joule.

37. Given below are two statements, one is labelled as Assertion (A) and the other is labelled as Reason (R). In light of the statements given below, choose the most appropriate answer from the options.

Assertion (A): In simple harmonic motion, the potential energy of the particle is maximum at the extreme positions.

Reason (R): At the extreme positions, the velocity of the particle is maximum. ***[BPSC AEDO]***

A. Both (A) and (R) are true and (R) is the correct explanation of (A)
B. Both (A) and (R) are true but (R) is not the correct explanation of (A)
C. (A) is false but (R) is true
D. (A) is true but (R) is false

Ans. D

Exp:

- SHM is a type of motion where a particle **moves back and forth around a mean (central) position**, like a spring or a pendulum (small oscillations). The key idea: **restoring force always pulls it back toward the centre**
- Mean position- central point (equilibrium), Extreme positions- farthest points from the centre, Velocity- speed of the particle, Potential energy (PE) - energy due to position.
- Potential energy is maximum at the extreme positions because the displacement from the mean position is maximum there.
- **Velocity in SHM:** At the mean position, the velocity is maximum. At the extreme position velocity becomes zero. Because at the extreme point, the particle stops for a moment before coming back.

BCW Bits:

- Total mechanical energy in SHM is constant (PE + KE = constant)
- When PE is maximum, KE is zero, and vice versa
- SHM is an example of periodic motion (repeats after equal time intervals)
- Acceleration is always directed towards the mean position, even at extremes

38. In an Alternating Current (AC) circuit, the phase relationship between current and voltage depends on the circuit elements. Match the entries in Column – I with the appropriate relations in Column – II. ***[BPSC AEDO]***

Column I Type of Circuit		**Column II Phase Relationship between Current and Voltage**	
i	Pure Resistance (R)	p	Current leads voltage by 90°
ii	Pure Capacitance (C)	q	Current and voltage are in same phase
iii	Pure Inductance (L)	r	Current and voltage may or may not be in same phase
iv	R – L – C Series Circuit	s	Current lags voltage by 90°

Select the correct matching option:

	i	ii	iii	iv
A.	p	s	q	r
B.	q	p	s	r
C.	q	s	p	r
D.	s	r	p	q

Ans. B

Exp:

- AC is a current that changes direction and magnitude with time (like household electricity). Because it's changing, current and voltage don't always peak together—the difference between them is called phase difference. The phase relationship depends on what's in the circuit: R, L, C. **Resistance (R)** → no delay, **Capacitor (C)** → current goes ahead, **Inductor (L)** → current falls behind.
- **Pure Resistance**: - No energy storage, no delay. Current and voltage are in the same phase, Phase difference = 0°; **Pure Capacitance (C) :-** Capacitor charges/discharges quickly, Current leads voltage by 90°; **Pure Inductance (L) :-** Inductor resists change in current, Current lags voltage by 90°.
- R-L-C Series Circuit:- Current and voltage may or may not remain in the same phase depending on the relative values of resistance, inductance and capacitance.

BCW Bits:

- In RLC circuits, when inductive and capacitive effects balance, it is called resonance
- At resonance, current and voltage become in phase again
- Impedance becomes minimum at resonance.
- Power factor depends on phase difference - important in real-life electricity use
- AC circuits are widely used because they are efficient for long-distance transmission

39. Given below are two statements, one is labelled as Assertion (A) and the other is labelled as Reason (R). In light of the statements given below, choose the most appropriate answer from the options.

Assertion (A): Gauss's law in magnetostatics and electrostatics are respectively,

$$\oint \vec{B} \cdot d\vec{S} = 0 \text{ and } \oint \vec{E} \cdot d\vec{S} = \frac{q_{enclosed}}{\varepsilon_0}$$

Reason (R): Magnetic monopoles do not exist, whereas electric monopoles (isolated charges) exist.

[BPSC AEDO]

A. Both (A) and (R) are true and (R) is the correct explanation of (A)
B. Both (A) and (R) are true but (R) is not the correct explanation of (A)
C. (A) is false but (R) is true
D. (A) is true but (R) is false

Ans. A

Exp: Gauss's law relates the net flux through a closed surface to the source enclosed within that surface.

For electrostatics:

$$\oint \vec{E} \cdot d\vec{S} = \frac{q_{enclosed}}{\varepsilon_0}$$

- Electric charges exist as isolated positive and negative charges (electric monopoles). Therefore, a closed surface can enclose a net electric charge, producing non-zero electric flux.

For magnetostatics:

$$\oint \vec{B} \cdot d\vec{S} = 0$$

- Magnetic field lines always form closed loops. Since isolated magnetic poles (magnetic monopoles) have never been observed, a closed surface cannot enclose a net magnetic charge. Hence, the net magnetic flux through any closed surface is always zero.
- Therefore, both the Assertion and the Reason are true, and the Reason correctly explains the Assertion.

BCW Bits:

- Gauss's law is one of the four Maxwell's equations.
- Magnetic field lines inside a magnet go south → north internally, forming loops.
- Scientists are still searching for magnetic monopoles, but none have been confirmed yet.
- These laws are very useful for solving problems with symmetry (sphere, cylinder, plane).

40. Which of the following statements is correct?
As the temperature increases, ***[BPSC AEDO]***

A. The resistivity of a semiconductor increases, whereas the resistivity of a conductor decreases.
B. The resistivity of a semiconductor decreases, whereas the resistivity of a conductor increases.
C. The resistivity of both semiconductor and conductor decreases.
D. The resistivity of both semiconductor and conductor increases.

Ans. B

Exp:

- **Resistivity** (ρ) measures how strongly a material opposes the flow of electric current. Higher resistivity means greater opposition to current flow.
- **Conductors** such as copper and aluminium contain a large number of free electrons. As temperature increases, the atoms of the conductor vibrate more vigorously. These vibrations increase collisions between electrons and atoms, reducing electron mobility. Therefore, the resistivity of a conductor increases with temperature.

- **Semiconductors** such as silicon and germanium have relatively few free charge carriers at low temperatures. As temperature increases, more electrons gain sufficient energy to move into the conduction band, increasing the number of charge carriers. Hence, current flows more easily and the resistivity decreases.
- Therefore, with increase in temperature:
  - **Conductor → Resistivity increases**
  - **Semiconductor → Resistivity decreases**

BCW Bits:

- Resistivity depends on the nature of the material and temperature.
- Conductors have a **positive temperature coefficient of resistance**.
- Semiconductors have a **negative temperature coefficient of resistance**.
- Thermistors are temperature-sensitive resistors based on semiconductor properties.
- Silicon and germanium are the most common semiconductors.
- At absolute zero, an intrinsic semiconductor behaves like an insulator.
- In superconductors, resistivity becomes zero below a critical temperature.

41. A particle is found to be at rest when observed from a reference frame S_1, and the same particle is found to move with a constant velocity when observed from another frame S_2. Which of the following statements is true? ***[BPSC AEDO]***

A. S_1 is an inertial frame, but S_2 is a non-inertial frame.
B. S_2 is an inertial frame, but S_1 is a non-inertial frame.
C. Both S_1 and S_2 are non-inertial frames.
D. Both S_1 and S_2 are inertial frames.

Ans. D

Exp:

- **An inertial frame is a frame where a body at rest stays at rest. A body in motion continues with constant velocity (no acceleration). In short, Newton's laws work normally.**
- If a particle is at rest in one frame (S_1) and moves with constant velocity in another frame (S_2), this implies that the two frames move with constant relative velocity.
- If one frame is inertial and the second frame moves at **constant velocity (no acceleration)** relative to it, then the second frame is **also inertial.**
- Only **acceleration** makes a frame non-inertial; constant velocity does **not** break inertial nature.

BCW Bits:

- Example:
  - Sitting in a moving train (constant speed) → you feel at rest.
  - The person outside sees you moving → both frames are inertial.
- Non-inertial frames involve:
  - Acceleration
  - Rotation
- In non-inertial frames, we introduce pseudo forces (like centrifugal force).
- Galileo's principle: laws of physics are the same in all inertial frames.

42. Consider the interaction forces between two protons separated by a distance of the order of 1 femtometer (10^{-15} m). The correct order of the strengths of the electromagnetic, gravitational and nuclear forces acting between them is: ***[BPSC AEDO]***

A. Gravitational > Electromagnetic > Nuclear
B. Electromagnetic > Nuclear > Gravitational
C. Nuclear > Gravitational > Electromagnetic
D. Nuclear > Electromagnetic > Gravitational

Ans. D

Exp:

- Between two protons, three main forces can act: Nuclear force (strong force), Electromagnetic force (electric repulsion), Gravitational force.
- Nuclear (Strong) Force: Strongest force in nature. Works only at a very short range (nucleus level). Holds protons and neutrons together. Even though protons repel electrically, this force overcomes that repulsion.
- Electromagnetic Force: Causes repulsion between two protons; strong, but weaker than nuclear force at nuclear scale.
- Gravitational Force: Extremely weak at the atomic level, almost negligible compared to other forces.

BCW Bits:

- Strong force is mediated by gluons.
- Electromagnetic force is mediated by photons.
- Gravity is weakest but acts over an infinite range.
- Without the strong force, atomic nuclei would not exist.
- Strong force becomes negligible beyond a few femtometers.
- This is why nuclei have limited size.

43. Two astronauts on the moon are standing 10 m apart. Astronaut A shouts toward Astronaut B and simultaneously flashes a light signal. Which of the following statements is correct? ***[BPSC AEDO]***

A. Both sound and light from Astronaut A will reach Astronaut B, but light will arrive slightly earlier.
B. Both sound and light from Astronaut A will reach Astronaut B at equal speed.
C. Neither sound nor light from Astronaut A will reach Astronaut B.

D. The light signal from Astronaut A will reach Astronaut B but not the sound.

Ans. D

Exp:

- The moon has no atmosphere, so it is essentially a vacuum (no air). This is the most important thing to remember. Sound needs a medium (like air) to travel.
- On Earth → sound travels through air; on the Moon → no air → no sound transmission. So, sound will NOT reach Astronaut B. Light does not need any medium; it can travel in a vacuum. That's how sunlight reaches Earth. So, light WILL reach Astronaut B.

BCW Bits:

- That's why in space, explosions are silent.
- Astronauts communicate using radio signals (electromagnetic waves)
- Speed of light = 3×10^8 m/s
- Sound speed depends on medium (~340 m/s in air)
- Light can travel through vacuum, but sound cannot.

44. A car moves along a straight road, covering a displacement x with a constant velocity v_1 and then another equal displacement x with a constant velocity v_2. What is the average velocity of the car for the entire journey? ***[BPSC AEDO]***

A. $\frac{v_1 + v_2}{2}$

B. $\sqrt{v_1 v_2}$

C. $\frac{v_1^2 + v_2^2}{v_1 + v_2}$

D. $\frac{2v_1 v_2}{v_1 + v_2}$

Ans. D

Exp: Average velocity = Total displacement / Total time

Total displacement:

$x + x = 2x$

Time taken in each part:

$$t_1 = \frac{x}{v_1} t_2 = \frac{x}{v_2}$$

Total time:

$$t_1 = \frac{x}{v_1} + \frac{x}{v_2}$$

Applying the formula:

$$v_{avg} = \frac{2x}{\frac{x}{v_1} + \frac{x}{v_2}}$$

Taking common in the denominator and cancelling it:

$$v_{avg} = \frac{2x}{x\left(\frac{v_1 + v_2}{v_1 v_2}\right)} v_{avg} = \frac{2v_1 v_2}{v_1 + v_2}$$

Therefore, the average velocity of the car is:

$$v_{avg} = \frac{2v_1 v_2}{v_1 + v_2}$$

BCW Bits:

- If distances are equal → harmonic mean, if times are equal → simple average
- This concept is frequently asked in Speed-time problems and physics + aptitude questions. Always check whether distance or time is equal before applying the formula.

45. If we shine green light on a tree, the tree will ______: ***[BPSC AEDO]***

A. will camouflage in the light

B. appear darker

C. appear the same as before

D. appear brighter

Ans. C

Exp:

- The green colour of most leaves is due to the pigment **chlorophyll**, which absorbs mainly red and blue light and reflects green light.
- An object appears coloured because it reflects certain wavelengths of light and absorbs others. A green leaf reflects green light and absorbs most of the red and blue components.
- When green light is shone on the tree, the leaves continue to reflect the incident green light. Since the reflected light reaching the observer's eyes is still green, the tree appears green as before.

  Green Leaf + Green Light → Reflection of Green Light → Green Appearance

  Therefore, the tree appears the same as before.

BCW Bits:

- **Chlorophyll** is the main green pigment in plants.
- Visible light ranges approximately from **400 nm to 700 nm**.
- Green light has a wavelength of about **495–570 nm**.
- White light contains all colours of the visible spectrum.
- A white object reflects most incident light and appears the colour of the illuminating light.
- A black object absorbs most incident light and appears dark.
- Red and blue light are most effective for photosynthesis.
- Colour perception depends on the light reflected from an object reaching the retina.

46. Arrange the following electromagnetic radiations with correct decreasing order of their associated frequencies from the code given below: ***[BPSC AEDO]***

(i) Infra-red radiation

(ii) Microwave radiation

(iii) Ultra-violet radiation

What is the correct relationship among them regarding frequencies?

A. (ii), (iii), (i) B. (iii), (i), (ii)

C. (iii), (ii), (i) D. (i), (ii), (iii)

Ans. B

Exp:

- Frequency and wavelength are inversely related. $c = \nu\lambda$. Where c = speed of light, ν = frequency, λ= wavelength. This means higher frequency → lower wavelength, lower frequency → higher wavelength. In the electromagnetic spectrum Ultra-violet rays have very high frequency and high energy. Infra-red rays have lower frequency than visible light but higher than microwaves. Microwaves have comparatively lower frequency and longer wavelength. So among the given options UV has the highest frequency, Infra-red comes next, Microwaves have the lowest frequency.

BCW Bits:

- Visible light lies between infrared and ultraviolet regions.
- Violet color has higher frequency than red color.
- Gamma rays have maximum penetrating power.
- X-rays are widely used in medical imaging.
- Infrared radiation is associated with heat.
- Microwaves are used in radar and microwave ovens.
- Radio waves are used in communication systems.
- Ozone layer absorbs most harmful ultraviolet radiation.
- Frequency is measured in hertz (Hz).
- Electromagnetic waves do not require a material medium for propagation.

47. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

[BPSC AEDO]

Assertion (A): At the highest position in a vertical circular motion, tension in the string is minimum.

Reason (R): At the highest point, the tension = centrifugal force − weight of body. This is the minimum value for tension throughout the motion.

In the light of the above statements, choose the most appropriate answer from the options given below:

A. Both (A) and (R) are correct and (R) is the correct explanation of (A)

B. (A) is correct but (R) is incorrect

C. (A) is incorrect but (R) is correct

D. None of the above is correct

Ans. B

Exp:

- In vertical circular motion, the tension in the string is minimum at the highest point of the circle.
- At the highest point, both tension and weight act toward the centre and together provide the required centripetal force.

$$T + mg = \frac{mv^2}{r}$$

Therefore,

$$T = \frac{mv^2}{r} - mg$$

- Since weight assists in providing the centripetal force, the string has to provide a smaller force. Hence, tension is minimum at the top.
- The Reason is incorrect because it explains the situation in terms of **centrifugal force**, which is a pseudo force and is not used in the standard inertial-frame analysis of vertical circular motion. The correct explanation is based on the centripetal force requirement.

BCW Bits:

- Centripetal force always acts toward the center of the circle.
- Circular motion requires continuous change in direction of velocity.
- Velocity is a vector quantity.
- Uniform circular motion has constant speed but changing velocity.
- Centrifugal force is a pseudo force observed in non-inertial frames.
- SI unit of force is Newton.
- Gravity acts vertically downward toward Earth's center.
- Banking of roads helps vehicles move safely on curves.
- Satellites revolve due to gravitational centripetal force.
- Angular velocity is measured in radian per second.

48. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

[BPSC AEDO]

Assertion (A): When a pendulum is taken to the Moon, its time period will decrease.

Reason (R): At the Moon, the value of gravitational constant 'g' will increase that in turn decreases the time period of the pendulum.

In the light of the above statements, choose the most appropriate answer from the options given below:

A. Both (A) and (R) are correct and (R) is the correct explanation of (A)

B. (A) is correct but (R) is incorrect

C. (A) is incorrect but (R) is correct

D. None of the above is correct

Ans. D

Exp: The time period of a simple pendulum is given by:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

where:

- T = time period,
- g = length of the pendulum,
- g = acceleration due to gravity.
- From the formula, the time period is inversely proportional to the square root of g. Therefore, if g decreases, the time period increases.
- **Assertion (A) is false.** On the Moon, the acceleration due to gravity is about one-sixth of that on Earth: $g_{Moon} \approx \frac{g_{Earth}}{6}$
- Hence, the pendulum oscillates more slowly and its time period increases, not decreases.
- **Reason (R) is also false.** The value of on the Moon does not increase; it decreases. Moreover, g represents **acceleration due to gravity**, whereas the **universal gravitational constant** is denoted by g.

BCW Bits:

- SI unit of acceleration due to gravity is m/s².
- Average value of g on Earth is about 9.8 m/s².
- The Moon's gravity is about one-sixth that of Earth.
- Time period of a pendulum depends on length and gravity.
- Mass of the pendulum bob does not affect time period.
- Galileo studied pendulum motion.
- Seconds pendulum has time period of 2 seconds.
- Universal gravitational constant G was measured by Cavendish.
- Weight decreases on the Moon but mass remains unchanged.
- Pendulums are used in clocks to measure time.

49. The direction of the magnetic field around a current-carrying wire can be determined by using: ***[BPSC AEDO]***

A. Ohm's law
B. Fleming's left-hand rule
C. Lenz's law
D. Right-hand thumb rule

Ans. D

Exp:

- According to the **Right-hand Thumb Rule**, if a straight current-carrying conductor is held in the right hand such that the thumb points in the direction of current, then the curled fingers indicate the direction of the magnetic field lines around the conductor.
- The magnetic field produced around a straight current-carrying wire consists of concentric circular field lines centered on the wire.
- The strength of the magnetic field increases with current: $B \propto I$
- Thus, a larger current produces a stronger magnetic field.
- Therefore, the direction of the magnetic field around a current-carrying conductor is determined using the **Right-hand Thumb Rule**.
- Other options are incorrect:
  - **Ohm's Law:** relates voltage, current, and resistance (IR).
  - **Fleming's Left-hand Rule:** gives the direction of force in an electric motor.
  - **Lenz's Law:** determines the direction of induced current in electromagnetic induction.

BCW Bits:

- Hans Christian Oersted discovered magnetic effect of current.
- Magnetic field lines never intersect each other.
- SI unit of magnetic field is tesla.
- Electromagnets work using current-carrying coils.
- Electric motors convert electrical energy into mechanical energy.
- Generators convert mechanical energy into electrical energy.
- Solenoids produce magnetic fields similar to bar magnets.
- Electromagnetic induction was discovered by Faraday.
- Current is measured in ampere.
- Transformers work on the principle of electromagnetic induction.

50. Which of the following laws support the fact that it is difficult to fix a nail on a freely suspended wooden frame? ***[BPSC AEDO]***

A. Pascal's law
B. Newton's second law of motion
C. Newton's third law of motion
D. Newton's law of inertia

Ans. C

Exp:

- **Newton's Third Law of Motion** states that for every action, there is always an equal and opposite reaction. This means that forces always exist in pairs; if Object A exerts a force on Object B, then Object B exerts an equal force back on Object A in the opposite direction.
- When you attempt to fix a nail into a wooden frame, the hammer exerts a large force (action) on the nail. For the nail to penetrate the wood, the wooden frame must provide a sufficient opposing reaction force to keep the nail from simply moving the entire frame forward.
- In the case of a freely suspended wooden frame, there is no rigid support behind it to provide this necessary opposing reaction force. Instead of the force being

utilized to overcome the internal friction of the wood (which would allow the nail to enter), the force causes the freely suspended frame to accelerate and move away.

- Because the reaction force provided by a suspended object is significantly less than that of a fixed object, the nail does not pierce the surface effectively. Hence, the statement regarding the difficulty of fixing a nail on a suspended frame is directly supported by Newton's third law.
- **Why the other options are incorrect: Newton's First Law** (Inertia) describes the frame's resistance to moving, but it doesn't explain the missing opposing force required for the nail to actually penetrate. **Newton's Second Law** (F = ma) calculates the acceleration of the swinging frame after it is hit, but doesn't address the penetration mechanics. **Pascal's law** is entirely unrelated, as it deals with pressure transmission in confined fluids.

BCW Bits:

- Inertia depends on mass of the object.
- More mass means greater inertia.
- SI unit of force is Newton.
- Momentum is the product of mass and velocity.
- Friction opposes motion between surfaces.
- Pascal's law is used in hydraulic brakes and lifts.
- Rockets move according to Newton's third law.
- Seat belts help due to the principle of inertia.
- Galileo laid the foundation for the concept of inertia.

51. When are the two vibrating particles said to be in the same phase? ***[BPSC AEDO]***

A. If the phase difference between them is an even multiple of π
B. If the path difference is an even multiple of (λ/2)
C. If the time interval is an even multiple of (T/2)
D. All of the above

Ans. D

Exp: Two vibrating particles are said to be in the **same phase** when they have the same displacement and are moving in the same direction at the same instant of time.

- The condition for the same phase is that the phase difference should be an even multiple of π:

$\Delta\phi = 2n\pi$

where $n = 0, 1, 2, 3, ...$

Phase difference and path difference are related by:

$$\Delta\phi = \frac{2\pi}{\lambda}\Delta x$$

If the path difference is an even multiple of $(\lambda/2)$,

$$\Delta x = 2n\left(\frac{\lambda}{2}\right) = n\lambda$$

Substituting in the above relation:

$$\Delta\phi = \frac{2\pi}{\lambda}(n\lambda) = 2n\pi$$

Hence, the particles are in the same phase.

Similarly, phase difference and time difference are related by:

$$\Delta\phi = \frac{2\pi}{T}\Delta t$$

If the time interval is an even multiple of $\left(\frac{T}{2}\right)$,

$$\Delta t = 2n\left(\frac{T}{2}\right) = nT$$

Therefore,

$$\Delta\phi = \frac{2\pi}{T}(nT) = 2n\pi$$

- Thus, the particles complete full cycles and return to the same phase.
- Hence, all the given conditions correspond to particles being in the same phase.

BCW Bits:

- Wavelength is represented by the symbol λ.
- SI unit of frequency is hertz (Hz).
- Frequency and time period are inversely related.
- Sound waves are mechanical waves.
- Light waves are electromagnetic waves.
- Constructive interference produces maximum intensity.
- Destructive interference produces minimum intensity.
- Amplitude determines loudness in sound waves.
- Frequency determines pitch of sound.

52. A galvanometer can be converted into a voltmeter by connecting it with which of these? ***[BPSC AEDO]***

A. Low resistance wire in series
B. Low resistance wire in parallel
C. High resistance wire in series
D. High resistance wire in parallel

Ans. C

Exp:

- A galvanometer is a sensitive instrument used to detect and measure small electric currents. To convert a galvanometer into a voltmeter, a high resistance is connected in series with it.
- A voltmeter is used to measure the potential difference between two points in a circuit. Ideally, a voltmeter should draw very little current from the circuit. For this reason, it must have very high resistance.
- When a high-resistance component is connected in series with a galvanometer, the total resistance becomes very large, allowing only a small current to pass through the instrument while measuring the voltage safely. The

relation between voltage, current, and resistance is given by Ohm's law: V = IR.

- A low resistance would allow a larger current to pass, which could damage the galvanometer and would not make an effective voltmeter.
- Voltmeter: To *create* one, connect a **high resistance in series** with a galvanometer. To *use* one, connect it in **parallel** across circuit components.
- Ammeter: To *create* one, connect a **low resistance (shunt) in parallel** with a galvanometer. To *use* one, connect it in **series** within the circuit.

BCW Bits:

- Galvanometer detects small electric currents.
- Ammeter measures electric current.
- Voltmeter measures potential difference.
- Shunt resistance is used with ammeters.
- Ideal voltmeter has infinite resistance.
- Ideal ammeter has zero resistance.
- Ohm's law was given by Georg Simon Ohm.

53. Stars twinkle because: ***[BPSC AEDO]***

A. Due to motion of earth
B. The distance of the stars from the Earth changes with time
C. The refractive index of the different layers of the Earth's atmosphere changes continuously
D. The intensity of light emitted by them changes with time

Ans. C

Exp:

- As light from a distant star enters Earth's atmosphere, it passes through several layers of air having different temperatures and densities. Since refractive index depends on density, the path of light bends continuously. This phenomenon is called **atmospheric refraction**.
- Because of this continuous refraction, the apparent position of the star changes slightly, and the apparent brightness of the star also fluctuates. As a result, the star appears to twinkle.
- Stars are very far away and behave like **point sources** of light. Even a small atmospheric disturbance can cause noticeable variation in brightness. Planets usually do not twinkle much because they appear as extended sources rather than point sources, so the refraction effects from different parts of the planet's disk average out.
- Twinkling of stars → atmospheric refraction. Blue color of the sky → scattering. Rainbow formation → dispersion plus refraction plus reflection. Mirage → total internal reflection.

BCW Bits:

- Refraction is the bending of light when it passes from one medium to another.
- Refractive index depends on the optical density of the medium.
- Total internal reflection occurs when light travels from denser to rarer medium.
- Rainbow formation involves dispersion of sunlight.
- The atmosphere contains layers with varying temperature and density.
- The Sun appears reddish during sunrise and sunset due to scattering.

54. The significance of Hubble constant is: ***[BPSC AEDO]***

A. It calculates luminosity of galaxies
B. It determines the mass of Black Holes
C. It estimates the age of stars
D. It measures the rate of expansion of the universe

Ans. D

Exp:

- The Hubble constant is the proportionality constant in Hubble's law: $\mathbf{v = H_0 d}$, where **v** is the recession velocity of a galaxy, $\mathbf{H_0}$ is the Hubble constant, and **d** is the distance of the galaxy. This relation shows that the farther a galaxy is, the faster it recedes.
- The significance of the Hubble constant is that it measures the **current rate of expansion of the universe**. This observation provided strong evidence for the expanding universe and became one of the foundations of Big Bang cosmology.

BCW Bits:

- Edwin Hubble provided evidence for the expanding universe.
- Galaxies moving away show redshift in their light spectrum.
- Cosmology studies the origin and evolution of the universe.
- The James Webb Space Telescope studies distant galaxies and cosmic evolution.
- Redshift indicates increasing distance between galaxies.

55. **Statement:** An object of mass M is falling freely and its acceleration remains constant.

Conclusion I: The velocity of the object increases uniformly.

Conclusion II: The acceleration of the object depends on its mass. ***[BPSC AEDO]***

Choose the correct conclusion of the given statement:

A. Only I conclusion is correct
B. Only II conclusion is correct
C. Both conclusions are correct
D. Both conclusions are not correct

Ans. A

Exp:
- When an object falls freely under gravity and its acceleration remains constant, its velocity increases uniformly with time. This happens because free fall motion is governed by constant gravitational acceleration, usually written as **g**.
- Near the Earth's surface, gravitational acceleration is approximately **9.8 m/s²**. This means the velocity of a freely falling body increases by about **9.8 m/s every second**, provided air resistance is neglected. Since acceleration is constant, the increase in velocity is uniform.
- The motion of a freely falling body is described by the equations of motion, such as $v = u + gt$ and $s = ut + 1/2\ gt^2$. These equations show that velocity changes at a steady rate when acceleration is constant.
- According to Newton's laws, the gravitational force on a body is $F = mg$. Also, from Newton's second law, $F = ma$. Equating both gives $ma = mg$, so $a = g$. This shows that acceleration due to gravity does **not** depend on the mass of the object.
- Therefore, Conclusion I is correct, but Conclusion II is incorrect.

BCW Bits:
- Free fall means motion under gravity alone without any other external force.
- The value of **g** slightly varies from place to place on Earth.
- Gravitational acceleration is maximum at the poles and minimum at the equator.
- On the Moon, gravity is about one-sixth of Earth's gravity.
- Air resistance is also called drag force.
- Velocity is a vector quantity; speed is a scalar quantity.
- Acceleration can be positive or negative depending on direction.
- Newton's First Law is also known as the Law of Inertia.
- Galileo's Leaning Tower of Pisa experiment demonstrated equal acceleration of falling bodies.
- Escape velocity is the minimum speed needed to escape Earth's gravitational pull.
- In vacuum, all bodies fall with the same acceleration regardless of mass.

56. Statement: A black body is an ideal absorber and emitter of radiation. ***[BPSC AEDO]***

Conclusion I: Radiation is not reflected from a black body.

Conclusion II: A black body emits more energy at high temperature.

Choose the correct conclusion of the given statement:

A. Only I conclusion is correct
B. Only II conclusion is correct
C. Both conclusions are correct
D. Both conclusions are not correct

Ans. C

Exp:
- A black body is an ideal physical object that absorbs all incident electromagnetic radiation falling on it, regardless of wavelength or angle of incidence. Since it absorbs all incoming radiation completely, it does not reflect radiation. In practice, such a body neither reflects nor transmits radiation.
- The statement also says that a black body is an ideal emitter. A black body emits thermal radiation depending only on its temperature, not on its material composition. As temperature increases, the total energy emitted by a black body increases rapidly.
- The total energy radiated per unit area per second by a black body is given by the **Stefan–Boltzmann law**: $E = \sigma T^4$. This shows that emission increases strongly with temperature. Therefore, a hotter black body emits more energy.
- Hence, **both Conclusion I and Conclusion II are correct**.

BCW Bits:
- A black body is a perfect absorber and perfect emitter.
- A black body does not reflect or transmit radiation.
- Black body radiation depends only on temperature.
- Stefan–Boltzmann law: $E = \sigma T^4$.
- Wien's displacement law says the peak wavelength decreases as temperature increases.
- The Sun behaves approximately like a black body.

57. Assertion (A): The electric field inside a charged metallic sphere is zero.

Reason (R): Charges reside only on the outer surface of a conductor. ***[BPSC AEDO]***

Choose the correct option:

A. Both A and R are true and R is the correct explanation of A
B. Both A and R are true, but R is not the correct explanation of A
C. A is true, but R is false
D. A is false, but R is true

Ans. C

Exp:
- Assertion (A) is correct because a charged metallic sphere is a conductor, and in electrostatic equilibrium the electric field inside a conductor becomes zero. If any internal electric field existed, it would exert force on the free electrons, making them move until the field is cancelled.
- This can also be understood using **Gauss's law**. If we imagine a Gaussian surface inside the metallic sphere, the electric field on that surface is zero, so the enclosed net charge must also be zero. This supports the fact that the interior of the conductor has no electric field.

- Reason (R) is false because the word **"only"** makes it too absolute. Excess charge does move to the outer surface for an isolated conductor, but charges do not always reside exclusively on the outer surface in every case.
- For example, if a conductor has an internal cavity containing a point charge, an equal and opposite charge is induced on the inner surface of the cavity. So charges may also appear on inner surfaces in such situations.
- Hence, **Assertion (A) is true, but Reason (R) is false**.

BCW Bits:

- In electrostatic equilibrium, electric field inside a conductor is zero.
- Excess charge on an isolated conductor lies on its outer surface.
- Gauss's law helps explain why the field inside a conductor is zero.
- Electric field lines are perpendicular to the surface of a conductor.
- Conductors allow free movement of electrons.
- Electrostatic shielding is based on this principle.
- A Faraday cage protects sensitive instruments from external electric fields.

58. In photoelectric effect, Einstein's equation is: ***[BPSC AEDO]***

$E = h\nu - \varphi$

I. φ represents work function of metal.
II. The electrons will not be emitted if $h\nu < \varphi$.
III. Increasing intensity of falling photons increases the kinetic energy of emitted electrons.

Which are correct?

A. I and III only
B. I and II only
C. II and III only
D. All

Ans. B

Exp:

- The given equation is Einstein's famous photoelectric equation, which explains the photoelectric effect using the quantum nature of light. Einstein's photoelectric equation is: **$h\nu = \varphi + K\_max$**, or equivalently: **$K\_max = h\nu - \varphi$**, where: h = Planck's constant, ν = frequency of incident radiation, $h\nu$ = energy of incident photon, **φ (phi)** = work function of the metal, K_max = maximum kinetic energy of emitted electron. This equation states that a part of the photon's energy is used to remove the electron from the metal surface, while the remaining energy appears as kinetic energy of the emitted electron.
- **Statement I is correct.** φ represents the **work function** of the metal — the minimum energy required to remove an electron from the surface. Different metals have different work functions depending on how strongly electrons are bound inside them.
- **Statement II is correct.** If the energy of the incoming photon is less than the work function, electrons cannot escape from the metal surface. Mathematically: **$h\nu < \varphi \Rightarrow$ No photoelectric emission**. This means there exists a **threshold frequency** below which photoelectric emission does not occur, no matter how intense the light is. This was a major observation that classical wave theory failed to explain.
- **Statement III is incorrect.** Increasing the intensity of light increases the **number of photons** striking the surface, which increases the **number of emitted electrons (photocurrent)**. However, it does **not** increase the kinetic energy of individual electrons. The kinetic energy of emitted electrons depends solely on the **frequency** of the incident light, not on its intensity. This is one of the key conclusions of Einstein's quantum theory of the photoelectric effect.

BCW Bits:

- The threshold frequency is the minimum frequency needed for photoelectric emission.
- The stopping potential is used to measure the maximum kinetic energy of photoelectrons.
- Photoelectric emission is almost **instantaneous** after light falls on the surface.
- Photoelectric emission depends on **frequency of light**, not merely intensity.
- Ultraviolet light is generally more effective than visible light because of higher frequency.
- Photoelectric effect supports the **particle nature of light**.
- Compton effect also provided evidence for photon theory.
- Solar cells convert light energy into electrical energy using photoelectric principles.
- Metals with **low work function** are preferred for photoelectric devices.
- Quantum mechanics originated from attempts to explain **black body radiation** and photoelectric effect.
- Einstein was awarded the **Nobel Prize in Physics in 1921** for explaining the photoelectric effect.

59. If electromagnetic waves propagate in vacuum medium: ***[BPSC AEDO]***

I. Electric and magnetic fields oscillate perpendicular to each other.
II. Electromagnetic waves require a material medium to propagate.
III. Electromagnetic waves carry energy and momentum.

Which are correct?

A. I and II   B. II and III
C. I and III   D. All

Ans. C

Exp:

- Electromagnetic waves are special types of waves that can travel even through vacuum, without requiring any material medium. They are produced by accelerating electric charges and consist of oscillating electric and magnetic fields. These waves include radio waves, microwaves, infrared rays, visible light, ultraviolet rays, X-rays, and gamma rays. The electric field and magnetic field oscillate **perpendicular to each other**, and both fields are also **perpendicular to the direction of propagation**. Therefore, electromagnetic waves are called **transverse waves**.
- The arrangement can be represented as: **E ⊥ B ⊥ Direction of propagation**. An important relation between the magnitudes of the two fields is: **E = cB**, where c = speed of light. This perpendicular arrangement is one of the most fundamental characteristics of electromagnetic waves. Therefore, **Statement I is correct**.
- **Statement II is incorrect**. Electromagnetic waves do **not** require a material medium for propagation. This is one of the biggest differences between electromagnetic and mechanical waves. Mechanical waves such as sound need a medium like air, water, or solid to travel. Electromagnetic waves, however, can propagate through vacuum because changing electric fields generate magnetic fields and vice versa. This was mathematically predicted by **James Clerk Maxwell** through his equations. The speed of electromagnetic waves in vacuum is: **$c = 3 \times 10^8$ m/s**, which is also the speed of light.
- **Statement III is correct**. Electromagnetic waves carry both **energy and momentum**. The energy transported allows sunlight to heat Earth, radio signals to transmit information, and X-rays to penetrate body tissues. Electromagnetic radiation also exerts **radiation pressure**, proving it carries momentum.
- The energy of electromagnetic radiation is related to frequency through Planck's relation: **$E = h\nu$**, where E = energy of photon, h = Planck's constant, ν = frequency. Higher-frequency electromagnetic waves carry greater energy.
- Electromagnetic waves form a continuous spectrum called the **electromagnetic spectrum**. The relation among speed, frequency, and wavelength is: **$c = \nu\lambda$**, where c = speed of light, ν = frequency, λ = wavelength. As frequency increases, wavelength decreases.

BCW Bits:

- Radio waves have the **longest wavelength** in the electromagnetic spectrum.
- Gamma rays have the **highest frequency and energy**.
- Visible light is only a tiny part of the electromagnetic spectrum.
- X-rays are widely used in medical imaging.
- Ultraviolet radiation can cause skin damage and tanning.
- Infrared radiation is mainly associated with heat.
- Maxwell is often called the **"Father of Electromagnetism."**
- Electromagnetic waves travel **slower in media** like water or glass than in vacuum.
- Microwaves are used in radar, satellite communication, and microwave ovens.
- Radiation pressure from sunlight affects comet tails and spacecraft propulsion concepts.
- Important relation: **E = cB** (in vacuum, electric field magnitude = c × magnetic field magnitude).

60. Match List-I (Modern Physics Concept) with List-II (Associated Term): ***[BPSC AEDO]***

List-I		List-II	
I.	De Broglie Hypothesis	1.	$\gamma \rightarrow e^- + e^+$
II.	Compton Effect	2.	Splitting of Spectral Lines
III.	Zeeman Effect	3.	Matter Waves
IV.	Pair Production	4.	Photon Scattering

Codes:

A. 1, 2, 3, 4 B. 2, 1, 4, 3
C. 3, 4, 2, 1 D. 4, 1, 2, 3

Ans. C

Exp:

- The **De Broglie Hypothesis** was proposed by Louis de Broglie in 1924. He suggested that particles such as electrons also possess wave nature, just like light. This idea introduced the concept of **matter waves** and became one of the foundations of quantum mechanics. The De Broglie wavelength is given by: **$\lambda = h/p$**, where: λ = wavelength, h = Planck's constant, p = momentum. This equation shows that every moving particle has an associated wavelength.
- The **Compton Effect** is associated with **photon scattering**. Arthur Compton observed that when X-rays collide with electrons, the scattered radiation has a longer wavelength than the incident radiation. This proved that light behaves as particles carrying momentum. The wavelength shift in Compton scattering is: **$\Delta\lambda = h/(m_e c) \times (1 - \cos\theta)$**, where h = Planck's constant, m_e = rest mass of electron, c = speed of light, θ = scattering angle. This effect strongly supported the particle nature of electromagnetic radiation.
- The **Zeeman Effect** refers to the **splitting of spectral lines** when atoms are placed in a magnetic field. It was discovered by Pieter Zeeman. Normally, atoms emit sharp

spectral lines, but under an external magnetic field, these lines split into multiple components due to interaction between the magnetic field and electron motion. This effect became important evidence for electron structure and quantum transitions in atoms.

- The fourth concept is **Pair Production**, where a high-energy gamma photon transforms into an electron and a positron near a nucleus. The reaction is: $\gamma \rightarrow e^- + e^+$, where e^- = electron and e^+ = positron. This process demonstrates conversion of energy into matter according to Einstein's mass-energy relation: $E = mc^2$. For pair production to occur, the gamma ray must possess energy of at least **1.022 MeV** (= 2 × 0.511 MeV), since the rest mass energy of one electron is 0.511 MeV.

BCW Bits:

- Positron is the antiparticle of electron and was discovered by **Carl Anderson**.
- Electron diffraction experiments (Davisson-Germer) experimentally confirmed De Broglie waves.
- Compton Effect cannot be explained by classical wave theory alone.
- Zeeman Effect is used in astrophysics to study magnetic fields of stars.
- Quantum mechanics describes behaviour of matter at atomic and subatomic scales.
- Wave-particle duality applies to both matter and radiation.
- The Compton wavelength of an electron = $h/m_ec \approx 2.43 \times 10^{-12}$ **m**.

61. A moving coil galvanometer can be converted into an ammeter by ________. ***[BPSC AEDO]***

A. High resistance in series
B. Low resistance in parallel
C. High resistance in parallel
D. Low resistance in series

Ans. B

Exp:

- A moving coil galvanometer is a very sensitive instrument used to detect and measure small electric currents. However, it cannot measure large currents directly because excessive current may damage its delicate coil. To convert a galvanometer into an ammeter, a **low resistance called a shunt resistance** is connected **in parallel** with the galvanometer.
- The purpose of the shunt is to allow most of the current to bypass the galvanometer coil. Only a small safe portion of the total current flows through the galvanometer, while the remaining current passes through the low-resistance shunt. This enables the instrument to measure large currents without damaging the galvanometer.
- The arrangement is represented as: $I = I_g + I_s$, where: I = total current, Ig = current through galvanometer, Is = current through shunt resistance. Since the shunt resistance is very low, most current naturally chooses the easier low-resistance path.
- A galvanometer has comparatively high resistance and is designed to respond even to tiny currents. An ammeter, on the other hand, should have **very low resistance** so that it does not significantly alter the circuit current while measuring it. An ammeter is always connected **in series** in a circuit. An **ideal ammeter has zero resistance**.
- The value of shunt resistance is determined using the relation: $I_g \times G = (I - I_g) \times S$, therefore: $S = (I_g \times G) / (I - I_g)$, where: G = galvanometer resistance, S = shunt resistance, Ig = current for full-scale deflection, I = total current to be measured. This equation comes from the fact that the potential difference across parallel branches remains the same.

BCW Bits:

- A shunt resistance is always very small in value.
- Galvanometer can detect direction of current as well as magnitude.
- Voltmeter is made by connecting **high resistance (multiplier resistance) in series**.
- Resistance in parallel decreases equivalent resistance.
- Manganin is commonly used in shunt resistors.
- To increase the range of an ammeter **n times**, shunt required: $S = G / (n - 1)$.
- Connecting an ammeter directly across a battery may damage it due to excessive current.

62. The relationship between decay constant (λ) and half-life (T½) is ________. ***[BPSC AEDO]***

A. $T_{1/2} = \lambda$ B. $T_{1/2} = 1/\lambda$
C. $T_{1/2} = 0.693/\lambda$ D. $T_{1/2} = \lambda/0.693$

Ans. C

Exp:

- This question shows the relationship between the half-life of a radioactive substance and its decay constant. It is one of the most important formulas in nuclear physics and radioactivity. Radioactive substances contain unstable atomic nuclei. These unstable nuclei spontaneously break down and emit radiation in the form of alpha particles, beta particles, or gamma rays. This process is called radioactive decay. The rate of radioactive decay depends on the number of undecayed nuclei present at a given time. The basic radioactive decay law is: $N = N_0e^{(-\lambda t)}$, where: N_0 = initial number of nuclei, N = number of nuclei remaining after time t, λ = decay constant, t = time. The decay constant represents the probability of decay per unit time. A larger value of λ means faster decay. Half-life is the time required for half of the radioactive nuclei in a sample to decay. If initially there are N_0 nuclei, after one half-life: $\mathbf{N = N_0/2}$. Substituting this into the decay equation and simplifying gives: $\mathbf{T_{1/2} = \ln 2 / \lambda}$. Since $\mathbf{\ln 2 \approx 0.693}$, therefore: $\mathbf{T_{1/2} = 0.693 / \lambda}$. This shows that half-life is inversely proportional to the decay constant. If the decay constant increases, half-life decreases.

- Radioactive isotopes are widely used in medical diagnosis, cancer treatment, industrial measurements, and dating ancient objects. For example, Carbon Dating uses the radioactive isotope Carbon-14 to estimate the age of fossils and archaeological remains. Carbon-14 has a half-life of about **5730 years**.
- The SI unit of decay constant is $\mathbf{s^{-1}}$ because it represents the probability of decay per second. The activity of a radioactive sample is given by: $\mathbf{A = \lambda N}$, where A is the **activity** of the sample, measured in **Becquerel (Bq)**, equal to the number of disintegrations per second.

BCW Bits:

- Ernest Rutherford made major contributions to the study of radioactive decay.
- Alpha particles are helium nuclei (^4_2He), consisting of two protons and two neutrons.
- Beta particles are high-speed electrons or positrons.
- Gamma rays are high-energy electromagnetic radiation.
- The SI unit of radioactivity is **Becquerel (Bq)**; older unit is Curie (Ci).

63. The Big Bang theory explains which of the following observations? ***[BPSC AEDO]***

A. The internal structure of stars
B. The uniform expansion of the universe
C. There is no cosmic background radiation in the universe
D. The motion of planets

Ans. B

Exp:

- The Big Bang Theory is the most widely accepted scientific explanation for the origin and evolution of the universe. According to this theory, the universe began from an extremely hot, dense, and tiny state around 13.8 billion years ago and has been expanding ever since.
- The strongest observational support for the Big Bang theory comes from the discovery that galaxies are moving away from one another. This means the universe is expanding uniformly in all directions. This observation was first strongly supported by Edwin Hubble, who observed that distant galaxies show redshift in their spectral lines.
- The relation between recessional velocity and distance is given by Hubble's Law: $v = H_0 d$, where: v = recessional velocity of galaxy , H_0 = Hubble constant, d = distance of galaxy, This equation shows that farther galaxies move away faster, indicating expansion of space itself.
- The Big Bang theory does not mean an explosion occurring at a particular point in empty space. Instead, it means that space itself has been expanding over time. If we imagine reversing this expansion backward, all matter and energy converge toward an extremely dense initial state called a singularity.
- Another important support for the Big Bang theory is the formation of light elements. In the early universe, conditions were hot and dense enough to form hydrogen, helium, and small amounts of lithium. This process is called **Big Bang nucleosynthesis**.
- The Big Bang theory also explains the existence of **cosmic microwave background radiation (CMBR)**, which is often called the afterglow of the Big Bang. Therefore, option C is incorrect because cosmic background radiation does exist.
- The Big Bang theory also explains why the universe appears homogeneous and isotropic on large scales. Galaxies are distributed relatively uniformly when viewed across enormous cosmic distances.
- Modern cosmology further suggests that the expansion of the universe is accelerating due to a mysterious component called **dark energy**. Observations from distant supernovae provided evidence for this accelerated expansion.

BCW Bits:

- Georges Lemaître first proposed the idea of the expanding universe.
- Redshift means increase in wavelength due to recession of galaxies.
- Blue shift occurs when objects move toward the observer.
- CMBR is often called the "afterglow" of the Big Bang.
- Dark matter does not emit light but exerts gravitational effects.

64. In a convex lens, if the object is placed at 2F (twice the focal length), the image formed will be – ***[BPSC AEDO]***

A. Real, at position 2F and same size as the object
B. Virtual, at position F and magnified
C. Real, at position 2F and magnified
D. Virtual, beyond position 2F and diminished

Ans. A

Exp:

- The correct answer is **Real, at position 2F and same size as the object**. In a convex lens, when an object is placed at twice the focal length (2F), the image is formed at 2F on the opposite side of the lens. The image produced is: Real, Inverted, Same size as the object. A convex lens is also called a **converging lens** because it converges parallel rays of light toward a focal point. The image formation by a convex lens depends on the position of the object relative to the focal length.

- The lens formula is: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$
- Where: f = focal length, v = image distance, u = object distance. When the object is placed at 2F, solving the lens equation gives the image distance also equal to 2F. The magnification produced by a lens is given by: m=v/u.
  Since v = +2f and u = −2f (New Cartesian Sign Convention): m = v/u = −1. The magnitude |m| = 1 means same size, and the negative sign confirms the image is **inverted**."
- This means the image size is equal to the object size. In a convex lens, when the object is placed beyond the focal point, refracted rays actually meet on the opposite side of the lens. Since real rays physically intersect, the image formed is real and can be obtained on a screen.
- The behavior of a convex lens changes depending on object position:
  - Beyond 2F → Image between F and 2F, real, inverted, diminished
  - At 2F → Image at 2F, real, inverted, same size
  - Between F and 2F → Image beyond 2F, real, inverted, magnified
  - At F → Image at infinity
  - Between F and lens → Image virtual, erect, magnified
- The human eye also contains a convex lens that helps focus light onto the retina. Defects like hypermetropia are corrected using convex lenses because they help converge light rays more strongly.

BCW Bits:

- Convex lenses are thicker at the center and thinner at the edges.
- Concave lenses always form virtual, erect, and diminished images.
- Real images can generally be obtained on a screen.
- Virtual images cannot be projected on a screen.
- Optical power of a lens is given by: P = 1/f, where f is in **metres (m)** and power is expressed in **Dioptres (D)**.

65. Consider the following truth table of a logic gate with A and B as inputs and Y as output. ***[BPSC WMO]***

Input A	Input B	Output Y
0	0	1
0	1	1
1	0	1
1	1	0

The above truth table pertains to

A. NOR gate B. OR gate
C. NAND gate D. AND gate

Ans. C

Exp: NAND Gate: A **NAND** (NOT-AND) gate is a universal logic gate that produces a LOW (0) output only if all its inputs are HIGH (1). If any input is LOW (0), the output is HIGH (1). It is effectively an AND gate followed by a NOT gate.

Truth Table:		
Input A	**Input B**	**Output Y**
0	0	1
0	1	1
1	0	1
1	1	0

NOR Gate: A **NOR** (NOT-OR) gate produces a HIGH (1) output only when all its inputs are LOW (0). If any input is HIGH (1), the output is LOW (0).

Input A	Input B	Output Y
0	0	1
0	1	0
1	0	0
1	1	0

OR Gate: An **OR** gate produces a HIGH (1) output if at least one of its inputs is HIGH (1). The output is LOW (0) only if all inputs are LOW (0).

Input A	Input B	Output Y
0	0	0
0	1	1
1	0	1
1	1	1

AND Gate: An **AND** gate produces a HIGH (1) output only if all its inputs are HIGH (1).

Input A	Input B	Output Y
0	0	0
0	1	0
1	0	0
1	1	1

66. Consider the following forces. ***[BPSC WMO]***

I. Electromagnetic force
II. Nuclear force
III. Gravitational force

Arrange the above forces in increasing order of their strength.

A. ii, i, iii B. iii, ii, i
C. i, iii, ii D. iii, i, ii

Ans. D

Exp:

Force	Definition	Relative strength	Range
Gravitational force	It is a universal force, which means every object experiences it due to every other object. It can be defined as the force of mutual attraction between any two objects by virtue of their masses. It is always attractive. It is responsible for the motion of the moon around the earth, motion of the earth, planets around the sun, motion of artificial satellites around the earth Motion of the bodies falling to the earth. It also plays a key role in formation and evolution of stars, galaxies, etc.	10^{-39}	Infinite
Weak nuclear force	It appears only in certain nuclear processes for example in the decay of a nucleus. It is stronger than the gravitational force but much weaker than the electromagnetic as well as strong nuclear forces.	10^{-13}	Very short, sub-nuclear size (approx 10^{-16} m)
Electromagnetic force	It can be defined as the force between charged particles. It acts over large distances and does not need any intervening medium. This force is much stronger than the gravitational force. It can be attractive as well as repulsive. It governs the structure of atoms and molecules.	10^{-2}	Infinite
Strong nuclear force	It is the strongest of all fundamental forces (100 times stronger than the electromagnetic force. It can be defined as the force which binds protons and neutrons in a nucleus. It is an attractive force. It is independent of charges and acts equally between a proton and a proton, a neutron and a neutron and a proton and a neutron.	1	Short, nuclear size (approx 10^{-15} m)

Therefore, increasing order of their strength results in the sequence: **Gravitational Force < Electromagnetic Force < Strong Nuclear Force**.

67. Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R). ***[BPSC WMO]***

Assertion (A): Speed is a scalar quantity.

Reason (R): A vector quantity is one that has both magnitude and direction.

In light of the above two statements, choose the most appropriate answer from the options given below:

A. (A) is correct, but (R) is not correct

B. Both (A) and (R) are incorrect

C. Both (A) and (R) are correct and (R) is the correct explanation of A.

D. (A) is not correct, but (R) is correct

Ans. C

Exp: In physics, the quantities are classified as scalars and vectors. The difference is that a direction is associated with a vector but not with a scalar.

- A scalar quantity is a quantity with magnitude only. It is specified completely by a single number, along with the proper unit. Examples: Length, Volume, Speed, Mass, Density, Temperature, Pressure, Energy, Work, Power, time, etc.
- A vector quantity is a quantity that has both magnitude and direction and obeys the triangle law of addition or equivalently the parallelogram law of addition. Examples: Displacement, Velocity, Electric Field, Magnetic Field, Acceleration, etc.
- Both are correct, and the Reason (R) correctly explains the Assertion (A); because vector quantities require both magnitude and direction, and speed lacks a direction, speed is classified as a scalar rather than a vector.

68. In astronomical calculations, the term light-year is used very often. The light-year stands for ***[BPSC WMO]***

A. The time it takes for the earth to complete one rotation around the sun

B. The time it takes for the sunlight to reach earth

C. The distance that light travels in a year

D. None of the above

Ans. C

Exp: Despite its name, a light-year is a unit of distance, not time, defined as the total distance a beam of light travels through a vacuum in one Earth year. Because light is the fastest phenomenon in the universe, traveling at a universal constant speed of approximately 3 times 10^8 m/s (671 million mph), this distance equates to roughly 6 trillion miles (9.7 trillion kilometers). While light takes only 8 minutes and 20 seconds to reach Earth from the Sun and 1.28 seconds from the Moon, light-years are used to measure vast interstellar distances, such as the 4.2 light-years to Proxima Centauri; for even greater distances, astronomers use the parsec, which equals 3.26 light-years, while the Astronomical Unit (AU) is used specifically for distances within our solar system.

69. The Cosmic Microwave Background Radiation (CMBR) refers to — ***[BPSC WMO]***

A. The cosmic rays that are emitted by the sun and other astrophysical objects
B. The residual thermal radiation from the Big Bang permeating the entire universe
C. The electromagnetic radiation emitted by stars in the microwave frequency range
D. The radiations emitted by the heaviest stars in the galaxy

Ans. B

Exp: Cosmic Microwave Background Radiation (CMBR) is the faint, uniform glow of microwave radiation that fills the entire universe. It is considered the "afterglow" of the Big Bang, representing the thermal energy left over from the early, hot, and dense state of the universe. As the universe expanded and cooled over billions of years, this intense primordial heat stretched into the microwave portion of the electromagnetic spectrum, providing a vital snapshot of the early universe.

70. The electrical solar panels work on the principle of the photoelectric effect. In the photoelectric effect, the kinetic energy of the emitted electrons depends upon the ***[BPSC WMO]***

A. Frequency of the incident photons
B. The intensity of the incident photons
C. Orbital energy of the emitted electrons
D. All of the above

Ans. A

Exp: The photoelectric effect is the process by which electrons are emitted from a material when it is exposed to light. According to the laws of the photoelectric effect, the maximum kinetic energy of the emitted photoelectrons is directly proportional to the frequency of the incident photons (as defined by Einstein's photoelectric equation: $K_{max} = h\nu - \Phi$), and is independent of the light's intensity. While intensity affects the total number of electrons emitted (the photoelectric current), it does not alter their individual kinetic energy.

This principle is fundamental to the operation of photoelectric cells—devices that convert light into electrical energy—which were first introduced by Wilhelm Hallwachs and experimentally verified by Heinrich Hertz. Albert Einstein was awarded the 1921 Nobel Prize in Physics for his theoretical explanation of this phenomenon, which now powers modern technologies like electrical solar panels.

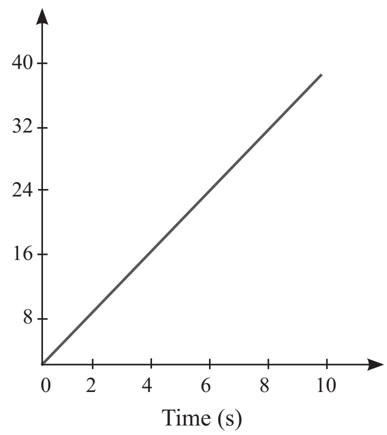
71. Tyndall effect is associated with ***[BPSC WMO]***

A. Emission of light
B. Both absorption and emission of light
C. Absorption of light
D. Scattering of light

Ans. D

Exp: The Tyndall effect is the phenomenon where a beam of light becomes visible as it passes through a colloidal solution or a fine suspension. This occurs because the particles in the colloid are large enough to deflect and scatter the light in different directions, making the path of the beam apparent. It is a classic example of light scattering, distinct from absorption or simple emission.

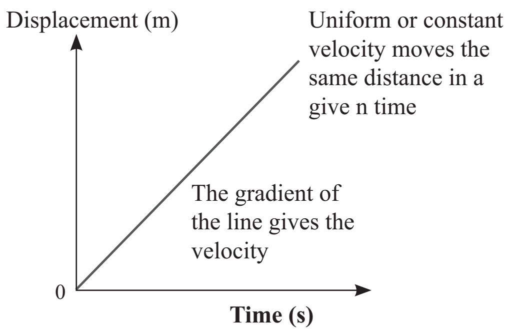
72. The displacement - time graph below shows: ***[BPSC WMO]***



A. An object in uniform motion
B. A stationary object
C. An object moving with constant acceleration
D. None of the above

Ans. A

Exp: In a displacement-time graph, the slope of the line represents the velocity of the object. A straight line starting from the origin with a constant positive slope indicates that the object covers equal displacements in equal intervals of time. Since the velocity (the rate of change of displacement) remains constant over time, the object is said to be in **uniform motion**.



Chemistry

1. pH value of 0.1N HCl solution is approximately: ***[71st BPSC]***

A. 1.0 B. 11.0 C. 10 D. 2.0

Ans. A

Exp: The pH (potential of hydrogen) is a scale used to specify the acidity or basicity of an aqueous solution. It is defined as the negative logarithm of the concentration of H+ ions:

- pH = -log10[H+]
- Given that Hydrochloric acid (HCl) is a strong acid that dissociates completely, the normality of this HCl solution (0.1 N) means the concentration of H+ ions is 0.1 (or 10^{-1}).
- pH = -log10(10^{-1}) = 1
- Therefore, the pH value is exactly 1.0. A pH value less than 7 is considered acidic, while a value greater than 7 is basic.
- A pH of 'a' simply means the concentration of H+ ions is 10^{-a}.

BCW Bits:

pH

- The pH scale was developed by a Danish scientist Soren Sorensen.
- It is a scale for measuring hydrogen ion concentration in a solution.
- The 'p' in pH stands for 'potenz' in German, meaning power.
- On the pH scale we can measure pH generally from 0 (very acidic) to 14 (very alkaline).

The pH of common substances:

Gastric Juices	About 1.2
Lemon Juice	About 2.2
Saliva before meal	About 7.4
Saliva after meal	About 5.8
Carrot juice	About 6
Pure water (Neutral Solution)	7
Blood	About 7.4
Milk of magnesia	About 10
Sodium Hydroxide solution	About 14

2. Which metal compound is used for treatment of carcinoma? ***[71st BPSC]***

A. Platinum B. Chromium
C. Iron D. Chlorine

Ans. A

Exp: **Cisplatin** [$Pt(NH_3)_2Cl_2$] is the primary **platinum-based metal compound** used in cancer chemotherapy to treat various carcinomas, including testicular, ovarian, lung, and bladder cancers.

- It acts by binding to DNA, causing cross-linking that interferes with repair mechanisms and triggers apoptosis in cancer cells. These compounds disrupt DNA replication and transcription.
- Besides cisplatin, other FDA-approved platinum-based drugs include carboplatin and oxaliplatin.
- Many hexavalent chromium [Cr(VI)] compounds are known human carcinogens, iron is used to treat anemia, and Chlorine is a non-metal and it is used as disinfectant for water purification.

3. Why oil is not stored in galvanized iron pot? ***[71st BPSC]***

A. It produce toxic compound
B. It reduced quality of oil
C. Corrosion observed
D. All of above

Ans. D

Exp: Oil shouldn't be stored in galvanized iron pots because the zinc coating reacts with acids or impurities in the oil, forming toxic zinc compounds, degrading oil quality (rancidity, bad taste), and corroding the pot, making it unsafe for food storage. This chemical reaction contaminates the oil with zinc, posing health risks, which is why stainless steel or food-grade plastics are preferred.

4. Which compound is used in contraceptive pills? ***[71st BPSC]***

A. Cholecalciferol B. Levonorgestrel
C. Venlafaxine D. Cetrizine

Ans. B

Exp: Levonorgestrel is a synthetic hormone belonging to the progestin class (a synthetic form of progesterone). It is a primary ingredient in many contraceptive pills. It works primarily by inhibiting ovulation, thickening cervical mucus to block sperm, and altering the uterine lining to prevent implantation.

BCW Bits:

- **Cholecalciferol:** Also known as Vitamin D_3, which is essential for calcium absorption and bone health, not for contraception.
- **Venlafaxine:** This is an antidepressant, used to treat mood disorders.
- **Cetrizine:** This is an antihistamine used to relieve allergy symptoms such as sneezing and runny nose.

5. Which compound is used for increase of octane rating? *[71st BPSC]*

A. Trimethyl hexane B. Tetramethyl Oxide
C. Tetraethyl lead D. Triethyl toluene

Ans. C

Exp: The quality of petrol is determined by its **anti-knock property**, which is expressed by the **octane number**. A higher octane number indicates greater resistance to engine knocking. Similarly, the quality of diesel is measured by its **cetane number**, which indicates the ease with which the fuel ignites under compression.

Note: **Tetraethyl lead (TEL), $Pb(C_2H_5)_4$**, was historically used as an anti-knock additive in petrol to significantly increase its octane rating and improve engine performance. However, its use has been discontinued in most countries because lead emissions are highly toxic, causing serious health and environmental problems, particularly neurological damage. Modern fuels use safer octane-enhancing additives instead.

BCW Bits:

- Petrol → Octane Number (Anti-knock quality)
- Diesel → Cetane Number (Ignition quality)
- Tetraethyl Lead (TEL) → Historical anti-knock additive in petrol
- $Pb(C_2H_5)_4$

6. Full form of PHBV biodegradable polymer: [71st BPSC]

A. Polyhydroxy butyl vaniline
B. Poly hydroxy butane veratric acid
C. Poly (3- Hydroxy butyrate -co-3 hydroxy valerate)
D. Poly hydroxy butyric acid veratric acid

Ans. C

Exp: PHBV: Poly beta-hydroxybutyrate– co-beta-hydroxy valerate. It is obtained by the copolymerization (more than one monomeric species is allowed to polymerize) of 3- hydroxybutanoic acid and 3- hydroxypentanoic acid. PHBV is a specialized biodegradable copolymer. Because it degrades safely in the environment via microbial action, it is heavily used in specialty packaging, orthopedic devices, and controlled drug release systems.

BCW Bits: Polymers

- **Definition:** The word polymer comes from two Greek words: Poly (meaning many) and mer (meaning unit). Polymers are high molecular mass macromolecules consisting of repeating structural units derived from simpler molecules called monomers. The process of forming these large molecules is called polymerization.
- **Types of Polymerization:**
  - **Addition (or chain growth) polymerization:** Monomers join together without losing any molecules (e.g., polythene).
  - **Condensation (or step growth) polymerization:** Monomers combine with the elimination of small molecules like water or alcohol (e.g., Nylon 6,6).
- **Classification by Source:**
  - **Natural Sources:** Polymers found in nature, such as proteins, cellulose, starch, and natural rubber.
  - **Synthetic Sources:** Man-made polymers, including plastics (polythene), synthetic fibers (Nylon 6, 6), and synthetic rubber (Buna-S).
- **Biodegradable Polymers:** These are polymers that contain functional groups similar to biopolymers, allowing them to be broken down naturally by microorganisms into non-toxic byproducts like water and carbon dioxide, eliminating long-term plastic pollution.
  - **Examples:**
    - **PHBV:** Used in specialty medical packaging, orthopedic devices, and controlled drug delivery systems.
    - **PLA (Polylactic Acid):** Derived from corn starch; widely used as the primary filament in **3D printing**, as well as in bio-plastic casings for consumer electronics.
    - **PGA (Polyglycolic Acid):** Known for high strength; used to make self-dissolving internal surgical staples and degradable tools in the oil and gas industry.
    - **PCL (Polycaprolactone):** Highly flexible with a low melting point; used for long-term time-release drug implants and rapid prototyping.
    - **PBAT:** Highly flexible and tough; used in commercial compostable shipping mailers and smart agricultural mulch films that are plowed directly into the soil.

Important Polymers:

Polymer	**Monomer**	**Uses**
Polypropene	Propene	Ropes, toys, pipes, fibres, etc.
Polystyrene	Styrene	Insulator, wrapping material, toys, radio and television cabinet.
Polyvinyl chloride (PVC)	Vinyl chloride	Raincoats, handbags, vinyl flooring, water pipes.
Urea-formaldehyde Resin	Urea Formaldehyde	Unbreakable cups and laminated sheets
Glyptal	Ethylene glycol Phthalic acid	Paints and lacquers.
Bakelite	Phenol Formaldehyde	Combs, electrical switches, handles of utensils and computer discs.
Kevlar	polyamides	Bullet proof jackets.

Natural Rubber	Isoprene	Medical devices, surgical gloves, tires etc.
Polyethene terephthalate	Ester	Responsible for the essence of fruits
Teflon	Tetrafluoroethylene	Non-stick pan

7. Which Compound is known as night glowing pigment from below? ***[71st BPSC]***

A. Europium dopped strontium aluminate
B. Copper Sulphide dopped zinc oxide
C. Barium sulphate dopped copper carbonate
D. Boron oxide dopped copper sulphate

Ans. **A**

Exp: Europium-doped strontium aluminate ($SrAl_2O_4$: Eu) is the compound widely recognized as a high-performance, modern night-glowing (phosphorescent) pigment. It is commonly used in safety signs, watches, and paints due to its ability to absorb light and emit a long-lasting, bright glow for up to 12 hours.

8. Which Material is used for smart film? ***[71st BPSC]***

A. Indium tin oxide B. Calcium carbonate
C. Zinc chloride D. Silica

Ans. **A**

Exp: **Indium Tin Oxide** (ITO) is a transparent conducting oxide essential for the functionality of smart films (also known as switchable privacy films or PDLC films). In smart films, ITO layers act as transparent electrodes that conduct an electric current to a layer of polymer-dispersed liquid crystals (PDLC). When voltage is applied, it causes the liquid crystals to align, switching the film from an opaque state to a transparent one.

- Besides smart films, it is the standard material for touch-screens, OLEDs, LCDs, and solar panels.

9. Who coined the word 'atom'? ***[BPSC ASO, Pre]***

A. E. Rutherford B. Democritus
C. John Dalton D. Thomson

Ans. **B**

Exp: Maharishi Kanada was the first one to give a theory on how atoms combine to form molecules and compounds, in his book **Vaisheshika Sutra**.

BCW Bits:

- The term 'atom' was coined by the ancient Greek philosopher Democritus around 400 B.C. He derived the word from the Greek term 'atomos', meaning 'uncuttable' or 'indivisible,' proposing that all matter is composed of small, indestructible, and indivisible particles.
- **John Dalton:** Proposed the first scientific **Atomic Theory (1808)**, stating that matter is composed of indivisible atoms and that all atoms of a given element are identical in mass and properties.
- **J.J. Thomson:** Discovered the **electron (1897)** using his cathode ray tube experiment and proposed the **"Plum Pudding" model**, proving that the atom is divisible and contains negatively charged particles.

10. The chemical behaviour of an atom is mostly determined by its ***[BPSC ASO, Pre]***

A. Mass number B. Valence electrons
C. Atomic weight D. Neutron count

Ans. **B**

Exp: The chemical behaviour of an atom is determined by the number and arrangement of **valence electrons**, which are the electrons located in the atom's outermost shell. These electrons are directly involved in forming chemical bonds with other atoms through losing, gaining, or sharing.

BCW Bits:

- Mass number, atomic weight, and neutron count primarily influence the physical properties and nuclear stability of an atom (such as the formation of isotopes) rather than its chemical behaviour.

11. Match the Columns. ***[BPSC ASO, Pre]***

Column A (Process/ Concept)		Column B (Definition / Example)	
i.	Sublimation	1.	Separation based on boiling point
ii.	Distillation	2.	Solid to gas directly
iii.	Evaporation	3.	Gas to liquid
iv.	Filtration	4.	Separation of solid from liquid
v.	Condensation	5.	Liquid to gas below boiling point

A. i-4, ii-5, iii-2, iv-l, v-3
B. i-3, ii-1, iii-5, iv-4, v-2
C. i-1, ii-2, iii-3, iv-4, v-5
D. i-2, ii-1, iii-5, iv-4, v-3

Ans. **D**

Exp: Matter around us exists in three distinct physical states: solid, liquid, and gas. These states are inter-convertible depending on physical conditions like temperature and pressure, which dictate the arrangement and kinetic energy of the constituent particles. In laboratory and industrial setups, understanding these phase transformations allows for the efficient separation of complex mixtures into their pure components using distinct physical properties like volatility, boiling point, and solubility.

i. **Sublimation (Matches 2):** A change of state directly from a solid to a gas without changing into a liquid state is called sublimation. For example, a mixture of sand and naphthalene can be separated by Sublimation because naphthalene balls sublimate directly.

ii. **Distillation (Matches 1):** It is used for the separation of components of a mixture containing two miscible liquids that boil without decomposition and have sufficient differences in their boiling points.

iii **Evaporation (Matches 5):** Evaporation is the change of a liquid into vapours at any temperature below its boiling point.

iv. **Filtration (Matches 4):** It is a physical separation technique that separates insoluble solid particles from a liquid or gas by passing the mixture through a porous filter medium (like filter paper) that traps the solids but allows the fluid to pass through, yielding a clear liquid called the filtrate and a solid residue.

v. **Condensation (Matches 3):** Condensation is the process through which vapour changes into liquid when it comes in contact with a cooler surface.

BCW Bits: Additional Separation Methods:

- **Inter-conversion of States:** The term refers to the change of matter from one state to another and back to its original state without any change in its chemical composition, occurring through phase transitions driven by temperature and pressure shifts—such as
  - **Melting** (Solid to Liquid),
  - **Freezing** (Liquid to Solid),
  - **Vaporization** (Liquid to Gas),
  - **Condensation** (Gas to Liquid),
  - **Sublimation** (Solid directly to Gas), and
  - **Deposition** (Gas directly to Solid).
- **Chromatography:** It is the technique used for the separation of those solutes that dissolve in the same solvent. It is used to separate colours in a dye, pigments from natural colours, and drugs from the blood.
- **Fractional Distillation:** To separate a mixture of two or more miscible liquids for which the difference in boiling points is less than 25 K, the fractional distillation process is used; for example, for the separation of different gases from the air, different factions from petroleum products, etc.
- **Expansion:** An increase in the volume of the substance while its mass remains the same. The anomalous expansion of water is an abnormal property of water whereby it expands instead of contracting when the temperature decreases from 4°C until it freezes at 0°C. This allows many aquatic life forms to survive through the winter.
- **Boiling** is the temperature at which the vapour pressure of liquids becomes equal to its atmospheric pressure.
- **Effervescence:** The formation of gas bubbles in liquid during a chemical reaction is called Effervescence. Bubbles occur as the gas escapes from the reaction.

12. Litmus, the pH indicator, is obtained from which of the following sources? ***[BPSC ASO, Pre]***

A. A lichen B. A brown alga
C. A red alga D. A fungi

Ans. A

Exp: Litmus is a water-soluble mixture of different dyes extracted from specific species of lichens, most notably *Roccella tinctoria*. Lichens are complex composite organisms arising from a symbiotic, mutually beneficial relationship between a fungus (mycobiont) and an alga or cyanobacterium (photobiont).

Other Options:

- **Option B (Brown Alga):** Known for storing food as mannitol or laminarin. They produce algin (a hydrocolloid) used as a thickening agent in the food and pharmaceutical industries.
- **Option C (Red Alga):** A commercial source of Agar-Agar and carrageenan, which are widely used as solidifying agents in microbial laboratory cultures, jellies, and ice creams.
- **Option D (Fungi):** Heterotrophic organisms with cell walls made of chitin. While many yield life-saving antibiotics (like Penicillium), they do not serve as indicators for pH tracking.

BCW Bits:

- **Acids:** pH value less than 7, higher concentration of H+ ions. The lower the value of pH stronger the acid.
- **Base:** pH value more than 7, higher concentration of OH- ions. The higher the value of pH stronger the Base.
- **Indicators:** Special types of substances which are used to test whether a substance is acidic or basic are called Indicators. The indicators change their colour when added to a solution containing an acidic or basic substance.
- **Litmus:** is the most commonly used natural indicator which is obtained from Lichens.
- When it is dissolved in an acidic solution it turns red and when added to a basic solution it turns blue. The litmus has purple (mauve) colour in distilled water. It is available in the form of a solution, strips of paper. Hence, only a basic solution can turn red litmus into blue colour.
- **China Rose:** It is also used as an indicator. It turns acidic solutions to dark pink or magenta and basic solutions to green.
- **Phenolphthalein:** gives a pink colour when the solution is basic. And when the solution is acidic, it remains colourless.

Note:

- An acid or base reacts with each other, to produce salt (which can be acidic, basic or neutral) and water in the process and the evolution of heat takes place, this reaction is called a **neutralization reaction.**

13. Manganese is present in the highest oxidation state in ***[BPSC ASO, Pre]***

A. MnO_2 B. $KMnO_4$
C. $Mn_2(CO)_{10}$ D. MnO_4^{2-}

Ans. B

Exp: An oxidation state is a hypothetical charge assigned to an atom in a compound, representing the number of electrons lost, gained, or shared during a bond, assuming all bonds are 100% ionic. It acts as a tool to track electron transfer.

Let the oxidation state of Manganese be 'x'. Then, on calculating the oxidation states for manganese (Mn) in the given compounds:

- **MnO_2 :** $x + 2(-2) = 0 \rightarrow x = +4$
- **$KMnO_4$:** $+1 + x +4 (-2) = 0 \rightarrow x = +7$
- **$Mn_2(CO)_{10}$:** $2x + 10 (0) = 0 \rightarrow x=0$ (neutral ligand)
- **MnO_4^{2-} :** $x + 4(-2) = -2 \rightarrow x = +6$

Therefore, Manganese exhibits its highest oxidation state, +7, in $KMnO_4$.

BCW Bits: Properties of Potassium Permanganate ($KMnO_4$):

- **Appearance:** Deep purple crystalline solid that dissolves in water to form intense pink or purple solutions.
- **Chemical Nature:** Powerful oxidizing agent that changes color as it reduces, acting as its own indicator in titrations.
- **Common Uses:** Utilized for water treatment (removing iron and hydrogen sulfide), as a disinfectant, and as a topical antiseptic.

14. Pyrolusite, Psilomelane and Braunite are the ores of the following mineral– ***[BPSC ASO, Pre]***

A. Manganese B. Iron Ore
C. Copper D. Bauxite

Ans. A

Exp: Pyrolusite, Psilomelane, and Braunite are the primary, major ores of Manganese (Mn). These minerals are important sources for extracting manganese, which is crucial for producing steel, alloys, batteries, and other chemical products.

- **Pyrolusite [MnO_2]:** The most common manganese ore used in dry cell batteries and also used by glass makers to remove green tints caused by iron impurities.
- **Psilomelane [$(Ba,H_2O)_2Mn_5O_{10}$]:** A hydrous manganese oxide mineral occurring as a dark, structural mass.
- **Braunite [$Mn^{2+}Mn_6^{3+}(SiO_{12})$]:** A silicate mineral of manganese.

BCW Bits: Major Ores of Other Options:

- **Option B (Iron Ore):** The principal ores of iron include **Hematite (Fe_2O_3)**, Magnetite (Fe_3O_4), Siderite ($FeCO_3$), and Iron Pyrites (FeS_2).
- **Option C (Copper):** Extracted primarily from **Copper Pyrites ($CuFeS_2$)**, Malachite ($CuCO_3.Cu(OH)_2$), and Cuprite (Cu_2O).
- **Option D (Bauxite):** Bauxite itself is the principal ore of **Aluminum** (consisting of hydrous aluminium oxides like $Al_2O_3.2H_2O$), named after the village Les Baux in France, where it was discovered.

15. Glass is best described as: ***[BPSC ASO, Pre]***

A. An alloy B. An eutectic mixture
C. A supercooled solution D. A gel

Ans. C

Exp: Glass is best described as a **supercooled solution** (or supercooled liquid) because it is an amorphous solid that lacks a definite crystalline structure. It is formed when a molten mixture of silicates is cooled rapidly, preventing the molecules from organizing into a regular geometric lattice.

- The atomic arrangement is disordered, similar to that of a liquid.
- It has extremely high viscosity, making it appear rigid like a solid, though it theoretically retains a slight ability to flow over vast periods of time.
- Unlike crystalline solids, glass softens gradually over a range of temperatures.

BCW Bits:

Option A (An alloy): A homogeneous mixture of two or more metals, or a metal and a non-metal (e.g., brass, bronze), combined to enhance properties like strength or corrosion resistance.

Option B (An eutectic mixture): A specific mixture of substances that melts and solidifies at a single, lowest possible temperature, lower than the melting point of any of its individual components.

Option D (A gel): A colloidal system in which a liquid is dispersed throughout a solid medium, forming a semi-solid structure (e.g., jelly, cheese, silica gel).

16. The Carbon—Carbon bond length is maximum in- ***[BPSC ASO, Pre]***

A. C_6H_6 B. $CH_2 = CH_2$
C. CH_3CH_3 D. $HC \equiv CH$

Ans. C

Exp: The length of a Carbon-Carbon (C-C) bond depends on the bond order (number of shared electron pairs) and the hybridization of the carbon atoms. Generally, as the bond order increases, the bond length decreases because the nuclei are pulled closer together.

- **Ethane (CH_3CH_3):** This molecule contains a single bond between two (sp^3) hybridized carbon atoms. Single bonds are the longest type of covalent carbon bonds, with a length of approximately (1.54 Å).
- **Benzene (C_6H_6):** Due to resonance, the C-C bonds in benzene are intermediate between a single and a double bond (bond order 1.5). The bond length is approximately (1.39 Å).
- **Ethene (CH_2=CH_2):** This molecule contains a double bond between two (sp^2) hybridized carbon atoms. Double bonds are shorter than single bonds, with a length of approximately (1.34 Å).
- **Ethyne ($HC \equiv CH$):** This molecule contains a triple bond

between two (sp) hybridized carbon atoms. Triple bonds are the shortest, with a length of approximately (1.20 Å).

17. Chloramine-T is an: ***[BPSC ASO, Pre]***

A. Analgesic B. Disinfectant
C. Antipyretic D. Antiseptic

Ans. B

Exp: **Chloramine-T (Sodium tosylchloramide)** is an organic chloride compound that acts as a powerful oxidizing agent. When dissolved in water, it slowly releases active chlorine and hypochlorous acid, giving it strong **virucidal, bactericidal, and fungicidal** properties. It is extensively used as a **heavy-duty disinfectant for water treatment, hospital sanitation, and biosecurity protocols to neutralize viral loads in poultry environments.**

BCW Bits: Classification of Everyday Medicinal Chemicals

The remaining options represent critical classes of pharmacological chemicals used in modern medical science:

Drug Class	Functional Definition	Common Chemical Examples
Disinfectant	Non-selective antimicrobial agents applied strictly to non-living surfaces (objects, floors, water) to destroy microbes, as they are toxic to living tissues.	Chloramine-T, 1% Phenol, Chlorine gas (0.2 to 0.4 ppm).
Antiseptic	Antimicrobial substances that kill or inhibit microorganisms and are safe to apply directly to living tissues (wounds, cuts, skin) without causing damage.	Dettol (Chloroxylenol + Terpineol), Savlon, Iodine tincture, 0.2% Phenol.
Analgesic	Medicinal compounds designed specifically to relieve pain by acting on the central or peripheral nervous system without causing a loss of consciousness.	Aspirin, Paracetamol, Morphine, Ibuprofen.
Antipyretic	Chemical substances used to reduce body temperature during a fever by acting on the hypothalamus.	Paracetamol, Aspirin.

18. Who discovered the process of fermentation? ***[BPSC ASO, Pre]***

A. Robert Koch B. Joseph Mister
C. Ernst Haeckel D. Louis Pasteur

Ans. D

Exp: In 1857, Louis Pasteur, a French chemist and microbiologist, discovered that fermentation is caused by living yeast and bacteria. His work demonstrated that these microorganisms convert sugar into alcohol and lactic acid, forming the basis of the germ theory of disease.

BCW Bits:

- Fermentation is an anaerobic metabolic process where microorganisms like bacteria, yeast, or fungi convert carbohydrates into acids, gases, or alcohol. It is widely used to produce foods (yogurt, bread), beverages (wine, beer), and industrial chemicals.
- **Option A (Robert Koch):** A pioneering German physician who is celebrated as one of the main founders of modern bacteriology. He discovered the specific causative agents of deadly diseases like **tuberculosis, cholera, and anthrax**, and formulated **Koch's Postulates**—a set of criteria used to establish a causal relationship between a microbe and a disease.
- **Option B (Joseph Meister):** The famous young French boy who, in 1885, was severely bitten by a rabid dog. He became the **first human being to receive Louis Pasteur's experimental rabies vaccine** and survived, proving the vaccine's real-world efficacy.
- **Option C (Ernst Haeckel):** An influential German zoologist and evolutionist who proposed the **Three-Kingdom Classification system** (introducing the kingdom *Protista* to group single-celled organisms). He also coined several fundamental terms used in biology today, including **Ecology, Phylum, and Phylogeny**.

19. Which of the following are present in higher amount in hard water? ***[BPSC ASO, Pre]***

A. Sodium and Manganese
B. Calcium and Sodium
C. Sodium and Magnesium
D. Calcium and Magnesium

Ans. D

Exp: Hard water is defined by its high concentration of dissolved multivalent mineral cations. The two most abundant and primary minerals responsible for water hardness are **Calcium** (Ca^{2+}) and **Magnesium** (Mg^{2+}). The presence of these ions prevents soap from lathering easily and leads to the formation of 'scum' (insoluble precipitates like calcium stearate).

- Sodium, on the other hand, does not cause hardness; in fact, **sodium ions are used in water softeners** to replace calcium and magnesium ions to make the water soft.

BCW Bits: Temporary vs. Permanent Hardness

- **Temporary Hardness:** Caused by calcium and magnesium **bicarbonates** [$Ca(HCO_3)_2$ and $Mg(HCO_3)_2$]. Easily removed by simple physical boiling or by adding lime (Clark's method).

- **Permanent Hardness:** Caused by calcium and magnesium **bicarbonates** [$Ca(HCO_3)_2$ and $Mg(HCO_3)_2$]. Easily removed by simple physical boiling or by adding lime (Clark's method).

20. The principal buffer present in human blood *[BPSC ASO, Pre]*

A. $CH_3COOH + CH_3COONa$
B. $NaH_2PO_4 + Na_2HPO_4$
C. $H_2CO_3 + HCO_3^-$
D. $H_3PO_4 + NaH_2PO_4$

Ans. C

Exp: A **buffer solution** is a mixture of **a weak acid and its conjugate base (or a weak base and its conjugate acid)** that resists drastic changes in pH when small amounts of an acid or a base are added to it.

- The primary mechanism for maintaining the pH of human blood within the narrow, life-sustaining range of 7.35 to 7.45 is the **bicarbonate buffer system**.
- It consists of **Carbonic acid (H_2CO_3, a weak acid) and the Bicarbonate ion (HCO_3^-, its conjugate base).**
- When excess hydrogen ions (acids) enter the bloodstream, they are neutralized by bicarbonate ions; when excess hydroxyl ions (bases) enter, they react with carbonic acid.

BCW Bits:

- According to the **Bronsted-Lowry theory,** a **conjugate acid** is formed when a base gains a proton (H+), while a **conjugate base** is formed when an acid loses a proton (H+).
- **Option A ($CH_3COOH + CH_3COONa$):** A mixture of acetic acid and sodium acetate. This is a classic **acidic laboratory buffer** used extensively in chemical research and industrial processes, typically maintaining a pH around 4.75.
- **Option B ($NaH_2PO_4 + Na_2HPO_4$):** The **Phosphate buffer system**. While it plays a minor role in blood plasma, it is the *principal intracellular buffer* inside body cells and plays a vital role in maintaining the pH of fluid in the kidney tubules.
- **Option D (H_3PO_4 + NaH2PO$_4$):** A mixture of phosphoric acid and sodium dihydrogen phosphate. It functions as a strong chemical buffer system in laboratory configurations but does not naturally occur as a primary biological buffer system in human blood.

21. Which of the following is used in rocket propellants? *[BPSC ASO, Pre]*

A. N_2H_4 B. O_2
C. H_2 D. All of the above

Ans. D

Exp: Rocket propellants generally consist of a fuel and an oxidizer.

- **N_2H_4 (Hydrazine):** A common liquid fuel.
- **O_2 (Liquid Oxygen):** The most widely used oxidizer in modern liquid-fueled rockets. It provides the necessary oxygen for combustion in the vacuum of space.
- **H_2 (Liquid Hydrogen):** A high-performance cryogenic fuel known for its high energy density.
- Because all three chemicals are standard components used in rocket propulsion, the **correct answer is All of the above.**

22. Bell-metal is an alloy of [BPSC ASO, Mains]

A. Copper and lead B. Copper and tin
C. Copper and zinc D. Copper and nickel

Ans. B

Exp: Alloys are metallic substances formed by combining two or more elements to enhance properties like strength, corrosion resistance, or conductivity. By modifying the atomic lattice of a base metal, engineers create materials with superior performance compared to pure metals.

- Bell-metal is a high-tin **bronze** (typically **80% copper and 20% tin**). This specific ratio creates a hard, elastic crystalline structure that produces the resonant, long-lasting "ringing" sound essential for bells and musical instruments.

Table of Essential Alloys:

Alloy Name	Principal Constituents	Distinctive Property	Primary Applications
Brass	Copper + Zinc	High malleability & acoustic properties	Musical instruments, low-friction fittings.
Bronze	Copper + Tin	Corrosion resistance & hardness	Statues, marine hardware, heavy-duty bearings.
Steel	Iron + Carbon	High tensile strength & durability	Infrastructure, skyscraper frames, automotive.
Duralumin	Aluminum + Copper	High strength-to-weight ratio	Aircraft structures, racing car components.
Amalgam	Mercury + Silver/Tin/ Zinc	Soft initially, hardens over time	Dental fillings, gold/silver extraction.
Solder	Tin + Lead (or Silver)	Low melting point	Electrical circuit bonding, plumbing.

Gunmetal	Copper + Tin + Zinc	High strength & easy to cast	Buttons, gears, heavy machinery parts.
Nitinol	Nickel + Titanium	Shape-memory & super elasticity	Medical stents, robotic actuators.
Inconel	Nickel + Chromium	Extreme heat & oxidation resistance	Rocket engines, gas turbine blades.
Mag-Lithium	Magnesium + Lithium	Lowest density structural metal	Ultra-light laptop chassis, satellites.
Liquid metal	Zirconium + Others	Amorphous (glass-like) structure	High-end electronics, wear-resistant gear.

23. Iodine can dissolve readily in ***[BPSC ASO, Mains]***

A. Water B. Carbon tetrachloride
C. Alcohol D. Potassium iodide

Ans. B

Exp: The solubility of a substance depends on the intermolecular forces between the solute and the solvent. Iodine exists as a diatomic, nonpolar molecule (I_2), which dictates its interactions with various liquid media according to the chemical rule: **"Like dissolves like."**

- Iodine (I_2) is a non-polar molecule that dissolves most readily in non-polar solvents like **carbon tetrachloride (CCl_4)**, forming a deep violet solution.
- Conversely, it has extremely low solubility in polar **water (H_2O)** because the weak iodine-water attractions cannot break water's strong hydrogen bonds.
- In **alcohol (C_2H_5OH)**, it is only moderately soluble, creating a **brown "Tincture of Iodine."**
- Finally, while it dissolves in **potassium iodide (KI)**, this occurs via a chemical reaction that forms a **triiodide complex (I_2 + KI —> KI_3)** rather than simple physical dissolution.

24. Which of the following is the least stable oxide of chlorine? [BPSC ASO, Mains]

A. Cl_2O B. ClO_2 C. Cl_2O_7 D. ClO_3

Ans. A

Exp: Oxides of chlorine are generally unstable, endothermic compounds because the bond energy between chlorine and oxygen is relatively low. Their stability typically increases with the oxidation state of the chlorine atom (from +1 to +7), although molecular structure and electron configuration play critical roles in their reactivity.

The stability of these oxides is determined by their thermal decomposition limits and their chemical structure:

- **Dichlorine Monoxide (Cl_2O):** In this compound, chlorine is in its **lowest oxidation state (+1)**. Because the stability of chlorine oxides typically follows the trend of increasing stability with higher oxidation states, **Cl_2O is considered the least stable among the common oxides.** It is a yellowish-brown gas that decomposes explosively even at low temperatures or under minor physical shock.
- **Dichlorine Heptoxide (Cl_2O_7):** With chlorine in the **+7** oxidation state, this is the **most stable oxide.** It is the anhydride of perchloric acid and is significantly less reactive than the other oxides.
- **Chlorine Dioxide (ClO_2) and Chlorine Trioxide (ClO_3):** While these are highly reactive and powerful oxidising agents, they **possess higher oxidation states** (+4 and +6/+7 in dimer form, respectively), which, according to the standard trend of chlorine oxides, makes them **more stable relative to the +1 state of Cl_2O.**

Stability Trend: Cl_2O (Least Stable) < ClO_2 < Cl_2O_6 (ClO_3) < Cl_2O_7 (Most Stable)

25. Which of the following is not an alkali?
[BPSC ASO, Mains]

A. NaOH B. KOH
C. NH_4OH D. C_2H_5OH

Ans. D

Exp: The alkali metals and their compounds are fundamental to inorganic chemistry. These elements, located in **Group 1** of the periodic table (excluding Hydrogen), include **Lithium (Li)**, **Sodium (Na)**, **Potassium (K)**, **Rubidium (Rb)**, **Caesium (Cs)**, and **Francium (Fr)**.

They are highly reactive because they possess a single valence electron, which they readily lose to form stable cations. When their oxides or hydroxides dissolve in water, they form **alkalis**—soluble bases that characterise much of the group's chemical behaviour.

An alkali is defined as a base that dissolves in water to produce **hydroxyl ions (OH-)**.

- **Sodium Hydroxide (NaOH):** A strong alkali; completely dissociates in water.
- **Potassium Hydroxide (KOH):** A strong alkali; high solubility and ionisation.
- **Ammonium Hydroxide (NH_4OH):** A weak alkali; partially dissociates in aqueous solution.
- **Ethanol (C_2H_5OH):** Although the formula ends in —OH, it is an **organic alcohol**, not an ionic hydroxide. In water, the covalent bond between the carbon and the oxygen does not break; therefore, it does not release OH- ions and remains chemically neutral.

The term "alkali" specifically refers to the soluble hydroxides of the **Alkali Metals** found in **Group 1** (s-block) of the periodic table.

Property	Chemical Detail	Industrial Use
Valency & Ionization	**[Noble Gas] ns¹**; Lowest 1st ionisation energy in their period.	High reactivity; never found free in nature.
Oxide Formation	Li forms **monoxide** (Li_2O); Na forms **peroxide** (Na_2O_2); K/Rb/Cs form **superoxides** (MO_2).	KO_2 is used in oxygen masks/ submarines to absorb CO_2 and release O_2.
Reactivity with Water	**$2M + 2H_2O$ —> $2MOH + H_2$ + Heat**. Reactivity increases down the group.	Explosive reaction; Li is steady, while K/Rb/Cs ignite spontaneously.
Hydration Enthalpy	**Li+ > Na+ > K+ > Rb+ > Cs+**; Smallest ion (Li+) attracts the most water molecules.	Explains why Lithium salts are mostly found as hydrates (e.g., $LiCl·_2H_2O$).
Ammonia Solutions	Dissolve in liquid NH_3 to form **ammoniated electrons** (Blue solution).	Powerful reducing agents used in organic synthesis.
Reducing Nature	Li is the **strongest reducing agent** in aqueous solution (highest negative electrode potential).	Essential for high-density Lithium-ion battery technology.
Flame Coloration	Electrons excite and emit specific wavelengths: **Li (Crimson), Na (Yellow), K (Lilac).**	Used in analytical chemistry (Flame Test) and pyrotechnics.
Carbonate Stability	**Li_2CO_3** decomposes easily; others are stable and solubility increases down the group.	Li_2CO_3 is used in glass manufacturing and psychiatry.

26. Which of the following is most acidic?

[BPSC ASO, Mains]

A. $HClO_4$ B. $HClO_2$ C. HclO D. $HClO_3$

Ans. A

Exp: Oxoacids are chemical compounds that contain oxygen, at least one other element (such as chlorine), and at least one hydrogen atom bonded to oxygen which can dissociate to release a proton.

- In the case of chlorine, these acids form a series where the central chlorine atom is bonded to varying numbers of oxygen atoms, directly impacting their chemical behavior. ur oxygen atoms, making **Perchloric acid ($HClO_4$)** one of the strongest known acids.

Chemical Analysis & Stability Trends:

The acidity of chlorine oxoacids follows the order: **HClO < $HClO_2$ < $HClO_3$ < $HClO_4$.** This trend is driven by two primary chemical factors:

- **Oxidization State and Electronegativity:** As more oxygen atoms are added to the central Chlorine atom, its oxidation state increases. This highly positive center draws electron density away from the **O—H bond**, polarizing it. The weaker the O—H bond, the more easily the acid releases a proton (H+).
- **Conjugate Base Stability (Resonance):** After losing a proton, the stability of the resulting ion determines the acid's strength. The **Perchlorate ion (ClO4-)** is exceptionally stable because the negative charge is delocalized (spread out) across four oxygen atoms via resonance. In contrast, the **Hypochlorite ion (ClO-)** has no resonance stabilization, making it a very weak acid.

Acid Name	Formula	Oxidation State	Acid Strength	Conjugate Base
Hypochlorous Acid	HClO	+1	Very Weak	ClO- (Hypochlorite)
Chlorous Acid	$HClO_2$	+3	Weak	ClO2- (Chlorite)
Chloric Acid	$HClO_3$	+5	Strong	ClO3- (Chlorate)
Perchloric Acid	**$HClO_4$**	**+7**	**Strongest**	**ClO_4- (Perchlorate)**

27. Which of the following is used as anaesthetic?

[BPSC ASO, Mains]

A. Ethyl acetate B. Ethyl alcohol
C. Ether D. Carbon tetrachloride

Ans. C

Exp: Anaesthetics are a category of medicinal drugs that cause a temporary loss of sensation or awareness. They are primarily used to facilitate surgical procedures by preventing pain and distress in patients.

- **Diethyl Ether [$(C_2H_5)_2O$]** is a pioneering inhalational anaesthetic that functions by depressing the central nervous system and interacting with nerve cell membranes to induce reversible analgesia and muscle relaxation.
- While it revolutionized 19th-century surgery, its extreme **flammability** and tendency to cause post-operative nausea have led to its replacement in modern medicine by safer, non-flammable halogenated ethers like **Isoflurane** and **Sevoflurane**.

While the other options are not suitable for anaesthesia, they play vital roles in industrial chemistry:

- **Substance Chemical Formula Primary Uses**
- **Ethyl Acetate** $CH_3COOC_2H_5$ A common **solvent** in nail polish removers, glues, and the decaffeination of coffee beans and tea leaves.

- **Ethyl Alcohol** C_2H_5OH Used as a **fuel (bio-ethanol)**, an industrial solvent, an antiseptic in hand sanitizers, and the primary component in alcoholic beverages.
- **Carbon Tetrachloride** CCl_4 Historically used in fire extinguishers and as a cleaning agent; now strictly regulated due to its extreme **toxicity** and role in ozone depletion.

28. Iron is a constituent of which of the following? ***[BPSC ASO, Mains]***

A. Chlorophyll B. Thymine
C. Protein D. Haemoglobin

Ans. D

Exp: **Haemoglobin** is a complex protein found in red blood cells that is responsible for transporting oxygen from the lungs to the rest of the body. Each haemoglobin molecule contains four **iron** atoms in the form of "heme" groups, which bind to oxygen molecules. This iron content is also what gives blood its characteristic red color. In contrast, **Chlorophyll** (the pigment in plants) contains **Magnesium** at its center rather than iron. While some proteins do contain iron (known as metalloproteins), iron is the fundamental, defining constituent required for the structure and function of haemoglobin.

29. Match List-I with List-II and choose the correct answer from the codes given below: ***[BPSC ASO, Mains]***

List-I		List-II	
a.	Magnesium oxide	1.	Amphoteric
b.	Phosphorus pentoxide	2.	Basic
c.	Aluminum oxide	3.	Strongly basic
d.	Sodium oxide	4.	Acidic

Codes:

A. a-4, b-2, c-3, d-1 B. a-3, b-4, c-1, d-2
C. a-2, b-4, c-1, d-3 D. a-1, b-2, c-3, d-4

Ans. C

Exp: Oxides are binary compounds formed by the reaction of oxygen with another element. The chemical nature of an oxide (whether it is acidic, basic, or amphoteric) depends primarily on the electronegativity of the element bonded to oxygen and its position in the periodic table.

Types of Oxides and Chemical Properties

Type of Oxide	Chemical Property	Common Examples	Typical Reaction
Basic Oxides	Formed by metals; react with water to form bases or with acids to form salt and water.	Na_2O, MgO, CaO	**$CaO + H_2O \rightarrow Ca(OH)_2$**
Acidic Oxides	Formed by non-metals; react with water to form oxyacids or with bases to form salt and water.	SO_2, CO_2, P_2O_5	**$CO_2 + H_2O \rightarrow H_2CO_3$**
Amphoteric Oxides	Exhibit both acidic and basic characteristics; react with both acids and bases.	Al_2O_3, ZnO, PbO	**$Al_2O_3 + 6HCl \rightarrow 2AlCl_3 +$ 3H2O**
Neutral Oxides	Do not react with either acids or bases to form salts.	CO, NO, N_2O	No salt-forming reaction.

This question identifies the periodic trend of Period 3 elements (and neighbors) where the nature of oxides shifts from **strongly basic** to **strongly acidic** as you move from left to right across the period.

- **Sodium Oxide (Na_2O):** As a Group 1 alkali metal oxide, it is **strongly basic**. It reacts vigorously with water to produce a strong alkali: **$Na_2O + H_2O \rightarrow 2NaOH$**.
- **Magnesium Oxide (MgO):** A Group 2 alkaline earth metal oxide. It is **basic** but less reactive than sodium oxide. It forms a weak base with water: **$MgO + H_2O \rightarrow$ Mg(OH)2**.
- **Aluminum Oxide (Al2O$_3$):** This is a classic **amphoteric** oxide. It can act as a base (reacting with HCl) or as an acid (reacting with NaOH to form sodium aluminate).
- **Phosphorus Pentoxide (P_2O_5 or P_4O_{10}):** A non-metal oxide that is highly **acidic**. It reacts violently with water to form phosphoric acid: **$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$**.

30. Hard water contains which of the following salts? ***[BPSC ASO, Mains]***

A. Calcium and sodium B. Magnesium and calcium
C. Barium and calcium D. Calcium and potassium

Ans. B

Exp: **Hardness of water** is primarily caused by the presence of dissolved bicarbonate, chloride, and sulfate salts of **calcium and magnesium**. These minerals prevent soap from lathering effectively by forming an insoluble "scum." Temporary hardness is caused by calcium and magnesium bicarbonates (which can be removed by boiling), while permanent hardness is caused by their chlorides and sulfates. Sodium salts (found in common salt) do not cause water hardness; in fact, sodium is often used in water softeners to replace the calcium and magnesium ions.

31. Which of the following is used in making baking soda? ***[BPSC ASO, Mains]***

A. Sodium carbonate
B. Bleaching powder
C. Sodium hydroxide
D. None of the above

Ans. A

Exp: Baking soda, chemically known as **Sodium Bicarbonate (NaHCO3)**, is a vital chemical used in the food industry, medicine, and fire extinguishers. While the Solvay process is used for large-scale industrial production, baking soda is manufactured directly by reacting **Sodium Carbonate (Na_2CO_3)** with carbon dioxide and water.

Methods of Preparation:

1. The production involves a carbonation process where a saturated solution of sodium carbonate (also known as washing soda) reacts with carbon dioxide gas. Because sodium bicarbonate is less soluble, it precipitates out as a solid.

   The chemical reaction is: $Na_2CO_3 + H_2O + CO_2 \rightarrow$ 2NaHCO+

2. The Solvay process is an elegant cycle where the primary raw materials are **Sodium Chloride (Brine)**, **Ammonia (NH3)**, and **Limestone ($CaCO_3$)**. The process involves several distinct chemical stages:

   1. **Formation of Ammonium Bicarbonate:** Ammonia reacts with carbon dioxide (sourced from heating limestone) in the presence of water to form ammonium bicarbonate.
      - $\mathbf{NH_3 + H_2O + CO_2 \rightarrow NH_4HCO_3}$
   2. **Formation of Sodium Bicarbonate (The Main Reaction):** When ammonium bicarbonate reacts with a concentrated brine solution (NaCl), sodium bicarbonate precipitates out because it is sparingly soluble in the presence of ammonium ions.
      - $\mathbf{NaCl + NH_4HCO_3 \rightarrow NaHCO_3 + NH_4Cl}$
   3. **Recovery of Ammonia (Efficiency Step):** To make the process economical, ammonia is recovered by reacting the byproduct ammonium chloride with calcium hydroxide (slaked lime).
      - $\mathbf{2NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2NH_3 + 2H_2O}$

Other Compounds:

Compound Name	**Chemical Formula**	**Key Facts**
Baking Soda	$\mathbf{NaHCO_3}$	The target product (Sodium Bicarbonate).
Washing Soda	$Na_2CO_3.10H_2O$	Formed by **heating** baking soda (**$2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$**).
Bleaching Powder	$CaOCl_2$	Used as a disinfectant; contains chlorine, unrelated to baking soda production.
Sodium Hydroxide	NaOH	Also known as Caustic Soda; produced via the Chlor-alkali process, not used to make baking soda.

32. Neutron was discovered by ***[BPSC DSO, Pre]***

A. Rutherford
B. Chadwick
C. Hahn and Strassman
D. Millikan

Ans. B

Exp: The **neutron was discovered by James Chadwick** in **1932** by bombarding beryllium atoms with alpha particles, resulting in the emission of an uncharged radiation consisting of particles with a mass slightly greater than that of a proton.

- **(Hahn and Strassman):** Discovered nuclear fission (the splitting of an atomic nucleus), which laid the foundation for nuclear energy and atomic bombs.
- **(Millikan):** Measured the elementary charge of an electron using his famous oil-drop experiment.

BCW Bits:

Name	Symbol Name	Absolute Charge/C	Relative Charge	Mass/kg	Discovered
Electron	e	$-1.60217663 \times 10^{-19}$	-1	$9.1093837 \times 10^{-31}$	J.J. Thomson
Proton	p	$+1.60217663 \times 10^{-19}$	+1	$1.6726216 \times 10^{-27}$	Ernest Rutherford
Neutron	n	0	0	$1.6749275 \times 10^{-27}$	James Chadwick

- **Atomic Neutrality:** Atoms are electrically neutral because they contain equal numbers of positively charged protons and negatively charged electrons.
- **The Nucleus:** Located at the center of the atom, it consists of protons and neutrons bound together by a strong nuclear force. It was discovered by Ernest Rutherford.
- **Atomic Mass Concentration:** Nearly all of an atom's mass is concentrated in the nucleus. The mass of an electron is negligible—only about 1/2000th the mass of a proton or neutron.
- **Proton vs. Neutron Mass:** Their masses are nearly equal, though the neutron is slightly heavier than the proton.
- **Atomic Mass Unit (amu):** The standard unit used to measure atomic mass, where 1 amu = 1.66×10^{-24} grams.

33. The 'Royal Water' which can dissolve Gold and Platinum is made up of a mixture of- ***[BPSC DSO, Pre]***

A. 1 : 3 mixture of conc. HNO_3 and HCl

B. 3 : 1 mixture of conc. HNO_3 and HCl
C. 1 : 3 mixture of conc.H_2SO_4 and HCl
D. 3 : 1 : 1 mixture of HNO_3, HCl, H_2SO_4

Ans. **A**

Exp: **'Royal Water'** (also known as **Aqua Regia**) is a highly corrosive, fuming liquid mixture of concentrated nitric acid (HNO_3) and concentrated hydrochloric acid (HCl), typically in a 1:3 molar ratio. It is freshly prepared by mixing 1 part concentrated nitric acid to 3 parts concentrated hydrochloric acid by volume. It is capable of dissolving noble metals like gold, platinum, and palladium, which do not dissolve in either acid alone.

34. Iron is galvanized by ***[BPSC DSO, Pre]***

A. Zn (Zinc) B. Cu (Copper)
C. Ag (Silver) D. Lead

Ans. **A**

Exp: **Galvanization** is the process of **applying a protective zinc coating to iron or steel**. Zinc is used because it is more reactive to oxygen than iron, meaning it oxidizes more quickly and acts as a sacrificial anode to protect the underlying metal from rusting.

- Corrosion results in the conversion of the metal into its chemically stable form, which can either be oxides, hydroxides, or sulphides. Rusting is a type of chemical change that results in the formation of iron oxides, which is an entirely new substance. This is an example of an oxidation reaction:
  - $Fe + O_2 + H_2O \rightarrow FeO$ and Fe_2O_3

Weight Changes During Rusting:

- Due to the addition of oxygen during rusting, we need to take into account the weight of oxygen now attached to the iron, which causes the overall weight of rusted iron to increase:
  - Rusted Iron Weight = Weight of Iron + Weight of Oxygen

BCW Bits:

Prevention from rusting:

- **Alloying –** alloying (mixing) the iron with other stable metals or alloys can slow down the process of rusting.
- **Galvanizing –** Applying a protective layer of metallic zinc on steel or iron to prevent rusting is called **Galvanization,** it is an inexpensive procedure. **Zinc is used** for galvanizing because (creates a sacrificial anode) zinc is more reactive to oxygen than iron, so it oxidizes more quickly and protects from rusting.
- **Coating and painting –** coating the surface of a metal object with a layer of either paint or varnish will work as a barrier between the surface and atmospheric oxygen making it immune to rusting.

35. Which among the following is not among the 'rare earth metals' that is used in manufacturing electrical and electronic components? ***[BPSC DSO, Pre]***

A. Scandium (Sc) B. Lanthanum (La)
C. Cerium (Ce) D. Radium (Ra)

Ans. **D**

Exp: **Radium** is an intensely **radioactive alkaline earth metal** (not a rare earth element). Discovered by Marie and Pierre Curie, it was historically used in self-luminous paints for watches and instrument dials, and historically in cancer radiation treatments, though it has largely been replaced due to its extreme toxicity.

BCW Bits:

- The **Rare Earth Elements** (REEs) consist of **17** specific elements: the 15 lanthanides (lanthanum to lutetium), plus scandium and yttrium. Rare earth metals are defined by their unique chemical and magnetic properties, which make them essential for high-tech electronics like smartphones, magnets, and display screens.

The 17 REEs are:

- Lanthanum (La), Cerium (Ce), Praseodymium (Pr), Neodymium (Nd), Promethium (Pm), Samarium (Sm), Europium (Eu), Gadolinium (Gd), Terbium (Tb), Dysprosium (Dy), Holmium (Ho), Erbium (Er), Thulium (Tm), Ytterbium (Yb), and Lutetium (Lu).
- **Scandium (Sc) and Yttrium (Y)**, are also included because they often occur in the same ore deposits and share similar chemical properties.

36. The Matchstick tips are loaded with ***[BPSC DSO, Pre]***

A. White Phosphorus B. Red Phosphorus
C. Phosphorus Trichloride D. Phosphorus Pentaoxides

Ans. **B**

Exp: The tip of Matchstick are coated with **Red Phosphorus**. Also Red Phosphorus is used on the side of the matchbox because it is stable, non-toxic, and safe to handle, whereas white phosphorus is highly poisonous and reactive.

- When struck, the heat from friction converts the red phosphorus to white phosphorus, which ignites.
- Safety matches were invented to replace dangerous white phosphorus matches.

Note: In modern safety matches, the matchstick tips are coated with potassium chlorate, sulfur, and antimony trisulfide, while the red phosphorus is located on the striking surface of the box.

37. Which chemical is most commonly used for artificial rain? ***[BPSC DSO, Pre]***

A. Sodium Nitrate B. Silver Nitrate
C. Silver Iodide D. Potassium Chlorate

Ans. **C**

Exp: Silver Iodide (AgI) is the most common chemical agent used for artificial rain, a process technically known as cloud seeding. It is highly effective because its crystalline structure is very similar to that of natural ice. When dispersed into clouds (via aircraft or rockets), it acts as an 'ice nucleus,' providing a surface for supercooled water droplets to freeze and coalesce. Once these droplets become heavy enough, they fall to the ground as precipitation (rain or snow).

BCW Bits:

- **Silver Nitrate:** Highly significant as the chemical used to make **election ink** (indelible ink applied to voters' fingers to prevent fraud) because it reacts with the skin to form a permanent dark stain. It is also used in medicine as a topical antiseptic and cauterizing agent.

38. Which variety of coal can be used to recognise the traces of plant material from which it is formed?

[BPSC DSO, Pre]

A. Bituminous B. Lignite
C. Anthracite D. Peat

Ans. D

Exp: Peat is the type of coal in which the deposit contains remarkably visible and recognizable traces of the original plant material, such as wood, moss, and roots.

- As the first stage of coalification, it has a low carbon content and has undergone minimal decomposition.
- It is considered the rawest form of coal, often resembling compressed, decaying vegetable matter. It consists of roughly (55 % - 60 %) carbon, with high moisture and impurities.

BCW Bits:

Other Coal Varieties by Carbon Content:

- **Option A (Bituminous):** Soft coal containing around 60% - 80% carbon. It is the most common variety used for electricity generation and making coke.
- **Option B (Lignite):** Known as brown coal, it carries about 40% - 55% carbon and has a high moisture content.
- **Option C (Anthracite):** The highest grade of coal, containing over 80% - 95% carbon. It is hard, glossy, and burns with the highest heat output and minimal smoke.

39. $H_2O + C \rightarrow H_2 + CO$

The above reaction is an example of: ***[BPSC AEDO]***

A. Ziegler–Natta process B. Water gas reaction
C. Aldol condensation D. Wacker process.

Ans. B

Exp:

- **Steam (H_2O)** is passed over **hot carbon (C)**, producing hydrogen **gas (H_2)** and **carbon monoxide (CO).** This mixture of **H_2 + CO** is called **water gas.**
- **(A) Ziegler–Natta process:** Used for making polymers like polyethene and polypropylene, involves catalysts; **(B) Aldol condensation:** Organic reaction involving aldehydes/ketones, produces larger organic molecules; **(C) Wacker process:** Used to convert ethylene into acetaldehyde, involves palladium catalyst.

BCW Bits:

- Water gas, also called "syngas" (synthesis gas), is an industrial fuel gas.
- It is used to manufacture methanol and synthetic fuels. The reaction is endothermic (requires heat)
- A related reaction is producer gas, which includes nitrogen along with CO

40. 'Superglue' is a polymer of which monomer?

[BPSC AEDO]

A. Sodium benzoate B. Acetic acid
C. Vinyl chloride D. Cyanoacrylate

Ans. D

Exp:

- Superglue is a very fast-acting adhesive—it sticks things almost instantly. Chemically, it is made from cyanoacrylate compounds. Monomer is a small molecule (building block), Polymer is a large molecule formed by joining many monomers. So, superglue is a polymer formed from cyanoacrylate monomers. Superglue works so fast because it reacts with moisture in air or on surfaces which triggers rapid polymerization, that's why it bonds quickly.
- Sodium benzoate is used as a food preservative. Acetic acid (CH_3COOH) Main component of vinegar. Vinyl chloride Used to make PVC (plastic pipes, etc.). Cyanoacrylates are the actual monomers used in superglue.

BCW Bits:

- Superglue was discovered accidentally during World War II research.
- It is used in the medical field for wound closure (skin adhesives).
- Bonds well with materials like plastic, metal, and rubber.
- Doesn't work well on smooth or oily surfaces.
- Heat and moisture can weaken the bond over time.
- Cyanoacrylate fumes can irritate eyes and nose.
- It sets faster in humid environments.

41. A supercritical fluid is formed whenever: ***[BPSC AEDO]***

A. A substance is heated above its critical temperature
B. A substance is completely burnt
C. A substance is heated far below its critical temperature
D. None of the above.

Ans. A

Exp:

- A supercritical fluid is a special state of matter where a substance behaves like both a liquid and a gas at the same time. It spreads like a gas but dissolves substances like a liquid. It exists beyond a certain point called the critical point.
- The critical temperature is the temperature above which a gas cannot be turned into liquid, no matter how much pressure you apply. It's a key condition for forming a supercritical fluid.

- Supercritical fluid forms under 2 conditions, Temperature above critical temperature, Pressure above critical pressure. Once both are crossed → liquid and gas phases merge supercritical fluid forms.

BCW Bits:

- Supercritical CO_2 is widely used in Decaffeination of coffee, Extraction of essential oils.
- It is considered an environmentally-friendly solvent.
- Used in dry cleaning industry as an alternative to chemicals.
- Supercritical fluids have high diffusion rates like gases.
- They can penetrate materials better than liquids.
- Used in pharmaceutical and food industries.
- Important in green chemistry applications.

42. Which of the following statements is incorrect about Erythromycin? ***[BPSC AEDO]***

A. It is a macrocyclic compound
B. It can be used as antibiotic
C. For legionellosis, it has good coverage
D. It cannot be produced in fermentation from a strain of the *Saccharopolyspora erythraea.*

Ans. D

Exp:

- **Erythromycin** is a commonly used **antibiotic** belonging to the **macrolide class.** Used to treat infections like Respiratory infections, Skin infections, Atypical pneumonia. **Macrocyclic compound** - Erythromycin is a macrolide antibiotic containing a large macrocyclic lactone ring **.**It inhibits bacterial protein synthesis and stops bacterial growth **.** Legionellosis (caused by *Legionella bacteria*). Erythromycin is effective because it works against **intracellular bacteria.**
- Erythromycin is often used as an **alternative to penicillin.**
- Macrolides bind to the **50S ribosomal subunit** and are considered **bacteriostatic antibiotics.**
- Newer macrolides include azithromycin, clarithromycin.
- It can cause **mild side effects, such as an upset stomach.**
- Antibiotic resistance to macrolides is a growing concern.

43. Barium appears primarily in ore(s) of: ***[BPSC AEDO]***

A. Barite only
B. Both barite and witherite
C. Witherite only
D. None of the above

Ans. B

Exp:

- **Minerals** are naturally occurring substances in the earth's crust, **Ores** are those minerals from which we can **economically extract metals.** Example: Iron -Hematite, Magnetite, Aluminium- Bauxite.
- **Barite-** Formula: **$BaSO_4$ (Barium sulphate),** Most important and widely used ore. Uses- Oil drilling, Paints, Rubber industry.
- **Important Ores & Minerals:** Iron- Hematite, Magnetite; Aluminium - Bauxite; Copper- Chalcopyrite; Zinc- Zinc blende; Lead- Galena; Mercury- Cinnabar; Barium- Barite, Witherite.
- Barite is very heavy, so it is used in **drilling mud to control pressure.**
- Witherite is less common than barite.
- Barium compounds are used in **X-ray imaging (e.g., the barium meal test).**
- Some barium salts are **toxic**, so handled carefully.
- Minerals form over **millions of years through geological processes.**
- Ore quality depends on **metal concentration (grade of ore).**
- Mining methods depend on depth and type of deposit.

44. With reference to conductors, consider the following statements: ***[BPSC AEDO]***

1. The valence band and the conduction band overlap each other.
2. The resistivity of a conductor decreases exponentially with temperature.
3. Conductors conduct electric current less than semiconductors and greater than insulators.

How many of the above three statements given above are correct?

A. Only second and third statement
B. Only second statement
C. Only first and second statement
D. Only first statement

Ans. D

Exp:

- In conductors, the valence band and conduction band either overlap or the conduction band is partially filled. Because of this overlap, electrons can move freely, which allows conductors to carry electric current easily. This is the main reason metals like copper, silver, and aluminium are good conductors of electricity. In band theory Conductors → overlapping bands; Semiconductors → small energy gap; Insulators → large energy gap.
- In conductors, resistivity generally increases with increase in temperature, not decreases. As temperature rises, metal atoms vibrate more strongly, causing greater collision with free electrons. This increases resistance and resistivity. For conductors $R \propto T$ approximately over a moderate temperature range. Conductors conduct electricity much better than semiconductors, while insulators conduct the least.

BCW Bits:

- Copper is widely used in electrical wiring.
- Silver is the best electrical conductor among common metals.
- In metals, free electrons are responsible for electrical conduction.
- Silicon and germanium are common semiconductors.

- Insulators have very high resistance.
- Band theory explains electrical behavior of solids.
- Superconductors show zero electrical resistance at very low temperatures.
- Ohm's law relates voltage, current, and resistance.
- Resistance is measured in ohm (Ω).
- Semiconductors are widely used in electronic devices.
- Diodes and transistors are semiconductor devices.

45. Which of the following metals can be used in the treatment of rheumatoid arthritis? ***[BPSC AEDO]***

A. Copper B. Gold
C. Iron D. Ruthenium

Ans. B

Exp:

- Certain compounds of Gold have been used in the treatment of rheumatoid arthritis. This therapy is known as chrysotherapy. Gold salts such as sodium aurothiomalate and auranofin were earlier used to reduce inflammation, pain, and joint damage in rheumatoid arthritis patients.
- Rheumatoid arthritis is an autoimmune disease in which the body's immune system attacks its own joints. This leads to swelling,stiffness,pain,and gradual joint deformity. Gold compounds helped suppress abnormal immune activity and inflammation. Although nowadays newer medicines like biologics and DMARDs are more commonly used, gold therapy remains an important historical and scientific example in medicinal chemistry.
- Copper is an essential trace element needed in the body for enzyme activity, iron metabolism,and formation of red blood cells. Iron is mainly associated with hemoglobin formation, oxygen transport, and treatment of anemia.
- Iron supplements are commonly used in iron-deficiency anemia, especially in women and children. Ruthenium is a rare transition metal mainly used in electronics, catalysts,and advanced chemical research. Some ruthenium compounds are being studied experimentally in medicinal chemistry, including cancer research.

BCW Bits:

- "Au" is the chemical symbol of gold.
- Gold is a noble metal and resists corrosion.
- Iron deficiency causes anemia.
- Copper is a good conductor of electricity.
- Platinum-based drug Cisplatin is used in cancer treatment.
- Autoimmune diseases occur when the immune system attacks body tissues.
- Rheumatoid arthritis commonly affects small joints first.
- Synovial joints are freely movable joints in the body.
- Transition metals often show catalytic properties.
- DMARD stands for Disease-Modifying Anti-Rheumatic Drug.

46. Which of the following is the strongest oxidising agent? ***[BPSC AEDO]***

A. I_2 B. Cl_2 C. F_2 D. Br_2

Ans. C

Exp:

- The correct answer is Fluorine (F_2) because fluorine is the strongest oxidising agent among all halogens. An oxidising agent is a substance that gains electrons itself, and causes oxidation of another substance. In simple words Strong oxidising agent = strong tendency to accept electrons. Among halogens, oxidising power decreases down the group: $F_2 > Cl_2 > Br_2 > I_2$. This happens because fluorine:
- has the highest electronegativity,very small atomic size, and very high tendency to gain electrons. Fluorine can oxidize almost all other halide ions. For example $F_2 + 2Br^- \rightarrow 2F^- + Br_2$ Here fluorine gains electrons from bromide ions and oxidizes them.

BCW Bits:

- Halogens belong to Group 17 of the periodic table.
- Fluorine is the most electronegative element and shows the highest oxidising power among all elements.
- Fluorine is highly reactive and pale yellow in color.
- Chlorine gas is greenish-yellow.
- Bromine is the only non-metal liquid at room temperature.
- Iodine sublimes to form violet vapors.
- Oxidation means loss of electrons.
- Reduction means gain of electrons.
- Bleaching powder contains chlorine.
- Hydrogen fluoride can etch glass due to reaction with silica.

47. The rate of diffusion of methane at a given temperature is twice that of an unknown gas. The molar mass of unknown gas is ____: ***[BPSC AEDO]***

A. 8 B. 4 C. 64 D. 32

Ans. C

Exp:

- The rate of diffusion of a gas is inversely proportional to the square root of its molar mass. $\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$ where: r1,r2 = rates of diffusion; M1,M2 = molar masses, in here Methane diffuses twice as fast as the unknown gas So: rCh4/r unknown = 2; Molar mass of methane (CH4): 12 + 4(1) = 16; **Applying Graham's Law:** 2 = under root M/16, Squaring both sides: 4 = M/16, therefore M = 64. Hence the molar mass of the unknown gas is: 64g/mol

BCW Bits:

- Graham's Law applies to diffusion and effusion of gases.
- Diffusion means mixing of gases due to random motion.
- Effusion means escape of gas through a tiny hole.

- Methane has molecular formula CH_4.
- Molar mass is expressed in g/mol.
- Hydrogen is the lightest gas.
- Temperature increases the kinetic energy of gas molecules.
- Lighter gases generally move faster than heavier gases.
- Avogadro's law relates equal volumes of gases to equal numbers of molecules.
- The ideal gas equation is PV = nRT.

48. Boron cannot form which one of the following anions? ***[BPSC AEDO]***

A. $B(OH)_4^-$ B. BH_4^- C. BO_2^- D. BF_6^{3-}

Ans. (D)

Exp:

- Boron belongs to the second period of the periodic table. Second-period elements have only 2s orbital, and 2p orbitals. They do not possess vacant d-orbitals in their valence shell. Because of this limitation, they cannot accommodate more than eight electrons around the central atom.
- In BF_6^{3-} boron would have to make six bonds with fluorine atoms. That would place 12 electrons around boron, meaning boron would need an expanded octet. But second-period elements like boron cannot expand their octet because they do not have d-orbitals available for bonding. Therefore, BF_6^{3-} cannot exist as a stable boron anion.
- Boric acid accepts hydroxide ion from water $B(OH)_3 + OH^- \rightarrow B(OH)^{-4}$, Here boron accepts an electron pair from hydroxide ion and forms $B(OH)^{4-}$. BF_4^- borohydride ion, ($NaBH_4$) are extremely important reducing agents. The ion has tetrahedral geometry and stable covalent bonding. BO_2^- metaborate ions are formed from boric acid and borates during dehydration reactions. Boron forms several oxyanions such as metaborate, tetraborate, orthoborate etc.
- Boron is electron-deficient but cannot expand octet. For example BF3 is electron deficient because boron has only six electrons around it. Yet boron still cannot expand beyond octet like phosphorus or sulfur.

BCW Bits:

- Boron belongs to Group 13 of the periodic table.
- Boron has atomic number 5.
- BF_3 is a planar molecule with sp^2 hybridization.
- Sodium borohydride is used in reduction reactions.
- Borax is an important boron mineral.
- Boric acid is weakly acidic in water.
- Second-period elements generally obey octet limitation.
- Fluorine is the most electronegative element.
- Expanded octet is common in third-period elements onward.
- SF6 has octahedral geometry.

49. Among the following, the maximum covalent character is shown by the compound: ***[BPSC AEDO]***

A. $AlCl_3$ B. $MgCl_2$ C. $SnCl_2$ D. $FeCl_2$

Ans. A

Exp:

- The correct answer is $AlCl_3$ because it shows the maximum covalent character among the given compounds. Greater the polarization of an anion by a cation, greater will be the covalent character. a **small and highly charged cation** strongly distorts the electron cloud of the anion,and this distortion increases covalent nature This distortion is called **polarization**.
- Aluminium chloride shows very high covalent character because Al3+ ion is very small, and carries high positive charge (+3). Due to its high charge density, aluminium ion strongly polarizes the chloride ion (Cl^-). As a result, electron sharing increases and the compound behaves more covalently. In fact, anhydrous $AlCl_3$ exists largely as a covalent compound and even forms dimeric structure Al_2Cl_6, This dimer formation itself is a strong indication of covalent behavior. molten $AlCl_3$ conducts electricity poorly compared to strongly ionic salts, which again supports its covalent nature.
- Magnesium chloride is more ionic, Although magnesium ion (Mg^{2+}) can polarize chloride ion to some extent, its charge density is lower than aluminium ion because charge is +2 instead of +3, and ionic size is comparatively larger. Therefore, polarization is weaker and covalent character is less.
- **Covalent character increases when** cation is small, cation charge is high, anion is large, and anion is easily polarizable. This is why $AlCl_3$ is highly covalent, while compounds like NaCl remain strongly ionic.

BCW Bits:

- Fajan's Rule explains partial covalent character in ionic compounds.
- Small cations have high polarizing power.
- Large anions are easily polarizable.
- $AlCl_3$ sublimes easily due to covalent nature.
- Ionic compounds usually have high melting points.
- Covalent compounds generally have lower electrical conductivity.
- Anhydrous $AlCl_3$ acts as a Lewis acid.
- Charge density means charge concentrated over small size.
- Polarization distorts electron cloud of an ion.
- Covalent bonding involves sharing of electrons.

50. Consider the following statements: ***[BPSC AEDO]***

Assertion (A): A carboxylic acid does not give the test for carbonyl group.

Reason (R): The carbonyl carbon of the carboxylic group is electron deficient and hence it is less susceptible to nucleophilic attack.

Select the correct answer using the codes given below:

A. (A) is false but (R) is true

B. Both (A) and (R) are true and (R) is not the correct explanation of (A)

C. (A) is true but (R) is false

D. Both (A) and (R) are true and (R) is the correct explanation of (A)

Ans. C

Exp:

- The **Assertion** is true Carboxylic acid does not give the usual tests for carbonyl compounds such as Schiff's test, Tollens' test, Fehling's test, or 2,4-DNP test. In aldehydes and ketones, the carbonyl carbon is highly reactive toward nucleophilic addition because the carbon atom carries significant partial positive charge. But in carboxylic acids, the carbonyl group is attached to an OH group. Due to resonance between oxygen atoms, electron density gets delocalized over the entire carboxyl group. This resonance stabilization reduces the typical reactivity of the carbonyl carbon toward nucleophilic addition reactions. The carboxyl group behaves differently from ordinary aldehydes and ketones. The resonance can be represented as: RCOOH↔RCOO−+H+ Because of this stabilization, carboxylic acids usually undergo nucleophilic substitution, rather than nucleophilic addition. This is the core concept behind the Assertion.
- The Reason says "The carbonyl carbon of the carboxylic group is electron deficient and hence it is less susceptible to nucleophilic attack."

  This statement is incorrect. Actually electron deficiency generally increases susceptibility toward nucleophilic attack, not decreases it.
- Nucleophiles are electron-rich species and they attack electron-deficient centers. So the logic given in the Reason is chemically wrong. The real reason for lower reactivity is resonance stabilization, electron donation by oxygen, and reduced electrophilic character of carbonyl carbon.
- Aldehyde and Ketone readily undergo nucleophilic addition reactions because carbonyl carbon is strongly electrophilic, and there is no strongly electron-donating −OH-OH−OH attached. That is why they easily give 2,4-DNP test, Schiff's test, Tollens' test etc. In carboxylic acids resonance stabilizes the structure, and −OH-OH−OH group changes the behavior of carbonyl carbon. Therefore, ordinary carbonyl tests generally fail.

BCW Bits:

- Carboxylic acids contain the functional group −COOH.
- Aldehydes contain the functional group −CHO.
- Ketones contain carbonyl group between two carbon atoms.
- 2, 4-DNP test is used for aldehydes and ketones.
- Tollens' reagent gives silver mirror test.
- Fehling's solution is used mainly for aldehydes.
- Carboxylic acids are acidic due to release of H+H^+H+ ion.
- Resonance stabilizes carboxylate ion.
- Electrophiles are electron-deficient species.

51. Which of the following is the sweetest sugar?

[BPSC AEDO]

A. Maltose B. Glucose C. Sucrose D. Fructose

Ans. D

Exp:

- Fructose is also called fruit sugar, because it is abundantly found in fruits, honey, and many natural sweet substances. Its sweetness is higher than glucose, sucrose, and maltose. Approximate sweetness order is Fructose>Sucrose>Glucose>Maltose, Fructose is a monosaccharide, and a ketohexose sugar. It is naturally present in fruits, honey, and nectar. Its high sweetness is the reason many processed foods use high-fructose sweeteners.
- Maltose is the least sweet, Maltose is also known as malt sugar. It is formed by the combination of two glucose molecules Glucose+Glucose→Maltose, Maltose is produced during breakdown of starch, especially in germinating grains. Maltose is a reducing sugar because it contains a free aldehyde group after ring opening.
- Glucose is less sweet than fructose and sucrose. Glucose is one of the most important biological sugars because it is the primary source of energy in living organisms, and circulates in blood as blood sugar. It is a monosaccharide and an aldohexose sugar. Plants produce glucose during photosynthesis $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$. glucose gives Tollens' and Fehling's test because it is a reducing sugar.
- Sucrose is ordinary table sugar used in homes. It is sweeter than glucose but less sweet than fructose. Sucrose is a disaccharide formed from glucose, and fructose. Glucose + Fructose → Sucrose, sucrose is a non-reducing sugar.
- Sugars are generally classified as **Monosaccharides.** Simple sugars that cannot be hydrolyzed further. Examples: glucose, fructose. **Disaccharides** formed by combination of two monosaccharides. Examples: sucrose, maltose, lactose,

BCW Bits:

- Carbohydrates are major energy-providing biomolecules.
- Sucrose is obtained mainly from sugarcane and sugar beet.
- Lactose is called milk sugar.
- Starch is a polysaccharide found in plants.

- Glycogen is called animal starch.
- Cellulose is the structural carbohydrate of plants.
- Reducing sugars can reduce Tollens' and Fehling's reagents.
- Honey contains large amounts of fructose and glucose.

52. Match List-I and List-II. Select the correct answer using the codes given below the list: ***[BPSC AEDO]***

List-I (Metal)		List-II (Name of the refining process)	
a.	Ni	1.	Van Arkel process
b.	Cu	2.	Electrolysis
c.	Zr	3.	Mond process
d.	Ge	4.	Zone refining

Codes:

	a	b	c	d
(A)	1	4	3	2
(B)	2	3	4	1
(C)	4	1	2	3
(D)	3	2	1	4

Ans. D

Exp:

- Nickel is purified by the Mond process. The Mond process is based on the formation of volatile nickel carbonyl. Nickel reacts with carbon monoxide at moderate temperature to form nickel tetracarbonyl $Ni+4CO \rightarrow Ni(CO)_4$. This nickel carbonyl is volatile and can be decomposed at higher temperature to obtain pure nickel $Ni(CO)_4 \rightarrow Ni + 4CO$. Mond process → mainly for nickel, Van Arkel process → mainly for titanium and zirconium.
- Copper is commonly refined by electrolytic refining. In this method impure copper acts as anode, pure copper sheet acts as cathode, and copper sulfate solution acts as electrolyte. When electric current passes pure copper gets deposited on cathode, impurities either dissolve or settle as anode mud. Electrolytic refining is widely used for metals like copper, silver, gold, zinc etc. Anode mud obtained during copper refining contains precious metals like silver and gold.
- Zirconium is purified by the Van Arkel process. It is used for obtaining ultra-pure metals such as zirconium, titanium. In this process impure metal reacts with iodine to form volatile iodide, which later decomposes on a hot tungsten filament to give pure metal. Example $Zr + 2I_2 \rightarrow ZrI_4$ Then decomposition $ZrI_4 \rightarrow Zr + 2I_2$.
- Germanium is purified by zone refining. This process is used mainly for semiconductors such as germanium, silicon. In zone refining a narrow heated zone moves slowly through the metal rod, impurities concentrate in the molten region, and finally move toward one end. This gives highly pure semiconductor material. The process is extremely important in the electronics industry because semiconductor devices require very high purity.
- **Different refining methods are chosen based on:** physical properties, chemical reactivity, volatility, and purity requirements of metals. For example volatile carbonyl → Mond process; volatile iodide → Van Arkel process; electrochemical behavior → Electrolysis; semiconductor purity → Zone refining.

BCW Bits:

- Metallurgy deals with extraction and purification of metals.
- Electrolysis involves chemical decomposition using electric current.
- Germanium and silicon are important semiconductors.
- Nickel is used in stainless steel and alloys.
- Zirconium is used in nuclear reactors due to corrosion resistance.
- Tungsten has very high melting point.
- Carbon monoxide forms metal carbonyl compounds.
- Semiconductors are essential in transistor and chip manufacturing.
- Copper is an excellent conductor of electricity.
- Impurities removed during refining are called gangue or slag depending on stage.

53. Which gas is more easily liquified among the following? ***[BPSC AEDO]***

A. NH_3 B. CH_4 C. CO D. H_2

Ans. A

Exp:

- A gas that can be liquified easily is the one having stronger intermolecular forces and a higher boiling point or critical temperature. Among the given gases, ammonia (NH_3) possesses strong intermolecular hydrogen bonding, making it much easier to liquify compared to methane (CH_4), carbon monoxide (CO), and hydrogen (H_2). Liquefaction of gases depends mainly on how strongly gas molecules attract each other. If intermolecular forces are strong, molecules can come together more easily when pressure is applied or temperature is lowered. The stronger the attraction, the easier the gas changes into liquid form. Ammonia molecules are polar and contain hydrogen atoms bonded to highly electronegative nitrogen atoms. This leads to **hydrogen bonding,** which is one of the strongest intermolecular forces found in ordinary molecular substances. Hydrogen bonding can be represented as: $N-H\cdots N$
- Because of this strong attraction, ammonia has a comparatively high boiling point and critical temperature. Therefore, it can be liquified relatively easily. The concept of **critical temperature** is extremely important here. Critical temperature is the highest temperature at which

a gas can be liquified by applying pressure alone. Gases with higher critical temperatures are easier to liquify.

- Approximate critical temperatures are: NH_3: very high among the given gases,CH_4: much lower, CO: lower, H_2: extremely low Thus, ammonia liquifies most easily.
- The main methods of liquefying gases are: Increasing pressure, lowering temperature. The relationship between pressure and volume is described by Boyle's law: PV=constant. At lower temperatures, molecular motion decreases, making liquefaction easier.
- Historically, gases like oxygen, nitrogen, and hydrogen were once called "permanent gases" because scientists could not liquify them initially. Later, advances by scientists like Thomas Andrews and James Dewar showed that every gas can be liquified if cooled below its critical temperature.
- Liquefied gases have huge industrial importance. Liquefied ammonia is widely used in refrigeration and fertilizer industries. Liquefied natural gas (LNG) is used for transport and storage of methane-rich fuel gases.

BCW Bits:

- Hydrogen bonding is stronger than ordinary dipole-dipole attraction.
- Water and ammonia both show hydrogen bonding.
- Ammonia is widely used as a refrigerant in cooling systems.
- Intermolecular forces strongly influence boiling point and liquefaction.
- Helium has the lowest boiling point among all elements.
- Liquefied petroleum gas (LPG) mainly contains propane and butane.
- Critical temperature above room temperature means easier liquefaction.
- Real gases deviate from ideal gas behavior at high pressure and low temperature.
- Van der Waals equation explains behavior of real gases.
- NH_3 is highly soluble in water due to polarity and hydrogen bonding.
- LNG stands for Liquefied Natural Gas.
- Refrigerators often use gases that can be easily liquified and vaporized repeatedly.

54. The iron obtained from the blast furnace is known as–

[BPSC AEDO]

A. Wrought Iron B. Pig Iron
C. Steel Iron D. Cast Iron

Ans. B

Exp:

- Pig iron is the crude form of iron obtained directly from a blast furnace during the extraction of iron from its ores. It contains a high percentage of carbon along with impurities such as silicon, sulfur, phosphorus, and manganese. Because of these impurities, pig iron is brittle and not directly suitable for most engineering purposes.
- Iron is mainly extracted from ores like hematite (Fe_2O_3) and magnetite (Fe_3O_4). In a blast furnace, iron ore is heated along with coke and limestone. Coke acts as a reducing agent, while limestone removes impurities by forming slag. The major reduction reaction is: $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$, Here: Carbon monoxide reduces iron oxide to metallic iron The molten iron collected at the bottom of the furnace is called pig iron. The name "pig iron" originated historically because molten iron from the furnace used to flow into branching sand molds resembling piglets suckling from a sow. The central channel looked like a mother pig, while side channels resembled piglets. Pig iron generally contains about 3–4% carbon. Due to this high carbon content, it is hard but brittle. It is therefore not suitable for direct use in tools or structures. Instead, it is further processed to produce wrought iron, cast iron, and steel.
- Steel is an alloy of iron containing controlled amounts of carbon and sometimes other elements like chromium or nickel. Cast iron is produced by remelting pig iron along with scrap iron and coke. It is not the direct product obtained from the blast furnace.
- The blast furnace itself is a tall cylindrical furnace where different reactions occur in different temperature zones. The raw materials added are: Iron ore → source of iron
- Coke → reducing agent and fuel, Limestone → flux. Limestone decomposes as: $CaCO_3 \rightarrow CaO + CO_2$, Then calcium oxide reacts with silica impurities: $CaO + SiO_2 \rightarrow CaSiO_3$. The product $CaSiO_3$ is slag, which floats over molten iron and can be separated easily. The different forms of iron mainly differ in carbon content: Pig iron → high carbon, Cast iron → slightly refined pig iron, Steel → controlled carbon, Wrought iron → very low carbon. Among these, steel is the most widely used because it combines strength and toughness effectively. Modern industries depend heavily on steel production for buildings, railways, bridges, automobiles, machinery, and defense equipment. India is one of the major steel-producing countries in the world.

BCW Bits:

- Steel Authority of India Limited (SAIL) is a major public sector steel producer.
- Jamshedpur in Jharkhand is famous for steel production.
- Wrought iron contains less than 0.1% carbon.
- Cast iron contains about 2–4% carbon.
- Stainless steel contains chromium for corrosion resistance.
- Slag is used in road construction and cement industries.
- Coke is obtained from destructive distillation of coal.
- Hematite is called "red ore" of iron.

- Magnetite is the richest iron ore magnetically.
- Blast furnaces operate continuously for long durations in large steel plants.

55. Dacron is synthesized by polymerization of *[BPSC WMO]*

A. Dimethyl terephthalate with adipic acid
B. Methyl methacrylate with styrene
C. Ethylene glycol with hexamethylene diamine
D. Ethylene glycol with dimethyl terephthalate

Ans. D

Exp: Dacron (also known as Terylene or polyethylene terephthalate) is a strong, wrinkle-resistant synthetic polyester synthesised via condensation polymerisation of **ethylene glycol (HO–CH_2–CH_2–OH)** and **dimethyl terephthalate** [**$C_6H_4(COOCH_3)_2$**]. (or alternatively, terephthalic acid), heated at high temperatures in the presence of a catalyst.

- $nHOCH_2CH_2OH + nC_6H_4(COOCH_3)_2 \rightarrow [-OCH_2CH_2OCOC_6H_4CO-]n$
- Due to its high strength, wrinkle resistance, and durability, Dacron is widely used in **textiles (clothing, curtains, bedsheets), fiberfill materials (pillows, jackets), industrial products (ropes, sails), and medical applications such as vascular grafts.**

56. Winkler method is used to determine *[BPSC WMO]*

A. Organic carbon
B. Elemental carbon
C. Dissolved oxygen
D. Biochemical Oxygen Demand

Ans. C

Exp: The **Winkler method** determines **dissolved oxygen (DO)** in water by converting oxygen into an equivalent amount of iodine, which is then titrated with **sodium thiosulfate**. Although DO measurements are used in **BOD** determination, the method directly measures **dissolved oxygen**, not BOD.

57. Water and ammonia molecules contain *[BPSC WMO]*

A. Water has sp2 while NH3 has sp3 hybridization and their bond angles are 109° and 107°, respectively
B. Both have sp3 hybridization and they have same bond angle 104.5°
C. Both have sp3 hybridization and bond angles 104.5° and 107°, respectively
D. Water has sp3 while NH3 has sp2 hybridization and their bond angles are 104.5° and 109°, respectively

Ans. C

Exp: The chemical structure, hybridisation, and bond angles of molecules are determined by the number of bonding pairs and lone pairs of electrons around the central atom, according to the VSEPR (Valence Shell Electron Pair Repulsion) theory.

- Water (H_2O) and ammonia (NH_3) both exhibit **sp^3 hybridisation** because their central atoms possess a steric number of 4, meaning that the total number of bonded atoms and lone pairs surrounding the central atom is four. This arrangement corresponds to an ideal tetrahedral electron-pair geometry with a bond angle of **109.5°**.

Water (H_2O):

..
H — O — H
..

In a water molecule, the central oxygen atom is attached to two hydrogen atoms and carries two lone pairs (2 bonded atoms + 2 lone pairs = steric number of 4). According to VSEPR theory, lone pairs repel more strongly than bonding pairs, and the significant repulsion between the two lone pairs compresses the ideal tetrahedral angle from 109.5° to **104.5°**.

Ammonia (NH_3):

..
N
/ | \
H H H

In an ammonia molecule, the central nitrogen atom is bonded to three hydrogen atoms and contains one lone pair of electrons (3 bonded atoms + 1 lone pair = steric number of 4). Since there is only one lone pair, the repulsive effect is less pronounced than in water, causing the bond angle to decrease only slightly from 109.5° to **107°**.

The order of electron-pair repulsion is:

- Lone Pair–Lone Pair>Lone Pair–Bond Pair>Bond Pair–Bond Pair
- As water contains two lone pairs whereas ammonia contains only one, the bond angle in H_2O is smaller than that in NH_3. Therefore, both molecules are **sp^3 hybridized**, with bond angles of **104.5°** and **107°**, respectively, making **option C correct**.

58. With reference to the thermal properties of the substances, consider the following statements. *[BPSC WMO]*

i. The temperature at which the solid and the liquid states of the substance are in thermal equilibrium with each other is called its melting point.
ii. The temperature at which the liquid and the vapour states of the substance coexist is called its boiling point.
iii. The change of state from solid to liquid is called melting and from liquid to solid is called fusion.

Which of the above statements are correct?

A. Only (i) and (iii) B. Only (ii) and (iii)
C. Only (i) and (ii) D. All three

Ans. C

Exp: The thermal properties of a substance are characterized by the temperatures at which changes of state occur under a given pressure.

- **Statement i is correct:** The melting point is the specific temperature at which a solid and its liquid phase coexist in thermal equilibrium. At this temperature, both states can exist together without any net change in their proportions unless heat is added or removed.
- **Statement ii is correct:** The boiling point is the temperature at which the liquid and vapor (gas) states of a substance coexist in thermal equilibrium. At this point, the vapor pressure of the liquid equals the external pressure surrounding the liquid.
- **Statement iii is incorrect:** The change of state from solid to liquid is called melting, the change of state from liquid to solid is called **freezing** (or solidification). The term **fusion** is another word for melting (solid to liquid), not liquid to solid.

Since only statements **(i)** and **(ii)** are correct, option C is the right choice.

59. Which concentration term is temperature independent? ***[BPSC WMO]***

1. Molarity
2. Molality
3. Normality
4. Mole fractions

A. Only 1 is correct
B. Both 2 and 4 are correct
C. 1 and 4 are correct
D. 1, 2 and 3 are correct

Ans. B

Exp: Concentration terms can be divided into two categories: volume-dependent and mass-dependent. Since volume changes with temperature (due to thermal expansion or contraction) while mass remains constant, any concentration term involving volume depends on temperature, whereas terms based solely on mass or moles are temperature independent.

Molality: Molality is defined as the number of moles of solute per kilogram of solvent. Since both moles and mass do not change with temperature, molality is temperature independent.

$$m = \frac{moles\ of\ solute}{kg\ of\ solvent}$$

Mole fractions: Mole fraction is the ratio of the number of moles of a particular component to the total number of moles in the solution. Since it depends entirely on the number of moles (which are mass-based), it is temperature independent.

$$x_{\mathrm{i}} = \frac{n_i}{\sum^{n}}$$

Molarity: Molarity is defined as the number of moles of solute per litre of solution. Because it involves volume, its value changes when the temperature changes. Thus, it is temperature-dependent.

$$M = \frac{moles\ of\ solute}{L\ of\ solution}$$

Normality: Normality is defined as the number of gram equivalents of solute per litre of solution. Like molarity, it relies on the volume of the solution, making it temperature dependent.

$$N = \frac{gram\ equivalents\ of\ solute}{L\ of\ solution}$$

Therefore, only **molality** and **mole fraction** are temperature independent, making **statements 2 and 4 correct** and **option B.** the correct answer.

60. Peeling of the ozone umbrella is due to ***[BPSC WMO]***

1. PAN
2. Freons
3. CO_2
4. NO

A. 2 and 4 are correct
B. Both 2 and 3 are correct
C. 1 and 4 are correct
D. 1, 3 and 4 are correct

Ans. A

Exp: The "ozone umbrella" refers to the protective ozone layer in the stratosphere that absorbs harmful ultraviolet (UV) radiation from the sun. The "peeling" of this umbrella means the depletion or thinning of this layer, which is primarily caused by certain ozone-depleting substances (ODS).

- **Freons:** Freons are Chlorofluorocarbons (CFCs). When released into the atmosphere, they drift up into the stratosphere where UV light breaks them down to release highly reactive chlorine atoms. A single chlorine atom can destroy thousands of ozone molecules. Therefore, Freons are a major cause of ozone depletion.
- **NO (Nitric Oxide):** Nitrogen oxides (NOx), including nitric oxide emitted by supersonic jets and industrial processes, act as catalysts in the stratosphere that actively destroy ozone molecules through chemical chain reactions. Therefore, NO is also a correct cause.
- **PAN (Peroxyacetyl Nitrate):** PAN is a secondary pollutant found in photochemical smog. It causes eye irritation and respiratory issues in the lower atmosphere (troposphere) but is not a primary driver for thinning the stratospheric ozone layer.
- **CO_2 (Carbon Dioxide):** Carbon dioxide is the primary greenhouse gas responsible for global warming and trapping heat in the lower atmosphere (troposphere). It does not directly destroy ozone molecules in the stratosphere.

Therefore, only **Freons (2)** and **NO (4)** are responsible for the depletion of the ozone layer, making **option A.** correct.

61. In the radioactive alpha decay, the daughter nucleus has- ***[BPSC WMO]***

A. Same number of protons as its parent nucleus

B. Two protons and two neutrons more than its parent nucleus
C. Same number of neutrons as its parent nucleus
D. Two protons and two neutrons fewer than its parent nucleus

Ans. D

Exp: The radioactive alpha decay is a nuclear process where an unstable atom's nucleus emits an alpha particle to become more stable. An alpha particle is identical to a helium nucleus, consisting of two protons and two neutrons. Because these particles are ejected from the parent nucleus, the resulting daughter nucleus loses those exact components.

The general nuclear equation for alpha decay is:

(A, Z) X -----> (A-4, Z-2) Y + (4, 2) alpha (or He)

Where:

- X is the parent nucleus
- Y is the daughter nucleus
- alpha is the emitted alpha particle (helium nucleus)
- A is the mass number (total protons and neutrons)
- Z is the atomic number (number of protons)

Two protons and two neutrons fewer: As the parent nucleus expels an alpha particle, its atomic number (Z) decreases by 2 (losing 2 protons), and its mass number A. decreases by 4 (losing 2 protons and 2 neutrons).

Therefore, **D** is correct, the daughter nucleus has exactly two protons and two neutrons fewer than its parent nucleus.

Biology

4

1. Hormones are secreted by: *[71st BPSC]*
   A. Reproductive glands
   B. Sebaceous glands
   C. Exocrine glands
   D. Endocrine glands

Ans. D

Exp: Hormones are chemical messengers secreted directly into the bloodstream by endocrine glands. Unlike exocrine glands such as sweat or salivary glands, which use ducts, endocrine glands are ductless.

- Hormones regulate different functions and activities in our body, including metabolism, growth, development, and reproduction.

Gland Type	Secretion Mechanism	Primary Secretions / Examples	Key Exam Facts
Endocrine Glands	**Ductless**; secretes directly into the bloodstream.	Hormones (e.g., Insulin, Thyroxine, Growth Hormone).	secrete hormones that control long-term, full-body (**systemic**) balance (**homeostasis**), energy production (**metabolism**), and growth. ◆ In this context, **systemic** means the effects spread throughout the entire body rather than staying in one local part. ◆ **Homeostasis** is the body's ability to maintain a stable, healthy internal environment (like regulating temperature or blood sugar), ◆ while **metabolism** is the chemical process used to turn food into energy.
Exocrine Glands	**With Ducts**; secretes onto epithelial surfaces.	Sweat, Saliva, Digestive enzymes, Sebum.	Works only in one specific spot instead of affecting the whole body (e.g., salivary glands, sweat glands).
Reproductive Glands	Mixed / Endocrine function.	Sex hormones (Estrogen, Progesterone, Testosterone).	Represent the core reproductive glands (female ovaries and male testes) responsible for regulating secondary sexual characteristics.
Sebaceous Glands	**Exocrine** mechanism via hair follicles.	Sebum (an oily, waxy matter).	Located in the skin; lubricates and waterproofs the skin and hair.

2. Which class of immunoglobulin is the first to be produced in response to an infection? *[71st BPSC]*
   A. IgA B. IgG C. IgM D. IgE

Ans. C

Exp: IgM is the first antibody that the body makes when it fights a new infection. It is a large molecule (a pentamer) that provides a quick, short-term defense before IgG is produced in larger quantities for long-term immunity.

- **Antibodies, also known as immunoglobulins**, are Y-shaped proteins created by the body's immune system to identify and destroy harmful invaders like bacteria and viruses (called pathogens). The human body makes five distinct classes of these antibodies, each assigned to a different role—some act as fast first-responders during a sudden illness, while others guard our body surfaces or trigger allergic reactions to protect us.

Immunoglobulin Class	Primary Role / Exam Fact	Key Facts
IgA	**Secretory Immunity**	Found heavily in body secretions like saliva, tears, sweat, and respiratory passages. It protects body surfaces that are exposed to the outside world.
IgG	**Long-Term Immunity**	The most abundant antibody in the blood. It takes over after IgM to provide long-term protection and is the only antibody that can cross the placenta from mother to baby.

IgM	**First Responder**	The very first antibody produced during an initial infection to deliver immediate, short-term defense.
IgE	**Allergies & Parasites**	Triggers the body's allergic reactions (like sneezing or hives) and helps defend against parasitic worm infections.

3. Which of the following is a nonsense codon? ***[71st BPSC]***

A. AUG B. UAA C. UUU D. UGC

Ans. B

Exp: A non-sense codon (or stop codon) signals the termination of protein synthesis during translation. There are three stop codons: UAA, UAG, and UGA.

- During protein synthesis, the genetic code is read in groups of three letters called **codons**. While most codons tell the cell which building block (amino acid) to add next to build a protein, a few special codons act as punctuation marks to tell the cell exactly where to start and stop the process.

Codon	Biological Role / Designation	Key Facts
AUG	**Start Codon / Methionine**	Acts as the green light. It signals the cell to begin building a protein chain and always codes for the amino acid Methionine.
UAA	**Stop Codon / Nonsense Codon**	Acts as a red light. It does not code for any amino acid; instead, it tells the cell to stop building and release the finished protein.
UUU	**Sense Codon / Phenylalanine**	A standard codon that specifically instructs the cell to add the amino acid Phenylalanine to the growing protein chain.
UGC	**Sense Codon / Cysteine**	A standard codon that specifically instructs the cell to add the amino acid Cysteine to the growing protein chain.

BCW Bits:

- **Alternative Names:** Scientists use historical nicknames for the three stop codons: **UAA** is called *Ochre*, **UAG** is called *Amber*, and **UGA** is called *Opal*.
- **Universal Language:** The genetic code is a universal biological language. The start codon (**AUG**) means "start" in almost every living organism on Earth, from tiny bacteria to humans.
- **Amino Acids:** These are the chemical building blocks that link together in chains to form functional proteins in living organisms.

4. Which metal compound is used for the treatment of carcinoma? ***[71st BPSC]***

A. Chromium B. Platinum

C. Iron D. Chlorine

Ans. B

Exp: Platinum-based chemotherapy drugs, most notably **Cisplatin [$Pt(NH_3)_2Cl_2$]**, are widely used to treat various types of cancers (carcinomas), such as testicular, ovarian, lung, and bladder cancers. These compounds work by binding directly to the DNA inside cancer cells, causing cross-linking. This damage stops the cancer cells from dividing and replicating, which ultimately triggers programmed cell death, a biological process known as **apoptosis**.

- In medical science and biochemistry, specific heavy metals and chemical elements are classified based on whether they treat diseases, act as sanitizers, or pose serious toxic risks to human health.

Element / Compound	Primary Biological Role	Simple Explanation & Exam Facts
Chromium	Industrial Metal / Carcinogen	Unlike therapeutic metals, chromium compounds are highly toxic and recognized as known cancer-causing substances (**carcinogens**) rather than treatments.
Platinum	Anti-Cancer Agent	Forms the core of chemotherapy drugs like Cisplatin, Carboplatin, and Oxaliplatin which disrupt tumor growth.
Iron	Essential Mineral	Iron is a non-toxic metal used primarily to synthesize hemoglobin and treat iron-deficiency **anemia**.
Chlorine	Non-Metal Halogen	Chlorine is a non-metal element commonly used as an antimicrobial **disinfectant** for water purification and sanitation.

BCW Bits:

- **Alternative Cancer-Treating Isotopes:** Beyond platinum compounds, radioactive forms of other elements are heavily utilized in targeted oncology:
  - **Cobalt-60:** A highly radioactive isotope that emits powerful gamma rays to selectively target and destroy malignant cancer cells during external radiation therapy.
  - **Iodine-131:** A key medical radioisotope widely prescribed to treat thyroid gland disorders and thyroid cancer.
  - **Phosphorus-32:** A radioactive isotope utilized in hematology to slow down the overproduction of abnormal blood cells.

5. Which compound is used in contraceptive pills? ***[71st BPSC]***

A. Levonorgestrel B. Cholecalciferol
C. Vanlaflexine D. Cetrizine

Ans. A

Exp: Levonorgestrel is a synthetic progestogen used in several types of birth control, including daily contraceptive pills and emergency contraceptive pills. It works by preventing ovulation and thickening cervical mucus to block sperm.

BCW Bits:

- **Cholecalciferol (Vitamin D3):** An essential nutrient that regulates the absorption of calcium and phosphorus, which is vital for building and maintaining strong bones.
- **Vanlafaxine (SNRI Antidepressant):** A prescription psychiatric medication that balances brain chemicals to treat major depressive disorder, generalized anxiety, and panic disorders.
- **Cetirizine (Antihistamine):** A common allergy medication that blocks histamine production in the body to relieve symptoms like sneezing, runny nose, and itchy, watery eyes.

6. Which one of the following terms characterizes the interaction between the herpes simplex virus and a human? [71st BPSC]

A. Symbiosis B. Parasitism
C. Endosymbiosis D. Endoparasitism

Ans. D

Exp: The interaction between the herpes simplex virus and a human is characterized as **endoparasitism**. In any parasitic relationship, one organism (the parasite) benefits by living at the direct expense of another organism (the host). Because a virus is an obligate intracellular entity that must live, replicate, and cause harm *inside* the host's actual cells, the prefix "endo-" (meaning internal) makes endoparasitism the most precise classification.

- In ecology and evolutionary biology, long-term interactions between different species are classified by where the organisms live and whether the relationship is mutually beneficial or harmful.
- **Symbiosis (Option A):** This is a broad umbrella term used to describe any close, long-term biological interaction between two different organisms. Depending on the setup, these relationships can be helpful, completely neutral, or harmful to the involved species.
- **Parasitism (Option B):** This represents a general harmful relationship where one species benefits directly by living at the expense of its host. This category is wide and includes parasites that live on the outside of a body (like ticks) as well as those living inside.
- **Endosymbiosis (Option C):** This describes a highly intimate, mutually beneficial relationship where one organism lives directly *inside* the body or cells of another, and **both organisms benefit** from the arrangement. A classic example is the mitochondria living inside human cells to produce energy.

BCW Bits:

- **Endo- vs. Ecto-:** In biological terms, **Endo** means *inside* (like endoparasites living in blood or cells), while **Ecto** means *outside* (like ectoparasites such as lice, fleas, or ticks living on the skin).
- **Obligate Parasite:** Viruses are considered obligate intracellular parasites, meaning they completely lack the machinery to reproduce on their own and are forced to hijack a host cell to survive.

7. The biological membranes are composed of: ***[71st BPSC]***

A. Lipids and proteins in varying combinations, which are specific to each species, cell type and organelle
B. Lipids only
C. Proteins only
D. None of the above

Ans. A

Exp: According to the Fluid Mosaic Model, cell membranes are dynamic structures primarily composed of lipids and proteins, along with a smaller amount of carbohydrates. The Lipid Bilayer provides the basic fluid structure. It consists mainly of phospholipids, cholesterol, and glycolipids, while the protein Diversity: Proteins act as the 'workhorses, ' serving as pumps, channels, receptors, and enzymes.

8. Xylem in plants is responsible for the following function: ***[71st BPSC]***

A. Transport of foods
B. Transport of water and dissolved minerals
C. Transport of oxygen and other gases
D. Transport of essential amino acids

Ans. B

Exp: Xylem transports water from roots to stems and leaves and it also transports nutrients. The best-known xylem tissue is wood. The term was introduced by **Carl Nägeli** in 1858.

- Xylem is unidirectional, while phloem is bidirectional.
- Xylem is located in the middle of the vascular bundle, while the phloem is located outside the vascular bundle. So, when the bark is removed or damaged, plants usually die as their phloem (which is present near the bark) gets damaged.

Note:

- Phloem is the vascular tissue that transports carbon (photosynthates) from the leaves to the basal parts of the plant. Phloem is also called **'Bast Tissue'.**
- **Cambium:** They are mainly present on the lateral sides of stems and roots (in between the permanent tissues). It provides secondary growth to the plants.

9. Who was first to use the characteristics of vascular tissues in the classification of plants? ***[71st BPSC]***

A. Engler and Prantl
B. Bentham and Hooker
C. A P de Condolle
D. More than one of the above

Ans. C

Exp: Augustin Pyramus de Candolle (A P de Candolle) was a Swiss botanist who revolutionized plant taxonomy. He was the first to divide the plant kingdom into two major structural groups based entirely on the presence or absence of conducting vessels: **Vasculares** (advanced plants possessing vascular tissues like xylem and phloem) and **Cellulares** (simpler, non-vascular plants consisting entirely of cells, such as mosses and liverworts).

- **Engler and Prantl (Option A):** Introduced the first major **Phylogenetic system**, arranging plants by evolutionary lineage from primitive **Gymnosperms** (naked seeds) to advanced **Angiosperms** (flowering plants).
- **Bentham and Hooker (Option B):** Published a **Natural system** based on multiple observable physical traits; it remains the global standard for arranging physical specimens in herbaria today.
- **A P de Candolle (Option C):** Pioneered structural classification using internal anatomy to split plants into vascular (*Vasculares*) and cellular (*Cellulares*) groups, and coined the term **"Taxonomy."**

10. The part which we eat in Mango is: ***[71st BPSC]***

A. Epicarp
B. Mesocarp
C. Endocarp
D. None of the above

Ans. B

Exp: A **drupe** is a fleshy fruit with a single stony seed. The fruit wall, known as the **pericarp**, is divided into three distinct layers:

- **Epicarp:** This is the outermost 'skin' of the fruit. It is usually leathery and protects the fruit. While edible in some varieties, it is typically peeled away.
- **Mesocarp:** This is the middle, fleshy, and succulent part, which we eat in Mango.
- **Endocarp:** This is the innermost layer.

BCW Bits:

- Fruit — Edible Part
- Mango — Mesocarp
- Coconut — Endosperm
- Apple — Thalamus (fleshy receptacle)
- Tomato — Pericarp and Placenta

11. In mycorrhizal association, the advantage of the plant is ***[71st BPSC]***

A. Protection
B. Food
C. A and B both
D. Increased mineral absorption and disease protection

Ans. D

Exp: Mycorrhiza is a symbiotic relationship between a fungus and plant roots. The fungus extends its hyphae into the soil, significantly increasing the surface area for mineral, especially phosphorus, and water absorption, while also providing a physical barrier and chemical defense against soil-borne pathogens (disease protection).

Mutualism

- It is defined as the interaction between two organisms of different species where the partners are benefited and none of the two are capable of living separately.
- This type of relationship is called a symbiotic relationship. Examples: Lichens, Mycorrhiza, Pollination and seed dispersal (the best example of plant-animal relationships).
- Lichen is a composite entity formed by algae (phycobiont) and fungus (mycobiont). The algae provide food to fungi while fungi provide shelter, water, and minerals to algae.
- Lichens are also known as 'Pollution indicators' as they can only grow in areas where the sulphur dioxide content is less in the environment.
- Litmus solution is extracted from the lichens which are used for acid, and base tests.

12. The species which are in danger of extinction are known as? ***[71st BPSC]***

A. Rare
B. Vulnerable
C. Endangered
D. Extinct

Ans. C

Exp: According to the IUCN Red List, **Endangered** species face a **very high risk** of extinction in the wild. **Rare** species have small, localized, or scattered populations that are naturally vulnerable but not yet endangered. **Vulnerable** indicates a high risk of extinction (one step below endangered), while **Extinct** means no living individuals remain.

13. Which of the following are phagocytic cells? ***[BPSC ASO, Pre]***

A. Neutrophils, macrophages
B. Neutrophils, mast cells
C. Mast cells, antibodies
D. Mast cells, macrophages

Ans. A

Exp: Phagocytic cells are a critical part of the innate immune system that protect the body by ingesting (eating) harmful foreign particles, bacteria, and dead or dying cells. Neutrophils and macrophages are the primary 'professional' phagocytes in the human body that perform this function to clear infections.

- **Mast cells** are non-phagocytic granulated cells residing in connective tissues that release chemical mediators like histamine to trigger inflammation and allergic responses, whereas **antibodies** are not cells at all, but rather Y-shaped protective proteins secreted by B-lymphocytes that specifically bind to and neutralize antigens to mark them for destruction by the actual phagocytic cells.

14. In which of the following type of cells the gap junctions are absent? ***[BPSC ASO, Pre]***

A. Brain cells B. Cardiac cells
C. Sperm cells D. Reproductive cells

Ans. C

Exp: Gap junctions are specialized intercellular connections that allow for direct chemical and electrical communication between adjacent cells. They are vital in the brain for neural signaling and in cardiac cells for synchronized heart contractions. Sperm cells are individual, free-moving cells that do not form stable tissue connections with neighbors, and therefore they lack gap junctions.

15. The principal buffer present in human blood ***[BPSC ASO, Pre]***

A. $CH_3COOH + CH_3COONa$
B. $NaH_2PO_4 + Na_2HPO_4$
C. $H_2CO_3 + HCO_3$
D. $H_3PO_4 + NaH_2PO_4$

Ans. C

Exp: The primary buffer system maintaining the pH of human blood is the bicarbonate buffer system. It consists of carbonic acid (H_2CO_3) as the weak acid and the bicarbonate ion (HCO_3) as the conjugate base. This system is crucial for neutralizing excess acids or bases to keep blood pH within the narrow survival range of 7.35 to 7.45.

16. Glycogen is primarily stored in which organs of human beings? ***[BPSC ASO, Pre]***

A. Brain and spinal cord
B. Kidneys and lungs
C. Liver and muscles
D. Skin and bones

Ans. C

Exp: Glycogen is the multi-branched polysaccharide that serves as the primary form of energy storage in humans. When the body has excess glucose, it is converted into glycogen and stored mainly in the liver (for systemic blood sugar regulation) and in skeletal muscles (for local energy use during physical activity).

17. Flowers without petals are pollinated by ***[BPSC ASO, Pre]***

A. Wind B. Insect
C. Butterflies D. Birds

Ans. A

Exp: Petals are primarily used by plants to attract pollinators like insects, butterflies, and birds through bright colors and scents. Flowers that lack petals or have very small, inconspicuous ones do not invest energy in attracting animal pollinators; instead, they generally rely on the wind to carry their pollen from the male part of the plant (Stamen: composed of anther and filament) to the female part of the plant (Pistil or Carpel: composed of stigma, style, and ovary).

- Insects, butterflies, and birds are agents of cross-pollination, which requires the help of such biotic and abiotic agents to transfer pollen grains from the anther of one flower to the stigma of another. Additionally, the study of these pollen grains is called **palynology**. **Saffron** is the stigma of the plant and **hing (Asafoetida)** is obtained from the roots of plants.

18. Edible portion of the coconut is ***[BPSC ASO, Pre]***

A. Cotyledon B. Embryo
C. Fruit wall D. Endosperm

Ans. D

Exp: In a coconut, the white fleshy part that we eat and the coconut water we drink are both forms of endosperm. The endosperm is a tissue produced inside the seeds of most flowering plants following fertilization, providing essential nutrients to the developing plant embryo.

19. What is the main reason that bryophytes are referred to as 'amphibians of the plant kingdom'? ***[BPSC ASO, Pre]***

A. They have a life cycle that includes both aquatic and terrestrial stages.
B. They have a strong tolerance to extreme temperatures.
C. They can live both on land and in water.
D. They require water for sexual reproduction.

Ans. D

Exp: While bryophytes live on land, they are called the 'amphibians of the plant kingdom' because they are not fully independent of water. Much like biological amphibians, bryophytes require a film of water for their flagellated sperm to swim to the egg for fertilization to occur. Without water, their sexual reproduction cycle cannot be completed.

20. Find the incorrect statement for algae. ***[BPSC ASO, Pre]***

A. Chlorella is used in space food
B. Algin is produced by algae
C. 'Agar-agar' is produced from Gracilaria
D. Mannitol is a food reserve of red-alga

Ans. D

Exp: Chlorella is a nutrient-rich alga used by space travelers, Algin is a hydrocolloid obtained from brown algae, and Agar-agar is derived from Gracilaria and Gelidium. Statement D is incorrect because Mannitol is the complex carbohydrate used as a food reserve in brown algae (Phaeophyceae), whereas red algae (Rhodophyceae) store food as Floridean starch.

21. What is the role of the enzyme RuBisCO in photosynthesis? ***[BPSC ASO, Pre]***

A. It fixes carbon dioxide to produce carbohydrate.
B. It regulates stomatal opening and closing for gaseous exchange.
C. It catalyzes the splitting of water molecules to generate oxygen.
D. It is involved in the electron transport chain to generate ATP.

Ans. A

Exp: RuBisCO (Ribulose-1, 5-bisphosphate carboxylase/ oxygenase) is the most abundant enzyme on Earth and plays a critical role in the Calvin cycle of photosynthesis. Its primary function is to catalyze the first major step of carbon fixation, where it attaches atmospheric carbon dioxide to a five-carbon sugar to eventually produce energy-rich carbohydrates like glucose.

- This occurs via an oxidation-reduction reaction where carbon dioxide is reduced to glucose and water is oxidized to release oxygen, as represented by chemical equation for photosynthesis:

  $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6(\text{glucose}) + 6O_2$

- Stomatal opening and closing are regulated by **potassium ions (K+)**, which alter the water pressure (turgor pressure) inside the surrounding guard cells to open or close the pore. The splitting of water molecules to release oxygen is called **photolysis of water**, a light-dependent process that occurs within the thylakoid membranes where chlorophyll absorbs sunlight. Finally, the generation of energy in the form of ATP is carried out during the light reactions by a specialized enzyme complex called **ATP synthase**, which acts like a molecular motor powered by a flow of hydrogen ions.

22. Which of the following statement is true regarding apical dominance in plants? ***[BPSC ASO, Pre]***

A. Root and shoot in their apex are dormant
B. Removal of apical buds promote lateral bud growth
C. Decrease in cytokinin increases apical dominance
D. Auxin stimulates lateral bud formation

Ans. B

Exp: Apical dominance is a phenomenon where the central stem of the plant grows more strongly than other side stems because the apical bud produces hormones like auxin that inhibit the growth of lateral (side) buds. When the apical bud is removed, the source of this inhibition is gone, allowing the lateral buds to grow and making the plant bushier.

23. In angiosperms, xylem is made up of ***[BPSC ASO, Pre]***

A. Fibers and tracheids
B. Sieve elements and tracheids
C. Vessels and fibers
D. Tracheids, vessels and fibers

Ans. D

Exp: Xylem is the complex permanent tissue in vascular plants responsible for transporting water and minerals. In angiosperms (flowering plants), the xylem consists of four main components:

- **tracheids** (elongated, dead cells with tapering ends that provide mechanical strength and water pathways),
- **vessels** (long, tube-like structures arranged end-to-end with open perforation plates that allow rapid, efficient water flow, which are absent in most gymnosperms),
- **xylem fibers** (thick-walled cells that offer structural support),
- and **xylem parenchyma** (the only living cells in the tissue, used for storage).
- Option B is incorrect because **sieve elements** (sieve tubes and companion cells) are specialized living components belonging exclusively to the **phloem**, which functions to transport synthesized organic food (carbohydrates) rather than water.

24. Why is grafting not possible in monocots? ***[BPSC ASO, Pre]***

A. Because they do not have venation
B. Because they are herbaceous
C. Because they have parallel venation
D. Because they lack cambium

Ans. D

Exp: Successful grafting requires the vascular cambium of the stock and scion to align and fuse so that the vascular tissues (xylem and phloem) can connect. Monocots lack a lateral meristem or vascular cambium; their vascular bundles are scattered throughout the stem rather than arranged in a ring, making it impossible for the tissues to join together as they do in dicots.

25. Which of the following is a medicinal plant used to treat a certain type of cancers? *[BPSC ASO, Pre]*

A. Himalayan Birch B. Himalayan Yew
C. Chir-pine D. Himalayan Oak

Ans. B

Exp: The Himalayan Yew (Taxus wallichiana) contains a chemical compound called Taxol in its needles and bark. Taxol is a powerful anti-cancer drug used extensively in the treatment of various malignancies, including breast, ovarian, and lung cancers.

26. The point at which the rate of photosynthesis is equal to the rate of respiration is called *[BPSC ASO, Pre]*

A. Respiratory quotient
B. Equivalence point
C. Photosynthetic quotient
D. Compensation point

Ans. D

Exp: The compensation point usually occurs at low light intensities like dawn or dusk, when the amount of carbon dioxide fixed during photosynthesis exactly matches the amount of carbon dioxide released during cellular respiration. At this specific point, there is no net exchange of gases between the plant and the environment.

27. Litmus, the pH indicator, is obtained from which of the following source? *[BPSC ASO, Pre]*

A. lichen B. A brown alga
C. A red alga D. A fungi

Ans. A

Exp: Litmus is a water-soluble mixture of different dyes extracted from various species of lichens, particularly Roccella tinctoris. It is one of the oldest forms of pH indicators, turning red under acidic conditions and blue under basic (alkaline) conditions.

28. Nymph is the young stage of *[BPSC ASO, Pre]*

A. Housefly B. Butterfly
C. Cockroach D. Beetle

Ans. C

Exp: Insects like cockroaches undergo incomplete metamorphosis, where the young stage resembles a miniature version of the adult but lacks wings and functional reproductive organs. This immature stage is called a nymph. In contrast, houseflies, butterflies, and beetles undergo complete metamorphosis, involving larval (maggot, caterpillar, or grub) and pupal stages.

29. Catharanthus roseus contains *[BPSC ASO, Mains]*

A. Vincristine B. Taxol
C. Aflatoxin D. Apigenin

Ans. A

Exp: *Catharanthus roseus*, commonly known as the **Madagascar Periwinkle** (Sadabahar), is a medicinal plant famous for producing alkaloids used in cancer treatment. It contains **Vincristine** and **Vinblastine**, which are potent chemotherapy medications used to treat various types of leukemia and lymphoma.

Substance	**Chemical Class**	**Source**	**Primary Fact**
Vincristine	Alkaloid	**Catharanthus roseus**	Anti-cancer drug; blocks cell division (mitosis).
Taxol	Diterpenoid	**Pacific Yew Tree** (*Taxus brevifolia*)	Also an anti-cancer drug, but stabilizes microtubules instead of preventing their formation.
Aflatoxin	Mycotoxin	**Aspergillus fungi**	Highly carcinogenic toxins found in contaminated crops like peanuts and corn.
Apigenin	Flavonoid	**Parsley, Chamomile, Celery**	A common dietary antioxidant with anti-inflammatory and sedative properties.

30. Xylem vessels help in water conduction commonly present in *[BPSC ASO, Mains]*

A. Gymnosperms
B. Angiosperms
C. Gymnosperms and Angiosperms
D. Pteridophytes and Angiosperms

Ans. B

Exp: The **Plant Kingdom** (Kingdom Plantae) is traditionally classified into five primary divisions based on their evolutionary complexity and structural differentiation. This phylogenetic system of classification was significantly advanced by the botanist **August Wilhelm Eichler** in 1883, who divided the kingdom into Cryptogamae (non-seeded) and Phanerogamae (seeded) plants.

Divisions of the Plant Kingdom:

- **Thallophyta:** Simple plants like Algae; no distinct roots, stems, or leaves.
- **Bryophyta:** "Amphibians of the plant kingdom" (e.g., Mosses); lack true vascular tissue.
- **Pteridophytes:** First terrestrial plants to possess vascular tissue (Xylem/Phloem), but do not produce seeds (e.g.,

Ferns).

- **Gymnosperms:** Naked-seeded plants; seeds are not enclosed in a fruit (e.g., Pines).
- **Angiosperms:** Flowering plants; seeds are enclosed within fruits. This group possesses the most advanced vascular systems.
- In the plant kingdom, **Angiosperms** (flowering plants) possess highly efficient water-conducting tissue, including **Xylem Vessels**. These are long, wide, tube-like structures formed by cells joined end-to-end with perforated walls. Their large diameter enables **rapid, high-volume water conduction**, a defining evolutionary characteristic of **Angiosperms**.
- While Pteridophytes and most **Gymnosperms** (like pines) do have xylem, their water conduction relies primarily on **tracheids** rather than vessels. These are narrower, elongated cells with tapering ends. While they also conduct water, they are less efficient than vessels.

31. Which one is used in spices? ***[BPSC ASO, Mains]***

A. Fungi B. Lichens
C. Mosses D. Ferns

Ans. B

Exp:

a. **Lichens** are complex life forms that represent a symbiotic partnership between two separate organisms: a **fungus** (the mycobiont) and an **alga** or cyanobacterium (the photobiont). The fungus provides the structural framework and protection, while the alga performs photosynthesis to provide food. This unique relationship allows lichens to survive in extreme environments where neither organism could thrive alone.
Common examples of lichens include **Parmotrema perlatum** (Stone Flower), **Cladonia rangiferina** (Reindeer Moss), **Usnea** (Old Man's Beard), and **Roccella tinctoria** (the source of Litmus dye).
A specific type of lichen, **Parmotrema perlatum** (commonly called "Stone Flower" or *"Pathar Phool"* in Hindi), is widely used **as a spice in Indian cuisine,** particularly in meat dishes and garam masala. It adds a distinct earthy aroma and flavor to the food. While some fungi (like mushrooms) are eaten as food, they aren't typically classified as spices.

b. **Fungi:** A kingdom of eukaryotic organisms (including yeasts, moulds, and mushrooms) that decompose organic matter. While some, like mushrooms, are consumed as vegetables or used for fermentation, they are not typically classified as dry spices.

c. **Mosses:** Small, non-vascular, flowerless plants that belong to the division **Bryophyta**. They typically form dense green clumps or mats in damp or shady locations and are not used in human culinary practices.

d. **Ferns:** Vascular plants belonging to the division **Pteridophyta** that reproduce via spores and have neither seeds nor flowers. While some young fern fronds (fiddleheads) are eaten as greens, they are not used as spices.

32. Most abundant enzyme on the earth is ***[BPSC ASO, Mains]***

A. Cytochrome oxidase B. PEP carboxylase
C. Indole oxidase D. Rubisco

Ans. D

Exp: Enzymes are biological catalysts that accelerate chemical reactions within living organisms. The distribution and abundance of an enzyme are often dictated by its role in essential life processes like respiration or photosynthesis.

- **Rubisco** (Ribulose-1, 5-bisphosphate carboxylase/ oxygenase) is the enzyme responsible for the first major step of carbon fixation in the Calvin cycle of photosynthesis. Because it is essential for almost all plant life and is relatively inefficient (meaning plants need a lot of it to function), it is produced in massive quantities, making it the most abundant protein and enzyme on Earth.

Other Options:

Enzyme	Biological Role	Key Fact
Cytochrome Oxidase	Found in mitochondria; the final enzyme in the **Electron Transport Chain**.	Essential for aerobic respiration, but present in much smaller quantities than Rubisco.
PEP Carboxylase	Used in **C4 and CAM plants** to capture CO2 in the cytosol.	Highly efficient and works better than Rubisco in hot, dry conditions, but is found in fewer plant species.
Indole Oxidase	Involved in the metabolism of indole compounds and the synthesis of **Auxins** (plant hormones).	Crucial for plant growth and development, but exists in trace amounts compared to metabolic enzymes.

33. With reference to Spirulina, consider the following statements: ***[BPSC ASO, Mains]***

(1) Protein-rich
(2) Prokaryote
(3) Heterocyst

Which of the statements given above is/are correct?

A. Only (1) B. All three
C. One and two D. None of the above

Ans. C

Exp: **Spirulina** is a nutrient-dense biomass of **Cyanobacteria** (blue-green algae) widely utilized as a superfood.

(1) Protein-rich: Spirulina is a highly nutritious blue-green algae (cyanobacteria) containing about **60–70%**

protein by dry weight, along with various vitamins and minerals.

(2) **Prokaryote:** Being a cyanobacterium, it is a **prokaryotic organism**, meaning it lacks a membrane-bound nucleus and other complex organelles.

(3) **Heterocyst:** Spirulina is a non-heterocystous cyanobacterium. Unlike organisms like *Nostoc* or *Anabaena*, it **does not possess heterocysts** (specialized cells for nitrogen fixation). It carries out photosynthesis but lacks these specific structures.

- Because Spirulina lacks heterocysts, statement (3) is incorrect, leaving **(1) and (2)** as the only valid choices.

34. Given below are two statements: one is labelled as Assertion (A) and other is labelled as Reason (R).

[BPSC ASO, Mains]

Assertion (A) : Global warming is caused due to increase in the environmental temperature.

Reason (R) : It is the result of increase in level of CO_2, deforestation and increase in methane production.

In light of the above statements, choose the most appropriate answer from the options given below.

A. Both (A) and (R) are correct and (R) is the correct explanation of (A)
B. (A) is correct, but (R) is not correct
C. (A) is not correct, but (R) is correct
D. None of the above

Ans. **C**

Exp: **Global warming** refers to the unusually rapid increase in Earth's average surface temperature over the past century, primarily due to the greenhouse gases released by people burning fossil fuels. This phenomenon is a specific aspect of climate change, characterized by an enhanced greenhouse effect where the atmosphere traps more heat than it radiates back into space.

- **Assertion (A) is Incorrect:** In chemistry, we distinguish between a **reaction** and its **result**. Global warming is the name of the "result" (the increase in temperature). To say that global warming is caused by rising temperatures is like saying "burning is caused by fire." The increase in temperature **is** the definition of the phenomenon, not the chemical trigger behind it.
- **Reason (R) is Correct:** This identifies the actual chemical reagents and catalysts.
- **Greenhouse Gases:** Molecules like **CO2** and **CH4** have specific molecular structures that allow them to vibrate and trap heat radiation.
- **Methane (CH4):** Chemically, methane is more potent than CO2 because its molecular bonds absorb a wider spectrum of thermal energy.
- **Deforestation:** This removes "Carbon Sinks"—biological systems that perform the chemical reaction of photosynthesis, removing CO2 from the atmosphere.

35. Arrange the following organisms in correct order of the evolution from the codes given below:

[BPSC ASO, Mains]

[Deleted by BPSC in official ans key]

(I) Phytophthora (II) Agaricus
(III) Alternaria (IV) Morchella

Codes:

A. (I), (II), (IV), (III) B. (II), (IV), (III), (I)
C. (I), (IV), (II), (III) D. (III), (II), (I), (IV)

Ans. **C**

Exp: Fungi are **eukaryotic heterotrophs** that absorb nutrients from organic matter. Unlike plants, they cannot make their own food. Their cell walls are unique because they are made of **chitin**, a tough structural carbohydrate.

- Modern fungal classification is based on the work of **Heinrich Anton de Bary** and **Robert Whittaker**. **De Bary** is known as the "father of modern mycology" because he proved that fungi are independent organisms capable of causing disease. **Whittaker** established the "Five Kingdom" system, placing Fungi in their own separate kingdom.
- The evolutionary hierarchy of fungi moves from simple aquatic forms to complex land-dwelling species:

(I) Phytophthora (Oomycetes): These are primitive "water molds." They are an ancestral link to algae because their cell walls contain **cellulose** instead of chitin.

(IV) Morchella (Ascomycota): These are "sac fungi." They represent an early stage of land evolution and produce spores in a sac called an **ascus**.

(II) Agaricus (Basidiomycota): These are advanced "club fungi" like mushrooms. They have complex structures and bear spores on club-shaped bases called **basidia**.

(III) Alternaria (Deuteromycota): These are highly specialized "imperfect fungi." They have evolved to prioritize fast asexual reproduction for better survival.

36. Match List-I with List-II and choose the correct answer from the codes given below: ***[BPSC ASO, Mains]***

List-I		List-II	
a.	IBA	1.	Seed dormancy
b.	GA3	2.	Root development
c.	ABA	3.	Fruit ripening
d.	Ethylene	4.	Plant growth

Codes:

A. a-2, b-4, c-1, d-3 B. a-4, b-1, c-2, d-3
C. a-3, b-2, c-4, d-1 D. a-1, b-3, c-2, d-4

Ans. **A**

Exp: Plant hormones, or phytohormones, are naturally occurring organic substances that influence physiological processes at very low concentrations. These regulators are essential for coordinating growth, development, and responses to environmental stimuli.

Functional Analysis of Phytohormones:

- The matching of these growth regulators is based on their primary roles in plant life cycles:
  - **IBA (Indole-3-butyric acid) → 2. Root development** IBA is a synthetic **Auxin** widely used in agriculture and horticulture. Its primary application is to initiate **adventitious root formation** in stem cuttings, making it the most common "rooting hormone."
  - **GA3 (Gibberellic Acid) → 4. Plant growth** Gibberellins like **GA3** are critical for **internodal elongation**, breaking seed dormancy, and increasing fruit size. By stimulating cell division and elongation, they directly increase the plant's overall height and biomass.
  - **ABA (Abscisic Acid) → 1. Seed dormancy.** Often called the **"stress hormone, "** ABA acts as a growth inhibitor. It prevents premature germination by inducing **seed dormancy** and triggers stomatal closure during water stress to prevent desiccation.
  - **Ethylene → 3. Fruit ripening:** Ethylene is the only **gaseous plant hormone.** It acts as a signalling molecule that initiates the **ripening process** in climacteric fruits, causing the breakdown of chlorophyll, softening of cell walls, and conversion of starches to sugars.

37. An animal that respires, but has no respiratory organs, is ***[BPSC ASO, Mains]***

A. Fish B. Frog
C. Earthworm D. Cockroach

Ans. C

Exp: Respiration is the biological process by which organisms exchange gases—specifically oxygen and carbon dioxide—to maintain metabolic functions. While many animals have evolved complex organs for this, some rely on direct diffusion through their body surfaces.

The animal kingdom utilizes different systems based on habitat and biological complexity:

- **Earthworm (Cutaneous Respiration):** Earthworms lack lungs, gills, or a tracheal system. They breathe entirely through their **skin**. Oxygen dissolves in the moisture on the skin's surface and diffuses directly into tiny blood vessels (capillaries). This is why earthworms must live in damp soil; if their skin dries out, they cannot absorb oxygen and will suffocate.
- **Fish (Branchial Respiration):** Fish possess specialised organs called **gills**. These are highly vascularized structures that extract dissolved oxygen from water as it passes over them.
- **Frogs (Amphibious Respiration):** Frogs are unique in that they use multiple methods. They utilize **lungs** (pulmonary respiration) on land, their **moist skin** (cutaneous respiration) in water, and even the lining of their mouth.
- **Cockroach (Tracheal Respiration):** Insects like cockroaches use a network of tubes called **tracheae**. Air enters through small openings on the body surface called **spiracles** and travels directly to the tissues, bypassing the need for a blood-based oxygen transport system.

38. Which of the following techniques is other than microscopy? ***[BPSC ASO, Mains]***

A. Plasmolysis B. Autoradiography
C. Maceration D. Chromatography

Ans. D

Exp: Biological research utilizes various physical and chemical techniques to study cells. While **Microscopy** focuses on visualizing structures that are too small for the naked eye, other laboratory techniques focus on the physical separation or chemical analysis of cellular components.

- **Plasmolysis:** This is the shrinking of the protoplasm away from the cell wall in a plant cell due to water loss (osmosis). This process is **directly observed under a microscope** to study cell membrane behavior and tonicity.
- **Autoradiography:** This technique uses radioactive isotopes to label specific molecules within a cell. The resulting "image" of where these molecules are located is typically **visualized using a microscope** or photographic film to track metabolic pathways.
- **Maceration:** This involves treating plant tissues with chemicals to dissolve the "middle lamella" (the cement between cells). This softens the tissue, allowing individual cells to be separated and **examined under a microscope** for anatomical study.
- **Chromatography:** Unlike the other options, chromatography is not a visualization tool. It is a technique used to **separate the components of a mixture** (such as plant pigments, proteins, or amino acids). It works by passing the mixture through a stationary phase (like paper or silica) using a mobile phase (a solvent). Components separate based on their solubility and molecular weight.
- While Plasmolysis, Autoradiography, and Maceration are either observed under the microscope or prepared for it, **Chromatography** stands alone as a chemical separation method.

39. Read the statements given below with regard to the functions performed by the Golgi apparatus: ***[BPSC ASO, Mains]***

I. Transports and chemically modifies the material contained within it.
II. Secretes mucin in the respiratory tract.
III. Secretes slime in the insectivorous plants.

Which of the following is the correct answer?
A. I is wrong, but II and III are correct
B. II is wrong, but I and III are correct
C. II and III are wrong, but I is correct
D. All are correct

Ans. D

Exp: The **Golgi apparatus** (also known as the Golgi complex or Golgi body) is a membrane-bound organelle found in eukaryotic cells. It consists of a series of flattened, stacked pouches called **cisternae**. Its primary role is to act as the "post office" of the cell, processing and directing the flow of proteins and lipids.

- The Golgi apparatus was discovered by **Camillo Golgi**, an Italian physician and biologist, in **1898.** He identified the structure while investigating the nervous system. Using a staining technique he developed called the **"black reaction"** (la reazione nera)—which involved silver nitrate—he observed a "reticular apparatus" (a net-like structure) inside the nerve cells of an owl and a cat.
- He shared the **Nobel Prize in Physiology or Medicine in 1906** with Santiago Ramón y Cajal for their work on the structure of the nervous system.Beyond the Golgi apparatus, he also discovered the **Golgi tendon organ** and **Golgi receptors** in muscles and tendons.

The Golgi apparatus is essential for both internal cell maintenance and external secretions:

(i) Transport and Modification: This is the most fundamental role of the Golgi. It receives raw proteins and lipids from the **Endoplasmic Reticulum (ER)**. Inside the Golgi, these molecules undergo chemical modifications—such as **glycosylation** (adding sugar chains to form glycoproteins)—before being sorted and packaged into vesicles for transport to their specific destinations.

(ii) Secretion of Mucin: In specialised cells called **goblet cells**, found in the lining of the respiratory and digestive tracts, the Golgi is exceptionally active. It synthesises and packages **mucin**. When mucin is released and mixed with water, it becomes **mucus**, which protects and lubricates the respiratory and digestive linings.

(iii) Secretion of Slime in Plants: In insectivorous plants (like the Venus Flytrap or Sundew), the Golgi apparatus in glandular cells is responsible for producing and secreting a **mucilaginous slime**. This sticky substance is used to trap insects and often contains digestive enzymes to break down the prey.

40. Colour blindness is found more in males than in females because ***[BPSC ASO, Mains]***
A. Males containing the single-affected X chromosome are colour blind
B. Heterozygous females are colour blind
C. Males having affected Y chromosomes are colour blind
D. Affected X-chromosome has a much higher affinity to the Y chromosome as compared to unaffected X chromosome

Ans. A

Exp: Inheritance of certain traits is determined by genes located on the sex chromosomes. Colour blindness is a classic example of a **sex-linked recessive** disorder, meaning the gene responsible for the condition is located on the **X-chromosome**.

- Colour blindness is a **sex-linked recessive trait** carried on the **X-chromosome**.
- **In Males (XY):** Males possess only one X-chromosome (inherited from the mother) and one Y-chromosome (inherited from the father). Because they lack a second X-chromosome, they do not have a "backup" gene to override a defective one. If their single X-chromosome carries the colour blindness gene, they **will** express the condition. This state is often referred to as **hemizygous**.
- **In Females (XX):** Females possess two X-chromosomes. Since the trait is recessive, a female only becomes colour blind if **both** of her X-chromosomes carry the defective gene. If she has only one affected chromosome, she is a **heterozygous carrier**; she has normal vision because the healthy gene on her second X-chromosome "masks" the defective one.
- Because males need only one copy of the defective gene, whereas females need two, the probability of males being affected is much higher.

41. The correct order of the following is ***[BPSC ASO, Mains]***
A. Mesozoic → Archaeozoic → Proterozoic
B. Palaeozoic → Archaeozoic → Cenozoic
C. Archaeozoic → Palaeozoic → Proterozoic
D. Palaeozoic → Mesozoic → Cenozoic

Ans. D

Exp: The history of Earth is divided into large spans of time called **Aeons**, which are further subdivided into **Eras**. The **Phanerozoic Aeon** represents the period of "visible life" and is the most recent aeon in Earth's history.

The sequence follows the evolution of life from ancient forms to modern biological diversity:

1. **Palaeozoic Era (Ancient Life):** Starting approximately 541 million years ago, this era saw the "**Cambrian Explosion**" of life. It is characterized by the rise of marine invertebrates, the first land plants, fishes, amphibians, and early reptiles.
2. **Mesozoic Era (Middle Life):** Known as the **"Age of Reptiles, "** this era is famous for the dominance of dinosaurs. It also saw the appearance of the first mammals, birds, and flowering plants. It ended with a mass extinction event about 66 million years ago.

3. **Cenozoic Era (Recent Life):** Often called the **"Age of Mammals, "** this is the current era. Following the extinction of dinosaurs, mammals and birds diversified and became the dominant land animals. It includes the evolution of humans and continues to the present day.

   The geological time scale demonstrates the chronological evolution of life, moving from ancient marine organisms to the dominance of mammals. This sequence confirms that the correct order of eras within the Phanerozoic Aeon is **Palaeozoic**, followed by **Mesozoic**, and ending with the **Cenozoic**.

42. Iodine deficiency in a frog tadpole will lead to ***[BPSC ASO, Mains]***

A. accelerated metamorphosis
B. development of a giant frog
C. inhibition of metamorphosis
D. development of a miniature frog

Ans. C

Exp: Metamorphosis is the biological process by which an animal physically develops after birth or hatching, involving a conspicuous and relatively abrupt change in the animal's body structure. In amphibians like frogs, this transition from an aquatic tadpole to a terrestrial adult is entirely regulated by the endocrine system.

- Metamorphosis in frogs (the transformation from a tadpole to an adult frog) is strictly controlled by the hormone **thyroxine**, which is secreted by the thyroid gland.
- **Iodine** is a critical component required for the synthesis of thyroxine. If there is an iodine deficiency in the water where the tadpole lives, the thyroid gland cannot produce enough hormone. As a result, the tadpole fails to undergo metamorphosis and remains in its larval (tadpole) stage, often continuing to grow in size without ever developing into an adult frog.

43. The part of the nephron that helps in active reabsorption of sodium is ***[BPSC ASO, Mains]***

A. Bowman's capsule
B. Distal convoluted tubule
C. Ascending limb of Henle's loop
D. Proximal convoluted tubules

Ans. D

Exp: The nephron is the functional unit of the kidney, responsible for filtering blood and forming urine. Reabsorption is a critical process in which the body recovers essential solutes, such as sodium, glucose, and amino acids, from the filtrate and returns them to the bloodstream to maintain homeostasis.

- The **Proximal Convoluted Tubule (PCT)** is the primary site for the bulk reabsorption of essential substances from the glomerular filtrate due to its high metabolic activity and structural adaptations:
- **Sodium Reabsorption:** Approximately **65–70% of filtered sodium** is reabsorbed here. This is an **active transport** process driven by the **Sodium-Potassium (Na+/K+) ATPase pump** on the basolateral membrane.
- **Water and Solute Recovery:** Because sodium is actively moved, water follows passively by **osmosis** (obligatory water reabsorption). Additionally, **100% of filtered glucose and amino acids** are normally reabsorbed in the PCT through secondary active transport.
- **Structural Adaptations:** The PCT cells are lined with a **"brush border" of microvilli**. This specialization increases the surface area exponentially, facilitating the massive exchange of solutes.
- **Bicarbonate Recovery:** The PCT plays a vital role in acid-base balance by reabsorbing about **80–90% of filtered bicarbonate (HCO3-)**.
- While several segments participate in electrolyte transport, the **Proximal Convoluted Tubule (PCT)** is the most significant site for bulk active sodium recovery. This process ensures that vital nutrients and the majority of filtrate volume are returned to circulation.

44. Which one of the following is the correctly matched pair of a product and the microorganism responsible for it? ***[BPSC ASO, Mains]***

A. Ethyl alcohol — Yeast
B. Cheese — Nitrobacter
C. Acetic acid — Lactobacillus
D. Curd — Azotobacter

Ans. A

Exp: Microbiology plays a pivotal role in the production of various food and chemical products. Many industrial processes rely on the metabolic activities of specific bacteria and fungi to convert raw materials into finished goods through fermentation and chemical transformation.

- Each pairing is determined by the unique way these microbes break down substances:
- **Ethyl Alcohol and Yeast:** This is the correct industrial pairing. **Saccharomyces cerevisiae**, commonly known as Brewer's yeast, performs **anaerobic fermentation**. It breaks down glucose into ethyl alcohol (ethanol) and carbon dioxide. This process is the foundation of the brewing and baking industries.
- **Cheese and Lactic Acid Bacteria:** The pairing in the option is incorrect because **Nitrobacter** is a nitrifying bacterium found in soil. Cheese production actually requires **Lactobacillus** or **Streptococcus**, which ferment lactose into lactic acid, causing milk proteins to coagulate into curd.
- **Acetic Acid and Acetobacter:** The pairing in the option is incorrect because **Lactobacillus** produces lactic acid, not acetic acid. Acetic acid (the main component of vinegar) is produced by **Acetobacter aceti**, which oxidizes ethyl alcohol into acetic acid in the presence of oxygen.

- **Curd and Lactobacillus:** The pairing in the option is incorrect because **Azotobacter** is a free-living, nitrogen-fixing bacterium used in biofertilizers. Curd formation is a biological process in which **Lactobacillus** bacteria convert milk lactose into lactic acid, denaturing milk proteins and creating a thick texture.
- Based on industrial microbiology, only **Ethyl alcohol — Yeast** is correctly matched. The other microorganisms listed either belong to the nitrogen cycle or are mismatched with their respective fermentation products.

45. Refractory period in a neuron is the period of complete inexcitability between ***[BPSC ASO, Mains]***

A. Polarization and depolarization
B. Depolarization and repolarization
C. Repolarization and polarization
D. None of the above

Ans. B

Exp: In neurobiology, the **Refractory Period** serves as a vital reset mechanism. It ensures that nerve impulses (action potentials) travel in only one direction and prevents the neuron from becoming overstimulated.

- **How the Neuron Locks Down?**
  - The state of "complete inexcitability" mentioned in the question refers to the **Absolute Refractory Period**. This occurs due to the specific behavior of voltage-gated ion channels:
  - **The Surge (Depolarization):** Voltage-gated sodium (Na+) channels open rapidly, allowing positive ions to flood the cell. At this stage, the channels are already "busy, " so a second stimulus cannot open them further.
  - **The Shutdown (Early Repolarization):** Once the peak of the action potential is reached, the sodium channels enter an **inactivated state** (locked). They cannot be reopened until the membrane potential returns to a near-resting level.
  - **The Biological Limit:** Because the Na+ channels are either already open or physically locked, the neuron is completely incapable of firing another impulse, regardless of how strong the next stimulus is.
  - The period of absolute or "complete" inexcitability occurs precisely between the start of **depolarization** and the transition through **repolarization**. This timing is fundamental for maintaining the rhythmic and directional flow of signals in the nervous system.

46. The pH value of human blood is ***[BPSC DSO, Pre]***

A. 7.35 to 7.45 B. 6.5 to 7.00
C. 7.5 to 7.8 D. 5.9 to 6.4

Ans. A

Exp: Human blood is slightly basic or alkaline. The body uses buffer systems to maintain the pH strictly between 7.35 and 7.45, as any significant change outside this narrow range can lead to severe medical complications.

BCW Bits:

- Karl Landsteiner, an Austrian scientist discovered the ABO blood group system in the year 1900.
- The two surface antigens A and B on the RBCs are chemicals that can induce an immune system during blood transfusion.
- Similarly, the plasma of different individuals contains two natural antibodies or proteins produced in response to antigens.
- Therefore, the blood of a donor has to be carefully matched with the blood of a recipient before any blood transfusion to avoid severe problems of clumping (destruction of RBC).
- The donor's compatibility is also shown in the table given below.

Blood Group	Antigens on rbcs	Antibodies in Plasma	Donor's Group
A	A	anti-B	A, O
B	B	anti-A	B, O
AB	AB	nil	AB, A, B, O
O	NIL	anti-A, B	O

- From the above-mentioned table, it is evident that group 'O' blood can be donated to persons with any other blood group and hence 'O' group individuals are called 'universal donors.
- Persons in the 'AB' group can accept blood from persons with AB as well as the other groups of blood. Therefore, such persons are called 'universal recipients.

47. Which of the following elements is not found in DNA and RNA? ***[BPSC DSO, Pre]***

A. Oxygen B. Nitrogen
C. Sulphur D. Phosphorus

Ans. C

Exp: DNA and RNA are composed of nucleotides, which consist of a pentose sugar (containing Carbon, Hydrogen, and **Oxygen**), a **Nitrogenous** base, and a **Phosphate** group (containing **Phosphorus** and Oxygen). **Sulphur** is absent in the chemical structure of both nucleic acids (though it is a key element found in certain amino acids that form proteins)

48. Colour blind and Haemophilia diseases of human being are due to ***[BPSC DSO, 2025]***

A. Dominant genes present on Y chromosome
B. Dominant genes present on X chromosome
C. Recessive genes present on Y chromosome
D. Recessive genes present on X chromosome

Ans. D

Exp: Both Hemophilia and Color blindness are X-linked recessive disorders. This means the defective gene is located on the X chromosome and the trait is recessive. These conditions are much more common in males because they have only one X chromosome.

49. Pathogenic bacteria secrets *[BPSC DSO, Pre]*

A. Antigen B. Antibiotic
C. Antibodies D. Interferons

Ans. A

Exp: Pathogenic bacteria produce or carry specific molecules called antigens that the host's immune system recognizes as foreign. This recognition triggers the body to produce antibodies to fight the infection. Antibiotics are produced by fungi or other bacteria, and interferons are produced by host cells in response to viruses.

50. Which cells of our body cannot respire anaerobically? *[BPSC DSO, Pre]*

A. Mature RBC B. Brain cells
C. Muscle cells D. Newly formed RBCs

Ans. B

Exp: Brain cells (neurons) have a very high metabolic rate and rely almost exclusively on aerobic respiration to function. They cannot survive or function through anaerobic pathways for more than a few minutes. In contrast, Mature RBCs only respire anaerobically because they lack mitochondria, and muscle cells can switch to anaerobic respiration during heavy exercise.

51. Which part of the plant is never grazed by grazing animals? *[BPSC DSO, Pre]*

A. Root B. Stem C. Bark D. Wood

Ans. D

Exp: Grazing animals typically consume the parts of the plant that are tender, accessible, and high in nutritional value. They often eat leaves, green stems, and even strip bark or pull up shallow roots. Wood, however, refers to the secondary xylem of a tree or shrub, which is highly lignified and extremely hard. Because wood is physically difficult to chew and lacks the digestibility and nutrient profile of other plant tissues, it is the one structural part of the plant that is fundamentally avoided by grazing animals.

52. Which one of the following is a modified stem? *[BPSC DSO, Pre]*

A. Pneumatophore of Mangrove plants
B. Cactus (Opuntia) Phylloclade
C. Cabbage
D. Turnip

Ans. B

Exp: In the Cactus (Opuntia), the leaves are reduced to spines to conserve water, and the stem becomes a flattened, green, fleshy structure called a phylloclade which performs photosynthesis. Pneumatophores and Turnips are modified roots, while Cabbage is a large terminal bud.

53. What is not required for seed germination? *[BPSC DSO, Pre]*

A. Moisture B. Oxygen
C. Temperature D. Minerals

Ans. D

Exp: Seed germination requires moisture to activate enzymes, oxygen for respiration, and an appropriate temperature. Minerals from the soil are not required for the initial germination process because the seed uses nutrients already stored in its cotyledons or endosperm.

54. Erosion of top soil and run off fertilisers and pesticides passing into water bodies are examples of *[BPSC DSO, Pre]*

A. Negative and positive pollution
B. Positive and negative pollution
C. Positive and positive pollution
D. Negative and negative pollution

Ans. A

Exp: In the environment, 'positive' and 'negative' in the context of soil pollution don't mean 'good' or 'bad.' Instead, they refer to whether we are **adding** something harmful or **removing** something vital. Soil erosion is considered 'negative pollution' because it involves the removal of useful components (topsoil) from the land. Run-off of fertilisers and pesticides is the addition of harmful chemicals into the ecosystem (and eventually water bodies). Therefore, it is Positive Pollution.

55. The top predators is most marine food chains of the community ecology are Nectons. Which does not include? *[BPSC DSO, Pre]*

A. Jelly Fish B. Sharks
C. Cuttle Fish D. Lobsters

Ans. A

Exp: Nectons (or Nekton) are aquatic animals that can swim and move independently of water currents. Sharks and Cuttlefish are classic examples. While lobsters are bottom-dwellers (Benthos), Jellyfish are technically classified as Plankton (specifically zooplankton) because they are mostly at the mercy of the water currents.

56. Ishihora test is associated with *[BPSC DSO, Pre]*

A. Kidney function
B. Colour Vision Test
C. Liver Function Test
D. Prostrate Gland Test

Ans. B

Exp: The **Ishihara test** is a diagnostic **Colour Vision Test** used to screen for red-green color blindness utilizing pseudoisochromatic plates filled with randomized colored dots that form a number or path visible only to normal vision.

- Kidney function is evaluated through blood creatinine or clearance tests, and the prostate gland is typically screened using a Prostate-Specific Antigen (PSA) blood test.
- Liver function is checked via enzyme panels measuring **ALT (Alanine Transaminase)** and **AST (Aspartate Transaminase)**; these are metabolic enzymes normally contained inside liver cells that leak into the bloodstream when the liver is damaged or inflamed, serving as key biomarkers for liver injury.

57. 10% law of flow of energy in the ecosystem was proposed by ***[BPSC DSO, Pre]***

A. Lindman B. Carl Mobius
C. Tansley D. Vent Hoff

Ans. A

Exp: Raymond Lindeman (spelled Lindman in the paper) proposed the 10% Law of Energy Transfer. It states that during the transfer of energy from one trophic level to the next, only about 10% of the energy is passed on, while the rest is lost as heat or used for metabolic processes.

BCW Bits:

- German zoologist **Carl Mobius** introduced the concept of **biocoenosis** (an interacting community of living organisms sharing a specific habitat like an oyster bed).
- English botanist **Arthur Tansley** coined the term **"ecosystem"** in 1935 to integrate this living community with its non-living physical environment.
- Dutch chemist **Jacobus Henricus van 't Hoff** formulated the **Van 't Hoff equation**
- (d(ln K)/dT = delta H / R * T^2), which mathematically describes how a chemical reaction's equilibrium constant (K) changes in response to shifts in absolute temperature (T).

58. In plants, water is mainly transported through which tissue? ***[BPSC AEDO]***

A. Phloem B. Xylem
C. Cortex D. Cambium

Ans. B

Exp: Plants need to move things internally mainly: **Water and minerals** (from roots to leaves), **Food (sugars)** (from leaves to other parts). This internal movement is called **transportation**.

- Plants have specialised tissues that do different jobs. For transport, there are two main tissues: **Xylem** - carries water and minerals; **Phloem** - carries food.

BCW Bits:

- Xylem **transports water and minerals** from roots to the rest of the plant; movement is **upward (unidirectional), and water movement occurs through the transpiration pull, Cohesion and adhesion of water molecules. Xylem mainly consists of tracheids and vessels ("Xylem = Water pipeline of plant")**
- Phloem transports **food (glucose, sugars)** prepared in leaves; movement can occur in both upward and downward directions. It is primarily responsible for the transport of food.
- Cortex is a **storage tissue,** found between outer layer and vascular tissues, and does not play a major role in transport.
- Cambium, a growth tissue, helps in **increasing thickness (secondary growth).** produces new xylem and phloem; it makes tissues but doesn't transport itself.

59. A plant cell accumulates ER-stress markers due to misfolded proteins. Which process is triggered? ***[BPSC WMO]***

A. RNA interference
B. Unfolded Protein Response (UPR)
C. GFP-induced autophagy
D. Nonsense-mediated decay

Ans. B

Exp: The endoplasmic reticulum (ER) is the cellular factory responsible for folding and assembling proteins. Under certain conditions, such as environmental stress or metabolic imbalances, proteins may fail to fold correctly, leading to their accumulation within the ER lumen. This state is known as **ER stress**.

- To prevent the damage caused by misfolded protein accumulation, the cell activates the **Unfolded Protein Response (UPR)**. This is a vital, highly conserved quality-control signaling pathway that acts as a damage-limitation system.
- The UPR functions like a crisis-management team for the cell, initiating three main strategies to restore balance:
- **Expansion of Capacity:** It increases the production of chaperone proteins, which assist other proteins in folding correctly.
- **Translation Control:** It temporarily slows down the synthesis of new proteins to prevent the ER from becoming further overwhelmed.
- **Clearing Damage:** If the stress is successfully managed, the cell resumes normal function. If the stress is too severe to fix, the UPR initiates programmed cell death (apoptosis) to prevent the accumulation of toxic proteins from harming the rest of the organism.

BCW Bits:

- **RNA interference:** A gene-silencing process triggered by double-stranded RNA.
- **GFP-induced autophagy:** An artificial process used in research to degrade GFP-tagged proteins; it is not a general stress response mechanism.

- **Nonsense-mediated decay:** A surveillance pathway that eliminates mRNAs containing premature stop codons.

60. Which of the following is the primary pigment involved in photosynthesis? ***[BPSC AEDO]***

A. Chlorophylla B. Chlorophyllb
C. Carotene D. Xanthophyll

Ans. A

Exp: Photosynthesis is the process by which **green plants prepare food** using **sunlight, Carbon dioxide (CO_2), and water (H_2O).** The end products are **glucose (food)** and **oxygen. Pigments** are light-absorbing substances present in plant cells that help capture solar energy for photosynthesis. Different pigments absorb different wavelengths of light.

- **Chlorophyll a is the primary pigment directly involved in the conversion of light energy into chemical energy. It is present in all photosynthetic plants and plays a central role in photosynthesis. Chlorophyll b is an accessory pigment that absorbs additional light energy and transfers it to chlorophyll a. Carotene, an accessory** pigment, gives an **orange colour** (like in carrots), helps in **absorbing light and protects plant cells. Xanthophyll, another** accessory pigment, gives a **yellow colour. It also helps in light absorption and protection.**

BCW Bits:

- Chlorophyll absorbs mainly **blue and red light**, reflects green (that's why plants look green)
- Accessory pigments help plants **utilize sunlight more efficiently.**
- These pigments are located in **chloroplasts**, specifically in structures called **thylakoids.**
- During autumn, breakdown of chlorophyll reveals carotene and xanthophyll pigments (yellow/orange leaves)

61. Given below are two statements, one is labelled as Assertion (A) and the other is labelled as Reason (R). In light of the statements below, choose the most appropriate answer from the options. ***[BPSC AEDO]***

Assertion (A): Transpiration helps in the ascent of sap.

Reason (R): Transpiration increases osmotic pressure in roots.

A. Both (A) and (R) are true and (R) is the correct explanation of (A)
B. Both (A) and (R) are true but (R) is not the correct explanation of (A)
C. (A) is false but (R) is true
D. (A) is true but (R) is false

Ans. D

Exp: Transpiration is the loss of water in the form of vapour from leaves, mainly through tiny pores called stomata present on leaves. Ascent of sap refers to the upward movement of water and minerals from roots to leaves through xylem. During transpiration, evaporation of water from leaves creates a suction force known as transpiration pull. This pull helps water move upward, thereby aiding the ascent of sap.

- However, transpiration does not increase osmotic pressure in roots. Hence, the Reason statement is false.

BCW Bits:

- The main mechanism for upward water movement is explained by the **cohesion-tension theory.**
- Water molecules stick together (**cohesion**) and to xylem walls (**adhesion**)
- This creates a continuous **water column from roots to leaves.**
- Transpiration pull is stronger than root pressure, especially in **tall trees.**
- On humid days, transpiration decreases, so water movement slows down.

62. Which of the following statements about glycolysis is correct? ***[BPSC AEDO]***

A. It occurs in mitochondria
B. It requires oxygen
C. It produces carbon dioxide directly
D. It takes place in cytoplasm

Ans. D

Exp:

- **Glycolysis** is the first step of cellular respiration—the process by which cells break down glucose to release energy. In simple words: **Glucose is broken down and energy is released.** It converts **glucose (6-carbon)** into **pyruvate (3-carbon).** It happens in **all living cells.** Glycolysis occurs in the cytoplasm of the cell. The cytoplasm is the jelly-like fluid inside the cell where many reactions happen. It does not require mitochondria.

BCW Bits:

- Glycolysis produces a small amount of energy: 2 ATP molecules
- It is the only stage of respiration common to both aerobic and anaerobic pathways
- In the absence of oxygen, pyruvate can convert into lactic acid (in muscles) or ethanol (in yeast)
- The word "glycolysis" literally means "breaking of sugar"
- Glycolysis is also known as the EMP pathway (Embden–Meyerhof–Parnas pathway).

63. Which of the following correctly represents the law of segregation? ***[BPSC AEDO]***

A. 9 : 3 : 3 : 1 B. 3 : 1
C. 1 : 1 D. 1 : 2 : 1

Ans. B

Exp: The Law of Segregation (also known as Mendel's First Law of Inheritance) states that during the formation of gametes, the two alleles for a specific trait separate (segregate) from each other, so that each gametic cell carries only one allele for each gene.

- When Mendel conducted a monohybrid cross (crossing two plants that differ in only one trait, such as tall vs. dwarf pea plants), he observed the following patterns across generations:
  1. F1 Generation (First Filial): Crossing a homozygous dominant tall plant (TT) with a homozygous recessive dwarf plant (tt) results in all heterozygous tall plants (Tt).
  2. F2 Generation (Second Filial): When these F_1 plants (Tt) are self-pollinated, the alleles segregate during gamete formation.

  Phenotypic and Genotypic Ratio in F2
  - **Phenotypic Ratio (3 : 1):** This represents the observable physical traits. Out of the 4 offspring combinations, 3 appear tall (TT, Tt, Tt) and 1 appears dwarf (tt). This is whyB. is the correct answer representing the standard Mendelian law of segregation ratio.
  - **Genotypic Ratio (1 : 2 : 1):** This represents the actual genetic makeup of the offspring—1 Homozygous Tall (TT) : 2 Heterozygous Tall (Tt) : 1 Homozygous Dwarf (tt).

BCW Bits:
- Mendel discovered this law using pea plant experiments.
- The law of segregation is also called the law of purity of gametes.
- It applies to monohybrid crosses.
- It explains why traits don't blend but remain distinct.
- Modern genetics explains it through separation of homologous chromosomes during meiosis.
- It is one of the basic principles of heredity.

64. Given below are two statements, one is labelled as Assertion (A) and the other is labelled as Reason (R). In light of the below given statements, choose the most appropriate answer from the options. ***[BPSC AEDO]***

Assertion (A): Corruption is a major threat to climate action.

Reason (R): It acts as a hindrance in reducing emissions and adapting to the unavoidable effects of global heating.

A. Both (A) and (R) are true and (R) is the correct explanation of (A)
B. Both (A) and (R) are true but (R) is not the correct explanation of (A)
C. (A) is false; (R) is true
D. (A) is true; (R) is false

Ans. A

Exp: Corruption means misuse of power or public resources for private gain. Climate action includes two big things: Mitigation → reducing greenhouse gas emissions (clean energy, efficiency, etc.), Adaptation → preparing for impacts (flood control, drought planning, resilient infrastructure). Global heating refers to rising global temperatures due to greenhouse gases.

- Corruption weakens climate mitigation by bypassing environmental regulations and delaying clean-energy projects. It also affects adaptation by diverting funds meant for climate-resilient infrastructure.

BCW Bits:
- Climate funds (like international aid) require transparency and accountability to be effective.
- Corruption can lead to "greenwashing"—projects appear eco-friendly on paper but aren't in reality.
- Strong institutions and e-governance can reduce corruption in climate programs.
- Public participation and community monitoring improve outcomes of climate projects.
- Many global frameworks emphasise good governance as key to climate success.

65. Who is known as the father of genetics? ***[BPSC AEDO]***

A. Watson and Crick B. Charles Darwin
C. Gregor Mendel D. Thomas Hunt Morgan

Ans. C

Exp: Genetics is the study of **Heredity** (how traits pass from parents to children) and **Variation** (why individuals differ). Example: eye color, height, blood group.

- **Gregor Mendel-**Did experiments on **pea plants,** Discovered basic laws of inheritance, Law of Dominance, Law of Segregation and Law of Independent Assortment; **James Watson** and **Francis Crick:** Discovered **double helix structure of DNA (1953); Charles Darwin:** Gave theory of **natural selection,** Explained evolution, not inheritance mechanism; **Thomas Hunt Morgan:** Worked on fruit flies, Proved that **genes are located on chromosomes.**
- Mendel is also called the **"Father of Modern Genetics",** His work was ignored initially and rediscovered around **1900.** Pea plants were chosen because they had **clear traits (tall/short, green/yellow).** Each trait is controlled by **genes (units of heredity).**
- Morgan's work helped physically link **genes to chromosomes.**
- DNA carries genetic information in all living organisms

  Genetics is important in:
  - Medicine
  - Agriculture
  - Biotechnology

66. Which of the following best explains the pressure flow hypothesis in phloem transport? ***[BPSC AEDO]***

A. Active transport of sugars from sink to source
B. Osmotic pressure difference between source and sink
C. Transport depends only on gravity
D. Movement of water only through xylem

Ans. B

Exp: Phloem transport: In plants, **phloem** carries **food (mainly sugars like sucrose)** from **Source** where food is made (leaves), **Sink** where it is used/stored (roots, fruits, growing parts).

- **Pressure Flow Hypothesis:** It is the process of how food moves in phloem, Sugar is loaded into phloem at the source (leaves) which increases osmotic pressure, Water enters from the xylem by osmosis, creating high turgor pressure , this pressure pushes food toward the sink. At the sink sugar is removed, pressure becomes low so movement happens due to a pressure difference.
- Phloem transport can move **both upward and downward**
- Requires energy only during **loading/unloading of sugars**
- Sucrose is the **main sugar transported**
- Xylem and phloem are closely linked in function
- Transport speed in phloem is relatively **slow compared to xylem flow**
- Aphid experiments helped prove phloem transport mechanism
- Pressure flow hypothesis was proposed by **Munch (1930).**

67. Haemoglobin carries oxygen in the blood and is ______ containing compound. ***[BPSC AEDO]***

A. Zinc B. Copper
C. Iron D. Magnesium

Ans. C

Exp: Haemoglobin is a protein present in red blood cells (RBCs). Its main job is very simple: to carry oxygen (O_2) from lungs to different parts of the body. Haemoglobin has a special part called the **heme group.** This heme group contains **iron (Fe).** Oxygen binds directly to this **iron atom.** When oxygen attaches, Oxyhemoglobin **forms.** When oxygen is released, it goes back to normal haemoglobin.

- Iron helps in **binding and releasing oxygen efficiently.** Without iron, oxygen transport will fail, leading to **anaemia (low hemoglobin).**
- Each hemoglobin molecule can carry **4 oxygen molecules**
- Blood appears **red because of hemoglobin**
- Carbon monoxide binds hemoglobin more strongly than oxygen → dangerous
- Iron deficiency is one of the **most common nutritional problems**
- Haemoglobin level is measured in **g/dL**
- Fetal hemoglobin has **higher oxygen affinity** than adult hemoglobin
- High altitude increases hemoglobin production in the body.

68. Fluid Mosaic Model of plasma membrane was proposed by: ***[BPSC AEDO]***

A. Watson and Crick B. Singer and Nicolson
C. Hooke D. Robertson

Ans. B

Exp: This model explains the **structure of the cell (plasma) membrane**. The membrane is **fluid** (flexible, not rigid), It is a **mosaic** (mix) of Lipids, Proteins. Imagine it like a sea **of lipids,** with **proteins floating in it.** The model was given by **S. J. Singer** and **Garth Nicolson** in 1972.
They explained that Proteins are not fixed, They move within the lipid layer.

- Phospholipids have: **Hydrophilic head (water-loving), Hydrophobic tail (water-repelling).**
- Membrane fluidity helps in Cell communication, Transport.
- Cholesterol helps regulate **membrane stability.**
- Proteins act as Channels, Receptors.
- The model replaced older rigid membrane ideas.
- Important for understanding Diffusion, Active transport.

69. Consider the following statements: ***[BPSC AEDO]***

i. Mendel is called the father of genetics.
ii. Law of segregation states that each gamete receives two alleles for a trait.
iii. In a monohybrid cross, F_2 phenotypic ratio is 3 : 1.
iv. The law of independent assortment works only for traits on different chromosomes.

Select the group of correct statements only:

A. i and ii B. i and iii
C. ii, iii and iv D. i, iii and iv

Ans. D

Exp: Segregation - **one allele per gamete,** Monohybrid - **3 : 1,** Independent assortment- **different chromosomes;**

- **Each gamete receives only one allele for a trait, not two.**
- Mendel worked in a monastery and was also a **mathematician**
- Traits are controlled by **genes located on chromosomes**
- Dominant traits mask **recessive traits**
- Test cross is used to determine **genotype**
- Dihybrid cross gives **9:3:3:1 ratio**
- Exceptions occur due to:

1. Linkage
2. Gene interaction

Modern genetics includes DNA and molecular biology.

70. What is the correct order of stages in incomplete metamorphosis in insects? ***[BPSC AEDO]***

A. Egg – Larva – Pupa – Adult
B. Egg – Larva – Adult
C. Egg – Nymph – Pupa – Adult
D. Egg – Nymph – Adult

Ans. D

Exp: Metamorphosis is the process by which an insect changes its body form during development. Insects do not always look the same throughout life. They pass through different stages before becoming adults.

- Types of metamorphosis in insects, mainly two types: Complete metamorphosis- Stages, Egg → Larva → Pupa → Adult Examples Butterfly, Mosquito, Housefly.
- Incomplete metamorphosis: Stages Egg → Nymph → Adult, No larva stage, No pupa stage. A nymph resembles a small adult but lacks fully developed wings and reproductive organs, As it grows and moults several times, it becomes an adult.

BCW Bits:

- Shedding of outer covering in insects is called **moulting (ecdysis)**
- Hormones control insect metamorphosis
- Juvenile hormone helps maintain immature stages
- Complete metamorphosis gives better adaptation because larvae and adults eat different food
- Incomplete metamorphosis is also called **hemimetabolous development**
- Complete metamorphosis is called **holometabolous development**
- Insects belong to phylum **Arthropoda**

71. Which scientist worked independently and reached a similar conclusion to Darwin's? ***[BPSC AEDO]***

A. Jean Lamarck
B. Gregor Mendel
C. Hugo de Vries
D. Alfred Russel Wallace

Ans. D

Exp: Alfred Russel Wallace independently developed the idea of natural selection, very similar to Charles Darwin. Both scientists reached almost the same conclusion: Species evolve over time. Nature selects organisms better adapted to survive.

- Darwin proposed the Theory of Evolution by Natural Selection. Organisms with favorable traits survive and reproduce more, Over long periods, species gradually change.
- In 1858, Wallace sent his ideas on natural selection to Darwin, surprising him because the explanation closely matched Darwin's own theory.
- Darwin later published *On the Origin of Species* in 1859.
- Jean-Baptiste Lamarck, Proposed Theory of inheritance of acquired characters, Example-Giraffe stretched neck - offspring got longer neck.
- Gregor Mendel Worked on heredity and pea plants Founder of genetics, not evolution by natural selection.
- Hugo de Vries Proposed mutation theory.

72. Which of the following statement(s) is/are true about Vitamin B_{12}? ***[BPSC AEDO]***

a. It is also known as cobalamin.
b. It is composed of cobalt atoms.
c. It consists of corrin rings.

Choose the correct option:

A. Only **a** is true statement
B. Only **b** and **c** are true statements
C. a, b and c are true statements
D. Only a and c are true statements

Ans. C

Exp: Vitamin B_{12} is a very important **water-soluble vitamin** needed for: Formation of red blood cells, Nervous system function, DNA synthesis; **Deficiency can cause:** Anaemia Weakness, Nerve problems. A corrin ring is a large organic ring structure. Similar to porphyrin rings found in hemoglobin, but slightly different. It holds the cobalt atom at the center.

- Sources of Vitamin B_{12}:- Mostly found in Milk, Eggs, Fish, Meat.Strict vegetarians may develop deficiency if intake is low. Importance:- It helps in Nerve insulation (myelin sheath), RBC formation, Cell division.
- Vitamin B_{12} deficiency can cause **pernicious anaemia.**
- Absorption of B_{12} requires **intrinsic factor** from stomach.
- Liver stores Vitamin B_{12} for several years.
- B_{12} deficiency may lead to tingling sensation in hands and feet.
- Cyanocobalamin is a commonly used supplement form of B_{12}.
- B_{12} is synthesized naturally by certain bacteria.
- Vegans are often advised to take B_{12} supplements.

73. Place these taxonomic classification systems in chronological order of their development: ***[BPSC AEDO]***

a. Whittaker's five-kingdom system
b. Linnaeus's artificial system
c. Bentham and Hooker's natural system
d. Engler and Prantl's phylogenetic system
e. APG system for flowering plants

A. b – c – d – a – e
B. a – b – c – d – e
C. a – c – b – d – e
D. b – d – c – a – e

Ans. A

Exp: Scientists gradually improved classification methods First based on simple visible characters, Later based on natural relationships, Then evolutionary relationships, Finally DNA/genetic data.

- **Linnaeus's Artificial System:-** Time, 18th century (around 1735), Artificial classification system, Based mainly on Number of stamens and pistils in flowers. It was called artificial Because It used only a few characters, Did not show true evolutionary relationship. But it was revolutionary for its time.
- **George Bentham** and **Joseph Dalton Hooker:-** Time- 19th century (1862–1883) Used many plant characters together, tried to show natural similarities. Very practical system, still important in herbarium classification.
- **Adolf Engler:-** Late 19th century Based on evolutionary relationships, Tried to show plant evolution (phylogeny). Hence called phylogenetic system.
- **Robert H. Whittaker:-** 1969 Divided organisms into:Monera, Protista, Fungi, Plantae, Animalia. Based on Cell structure, Mode of nutrition, Organization level.
- **Angiosperm Phylogeny Group: Modern era (from 1998 onwards) uses DNA sequencing and Molecular** biology. Most modern system for flowering plants.

BCW Bits:

- Linnaeus introduced **binomial nomenclature.**
- Taxonomy helps in scientific naming and identification.
- Herbarium = collection of preserved plant specimens.
- APG classification is continuously updated with genetic research.
- Phylogeny means evolutionary history of organisms.
- Whittaker separated fungi from plants due to different nutrition modes.
- Modern taxonomy combines:Morphology, Anatomy, Genetics, Biochemistry.

74. With reference to glucose metabolism and cellular respiration, consider the following statements: ***[BPSC AEDO]***

- **Statement 1:** The conversion of succinate to fumarate in the TCA cycle is unique because it is the only step catalyzed by an enzyme embedded in the inner mitochondrial membrane and directly links the TCA cycle to the electron transport chain.
- **Statement 2:** Oxidative phosphorylation is the only stage of cellular respiration where oxygen is directly consumed, and all ATP generated in this stage is produced by substrate-level phosphorylation.
- **Statement 3:** In glycolysis, the enzyme phosphofructokinase-1 (PFK-1) is allosterically inhibited by high concentrations of ATP and citrate, making it a key regulatory point for both glycolysis and the TCA cycle.

Which of the statements given above are correct?

A. All statements are correct
B. Statement 1 and Statement 3 only
C. Statement 3 only
D. Statement 1 and Statement 2 only

Ans. B

Exp: Cellular respiration is the process by which cells break down glucose to release energy in the form of ATP. Main stages: Glycolysis, Link reaction, TCA/Krebs cycle, Electron transport chain.

- Glycolysis Occurs in Cytoplasm Produces Pyruvate, ATP, NADH; TCA Cycle Occurs in Mitochondrial matrix, Oxidize acetyl-CoA completely Produces NADH, $FADH_2$, ATP/GTP; Oxidative Phosphorylation Occurs in Inner mitochondrial membrane Most ATP is formed here.
- The conversion of succinate to fumarate in the TCA cycle is catalyzed by succinate dehydrogenase, which is the only TCA cycle enzyme embedded in the inner mitochondrial membrane. This enzyme also functions as Complex II of the electron transport chain, directly linking the TCA cycle with oxidative phosphorylation.
- Oxygen is directly consumed during oxidative phosphorylation as the final electron acceptor in the electron transport chain. However, the ATP formed during this stage is produced mainly by chemiosmosis (oxidative phosphorylation), not by substrate-level phosphorylation. Substrate-level phosphorylation occurs only in glycolysis and certain steps of the TCA cycle.
- Phosphofructokinase-1 (PFK-1) is a major regulatory enzyme of glycolysis. High ATP and citrate levels inhibit PFK-1, signaling that the cell already has sufficient energy. Thus, it acts as an important regulatory link between glycolysis and the TCA cycle.

BCW Bits:

- Glycolysis can occur even without oxygen
- Mitochondria are called the "powerhouse of the cell"
- NADH and $FADH_2$ carry high-energy electrons
- ATP synthase is sometimes called a molecular turbine
- Krebs cycle is named after Hans Krebs
- One glucose molecule yields about 30–32 ATP molecules
- Cyanide poisoning blocks the electron transport chain
- Brown fat cells produce heat by uncoupling oxidative phosphorylation

75. The realized niche of an organism is: ***[BPSC AEDO]***

A. the life pattern that the organism actually assumes
B. the habitat of a species within a community resulting from clumping
C. the habitat that exists in nature as opposed to the ideal
D. the area a species can occupy in the face of exploitative competition

Ans. D

Exp: A niche means the role and position of an organism in an ecosystem. It includes What it eats, Where it lives, How it survives, How it interacts with other organisms.

Fundamental niche is the total possible area and conditions where a species *could* survive if there were No competition, No predators, No restrictions. It is the organism's ideal ecological space. Realized niche is the actual area the species occupies in nature after facing Competition, Predators, Environmental pressure. So the realized niche is usually smaller than the fundamental niche.

- Due to competition and environmental pressures, organisms usually occupy a smaller area than their total potential range. This actual occupied range is called the realized niche.

BCW Bits:

- The term "niche" was developed mainly by ecologist Hutchinson.
- No two species can occupy exactly the same niche for long (competitive exclusion principle).
- Predators also influence the realized niche.
- Invasive species can alter the realized niche of native organisms.
- Ecological niche includes bothBiotic factors, Abiotic factors.
- Resource partitioning helps species coexist.
- Realized niche may change due to climate change.

76. In the Z-scheme of photosynthesis, which of the following electron flow combinations are NOT correct?

[BPSC AEDO]

(i) P680 → PQA → PQB → Cyt b6f → Pheo → PC → P700
(ii) P700 → A0 → A1 → FeSX → FeSA → FeSB → Fd
(iii) P680 → Pheo → PQA → PQB → Cyt b6f → PC → P700
(iv) P700 → A1 → A0 → FeSA → FeSB → FeSX → Fd

A. (iii) and (iv) B. (i) and (iv)
C. (ii) and (iii) D. (i) and (ii)

Ans. B

Exp: It explains how electrons move through **Photosystem II (PSII)** and **Photosystem I (PSI)** during the light reaction of photosynthesis. In the Z-scheme, electrons first move through PSII and then reach PSI through an electron transport chain. The correct electron flow in **Photosystem II** is:

- P680 → Pheo → PQA → PQB → Cyt → b6f → PC → P700
- So statement **(iii)** is actually correct because it follows the proper sequence. statement **(i)**. It places Pheo directly after Cyt b6f, which is incorrect. Pheophytin (Pheo) should come immediately after P680, not later in the chain. Hence statement (i) is incorrect. For **Photosystem I**, the correct sequence is:
- P700 → A0 → A1 → FeSX → FeSA → FeSB → Fs
- Statement **(ii)** follows this order correctly, statement **(iv)** wrongly interchanges A0 and A1 and also disturbs the iron-sulfur center sequence. Therefore statement (iv) is incorrect.
- The **Z-scheme** represents the electron flow during the light reaction in chloroplasts. It is called the Z-scheme because the energy graph of electrons looks like the English letter "Z". **Photosystem II (PSII)-** Reaction center: **P680,** Splits water molecules, Releases Electrons, Protons, Oxygen; **Photosystem I (PSI):-** Reaction center: **P700,** Produces high-energy electrons, Ultimately helps in formation of **NADPH.**
- **Important Electron Carriers:- Pheophytin (Pheo)** → first electron acceptor in PSII; **Plastoquinone (PQ)** → mobile carrier; **Cytochrome b6f** → proton pumping complex; **Plastocyanin (PC)** → transfers electrons to PSI; **Ferredoxin (Fd)** → final electron carrier in PSI.

BCW Bits:

- The Z-scheme was proposed by Hill and Bendall.
- Light reaction occurs in the thylakoid membrane.
- P680 absorbs light at a wavelength of 680 nm.
- P700 absorbs light at a wavelength of 700 nm.
- The oxygen released during photosynthesis comes from water.
- NADPH and ATP formed in light reactions are used in the Calvin cycle.
- Cyclic photophosphorylation involves only PSI.
- Non-cyclic photophosphorylation involves both PSI and PSII.
- Ferredoxin helps in $NADP^+$ reduction.
- Water splitting complex is also called Oxygen Evolving Complex (OEC).

77. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

[BPSC AEDO]

Assertion (A): Sieve tube elements in phloem are living cells but lack nuclei at maturity.

Reason (R): Companion cells control the metabolic activities of sieve tube elements through plasmodesmata, ensuring the continuous pressure gradient necessary for phloem loading and unloading of sugars.

In light of the above statements, choose the most appropriate answer from the options given below:

A. Both (A) and (R) are correct and (R) is the correct explanation of (A)
B. (A) is correct but (R) is not correct
C. Both (A) and (R) are not correct
D. None of the above

Ans. A

- Exp: Assertion (A): Sieve tube elements are the primary conducting cells of the phloem tissue. During their development, they undergo a specialized process where they lose their nucleus, ribosomes, and vacuoles. This loss is a functional adaptation; by eliminating these internal structures, the cell creates a hollow, low-resistance pathway that allows for the efficient bulk flow of sugars and organic nutrients throughout the plant. Despite this lack of a nucleus at maturity, the cell remains physiologically active and is classified as a living cell.

- Reason (R): Because sieve tube elements lack the machinery required to synthesize proteins and maintain metabolic homeostasis, they cannot survive independently. They are physically connected to specialized parenchyma cells known as companion cells via numerous cytoplasmic channels called plasmodesmata. The companion cells contain a dense cytoplasm, a prominent nucleus, and abundant mitochondria. They perform the vital metabolic tasks for the sieve tube elements, including protein synthesis, cellular repair, and the active transport of sugars into and out of the sieve tubes (phloem loading and unloading).
- The structural simplification of the sieve tube element(A) is only viable because of the metabolic support provided by the companion cells (R). Thus, the reason accurately explains the biological necessity for the unique anatomical state of the sieve tube elements.

BCW Bits:

- Phloem is a complex permanent tissue that transports Sugars, Amino acids, Other organic nutrients. Main components are Sieve tube elements, Companion cells, Phloem parenchyma, Phloem fibres.
- **Sieve Tube Elements** has Long tubular cells, Lack nucleus at maturity, Have sieve plates at ends, Conduct food materials.
- **Companion Cells** contain Living nucleated cells, Closely attached to sieve tubes. It helps in ATP supply, Protein synthesis, Phloem loading and unloading.
- **Pressure Flow Hypothesis-** Sugars move in phloem due to Osmotic pressure differences and Source-to-sink transport, This creates a pressure gradient for translocation.
- Phloem transport is bidirectional.
- Xylem mainly transports water and minerals.
- Sieve plates contain pores for movement of sap.
- Companion cells are absent in gymnosperms.
- Gymnosperms have albuminous cells instead.
- Translocation in the phloem requires energy (an active process).
- Plasmodesmata are cytoplasmic connections between plant cells.
- Munch proposed the Pressure Flow Hypothesis.
- Mature sieve tube elements contain very little cytoplasm. Phloem fibres are dead sclerenchymatous cells.

78. Assertion (A): The mtDNA haplogroups M31 and M32 observed in the Onge and Great Andamanese populations represent ancient maternal lineages that have undergone in situ evolution on the Andaman Islands, isolated since the initial Out-of-Africa coastal migration approximately 50–70 thousand years ago.

Reason (R): Complete mtDNA sequencing revealed that the coding region mutations defining M31 and M32 do not overlap with those of known Indian, East Asian, or Oceanian haplogroups M, N, or R, and surveys of over 6, 500 mainland Indian mtDNA sequences confirmed the absence of these specific mutations, indicating phylogenetic independence driven by genetic drift in small, isolated populations. ***[BPSC AEDO]***

Options:

A. (A) is false, but (R) is true

B. Both (A) and (R) are true, but (R) is not the correct explanation of (A)

C. Both (A) and (R) are true, and (R) is the correct explanation of (A)

D. (A) is true, but (R) is false

Ans. C

Exp: Assertion (A): The Onge and Great Andamanese populations are known for their long-term isolation. Genetic studies have identified haplogroups M31 and M32 as unique to these populations. The assertion correctly identifies these as ancient maternal lineages that branched off during the early migration out of Africa. Their distinct genetic profile is the result of thousands of years of in situ (local) evolution, essentially trapping these lineages on the islands for approximately 50, 000 to 70, 000 years.

Reason (R): This statement provides the scientific evidence for the Assertion. By conducting complete mitochondrial DNA (mtDNA) sequencing, researchers compared the genetic markers of Andamanese populations against vast databases (over 6, 500 samples) of mainland Indian, East Asian, and Oceanian populations. The lack of overlap confirms that M31 and M32 are not recent arrivals or branches shared with neighboring mainland groups. Their unique mutations and the statistical absence in mainland databases confirm that these lineages evolved independently, a process significantly accelerated by genetic drift—the change in gene frequency that occurs in small, isolated groups where no new genetic material is introduced from outside.

BCW Bits:

- Onge, Jarwa, Sentinelese and Great Andamanese are important tribes of Andaman Islands.
- Sentinelese remain one of the world's most isolated tribes.
- mtDNA studies are widely used in anthropology and evolutionary biology.
- Y-chromosome analysis traces paternal lineage.

- Andaman tribes are often studied to understand early human migration.
- Haplogroup M is one of the oldest Asian maternal lineages.
- Island populations usually show stronger genetic drift.
- Founder effect is another important concept in population genetics.
- DNA sequencing helps reconstruct evolutionary history.
- Coastal migration theory is also called the "Southern Route Migration" theory.

79. Consider the following statements about Dr. Lalji Singh and his research: ***[BPSC AEDO]***

1. Dr. Lalji Singh is widely regarded as the "Father of DNA Fingerprinting in India".
2. He independently invented the DNA fingerprinting technique in 1984, ahead of global developments.
3. His research extensively applied DNA fingerprinting to wildlife conservation, including ancestry studies on tiger, peacock and elephant populations in India.

How many of the statements given above are correct?

A. Only one B. Only two
C. All three D. None

Ans. B

Exp: **Statement 1 is correct, Lalji Singh** is popularly known as the **"Father of DNA Fingerprinting in India."** He played a major role in developing and applying DNA fingerprinting technology in India, especially in forensic science, criminal investigations, paternity disputes, and biodiversity studies.

- **Statement 2** is incorrect. The original DNA fingerprinting technique was developed by British scientist **Alec Jeffreys** in 1984 in the United Kingdom. Dr. Lalji Singh did not invent the technique before global developments.
- **Statement 3** is correct. Dr. Lalji Singh's research was not limited to forensic science. He also used molecular biology and DNA fingerprinting in Wildlife conservation, Species identification, and Evolutionary and ancestry studies. His work contributed to studies involving Indian wildlife such as Tigers, Peacocks, and elephants. This helped in biodiversity conservation and genetic analysis of endangered species.
- DNA fingerprinting is a technique used to identify individuals based on unique DNA patterns. Since every person's DNA is different (except identical twins), it became extremely important in Criminal investigations, Paternity testing, Identification of dead bodies, Biodiversity research.
- Dr. Lalji Singh was one of India's top molecular biologists. He established advanced DNA research in India and helped popularize forensic DNA analysis in Indian courts and scientific institutions. He also served as Vice Chancellor of BHU, Scientist at CCMB (Centre for Cellular and Molecular Biology).
- **Wildlife Genetics** DNA techniques are now widely used in Tracking poaching, Identifying animal species, Conservation breeding, Population genetics. This is especially useful for endangered animals.

BCW Bits:

- DNA fingerprinting is also called DNA profiling.
- CCMB is located in Hyderabad.
- Lalji Singh was awarded the Padma Shri.
- Mitochondrial DNA is inherited maternally.
- India's first DNA fingerprinting-based criminal cases gained attention in the 1990s.

80. Read the following Assertion (A) and Reason (R) carefully and choose the correct option: ***[BPSC AEDO]***

Assertion (A): In eukaryotic cells, the majority of ATP produced during aerobic respiration is generated through oxidative phosphorylation in the mitochondria.

Reason (R): Oxidative phosphorylation relies on a proton gradient across the inner mitochondrial membrane, established by the electron transport chain, which drives ATP synthesis via chemiosmosis.

Options:

A. (A) is false, but (R) is true
B. Both (A) and (R) are true, but (R) is not the correct explanation of (A)
C. (A) is true, but (R) is false
D. Both (A) and (R) are true, and (R) is the correct explanation of (A)

Ans. D

Exp: **Assertion (A):** In eukaryotic aerobic respiration, the cell breaks down glucose to maximize energy output. While a small amount of ATP is produced during earlier stages like glycolysis and the Krebs cycle (via substrate-level phosphorylation), the vast majority of the total ATP yield—roughly 26 to 28 molecules out of the theoretical total of 30–32—is generated during the final stage: oxidative phosphorylation. This process takes place exclusively within the mitochondria, confirming the assertion is correct.

- **Reason (R):** This statement accurately describes the underlying mechanism of oxidative phosphorylation. The process involves two major components:
  1. **Electron Transport Chain (ETC):** A series of protein complexes in the inner mitochondrial membrane that pump protons (H+ ions) into the intermembrane space, creating a steep electrochemical gradient.The largest amount of ATP is produced by the electron transport chain (ETC)
  2. **Chemiosmosis**: The energy stored in that proton gradient is then used as a "force" to drive the enzyme

ATP synthase, which acts like a molecular turbine to synthesize ATP from ADP and inorganic phosphate.

BCW Bits:

- **Aerobic respiration** is the complete breakdown of glucose in the presence of oxygen to release energy. Main stages are, Glycolysis, Link reaction, TCA/Krebs cycle, Electron Transport Chain (ETC).
- **Oxidative Phosphorylation** is the final stage of respiration where Oxygen acts as the final electron acceptor, Most ATP is generated.
- **Electron Transport Chain** is present on Inner mitochondrial membrane. It transfers electrons, creating proton gradients.
- **Chemiosmosis is ATP production via** proton flow across a membrane. This concept was proposed by **Peter Mitchell.**
- Mitochondria are called the "powerhouse of the cell".
- Glycolysis occurs in cytoplasm.
- TCA cycle occurs in mitochondrial matrix.
- ATP synthase is also called Complex V.
- Oxygen is the final electron acceptor in ETC.
- NADH produces more ATP than $FADH_2$.
- Cyanide poisoning blocks the electron transport chain.
- One glucose molecule produces about 30–32 ATP.
- Anaerobic respiration produces much less ATP than aerobic respiration.

81. A plant cross-section shows continuous pericycle around the stele in a dicot root. What function does this indicate? ***[BPSC WMO]***

A. Continual formation of vascular cambium
B. Bacterial symbiosis
C. Secondary growth via lateral roots
D. Mycorrhizal association

Ans. C

Exp: The pericycle is a layer of meristematic cells situated between the endodermis and the central vascular cylinder (stele) in plant roots. This tissue is crucial for two primary developmental processes:

1. **Lateral Root Initiation:** Being a meristematic tissue, the pericycle is the site where new lateral roots originate. These cells undergo active division to push through the outer root tissues, facilitating the branching of the root system.
2. **Secondary Growth:** In dicot roots, the pericycle plays an essential role in thickening the root. It contributes to the formation of the **vascular cambium** (which produces secondary xylem and phloem) and the **cork cambium** (phellogen), enabling the plant to increase in girth and develop woody tissue.

BCW Bits:

- **Continual formation of vascular cambium:** While the pericycle participates in forming the vascular cambium, this is a component of the broader process of secondary growth. "Continual formation" is not the accurate functional designation for this anatomical feature.
- **Bacterial symbiosis:** This process typically involves the development of specialised root nodules (e.g., in nitrogen-fixing legumes). It is an ecological interaction rather than a primary structural outcome of the pericycle.
- **Mycorrhizal association:** This refers to the mutualistic relationship between roots and fungi, characterised by the formation of mantles or arbuscules. It is an external fungal interaction, not a function of the internal pericycle structure.

82. The forensic analysis of the 282 skeletons unearthed from an abandoned well in Ajnala, Punjab, in 2014, primarily confirmed their identity as Indian soldiers from the 1857 Sepoy Mutiny through which key combined evidence? ***[BPSC AEDO]***

A. Cranial measurements and carbon isotope values suggesting local Punjabi ancestry
B. Mitochondrial DNA haplogroups and strontium isotope ratios indicating Gangetic plain origins
C. Dental caries patterns and skeletal trauma from Partition-era violence
D. Radiocarbon dating and the presence of European-style weapons

Ans. B

Exp: The discovery of 282 skeletons in an abandoned well at Ajnala, Punjab, in 2014, sparked significant investigation. Initially, there was debate about whether these remains belonged to victims of the 1947 Partition or soldiers from the 1857 Indian Rebellion. Forensic scientific methods were required to establish the timeline and the geographic origin of the individuals.

- **Genetic Evidence (mtDNA):** Researchers performed mitochondrial DNA (mtDNA) analysis to identify the maternal lineages of the individuals. The haplogroups identified were consistent with populations from North India, particularly those from the Gangetic plain (the region including present-day Uttar Pradesh and Bihar). This was crucial because the 26th Native Bengal Infantry—the unit associated with the 1857 mutiny—was largely recruited from these areas.
- **Isotopic Evidence (Strontium):** Strontium isotope analysis (87Sr/86 Sr ratios) is used to track the geographic origin of an individual based on the soil and water they consumed during their childhood (as these minerals are stored in tooth enamel). The ratios found in the teeth of the skeletons matched the geological profile of the Gangetic plain, not the local region of Punjab. This confirmed that

the individuals were "migrants" who had moved from the East to the Punjab region shortly before their death.

So from the combination of mtDNA (revealing regional ancestry) and strontium isotope analysis (revealing childhood geographic origin) provided definitive scientific proof that these individuals originated from the Gangetic plain, aligning with the historical records of the 1857 mutinous soldiers, rather than the local population of Punjab during the 1947 Partition.

83. The 'Study of alien life' is known as: ***[BPSC AEDO]***

A. Zoology B. Biochemistry
C. Xenobiology D. None of the above

Ans. C

Exp: **Xenobiology:** This term is derived from the Greek word xenos (meaning "stranger" or "foreigner") and biology. It is a branch of biology that specifically explores the potential existence and characteristics of life outside of Earth. It covers the study of how life might evolve, function, alien, non-terrestrial or be detected on other planets or moons.

- **Zoology:** This is the study of animals, focusing on their behavior, structure, physiology, classification, and distribution on Earth. It does not focus on extraterrestrial life.
- **Biochemistry:** This is the study of chemical processes within and relating to living organisms. It is a fundamental tool for understanding life, but it is not restricted to the study of alien life.

84. CRISPR is a technology of – ***[BPSC AEDO]***

A. Gene editing B. Laser binding
C. Colour mixing D. Road safety

Ans. A

Exp: Gene editing is the correct answer as CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats) is a groundbreaking innovation in biotechnology that is used for editing genes with high precision.

- It allows scientists to cut, remove, replace, or modify specific sections of DNA inside living organisms.
- Due to its accuracy, low cost, and efficiency, CRISPR has transformed modern genetics, biotechnology, medicine, and agricultural research.
- The technology works together with a protein called **Cas9**, which acts like molecular scissors.
- Scientists design a guide RNA that directs Cas9 to a specific DNA sequence. Cas9 then cuts the DNA at that location, allowing researchers to delete faulty genes or insert corrected genetic material. This is why CRISPR is often described as a **"genetic scissors"** technology.
- CRISPR was originally discovered as a natural defense mechanism in bacteria. Bacteria use CRISPR systems to protect themselves against viral infections. When a virus attacks, bacteria store fragments of viral DNA in their genome. If the virus attacks again, the CRISPR-Cas system recognizes and destroys the invading viral DNA. Scientists later adapted this natural mechanism into a powerful gene-editing tool.
- Laser binding: This term is not associated with biological sciences. While lasers are used in medicine (such as in eye surgery or dermatology), "laser binding" does not describe a process of gene manipulation or molecular interaction involved in genetic engineering.
- Colour mixing: This is a concept related to physics, art, or design, dealing with the combination of pigments or light wavelengths to create different hues. It has no relevance to molecular biology or DNA manipulation.
- Road safety: This refers to the methods and measures used to prevent accidents on roads and protect users of the road system. It is a field related to civil engineering and public policy, which is completely unrelated to the cellular and genetic processes where CRISPR is applied.

BCW Bits:

- Cas9 is the most commonly used CRISPR-associated enzyme.
- CRISPR technology can edit genes in plants, animals, and microorganisms.
- Gene therapy aims to treat diseases by correcting defective genes.
- The Human Genome Project mapped the complete human genetic sequence.
- DNA stands for Deoxyribonucleic Acid.
- RNA acts as a messenger in protein synthesis.
- Genetic engineering is widely used in production of insulin and vaccines.
- Bt cotton is an example of a genetically modified crop.
- Bioethics deals with ethical issues arising from biotechnology and medicine.
- "Designer babies" is a controversial term linked with human germline gene editing.

85. Consider the following statements: ***[BPSC AEDO]***

1. The Troposphere contains almost all weather phenomena
2. The stratosphere is mostly free of clouds and has the ozone layer
3. The Mesosphere protects Earth by burning meteoroids

Which statement(s) is/are correct?

A. 1 and 2 only B. 2 only
C. 1 and 3 only D. 1, 2 and 3

Ans. D

Exp: The Earth's atmosphere is divided into different layers on the basis of temperature variation. Starting from the Earth's surface upward, the major layers are:Troposphere, Stratosphere, Mesosphere, Thermosphere, Exosphere. Each layer has distinct characteristics and functions.

- **The troposphere** contains almost all weather phenomena. This is the lowest atmospheric layer and extends roughly up to 8 km at the poles and about 18 km at the equator. Clouds, rainfall, storms, winds, fog, cyclones, and almost all weather-related activities occur in this layer because it contains most of the atmosphere's water vapor and dust particles. Temperature generally decreases with height in the troposphere. The average rate of decrease is called the normal lapse rate: 6.5∘C per 1000 m. This layer is extremely important for life because it contains most atmospheric gases necessary for living organisms.
- The **Stratosphere,** located above the troposphere, is comparatively stable and mostly free from clouds and turbulence. This is why jet aircraft often fly in the lower stratosphere for smoother travel. The stratosphere contains the famous **ozone layer,** which absorbs harmful ultraviolet (UV) radiation coming from the Sun. Ozone molecules absorb UV rays and protect life on Earth from genetic damage, skin cancer, and harmful radiation effects. The ozone formation reaction can be simplified as: O2+UV→O+O followed by: O+O2→O3.
- The **Mesosphere** lies above the stratosphere and is the layer where most meteoroids burn up due to friction with atmospheric particles. Because of this protective role, the mesosphere helps shield Earth from frequent meteor impacts. When meteoroids enter Earth's atmosphere at high speed, intense friction generates enormous heat, causing them to burn and produce streaks of light commonly called "shooting stars." Most meteoroids disintegrate completely before reaching Earth's surface. The mesosphere is also the coldest atmospheric layer, where temperatures can drop below -90 degree celsius.

BCW Bits:

- The boundary between troposphere and stratosphere is called the tropopause.
- Temperature increases with height in the stratosphere due to ozone absorption of UV rays.
- The mesopause separates mesosphere and thermosphere.
- Auroras are mainly visible in the thermosphere.
- The ionosphere overlaps mainly with the thermosphere and helps radio communication.
- The exosphere is the outermost atmospheric layer.
- Commercial airplanes usually fly in the lower stratosphere.
- Ozone depletion is mainly caused by chlorofluorocarbons (CFCs).
- Greenhouse gases include carbon dioxide, methane, and water vapor.
- Atmospheric pressure decreases with altitude.

86. The 'Suicidal Bags' of the cell are called–

[BPSC AEDO]

A. Ribosomes B. Lysosomes
C. Golgi Bodies D. Mitochondria

Ans. B

Exp: Lysosomes are often referred to as the "suicidal bags" or "digestive bags" of the cell. They contain powerful digestive enzymes capable of breaking down almost any biological material. If a cell becomes damaged, diseased, or reaches the end of its life, the membrane of the lysosome may rupture, releasing these enzymes into the cell. This causes the cell to digest its own components, effectively leading to the cell's destruction. This self-destruction process is a vital part of cell recycling and removing unwanted waste.This self-destruction process is called **autolysis**.

- These hydrolytic enzymes function best in acidic conditions and digest proteins, fats, carbohydrates, nucleic acids, and worn-out organelles. Therefore, lysosomes act as the waste disposal and recycling system of the cell. The enzymes inside lysosomes are collectively known as acid hydrolases.
- This digestion process helps maintain cellular cleanliness and proper functioning. Lysosomes perform several important functions:
  - Digestion of foreign materials like bacteria
  - Destruction of worn-out cell organelles,
  - Recycling of cellular materials, Participation in programmed cell death.
- **Ribosomes:** These are the "protein factories" of the cell. Their primary function is to read genetic instructions and assemble amino acids into proteins. They are not involved in cell digestion or self-destruction.
- **Golgi Bodies:** These act as the "post office" or shipping department of the cell. They are responsible for packaging, modifying, and sorting proteins and lipids into vesicles to be sent to their correct destinations inside or outside the cell.
- **Mitochondria:** Often called the "powerhouse of the cell, " mitochondria are responsible for generating most of the cell's supply of adenosine triphosphate (ATP), which is used as a source of chemical energy.

BCW Bits:

- Lysosomal enzymes work best in acidic pH.
- Mitochondria contain their own DNA and ribosomes.
- Ribosomes are found in both prokaryotic and eukaryotic cells.
- Golgi apparatus is abundant in secretory cells.
- Vacuoles are generally larger in plant cells than in animal cells.
- Endoplasmic reticulum helps in transport within the cell.
- Smooth ER synthesizes lipids; rough ER synthesizes proteins.

- Cell theory was proposed by Matthias Schleiden and Theodor Schwann.
- The nucleus was discovered by Robert Brown.

87. Which of the following enzymes is primarily responsible for the removal of RNA primers during DNA replication in prokaryotes? ***[BPSC AEDO]***

A. DNA polymerase I B. DNA ligase
C. Helicase D. Primase

Ans. A

Exp: DNA replication is the process by which a cell copies its genetic material before cell division. According to the semi-conservative model proposed by James Watson and Francis Crick, each new DNA molecule contains one old strand and one newly synthesized strand.

- DNA polymerase I is the correct answer as during DNA replication, RNA primers are placed on the DNA strand to initiate the synthesis process. Once the main replication machinery has done its job, these RNA segments must be removed and replaced with DNA. DNA polymerase I is unique because it possesses a specialized "exonuclease" activity that allows it to identify and remove RNA nucleotides, replacing them with the correct DNA nucleotides in one continuous process.
- **DNA ligase:** This enzyme acts as the "glue." Once DNA polymerase I has replaced the RNA primer with DNA, there is still a tiny break in the sugar-phosphate backbone of the DNA strand. DNA ligase seals this "nick" to ensure the DNA strand is continuous and unbroken.
- **Helicase:** This is the "unzipping" enzyme. It travels ahead of the replication machinery, breaking the hydrogen bonds between the two strands of the DNA double helix to create a "replication fork" so that the template strands are accessible.
- **Primase:** This enzyme is the "starter." Before DNA synthesis can begin, Primase builds a short stretch of RNA (the RNA primer) on the template strand. This primer provides the necessary starting point for DNA polymerase to begin adding new DNA nucleotides.

BCW Bits:

- DNA polymerase can add nucleotides only in the: 5'-->3' direction.
- Arthur Kornberg discovered DNA polymerase I.
- DNA polymerase III is the main replication enzyme in prokaryotes.
- DNA polymerase I has both polymerase and exonuclease activities.
- Okazaki fragments were discovered by Reiji Okazaki.
- Helicase breaks hydrogen bonds between DNA strands.
- Single-strand binding proteins stabilize unwound DNA strands.
- Topoisomerase prevents supercoiling during replication.
- RNA primers are usually 5–10 nucleotides long in prokaryotes.
- Replication begins at a specific site called the origin of replication.
- Telomerase is important in eukaryotic chromosome replication.
- DNA replication occurs during the S-phase of the cell cycle.

88. Which of the following is considered to be the tallest known living organism on the earth? ***[BPSC AEDO]***

A. Sausage tree B. Sequoia tree
C. Banyan tree D. Jojoba tree

Ans. B

Exp: The tallest known living organisms on Earth are the giant **Sequoia** and **Coast Redwood** trees found mainly in California of the United States. These trees belong to the redwood group and are famous for their extraordinary height, longevity, and massive trunk size. here are mainly three famous redwood relatives: Coast Redwood-Tallest trees; Giant sequoia-largest by volume;Dawn Redwood-Deciduous Redwood. The tallest known living tree on Earth is a coast redwood named:**Hyperion** It is more than 115 meters tall. Because of such extraordinary height, sequoias/redwoods are considered the tallest living organisms.

- These trees grow in regions having cool climate, abundant moisture, coastal fog, fertile soil. Their height is supported by thick bark, strong wood, efficient water transport system. Some sequoias can live for 2000–3000 years making them among the oldest living organisms as well.
- Tallest Organism- Coast Redwood / Sequoia; Largest by Volume Giant Sequoia ("General Sherman" tree); Largest Canopy Spread Banyan Tree.
- Sequoias are ecologically important because they:store massive amounts of carbon, support biodiversity, resist insects and fire due to thick bark, help regulate climate. Due to climate change and forest fires, conservation of redwood forests has become an important environmental issue.

BCW Bits:

- The tallest known tree "Hyperion" is a coast redwood.
- Sequoias are found mainly in California.
- Giant Sequoia scientific name: Sequoiadendrongiganteum
- Coast Redwood scientific name: Sequoiasempervirens
- Banyan tree is India's national tree.
- Great Banyan Tree is famous for massive canopy spread.
- Redwood forests are important carbon sinks.
- Some sequoias survive forest fires because of thick bark.
- Jojoba oil is used in cosmetics and industry.
- Prelims shortcut:

- Tallest → Sequoia
- Widest spread → Banyan

89. Explosive seed pods that release seeds with very high velocity and send them off to large distances are observed in which of the following? ***[BPSC AEDO]***

A. Salvia B. Coconut
C. Impatiens D. Dandelions

Ans. C

Exp: Impatiens is one of the best-known examples of plants that disperse seeds through an **explosive mechanism**, also called **ballistic dispersal** or **explosive dehiscence**. When the seed pod matures, tension builds up in the fruit wall.

- Even a slight touch, vibration, or natural drying causes the pod to burst suddenly, throwing seeds away at considerable speed and distance.
- The fruit wall stores elastic energy as it matures. Once triggered, this stored energy is released rapidly, causing the pod to split and curl backward.
- The seeds are ejected outward in different directions. Because the plant appears to "explode" when touched, some species of Impatiens are commonly known as **touch-me-not** or **balsam** plants.
- This mechanism ensures that seeds land away from the parent plant, increasing their chances of successful germination.
- **Salvia:** While Salvia plants produce seeds, their dispersal mechanism is not primarily explosive. They often rely on other methods, such as falling to the ground near the parent plant or being dispersed by wind or insects.
- **Coconut:** Coconuts are famous for hydrochory, which is dispersal by water. Their fibrous outer husk allows them to float across oceans to reach new, distant shores. They are not designed for explosive propulsion.
- **Dandelions:** Dandelions utilize anemochory, or wind dispersal. Their seeds are attached to feathery structures called pappus, which act like tiny parachutes. This allows them to be carried long distances by the wind rather than being shot out by an explosive mechanism.

BCW Bits:

- Seed dispersal by wind is called **anemochory**.
- Seed dispersal by water is called **hydrochory**.
- Seed dispersal by animals is called **zoochory**.
- Cotton and maple are classic examples of wind-dispersed seeds.
- Lotus and coconut are examples of water-dispersed seeds.
- Xanthium and Urena possess hooked fruits that cling to animal fur.
- Fruits develop from the ovary after fertilization.
- Seeds develop from ovules after fertilization.
- Germination requires suitable moisture, oxygen, and temperature.
- The touch-me-not plant (*Impatiens*) should not be confused with *Mimosa pudica*, which is famous for folding leaves when touched.

90. Plants are also associated with the colonial British rule in India and have been linked to some revolts on Indian soil. Choose the one that is also associated with one of the great freedom fighters ***[BPSC AEDO]***

A. *Pyrus malus* B. *Indigofera tinctoria*
C. *Azadirachta indica* D. *Ficus indica*

Ans. B

Exp: Indigofera tinctoria is the Correct answer : This plant is the source of natural Indigo dye, which was at the center of the famous Indigo Revolt (Neel Vidroha) of 1859–1860 in Bengal. The British planters forced Indian farmers to grow indigo instead of food crops, often through brutal oppression and exploitative contracts. The struggle was highlighted in the play Neel Darpan by Dinabandhu Mitra. This movement is deeply linked to the freedom struggle and historical figures like Mahatma Gandhi, who later launched his first major Satyagraha in India at Champaran (1917) specifically to support the indigo farmers against the oppressive "Tinkathia" system.

- **Pyrus malus:** This is the scientific name for the common apple. While apple cultivation in India was encouraged in some hilly regions during British rule (like in Himachal Pradesh and Kashmir), it is not associated with any specific freedom struggle or revolt.
- **Azadirachta indica:** Commonly known as the Neem tree. While it is a sacred and highly respected tree in Indian culture for its medicinal properties, it is not historically linked to any specific revolt against British colonial rule.
- **Ficus indica:** The Banyan tree. While the Banyan is iconic to the Indian landscape and has been a site for community gatherings and historic protests, it is not specifically defined as a crop or plant linked to a singular "revolt" in the way Indigo was.

91. This person is known as the Father of Indian Botany for his enormous contributions to the field. Choose the correct option: ***[BPSC AEDO]***

A. Panchanan Maheshwari B. Salim Ali
C. William Roxburgh D. Robert Hooker

Ans. C

Exp: William Roxburgh is widely known as the **Father of Indian Botany** due to his pioneering work in studying, classifying, documenting, and describing the flora of India. His contributions laid the foundation of systematic botanical research in the Indian subcontinent and greatly advanced the scientific understanding of Indian plant diversity.

- He was a Scottish surgeon and botanist who worked for the British East India Company during the late 18th and early 19th centuries. While serving in India, he conducted extensive surveys of Indian vegetation and collected thousands of plant specimens from different regions.

- One of his most important contributions was the monumental work **Flora Indica**, which described a large number of Indian plant species. His detailed observations, illustrations, and taxonomic descriptions became valuable references for later botanists. Because of this extensive documentation, he is regarded as the founder of modern botanical studies in India.
- Panchanan Maheshwari is known as the **Father of Modern Plant Embryology in India**. His research on plant embryology, fertilization, and tissue culture made him famous.
- Salim Ali was an eminent ornithologist, not a botanist. He is popularly known as the **Birdman of India** for his extensive studies on Indian birds and avian ecology.
- Robert Hooke was a pioneering English scientist best known for discovering and naming the cell while examining cork under a microscope in 1665. He made important contributions to biology and microscopy, but he is not associated with the title Father of Indian Botany.

BCW Bits:

- Carolus Linnaeus is known as the Father of Taxonomy.
- Binomial nomenclature follows the format: Genus+SpeciesGenus + SpeciesGenus+Species
- The first word in a scientific name denotes the genus; the second denotes the species.
- Robert Hooke coined the term "cell" in 1665.
- Jagadish Chandra Bose demonstrated sensitivity and response in plants.
- Botanical Survey of India was established in 1890.
- Herbarium is a collection of preserved plant specimens used for scientific study.
- India is one of the 17 mega-diverse countries of the world.
- The Indian Botanic Garden at Howrah is famous for its historic plant collections and the Great Banyan Tree.

92. The phenomenon of production of whole organism from a cell by providing a suitable nutrient medium is called as: ***[BPSC AEDO]***

A. Apospory B. Totipotency
C. Vegetative propagation D. Pluripotency

Ans. B

Exp: Totipotency is the unique biological ability of a single plant cell to divide, differentiate, and regenerate into an entire, fully functional organism. When a scientist places a small piece of plant tissue (an "explant") into a laboratory dish containing a sterile, hormone-rich nutrient medium, the cells are "reprogrammed" to grow into a complete new plant. This principle is the foundation of plant tissue culture and biotechnology.

- This principle has enormous practical applications: Micropropagation of disease-free plants, Production of genetically identical clones, Conservation of rare and endangered species, Germplasm preservation, Genetic engineering and transgenic plant development.
- **Apospory:** This refers to the development of a gametophyte (the sexual phase of a plant's life cycle) directly from the vegetative cells of a sporophyte, without the formation of spores. It is a form of asexual reproduction, but it does not involve the regeneration of a whole organism from a single cell via nutrient media.
- **Vegetative propagation:** This is a natural (or artificial) form of asexual reproduction where new plants grow from parts of the parent plant, such as stems, roots, or leaves (e.g., growing a potato from an "eye"). While it results in a whole organism, it is not the process of growing an organism from a single isolated cell in a nutrient medium.
- **Pluripotency:** This term is typically used in animal biology (specifically with stem cells). A pluripotent cell can give rise to many different types of cells in the body (like skin, muscle, or nerve cells), but it usually cannot form a complete, independent organism on its own.

BCW Bits:

- Gottlieb Haberlandt proposed the concept of cellular totipotency in 1902.
- An explant is the plant tissue used to initiate tissue culture.
- Callus is an unorganized mass of dividing plant cells.
- Auxins generally promote root formation, while cytokinins promote shoot formation.
- Micropropagation enables rapid multiplication of elite plant varieties.
- Tissue culture requires sterile (aseptic) conditions to prevent contamination.
- Somatic embryogenesis is the formation of embryos from somatic cells.
- Banana, orchid, sugarcane, and potato are commonly propagated through tissue culture.
- Pluripotent stem cells can generate many tissue types but not an entire organism.
- Totipotency is most prominently expressed in plant cells compared with animal cells.

93. Scientists have termed one of the following as Green Gold based on its economic and ecological importance: ***[BPSC AEDO]***

A. Sal B. Coffee C. Bamboo D. Rice

Ans. C

Exp: Bamboo is popularly known as **"Green Gold"** because of its enormous economic value, rapid growth rate, wide industrial applications, and significant ecological benefits. It is one of the most versatile plant resources available to humanity and plays an important role in rural livelihoods, environmental conservation, and sustainable development.

- Many bamboo species attain heights comparable to trees, botanically they are giant grasses. Its extensive root system binds soil particles together and reduces erosion, especially in hilly regions.
- The ecological importance can be summarized as: Bamboo Growth → Carbon Absorption + Soil Conservation + Habitat Support → Environmental Sustainability. Because it combines economic productivity with environmental conservation, scientists and policymakers often describe it as Green Gold.

BCW Bits:

- Bamboo belongs to the **Poaceae (Grass) family**, not the tree category.
- India possesses one of the world's richest bamboo resources.
- Major bamboo-producing states include Assam, Mizoram, Tripura, and Arunachal Pradesh.
- Bamboo flowering is often rare and may occur after several decades in many species.
- Bamboo shoots are consumed as food in several northeastern states of India.
- Bamboo can be used to manufacture biodegradable products as alternatives to plastic.
- It is considered an excellent carbon sink because of its rapid biomass accumulation.
- The National Bamboo Mission operates under the Ministry of Agriculture and Farmers Welfare.
- Bamboo-based agroforestry is promoted for sustainable land management.
- Bamboo is sometimes called the "Poor Man's Timber" because of its versatility and affordability.

94. Which statement correctly describes a zoonotic disease? ***[BPSC AEDO]***

A. It spreads only through contaminated food
B. It is caused by plant pathogens
C. It is a disease shared between vertebrate animals and humans
D. It is a disease confined to wildlife species only

Ans. **C**

Exp: **It is a disease shared between vertebrate animals and humans**. A **zoonotic disease (zoonosis)** is an infectious disease that can naturally be transmitted between vertebrate animals and humans. The causative agent may be a virus, bacterium, parasite, fungus, or other pathogen. These diseases are of enormous public health importance because a large proportion of emerging infectious diseases in humans originate from animals.

- The word **"zoonosis"** comes from the Greek words *zoon* (animal) and *nosos* (disease). The concept does not mean that every animal disease is zoonotic. A natural transmission may occur through direct contact, bites, contaminated food, body fluids, vectors such as mosquitoes and ticks, or environmental contamination.
- Many important human diseases are zoonotic. Examples include:
  - Rabies → Dogs, bats and other mammals
  - Nipah → Fruit bats
  - Bird Flu (H5N1) → Birds
  - Swine Flu (H1N1) → Pigs (historically associated)
  - Plague → Rodents
  - Kyasanur Forest Disease → Small mammals and monkeys
  - Japanese Encephalitis → Pigs and water birds

BCW Bits:

- About **60% of known human infectious diseases** are estimated to be zoonotic.
- Approximately **75% of emerging infectious diseases** in humans originate from animals.
- A **reservoir host** harbors a pathogen and serves as its long-term source in nature.
- A **vector** transmits a pathogen between hosts (e.g., mosquito, tick, sandfly).
- Rabies is one of the oldest known zoonotic diseases.
- Fruit bats are important reservoirs for several viral diseases, including Nipah.
- The **One Health** concept integrates human, animal, and environmental health.
- Not all vector-borne diseases are zoonotic, and not all zoonotic diseases require vectors.
- Livestock, pets, wildlife, and birds can all participate in zoonotic disease transmission cycles.

95. Type I (Immediate) hypersensitivity is mediated primarily by: ***[BPSC AEDO]***

A. IgA B. IgG C. IgM D. IgE

Ans. **D**

Exp: Type I hypersensitivity, also known as **immediate hypersensitivity** or **allergic hypersensitivity**, is primarily mediated by IgE antibodies. This type of immune response occurs very rapidly—often within seconds to minutes after exposure to an allergen—and is responsible for common allergic conditions such as hay fever, allergic rhinitis, asthma, urticaria (hives), food allergies, and even life-threatening anaphylactic shock.

- **Antibodies**, also called **immunoglobulins**, are proteins produced by B-lymphocytes and plasma cells in response to foreign substances known as antigens. The major classes of antibodies in humans are:
  - IgG
  - IgA

- IgM
- IgE
- IgD

- Each class performs a specialized function in immunity:
  - **IgE is specifically associated with allergic reactions and defense against parasitic worms**. The mechanism of Type I hypersensitivity begins when a person is exposed to an allergen such as pollen, dust mites, animal dander, certain foods, insect venom, or drugs. During the first exposure, the immune system produces IgE antibodies against that allergen. These IgE molecules attach to mast cells and basophils.
  - During subsequent exposure, the allergen binds to IgE molecules already attached to mast cells, triggering the release of inflammatory chemicals such as histamine, leukotrienes, and prostaglandins.
  - Histamine is largely responsible for symptoms such as sneezing, itching, swelling, redness, watery eyes, bronchoconstriction, and increased mucus secretion.
  - IgA is mainly associated with **mucosal immunity**. It is found in saliva, tears, breast milk, respiratory secretions, and intestinal secretions. Its primary function is to prevent pathogens from entering body tissues through mucosal surfaces.
  - IgG is the most abundant antibody in blood and tissues. It provides long-term immunity, neutralizes toxins and pathogens, and can cross the placenta to provide passive immunity to the fetus.
  - IgM is the first antibody produced during a primary immune response. It is highly effective in activating the complement system.

BCW Bits:

- IgG is the most abundant immunoglobulin in human serum.
- IgM is the first antibody produced during primary infection.
- IgA is the principal antibody in secretions such as saliva and tears.
- IgD mainly functions as a receptor on B-lymphocytes.
- Mast cells contain granules rich in histamine and heparin.
- Basophils are the least abundant white blood cells.
- Anaphylaxis may be triggered by insect stings, drugs, foods, or latex.
- Complement activation is strongest with IgM.
- Maternal IgG crosses the placenta and provides passive immunity to the fetus.
- Type IV hypersensitivity (e.g., tuberculin test, contact dermatitis) is mediated by T-lymphocytes rather than antibodies.

96. Which of the following is not a Mendelian law of inheritance? ***[BPSC AEDO]***

A. Law of Dominance
B. Law of Segregation
C. Law of Independent Assortment
D. Law of Purity of Gametes

Ans. D

Exp:

The three classical Mendelian laws are:

1. **Law of Dominance**
2. **Law of Segregation**
3. **Law of Independent Assortment**

- These three laws were proposed by Gregor Mendel based on his experiments on pea plants (*Pisum sativum*).
- The **Law of Dominance** states that when two contrasting alleles are present together, only one expresses itself in the F_1 generation while the other remains masked. For example, in pea plants, tallness (T) is dominant over dwarfness (t). Thus: $TT \times tt \rightarrow Tt$, All F_1 offspring become tall because the dominant allele masks the recessive allele. The **Law of Segregation** states that the two alleles of a gene separate during gamete formation, so each gamete receives only one allele. This is Mendel's most fundamental principle and is universally valid. The segregation process can be represented as: $Tt \rightarrow T+t$
- This explains why parental traits reappear in later generations. The **Law of Independent Assortment** states that alleles of different genes assort independently during gamete formation, provided the genes are not linked. Mendel discovered this through dihybrid crosses. For example: $RrYy \rightarrow RY, Ry, rY, ry$; This law explains the generation of new combinations of traits in offspring.
- The term **"Law of Purity of Gametes"** is actually another name commonly used for the **Law of Segregation**, especially in older Indian school textbooks and many competitive-exam guidebooks. According to this interpretation, gametes are "pure" because they contain only one allele of a gene and not a mixture of alleles. This creates a conceptual complication. Scientifically: Law of Purity of Gametes = Law of Segregation

BCW Bits:

- Mendel is known as the "Father of Genetics.
- Mendel conducted experiments on approximately 29, 000 pea plants.
- Pea plants were chosen because they show clear contrasting traits and self-pollination.
- The F_1 generation is usually uniform due to dominance.
- Monohybrid crosses typically produce a 3:1 phenotypic ratio in F_2.
- Dihybrid crosses produce the famous 9:3:3:1 phenotypic ratio.
- Linked genes do not follow independent assortment completely.

- Mendel's work was rediscovered independently by Hugo de Vries, Carl Correns, and Erich von Tschermak in 1900.
- The term "gene" was introduced by Wilhelm Johannsen.
- Incomplete dominance and codominance are exceptions to the simple Law of Dominance.
- ABO blood group inheritance is a classic example of multiple alleles and codominance.

97. In the DNA double helix model proposed by Watson and Crick, the two strands of DNA run: ***[BPSC AEDO]***

A. Parallel B. Anti-parallel
C. Circular D. Perpendicular

Ans. B

Exp: Anti-parallel is the correct answer as the DNA double helix consists of two complementary strands held together by hydrogen bonds between nitrogenous bases.

- Watson and Crick identified that these strands run in opposite directions, a structure known as anti-parallel. Specifically, one strand runs in the 5' to 3' direction, while the opposite strand runs in the 3' to 5' direction.
- This orientation is chemically and structurally vital. It allows the molecule to maintain a consistent shape and width, ensuring that genetic information is stored securely and can be accurately copied during cell division.
- The nitrogenous bases (Adenine, Thymine, Cytosine, and Guanine) must pair specifically with their partners, the anti-parallel arrangement provides the perfect geometry to allow these bonds to form effectively while keeping the sugar-phosphate backbone on the outside.
- Parallel: If the strands were parallel (both running 5' to 3'), the chemical bases would not be able to align correctly to form the stable hydrogen bonds that hold the "ladder" together.
- Circular: While some specialized DNA (like that in bacteria or mitochondria) is circular, the classic Watson-Crick model describes the structure of long, linear DNA molecules found in chromosomes.
- Perpendicular: This would imply the strands cross at 90-degree angles, which does not reflect the helical (spiral) structure of the DNA molecule.

BCW Bits:

- DNA was first isolated by Friedrich Miescher in 1869; he called it "nuclein."
- RNA usually exists as a single-stranded molecule.
- In RNA, Uracil replaces Thymine.
- DNA replication is called semi-conservative because each new DNA molecule contains one old and one new strand.
- Human DNA contains about 3 billion base pairs.
- Mitochondria possess their own circular DNA.
- Histone proteins help package DNA into chromosomes in eukaryotes.
- The Human Genome Project officially ended in 2003.
- A-T base pairs have 2 hydrogen bonds; G-C pairs have 3 hydrogen bonds, making G-C-rich DNA more stable.
- B-DNA is the most common form of DNA found under normal cellular conditions.

98. Who popularized *Drosophila* as a genetic model organism? ***[BPSC AEDO]***

A. Gregor Mendel B. Rosalind Franklin
C. Thomas Hunt Morgan D. James Watson

Ans. C

Exp: Thomas Hunt Morgan is the scientist who famously utilized Drosophila melanogaster (the common fruit fly) in his laboratory at Columbia University in the early 20th century. By studying thousands of flies, he provided experimental evidence that genes are located on chromosomes and identified the phenomenon of sex-linked inheritance. His work turned the fruit fly into the gold standard for genetic research, and he was awarded the Nobel Prize in Physiology or Medicine in 1933 for his contributions.

- Morgan's work strongly supported the **Chromosomal Theory of Inheritance**, proposed earlier by Walter Sutton and Theodor Boveri. According to this theory, genes are carried on chromosomes, and chromosomes behave during meiosis in a way that explains Mendel's laws. Morgan's experimental evidence transformed genetics from a theoretical subject into an experimental science.
- **Gregor Mendel:** Often called the "father of genetics, " Mendel worked with pea plants (Pisum sativum) to discover the fundamental laws of inheritance. He did not use fruit flies.
- **Rosalind Franklin:** She was a brilliant chemist and X-ray crystallographer whose work was instrumental in capturing the DNA X-ray diffraction images (specifically "Photo 51"). This image was critical for understanding the helical structure of DNA, but she is not associated with Drosophila research.
- **James Watson:** He is famous for being one of the co-discoverers of the DNA double helix structure, working alongside Francis Crick. While he used molecular biology techniques, he is not credited with popularizing the fruit fly as a model organism.

BCW Bits:

- *Drosophila melanogaster* has only 4 pairs of chromosomes, making genetic study easier.
- The white-eyed mutation in fruit flies helped prove sex-linked inheritance.
- Mendel's laws were rediscovered independently by de Vries, Correns, and von Tschermak around 1900.
- The unit of heredity is called a gene.
- Genes are located on chromosomes at specific positions called loci.

- Crossing over occurs during meiosis and increases genetic variation.
- The X chromosome carries many more genes than the Y chromosome.
- Morgan worked in the famous "Fly Room" at Columbia University.
- Fruit flies share a significant percentage of disease-related genes with humans.
- Barbara McClintock later discovered "jumping genes" or transposons in maize genetics.

99. A trait that increases an organism's chance of survival is called: ***[BPSC AEDO]***

A. Mutation B. Adaptation
C. Variation D. Hybrid

Ans. B

Exp: An adaptation is a specific physical trait, behavior, or internal process that has evolved over time to help an organism survive and reproduce in its particular environment. These traits are "selected for" because they provide a clear advantage—such as better camouflage, more efficient hunting, or tolerance to extreme temperatures—allowing individuals with these traits to survive and pass them on to the next generation.

- The concept of adaptation is closely linked with the theory of evolution given by Charles Darwin. Darwin explained that organisms show variations, and nature selects those variations that are useful for survival. This idea is popularly known as "survival of the fittest". Here, "fittest" does not mean strongest; it means best suited to the environment. For example, the thick fur of polar bears helps them survive in extremely cold regions, while the long neck of giraffes helps them reach leaves on tall trees.
- Mutation: This is a permanent change or "error" in the DNA sequence of an organism. Mutations may be beneficial, harmful, or neutral. While mutations can lead to adaptations, a mutation itself is just a change in the genetic code.
- Variation: This refers to the natural differences that exist between individuals of the same species (for example, slight differences in height or coat color). Variation is the "raw material" upon which natural selection acts, but the variation itself is just the state of being different, not necessarily an improvement for survival.
- Hybrid: This is the offspring resulting from combining the qualities of two organisms of different breeds, varieties, or species. While hybrids can sometimes exhibit "hybrid vigor" (increased strength or resilience), being a hybrid is not a general definition for a survival trait.

BCW Bits:

- The term "survival of the fittest" was coined by Herbert Spencer, not Darwin.
- Mimicry is a type of adaptation where one organism resembles another for protection.
- Camouflage helps organisms blend with surroundings to avoid predators.
- Vestigial organs are reduced structures inherited from ancestors, like the human appendix.
- Adaptive radiation means evolution of different species from a common ancestor to occupy different habitats.
- Industrial melanism in peppered moths is a famous example of natural selection.
- Antibiotic resistance in bacteria is considered an example of evolutionary adaptation.
- Homologous organs indicate common ancestry, while analogous organs indicate similar adaptation.
- Xerophytes are plants adapted to dry conditions; hydrophytes are adapted to aquatic habitats.
- Genetic variation is essential for evolution; without variation, natural selection cannot occur.

100. Temporary parasites are: ***[BPSC AEDO]***

A. Visit host only for feeding
B. Live on the surface of host's body
C. Completes its life cycle inside the host
D. Free living but become parasitic if opportunity arises

Ans. A

Exp: Parasites are organisms that Live **on or inside a host,** Take **nutrition from the host.**

- Temporary parasites visit the host only for feeding and do not remain permanently attached. Example Mosquito, Bed bug.
- **Types of parasites: Temporary parasite - only visits the host; Permanent parasite - lives on the host; Ectoparasite - lives outside the body; Endoparasite - lives inside the body; Facultative parasite** - can live both ways.
- Mosquito is a temporary parasite but also a **disease vector that spreads illnesses such as malaria and dengue.**
- Parasites can affect Humans, Animals, Plants.
- Some parasites show **host specificity.**
- Parasitism is a type of **symbiotic relationship.**
- Adaptations include piercing mouthparts.
- Study of parasites is called **Parasitology.**

101. Which of the following animals shows metamerism? ***[BPSC WMO]***

A. Amoeba B. Earthworm
C. Hydra D. Planaria

Ans. B

Exp: Metamerism is a biological phenomenon characterised by the repetition of homologous body segments, known as metameres, along the longitudinal axis of an animal's body. Earthworms (members of the phylum Annelida) exhibit true metamerism, where the body is divided into a series of similar, ring-like segments internally and externally. In contrast, Amoeba, Hydra, and Planaria do not possess this segmented body structure.

102. Which of the following is the correct sequence in taxonomic hierarchy from highest to lowest? ***[BPSC WMO]***

A. Order → Class → Family → Genus
B. Family → Order →Phylum → Class
C. Class → Phylum → Order → Genus
D. Phylum → Class → Order → Family

Ans. **D**

Exp: In the taxonomic hierarchy, classification levels are organised from the broadest category to the most specific. The full sequence from highest to lowest is: Domain, Kingdom, **Phylum**, **Class**, **Order**, **Family**, Genus, and Species.

BCW Bits:

1. **Phylum:** A major category of classification that groups organisms based on a shared fundamental body plan or structural organization. e.g.: **Chordata** (animals possessing a notochord or spinal cord).
2. **Class:** A more specific division within a phylum, grouping organisms that share significant biological characteristics and evolutionary history. e.g.: **Mammalia** (animals that have hair and produce milk).
3. **Order:** A rank that groups related families based on shared physiological traits and specialized behaviours. e.g.: **Carnivora** (mammals primarily adapted for a meat-based diet).
4. **Family:** A taxonomic group containing one or more related genera that share a common ancestral lineage. e.g.: **Canidae** (the dog family, including wolves, foxes, and jackals).
5. **Genus:** A low-level taxonomic rank used to group closely related species that share a very recent common ancestor; it forms the first part of an organism's scientific name. e.g.: ***Canis*** (the genus containing dogs, wolves, and coyotes).

103. In a transverse section of dicot stem treated with phloroglucinol-HCl, crimson-stained regions correspond to ***[BPSC WMO]***

A. Phloem sieve tubes
B. Parenchyma of pith
C. Collenchyma cells in cortex
D. Primary xylem vessels (lignified)

Ans. **D**

Exp: Phloroglucinol-HCl is a chemical reagent commonly used in plant histology to stain lignin, a complex organic polymer that provides structural support and waterproofing to cell walls. When a transverse section of a dicot stem is treated with this reagent, the highly lignified cell walls—such as those found in xylem vessels and fibres—react to produce a distinct crimson or red-pink colouration. In contrast, non-lignified cells like parenchyma, phloem, and collenchyma do not stain crimson, allowing for the clear identification of vascular tissues.

104. A FRAP (Fluorescence Recovery After Photobleaching) study shows complete recovery of GFP-tagged membrane protein in 15 s at 37°C and 45 s at 10°C. What is the most likely reason for slower recovery at low temperature? ***[BPSC WMO]***

A. Lower transcription rate of GFP gene
B. Photobleaching efficiency changes
C. Decreased endocytosis
D. Reduced lateral membrane fluidity

Ans. **D**

Exp: FRAP is a technique used to measure the mobility of molecules within a cell membrane. The rate of fluorescence recovery depends primarily on the lateral diffusion of unbleached GFP-tagged proteins into the bleached area. According to the fluid-mosaic model, membrane fluidity is temperature-dependent; as temperature decreases, the lipid bilayer becomes more viscous and ordered, which restricts the movement of membrane proteins. Consequently, the lateral diffusion rate slows down at lower temperatures, leading to a longer recovery time for the fluorescent signal.

105. For DNA-based microbial community surveys, why is metagenomic sequencing superior to culture-dependent methods when estimating true diversity? ***[BPSC WMO]***

A. It retrieves functional genes from uncultured taxa
B. It eliminates need for bioinformatic analysis
C. It guarantees longer read lengths
D. It avoids PCR amplification errors

Ans. **A**

Exp: Metagenomic sequencing is superior to culture-dependent methods because the vast majority of microorganisms in most natural environments are "unculturable" in a laboratory setting. Culture-dependent techniques rely on growing organisms on specific media, which inherently biases result toward fast-growing or easily cultivated species, leaving the "great plate count anomaly" largely unresolved. Metagenomics, by contrast, involves extracting and sequencing total DNA directly from an environmental sample, allowing researchers to capture the genetic material and functional potential of the entire community, including the massive proportion of taxa that cannot currently be grown in the lab.

106. While curating insect specimens, you encounter a wing venation pattern intermediate between two established genera. According to the Biological Species Concept, what additional evidence is crucial before erecting a new genus? ***[BPSC WMO]***

A. Preservation of holotype in a recognized museum
B. Ability (or inability) to interbreed and produce fertile offspring
C. Geographic range and habitat overlap
D. Distinct karyotype counts alone

Ans. B

Exp: The **Biological Species Concept** defines a species based on the ability of individuals to interbreed in nature and produce viable, fertile offspring, and their reproductive isolation from other such groups. While morphological characteristics like wing venation are traditional tools for identifying and classifying organisms, they can be misleading due to phenotypic plasticity or convergent evolution. To formally validate a new taxonomic distinction under this concept, evidence of reproductive isolation—demonstrating that the populations do not interbreed or produce fertile offspring—is considered the most crucial biological standard.

107. Which metal ion is present in vitamin B12? ***[BPSC WMO]***

A. Mg (II) B. Fe (III) C. Zn (II) D. Co (III)

Ans. D

Exp: Vitamin B12, also known as cobalamin, is a complex organometallic compound characterized by a corrin ring structure similar to that of a porphyrin. At the centre of this ring system, a cobalt ion is coordinated to four nitrogen atoms. In its biologically active state, the metal is typically found in the **cobalt (III)** oxidation state. This ion is essential for the vitamin's function as a cofactor in critical enzymatic reactions, such as the synthesis of methionine and the conversion of methyl malonyl-CoA to succinyl-CoA.

108. Fluid mosaic model is proposed for ***[BPSC WMO]***

A. Plasma membrane B. Ribosome
C. DNA structure D. Chloroplast

Ans. A

Exp: The **fluid mosaic model**, proposed by S.J. Singer and Garth L. Nicolson in 1972, describes the structure of the cell membrane. It depicts the membrane as a "mosaic" of various components—primarily phospholipids, cholesterol, and proteins—that are not static but move laterally within the lipid bilayer, giving it a "fluid" character. While other organelles like chloroplasts are bounded by membranes that share similar fluid properties, the model was specifically formulated to explain the dynamic molecular architecture of the **plasma membrane**.

109. A researcher supplies 13C-labeled CO2 for 60 s to an illuminated C3 leaf, then analyzes metabolites. The earliest radiolabeled compound is 3-phosphoglycerate. These supports ***[BPSC WMO]***

A. Photorespiration predominance
B. PEP carboxylase fixation in mesophyll
C. Malate-aspartate shuttle activity
D. The Calvin-Benson cycle as the primary carboxylation pathway

Ans. D

Exp: In C3 plants, carbon fixation occurs via the Calvin-Benson cycle, where the enzyme RuBisCO catalyses the carboxylation of ribulose-1, 5-bisphosphate using CO_2. This reaction immediately produces two molecules of **3-phosphoglycerate** (3-PGA), which is a stable, three-carbon compound. Because 3-PGA is the first stable intermediate formed in this pathway, identifying it as the earliest radiolabelled compound after exposing a leaf to 13C-labelled CO_2 provides direct evidence that the plant is utilising the Calvin-Benson cycle for primary carbon fixation. Options involving PEP carboxylase (typical of C4 or CAM plants) or photorespiration pathways would result in different primary labelled products, such as malate or aspartate.

110. Two insect communities have identical species richness but different evenness. Which index would best capture this difference? ***[BPSC WMO]***

A. Shannon diversity index
B. Simpson's dominance index
C. Marg alef's index
D. Species richness

Ans. A

Exp: Ecological indices are mathematical tools used to quantify the biodiversity of a habitat. The **Shannon diversity index** (H') uniquely incorporates both species richness and the relative abundance of individuals, making it the most effective tool for detecting differences in evenness between communities with identical richness.

BCW Bits:

- **Simpson's dominance index:** Focuses on the probability that two randomly selected individuals belong to the same species, making it more sensitive to dominant species than overall community evenness.
- **Marg alef's index:** Measures species richness relative to the total number of individuals but does not account for their proportional distribution.
- **Species richness:** Simply counts the number of species present, providing no information regarding how individuals are distributed across those species.

111. A cell with extensive smooth ER and few lysosomes likely specializes in ***[BPSC WMO]***

A. Detoxification of xenobiotics
B. Protein secretion

C. Glycolysis
D. Phagocytosis

Ans. A

Exp: The smooth endoplasmic reticulum (SER) is the primary site for the metabolic processing and detoxification of drugs, alcohol, and other foreign chemical compounds (xenobiotics). Cells specialized in this function, such as hepatocytes (liver cells), contain an extensive network of SER. Conversely, these cells would require fewer lysosomes, as those organelles are primarily involved in intracellular digestion and waste disposal, rather than chemical detoxification.

BCW Bits:

- **Protein secretion:** Relies heavily on the *rough* ER and Golgi apparatus, not the smooth ER.
- **Glycolysis:** Occurs in the cytoplasm, not within the endoplasmic reticulum.
- **Phagocytosis:** Requires an abundance of lysosomes to break down ingested material, making it inconsistent with the "few lysosomes" description.

112. A probiotic strain reduces serum cholesterol by deconjugating bile acids. What enzymatic activity underlies this effect? ***[BPSC WMO]***

A. Urease B. Catalase
C. beta-galactosidase D. Bile-salt hydrolase

Ans. D

Exp: Bile-salt hydrolase (BSH) is the specific enzyme responsible for the deconjugation of bile acids. By catalysing the hydrolysis of the amide bond that links the bile acid to an amino acid (typically glycine or taurine), this enzyme releases free bile acids. Because these deconjugated bile acids are less efficiently reabsorbed by the intestine and are more readily excreted in faeces, the body must utilise more serum cholesterol to synthesise new bile acids, ultimately leading to a reduction in total serum cholesterol levels.

BCW Bits:

- **Urease:** Breaks down urea into ammonia and carbon dioxide; involved in nitrogen metabolism.
- **Catalase:** Facilitates the decomposition of hydrogen peroxide into water and oxygen; primarily a protective antioxidant enzyme.
- **beta-galactosidase:** Hydrolyses lactose into glucose and galactose; unrelated to bile acid metabolism.

113. Minamata disease is caused by ***[BPSC WMO]***

A. Mercury poisoning B. Sb (III) poisoning
C. Cadmium poisoning D. As (III) poisoning

Ans. A

Exp: Minamata disease is a severe neurological syndrome caused by severe mercury poisoning. It was first discovered in Minamata City, Japan, in 1956, after industrial wastewater containing methylmercury was discharged into Minamata Bay. The mercury bioaccumulated in fish and shellfish, which were then consumed by the local population, leading to mercury toxicity characterised by ataxia, sensory disturbances, and impaired cognitive function.

114. You discover a marine annelid with paired, leaf-like parapodia rich in capillaries. Predict its primary function and justify. ***[BPSC WMO]***

A. Locomotion because setae anchor substrate
B. Sensory reception via mechanoreceptors
C. Gas exchange because surface area and blood supply are high
D. Gamete transfer because parapodia lie near gonopores

Ans. C

Exp: Marine annelids, particularly polychaetes, often utilise parapodia—lateral, fleshy appendages—for various tasks. When these structures are modified to be leaf-like and are heavily vascularized with capillaries, they function as gills. The increased surface area, combined with an extensive blood supply, facilitates efficient diffusion of oxygen and carbon dioxide, which is essential for respiration in an aquatic environment.

115. A student compares cross-sections of C3 and C4 leaves. Which measurable parameter most reliably distinguishes Kranz anatomy? ***[BPSC WMO]***

A. Bundle sheath cell chloroplast area relative to mesophyll
B. Stomatal density on adaxial surface
C. Palisade thickness
D. Cuticle thickness

Ans. A

Exp: Kranz anatomy, the defining feature of C4 plants, is characterized by a wreath-like arrangement of bundle sheath cells around leaf veins. These cells are notably rich in large, specialized chloroplasts, providing a distinct contrast to the surrounding mesophyll—a structural ratio not found in C3 plants.

BCW Bits:

- Stomatal density, palisade thickness, and cuticle thickness are primarily environmental adaptations rather than consistent anatomical traits that reliably distinguish C4 Kranz anatomy from C3 leaves.

116. Ribosomes are the site of ***[BPSC WMO]***

A. RNA synthesis B. Lipid synthesis
C. Protein synthesis D. DNA replication

Ans. C

Exp: Ribosomes are complex cellular structures composed of ribosomal RNA (rRNA) and proteins. They serve as the primary site where the genetic information carried by messenger RNA (mRNA) is translated into the specific amino acid sequences that form functional proteins.

BCW Bits:

- **RNA synthesis:** Primarily occurs in the nucleus (transcription).
- **Lipid synthesis:** Occurs in the smooth endoplasmic reticulum.
- **DNA replication:** Takes place in the nucleus during the S phase of the cell cycle.

117. Nitrogenase in bacteria is oxygen sensitive. Which bacterial adaptation favours nitrogen fixation? ***[BPSC WMO]***

A. Aerobic photosynthesis
B. Production of extracellular protease
C. Elevated respiration rate
D. Spore formation

Ans. **C**

Exp: Nitrogenase is the enzyme responsible for nitrogen fixation—converting atmospheric nitrogen into ammonia. However, this enzyme is irreversibly inactivated by the presence of oxygen. This creates a physiological dilemma for aerobic bacteria: they require oxygen for energy production, yet they must protect their nitrogenase from it to function.

Bacterial Adaptation: High Respiration

- To resolve this, many aerobic nitrogen-fixing bacteria employ an **elevated respiration rate**. By increasing their metabolic activity, these bacteria rapidly consume oxygen in their immediate environment. This effectively creates a localized, oxygen-depleted (anaerobic) microenvironment around the nitrogenase enzyme, shielding it from oxidative damage while the rest of the cell remains aerobic.

BCW Bits:

- **Aerobic photosynthesis:** This process produces oxygen, which would inhibit rather than Favor nitrogen fixation.
- **Production of extracellular protease:** This is involved in protein degradation and nutrient acquisition, not in protecting nitrogenase from oxygen.
- **Spore formation:** This is a survival mechanism for harsh conditions, but it is not a direct adaptation for protecting nitrogenase during active nitrogen fixation.

118. D-mannose is C-2 epimer of ***[BPSC WMO]***

A. D-erythrose B. D-galactose
C. D-fructose D. D-glucose

Ans. **D**

Exp: Epimers are a specific type of stereoisomer that differ in their configuration at only one chiral center. Both D-mannose and D-glucose are aldohexoses—six-carbon sugars featuring an aldehyde group. They are defined as **C-2 epimers** because their structural configurations are identical at every chiral carbon except for the one at position 2 (C-2). In a standard Fischer projection, the hydroxyl group at C-2 points to the right in D-glucose, while it points to the left in D-mannose.

BCW Bits:

- **D-erythrose:** An aldotetrose, which has only four carbons, making it structurally distinct from the hexose sugars.
- **D-galactose:** A C-4 epimer of D-glucose, not a C-2 epimer.
- **D-fructose:** A ketohexose (containing a ketone group), whereas D-mannose is an aldohexose (containing an aldehyde group).

119. A new bacterial strain displays <95% 16S rRNA sequence identity compared to known species. Which is the most reliable next step to determine if it's a novel species? ***[BPSC WMO]***

A. Check GC content
B. Run a BLAST search
C. Observe morphology via electron microscopy
D. Perform Average Nucleotide Identity (ANI) analysis

Ans. **D**

Exp: When a bacterial isolate shows less than 98.7%–99% 16S rRNA sequence identity to known species, it indicates that the organism likely belongs to a distinct lineage. However, 16S rRNA analysis lacks the resolution to definitively confirm a novel species. **Average Nucleotide Identity (ANI)** has emerged as the modern gold standard for this purpose.

- ANI provides a comprehensive genomic measurement of sequence similarity across the entire genome. A threshold of approximately 95%–96% ANI is widely recognized as the genomic equivalent to the traditional 70% DNA-DNA Hybridization (DDH) benchmark, which historically defined bacterial species boundaries.

BCW Bits:

- **GC Content:** While informative for broad phylogenetic grouping, GC content is not a reliable metric for species delineation, as it can vary significantly within a species and remain similar across unrelated taxa.
- **BLAST Search:** BLAST is an excellent preliminary screening tool for comparing sequences against databases, but it lacks the necessary specificity and rigorous statistical framework required to formally validate a new species.
- **Morphology via Electron Microscopy:** Morphological characteristics are often subject to phenotypic plasticity (changes based on environment) and convergent evolution. Consequently, they are considered supportive evidence rather than primary diagnostic criteria for classifying novel species.

Ancient History

1. Which one of the following is not related to the Rigvedic period? *[71st BPSC]*
   A. Ghosha B. Gargi
   C. Apala D. Vishvavara

Ans. B

Exp: Rigveda was composed in the early vedic period (c. 1500–1000 BCE), while Gargi belonged to the later Vedic and Upanishad period (c. 800–500 BCE). She was a renowned philosopher and scholar mentioned in the Brihadaranyaka Upanishad, famous for her intellectual debate with the sage Yajnavalkya in the court of King Janaka.

- **Ghosha:** She was a prominent female seer (Rishika) of the Rigvedic period who composed hymns in the tenth Mandala (book) of the Rigveda.
- **Apala:** She was a scholar and poet of the Rigvedic period; the Rigveda (Mandala 8, Hymn 91) is attributed to her.
- **Vishvavara:** She was a notable female philosopher and composer of hymns during the Rigvedic period, specifically associated with the 5th Mandala.

2. Match List-I with List-II. *[71st BPSC]*

List-I Event/Place		(List-II) Connected with Person	
a.	Chola Gangam Lake	1.	Chandragupta Maurya
b.	Appointment of Dharma Mahamatra	2.	Harshavardhan
c.	Assembly at Prayag	3.	Rajendra I
d.	Sudarshan Lake	4.	Ashoka

Select the correct answer using the codes given below:

	a	b	c	d
A.	3	4	2	1
B.	1	4	2	3
C.	3	4	1	2
D.	2	3	1	4

Ans. A

Exp: **Chola Gangam Lake** (Ponneri Lake) was built by Chola King Rajendra I, to mark his victorious northern campaign.

- The **Prayag Assembly (Maha Moksha Parishad)** was a 75-day charitable and religious gathering organised by Emperor Harshavardhana every five years at the confluence of the Ganga and Yamuna in Allahabad. The Chinese traveller Hiuen Tsang (Xuanzang) witnessed the sixth such assembly in 643 A.D.

BCW Bits

- The Kannauj assembly was conved by Harsha to pay homage to Hiuen Tsang.
- **Sudarshan Lake** was constructed during the reign of Chandragupta Maurya by Pushyagupta, the provincial governor of chandragupta Maurya in Girnar Hill in Gujarat.
- A new officer called **'Dhamma-mahamatra'** was appointed by Ashoka for the promotion of 'Dhamma'.

3. Consider the following dynasties of Ancient India: *[71st BPSC]*
   1. Shunga 2. Maurya
   3. Kushana 4. Kanva

   Which of the following is the correct chronological order of their rule?
   A. 2-1-4-3 B. 1-2-3-4 C. 2-3-1-4 D. 2-1-3-4

Ans. A

Exp: The timeline of these ancient Indian dynasties is:

- **Maurya Dynasty** (c. 322–185 BCE): Founded by Chandragupta Maurya, this was the first major empire to unify most of India.
- **Shunga Dynasty** (c. 185–73 BCE): Established by Pushyamitra Shunga after he assassinated the last Maurya ruler, Brihadratha.
- **Kanva Dynasty** (c. 73–28 BCE): Founded by Vasudeva Kanva, who overthrew the last Shunga king, Devabhuti.
- **Kushana Dynasty** (c. 1st–3rd century CE): A later dynasty that reached its peak under Kanishka in the 2nd century CE, long after the decline of the Kanvas.

4. Match List-I with List-II. *[71st BPSC]*

List-I (Places in Ancient India)		List-II (Presently in which State)	
(a)	Karnasuvarna	1.	Maharashtra
(b)	Pragjyotish	2.	Gujarat
(c)	Girnar	3.	West Bengal
(d)	Pratishthan	4.	Assam

Select the correct answer using the codes given below:

	a	b	c	d
A.	4	3	1	2
B.	3	4	2	1
C.	3	2	4	1
D.	4	2	1	3

Ans. B

Exp: **Karnasuvarna:** Located near Murshidabad (West Bengal), was the capital of the Gauda Kingdom under King Shashanka.

- **Pragjyotish:** The ancient name for the region around modern-day Guwahati (Assam).

- **Girnar:** A hill near Junagarh (Gujarat), famous for Ashokan rock edicts.
- **Pratishthan:** Known today as Paithan (in Maharashtra). It was the capital of the Satavahana dynasty.

5. With reference to the Harappan Civilisation, consider the following statements. ***[BPSC ASO, Pre]***

1. Harappan script was an exact replica of Egyptian and Mesopotamian script, indicating their cultural links.
2. The usage of standard weights and measurements throughout the civilisation represented the uniformity of the Harappan culture.
3. The Harappan pottery was mostly handmade, as they were unaware of the potter's wheel.

How many of the statements given above are correct?

A. All three B. Only two
C. Only one D. None of the above

Ans: C

Exp: The Harappan script is unique and remains undeciphered. It was an indigenous pictographic script, not an 'exact replica' of Egyptian (hieroglyphs) or Mesopotamian (cuneiform) scripts. Hence, **Statement 1 is incorrect**.

- The Harappan Civilisation was remarkably unified, evidenced by a standardised system of weights and measures across all major sites. Weights were typically made of chert and followed a Precise binary and decimal system, which facilitated regulated trade and administrative control. Hence, **Statement 2 is correct**.
- Harappan pottery was primarily wheel-made, not mostly handmade. They were well aware of the fast potter's wheel, which they used to produce vast quantities of specialised pottery, including decorated and glazed types. Hence, **Statement 3 is incorrect.**

6. During the Gupta Empire, the term 'UPARIKARA' was used for– ***[BPSC ASO, Pre]***

A. It was a voluntary offering by people to the king.
B. An extra tax levied on all subjects.
C. King's customary share of the production normally amounts to l/6'h of the production.
D. Periodic supplies of fruits, firewood, flowers, etc.

Ans. B

Exp: During the Gupta Empire, the term **UPARIKARA** was used for an **extra or additional tax** levied on all subjects, often specifically imposed on temporary tenants. It was distinct from the regular land tax (Udranga or Bhaga) and helped fund the state's administrative and infrastructure needs.

- It was often applied to cultivators who lacked permanent proprietary rights to the soil, specifically targeting temporary tenants.

7. Which famous ancient Indian physician is considered one of the principal founders of Ayurveda and authored the 'Samhita'? ***[BPSC ASO, Pre]***

A. Charaka B. Sushruta C. Patanjali D. Aryabhata

Ans. A

Exp: **Charaka,** born about **300 BC,** was the court physician of the famous king Kanishka of the Kushan Empire. He was the renowned author of the oldest surviving text in Ayurveda, the **Charaka Samhita.** He is regarded as the **father of Ayurveda,** or **Ayurvedic medicine.** Charaka Samhita, a treatise on medication describes a wide range of ailments and explains their treatment.

- Brahmagupta (AD 628) was the first mathematician to provide the formula for the area of a **cyclic quadrilateral.** His contributions to geometry are significant.
  1. He is the first person to discuss the method of finding a cyclic quadrilateral with rational sides.
  2. His celebrated theorems on the circumdiameter of a triangle and on obtaining the diagonals of a cyclic quadrilateral are now well known.
  3. Brahmagupta's best-known work, the Brahma-sputa-Siddhanta (Correctly Established Doctrine of Brahma), was written at Bhinmal, a town in the Jalore district of Rajasthan, India. It has 25 chapters which contains results.
  4. It contains ideas including a good understanding of the mathematical role of zero, rules for manipulating both negative and positive numbers, a method for computing square roots, methods of solving linear and some quadratic equations, and rules for summing series, the Brahmagupta's identity, and the Brahmagupta's theorem.

8. Which of the following sites has yielded a short inscription written in Harappan script with huge alphabets? ***[BPSC DSO, Pre]***

A. Banawali B. Kalibangan
C. Mohen jo-daro D. Dholavira

Ans. D

Exp: The Harappan site of Dholavira, located in the Rann of Kutch (Gujarat), yielded a unique finding, known as the Dholavira Signboard. This is a short inscription comprising ten large-sized signs (alphabets) of the Harappan script.

- Each sign or letter is approximately 37cm (15 inches) high, which is exceptionally large compared to the tiny script typically found on seals.
- The characters were made out of pieces of milk-white gypsum and were originally inlaid on a large wooden board that was about 3 meters (9.8 feet) long.

- The signboard was found in one of the side rooms of the northern gateway of the citadel, suggesting it was prominently displayed to be seen from a distance.

9. The Kalinga war is described in which Rock-Edict of Ashoka? ***[BPSC DSO, Pre]***

A. Rock Edict XIII B. Rock Edict XI
C. Rock Edict IX D. Rock Edict XII

Ans. A

Exp: **Ashoka,** the son of **Bindusara,** was crowned in the year 269 B.C and fought the Kalinga war in the year **261 B.C.**

- Ashoka is considered the greatest of the Mauryan rulers. It was after the Kalinga War (the only major war fought by him) that Ashoka gave up the policy of **physical occupation (Bherighosha)** in favour of **cultural conquest (Dhammaghosha).** Ashoka's dhamma was neither a new religion nor a new political philosophy. Rather, it was **a way of life**, a code of conduct, and a set of principles to be adopted and practised by the people at large. (Dhamma is a Prakrit form of the Sanskrit word Dharma).
- Dhamma was propagated through **Ashoka's inscriptions,** which are called **Dhammalipi.** The broad objective was to Preserve the social order. He believed in religious tolerance and ordained that people should obey their parents, pay respect to the Brahmanas and Buddhist monks and show mercy to slaves and servants. Above all, the Dhammalipi asks people to show their firm devotion **(Dridha bhakti)** or loyalty to the king. He sent ambassadors of peace to the Greek kingdoms in West Asia and Greece. He also appointed **Dhamma-mahamatras** to propagate Dhamma among various social groups, including women.
- **Ashoka,** the great missionary ruler in the history of ancient India, is generally referred to as **Devnamapriya** (meaning beloved of God) or **Priyadarshi (**meaning pleasing appearance) in his inscriptions. He was mentioned as Ashoka in **Maski, Brahmagiri** (Karnataka), **Gujjara** (MP), and **Nettur** (Andhra Pradesh) inscriptions.

Ashoka's Important Edicts:

Edicts	Description
1st Rock Edict	Banned Animal Killings
7th Pillar Edict	Largest Edict
Rock Edict-12	Taught Religious Tolerance
Edict-13	Kalinga war
Rock Edict-14	Purpose of rock edicts

Ashoka's Inscriptions:

- The language used **was Prakrit,** a script that used **Greek, Kharosthi,** and **Brahmi.**
- The maximum inscriptions of Ashoka were written in the **Prakrit** language.
- **Rummindei Inscription, Nepal,** mentions **Ashoka's** visit to **Lumbini**
- **The Bhabhru Pillar** Inscription in Rajasthan declared him as **Piyadassi Raja Magadha.**
- **The Sarnath Pillar Inscription called** him **'Dharma-Ashoka'.**

10. Which of the following inscriptions of Samudragupta describes him as a 'Param Bhagavat'? ***[BPSC DSO, Pre]***

A. Allahabad Pillar Inscription
B. Eran Pillar Inscription
C. Bhitri Inscription
D. Mahrauli Inscription

Ans: A

Exp: The **Allahabad Pillar Inscription** (also known as the Prayag Prashasti) was composed by Harisena, the court poet of Samudragupta. It describes Samudragupta as a **'Param Bhagavat'** (a suPreme devotee of Lord Vishnu), highlighting his strong adherence to Vaishnavism.

- Besides 'Param Bhagavat', the inscription refers to him as **Kaviraja** (King of Poets) and **Lichchhavi-dauhitra** (grandson of the Lichchhavis).
- Also, the Nalanda Copper Plate Inscription and the Gaya Copper Plate Inscription explicitly list **'Param Bhagavat'** as one of his titles.

11. Which one of the following is an example of rock-cut temple architecture? ***[BPSC DSO, Pre]***

A. Kailasanath Temple at Kanchi
B. Rajaraja Temple of Thanjavur
C. Jagannatha Temple of Puri
D. Kailasha Temple of Ellora

Ans. D

Exp: The Kailasha Temple (Cave 16) at Ellora (Maharashtra) is the most famous example of monolithic rock-cut architecture in the world. Unlike traditional structural temples, it was carved from top to bottom out of a single massive basalt cliff face. It was commissioned by the Rashtrakuta King, Krishna I, in the 8th century CE.

- Kailasanath Temple at Kanchi is a structural temple built using blocks of sandstone and granite foundations, rather than being carved from a single rock.
- The Rajaraja Temple of Thanjavur, also known as the Brihadeshwara Temple, is a massive structural temple built using granite blocks by the Chola Emperor Rajaraja
- Jagannatha Temple of Puri is a structural temple built in the Kalinga style of architecture (Rekha Deula) by King Anantavarman Chodaganga Deva of the Eastern Ganga dynasty, roughly around 1135 AD.

12. With reference to the burial practices of the Harappans, consider the following statements:

1. The Harappans practised three types of burials: complete burial, fractional burial and cremation burial.
2. They placed the dead in a North-South direction, often accompanied by grave goods such as pottery and tools.
3. Joint burials have been found at Harappan sites like Lothal.

Which of the statements given above are correct?

[BPSC AEDO]

A. 1 and 2 only B. 2 and 3 only
C. 1 and 3 only D. 1, 2 and 3

Ans. D

Exp:

- The Indus Valley Civilisation (IVC) (c. 2600–1900 BCE) was one of the earliest urban civilisations in the world. Major sites: Harappa, Mohenjo-daro, Lothal, Dholavira, Kalibangan. Known for: Town planning (grid pattern), Drainage system, Trade and crafts.Types of burial: Archaeologists have found-Complete burial - whole body buried, Fractional burial - only part of the remains buried, which involved burying only selected bones or skeletal remains after the decomposition or exposure of the body.
- Bodies were usually laid in the North–South direction. Graves often contained pottery, Ornaments, and tools— This may indicate belief in afterlife practices.
- Joint burials: At some sites, such as Lothal, joint burials (two skeletons together) have been found.

BCW Bits:

- Harappans did **not build large tombs or pyramids** like the Egyptians.
- No clear evidence of **religious texts or priest class dominance**
- Cemeteries like **R-37 (Harappa)** give most burial evidence.
- Some graves show **social differences (rich vs simple burials)**
- Skeleton studies show people were generally **healthy but had some diseases.**
- No strong evidence of **warfare or weapons in graves**
- Burial practices were **simple and standardised**, showing social equality.

13. Which of the following statements is true about the role of 'Bhandagarika' during ancient Indian Administration?

[BPSC AEDO]

A. They were officials responsible for military training.
B. They kept records of transactions and conventions of guilds, as mentioned in the Nigrodha Jataka.
C. They managed the royal treasury and minting of coins.
D. They served as chief tax collectors in the Mauryan Empire.

Ans. C

Exp:

- **Bhandagarika literally means "keeper of the store/ treasury".** An official responsible for treasury and storage of wealth looked after Royal funds, Precious items, sometimes connected with coin management/ minting activities.
- **Nigrodha Jataka**, A Buddhist Jataka story, gives insight into Social life and guild systems. Mentions merchant organisations, not treasury officers.
- **Minting of coins:** Coins were often punch-marked in ancient India. Treasury officials ensured Proper storage and circulation control.
- **Mauryan administration (important roles):** Samaharta - Chief tax collector; Sannidhata - Treasurer (closely related role); Amatyas - Officials; Bhandagarika is similar in nature to treasury roles.

BCW Bits:

- Ancient Indian administration had specialised officials for each function.
- The Treasury was crucial because kingdoms depended on taxes and trade revenue.
- Jataka tales are valuable sources for understanding economic life.
- Guilds (Shrenis) acted like early trade unions or corporations.
- The Mauryan Empire had one of the most organised administrative systems.
- Coins were made of silver, copper, and sometimes gold.
- Financial officers had to ensure no misuse of state wealth.

14. During the time of Nahapana, what was the exchange rate between Suvarna and Karshapana in Maharashtra?

[BPSC AEDO]

A. 1: 10 B. 1: 15 C. 1: 35 D. 1: 25

Ans. C

Exp:

- Nahapana was a ruler of the Western Kshatrapas (Shakas) around the 1st–2nd century CE. Ruled parts of western India (Maharashtra, Gujarat), known mainly from Coins and Inscriptions (like Nasik caves). His period gives us important clues about trade and the monetary system.
- Suvarna was a high-value gold coin used for major transactions, while Karshapana was a silver/copper coin commonly used in daily trade. Historical evidence (especially from inscriptions and coin studies) suggests 1 Suvarna = 35 Karshapanas. This reflects a structured monetary system and an active trade economy.

- Contemporary ruler:- Gautamiputra Satakarni (Satavahana ruler) defeated Nahapana and reissued his coins.
- There was standardisation in currency, trade (especially in western India) was quite advanced, and the economy was monetised, not just barter-based.

BCW Bits:

- Western India was connected to Roman trade routes.
- Gold coins were often linked with foreign trade (Roman gold inflow).
- Karshapana coins were often punch-marked in earlier periods.
- Nasik inscriptions provide details about donations and the economy.
- Trade guilds played a major role in currency circulation.
- Coin-based economy shows rise of urban centres.

15. The ancient evidence of silver in India is found in the: ***[BPSC AEDO]***

A. Silver Punch-marked coins
B. Chalcolithic culture of Western India
C. Vedic text
D. Harappan culture

Ans. D

Exp:

- The correct answer is the **Indus Valley Civilisation or Harappan culture**. Archaeological excavations from Harappan sites have provided some of the earliest evidence of silver use in India. Silver ornaments, vessels, and other objects have been discovered from sites associated with the Harappan Civilisation. This shows that the Harappans had knowledge of metallurgy and were familiar with Precious metals. Silver Punch-marked coins are incorrect because punch-marked coins belong to a much later historical period, mainly around the Mahajanapada and Mauryan age. These coins are important as some of the earliest coins of India, but they are not the earliest evidence of silver itself. The Chalcolithic culture of Western India is also incorrect. Chalcolithic cultures mainly rePresent the Copper-Stone Age, where copper tools were common along with stone implements. Though some metal use existed, the earliest clear evidence of silver in India is traced further back to the Harappan Civilisation. Vedic literature does mention metals, including gold and silver, but literary references are not considered the earliest archaeological evidence.
- The Harappans were highly advanced in urban planning, metallurgy, drainage systems, bead-making, and trade networks. Their use of metals included copper, bronze, gold, silver, and lead.

BCW Bits:

- Harappan Civilisation is also called the Indus Valley Civilisation.
- The Harappan Civilisation belonged to the Bronze Age.
- Harappa is located in Present-day Pakistan.
- Mohenjo-daro is famous for the Great Bath.
- Dholavira is located in Present-day Gujarat.
- Lothal is famous for its dockyard.
- Harappan script remains undeciphered.
- Punch-marked coins are among the earliest coins of India.
- The Chalcolithic Age means the Copper-Stone Age.
- Iron became prominent in the later Vedic period.

16. Chand-Pradyota was the ruler of which Ancient Republic? ***[BPSC AEDO]***

A. Vajji B. Anga C. Avanti D.Kasi

Ans. C

Exp:

- **Pradyota**, also known as Chand-Pradyota, was the ruler of Avanti. Avanti was one of the sixteen Mahajanapadas and was located in the western part of India, mainly in the Malwa region of Present-day Madhya Pradesh. Its important capital was Ujjayini (Ujjain), which later became an important political, trade, and cultural centre. Pradyota was considered a powerful ruler and is mentioned in Buddhist and Jain texts. During this period, several Mahajanapadas were competing for political dominance in northern India.
- **Vajji** was a republican confederacy rather than a monarchy. Its capital was Vaishali, and it was associated with the Lichchhavis. **Anga** was located in Present-day Bihar and had Champa as its capital. Anga was later annexed by Magadha under Bimbisara. **Kashi had Varanasi** as its capital and was one of the early powerful states.
- The Mahajanapada period is important because it marked the transition from tribal political systems to organised kingdoms and republics in ancient India.

BCW Bits:

- There were 16 Mahajanapadas in ancient India as per Buddhist literature Anguttara Nikaya and jain text Bhagwati sutra.
- Magadha later became the most powerful Mahajanapada.
- Bimbisara and Ajatashatru were important rulers of Magadha.
- Vaishali is considered one of the world's earliest republics.
- Ujjain was an important trade centre in ancient India.
- Buddha and Mahavira lived during the Mahajanapada period.
- The 6th century BCE is known for the rise of new religions.
- Pataliputra later became the capital of Magadha.
- Anga was famous for trade and commerce.
- The Mahajanapada period Preceded the Mauryan Empire.

17. The Ancient University of Vikramshila was founded by which ruler of the Pala dynasty? ***[BPSC AEDO]***

A. Ramapala B. Devapala
C. Gopala D. Dharmapala

Ans. D

Exp:

- Dharmapala founded the famous Vikramshila University. Vikramshila was one of the most important centres of Buddhist learning in ancient India and was located in Present-day Bihar near Bhagalpur. It became especially famous for higher studies in Buddhist philosophy, logic, tantra, and grammar. The university attracted students and scholars from different parts of Asia, including Tibet.
- The Pala rulers were major patrons of Buddhism, particularly Mahayana Buddhism. Dharmapala played a significant role in promoting education and monastic learning. Along with Vikramshila, the Palas also supported other great educational centres like Nalanda and Odantapuri. Palas were among the last major Buddhist rulers of northern India, and their patronage helped Preserve Buddhist scholarship for centuries.

BCW Bits:

- Nalanda University was one of the oldest residential universities in the world.
- Xuanzang and I-Tsing studied at Nalanda.
- The Pala dynasty mainly ruled Bengal and Bihar.
- Gopala was elected by local chiefs according to some historical accounts.
- Odantapuri was another important Buddhist learning centre.
- Bakhtiyar Khalji destroyed several Buddhist institutions in Bihar.
- Ancient universities functioned mainly through monasteries.
- Bihar was a major centre of Buddhist learning in ancient India.
- The Sena dynasty succeeded the Palas in Bengal.

18. The first capital of the Magadha Kingdom before Pataliputra was: ***[BPSC AEDO]***

A. Nalanda B. Rajgir C. Vaishali D. Champa

Ans. B

Exp:

- Before Pataliputra became the capital, the first capital of the Magadha kingdom was Rajgir, also known in ancient times as Rajagriha.
- Rajgir was surrounded by hills, which made it naturally protected from enemy attacks. This geographical advantage helped early Magadha rulers strengthen their kingdom. Important rulers like Bimbisara and Ajatashatru were associated with Rajgir during the early expansion of Magadha.
- Later, the capital was shifted to Pataliputra because of its strategic location near major rivers like the Ganga, Son, Gandak, and Punpun. River connectivity improved trade, communication, administration, and military movement. This helped Magadha grow into the most powerful kingdom in northern India.
- Rajgir → early Magadha capital, Pataliputra → Mauryan capital, Vaishali → Vajji republic, Nalanda → ancient university.

BCW Bits:

- Magadha became the most powerful Mahajanapada in ancient India.
- Bimbisara belonged to the Haryanka dynasty.
- Ajatashatru s/o Bimbisara fortified Pataligrama, which later became Pataliputra.
- The Nanda and Maurya dynasties later ruled from Pataliputra.
- Megasthenes(ambassador in the court of chandragupta Maurya)called Pataliputra "Palibothra."
- The First Buddhist Council was held at Rajgir.
- Rajgir is associated with both Buddhism and Jainism.
- Mahavira and Buddha both visited the Magadha region.
- The fertile Gangetic plains helped Magadha expand economically.
- Iron ore deposits near Magadha helped in military growth.

19. The Palas of Bengal and Bihar were patrons of: ***[BPSC AEDO]***

A. Vaishnavism B. Shaivism
C. Buddhism D. Jainism

Ans. C

Exp:

- The Pala dynasty was a major patron of Buddhism, especially Mahayana Buddhism. The Pala rulers ruled large parts of Bengal and Bihar between the 8th and 12th centuries CE. They played a major role in Preserving and promoting Buddhism at a time when it was declining in many other parts of India. Under their patronage, famous Buddhist centres like Nalanda, Vikramshila, and Odantapuri flourished as important international centres of learning in Bihar.
- Dharmapala and Devapala were among the most important rulers of the dynasty. Dharmapala is especially remembered for founding Vikramshila University. Buddhist scholars from Tibet, Nepal, and Southeast Asia came to study in these institutions. The Palas are not mainly known for patronage of Vishnu worship, though Hindu practices continued among the population.

- Palas → Buddhism, Rashtrakutas → Jain patronage, Cholas → Shaivism, Guptas → Vaishnavism.
- The Pala period is also important because it contributed significantly to Buddhist art and sculpture. The famous Pala School of Art influenced Buddhist artistic traditions in Tibet and Southeast Asia.

BCW Bits:

- Gopala founded the Pala dynasty.
- The word "Pala" means protector.
- The Pala dynasty mainly ruled eastern India.
- Atisha Dipankara was a famous Buddhist scholar associated with the Pala period.
- Mahayana Buddhism emphasised Bodhisattva ideals.
- Pala art is known for bronze sculptures and manuscript paintings.
- The Sena dynasty succeeded the Palas in Bengal.
- Bakhtiyar Khalji later destroyed major Buddhist institutions in Bihar.

20. Which of the following Gods is referred to as 'Ritasya Gopa' in the Rigveda? ***[BPSC AEDO]***

A. Varuna B. Marut C. Indra D. Agni

Ans. A

Exp:

- The term **"Ritasya Gopa"** means **"Protector or Guardian of Rita."** Here, the word **Rita** is extremely important. In Rigvedic thought, Rita referred to the **cosmic order**, natural law, and moral order that governed the universe. It was believed that everything in the universe — seasons, sunrise, rainfall, truth, and even social harmony — functioned according to Rita. Among the Rigvedic gods, Varuna was considered the protector of this cosmic order. Therefore, he was called **"Ritasya Gopa."** Varuna was associated with truth, morality, law, and universal order. He was believed to punish those who violated moral and cosmic laws. Over time, however, the importance of Varuna declined, and gods like Indra became more dominant in later Vedic worship.
- The Maruts were storm gods associated with thunder, lightning, and strong winds. They are usually linked with warfare and natural forces. Indra was the most celebrated god in the Rigveda and was known as the god of rain, thunder, and war. He was famous for defeating the demon Vritra and releasing rivers. Although Indra was the chief deity in many hymns. Agni was the god of fire and acted as a mediator between humans and gods because offerings in yajnas were made through fire. Agni occupied a very important place in Vedic rituals.

BCW Bits:

- The Rigveda is the oldest Veda.
- Indra has the maximum number of hymns in the Rigveda.
- Agni acted as the messenger between the gods and humans.
- Soma was both a sacred drink and a deity in the Rigvedic tradition.
- Gayatri Mantra is found in the Rigveda.
- Rita later evolved into the broader concept of Dharma.
- Varuna was associated with water bodies and cosmic law.
- Early Vedic religion did not involve temple worship.
- Yajna was the central ritual practice of Vedic society.
- The Rigvedic society was mainly pastoral in nature.

21. Which area did Alexander grant to King Pururaj? ***[BPSC AEDO]***

A. Between Bias and Jhelum
B. Between Sindh and Jhelum
C. Between Ravi and Jhelum
D. Between Jhelum and Chenab

Ans. C

Exp:

- Alexander the Great invaded northwestern India in 326 BCE after conquering Persia and regions of Central Asia. At that time, northwestern India was divided into many small kingdoms and tribal republics. One of the most powerful rulers in this region was King Porus, who is referred to in Indian sources as **Pururaj** or **Puru**. The famous **Battle of Hydaspes** was fought between Alexander and Porus on the banks of the Jhelum River, which was called Hydaspes by the Greeks. Although Porus was defeated, he fought very bravely and impressed Alexander with his courage and leadership. According to Greek accounts, when Alexander asked Porus how he wished to be treated, Porus replied, "Treat me like a king." Alexander was impressed by this answer and restored his kingdom. Not only this, Alexander also granted him additional territory. The area granted to Porus was the land **between the Ravi and Jhelum rivers**. This region formed an important part of Punjab. In ancient times, many kingdoms were identified according to the river regions they controlled. Since Punjab literally means "land of five rivers," .
- Alexander never entered the Gangetic plains. His army became exhausted and refused to advance beyond the Beas River due to fear of the powerful Nanda Empire of Magadha.

BCW Bits:

- The ruler of Taxila, Ambhi, supported Alexander.
- Alexander died in Babylon in 323 BCE.
- Megasthenes later came to India during Mauryan rule.
- Chandragupta Maurya defeated Seleucus Nicator, Alexander's successor in the east.
- Punjab means "Land of Five Rivers."

- Alexander founded several cities named Alexandria during his campaigns

22. Ashoka's high officers went on 'Nirikshatan' for administrative visit in Ujjaini and Taxila in every ______ years. ***[BPSC AEDO]***

A. 2 B. 3 C. 4 D. 5

Ans. D

Exp:

- Ashoka ruled over a vast empire that stretched from Present-day Afghanistan to parts of southern India. Managing such a huge empire was not easy, so the Mauryas developed a strong administrative system with provinces, officers, spies, revenue officials, and periodic inspections. The term **"Nirikshatan"** refers to official inspection tours or administrative visits conducted by high officers. These inspections were meant to check whether provincial administration was functioning properly, whether officials were following royal orders, and whether the people were being treated fairly. According to Ashokan inscriptions, especially the edict-related administrative references, high officers were sent for inspection tours to important provincial centers like Ujjain and Taxila every **five years**.
- The Mauryan Empire was divided into provinces for easier administration. Important provincial capitals included Taxila in the northwest,Ujjain in the west,Tosali in the east,Suvarnagiri in the south. These provinces were often governed by princes or trusted officials. A very important feature of Ashoka's administration was his concern for public welfare and moral governance. After the Kalinga War, Ashoka adopted the policy of **Dhamma** and emphasized ethical administration. He appointed special officers called **Dhamma Mahamatras** to spread moral values and look after the welfare of people.

BCW Bits:

- Ashoka was the grandson of Chandragupta Maurya.
- Kalinga War was fought around 261 BCE.
- Ashoka adopted Buddhism after the Kalinga War.
- Dhamma was Ashoka's moral code, not a separate religion.
- Ashokan inscriptions were written mainly in Prakrit language and Brahmi script.
- The script of Ashokan inscriptions was deciphered by James Prinsep.
- Pataliputra was the capital of the Mauryan Empire.
- Taxila was an important center of learning and administration.
- The Lion Capital of Sarnath is India's national emblem.

23. Which one of the following scholars was associated with King Kanishka? ***[BPSC AEDO]***

A. Vasubandhu B. Dignaga
C. Buddhaghosh D. Ashvaghosh

Ans. D

Exp:

- Kanishka was a famous ruler of the Kushan dynasty who ruled during the 2nd century CE. His empire extended over large parts of Central Asia, Afghanistan, Pakistan, and northwestern India. He is especially remembered for promoting **Mahayana Buddhism** and for encouraging art, literature, trade, and scholarship. The scholar associated with Kanishka was Ashvaghosha. Ashvaghosha was a famous Buddhist philosopher, poet, and dramatist. He is regarded as one of the earliest Sanskrit playwrights and an important intellectual figure in Buddhist literature.
- Ashvaghosha is best known for writing the famous work Buddhacharita, which describes the life of Gautama Buddha in poetic form. He also wrote works like *Saundarananda*. His writings played an important role in spreading Buddhist philosophy in literary form.
- Kanishka's court included several learned scholars and Buddhist thinkers. During his reign, the **Fourth Buddhist Council** was held in Kashmir, probably at Kundalvana. This council was closely associated with Mahayana Buddhism. Sanskrit began to gain importance in Buddhist literature during this period.
- Vasubandhu was associated with Buddhist philosophy, especially the Yogachara school, Dignaga was a Buddhist logician and philosopher of a later era known for contributions to Buddhist logic and epistemology. Buddhaghosa was a Theravada Buddhist commentator associated mainly with Sri Lanka. He is famous for the work *Visuddhimagga* and lived centuries after Kanishka.
- Under Kanishka, the famous **Gandhara School of Art** flourished. This school showed strong Greek influence and is known for the first large-scale human images of Buddha.

BCW Bits:

- Kanishka is often called the "Second Ashoka."
- Capital of Kanishka was Purushapura (modern Peshawar).
- Mahayana Buddhism flourished during Kushan rule.
- Gandhara art showed strong Greek-Roman influence.
- Gandhara art popularized Buddha images in human form.
- Mathura School of Art also developed during the Kushan period.
- Kushans promoted Silk Route trade between India and Central Asia.
- Kanishka issued gold coins with Greek and Iranian deities.
- The Kushans were originally from Central Asia.

24. To which religion/sect Ashoka granted Barabar caves situated in Bihar? ***[BPSC AEDO]***

A. Buddhism B. Jainism C. Ajivakas D. Hinduism

Ans. C

Exp:

- The famous Barabar Caves located in Bihar were granted by Ashoka to the **Ajivaka sect**. The Barabar caves are considered the **oldest surviving rock-cut caves in India**. They are located in Present-day Jehanabad district of Bihar and are famous for their highly polished interior walls, which reflect the famous Mauryan polish seen in Ashokan pillars as well. These caves were donated to the Ajivakas through inscriptions engraved during Ashoka's reign. Some caves were also later dedicated by Ashoka's grandson Dasharatha Maurya to the same sect. The **Ajivakas** were an ancient heterodox religious sect that emerged around the same time as Buddhism and Jainism. Their founder is generally considered to be **Makkhali Gosala** who was a contemporary of both Mahavira and Buddha. The Ajivakas believed strongly in **Niyati** or fate/ determinism. According to their philosophy, everything in life was Predetermined, and human effort could not change destiny. This belief made their doctrine very different from Buddhism and Jainism, both of which emphasized karma and individual effort.
- Ashoka became a follower of Buddhism after the Kalinga War, the Barabar caves were not donated to Buddhist monks. Ashoka did support Buddhism greatly by building stupas and sPreading Dhamma, but these specific caves were associated with the Ajivakas. Jainism was another important heterodox religion of ancient India founded by Mahavira .
- The caves were cut from hard granite rocks and have smooth mirror-like interiors. This advanced stone-cutting skill later influenced the development of Buddhist rock-cut cave architecture in western India, such as Ajanta and Karle caves.

BCW Bits:

- Barabar Caves are the oldest surviving rock-cut caves in India.
- The caves are located in Jehanabad district of Bihar.
- Mauryan polish is a major feature of Barabar caves.
- Makkhali Gosala founded the Ajivaka sect.
- Ajivakas believed in determinism or fate (Niyati).
- Ashoka adopted Buddhism after the Kalinga War.
- Ashokan inscriptions were mainly written in Prakrit language.
- Brahmi script was deciphered by James Prinsep.
- Dasharatha Maurya also donated caves to Ajivakas.
- Mauryan period marks the beginning of monumental stone architecture in India.

25. Which of the following rulers merged Avanti in the Magadha Empire after the victory? ***[BPSC AEDO]***

A. Shisunag B. Kalashok
C. Bimbisar D. Ajatshatru

Ans. A

Exp:

- Among the sixteen Mahajanapadas, Magadha gradually became the most powerful. Its rise was not sudden. Several rulers from different dynasties expanded the kingdom through wars, diplomacy, marriage alliances, and annexation of neighboring states. The kingdom of Avanti was one of the strongest rivals of Magadha. Avanti was located in western-central India with Ujjain as an important center. It was politically and commercially significant because it controlled trade routes connecting northern and western India.The ruler who finally defeated Avanti and merged it into the Magadhan Empire was Shishunaga. This victory was extremely important because after the annexation of Avanti, Magadha became almost supreme in northern India. One of its biggest rivals disappeared, and Magadha's political dominance became much stronger.
- Kalashoka, was the successor of Shishunaga and is mainly remembered for the **Second Buddhist Council** held at Vaishali. Bimbisara, played a major role in the rise of Magadha. Bimbisara expanded Magadha through diplomacy and conquest, especially annexing Anga . Ajatashatru fought against Kosala and the Vajji confederacy and strengthened Magadha greatly, but the final annexation of Avanti happened later under Shishunaga.

BCW Bits:

- Magadha became the most powerful Mahajanapada.
- Capital of early Magadha was Rajgir.
- Pataliputra later became the capital of Magadha.
- Bimbisara belonged to the Haryanka dynasty.
- Ajatashatru built the fort of Pataligrama.
- Avanti's important capital was Ujjain.
- Nanda dynasty succeeded the Shishunaga dynasty.
- Mahapadma Nanda is called the "first historical emperor of India" in some sources.
- Iron deposits near Magadha helped in military expansion.
- River Ganga played a major role in Magadha's trade and communication.

26. Which of the following is a characteristic of Mandala IX of the RigVeda? [BPSC AEDO]

A. It is attributed to the Kanva clan and Angirasa poets.
B. It contains hymns dedicated to Soma Pavamana, focusing on the sacred purification process.
C. It contains the Nadi Sukta, which praises rivers as sacred entities.
D. It features philosophical speculation about the creation of the universe.

Ans. B

Exp:
- The Rig Veda is the oldest Vedic text and one of the earliest literary works of India. It is divided into 10 Mandalas (books). Each Mandala has its own pattern—some are based on rishi families, some on themes.
- Mandala 9 is unique because it is entirely dedicated to Soma. Specifically, it is dedicated to Soma Pavamana (purified Soma). Hymns describe: the extraction of Soma juice, its purification (filtering process), and its ritual and religious significance.
- Therefore, Mandala 9 is more about **ritual process and symbolism** rather than general gods or philosophy.

BCW Bits:
- Mandala 9 - Soma (purification, ritual), Mandala 10 - Philosophy, creation, rivers
- Rig Veda has 1028 hymns (suktas) in total.
- Mandalas 2–7 are called “family books” (each linked to one rishi family)
- Mandala 1 & 10 are considered later additions.
- The Nadi Sukta and Nasadiya Sukta (Creation Hymn) are also found in Mandala 10
- Soma is not just a drink—it is also treated like a divine entity (deity) in Vedic thought.

27. Which of the following statements about the play *Karpuramanjari* is correct? ***[BPSC AEDO]***

A. It was written in Sanskrit by Bhoja to please his wife.
B. It was written by Rajasekhara in Sauraseni Prakrit to please his wife Avantisundari.
C. It was composed during the reign of Nagabhata II.
D. It is a famous work by Kalidasa that praises the Gupta kings.

Ans. B

Exp:
- *Karpuramanjari* is a classical Indian drama associated with Rajasekhara, a court poet of the Gurjara-Pratihara period (9th–10th century CE). It is written entirely in Prakrit, specifically Sauraseni Prakrit, not Sanskrit. It was composed to please his wife Avantisundari, who encouraged his literary work.
- **Rajasekhara,** A **court poet** of the **Gurjara-Pratihara rulers, lived** around the **9th–10th century CE.** Known for literary works like *Karpuramanjari and Kavyamimamsa, he* wrote in both **Sanskrit and Prakrit.**
- **Sauraseni Prakrit,** A **popular spoken language form** in ancient India. Used in dramas to rePresent **common people’s speech.** *Karpuramanjari* is special because it is written fully in this language.
- **Nagabhata II,** A ruler of the **Gurjara-Pratihara dynasty, reigned** in the **8th–9th century CE.**
- **Kalidasa, one** of the greatest Sanskrit poets. Famous works: *Abhijnanasakuntalam, Meghaduta, Raghuvamsha; lived in the **Gupta period**.*

BCW Bits:
- Rajasekhara described himself as a “poet of poets” in *Kavyamimamsa.*
- His works give insight into court culture and literary theory.
- Prakrit languages were often used in dramas for female and common characters.
- Gurjara-Pratiharas were key rulers in North India resisting Arab invasions.
- Kalidasa is often linked with Chandragupta II (Vikramaditya)
- Bhoja (Paramara ruler) was known as a scholar-king and patron of arts.
- Sanskrit dramas usually mix Sanskrit + Prakrit, but this play is unique.
- Rajasekhara’s time reflects a transition phase in classical literature.

Medieval History

1. Who among the following rulers, ruled for the maximum period? *[71st BPSC]*
   A. Alauddin Khilji B. Muhammad bin Tughluq
   C. Bahlol Lodi D. Balban

Ans. C

Exp: Among the given rulers, **Bahlol Lodi** had the longest reign, spanning **38 years**. He was the founder of the Lodi dynasty and ruled the Delhi Sultanate from 1451 to 1489.
- **Muhammad bin Tughluq** ruled during 1325–1351 (approx. 26 years)
- **Balban** reigned during 1266–1287 (approx. 21 years)
- **Alauddin Khilji** ruled from 1296 to 1316 (approx. 20 years)

  [***Note:*** Among all the rulers of Delhi Sultanate, Bahlol Lodi ruled for the maximum year (38 years), followed by Firoz Shah Tughlaq (37 years)]

2. Which one of the following is not written by Tulsidas? *[71st BPSC]*
   A. Geetavali B. Vinay Patrika
   C. Sahitya Lahari D. Kavitavali

Ans. C

Exp: Tulsidas, a 16th-century saint and poet, is primarily known for his devotion to Lord Rama and authored several significant works in both the Awadhi and Braj languages. His major works include – Geetavali, Vinay Patrika, Kavitavali and Ramcharitmanas.
- **Sahitya Lahari** was not written by Tulsidas; it is a famous work attributed to the Bhakti poet Surdas.
- Surdas, a contemporary of Tulsidas, is well-known for his devotional compositions dedicated to Lord Krishna. His notable works include Sursagar and Sur Saravali.

3. Match List-I with List-II. *[71st BPSC]*

List-I (Father)		List-II (Son)	
a.	Babar	1.	Salim
b.	Akbar	2.	Khusrau
c.	Shah Jahan	3.	Kamran
d.	Jahangir	4.	Murad

Select the correct answer using the codes given below:

	a	b	c	d
A.	4	1	3	2
B.	3	1	4	2
C.	3	1	2	4
D.	4	2	3	1

Ans. B

Exp:
- Kamran Mirza was one of the sons of Babar and brother to Humayun.
- Prince Salim, who later became Emperor Jahangir, was the son of Akbar.
- Prince Murad Bakhsh was one of the sons of Shah Jahan.
- Prince Khusrau Mirza was the son of Jahangir.

4. Match List-I with List-II. *[71st BPSC]*

List-I (Building)		List-II (Built by)	
a.	Kabuli Bagh Mosque	1.	Qutb ud-Din Aibak
b.	Adhai Din ka Jhopra	2.	Nur Jahan
c.	Allahabad fort	3.	Akbar
d.	Itmad-ud-Daulah Mausoleum	4.	Babar

Select the correct answer using the codes given below:

	a	b	c	d
A.	4	1	3	2
B.	2	1	4	3
C.	4	1	2	3
D.	1	4	3	2

Ans. A

Exp: The **Kabuli Bagh Mosque** in Panipat (Haryana), was built in 1527 by Babur to celebrate his victory over Ibrahim Lodi at the First Battle of Panipat in 1526. Named after his wife, Mussammat Kabuli Begum, it represents early Mughal architecture, featuring a central dome, courtyard, and a surrounding garden.
- **Adhai Din Ka Jhopra** is a historical mosque in the city of Ajmer (Rajasthan), built (1192-1199) by Qutb-ud-Din-Aibak.
- The **Allahabad Fort** was constructed by the Mughal Emperor Akbar in 1583.
- **Tomb of Itmad-ud-Daulah** is a Mughal mausoleum in the city of Agra, commissioned (1622-1628) by Nur Jahan, the wife of Jahangir, for her father Mirza Ghiyas Beg. It is the first Mughal structure to extensively feature **pietra dura** (semi-precious stone inlay) and white marble, serving as a direct precursor to the Taj Mahal.
  - Known as the 'Baby Taj' or 'Jewel Box'.

5. With reference to the Vijayanagara Empire, consider the following statements. *[BPSC ASO, Pre]*
   1. The Vijayanagara Empire was founded by Harihara – I and Bukka - I.
   2. Krishnadevaraya, the greatest ruler of the Tuluva dynasty, authored 'Amuktamalyada' in Telugu.

3. The city of Hampi, the capital of Vijayanagara, was sacked after the Battle of Talikota.

How many of the statements given above are correct?

A. All three B. Only one and three
C. Only one D. None of the above

Ans. A

Exp: The Vijayanagara Empire was founded in 1336 CE by two brothers, **Harihara I** and **Bukka Raya I**. They were initially feudatories of the Kakatiyas or the Hoysalas and founded the Sangama dynasty.

- Krishnadevaraya (reigned 1509–1529 CE) was the most prominent ruler of the Tuluva dynasty. He was a great patron of literature and authored the epic poem '**Amuktamalyada'** in Telugu, which is considered a masterpiece of the era.
- After the **Battle of Talikota in 1565 CE**, where the Vijayanagara empire was defeated by a coalition of the Deccan Sultanates, the capital city of Hampi was systematically sacked, looted, and destroyed.

6. Consider the following statements regarding Mughal administration. ***[BPSC ASO, Pre]***

1. The Mansabdari system was a unique administrative system that integrated military and civil services.
2. Akbar introduced the 'Dahsala system' of land revenue assessment.
3. The 'Jagirdari' system involved granting land assignments to Mansabdars in lieu of cash salaries.

How many of the statements given above are correct?

A. All three B. Only one and two
C. Only one D. None of the above

Ans. A

Exp: The Mansabdari system, introduced by Akbar in 1571, was the backbone of Mughal administration. It was a unique, single service that integrated both military and civil services, where all officials—regardless of their role—held a 'mansab' (rank) that defined their status and responsibilities.

- Akbar introduced the 'Dahsala system' (also known as the Zabti system) of land revenue assessment in 1580. Developed by his finance minister, Raja Todar Mal, it calculated revenue based on the average yield and prices of various crops over the preceding ten years to provide stability and accuracy in taxation.
- While some officers were paid in cash (*Naqdi*), most Mansabdars were granted *Jagirs* (land revenue assignments). These grants allowed them to collect revenue from specific territories in lieu of their cash salaries to maintain their troops and household.

7. Which is not associated with Balban? ***[BPSC DSO, Pre]***

A. Chahalgani Turks B. Blood and Iron Policy
C. Mamaluk dynasty D. Transfer of capital

Ans. D

Exp: Ghiyasuddin Balban, a prominent ruler of the **Mamluk (Slave) dynasty**, was a powerful administrator known for his autocratic style of governance and ruthless policies to restore the prestige of the crown. He never shifted the capital of the Delhi Sultanate. The historic transfer of the capital from Delhi to Devagiri (which was renamed Daulatabad) took place much later, in 1327 AD, under the reign of Sultan **Muhammad bin Tughlaq** of the Tughlaq dynasty.

About Balban

- He took the throne after the death of **Nashiruddin Mahmud.**
- He declared himself as the **Shadow of God,** His theory of kingship states that the **king was the representative of God** on the earth and Kingship was a divine institution. He declared this to make the nobles believe that he got the crown or the Kingship not through their mercy but by the mercy of God.
- Balban reorganized the war office as **Diwan-e-Arz** and heightened the status of **Ariz-i-Mumalik,** his chief of staff.
- **Barids** were **intelligence officers** in his court. Balban suppressed 'Corps of Forty' or ***'Turkan- e- Chahalgani'.***
- He introduced Persian traditions in his court as ***Sijda*** (bowing down to the ground to touch it with your forehead), ***Paibos*** (Kissing the feet of the Sultan) and ***Zaminbos*** (Kissing the ground in front of the Sultan's throne).
- Balban introduced **Navroz** (Persian new year).

8. Consider the following statements regarding Sher Shah's administration: ***[BPSC DSO, Pre]***

I. Sher Shah's empire was divided into forty seven sarkars. Chief Shiqdar (law and order) and Chief Munsif (judge) were the two officers in charge of the administration in each sarkar.

II. Each sarkar was divided into several parganas.

Which of the following statement(s) is/are correct?

A. Only I B. Only II
C. Both I and II D. Neither I nor II

Ans. C

Exp: Shah Suri established a highly centralized and efficient administrative framework. For smooth governance, his entire empire was systematically divided into **forty-seven Sarkars** (districts). Each Sarkar was managed by two top-tier officials: the ***Chief Shiqdar*** (responsible for maintaining law and order) and the ***Chief Munsif*** (responsible for judiciary matters and land assessment). Furthermore, each Sarkar was subdivided into smaller administrative units called **Parganas** (sub-districts).

BCW Bits:

Sher Shah Suri, originally named Farid, was the son of Hasan Khan, the jagirdar of Sasaram. He was the founder of the Sur Empire in India.

- **Revenue Administration:**
  - The head of revenue and finance administration was called the ***Diwan-i-Wazarat*** or the *Wazir*.
  - The land was classified into **three categories** (good, bad and middling) for calculating the revenue based on the yield and measurement of land.
  - Schedules of rates were put up that fixed the revenue of land in terms of cash.
  - ***'Pattas'*** were issued to the peasants and '***Qabuliyats'*** were received from them.
  - He had established a famine relief fund which was maintained by collecting two and a half seers per bigha from the peasants.
  - During Sher Shah Suri's reign, significant measures were taken to standardize the coins made of gold, silver, and copper. Additionally, the introduction of standardized weights and measures was implemented.
  - Sher Shah Suri introduced silver ***Rupiya***.
  - **Sher Khan or Sher Shah became the virtual ruler of South Bihar in 1530.**
  - Each sarkar was further divided into various Parganas and was in charge of various officers.
    - Shiqdar – Military Officer
    - Amin – Land Revenue
    - Fotedar – Treasurer
    - Karkuns – Accountant
- **Battle Fought by Sher Shah:**
  - **Battle of Surajgarh-**1534, defeated combined forces of Lohani Chief of Bihar and Mohamud Shah of Bengal at Surajgarh (Munger) and after this battle, Sher Shah took the title of '***Hazrat-e-Allah'*** i.e. Dignity or power of Allah.
  - **Battle of Chausa**-1539, defeated Humayun.
  - **Battle of Kannauj**-1540, defeated Humayun.
  - **Battle of Kalinjar**-1545, his last battle against the Rajputs of Mahoba in the Bundelkhand fort.
- **Sur Architecture**
  - Rohtas Fort (UNESCO World Heritage Site, Pakistan)
  - Rohtasgarh Fort in Bihar (Sasaram)
  - Purana Quila (Delhi) called the Qila-i-Kuhna mosque was built in 1541
  - Sher Shah's tomb is located in **Sasaram.**

Note:

- Sher Shah introduced the land revenue system similar to Ryotwari system in most of the places, where the state kept direct relations with the peasants for the assessment and collection of land revenue.

9. Which among the following is not true about the Mughal Mansabdari system? ***[BPSC DSO, Pre]***

A. They were paid in cash as well as Jagirs.
B. They had to maintain armed forces.
C. They could not be transferred.
D. They could be given conditional or mashrut mansab.

Ans. C

Exp: The Mansabdari system can be traced back to Changez Khan. He organized his army on a decimal basis, with the lowest unit being ten and the highest ten thousand. Babur was the first to introduce the Mansabdari system to North India.

- However, it was formally introduced by the Mughal emperor **Akbar** in 1571 AD.
- Mansabdar means a rank holder or an officer. Akbar attempted to integrate the three major functionaries; the nobility, the armed forces, and the bureaucracy into a common pool of administration.
- There was dual representation of mansab i.e. ***Zat*** (personal rank and pay status) and ***Sawar*** (number of horsemen to maintain).

Note:

- Akbar's Mansabdari system or his reign was highly centralised.
- The Mansab (office/rank of dignity) was not hereditary.
- Mansabdars were frequently transferred from one region to another or between different branches of administration (civil to military and vice versa) to prevent them from establishing local power bases.

10. Who among the following catered to the needs of the Royal household during the Mughal period? ***[BPSC DSO, Pre]***

A. Diwan – i – Khalsa　B. Wakil – i - Mutalaq
C. Mir Saman　D. Mir Bakshi

Ans. C

Exp: Mir Saman (also known as *Khan-i-Saman*) was the officer in charge of the imperial household, responsible for the supply of goods, provisions, and the management of royal workshops (karkhanas) during the Mughal period. He handled the procurement, storage, and maintenance of items for the palace, including furniture, textiles, and precious objects.

- Mir Bakshi was head of the military department.
- Wakil-i-Mutalaq was the Prime Minister.
- Diwan-i-Khalsa was involved in revenue administration.

11. With reference to the Karkota dynasty, consider the following statements: ***[BPSC AEDO]***

1. Durlabhavardhana, also known as Prajnaditya, was the founder of the dynasty.

2. Lalitaditya Muktapida defeated the Turks and built the Martand Sun Temple at Anantnag.
3. Vajraditya's reign was marked by territorial expansion into the Deccan.
4. Jayapida imposed excessive taxation, which weakened his rule.

Which of the above statements are correct?

A. 1 and 2 only B. 3 and 4 only
C. 1, 2 and 4 only D. 2 and 4 only

Ans. C

Exp: The Karkota dynasty ruled Kashmir roughly from the 7th to the 9th century CE. It is considered one of the most powerful dynasties of early medieval Kashmir. The dynasty witnessed military expansion along with significant cultural and architectural development.

Important rulers of the dynasty:

- **Durlabhavardhana –** Founder, also known as **Prajnaditya**, came to power after a marriage alliance with an earlier ruling family.
- **Lalitaditya Muktapida –** His reign marked the peak of Karkota power; he conducted **military campaigns (including against the Turks).** Built the famous **Martand Sun Temple (Anantnag).**
- **Vajraditya –** Successor, witnessed gradual decline; Vajraditya was **not known for major expansion**, especially not into the Deccan. In fact, after Lalitaditya, the dynasty **began to weaken.**
- **Jayapida –** Jayapida started as a capable ruler. Later, he imposed heavy taxes; these policies caused public dissatisfaction, and his rule weakened.

Source: Rajatarangini

Karkota period : mix of **political power + cultural richness**

Karkota rulers were primarily Hindu, but they also patronised Buddhism.

12. Dipankar Srijnana was the Buddhist scholar associated with: ***[BPSC AEDO]***

A. Vikramshila University B. Odantpuri University
C. Nalanda University D. None of the above

Ans. A

Exp:

- Dipankar Srijnana, widely known as Atisha, was a renowned Buddhist monk, philosopher, and teacher of medieval India. He played a crucial role in the revival of Buddhism in Tibet. Born in Vikrampur (982 CE), Atisha belonged to a royal family but later embraced Buddhism. Studied at the famous Nalanda University and Vikramashila University and mastered Buddhist philosophy, logic, and ethics.
- Invited by the king of Tibet to reform Buddhism,
- Travelled to Tibet around 1042 CE. Played a key role in purifying corrupted practices and establishing disciplined monastic traditions.
- Associated with the development of the Kadampa tradition.
- Authored *Bodhipathapradipa* (*"Lamp for the Path to Enlightenment"*), a concise guide to the Buddhist path.

BCW Bits:

- Vikramshila was especially known for Tantric Buddhism studies.
- Vikramshila University was founded by the Pala ruler Dharmapala.
- These universities (Nalanda, Vikramshila, Odantpuri) were destroyed during Turkish invasions (Bakhtiyar Khilji)

13. With reference to Akbar's Navratnas, consider the following pairs: ***[BPSC AEDO]***

	Person	—	Title/Contribution
1.	Faizi	—	A Persian poet and translator
2.	Abul Fazal	—	Author of Ain-i-Akbari
3.	Raja Todar Mal	—	Home Minister
4.	Raja Man Singh	—	Chief of Army Staff
5.	Tansen	—	Great musician and cultural figure
6.	Abdul Rahim Khan-i-Khana	—	Finance Minister

Which of the pairs given above are correct?

A. 1, 2, 4 and 6 only B. 1, 2, 4 and 5 only
C. 1, 2, 3, 4, 5 and 6 D. 3 and 6 only

Ans. B

Exp:

- **Faizi -** A Persian poet and scholar, Translated Sanskrit works into Persian;
- **Abul Fazl -** Wrote **Ain-i-Akbari** and **Akbarnama,** Close advisor of Akbar;
- **Raja Todar Mal -** Known for **revenue reforms (Zabt system),** He was basically **Finance Minister**;
- **Raja Man Singh I -** One of Akbar's top **military commanders,** Led campaigns for the empire;
- **Tansen:-** One of the greatest **classical musicians,** Associated with **Dhrupad style;**
- **Abdul Rahim Khan-i-Khana:-** Known for Hindi dohas and Persian works, A noble and scholar.

BCW Bits:

- Abul Fazl helped shape Akbar's idea of ***Sulh-i-Kul*** **(universal tolerance)**
- Faizi was Abul Fazl's brother.

- Todarmal's revenue system became the base for later British land systems.
- Tansen was one of the Navratnas (nine gems) of Akbar.
- Rahim was also a great patron of literature and culture.
- Raja Man Singh played a role in expansion into Bengal and beyond.

14. With reference to Trilochana-Pala, which of the following statements are correct? ***[BPSC AEDO]***

1. He was titled Maha-Mandaleshvara in copper plate inscriptions.
2. His inscriptions record the donation of Ekallahara village to a Brahmin.
3. The mythical origin of his dynasty is traced to a vessel called Chuluka.

Select the correct answer using the code given below:

A. 1 and 2 only B. 2 and 3 only
C. 1, 2 and 3 D. 1 and 3 only

Ans. C

Exp: Trilochana-Pala was a prominent ruler of the **Chalukyas of Lata** branch in the 11th century CE. In his copper-plate inscriptions (such as the Surat plates), he explicitly bears the title ***Maha-Mandaleshvara***, indicating his status as a powerful regional lord or feudatory.

- His 1050 CE copper-plate inscription (the Eklahare plates) specifically records the religious donation of Ekallahara village to a Brahmin named Taraditya, to support religious rites and scholarship.
- Like other branches of the Chaulukya/Chalukya lineage, his dynasty traced its mythical, divine ancestry back to a progenitor created from a "***Chuluka***" (a small vessel or the hollow of a palm filled with water) belonging to the creator deity Brahma (Virinchi).

15. The Fourth Buddhist Council was held during the reign of: ***[BPSC AEDO]***

A. Menander B. Ashoka
C. Harshavardhana D. Kanishka

Ans. D

Exp: The **Fourth Buddhist Council** was held during the reign of **Kanishka**, the famous ruler of the **Kushan dynasty**. The council is believed to have been held in Kashmir, especially at Kundal-vana, during the 1st–2nd century CE. The council is closely associated with the rise of Mahayana Buddhism. Buddhist scholars compiled commentaries on sacred texts, Sanskrit gained importance in Buddhist literature, and Buddhism spread further into Central Asia and China through the Silk Route.

- The Third Buddhist Council was held at Pataliputra under the guidance of Moggaliputta Tissa during Ashoka's reign. Ashoka played a major role in spreading Buddhism to Sri Lanka and other regions.
- **Menander**, was an Indo-Greek ruler associated with Buddhism and is mentioned in the famous text Milinda Panha.
- **Harshavardhana**, also supported Buddhism and organized religious assemblies, but no Buddhist Council is associated with his reign.

BCW Bits:

Buddhist council	Held at	Presided by	Ruler/ patronage by	Outcome
1st (483 BC)	Sattapani cave (saptaparni cave Rajgriha	Mahakasyapa	Ajatshatru	Compilation of Vinaya Pitaka(upali) and Sutta Pitaka (Ananda)
2nd (383 BC)	Vaishali	Sabakami	Kalasoka	Split into two groups Mahasanghikas & sthavir-vadins
3rd (250 BC)	Pataliputra	Mogaliputta Tissa	Ashoka	Compilation of Abhidhamma Pitaka
4th (72 AD)	Kunndalvan (Kashmir)	Vasumitra deputed by Ashwaghosha	Kanishka	Split into Hinyana and Mahayana.

- The Gandhara School of Art flourished during Kushan rule.
- Buddha images became popular during the Kushan period.
- The Silk Route helped spread Buddhism to Central Asia and China.
- Buddhism originated in the 6th century BCE.

16. Which religion had Rashtrakuta protection? ***[BPSC AEDO]***

A. Shakta B. Jain
C. Shaiva D. Buddha

Ans. B

Exp: The Rashtrakutas were known for giving strong patronage to **Jainism**. The Rashtrakutas ruled large parts of the Deccan region between the 8th and 10th centuries CE. Their capital was **Manyakheta**. Although many Rashtrakuta rulers followed Hindu traditions, they gave significant patronage to Jain scholars, monks, and institutions.

- King Amoghavarsha I was a devout patron of Jainism (Digambara sect) and is believed to have converted to the faith.
- Jain scholars such as Jinasena and Gunabhadra flourished under their rule, and Jain literature in Sanskrit and Kannada flourished.

However, many Rashtrakuta rulers also worshipped Shiva and supported Hindu temples. Many early rulers were devoted to Shaivism (such as building the famous Kailasa Temple), Jainism received widespread, formal royal patronage and patronage of its scholars during this period.

- By the Rashtrakuta period, **Buddhism** had already declined considerably in many parts of India compared with earlier periods such as the Mauryan and Kushan eras.
- The Rashtrakutas are also famous for constructing the Kailasa Temple at Ellora, one of the greatest achievements of rock-cut architecture.

BCW Bits:

- The Rashtrakuta dynasty was founded by **Dantidurga** (also known as Dantivarman). He established the empire in 753 CE after overthrowing the last Badami Chalukya king, Kirtivarman II.
- Ellora caves contain Hindu, Buddhist, and Jain monuments.
- They were major promoters of Kannada, which thrived alongside Sanskrit at the royal court. King Amoghavarsha I authored the ***Kavirajamarga***, the earliest available Kannada text on poetics.
- The tripartite struggle involved the Palas, the Pratiharas, and the Rashtrakutas.
- The capital was plundered by the Paramaras in 972 CE. The dynasty finally collapsed in 982 CE when the last ruler, Indra IV, committed ritual fasting to death *(Sallekhana)*.

17. In which reign Mongols were seen on the Bank of Indus river for the first time? ***[BPSC AEDO]***

A. Razia B. Iltutmish
C. Qutbuddin Aibak D. Balban

Ans. B

Exp:

- **The Mongols** were first seen on the banks of the Indus River during the reign of **Iltutmish**. This incident occurred when the Mongol ruler Genghis Khan pursued Jalal-ud-din Mangbarni, the ruler of Khwarazm, to the Indus frontier during the reign of Iltutmish.
- **Iltutmish** handled the situation very carefully. He avoided providing direct support to Jalal-ud-din because doing so could have provoked a Mongol attack on India. This diplomatic policy helped the Delhi Sultanate avoid immediate destruction from the Mongols.
- **Razia Sultan's reign** came later and is mainly remembered for internal political struggles rather than the first Mongol contact. She was the first and only woman ruler of the Delhi Sultanate.
- **Iltutmish** → first Mongol appearance,
- **Balban** → strong frontier defence against Mongols,
- **Alauddin Khalji** → repeated Mongol invasions and military reforms.

BCW Bits:

- Iltutmish is regarded as the real consolidator of the Delhi Sultanate.
- Iltutmish was the first Muslim ruler to govern an independent empire from Delhi, shifting the capital there from Lahore.
- He introduced the standard silver *tanka* and the copper *jital* coins.
- To secure his throne and prevent rebellions, he established the *Turkan-i-Chahalgani*—a trusted council of 40 loyal slave nobles who held high offices.
- Iltutmish introduced the Iqta system systematically.
- He is credited with completing the construction of the Qutub Minar, whose foundation was laid by his predecessor, Aibak.
- Qutbuddin Aibak appointed Iltutmish as governor of Badaun
- Alauddin Khalji built the Siri Fort in Delhi.
- The Delhi Sultanate began in 1206 CE.

18. Who was known as Hazar-Dinari? ***[BPSC AEDO]***

A. Qutbuddin Aibak B. Malik Kafur
C. Sikandar Lodi D. Balban

Ans. B

Exp: **Malik Kafur** was known as **"Hazar-Dinari",** because he was purchased for 1,000 dinars. The title "Hazar-Dinari" literally means "worth one thousand dinars."

- Malik Kafur, originally a slave reportedly purchased for one thousand dinars, later rose to become one of Alauddin Khalji's most trusted generals.
- **Malik Kafur** is especially famous for leading military campaigns into South India.
- Under Alauddin Khalji's orders, he attacked kingdoms like Devagiri, Warangal, Dwarasamudra, and Madurai. These expeditions brought enormous wealth to the Delhi Sultanate and expanded Khalji influence deep into the Deccan and South India.
- **Qutb ud-din Aibak** was known as "Lakh Baksh" because of his generosity.
- **Sikandar Lodi**, who belonged to the Lodi dynasty and is remembered for administrative reforms and founding Agra.
- **Balban** is known for his strict rule and the policy of "Blood and Iron."

BCW Bits:

- Malik Kafur (died 1316) was an enslaved eunuch who rose to become the supreme military commander and chief minister of the Delhi Sultanate under Alauddin Khalji.
- He was initially captured in 1299 during the invasion of Gujarat by Alauddin's general, **Nusrat Khan**.
- **Devagiri** was ruled by Yadavas, **Warangal** by Kakatiyas, **Dwarasamudra** by Hoysalas and **Madurai** by Pandyas.

19. Who built Gujari Mahal? ***[BPSC AEDO]***

A. Akbar B. Man Singh
C. Tej Karan D. Suraj Sen

Ans. B

Exp: **Raja Man Singh Tomar** built the famous Gujari Mahal, located inside the Gwalior Fort complex in present-day Madhya Pradesh.

- It was built in the 15th century by Raja Man Singh Tomar for his queen Mrignayani, who belonged to the Gujjar community, due to which it came to be known as 'Gujari Mahal'.
- The palace is famous for its architecture and water management system. According to popular accounts, Queen Mrignayani wanted a separate palace with an independent water supply, and Man Singh constructed Gujari Mahal to fulfill her wish.
- **Akbar** built several important structures like Fatehpur Sikri and Agra Fort, but Gujari Mahal was built much earlier during the Tomar rule.

BCW Bits:

- Gwalior Fort is called the "Gibraltar of India", due to its strategic, highly elevated location and its seemingly impregnable defensive architecture.
- Raja Man Singh Tomar was a monumental patron of Hindustani classical music—specifically the *Dhrupad style*—and the builder of the iconic **Man Mandir Palace.**
- He compiled the definitive musical treatise ***Man Kautuhal*** and provided a safe haven for legendary musicians.
- His royal music school trained some of the greatest musicians in medieval Indian history, including the legendary **Tansen** (who later became one of the Navaratnas in Emperor Akbar's court).

20. Which Mughal Emperor is known by the name of "Rangila"? ***[BPSC AEDO]***

A. Rafi-ud-Daulah B. Rafi-ud-Darajat
C. Muhammad Shah D. Farrukh Siyar

Ans. C

Exp: Muhammad Shah was popularly known as **"Muhammad Shah *Rangila*."** The word "*Rangila*" means colorful, pleasure-loving, or fond of luxury and entertainment. Muhammad Shah received this title because he was known for his interest in music, dance, art, and courtly pleasures.

- During his reign, Mughal court culture, painting, and music flourished, but politically the empire became weak and unstable.
- His reign is especially remembered for the **invasion by Nadir Shah** in 1739 CE, who looted Delhi and carried away the Peacock Throne and the Kohinoor diamond.
- **Farrukhsiyar**, is known for granting trading privileges to the British East India Company through the Farman of 1717. Later British historian J.A. Orme termed this Farman as the "Magna Carta of the East India Company", as it authorized the Company to issue its own trade passes, known as ***Dastaks***.

BCW Bits:

- Muhammad Shah's original name was **Roshan Akhtar**.
- Despite his administrative failures, he was a massive patron of music and culture. He wrote poetry and musical compositions under the pen name ***Sadrang***.
- Muhammad Shah (1719–1748) ascended the throne at the age of 17 with the Sayyid brothers (known as ***"King Makers"***) as regents until 1720.
- To restore his de facto power, Muhammad Shah arranged for the brothers to be killed with the help of Nizam-ul-Mulk.
- During his reign, the empire broke apart. Powerful nobles founded independent kingdoms, laying the foundation for **Hyderabad** (founded by Nizam-ul-Mulk), **Awadh**, and **Bengal.**
- In 1739, the Persian emperor **Nadir Shah** invaded India and utterly destroyed the Mughal army at the Battle of Karnal in just a few hours.
- Nadir Shah looted the imperial treasury entirely, took away the legendary **Peacock Throne** and the famed ***'Koh-i-Noor'.***

21. On which victory Alauddin Khilji was granted governorship of Awadh in addition to Kara? ***[BPSC AEDO]***

A. Chittor B. Gujarat C. Bhilsa D. Devagiri

Ans. C

Exp:

- Before becoming Sultan, Alauddin Khilji was known as **Ali Gurshasp** and was the nephew as well as son-in-law of Jalaluddin Khalji, the founder of the Khalji dynasty. Jalaluddin trusted him and in 1292 CE appointed him as the governor of Kara, an important province near present-day Uttar Pradesh.
- Alauddin gained first major military fame after his successful expedition against **Bhilsa** (M.P) in 1293 CE. Bhilsa was a prosperous town and an important center of wealth and temples.

- Alauddin's success in this campaign impressed Jalaluddin Khalji greatly. As a reward for this victory, Alauddin was granted the governorship of **Awadh in addition to Kara**. This event is important because it strengthened Alauddin's political and military position.
- The additional resources and power he gained later helped him organize the famous Devagiri expedition and eventually seize the Delhi throne.
- The Devagiri expedition was a turning point in Alauddin's career. Devagiri, ruled by the Yadavas, was immensely wealthy. Alauddin secretly attacked it and brought enormous treasure back to Kara.
- Using this wealth and military support, he later assassinated Jalaluddin Khalji and became Sultan in 1296 CE.

BCW Bits

- Alauddin Khalji reigned during 1296–1316 CE, and assumed the title of ***Sikandar-i-Sani*** (meaning "The Second Alexander").
- Alauddin Khalji's formidable military expansion was driven by a core group of trusted commanders - **Malik Kafur, Ulugh Khan, Nusrat Khan** and **Zafar Khan**.
- Malik Kafur led Alauddin's southern campaigns.
- Alauddin introduced market control and price control policies.
- He created a dedicated revenue department, the ***Diwan-i-Mustakhraj***, to track and collect tax arrears.
- He instituted a branding system for horses ***(Dagh)*** and maintained descriptive rolls of soldiers ***(Chehra)*** to prevent corruption, fraud, and false musters.
- He maintained a large standing army to resist Mongol invasions. He established an efficient espionage system with spies (***Barids***).
- Amir Khusrau lived during Alauddin Khalji's reign.
- Chittor campaign is linked with the legend of Rani Padmini.
- Alauddin was the first Sultan to adopt the policy of separating religion from politics to some extent.

22. Under which Sultan of Delhi, Muhammad Bin Tughlaq had worked before becoming the ruler? ***[BPSC AEDO]***

A. Balban B. Alauddin Khilji
C. Kaiqubad D. Khusrau Shah

Ans. D

Exp:

- Before becoming Sultan, Muhammad bin Tughlaq was known as **Jauna Khan** or Ulugh Khan. He was the son of Ghiyasuddin Tughlaq, the founder of the Tughlaq dynasty.
- Muhammad Bin Tughlaq served as ***Amir-i-Akhur*** (Master of the Stables/Horses) under **Sultan Khusrau Shah.** He held this position during Khusrau Shah's brief reign in 1320, just before his father (Ghiyasuddin Tughlaq) overthrew the Sultan to establish the Tughlaq dynasty.
- After the death of Alauddin Khalji, the Khalji dynasty became weak. Court conspiracies, assassinations, and succession struggles increased. Eventually, Khusrau Shah captured power after killing Mubarak Shah Khalji.
- At that time, Ghazi Malik, who was governor of Dipalpur, revolted against Khusrau Shah. His son Jauna Khan supported him in these military efforts.
- After defeating Khusrau Shah in 1320 CE, Ghazi Malik became Sultan with the title **Ghiyasuddin Tughlaq**, and led the foundation of Tughlaq dynasty.
- Muhammad bin Tughlaq himself later became one of the most controversial rulers in Indian history. He was highly intelligent and ambitious but many of his policies failed because of poor implementation.

BCW Bits:

Muhammad Bin Tughlaq :

- Despite being the Sultan of a Muslim empire, he **celebrated Hindu festivals** (like Holi), appointed Hindus to high administrative posts, and built shelters for Hindu saints.
- The famous Moroccan traveler **Ibn Battuta** visited his court and served as the *Qazi* (judge) of Delhi for eight years, documenting the Sultan's reign.
- Highly educated but often called the **"wisest fool"** due to his radical, miscalculated projects. He is most famous for five ambitious policies that were brilliant in concept but disastrous in implementation:
  - Transfer of Capital from Delhi to Daulatabad (1327).
  - Introduction of Token Currency made of copper and brass (1330)
  - Increased land taxes in the fertile Doab region
  - Khurasan Expedition
  - Karachil Expedition
- Firoz Shah Tughlaq was famous for building numerous canals for irrigation, establishing the city of Firozabad, Jaunpur, and creating hospitals and employment bureaus.
- Tughlaq architecture used sloping walls and massive structures.
- Ghiyasuddin Tughlaq built Tughlaqabad Fort near Delhi.
- The final blow came in **1398 CE,** when the Central Asian conqueror Timur invaded India, captured and sacked Delhi, and effectively shattered the authority of the Sultanate. The dynasty limped on until the last ruler, Nasir-ud-din Mahmud Tughlaq, was deposed in 1414 CE, clearing the way for the Sayyid dynasty.
- The chronology of Delhi Sultanate dynasties : Slave, Khalji, Tughlaq, Sayyid, and Lodi.

23. Who led the Mughal army in the Battle of Haldighati? ***[BPSC AEDO]***

(1) Bhagwan Das (2) Man Singh
(3) Akbar (4) Asaf Khan

Select the correct answer using the following code:

A. Only 3 B. Only 2
C. Both 2 and 4 D. 1, 2 and 4

Ans. C

Exp:

- The Battle of Haldighati was fought in **1576 CE** between the forces of Maharana Pratap of Mewar and the Mughal army of Akbar. The battle took place near Haldighati Pass in present-day Rajasthan.
  - The name "Haldighati" comes from the yellow-colored soil of the region, which resembles turmeric (*haldi*).
- The Mughal army in this battle was led mainly by **Man Singh I,** the Kachhwaha Rajput ruler of Amber and one of Akbar's most trusted generals. Along with him, **Asaf Khan** also played an important role in leading the Mughal forces.
- Maharana Pratap fought bravely with the support of his loyal chiefs like **Hakim Khan Sur** and Bhil warriors. His famous horse **Chetak** became legendary in Rajput history and folklore.
- Although the Mughal forces gained control of the battlefield, Maharana Pratap was not captured and he continued resistance against the Mughals for many years. Thus, many historians were of the view that battle was not a complete political victory for the Mughals.
- Raja Bhagwant Das was an important Rajput noble in Akbar's court and belonged to the Amber royal family, but he did not lead the Mughal army in the Battle of Haldighati.
- Akbar himself did **not** fought directly in the Battle of Haldighati.
- Unlike earlier Delhi Sultans, Akbar followed a policy of alliance and cooperation with many Rajput rulers. Several Rajput kingdoms accepted Mughal suzerainty and became part of the Mughal nobility. However, Mewar under Maharana Pratap strongly resisted Mughal authority.

BCW Bits:

- Capitals of Mewar during Maharana Pratap's reign were:
  - **Gogunda**: Temporary capital and site of Pratap's coronation
  - **Kumbhalgarh**: Adopted as his seat of government for a period.
  - **Chavand**: Established as the new capital in 1585.
- Chittor was captured by Akbar before the Battle of Haldighati.
- **Raja Bhagwant Das**, the Kachwaha ruler of Amber, was a powerful general for the Mughal Empire under Emperor Akbar. He played a vital role in consolidating Mughal-Rajput relations.
  - His sister, *Mariam-uz-Zamani* (historically referred to as **Jodha Bai**), was Akbar's chief consort and mother of Emperor Jahangir.
  - His eldest son, **Raja Man Singh I**, became one of the most prominent "Navaratnas" (Nine Jewels) of Akbar's court.
  - His daughter, **Man Bai** (Shah Begum), was married to Prince Salim (later Emperor Jahangir) in 1585.

24. In which field terminology 'Rai' was connected during Sher Shah's period? ***[BPSC AEDO]***

A. Judiciary B. Revenue C. Army D. Religion

Ans. B

Exp: Sher Shah Suri ruled for a short period, from 1540 to 1545 CE, but his administrative reforms were so effective that historians consider him one of the greatest rulers of medieval India.

- He reorganized revenue collection, roads, currency, military administration, and governance in a systematic way.
- During Sher Shah's reign, the term **"Rai"** was related to the **revenue system**. He instituted a system where land was categorized by fertility (good, middling, bad). The "Rai" was the average produce of these categories, and the state took about a third of this as tax.
- Sher Shah tried to standardize revenue assessment so that the state could collect tax fairly and efficiently. His administration carefully measured land and estimated produce before fixing revenue demand.
- Sher Shah's revenue reforms were based on **direct contact between the state and cultivators.** He ordered land measurement and classification according to fertility.
- The state share was generally fixed around one-third of the produce. Revenue could be paid either in cash or kind, depending on local conditions.
- Two important documents connected with Sher Shah's revenue administration were:
  - **Patta** → a document given to cultivators mentioning land area and revenue demand,
  - **Qabuliyat** → an agreement by the cultivator accepting the revenue terms.
- Sher Shah's administration is important because it created a more organized state structure. He improved roads like the famous **Grand Trunk Road,** introduced a strong currency system with the silver ***"Rupiya,"*** and strengthened communication through *sarais* and postal arrangements.
- Even though the Sur Empire lasted for a short time, Sher Shah's reforms had a long-lasting impact on Indian administration and later Mughal governance.

BCW Bits:
- Sher Shah Suri defeated Humayun at the **Battle of Chausa (1539)** and **Battle of Kannauj (1540)**.
- Sher Shah died on **May 22, 1545**, during the siege of the **Kalinjar fort**, when a gunpowder explosion backfired on him.
- He is buried in his hometown of **Sasaram (Bihar)**, inside a magnificent, red-sandstone mausoleum built in the middle of a lake.

25. With reference to the Guhilots/Sisodiyas, Rathors and Kachwahas, consider the following statements: ***[BPSC AEDO]***

1. The Guhilots ruled over Mewar with Chittor as their capital.
2. Sawai Jai Singh of the Rathor dynasty was a high-ranking noble in the Mughal court.
3. Man Singh of the Kachwahas was a trusted commander of Akbar.
4. Maharana Pratap of the Sisodiyas fought against Akbar at the Battle of Haldighati.
5. Jaswant Singh of the Kachwahas was also an astronomer.

Which of the above statements is/are correct?

A. 1, 2, 3 and 5 only
B. 2, 3 and 4 only
C. 1, 3, 4 and 5 only
D. 1, 3 and 4 only

Ans. D

Exp:
- **Guhilots/Sisodiyas-** The Guhilots (who later branched into the Sisodiyas) ruled the region of Mewar and made Chittor (Chittorgarh) their primary, long-term strategic capital.
  - **Rathors** - Ruled Marwar (Jodhpur region),
  - **Kachwahas**- Ruled Amber/Jaipur. These clans were important during the Mughal period; Kachwahas had a close alliance with the Mughals, especially under Akbar.
- **Sawai Jai Singh II** belonged to the Kachwaha **dynasty (Amber/Jaipur)**, not the Rathors. He was a distinguished Mughal noble and renowned astronomer.
- **Raja Man Singh,** belonged to the Kachwaha clan of Amber and was one of Akbar's Navratnas. He was a trusted general of Akbar and played a key role in Mughal expansion**.**
- **Maharana Pratap** was a Sisodiya ruler of Mewar who famously fought against Akbar's imperial army (led by Man Singh) at the **Battle of Haldighati** in **1576.** Sisodiyas of Mewar are known for their resistance to Mughal rule
- **Jaswant Singh** belonged to the **Rathor dynasty (Jodhpur),** and it was Sawai Jai Singh II, the Kachwaha ruler of Jaipur, who was renowned for astronomy.

BCW Bits

Sawai Jai Singh II
- Due to his keen interest in astronomy, Sawai Jai Singh II built five astronomical observatories (***Jantar Mantar***) across India, between 1724 and 1730.
- He built Jantar Mantar at five locations – **Jaipur, Delhi, Ujjain, Varanasi** and **Mathura**, of which the Jantar Mantar in Jaipur is the largest and is included in the UNESCO World Heritage Centre .
- In 1727, with the help of the architect Vidyadhar Bhattacharya from Bengal, he founded the city of **Jaipur**. Seeing his extraordinary qualities, **Aurangzeb** gave him the title of ***'Sawai'***, which means one and a quarter times superior to his contemporaries.
- Under his supervision, a precise table of stars called ***'Zij Mohammad Shahi'*** was prepared, and he himself composed an astrological treatise called ***'Jaisimha Karika'***.

Modern History

1. At which Viceroy was a bomb thrown on the occasion of his state entry into Delhi? ***[71st BPSC]***

A. Lord Curzon B. Lord Minto

C. Lord Hardinge D. Lord Wavell

Ans. C

Exp: A bomb was thrown at Viceroy **Lord Hardinge** on 23 December 1912, during his ceremonial state entry into Delhi. The attack occurred while he was riding an elephant through Chandni Chowk in Delhi. Lord Hardinge was wounded but survived. This attack was the part of the Delhi-Lahore Conspiracy led by Rash Behari Bose. Other key figures involved in the conspiracy included Basant Kumar Biswas.

Note: The event marked the transfer of the capital of British India from Calcutta to Delhi.

2. Who assumed the command of the Indian National Army and gave famous battle cry, 'Chalo Delhi'? ***[71st BPSC]***

A. Rash Behari Bose B. Mohan Singh

C. Shah Nawaz Khan D. Subhash Chandra Bose

Ans. D

Exp:

INA or Azad Hind Fauj (Rangoon and Singapore)

- The Indian National Army (INA), also known as the Azad Hind Fauj, was a military force formed during World War II with the objective of securing India's independence from British rule.
- While initially conceived by Capt. Mohan Singh in 1942, it was revived and significantly expanded under the charismatic leadership of **Subhas Chandra Bose.**
- Bose, having escaped India and sought Axis support, took command of the INA in 1943.
- He established two primary headquarters for the INA: one in **Singapore** and the other in **Rangoon (present-day Yangon), Burma.** From these locations, Bose launched military campaigns, famously **giving the call 'Chalo Delhi'** (March to Delhi) and hoping to march into India alongside Japanese forces to liberate the country.
- The INA symbolized a powerful assertion of Indian nationalism and a willingness to fight for freedom, even if it meant allying with wartime enemies of Britain.

3. Which Indian leader for the first time used the 'safety-valve theory' to attack the Moderates in the Congress? ***[71st BPSC]***

A. Bipin Chandra Pal B. Lala Lajpat Rai

C. Bal Gangadhar Tilak D. Dadabhai Naoroji

Ans. B

Exp: **Lala Lajpat Rai** used the **safety valve** theory to attack the moderates in Congress. As per this theory, A.O. Hume formed the INC with the idea that it would prove to be a safety valve for releasing the growing discontent among Indians.

BCW Bits:

- Lala Lajpat Rai described this theory first in an article published in **'Young India'.** This theory was well supported by extremist leaders.
- This theory has some support from the autobiography written by **William Wedderburn,** published in **1913.**
- **Rajani Palme Dutt (R.P.Dutt)** was a historian who believed that the establishment of Congress was a **conspiracy** to neutralize the resentment against the British Raj.

4. 'We shall either free India or die in the attempt...' who said it? ***[71st BPSC]***

A. Jawaharlal Nehru B. Mahatma Gandhi

C. Abul Kalam Azad D. V.D. Savarkar

Ans. B

Exp: Mahatma Gandhi said, 'We shall either free India or die in the attempt,' during his famous 'Do or Die' speech. He delivered this clarion call on 8 August 1942, at the Gowalia Tank Maidan in Bombay (now August Kranti Maidan, Mumbai) to launch the Quit India Movement, urging for immediate independence from British rule.

5. Which Act provided for the partition of India into two new dominions India and Pakistan? ***[71st BPSC]***

A. Govt. Of India Act, 1919

B. Govt. Of India Act, 1935

C. Indian Independence Act, 1947

D. None of the above

Ans. C

Exp: **Indian Independence Bill was:**

- Presented on 4 July 1947 in British Parliament.
- Passed by the House of Commons on 15 July 1947.
- Passed by the House of Lords on **16 July 1947.**
- Got assent of the Crown on 18 July 1947.

After all these progressions, the Indian Independence Bill becomes the Indian Independence Act. As per this act, two nations with Dominion States/ Nations were formed and the deadline to fulfill all the necessary augmentations was 15 August 1947.

6. Which Commission accentuated the discontent in all the political groups and parties for not associating any Indian member in it? ***[71st BPSC]***

A. Welby Commission
B. Cripps Mission
C. Simon Commission
D. Cabinet Mission

Ans. C

Exp: The **Government of India Act, 1919** contained a provision that a commission would be appointed ten years from the date to study the progress of the governance scheme and suggest new steps. However, a commission was constituted 2 years early as the Conservative government in power in Britain feared defeat in the coming elections.

- Therefore, a seven-member **Indian Statutory Commission, popularly known as the Simon Commission** (after the name of its chairman, Sir John Simon), was set up by the British government under Stanley Baldwin's prime ministership on 8 November 1927.
- Simon Commission arrived in Bombay (India) on 3 February 1928.
- The task of the commission was to recommend to the British government whether India was ready for further constitutional reforms and along what lines.
- **All its members were white** which angered the Indians as foreigners would discuss and decide upon India's fitness for self-government.
- The Congress session in **Madras (December 1927)** meeting under the presidency of **M.A. Ansari** decided to boycott the commission.

7. 'The Young India', a Nationalist paper, was started by ***[71st BPSC]***

A. Annie Besant
B. Mahatma Gandhi
C. Bal Gangadhar Tilak
D. Maulana Abul Kalam Azad

Ans. **Deleted**

Exp: BPSC has deleted the question because the correct name was 'Young India' and not 'The Young India'.

- Young India was a weekly English journal published by **Mahatma Gandhi** from 1919 to 1931. He used the publication to spread his philosophy of non-violence (Satyagraha) and to urge Indians to plan and organize for independence from British rule.
- It ran from 1919 to 1931 and was published out of Ahmedabad.
- The paper focused on social and economic issues, Swaraj (self-rule), and advocated for the removal of untouchability.
- In 1922, Gandhi was indicted by the British for publishing seditious articles in Young India, which ultimately led to his first major imprisonment in India.
- After the Young India journal concluded in 1931, Gandhi went on to publish the Harijan newspaper starting in 1933.

8. Which one of the following is not correctly matched? ***[71st BPSC]***

	Person	- Associated with
A.	James Augustus Hicky	- Calcutta Chronicle
B.	Dayananda Saraswati	- Satyarth Prakash
C.	Ramaswamy Naicker	- Justice Party
D.	Jyotirao Phule	- Satyashodhak Samaj

Ans. A

Exp: **James Augustus Hicky** is famously associated with **Hicky's Bengal Gazette** (also known as the Original Calcutta General Advertiser), which was the first newspaper published in India in 1780. The Calcutta Chronicle was a separate publication started later by different individuals (Daniel Stuart and Joseph Cooper in 1786).

- **Swami Dayananda Saraswati** authored **Satyarth Prakash** in 1875 to advocate for Vedic principles and social reform.
- **E.V. Ramaswamy (Periyar)** was a prominent leader of the **Justice Party** and later transformed it into the Dravidar Kazhagam.
- **Jyotirao Phule founded the Satyashodhak Samaj** (Truth Seekers' Society) in 1873 in Pune to fight for the rights of lower castes and women.

9. Name the Principal of M.A.O. College, Aligarh, who encouraged Muslim Communalism in the last decades of the 19th Century? ***[71st BPSC]***

A. Dunlop Smith
B. Minto
C. William Morris
D. Theodore Beck

Ans. D

Exp: **Theodore Beck** (1859–1899), the Principal of the Muhammadan Anglo-Oriental (MAO) College in Aligarh (appointed in 1883), encouraged Muslim communalism in the late 19th century. He promoted a separate political identity for Muslims, urging them to stay away from the Indian National Congress.

- He sought to align Indian Muslims with British interests rather than the Indian nationalist movement.
- He played a key role in shaping the political outlook of the Aligarh movement, promoting, in particular, the idea of Anglo-Muslim unity and, in turn, communal separation.

10. Who raised the cry 'Divide and Quit'? ***[71st BPSC]***

A. M. A. Ansari
B. M. A. Jinnah
C. Jawaharlal Nehru
D. Muhammad Iqbal

Ans. B

Exp: The **'Divide and Quit'** slogan was given by Muhammad Ali Jinnah, leader of the All-India Muslim League, during the1943 Karachi session. It served as a counter-slogan to the 1942 'Quit India' movement, demanding that the British should partition India to create Pakistan, before departing.

- It was a direct response to the INC's 'Quit India' movement, reflecting the Muslim League's demand for a separate nation.

11. Who got an opportunity to become Professor of Indian Philosophy in the University of California in 1911? ***[71st BPSC]***

A. Sohan Singh Bhakna
B. Virendranath Chattopadhyay
C. Har Dayal
D. Shyamji Krishna Varma

Ans. C

Exp: Har Dayal got an opportunity to become Professor of Indian Philosophy at the University of California in 1911. He was a prominent Indian revolutionary and scholar, who played a crucial role in the Ghadar Movement for India's independence. His academic pursuits led him to this position in California, USA.

BCW Bits:

- Sohan Singh Bhakna was the founder of Ghadar Party.
- Shyamji Krishna Varma is known for founding India House in London.

12. Which of the following statements about Subhas Chandra Bose are correct? ***[71st BPSC]***

1) He was elected President of the Indian National Congress at Tripuri session of 1938 A.D.
2) He escaped from home interment in January 1941 A.D.
3) On 21 October 1943 A.D, he constituted the Provincial Government of India in Singapore.

A. Only l and 2 B. Only 2 and 3
C. Only l and 3 D. 1, 2 and 3

Ans. Deleted

Exp: BPSC has deleted this question due to Factual error. The term 'PROVINCIAL' Government is incorrect. However the correct term is 'PROVISIONAL'. Thus statement 1 and 3 are both wrong. No Option for Option 2 alone is provided.

BCW Bits:

- Subhash Chandra Bose was elected as the President of INC at its Haripura Session (1938) and Tripuri Session (1939) but resigned from Tripuri due to differences with Gandhiji. He founded the Forward Bloc (1939) at Calcutta.
- Netaji escaped from his residence at Elgin Road on 16 January 1941, while under house arrest by the British police.
- On 21 October 1943, Netaji Subhas Chandra Bose announced the formation of the Azad Hind Government, also known as India's first independent provisional government, which started functioning from Singapore.

13. Consider the following events of Modern Indian History: ***[71st BPSC]***

1. Death sentence to Raja Nand Kumar.
2. Treaty of Salbai
3. Battle of Wandiwash
4. Death of Tipu Sultan

Which of the following is the correct chronological order of the above events?

A. 2-3-4-1 B. 3-1-2-4
C. 2-4-1-3 D. 3-2-1-4

Ans. B

Exp: **Battle of Wandiwash (1760)** was a decisive battle during the Third Carnatic War where British forces under Sir Eyre Coote defeated the French, ending French political ambitions in India.

- **Death sentence to Raja Nand Kumar (1775):** Raja Nand Kumar was tried for forgery and executed in Calcutta under English law, a case often cited as an instance of early colonial judicial overreach.
- **Treaty of Salbai (1782):** This treaty ended the First Anglo-Maratha War, establishing peace between the British East India Company and the Maratha Empire for nearly twenty years.
- **Death of Tipu Sultan (1799):** Tipu Sultan, the ruler of Mysore, was killed during the Fourth Anglo-Mysore War while defending his capital, Seringapatam.

14. The First Anglo-Mysore War, fought between the British and Hyder Ali from 1767 to 1769, concluded with which of the following agreements? ***[BPSC ASO, Pre]***

A. Treaty of Srirangapattanam
B. Treaty of Salbai
C. Treaty of Madras
D. Treaty of Mangalore

Ans. C

Exp:

Anglo-Mysore War

Anglo-Mysore War	Command of Mysore	Result
First **(1767-69)**	Haider Ali	English forces were defeated by Hyder Ali, leading to the signing of **Treaty of Madras** (4 April 1769).
Second **(1780-84)**	Hyder Ali died in 1782, war was carried on by Tipu Sultan	Tired with inconclusive war, both side opted for peace through the **Treaty of Mangalore** (March, 1784)
Third **(1790-92)**	Tipu Sultan	Tipu was defeated and had to sign the **Treaty of Seringapatam** (1792)

Fourth **(1799)**	Tipu Sultan	Lord Wellesley became the governor general and declared a war on Tipu Sultan alleging his growing friendship with the French. Tipu Sultan laid down his life fighting bravely.

15. Which of the following Government of India Act / Indian Council Act brought the three separate presidencies (Madras, Bombay and Bengal) into a common system? ***[BPSC ASO, Pre]***

A. Regulating Act, 1773
B. Indian Councils Act, 1861
C. Indian Councils Act, 1892
D. Government of India Act, 1858

Ans. **B**

Exp: The **Indian Councils Act of 1861** is recognized for bringing the three separate presidencies of Madras, Bombay, and Bengal into a common system of administration and law-making. This Act provided a unified administrative foundation by bringing the three presidencies under a common system, while simultaneously initiating legislative decentralization.

- Introduced by Lord Canning, this system transformed the executive council into a cabinet-like structure, creating a standardized administrative framework across British India.
- It restored the legislative powers of the Bombay and Madras presidencies, which had been taken away by the Charter Act of 1833, allowing them to make laws for their respective territories under a shared central authority.

BCW Bits:

- Regulating Act, 1773 was the first step toward centralization by making Bombay and Madras subordinate to the Governor-General of Bengal, but it did not create the 'common system' of modern administration or legislative participation characterized by the 1861 Act.

16. The famous Peacock Throne of Shah Jahan was taken away in 1739 by: ***[BPSC ASO, Pre]***

A. Mongol invader Chengiz Khan
B. Afghan invader Ahmed Shah Abdali
C. British East India Company
D. Persian invader Nadir Shah

Ans. **D**

Exp: The famous Peacock Throne (Takht-i-Taus), of Shah Jahan was taken away in 1739 by the Persian invader **Nadir Shah.** Following his victory over Mughal Emperor Muhammad Shah at the Battle of Karnal, Nadir Shah sacked Delhi and seized immense wealth, including the Peacock Throne and the Koh-i-Noor diamond.

17. Which institutions were founded under the Orientalist influence? ***[BPSC ASO, Mains]***

(1) Calcutta Madrassa (1781)
(2) Asiatic Society of Bengal (1784)
(3) Sanskrit College in Banaras (1794)

A. Only one
B. Only (1) and (2)
C. Only (2) and (3)
D. (1), (2) and (3)

Ans. **D**

Exp: Institutions founded under the **Orientalist influence** during the colonial period were primarily designed to study and preserve traditional Indian, Arabic, and Persian literature, laws, and cultures. The most prominent institutions include:

- **Calcutta Madrassa (1781):** Founded by Warren Hastings to promote the study of Arabic, Persian, and Islamic law.
- **Asiatic Society of Bengal (1784):** Founded by Sir William Jones in Kolkata to encourage research on the history, culture, literature, and sciences of Asia.
- **Sanskrit College in Banaras (1794):** Founded by Jonathan Duncan to encourage the study of ancient Sanskrit texts and Hindu law.

18. Which ideology is said to have brought fundamental changes in British Indian administration in the early 19th Century? ***[BPSC ASO, Mains]***

A. Romanticism
B. Mercantilism
C. Utilitarianism
D. Positivism

Ans. **C**

Exp: In the early 19th Century, the British changed how they governed India because of the idea of **Utilitarianism**. This philosophy, led by thinkers like Jeremy Bentham, believed that laws and government should focus on doing "the greatest good for the greatest number of people." Instead of just following old Indian traditions, they introduced new, strict rules and social changes to make the administration more "efficient" and modern. It led to the codification of Indian laws and the introduction of a professional bureaucracy.

BCW Bits:

- **Romanticism:** An intellectual movement that valued **emotion, tradition, and the glorification of the past**. In India, it was reflected in the work of **Sir William Jones**, who founded the***Asiatic Society of Bengal*** and **translated** the ***Abhijnanashakuntalam***, advocating for the preservation of ancient Sanskrit heritage.
- **Mercantilism:** An economic theory where the state regulates the economy to ensure a favourable balance of trade. **Thomas Mun**, through his work ***England's Treasure by Forraign Trade***, advocated for these policies, which initially saw the East India Company focus on trade monopolies and bullion accumulation.
- **Positivism:** A philosophical system founded by **Auguste Comte** (author of ***The Course on Positive Philosophy***).

It asserted that authentic knowledge is based on **scientific verification and empirical evidence**, eventually influencing 19th-century governance to be treated as a rigorous "social science."

19. Match List-I (Organization) with List-II (Founder): ***[BPSC ASO, Mains]***

(a)	National Indian Association	1.	Mary Carpenter
(b)	Indian Society	2.	Ananda Mohan Bose
(c)	East Indian Association	3.	Dadabhai Naoroji
(d)	India League	4.	Sisir Kumar Ghosh

Codes:
A. a-2, b-4, c-3, d-1
B. a-2, b-4, c-1, d-3
C. a-3, b-4, c-2, d-1
D. a-1, b-2, c-3, d-4

Ans. D

Exp: The **National Indian Association** was established by **Mary Carpenter** in 1870 to promote social reform and education.

- The **Indian Society** was founded in London by **Ananda Mohan Bose** in 1872 to foster unity among Indian students.
- **Dadabhai Naoroji**, known as the "Grand Old Man of India," founded the **East Indian Association** in 1866 to provide information on Indian grievances to the British public.
- The **India League** was started by **Sisir Kumar Ghosh** in 1875 with the specific goal of stimulating a sense of nationalism and political consciousness among the middle classes.

20. Consider the following events and arrange them in chronological order: ***[BPSC ASO, Mains]***
I. Direct Action Day,
II. The Cabinet Mission,
III. The Wavell Plan,
IV. Attlee's Announcement.
A. I, III, II and IV
B. III, II, I and IV
C. III, I, II and IV
D. II, III, I and IV

Ans. B

Exp:

- **The Wavell Plan (June 1945), proposed by Lord Wavell at the Shimla Conference,** aimed to reconstruct the Executive Council with equal representation for Caste Hindus and Muslims to move toward self-governance. However, it collapsed when **Muhammad Ali Jinnah** insisted that the Muslim League be the sole representative of all Indian Muslims, which was rejected by the congress.
- **The Cabinet Mission (March–June 1946):** A high-level mission comprising **Pethick-Lawrence, Stafford Cripps, and A.V. Alexander** was sent to discuss the transfer of power. It proposed a three-tier federal structure and rejected the explicit demand for a separate Pakistan.
- **Direct Action Day (August 16, 1946):** The Cabinet Mission's rejection of a sovereign Pakistan led the Muslim League to withdraw its acceptance and launch **Direct Action Day (August 16, 1946)**. This "Day of Destiny" triggered the horrific **"Great Calcutta Killings,"** effectively ending the possibility of a unified state.
- **Attlee's Announcement (February 20, 1947):** British PM **Clement Attlee** made his historic **Announcement on February 20, 1947**, declaring a firm deadline of June 30, 1948, for the British withdrawal, which accelerated the final partition process under Lord Mountbatten.

21. Consider the following statements about the use of Swadeshi goods: ***[BPSC ASO, Mains]***
(1) Gopal Hari Deshmukh, better known as Lokahitawadi of Bombay had urged the necessity of using indigenous goods.
(2) In 1873, Bholanath Chandra preached the establishment of indigenous banks, companies, mills and factories, and denounced the practice of preferring foreign goods to home-made manufacturing.
(3) Rabindranath started Swadeshi Bhandar in 1897.
(4) Sarala Devi started Lakshmi Devi Bhandar in 1903.
A. (1), (2) and (3)
B. (1), (2), (3) and (4)
C. (2) and (3)
D. (3) and (4)

Ans. B

Exp: The movement for economic self-sufficiency was shaped by diverse leaders:

- **Gopal Hari Deshmukh (Lokahitawadi)**: A pioneer of the rationalist movement in Maharashtra, Deshmukh utilized his weekly publication, ***Prabhakar*****,** to write **"*Shatapatre*"** (100 Letters). He was among the earliest to argue that India's poverty was a direct result of the drain of wealth and to urge the adoption of indigenous goods as a religious and national duty, long before the formal Swadeshi Movement of 1905.
- **Bholanath Chandra:** In 1873, writing for ***Mookerjee's Magazine*** *(edited by* ***Sambhu Chandra Mookerjee*** *and published in Calcutta)*, Chandra formulated a systematic economic critique of colonial rule. He advocated for "non-consumption" of British goods and preached the necessity of establishing **indigenous banks and mills**. His ideas provided the intellectual blueprint for "Economic Boycott," a strategy later adopted during the anti-partition agitation.
- **Rabindranath Tagore:** In 1897, he established the **Swadeshi Bhandar** in Calcutta to promote the sale of

indigenous products. He believed that the Swadeshi spirit should manifest in the revival of rural industries and self-reliance (*Atmashakti*), bridging the gap between cultural identity and economic independence.

- **Sarala Devi Chaudhurani**: She founded the **Lakshmi Devi Bhandar**(Lakshmi Bhandar) in 1903 to popularise Swadeshi goods among women. She also organised the ***Birastami Vrata* and *Pratapaditya Utsav*** to instill courage and a sense of pride in Indian heritage, linking physical culture with the economic boycott of foreign cloth.
- Together, these initiatives transformed the use of Swadeshi goods from a simple consumer choice into a powerful tool of political and economic resistance against colonial rule.

22. Given below are two statements: one labelled as Assertion (A) and the other labelled as Reason(R): ***[BPSC ASO, Mains]***

Assertion (A): Rajnarain Bose is regarded as a Pioneer of Nationalism in Bengal.

Reason (R): He founded the Hindu Mela in 1867.

In the context of the above two statements, which one of the following is correct?

A. Both (A) and (R) are true, and (R) is the correct explanation of (A)
B. Both (A) and (R) are true, but (R) is not the correct explanation of (A)
C. (A) is true, but (R) is false
D. (A) is false, but (R) is true

Ans. C

Exp: **Rajnarain Bose** is widely regarded as a "Pioneer of Nationalism in Bengal" for his tireless efforts to promote indigenous culture and national consciousness long before formal political movements took root. However, while **he provided ideological inspiration**, the **Hindu Mela** (originally the Chaitra Mela) was **actually founded** in 1867 **by Nabagopal Mitra**. Bose served as the intellectual mentor rather than the executive founder.

- Beyond the Mela, his legacy is rooted in his 1866 prospectus for a Society for the Promotion of National Feeling, his leadership in the Adi Brahmo Samaj, and his role as a mentor to revolutionaries like his grandson, Sri Aurobindo Ghosh.
- Nabagopal Mitra, popularly known as "National Mitra" for his relentless devotion to the national cause, was a multifaceted pioneer who founded the National Press, National Paper, and National Gymnasium. His vision for the Hindu Mela was to promote self-reliance through the display of indigenous industrial products and the revival of traditional Indian physical culture and martial arts.

23. Given below are two statements: one labelled as Assertion (A) and the other labelled as Reason(R): ***[BPSC ASO, Mains]***

Assertion (A): Lala Lajpat Rai argued in 1916 that the Congress was a "Product of Lord Dufferin's brain."

Reason (R): He believed that the Congress was intended to protect the British Empire's interests first and India's interests only secondarily.

In the light of the above statements, choose the most appropriate answer from the options given below?

A. Both (A) and (R) are true and (R) is the correct explanation of (A)
B. Both (A) and (R) true, but (R) is not the correct explanation of (A)
C. (A) is true, but (R) is false
D. (A) is false, but (R) is true

Ans. A

Exp: In 1916, through his work Young India, **Lala Lajpat Rai** popularised the "Safety Valve" theory, arguing that the Indian National Congress was a **"Product of Lord Dufferin's brain."** He believed the British facilitated the creation of the Congress not to help India achieve freedom, but to provide a controlled outlet for growing political discontent. According to Rai, the organisation was intended to protect the interests of the **British Empire** by preventing a violent uprising, ensuring that Indian political activity remained secondary to imperial stability.

- Hence, Both the assertion and the reason are accurate, and the reason provides the correct logic for the assertion.
- Beyond the Safety Valve theory, other significant interpretations regarding the birth of the Congress include:
  - **Lightning Conductor Theory:** Proposed by **Gopal Krishna Gokhale**, this theory argues that the early nationalists used A.O. Hume as a "lightning conductor." They believed that if an Indian had started such a massive political organisation, the British would have suppressed it immediately; whereas by having a former British official lead it, the movement gained a protective cover.
  - **Conspiracy Theory:** Championed by **R.P. Dutt** in his work *India Today*, this Marxist interpretation suggests that the INC was a result of a secret plan by the British to "thwart" the rising popular discontent and protect capitalist interests.

24. Consider the following statements related to the Quit India Movement: ***[BPSC ASO, Mains]***

(1) The Congress was not declared illegal.
(2) British authority disappeared in the parts of Uttar Pradesh, Bihar, West Bengal, Orissa, Andhra Pradesh, Tamil Nadu and Maharashtra.

(3) Ballia in Eastern Uttar Pradesh, Tamluk in Midnapore district of Bengal and in Satara district of Bombay, the revolutionaries set up 'parallel governments.'

(4) The students, workers and peasants provided the backbone of the 'revolt' while the upper classes and the bureaucracy remained loyal to the government.

Which of the statements given above is/are correct?

A. (1), (2), (3) and (4) B. (2), (3) and (4)

C. (2) and (3) D. (3) and (4)

Ans. B

Exp: The **Quit India resolution** was formally passed by the **All India Congress Committee (AICC)** on **August 8, 1942**, during its historic session at **Gowalia Tank Maidan** (now known as August Kranti Maidan) in Bombay. Immediately following the passage of the "Quit India" resolution on August 8, 1942, the British government launched **Operation Zero Hour**. Under this operation, the **Indian National Congress was declared an illegal association**, and its top leadership was arrested overnight. Despite the crackdown, the movement became a massive spontaneous revolt:

- British authority collapsed in several regions across UP, Bihar, West Bengal, and Odisha as the public took control.
- Parallel governments were established in places like Ballia (Chittu Pandey), Tamluk (Jatiya Sarkar), and Satara (Prati Sarkar under Nana Patil and Y.B. Chavan), functioning independently of British rule for varying periods.
- The "backbone" of this struggle was the youth, workers, and peasantry, who engaged in large-scale strikes and sabotage of colonial infrastructure, making it one of the most formidable challenges to British sovereignty.

25. Who founded the Brahmo Samaj? ***[BPSC ASO, Mains]***

A. Swami Dayanand

B. Gopal Krishna Gokhale

C. Raja Ram Mohan Roy

D. Jawaharlal Nehru

Ans. C

Exp: The **Brahmo Samaj** was a pioneering monotheistic reformist movement established in **1828** by **Raja Ram Mohan Roy**, at Calcutta. It played a pivotal role in the **Indian Renaissance** by advocating the worship of one formless God and promoting rational thought. **Building on this foundation of social reform,** the movement became the primary vehicle for challenging deep-rooted orthodoxies in 19th-century India.

- **Raja Ram Mohan Roy**, often hailed as the **"Father of Modern India,"** founded the *Brahmo Sabha* in August 1828. His objective was to synthesize Western rationalism with the monotheistic teachings of the Upanishads.
- **As a direct result of his advocacy through the Samaj,** the **Sati system** was formally abolished by Lord William Bentinck in **1829**, marking a landmark victory for human rights in India.
- The movement rejected idol worship and the caste system. **Instead, it focused on** universal brotherhood and the belief that no scripture is infallible, encouraging followers to use reason as their guide.
- Following Roy's death in 1833, leadership passed to **Debendranath Tagore**. The movement eventually expanded further under **Keshab Chandra Sen**, though internal differences later led to the creation of the *Brahmo Samaj of India* and the *Adi Brahmo Samaj*.

BCW Bits:

- **Swami Dayanand:** Founded the **Arya Samaj** in 1875, promoting the "Back to the Vedas" slogan and the Shuddhi movement.
- **Gopal Krishna Gokhale:** Established the **Servants of India Society** in 1905 to promote social and human development through education.
- **Jawaharlal Nehru:** Served as the first Prime Minister of India and was the primary architect of the modern, secular Indian nation-state.

26. The Permanent Settlement was introduced in 1793 by Lord Cornwallis in: ***[BPSC DSO, Pre]***

A. Bengal and Bihar

B. Bengal, Bihar and Orissa

C. Bengal, Bihar and Madras

D. Bengal and Bombay

Ans. B

Exp: Zamindari or Permanent settlement was introduced by **Cornwallis** in **1793** through the Permanent Settlement Act.It was introduced in the provinces of **Bengal, Bihar, Orissa** and **Benaras**. It comprised 19% of British India.

- If the Zamindars failed to pay the assessed revenue by the exact moment of sunset on a predetermined fixed date, their estates were immediately confiscated and auctioned off to recover the dues. Hence it is also called **'Sunset Law'.**
- Zamindars were recognized as the owners of the lands.
- **1/11** of the share belonged to Zamindars and **10/11** of the share belonged to the East India Company.

27. With reference to Indian history, consider the following statements: ***[BPSC DSO, Pre]***

1. The Dutch established their factories warehouses on the east coast on lands granted to them by Gajapati rulers.
2. Alfonso de Albuquerque captured Goa from the Bijapur Sultanate.
3. The English East India Company established a factory at Madras on a plot of land leased from a representative of the Vijayanagara empire.

Which of the statements given above are correct?

A. 1 and 2 only B. 2 and 3 only

C. 1 and 3 only D. 1, 2 and 3

Ans. B

Exp: The **Gajapati dynasty** ruled the Odisha region roughly from 1434 to 1541. However, the Dutch East India Company was not formed until 1602 and they established their first Indian factory at Masulipatnam in 1605. By the time the Dutch arrived on the east coast, the Gajapati rule had already ended; they instead obtained land from rulers like the Sultan of Golconda or local Nayakas. **Hence Statement 1 is incorrect.**

- **Alfonso de Albuquerque,** the second Portuguese governor of India, captured Goa from the Bijapur Sultanate (ruled by the Adil Shahi dynasty) in 1510. This victory established Goa as the headquarters of the Portuguese. Hence **Statement 2 is correct**.
- In 1639, the English East India Company secured a lease for a strip of land in Madras from **Damarla Venkatadri Nayaka,** a local chieftain and representative of the Vijayanagara Empire (specifically the Aravidu dynasty ruling from Chandragiri). Later, this site became Fort St. George. Hence **Statement 3 is correct.**

28. Badruddin Tyabji is known for which of the following contributions during the Indian National Movement? ***[BPSC DSO, Pre]***

A. He was the founder of the forward Bloc.

B. He was the first Muslim President of the Indian National Congress and worked actively for Hindu-Muslim unity.

C. He launched the Aligarh Movement for Muslim educational upliftment.

D. He presided over the Lahore Session of the INC in 1929, where Poorna Swaraj was declared.

Ans. B

Exp: **Badruddin Tyabji** was the first Muslim president of the INC who presided over the third Session in Madras (1887), and he worked actively for Hindu-Muslim unity.

- **Subhash Chandra Bose** founded the Forward Bloc in 1939 after resigning from the Indian National Congress.
- **Sir Syed Ahmad Khan** launched the Aligarh Movement in 1875 to modernize education for Muslims in India.
- **Jawaharlal Nehru** presided over the Lahore Session of the INC in 1929, where INC passed the landmark **Purna Swaraj** (complete independence) resolution.

29. Which of the following events marked the formal launch of the Non-Cooperation Movement by the Indian National Congress? ***[BPSC DSO, Pre]***

A. Jallianwala Bagh Massacre

B. Chauri Chaura Incident

C. Nagpur Session of Congress, 1920

D. Rowlatt Act Protests

Ans. C

Exp: The Non-Cooperation Movement was launched on **1 August 1920** by Mahatma Gandhi after the Jallianwala Bagh Massacre. The launch of Non-cooperation was decided in the **Nagpur session of the INC in 1920.**

The movement started with:

- Surrendering of the titles.
- Boycotting government schools and colleges.
- Hindi was adopted as the lingua franca for the whole country during the movement
- Hand-spinning and weaving were emphasized greatly.
  - During the movement, the **Chauri Chaura (in present-day Uttar Pradesh)** incident took place on 4 February 1922, due to which Mahatma Gandhi suspended NCM by calling a meeting of the congress working committee at Bardoli on 11 February 1922. He called the incident a 'Himalayan blunder'.

30. Which of the following leaders was prominently associated with the Champaran Satyagraha (1917) in Bihar, marking Mahatma Gandhi's first active involvement in Indian mass movements? ***[BPSC DSO, Pre]***

A. Rajendra Prasad B. Jayaprakash Narayan

C. Shri Krishna Sinha D. Anugrah Narayan Sinha

Ans. A

Exp: Champaran Satyagraha (1917) in Bihar was Mahatma Gandhi's first civil disobedience movement in India, initiated to address the grievances of farmers against European indigo planters.

- Under the prevailing **Tinkathia system**, peasants were legally forced to cultivate indigo on 3/20th of their total land.
- To offset financial losses caused by the introduction of cheaper German synthetic dyes, planters imposed high rents, levied illegal dues, and forced farmers to sell indigo at fixed, sub-market prices.
- Triggered by a request from a local man named **Raj Kumar Shukla**, Gandhi traveled to Champaran accompanied by leaders including **Rajendra Prasad, Mazhar-ul-Haq, Mahadev Desai, Narhari Parekh, and J.B. Kripalani.**
- When local authorities ordered Gandhi to leave the district immediately, he defied the order and accepted the legal penalties, establishing passive resistance as a practical political method.
- This non-violent non-compliance forced the administration to permit an official inquiry. The government subsequently appointed the Champaran Inquiry Committee, presided over by **F.G. Sly, with Gandhi** serving as a member.

- Following the submission of the committee's final report on October 4, 1917, the Tinkathia system was legally abolished through the **Champaran Agrarian Act on March 4, 1918.** The settlement also successfully reduced peasant rents and secured a partial refund of the illegally collected dues from the planters.

  [***Note:*** *Anugrah Narayan Sinha was indeed a core participant who accompanied Mahatma Gandhi in Champaran alongside Rajendra Prasad. However, his contribution was not as prominent as Rajendra Prasad. Hence, BPSC has considered Rajendra Prasad as the correct answer.*]

31. With reference to the Age of Consent Act. 1891 consider the following statement: ***[BPSC DSO, Pre]***

1. It was Behramji Malabari, a Parsi reformer from Bombay, who advocated for this legislation.
2. The Act was supported by the extremist wing led by Bal Gangadhar Tilak.

Which of the above statements is/are correct?

A. Only 2 B. Neither 1 nor 2
C. Only 1 D. Both 1 and 2

Ans. C

Exp: Behramji Malabari, a Parsi reformer from Bombay, was the primary advocate who pushed for the Age of Consent Act, 1891, which raised the age of consent for girls from 10 to 12 years.

- The extremist wing led by Bal Gangadhar Tilak strongly opposed the Act, viewing it as colonial interference in Hindu social and religious customs.

  Hence only Statement 1 is correct.

BCW Bits:

- The **Age of Consent Act, 1891** (also known as **Act X of 1891**) was enacted, which raised the age of girls, married and unmarried ladies for the consent of sexual intercourse, to twelve years from earlier ten years.
- The sufferings of two young girls Rukhmabai (One of the first practicing women doctors in the Colonial India) and Phulmoni Dasi led to this act.

32. Match List-I with List-II: ***[BPSC DSO, Pre]***

List-I (Real Name)		List-II (Popular Name)	
(a)	Mulshankar	(1)	Ramakrishna Paramahamsa
(b)	Narendra Nath Datta	(2)	Bhagini Nivedita
(c)	Gadadhar Chattopadhyay	(3)	Swami Dayanand Saraswati
(d)	Margaret Elizabeth Noble	(4)	Swami Vivekanand

Choose the correct answer from the options given below:

A. (a)-(l), (b)-(2), (c)-(3), (d)-(4)
B. (a)-(2), (b)-(1), (c)-(4), (d)-(3)
C. (a)-(3), (b)-(4), (c)-(1), (d)-(2)
D. (a)-(3), (b)-(4), (c)-(2), (d)-(1)

Ans. C

Exp: Mulshankar is popularly known as Swami Dayanand Saraswati, the founder of the Arya Samaj. He is sometimes referred to as the 'Martin Luther of Hinduism'.

- **Narendra Nath Datta** is popularly better known as Swami Vivekananda, who took Hindu philosophy to the West. He founded the Ramakrishna Mission.
- **Gadadhar Chattopadhyay** is popularly known as Ramakrishna Paramahansa. He served as a priest at the Dakshineswar Kali Temple, and was the guru of Swami Vivekananda.
- **Margaret Elizabeth Noble** is popularly known as Bhagini Nivedita (Sister Nivedita), an Irish-born teacher, author, and social activist, who became a dedicated disciple of Swami Vivekananda in 1895 and moved to India in 1898 for social work.

33. Which one of the following statements is not true? ***[BPSC DSO, Pre]***

A. Annie Besant of the Madras Theosophical society asserted that in ancient times Hindu women were educated and moved freely in society.
B. The Arya Samaj of North India strongly supported the cause of female education.
C. The Hunter Commission Report of 1882 recommended more liberal grants-in-aid for girls' schools than for boys and special scholarships and prizes for girls.
D. Dhondo Keshav Karve was against the education of high-caste Hindu widows.

Ans. D

Exp: Dhondo Keshav Karve (1858–1962), widely known as *Maharshi Karve*, was a towering social reformer in India who dedicated his life to the empowerment, education, and remarriage of Hindu widows.

- His pioneering work successfully dismantled orthodox social norms and established long-lasting institutions for women's welfare.
- Challenging the strict orthodox traditions of the era, he openly advocated for **widow remarriage** and boldly set an example by marrying a widow himself in 1893 after the passing of his first wife.
- He founded the **Widow Marriage Association** in 1893.
- To provide shelter and education for ostracized widows, he opened the **Hindu Widows Home** (later renamed ***Hingne Stree Shikshan Samstha***) in 1896 in Hingne, near Pune and subsequently established an orphanage for girls, the ***Anath Balikashram***.

- Recognizing that education was the key to financial independence, he established SNDT Women's University in 1916.It was the very **first women's university in India**.
- Maharshi Dhondo Keshav Karve was awarded the **Bharat Ratna** in 1958.

34. Which of the following statements about Bhaga Jatin is correct? ***[BPSC DSO, Pre]***

A. He led the Bardoli Satyagraha in 1928.

B. He was associated with the Jugantar revolutionary group and died fighting British forces at Balasore in 1915.

C. He was the first Indian to become President of the Indian National Congress.

D. He launched the Khilafat Movement in Bengal.

Ans. B

Exp: Jatindranath Mukherjee, popularly known as **Bagha Jatin**, was a key leader of the Jugantar group, a secret revolutionary organization in Bengal. He played a pivotal role in the **'German Plot' (Zimmerman Plan)** during World War I, attempting to secure German arms for an armed insurrection against British rule in India.

- On **9 September 1915,** Jatin and four associates fought a legendary trench battle against a heavily armed British force in Balasore (Odisha). He was severely wounded and died on 10 September 1915.

BCW Bits:

- The **Bardoli Satyagraha of 1928** was led by Sardar Vallabhbhai Patel.
- The first Indian to become the President of the Indian National Congress was **Womesh Chandra Bonnerjee** (1885).
- The Khilafat Movement (1919–1924) was launched by the **Ali brothers (Shaukat Ali and Mohammad Ali)** to support the Ottoman Caliphate.

35. Whose 'open letters' of the Indian economy to Lord Curzon forced the British Indian government to publish an official rebuttal and also initiate reformatory economic measures? ***[BPSC DSO, Pre]***

A. M. G. Ranade B. G. V. Joshi

C. R. C. Dutt D. Lala Lajpat Rai

Ans. C

Exp: **Romesh Chunder Dutt (R.C. Dutt),** a prominent Indian economist, authored a series of 'Open Letters' to Lord Curzon in 1900. These letters, titled ***Open Letters to Lord Curzon on Famines and Land Assessments in India***, criticized British land revenue policies as a cause of poverty and famine, forcing an official rebuttal and sparking land assessment reforms.

- The letters focused on the excessive land tax, the destruction of Indian industries, and the government's responsibility regarding widespread famines.
- The intensity of the critique compelled the British administration under Lord Curzon to issue an official response and implement reforms, including revising land revenue policies.

36. Which of the following newspapers made vigorous denunciation of the Indigo planters and supported the case of cultivators during 'Indigo riots'? ***[BPSC DSO, Pre]***

A. Hurkaru B. Hindu Patriot

C. Friend of India D. Hindustan

Ans. B

Exp: The **Hindu Patriot**, an English weekly published in Kolkata, played a pivotal role during the Indigo Revolt **(*Nil Bidroho*)** of 1859–1860 in Bengal. Under the courageous editorship of **Harish Chandra Mukherjee**, the newspaper became a powerful voice for the exploited cultivators (ryots).

- The newspaper fearlessly exposed the atrocities, coercion, and violence used by British indigo planters to force farmers into unprofitable cultivation.
- Mukherjee's advocacy was instrumental in pressuring the British colonial government to appoint the Indigo Commission in 1860, which eventually recognized the exploitative nature of the system and recommended reforms.

37. 'When the two-nation theory was gaining momentum, who taunted the Morley-Minto reforms by saying that the real originator of Pakistan was not Iqbal or Jinnah but Minto'? ***[BPSC DSO, Pre]***

A. C. Rajagopalachari

B. Dr. Rajendra Prasad

C. Govind Ballabh Pant

D. Jawaharlal Nehru

Ans. B

Exp: **Dr Rajendra Prasad** made this observation to critique the Indian Councils Act of 1909, commonly known as the Morley-Minto Reforms, because it introduced a separate electorate system for Muslims for the first time in British India.

- By implementing this divisive electoral framework, the reforms laid the structural groundwork for communal segregation, which made Rajendra Prasad to argue that the real originator of Pakistan was neither Muhammad Iqbal nor Muhammad Ali Jinnah, but the Viceroy Lord Minto, whom he considered the "Father of Pakistan."
- At the time of these structural changes, John Morley served as the Secretary of State for India while Lord Minto acted as the Viceroy.

- Beyond the implementation of separate electorates, the 1909 reforms expanded Indian participation in governance, notably leading to the appointment of Satyendra Prasad Sinha as the first Indian law member in the Governor-General's Executive Council.
- Ultimately, the Morley-Minto Reforms proved to be the most short-lived of all constitutional measures initiated by the British Government.

38. Which of the following is not correct matched? ***[BPSC DSO, Pre]***

A. Azad Muslim Conference - Allah Bakhsh
B. Khaksar Party - Allama Mashriki
C. Krishak Praja Party - Abdul Haqq
D. Khudai Khidmatgar - Abdul Gaffar Khan

Ans. C

Exp: The **Krishak Praja Party** (originally Nikhil Banga Praja Samiti) was founded in 1929 in Bengal. It was famously led by A.K. Fazlul Huq (popularly known as Sher-e-Bangla), not Abdul Haqq. The party represented the interests of tenant farmers against the landed gentry.

- The All India Azad Muslim Conference was established/ led by Allah Bakhsh Soomro (Chief Minister of Sindh) to oppose the partition of India.
- The Khaksar movement was founded in 1931 by Inayatullah Khan Mashriqi (also known as Allama Mashriqi).
- Khudai Khidmatgar - This non-violent resistance movement in the North-West Frontier Province was founded by Khan Abdul Ghaffar Khan (the 'Frontier Gandhi').

39. The 'Spanish Flu' first reached to which city of India in 1918? ***[BPSC DSO, Pre]***

A. Calcutta B. Bombay C. Lahore D. Poona

Ans. B

Exp: The 1918 'Spanish Flu' pandemic first reached India in Bombay (now Mumbai) during May/June 1918, entering via the port city on ships carrying soldiers returning from World War I.

- The first cases were reported around 29 May 1918, with significant, widespread infection identified by early June, 1918.
- The virus arrived in Bombay, often referred to as 'Bombay Fever' in its early stages.
- This pandemic was exceptionally deadly in India, claiming an estimated 12-17 million lives.

40. Arrange chronologically the following Famine Commissions: ***[BPSC DSO, Pre]***

(a) McDonnell (b) Strachey
(c) Lyell (d) Campbell

Select the correct answer from the codes given below:

A. (c), (a), (d), (b) B. (b), (c), (a), (d)
C. (d), (b), (c), (a) D. (a), (c), (d), (b)

Ans. C

Exp: The Famine Commissions in British India were established in response to major food crises to formulate relief policies and administrative codes. Their chronological timeline is as follows:

- **Campbell Commission (1866-67):** Formed under George Campbell following the devastating Orissa Famine of 1866.
- **Strachey Commission (1878-80):** Established under Sir Richard Strachey after the Great Famine of 1876–78; it laid the groundwork for the 1883 Famine Code.
- **Lyell Commission (1896-98):** Appointed under Sir James Lyall after the famine of 1896-97 to review existing relief policies.
- **McDonnell Commission (1900-01):** Appointed by Lord Curzon under Sir Anthony McDonnell after the 1899-1900 famine to refine famine management and institutionalize relief measures.

41. It was a secret society founded by Vinayak Damodar Savarkar (VD Savarkar) and his brother Ganesh Damodar Savarkar (Babarao Savarkar) in 1904 to overthrow British rule in India. It established connections with Indian revolutionaries in London and had links with the India House group, fostering international support for the Indian independence movement.

Which of the following revolutionary organisations is described in the above passage? ***[BPSC AEDO]***

A. Anushilan Samiti B. Abhinav Bharat
C. Ghadar Party D. Jugantar Party

Ans. B

Exp: Abhinav Bharat Society (also known as the Young India Society) was an Indian independence secret society founded by Vinayak Damodar Savarkar and his brother Ganesh Damodar Savarkar in 1904. It was initially started as ***Mitra Mela*** in Nasik in 1899 before expanding and changing its name. The society aimed to overthrow British rule through armed rebellion and established international connections with the India House group in London, where Savarkar later lived.

- **India House**(founded by **Shyamji Krishna Varma** in 1905) in London became a center of Indian revolutionary activity,Savarkar stayed there and connected Indian revolutionaries globally. It helped spread anti-British ideas internationally.
- **Anushilan Samiti,** established around 1902–1906 in Calcutta, founded by Pramathanath Mitra, strongly influenced by the nationalist movement after the Partition of Bengal 1905;

- **Ghadar Party,** founded in **1913,** Place: **San Francisco (USA),** Headquarters: **Yugantar Ashram;**
- **Jugantar Party, established** around 1906 in Calcutta, Originated from the revolutionary activities linked with Anushilan Samiti, Name derived from the newspaper Jugantar.

BCW Bits:

- Savarkar wrote **"The Indian War of Independence, 1857"**, which inspired revolutionaries
- Abhinav Bharat played a role in spreading **militant nationalism**, especially among youth

42. With reference to the White Mutiny, consider the following statements: ***[BPSC AEDO]***

1. It posed a serious threat to British control in India, similar to the Revolt of 1857.
2. It was primarily caused by the grievances of Indian sepoys who felt mistreated by British officers.

Which of the statements given above is/are correct?

A. 1 only B. 2 only
C. Neither 1 nor 2 D. Both 1 and 2

Ans. **C**

Exp:

- The "White Mutiny" (1859–60) was a protest by European (white) soldiers of the East India Company.After the Revolt of 1857, the British government took over India (1858).Company troops were to be transferred to the Crown's army. Many European soldiers refused transfer without consent, demanded bounty/compensation, objected to changes in service conditions. Thus, it was a protest by European soldiers against British military policies
- The 1857 Revolt—often called India's First War of Independence—was a massive, widespread uprising against the oppressive rule of the British East India Company. The introduction of the Enfield Rifle sparked the movement. On May 10, 1857, sepoys in Meerut rebelled, marched to Delhi, and declared the last Mughal Emperor, Bahadur Shah Zafar, as their leader.

BCW Bits:

- **The White Mutiny is also called the "European Mutiny"**
- It led to administrative tightening after 1857, especially over the army.
- The British realised the risk of discontent even within European ranks.
- Post-1857 reforms ensured greater Crown control over the military.
- The episode shows how pay, contracts, and service terms can trigger unrest even without political motives.
- It did not involve civilian participation or princely states.

43. The British East India Company, initially a trading entity, gradually established its control over Indian territories through various means. This policy was implemented by Lord Dalhousie, leading to widespread resentment among Indian rulers. ***[BPSC AEDO]***

Assertion (A): The Doctrine of Lapse was implemented to expand British territories.

Reason (R): It allowed the British to annex states with no direct male heir.

Choose the correct option:

A. Both (A) and (R) are true and (R) is the correct explanation of (A)
B. Both (A) and (R) are true but (R) is not the correct explanation of (A)
C. (A) is false but (R) is true
D. (A) is true but (R) is false

Ans. **A**

Exp: The British East India Company started as a trading body (1600) but gradually took political control by:

- Winning Battle of Plassey (1757) and Battle of Buxar (1764)
- Using treaties and alliances (Subsidiary Alliance),
- Annexing territories.

By the mid-19th century, it was effectively ruling large parts of India.

- **Doctrine of Lapse-** Introduced and enforced by Dalhousie, it had the provision that if a ruler died without a natural (biological) male heir, his kingdom would "lapse" to the Company. Adopted heirs were not recognised.
  - It was aimed to expand British territory without war.
  - States annexed under this policy: Satara (1848), Jhansi (1853), Nagpur (1854), Sambalpur (1849), Baghat (1850), Udaipur (1852).
  - This deeply angered Indian rulers, especially leaders like Rani Lakshmibai.
  - This policy violated Indian traditions of adoption rights, created distrust among princely states. It became one of the major cause behind the Revolt of 1857.

BCW Bits:

- Lord Dalhousie won the **Second Anglo-Sikh War** (1849) to annex the entirety of Punjab and conquered lower Burma in the **Second Burmese War** (1852).
- Father of Indian Railways: In 1853, he launched the **first passenger railway** line in India, running between Bombay (Mumbai) and Thane.
- He introduced the **electric telegraph network** and reformed the postal system, introducing uniform postage rates with the Post Office Act of 1854.
- He implemented **Charles Wood's Dispatch** in 1854, which established universities in Calcutta, Bombay, and Madras.

- He supported the **Hindu Widows' Remarriage Act** of 1856, which legalized the remarriage of Hindu widows.
- He founded the **Public Works Department (PWD)**, overseeing the completion of massive projects like the **Ganga Canal** and the **Grand Trunk Road**.
- The popular Himachal Pradesh **hill station of Dalhousie** is named in his honor.

44. Consider the following statements: ***[BPSC AEDO]***

Statement I: The Poligars' Revolt was one of the earliest uprisings against British rule in southern India, driven by the imposition of heavy land revenue demands on the Poligars.

Statement II: The Poligars were traditionally responsible for collecting land revenue in the regions they controlled and the British sought to undermine this authority by enforcing direct revenue collection.

Which one of the following is correct in respect of the above statements?

A. Both Statement I and Statement II are correct, and Statement II is the correct explanation for Statement I
B. Both Statement I and Statement II are correct, and Statement II is not the correct explanation for Statement I
C. Statement I is incorrect, but Statement II is correct
D. Statement I is correct, but Statement II is incorrect

Ans. A

Exp:

- The Poligars (Palaiyakkarars) were local chiefs in parts of Tamil Nadu (especially Tirunelveli–Madurai region). They controlled small territories, collected land revenue, and maintained local military forces.
- The **Poligar Revolt** (or Palaiyakkarar Wars) was a series of armed uprisings between 1799 and 1805 in southern India (primarily in modern-day Tamil Nadu). It was fought by local feudal chieftains (Poligars) against the British East India Company, triggered by oppressive taxation policies and British attempts to dismantle traditional regional autonomy.
- British East India Company removed Poligars' authority by trying to collect revenue directly. Disarmament orders and loss of autonomy added to resentment. These revolts are among the earliest anti-British resistances in South India.
- Key figures: Veerapandiya Kattabomman, Marudu Brothers.

BCW Bits:

- The conflicts are often called the "Polygar Wars" (First & Second)
- The British used military campaigns and fort demolitions to suppress them.
- These uprisings preceded larger rebellions like Vellore (1806) and 1857
- Highlighted early resistance to centralised colonial revenue systems.
- Many forts (e.g., Panchalankurichi) became symbols of resistance.

45. In 1917, Mahatma Gandhi led a movement which marked his first major involvement in the Indian independence movement. This movement was aimed at addressing the grievances of indigo planters suffering under oppressive policies. ***[BPSC AEDO]***

Assertion (A): The Champaran Satyagraha was Gandhi's first major mass movement in India.

Reason (R): The movement aimed at resolving the issues faced by the indigo planters.

Choose the correct option:

A. (A) is true but (R) is false
B. Both (A) and (R) are true but (R) is not the correct explanation of (A)
C. (A) is false but (R) is true
D. Both (A) and (R) are true and (R) is the correct explanation of (A)

Ans. D

Exp:

- Mahatma Gandhi returned from South Africa in 1915. He chose local agrarian issues to connect with common people and challenge British policies through non-violent protest.
- Champaran Satyagraha (1917): -Took place in Champaran (Bihar), marked Mahatma Gandhi's first major mass movement and first successful satyagraha in India.
- Issue: peasants were forced to cultivate indigo under the oppressive Tinkathia system by British planters.
- Indigo Plantation Problem (core issue):- Farmers had to grow indigo on part of their land. They got low returns but faced high pressure. Couldn't freely grow food crops. This created deep resentment among peasants

BCW Bits:

- Gandhi was invited by Raj Kumar Shukla, a local farmer.
- The movement demonstrated Gandhi's method of fact-finding before agitation.
- It helped Gandhi gain the confidence of the rural masses and marked the beginning of Gandhian mass politics in India.

46. Consider the following statements regarding the Satyashodhak Samaj: ***[BPSC AEDO]***

1. It was primarily aimed at challenging the Brahminical dominance and promoting equality for lower castes.
2. It worked towards universal education for women and lower castes.

3. The Samaj opposed the idea of untouchability as a necessary social custom.

How many of the statements given above is/are correct?

A. Only one B. Only two
C. None of these D. All three

Ans. B

Exp:

- The Satyashodhak Samaj (Society of Truth Seekers) was founded in **1873** by **Jyotirao Govindrao Phule** in Pune (Maharashtra). The organization emerged during a period when Indian society was deeply divided on the basis of caste, and social privileges were concentrated in the hands of upper castes, particularly Brahmins.
- Phule believed that the lower castes, peasants, laborers, and women had suffered centuries of exploitation and social discrimination. Therefore, the main objective of the Samaj was to establish a society based on equality, social justice, and human dignity.
- One of the central aims of the Satyashodhak Samaj was to challenge the monopoly of Brahmins over religion, education, and social status.
- The organization encouraged *Shudras* and *Atishudras* (*Dalits*) to become aware of their rights and reject social oppression. Thus, opposition to Brahminical dominance and promotion of social equality formed the very foundation of the movement.
- Along with his wife, Savitribai Phule, he established schools for girls and children from lower castes at a time when such efforts faced severe resistance from orthodox society.
- The Satyashodhak Samaj promoted education as a tool of empowerment and advocated educational opportunities for women, Dalits, and other marginalized groups.
- The Samaj never supported untouchability; rather, it fought against it. Untouchability was viewed as an unjust and inhuman practice that deprived large sections of society of basic dignity. Phule criticized caste discrimination and worked for social integration.

BCW Bits:

- Phule's famous work "***Gulamgiri***" criticized caste oppression and social inequality.
- Savitribai Phule is regarded as one of the first female teachers of modern India.
- Satyashodhak Samaj emphasized social equality over religious ritualism. The Samaj conducted marriage ceremonies without Brahmin priests; these were known as Satyashodhak marriages.
- Phule described lower castes as the original inhabitants of India and challenged the Aryan superiority theory.
- The movement significantly influenced later non-Brahmin and anti-caste movements in western India.
- Phule established the ***Balhatya Pratibandhak Griha*** for widows and abandoned children.
- Phule is often called the Father of Social Revolution in Maharashtra.
- Jyotiba Phule – Satyashodhak Samaj (1873),
- Dayanand Saraswati – Arya Samaj (1875),
- Raja Rammohan Roy – Brahmo Samaj (1828).

47. Given below are two statements, one is labelled as Assertion (A) and other is labelled as Reason (R). In light of the below given statements, choose the most appropriate answer from the options. ***[BPSC AEDO]***

Assertion (A): Since 2021, November 15th is celebrated as Janjatiya Gaurav Diwas to honour the contribution of tribal communities in the Indian Freedom struggle.

Reason (R): It is celebrated to mark the birth anniversary of Birsa Munda, one of the major tribal freedom fighters.

A. Both (A) and (R) are true, and (R) is the correct explanation of (A)
B. Both (A) and (R) are true, but (R) is not the correct explanation of (A)
C. (A) is false, (R) is true
D. (A) is true, (R) is false

Ans. A

Exp: The Government of India officially declared November 15th as ***Janjatiya Gaurav Diwas*** in 2021. This specific date was chosen because it marks the birth anniversary of the revered tribal freedom fighter Birsa Munda (born in 1875), who led the historic Ulgulan (Munda Rebellion) against British colonial rule and is celebrated for his immense contributions to India's independence movement.

BCW Bits:

- Birsa Munda is called "Dharti Aba" (Father of Earth)
- Tribal resistance started much before 1857
- **Tilka Manjhi,** One of the earliest tribal rebels (18th century) fought against British oppression in Bihar region.
- Major tribal uprisings included the Santhal rebellion, Kol uprising and Munda rebellion .

48. With reference to the French East India Company's activities in India, consider the following statements: ***[BPSC AEDO]***

1. The *Compagnie des Indes Orientales* was founded by Louis XIV in 1664 and was granted a 50-year monopoly on French trade in the Indian and Pacific Oceans.
2. The company established Pondicherry in 1674 as the central hub of French power in India.
3. The Treaty of Ryswick in 1697 allowed the French to retain their territories in India without any interference from the Dutch.

4. The French East India Company successfully established permanent colonies in Madagascar during its early years.

How many of the statements given above is/are correct?

A. Only one B. Only two
C. All four D. Only three

Ans. D

Exp: The ***Compagnie des Indes Orientales***(French East India Company) was established on September 1, 1664, by **King Louis XIV** under the active planning of his finance minister, **Jean-Baptiste Colbert**. The Crown granted it a 50-year trade monopoly in both the Indian and Pacific Oceans to actively compete against the commercial hegemony of the British and the Dutch.

- In 1674, François Martin acquired a site from the Governor of Valikondapuram (under the Sultan of Bijapur). This critical settlement grew into **Pondicherry** (modern Puducherry), becoming the ultimate administrative capital and central hub of French influence across India.
- During the Nine Years' War, the Dutch captured Pondicherry from the French in 1693. The conclusion of the conflict via the **Treaty of Ryswick** in September 1697 forced the Dutch to entirely restore the captured territory back to French control, letting them operate their commercial centers free of Dutch occupation.
- The French repeatedly tried to build a permanent colony at **Fort Dauphin in Madagascar** to serve as a strategic base for Indian ocean maritime trade. However, due to severe outbreaks of tropical disease, intense local tribal hostility, and administrative mismanagement, the settlement was a major failure and had to be completely abandoned by 1674.

BCW Bits:

- Dupleix was a key French governor in India,
- Pondicherry remained under French control until **1954 (de facto transfer).**
- French influence is still visible in architecture and culture.
- French also had settlements in Chandernagore, Mahe, Karaikal.

49. Arrange the following types of industries and mills installed in India for the first time during the British era in chronological order. ***[BPSC AEDO]***

1. First Cotton Mill
2. First Jute Mill
3. First Paper Mill
4. First Steel Plant

Select the correct answer using the code given below:

A. 1 – 2 – 3 – 4 B. 1 – 3 – 2 – 4
C. 2 – 3 – 1 – 4 D. 2 – 1 – 4 – 3

Ans. A

Exp:

- India's first commercially successful **cotton mill**, the Bombay Spinning and Weaving Company, was established in **1854** in Tardeo, **Bombay (Mumbai).** Founded by Parsi entrepreneur Cowasjee Nanabhoy Davar, it marked the birth of India's modern textile industry. Its operations began on February 7, 1856.
- First **jute mill** was established near **Kolkata (Rishra)** in **1855.** The jute industry grew rapidly because Raw jute was easily available in Bengal. The British needed jute bags for trade.
- The modern **paper mill** was established later, around **1870.** Paper demand increased because of Administration ,Education,Printing press expansion.
- **First steel plant**, Tata Iron and Steel Company Tata Steel plant established at Jamshedpur, Year- **1907.** This was a major milestone in Indian industrialization.

50. Kuka movement was organized by: ***[BPSC AEDO]***

A. Guru Gobind Singh
B. Guru Nanak
C. Guru Ram Singh
D. Guru Ram Das

Ans. C

Exp: The **Kuka Movement** was organized by **Guru Ram Singh**. The Kuka Movement, also known as the Namdhari Movement, began in Punjab during the 19th century. It started as a religious reform movement among Sikhs but gradually developed political and anti-British characteristics.

- Guru Ram Singh aimed to reform Sikh society by opposing caste discrimination, intoxication, and extravagant rituals while also promoting simplicity, discipline, and resistance to British rule.
- Followers wore simple white clothes and believed in strict moral conduct. Over time, the British started seeing the movement as a threat because it encouraged resistance and independent social organization.
- The movement became more aggressive after cow slaughter issues and growing resentment against British policies. In 1872, the British brutally suppressed the Kukas, and many followers were executed.

BCW Bits:

- **Guru Gobind Singh** founded the Khalsa in 1699 and militarized the Sikh community. **Guru Nanak**, was the founder of Sikhism and lived much earlier during the 15th–16th century. **Guru Ram Das** is mainly known for founding Amritsar.
- The **Kuka Movemen**t is one of the early organized movements against British authority before the rise of large-scale nationalist politics.

51. The Pagalpanthi rebellion was indeed a rebellion of: ***[BPSC AEDO]***

A. Koliya B. Garos

C. Gonds D. Bhils

Ans. B

Exp:

- **The Pagalpanthi** rebellion was associated mainly with the Garo people tribe.The movement began in the early 19th century in parts of present-day northern Bangladesh and Meghalaya under **Karam Shah** and later under **Tipu Shah**. The movement had both religious and political characteristics. Over time, it turned into a rebellion against oppressive taxation, exploitation by zamindars, and British interference in tribal life.
- **The Garos and Hajongs** formed an important support base of this movement. The movement opposed oppressive British revenue policies and reflected tribal resistance against economic exploitation and loss of traditional autonomy.

BCW Bits:

- **Kolis** were associated with western India.
- The **Gonds** were linked with central Indian tribal regions and had separate resistance movements against British expansion.
- The **Bhils** were mainly active in western and central India.

52. 'The Swadesh Vahini' was edited by: ***[BPSC AEDO]***

A. Motilal Nehru

B. C.N. Mudaliar

C. K. Ramakrishna Pillai

D. C.V. Raman Pillai

Ans. C

Exp:

- K. Ramakrishna Pillai was the editor of *Swadesh Vahini*. He was a well-known journalist, writer, and nationalist from Kerala. He became famous for his fearless criticism of the Travancore administration and British policies. Through his writings, he tried to spread political awareness and nationalist ideas among people.
- During the Indian freedom struggle, newspapers and journals were vital tools for mobilizing public opinion, spreading nationalist ideologies, and uniting the masses against colonial rule.
  - **Young India, Harijan, Indian Opinion, Navjivan**: Mahatma Gandhi
  - **Kesari, Mahratta**: Bal Gangadhar Tilak
  - **Bande Mataram**: Bipin Chandra Pal (later edited by Aurobindo Ghosh)
  - **Amrita Bazar Patrika**: Sisir Kumar Ghosh and Motilal Ghosh.
  - **The Hindu**: G. Subramania Iyer
  - **The Bengalee**: Edited by Surendranath Banerjee
  - **Hindoo Patriot**: Started by Harish Chandra Mukherjee
  - **Sambad Kaumudi**: Raja Ram Mohan Roy
  - **Free Hindustan:** Taraknath Das
  - **Al-Hilal:** Founded by Maulana Abul Kalam Azad
  - **The Bombay Chronicle:** Founded by Pherozshah Mehta
  - **The Tribune:** Started by Sardar Dayal Singh Majithia in Lahore.

BCW Bits:

The major legislative measures used to curb the press include:

- Censorship of Press Act, 1799: Enacted by Lord Wellesley
- Licensing Regulations, 1823
- Licensing Act, 1857
- Vernacular Press Act, 1878: Also known as the "Gagging Act," it was introduced by Lord Lytton. Later repealed by Lord Ripon in 1881.
- Newspaper (Incitement To Offences) Act, 1908
- Indian Press Act, 1910
- Indian Press (Emergency Powers) Act, 1931

53. In which year was the 'Native Marriage Act' passed? ***[BPSC AEDO]***

A. 1876 B. 1872

C. 1874 D. 1870

Ans. B

Exp:

- The Native Marriage Act, also popularly known as the Civil Marriage Act, was passed in the year **1872.**This landmark legislation was enacted primarily due to the persistent efforts of the renowned social reformer **Keshab Chandra Sen** and the **Brahmo Samaj.**
- The act was passed during British rule to provide a legal framework for civil marriages outside traditional religious customs. It was closely associated with reformist ideas promoted by organizations like the Brahmo Samaj. The law allowed inter-caste and inter-community marriages under certain conditions.
- One important feature of the Act was that people marrying under it had to renounce their respective religions, which made the law controversial and limited its acceptance at that time. Even then, it was considered an important step toward social reform and individual freedom in marriage.
- The Act reflected the growing influence of reform movements in 19th-century India. Reformers were questioning rigid caste practices, child marriage, restrictions on widow remarriage, and social inequality. The British government, under pressure from reformers, passed several laws related to social reform during this period.

BCW Bits:

- Widow Remarriage Act was passed in 1856.
- Raja Rammohan Roy opposed the practice of Sati, which was abolished in 1829 by Lord William Bentinck.
- Ishwar Chandra Vidyasagar strongly supported widow remarriage.
- Hindu Widow's Remarriage Act(also called Act XV of 1856), was introduced and drafted by Lord Dalhousie and passed by Lord Canning.
- Age of Consent Act was passed in 1891.
- Child Marriage Restraint Act was passed in 1929.

54. Where was the Ghadar Party established? ***[BPSC AEDO]***

A. England B. USA C. France D. Germany

Ans. B

Exp:

- The Ghadar Party was founded in 1913 at San Francisco (USA), mainly by Indian immigrants, especially Punjabis living abroad. The movement aimed to overthrow British rule in India through revolutionary activities and armed struggle.
- One of the major leaders associated with the party was Lala Har Dayal.
- Other important revolutionaries included Sohan Singh Bhakna and Kartar Singh Sarabha.
- The party published a newspaper called *Ghadar*, which spread revolutionary ideas among Indians living overseas.
- The word "Ghadar" means revolt or rebellion. The organization believed that British rule in India could not be ended only through petitions or constitutional methods and therefore promoted revolutionary action.
- Shyamji Krishna Varma established India House in London, which became a center for Indian revolutionaries.

BCW Bits:

- The Komagata Maru incident influenced revolutionary nationalism among overseas Indians.
- The Ghadar movement gained momentum during World War I.
- Rash Behari Bose later associated with revolutionary activities abroad.
- Many Ghadar revolutionaries were prosecuted in the Lahore Conspiracy Case.

55. When did the Congress Party pass the proposal of independence of India for the first time? ***[BPSC AEDO]***

A. 1935 B. 1915 C. 1942 D. 1929

Ans. D

Exp:

- The Indian National Congress passed the proposal of complete independence, known as "Purna Swaraj," for the first time in 1929. This historic resolution was passed during the Lahore Session of INC held under the presidency of Jawaharlal Nehru. It officially abandoned the demand for "dominion status" and declared ***Purna Swaraj*** (complete independence) as India's ultimate goal.
- Under the presidency of **Jawaharlal Nehru,** the Congress passed the historic Purna Swaraj resolution on December 19, 1929.
- On the midnight of **December 31, 1929**, Nehru hoisted the newly adopted tricolor flag of complete freedom on the banks of the **Ravi River** in Lahore.
- The session declared January 26, 1930, as the first Independence Day, urging Indians to take the pledge of total self-rule.
- The session authorized the Congress Working Committee to launch a Civil Disobedience Movement, cementing a shift toward active mass resistance.
- Congress members in central and provincial legislatures were asked to resign, and it was decided to boycott the Round Table Conference.

BCW Bits:

- The 1931 Karachi Session of INC, presided over by Sardar Vallabhbhai Patelendorsed the Gandhi-Irwin Pact and officially passed the historic Karachi Resolutions on Fundamental Rights and the National Economic Programme.
- The session selected Mahatma Gandhi to solely represent the INC at the Second Round Table Conference in London.

56. In which year the famous Gandhi-Irwin Pact took place? ***[BPSC AEDO]***

A. 1932 B. 1930 C. 1931 D. 1929

Ans. C

Exp:

- The pact was signed between Mahatma Gandhi and Lord Irwin in March 1931. Under this agreement, Gandhi agreed to suspend the Civil Disobedience Movement and participate in the Second Round Table Conference in London. In return, the British government agreed to release many political prisoners, return confiscated properties in certain cases, and allow peaceful picketing and manufacture of salt for personal use in coastal areas.
- The pact marked a temporary compromise between the Congress and the British government. However, many nationalists were unhappy with it because revolutionary prisoners like Bhagat Singh were not given relief under the agreement.

BCW Bits:

- In 1932, the Communal Award was announced by Ramsay MacDonald and later the Poona Pact was signed between Mahatma Gandhi and Dr. B.R. Ambedkar.

- Dandi March started from Sabarmati Ashram.
- The First Round Table Conference was held in 1930.
- Mahatma Gandhi attended only the Second Round Table Conference.
- Dr. B.R. Ambedkar actively participated in all three Round Table Conferences held in London between 1930 and 1932.He attended these conferences as a representative of the Depressed Classes (Dalits).

57. All India Depressed Classes League was established by: ***[BPSC AEDO]***

A. M. Jyotiba Phule B. Babu Jagjivan Ram
C. N.S. Kajrolkar D. Dr. B.R. Ambedkar

Ans. B

Exp: The All-India Depressed Classes League was founded by **Babu Jagjivan Ram** in 1935.It was an organization dedicated to attaining equality for untouchables and advocating for the social, educational, and political rights of the Depressed Classes.

- Babu Jagjivan Ram (1908–1986), popularly known as ***Babuji***, was a towering Indian freedom fighter, social reformer, and statesman from Bihar.
- He holds the record for the **longest-serving Union Cabinet Minister** in Indian history (35 years) and is celebrated as a champion of Dalit rights and social justice.
- He founded the All India Depressed Classes League in 1935 to fight for untouchability eradication and equal rights.
- He appeared before the **Hammond Commission** in Ranchi to demand voting rights for Dalits.
- He actively participated in the Civil Disobedience and Quit India movements, acting as a crucial bridge between Dalit communities and the Indian National Congress.
- As India's **first Labour Minister** in the Interim Government, he enacted landmark legislations like the Minimum Wages Act and the Employees' State Insurance (ESI) scheme.
- He served as **Defence Minister** during the crucial **Indo-Pak War of 1971**, which led to the creation of Bangladesh.
- As **Agriculture Minister**, he played a vital role in steering the **Green Revolution**, making India self-sufficient in food grains, and averting disaster during the 1974 drought.
- He also served as the **Deputy Prime Minister** of India in 1979.

BCW Bits:

- **Jyotiba Phule** was a pioneering social reformer from Maharashtra who worked against caste discrimination and for women's education. He founded the **Satyashodhak Samaj**in 1873 and strongly criticized Brahmanical social hierarchy.
- **B. R. Ambedkar,** founded organizations like the Bahishkrit Hitakarini Sabha (1924), Depressed Classes Association (1930), Independent Labour Party (1935) and.Scheduled Castes Federation (1942). He led **Mahad Satyagraha (1927)** and **Temple Entry Movement (1930).** Also he was key architect of **Poona Pact (1932).**

58. The Pakistan Resolution was drafted by: ***[BPSC AEDO]***

A. Fazlul Haque B. Sikandar Hayat Khan
C. Muhammad Ali Jinnah D. Rahmat Ali

Ans. B

Exp: While the historic document—originally called the Lahore Resolution (Pakistan Resolution)—was presented by the prominent Bengali leader A.K. Fazlul Haque, its primary draft was written by Sir Sikandar Hayat Khan, who served as the Premier of Punjab and played a major role in the committee that designed it.

- The resolution was passed during the Lahore Session of the All India Muslim League in March 1940. It demanded the creation of "independent states" for Muslims in the north-western and eastern Muslim-majority areas of British India.
- Though the word "Pakistan" was not directly used in the resolution itself, it later became famous as the Pakistan Resolution.
- Muhammad Ali Jinnah, was the most prominent leader of the Muslim League and strongly advocated the two-nation theory.

BCW Bits:

- The Lahore Session was held at Minto Park (now Iqbal Park), Lahore.
- The All India Muslim League was founded in 1906.
- Rahmat Ali coined the term **"Pakistan."**
- The two-nation theory became central to Muslim League politics.

59. In which year was the Indian National Army founded? ***[BPSC AEDO]***

A. 1943 B. 1941 C. 1942 D. 1940

Ans. C

Exp:

- The Indian National Army (INA), also known as **Azad Hind Fauj,** was founded in 1942. The INA was initially formed by Mohan Singh with the support of Japanese intelligence officer **Iwaichi Fujiwara,** during World War II. It was mainly created using Indian prisoners of war captured by the Japanese after the fall of Singapore.
- Later, the INA gained massive popularity and energy under the leadership of Subhas Chandra Bose, who reorganized and revived it in 1943. He took command and gave INA a stronger political and military direction.
- Subhas Chandra Bose established the Provisional Government of Free India (Azad Hind Government) in

Singapore in October 1943, and used slogans like "*Jai Hind*", "Give me blood, and I will give you freedom". The INA fought alongside Japanese forces in the Imphal and Kohima campaigns against the British.

- ***Azad Hind Bank***, also known as the Bank of Independence, was established on April 5, 1944, in Rangoon (now Yangon), Burma, by Netaji Subhas Chandra Bose. It served as the treasury for the Provisional Government of Azad Hind and financed the Indian National Army (INA) during the freedom struggle.

BCW Bits:

- **Colonel Shaukat Malik** of the INA hoisted the Indian tricolour flag for the first time on the Indian mainland in Moirang, Manipur, on April 14, 1944.
- "Delhi Chalo" became an important INA slogan.
- Captain Lakshmi Sahgal led the Rani Jhansi Regiment.
- The INA trials were held at Red Fort in Delhi.

60. Who was the Prime Minister of England when India attained independence? ***[BPSC AEDO]***

A. W. Churchill B. Clement Attlee
C. Harold Macmillan D. None of these

Ans. B

Exp:

- When India attained independence on 15 August 1947, Clement Attlee was the Prime Minister of Britain. Clement Attlee belonged to the Labour Party and became Prime Minister after the British general elections of 1945.
- His government played a major role in deciding to transfer power to India. After World War II, Britain was facing severe economic and political difficulties, and there was growing pressure from Indian nationalist movements as well.
- During Attlee's tenure the Cabinet Mission was sent to India in 1946, Lord Mountbatten was appointed the last Viceroy, and finally the Indian Independence Act was passed in 1947.

BCW Bits:

- **Winston Churchill** was one of the famous British Prime Ministers. He led Britain during much of World War II. However, Churchill strongly opposed granting independence to India.
- **Harold Macmillan**, ruled much later during the late 1950s and early 1960s. He is associated more with decolonization in Africa and the "Wind of Change" speech
- The Labour Party was generally more supportive of Indian self-government than the Conservatives.
- The partition of India led to massive migration and communal violence.
- The Constituent Assembly started functioning in 1946.

61. The Indian Independent Act, 1947 came into force on: ***[BPSC AEDO]***

A. August 14, 1947 B. July 10, 1947
C. July 18, 1947 D. July 4, 1947

Ans. C

Exp:

- The Indian Independence Act 1947 received royal assent and came into force on **18 July 1947.** This Act was passed by the British Parliament to end British rule in India and provide for the partition of British India into two independent dominions — India and Pakistan. The Act was based largely on the Mountbatten Plan announced earlier in June 1947.
- The law provided for partition of India, creation of India and Pakistan, end of British suzerainty over princely states, and transfer of power to Indian hands. It also abolished the office of the Secretary of State for India and gave legislative sovereignty to the new dominions.
- June 3, 1947 → Mountbatten Plan,
- July 4, 1947 → Independence Bill introduced,
- July 18, 1947 → Indian Independence Act passed,
- August 14–15, 1947 → Pakistan and India became independent.

BCW Bits:

- Sardar Vallabhbhai Patel played a major role in the integration of princely states.
- The Radcliffe Line became the boundary between India and Pakistan. Sir Cyril Radcliffe headed the Boundary Commission.
- India adopted its Constitution on 26 November 1949.

62. Which one of the following is not correctly matched? ***[BPSC AEDO]***

(Institutions – Year of establishment)
A. Sanskrit College at Benaras – 1791
B. Calcutta Hindu College – 1798
C. Calcutta Madarsa – 1781
D. Fort William College – 1800

Ans. B

Exp:

- Hindu College of Calcutta became one of the most important centers of modern education in colonial India. It played a major role in spreading Western education, liberal thought, and intellectual awakening in Bengal. Later, it developed into what is now known as Presidency University. Personalities associated with the Bengal Renaissance were closely linked with this institution.
- **Sanskrit College** at Benaras was established in **1791** by Jonathan Duncan. The aim was to promote the study of Hindu law, Sanskrit literature, and traditional Indian learning.

- **Calcutta Madrasah** was established in **1781** by Warren Hastings. It was created mainly for the study of Islamic law and Persian-Arabic education so that British administrators could better understand Muslim legal traditions.
- **Fort William College** was established in **1800** by Lord Wellesley. Its purpose was to train East India Company civil servants in Indian languages, administration, and local laws.
  - In the early phase, many British officials supported Oriental education, meaning promotion of Sanskrit, Persian, and Arabic learning. That is why institutions like Calcutta Madarsa and Sanskrit College were established. Later, a major debate emerged over whether Indians should be educated in traditional Indian learning or Western English education. The Anglicists eventually gained dominance, especially after Thomas Macaulay presented the famous Macaulay Minute in 1835.

BCW Bits:
- Hindu College was founded in 1817.
- English Education Act was passed in 1835.
- Fort William College promoted translation of many Indian texts into English.
- Warren Hastings is associated with Oriental learning policies.

63. Who of the following were sent to the court of the Shah of Persia (Iran) as British envoy during Lord Wellesley's period from India? ***[BPSC AEDO]***

(1) John Malcolm
(2) Mehdi Ali Khan
(3) Raja Shiv Prasad

(Use the following code for the selection of correct answer)

A. Only 1 B. Only 2
C. Both 1 and 2 D. 1, 2 and 3

Ans. C

Exp:
- Lord Wellesley served as Governor-General from 1798 to 1805. During this period, Britain was worried that Napoleon might try to attack India through Persia and Afghanistan. To prevent French influence in West Asia, Wellesley strengthened diplomatic relations with the Shah of Persia.For this purpose, envoys were sent from India to the Persian court. Among them**, John Malcolm** and **Mehdi Ali Khan** were important representatives.
- **John Malcolm** was a British officer, diplomat, and historian who played a major role in Anglo-Persian relations. He was sent to Persia to secure political cooperation against possible French influence. His mission was part of Britain's broader strategic policy to protect India.
- **Mehdi Ali Khan** also participated in diplomatic communication with Persia during this period. Such diplomatic missions were important because the British East India Company increasingly acted not just as a trading body but as a political and imperial power managing international relations.

BCW Bits:
- **Lord Wellesley** introduced the Subsidiary Alliance system, which forced Indian rulers to accept British troops.
- Hyderabad was the first state to accept Subsidiary Alliance.
- Tipu Sultan was defeated during Wellesley's period in the Fourth Anglo-Mysore War.
- Anglo-French rivalry strongly influenced early British expansion in India.
- **Raja Shiv Prasad** (1823–1895), widely known by the title ***Sitara-e-Hind*** ("Star of India"), was a prominent 19th-century Indian scholar, linguist, and historian, who played a foundational role in the promotion of Hindi education, literature, and the Nagari script during British rule. To counter the rising popularity of the INC, he proposed creating a pro-British organization like 'United Indian Patriotic Association' and 'Indian Loyal Association'.

64. Under whose Chairmanship an irrigation commission was set up in 1901? ***[BPSC AEDO]***

A. Sir Anthony Macdonnell
B. Sir Colin Scott Moncrieff
C. Sir Andrew Fraser
D. Lord Kitchener

Ans. B

Exp: The Irrigation Commission was set up in **1901,** under the chairmanship of Colin Scott-Moncrieff.
- During the 19th century, India suffered repeated famines due to irregular monsoons, droughts, and exploitative colonial policies. After facing severe famines, the British government realized that irrigation development was necessary both for increasing agricultural production and for securing land revenue.
- The commission under Sir Colin Scott-Moncrieff was established to examine irrigation potential and suggest ways to expand irrigation facilities in India. It emphasized the construction and improvement of canals, water management systems, and irrigation infrastructure.
- This was part of a broader colonial approach toward "protective irrigation" and "productive irrigation."
  - Protective irrigation aimed to reduce famine risk in drought-prone areas.
  - Productive irrigation aimed at increasing agricultural output and government revenue. Many important canal

projects developed during British rule were connected with this larger irrigation policy framework.

BCW Bits:

- The Great Famine of 1876–78 severely affected southern and western India.
- Canal irrigation expanded greatly in Punjab under British rule.
- British irrigation policy had both economic and strategic importance.
- Railways and canals were major infrastructure projects under colonial rule.
- Famine Commissions were also appointed repeatedly during British rule.
- Colonial land revenue policies often increased peasant hardship despite irrigation expansion.

65. On whom was an attempt to murder made at Muzaffarpur in 1908? ***[BPSC AEDO]***

A. William Curzon Wyllie B. Mr. Jackson
C. Mr. Kingford D. Lord Harding

Ans. **C**

Exp: On April 30, 1908, revolutionaries **Khudiram Bose** and **Prafulla Chaki** threw a bomb at a carriage in Muzaffarpur (Bihar), attempting to assassinate **Douglas Kingsford**, the Chief Presidency Magistrate of Calcutta. Kingsford was the intended target due to the harsh and repressive sentences he had previously handed down to Indian nationalists. However, the bomb mistakenly struck a different carriage, killing two British women (Wife and daughter of Pringle Kennedy), while Kingsford survived the attack.

- The two young revolutionaries — Khudiram Bose and Prafulla Chaki, were associated with the revolutionary organization, **Jugantar**, which believed that armed struggle was necessary to overthrow British rule.
- Later, Prafulla Chaki shot himself to avoid arrest when cornered by police, while Khudiram Bose was arrested after walking several miles barefoot.
- Young Khudiram Bose (around 18 years old) was executed on 11 August 1908.

BCW Bits:

- **Sir William Curzon Wyllie** (1848–1909) was a British Indian Army officer and colonial administrator, who was assassinated on July 1, 1909, by Madan Lal Dhingra in London**.**
- **Nashik conspiracy case: Mr. Jackson** (A.M.T. Jackson), the Collector of Nashik, was assassinated on December 21, 1909, by revolutionary Anant Kanhere in Maharashtra. This was connected with revolutionary activities of the Abhinav Bharat organization founded by Vinayak Damodar Savarkar.
- **Lord Harding**, the Viceroy of India on whom a bomb was thrown in Delhi in 1912 during a ceremonial procession after the transfer of capital from Calcutta to Delhi. This is known as the **Delhi Conspiracy Case**.
- The famous **Alipore Bomb Case (1908)** emerged after revolutionary crackdowns following the Muzaffarpur incident.

66. Who was the first President of Indian National Conference held in December 1883? ***[BPSC AEDO]***

A. Surendranath Banerjee
B. Anand Mohan Bose
C. W.C. Banerjee
D. M.G. Ranade

Ans. **None of the Above (Correct Ans. Ramtanu Lahiri)**

Exp: The Indian National Conference was a prominent precursor to the Indian National Congress, founded in 1883 in Kolkata by leaders Surendranath Banerjee and Anand Mohan Bose. It successfully united delegates from across the country to discuss civil service reforms and national policies before ultimately merging with the Congress in 1886.

- The first Indian National Conference, was held in Calcutta at the **Albert Hall** from December 28 to 30, 1883.
- The first National Conference organized by Indian Association in Kolkata between December 28 and 30, 1883, was presided over by **Ramtanu Lahiri**[#]**,** a veteran of Bengal Renaissance.*[#Source: PIB]*
- The Indian National Conference emerged from the activities of the **Indian Association**, which had been founded in 1876 by Surendranath Banerjee and Anand Mohan Bose in Bengal.
- The second session of Indian National Conference was held in 1885, around the same time INC was formed.
- **Ananda Mohan Bose** (1847–1906) was a pioneering Indian academic, lawyer, social reformer, and key leader of the early nationalist movement. He was the first Indian ***Wrangler*** (a top-tier first-class honors math graduate) at Cambridge University in 1874. He presided over the **Madras session** of the Indian National Congress in 1898. He laid the foundation stone for **Federation Hall** in Kolkata to promote Hindu-Muslim unity.
- Sir Surendranath Banerjee (1848–1925), also known as ***Rashtraguru***, was a foundational Indian nationalist, educator, and key figure of the moderate phase of the Indian independence movement. He started editing the '***The Bengalee'*** newspaper in 1879.

BCW Bits:

- In 1883, Surendranath Banerjee became the **first Indian journalist** to be imprisoned after being arrested for contempt of court.
- He served as **INC President twice** (Pune in 1895 and Ahmedabad in 1902).

- He supported the Montagu-Chelmsford Reforms (1919) and left the INC to form the **Indian National Liberation Federation**.
- He passed the prestigious **Indian Civil Service (ICS) examination** in 1869 but was disqualified due to age discrepancies. He cleared it again in 1871, only to be dismissed shortly after over a minor procedural error, an injustice that fueled his political drive.
- He taught English for nearly 37 years and founded **Ripon College**, which is now known as Surendranath College.
- He penned the notable memoir, ***A Nation in Making***.
- He also served as the **Minister for Local Self-Government** in Bengal from 1921 to 1924, and was knighted by the British in 1921.

67. Which Indian freedom fighter said, "We will not achieve any success in our labour if we croak once a year like a frog"? ***[BPSC AEDO]***

A. Bipin Chandra Pal B. Lala Lajpat Rai
C. Aurobindo Ghosh D. Bal Gangadhar Tilak

Ans. D

Exp:

- Tilak believed that merely holding annual meetings and making petitions would never force the British to grant rights to Indians. The phrase "croak once a year like a frog" was a sarcastic attack on the passive politics of the Moderates. According to Tilak, political struggle required continuous public mobilization, mass awakening, boycott movements, and assertive nationalism.
- Moderates like Dadabhai Naoroji, Gopal Krishna Gokhale, and Pherozeshah Mehta, believed in constitutional methods such as petitions, prayers, memorandums, and discussions with British authorities. They had faith in British justice and thought reforms could gradually be achieved through peaceful persuasion.
- Whereas Extremists like **Lal-Bal-Pal (**Lala Lajpat Rai, Bal Gangadhar Tilak, Bipin Chandra Pal) supported direct confrontation, mass mobilization, Swadeshi (promoting indigenous industries), and the boycott of British goods and institutions.

BCW Bits:

- Bal Gangadhar Tilak is famously known as the **"Father of Indian Unrest".** This title was given to him by the British journalist and author Sir Valentine Chirol in his 1910 book ***Indian Unrest,*** due to Tilak's relentless efforts to mobilize the masses and his fierce opposition to British colonial rule.
- Tilak (popularly called **"Lokmanya Tilak**) used newspapers like **Kesari** (Marathi) and **Mahratta** (English) to spread nationalist ideas among ordinary people.
- Tilak wrote **Gita Rahasya** and **The Arctic Home in the Vedas**
- Tilak organized **Ganapati Festival** and **Shivaji Festival**, for mass political awakening.
- **Lala Lajpat Rai** was known as the "Punjab Kesari" and played a major role protests against the Simon Commission. He was injured during a police lathi charge in 1928 and later died from those injuries.
- Lala Lajpat Rai founded the **Servants of the People Society** (*Lok Sevak Mandal*) in November 1921 in Lahore, with Mahatma Gandhi inaugurating the society.
- Moderates are often called the "**Political Mendicants**" by extremist critics because of their petition-based politics.
- Aurobindo Ghosh was associated with the revolutionary newspaper **Bande Mataram**.
- Bipin Chandra Pal is known as the "Father of Revolutionary Thoughts in India."

68. Which Viceroy's period was called for "Missions, Omissions and Commissions"? ***[BPSC AEDO]***

A. Lytton B. Curzon
C. Irwin D. Rippon

Ans. B

Exp: Lord Curzon's seven-year tenure in India (1899–1905) was famously characterized as a period of **"Missions, Omissions, and Commissions"** by the prominent Indian nationalist leader Gopal Krishna Gokhale.

- **Missions:** He dispatched aggressive diplomatic and military missions abroad, most notably the Younghusband expedition to Tibet (1903–1904).
- **Omissions:**His policies deliberately ignored or omitted the political aspirations and rights of the Indian people, notably culminating in the highly unpopular Partition of Bengal (1905).
- **Commissions:** He set up an unprecedented number of official committees and commissions to overhaul administration, including the Universities Commission (1902), the Police Commission (1902), and the Irrigation Commission (1901).

BCW Bits:

- Introduced Calcutta Corporation Act (1899) - Reduced the number of elected Indian members in the municipal corporation.
- Official Secrets Act (1904) - Curtailed the freedom of the press and gave the government power to censor and punish nationalist publications deemed seditious.
- Passed the Indian Universities Act of 1904, which brought universities under strict government control.
- Established the **Police Commission (1902)**, creating a centralized police training system, raising officer salaries, and introducing the Provincial Police Service.
- Established the **Agricultural Research Institute at Pusa (Bihar)**.

- Enacted the **Ancient Monuments Preservation Act (1904)** to protect India's historical sites and officially created the **Archaeological Survey of India (ASI).**
- Made the British pound legal tender in India and fixed its exchange rate.
- Created the **North-West Frontier Province (NWFP)** to better manage borders and check Russian expansion in Central Asia.

69. Consider the following statements regarding the Lucknow Pact 1916: ***[BPSC AEDO]***

1. This was marked for re-union of Moderate and Extremists parties
2. B. G. Tilak and Annie Besant dominated the Lucknow session
3. The Congress and League pact was done for United Scheme of Constitutional Reforms
4. Feroz shah Mehta participated in this session

Select the correct statements from the above:

A. 1 and 2 B. 2, 3 and 4
C. 1, 2 and 3 D. All the above

Ans. C

Exp: During the **Lucknow Session of 1916** the Moderates and Extremists wing within the Indian National Congress reunited. Also the historic agreement between the Congress and the Muslim League, known as the **Lucknow Pact** came into force.

- After the **Surat Split of 1907**, the Congress got divided into two groups: **Moderates** (led by leaders like Gopal Krishna Gokhale and Pherozeshah Mehta) and **Extremists (**led by Bal Gangadhar Tilak, Bipin Chandra Pal, and Lala Lajpat Rai). At the Lucknow Session, Moderates and Extremists reunited within Congress. This reunion was the outcome of the efforts of Bal Gangadhar Tilak and Annie Besant. The Lucknow Session was strongly influenced by Tilak and Mrs. Besant.
- The **Lucknow Pact** represented an agreement between the **Indian National Congress** and the **All India Muslim League** regarding constitutional reforms and political representation. Both organizations jointly demanded greater self-government and constitutional reforms from the British government.
- Under the pact, Congress accepted the Muslim League's demand for **separate electorates** for Muslims. In return, the Muslim League agreed to cooperate with Congress in demanding constitutional reforms. This was the first major instance of Hindu-Muslim political cooperation at the all-India level.
- Ferozeshah Mehta did not attend the historic Lucknow session of the INC in 1916, as he passed away in November 1915, prior to the session. Also Gopal Krishna Gokhale passed away earlier that year.

BCW Bits:

- The Lucknow Session of INC (**1916),**was presided by Ambica Charan Mazumdar.
- Separate electorates for Muslims were first introduced through the **Morley-Minto Reforms (1909)**.
- Sarojini Naidu famously described Muhammad Ali Jinnah as the **"Ambassador of Hindu-Muslim Unity"**, for his role in bridging the gap between the INC and the Muslim League.
- The August Declaration of 1917 was announced by Edwin Montagu.

70. In which month of 1905 the 'Partition of Bengal' was declared? ***[BPSC AEDO]***

A. July B. August
C. September D. October

Ans. A

Exp: The Partition of Bengal, implemented by Viceroy Lord Curzon on October 16, 1905, divided the Bengal Presidency into two provinces:

- a Hindu-majority West Bengal (Included Bengal, Bihar, and Odisha. Calcutta remained the capital.)
- a Muslim-majority Eastern Bengal and Assam (Dhaka as the capital.)

Though officially justified for administrative efficiency, it was a "divide and rule" strategy to curb the rising Indian nationalist movement.

Announcement Date: July 19, 1905
Effective Date: October 16, 1905
Annulment Date: December 1911

- **Official declared reason:**To reduce the immense administrative burden of the massive Bengal Presidency.
- **Real (Political):** Bengal was the epicenter of Indian nationalism. The British sought to divide Bengalis linguistically (reducing them to a minority in their own province) and divide Hindus and Muslims to weaken the freedom struggle.

BCW Bits:

- October 16, 1905, was observed as a day of mourning.
- Rabindranath Tagore composed the song "Amar Shonar Bangla" and suggested the tying of rakhi to symbolize unity. The slogan "***Vande Mataram"*** became a powerful nationalist cry.
- Indians actively boycotted British goods, schools, and institutions while promoting indigenous (Swadeshi) products and education.
- National Council of Education was established during the Swadeshi Movement to promote national education.
- The movement also gave a major boost to revolutionary organizations like **Anushilan Samiti** and **Jugantar**.

- The intensity of the movement caused a rift in the Indian National Congress, leading to a split between Moderates and Extremists at the 1907 Surat Session.
- Due to intense, ongoing political protests and the success of the Swadeshi and boycott movements, the British government annulled the partition in 1911 under Viceroy Lord Hardinge. At the same time, the capital of British India was shifted from Calcutta (modern Kolkata) to Delhi to distance the government from Bengali political intellectuals.

71. In which language the first issue of 'Gadar' magazine was published? ***[BPSC AEDO]***

A. Gurmukhi B. Hindi C. Urdu D. English

Ans. **C**

Exp: The first issue of the Gadar (or *Hindustan Ghadar*) weekly newspaper was originally published in the **Urdu** language on November 1, 1913.Shortly after, a Gurmukhi (Punjabi) edition was released on December 9, 1913, and it was eventually published in other languages as well.

- The Ghadar Movement emerged mainly among Indian immigrants, especially Punjabis, living in the United States and Canada. The movement formally began with the establishment of the **Ghadar Party in 1913** at San Francisco (United States).
- Originally formed as the ***Hindi Association of the Pacific Coast*** in Portland, Oregon. It was renamed the Ghadar Party in July 1913, with headquarters established at **Yugantar Ashram** in San Francisco.
- To spread revolutionary ideas, the party started publishing the newspaper **'Gadar'** on **1 November 1913.**
- The word **"Gadar"** itself means **revolt**, **rebellion**, or **mutiny**. The title was deliberately chosen to remind Indians of the Revolt of 1857, which revolutionaries considered the first major war against British rule.
- The Ghadar Movement became active during the **First World War (1914–1918)**.
- Important revolutionaries - Kartar Singh Sarabha, Rash Behari Bose and othes..

BCW Bits:

- **Baba Sohan Singh Bhakna**: Founder and first president.
- **Lala Har Dayal:** Prominent intellectual, general secretary, and main propagandist.
- Other Leaders: **Kartar Singh Sarabha**, **Bhai Parmanand**, and **Abdul Hafiz Mohamed Barakatullah**.
- Ghadar revolutionaries attempted military uprisings in India during World War I.
- Rash Behari Bose later played an important role in revolutionary activities in Japan and the formation of the Indian Independence League.
- The British suppressed the movement through the **Lahore Conspiracy Cases**.

72. Who hired the 'Camagata Maru' ship to go to Vancouver? ***[BPSC AEDO]***

A. Gurdit Singh B. Kartar Singh Sarabha
C. Sohan Singh D. Balwant Singh

Ans. **A**

Exp:

- The 'Komagata Maru / Camagata Maru ' ship was chartered (hired) in 1914 by **Gurdit Singh Sandhu**, a wealthy Sikh businessman from Singapore. He hired the Japanese steamship to transport **376 passengers**—primarily Sikhs, along with some Muslims and Hindus—from **Hong Kong to Vancouver**(Canada).
- The primary objective of this voyage was to challenge Canada's discriminatory **'Continuous Journey Regulation (1908)'**, an immigration law designed to block Asians from entering the country. Upon reaching Vancouver on May 23, 1914, the Canadian authorities refused to let most of the passengers disembark, forcing the ship to eventually return to India.
- When the Ship returned near Calcutta at **Budge Budge** on September 27, 1914, British authorities tried to arrest and control the passengers because they suspected revolutionary connections with the Ghadar Party.
- A violent clash took place between passengers and police, resulting in deaths and arrests. This further intensified anti-British feelings.

BCW Bits:

- According to the Continuous Journey Regulation, immigrants could enter Canada only if they arrived directly from their country of origin without any stop in between.
- The Komagata Maru ship was renamed ***Guru Nanak Jahaz***.
- The Shore Committee: The local South Asian community united to raise funds and provide legal aid, but the British Columbia Court of Appeal ultimately ruled against the passengers.

73. What was the age of Mahatma Gandhi when he returned to India from South Africa? ***[BPSC AEDO]***

A. 42 B. 43 C. 44 D. 45

Ans. **D**

Exp: Mahatma Gandhi was born in Porbandar, Gujarat, on **October 2, 1869.** He travelled to South Africa in **1893** to work as a lawyer, where he lived for 21 years and developed the Satyagraha philosophy. He returned to India from South Africa on **January 9, 1915** to join the freedom struggle. Mahatma Gandhi was **45 years** old when he permanently returned to India in 1915 (exactly 45 years, 3 months, and 7 days old).

- Mahatma Gandhi went to South Africa in 1893 on a one-year contract to serve as a legal adviser for **Dada Abdullah Jhaveri,** a wealthy Gujarati merchant. Expected to stay only briefly, he ended up remaining for 21 transformative years after encountering severe racial discrimination.
- Once while travelling from Durban to Pretoria for the lawsuit, Gandhi was thrown off a train in **Pietermaritzburg** at the demand of White passengers, despite holding a valid first-class ticket. This humiliation that awakened his social and political consciousness.
- During his stay in South Africa from **1893 to 1915**, Gandhi developed the concept of **Satyagraha**, which means "truth-force" or "soul-force." He established organizations such as the **Natal Indian Congress** and launched several non-violent campaigns.
- When Gandhi returned to India, Gopal Krishna Gokhale advised him to travel across India and understand Indian society before entering active politics.

BCW Bits:

- ***Pravasi Bharatiya Divas*** is celebrated in India on January 9 to commemorate the return of Mahatma Gandhi from South Africa to Mumbai on January 9, 1915.

74. Who of the following died on the day of commencement of Non-cooperation Movement? ***[BPSC AEDO]***

A. Madan Mohan Malviya
B. Aurobindo Ghosh
C. Bal Gangadhar Tilak
D. Annie Besant

Ans. C

Exp: Bal Gangadhar Tilak passed away on **August 1, 1920**, which was the exact same day Mahatma Gandhi formally launched the Non-Cooperation Movement across India. Tilak's passing marked the end of the Extremist Era and the transition into the Gandhian Era in the Indian independence struggle.

- Tilak's famous slogan, *"Swaraj is my birthright and I shall have it,"* became a rallying cry for Indian nationalism.
- He passed away in the early hours of August 1, 1920, following a brief illness, while staying at the **Sardar Griha guest house** near Crawford Market in Bombay.
- His funeral procession was attended by an estimated 200,000 people, marking it as one of the largest turnouts in Indian history at the time.
- Maulana Shaukat Ali was one of the prominent Muslim leaders who shouldered the bier during the final journey of Lokmanya Tilak. This moment served as a powerful, enduring symbol of the Hindu-Muslim unity movement that defined the era.
- Due to the massive crowds, he was not cremated at a traditional crematorium. Instead, the final rites took place at Chowpatty Beach.
- Tilak was cremated in a seated, cross-legged position ***(Padmasana)***—a rare honor usually reserved only for saints.
- Prominent Attendees: Leaders such as Mahatma Gandhi and Jawaharlal Nehru were present to join his funeral procession and pay their final respects.
- Following his death, Mahatma Gandhi announced the Tilak Swaraj Fund in 1921. It was launched exactly on his first death anniversary with the goal of raising ₹1 crore to support the Non-cooperation Movement and resist British rule.

BCW Bits:

- The Non-Cooperation Movement was launched on **1 August 1920**, by Gandhi in response to multiple issues that had deeply angered Indians, such as the **Rowlatt Act (1919)**, **Jallianwala Bagh Massacre (1919)** at Amritsar, and the **Khilafat Movement**.
- The movement was officially approved at the **Nagpur Session of Congress (1920)**.
- Gandhi returned the title **"Kaiser-i-Hind"** during the Non-Cooperation Movement.
- The movement was withdrawn after the **Chauri Chaura Incident (1922)**.
- Annie Besant was the first woman President of the Indian National Congress in **1917**.
- The Khilafat leaders included Mohammad Ali Jauhar and Shaukat Ali.

75. Which date is observed as 'Direct Action Day' by the Muslim League in 1946? ***[BPSC AEDO]***

A. 14 August
B. 15 August
C. 16 August
D. 18 August

Ans. C

Exp: The All-India Muslim League, under the leadership of Muhammad Ali Jinnah, observed **16 August 1946** as **'Direct Action Day'.** Following the failure of the Cabinet Mission Plan to reach a consensus for a unified India, Jinnah rejected the proposal and called for a nationwide hartal (general strike) to press their demand for a separate Muslim homeland, Pakistan. This day triggered widespread communal violence, particularly in Calcutta (now Kolkata), earning it the historical moniker **"The Great Calcutta Killings"**.

- By the mid-1940s, there was a political deadlock between INC and the Muslim League, on the demand of a separate nation based on the Two-Nation Theory.
- In 1946, the British Cabinet Mission Plan proposed a **united federal India**, but it collapsed due to

disagreements with the Indian National Congress over provincial grouping.

- As the negotiations remained unfruitful, the League declared 16 August 1946 as Direct Action Day. Intended as a political demonstration, it triggered catastrophic communal violence known as the Great Calcutta Killings. Thousands died, and riots spread to **Noakhali** (Bengal), Bihar, and Punjab, destroying Hindu-Muslim trust.
- The violence proved that coexistence within a single framework was impossible, which later paved the path for the Partition of India and Pakistan in August 1947.

BCW Bits:

- The Cabinet Mission arrived in Delhi on **March 24, 1946.** Its members were Pethick-Lawrence, Stafford Cripps, and A. V. Alexander.
- Originally known as the **Lahore Resolution**, (also referred to as the **Pakistan Resolution)** was adopted by the All-India Muslim League at its three-day general session in Lahore between March 22 and March 24, 1940.

76. Consider the following statements: ***[BPSC AEDO]***

Statement I: The Zamindari Association, also known as Landholder's Society, was established in 1838 to protect the interests of Zamindars in colonial India.

Statement II: The Zamindari Association raised its voice for the problems faced by common people and advocated for reforms benefiting the rural population.

Which one of the following is correct in respect of the above statements?

A. Both Statement I and Statement II are correct, and Statement II is the correct explanation for Statement I

B. Both Statement I and Statement II are correct, and Statement II is not the correct explanation for Statement I

C. Statement I is incorrect, but Statement II is correct

D. Statement I is correct, but Statement II is incorrect

Ans. D

Exp:

- The Zamindari Association—also called the Landholders' Society—was formed in 1838 in Bengal.To protect the interests of zamindars (landlords) under British rule, especially regarding Land revenue policies, Legal rights over estates, and relations with the colonial government. Prominent Bengali elites such as Dwarkanath Tagore were associated with it. These were wealthy landholders, not peasant representatives. So the core purpose was clearly zamindar-focused, not mass welfare.

BCW Bits:

- It was one of the **earliest political associations** in India before the INC (1885).
- It used **petitions and legal methods**, not mass movements.
- Later, it inspired the formation of the **British Indian Association (1851).**
- These early groups were called **"elite politics"**, not nationalist mass politics.
- Mass-based political mobilisation developed laterwith leaders like **Dadabhai Naoroji and Congress.**
- Zamindars benefited from the **Permanent Settlement (1793)**, so they wanted to protect that system.
- These associations show the **early stage of political consciousness** in India.

77. Lord Dalhousie's approach to constitutional and educational reforms reflected which of the following broader trends in British colonial policy during the mid-19th century? ***[BPSC AEDO]***

A. The gradual devolution of political power to Indian representatives.

B. The consolidation of imperial control through limited liberal reform.

C. The transition to complete self-governance under the Charter Act of 1853.

D. The prioritization of missionary activities over administrative efficiency.

Ans. B

Exp:

- Lord Dalhousie was a key British administrator who led to the expansion of British territory, Administrative modernization, Limited reforms in governance and education. His approach to reforms included:
  - **Constitutional reforms:** Linked with Charter Act of 1853, Introduced Open competition for civil services (ICS),Separation of legislative and executive functions. Though these reforms sounds progressive, but real power remained with the British, Indians had very little participation.
  - **Educational reforms:** Promoted modern education system, Supported English education, Laid foundation for universities (1857 later). Again, reform was useful—but also aimed at creating a class of educated Indians loyal to British rule.
  - **Charter Act 1853:** Last Charter Act, It introduced open competition for civil services, No fixed tenure for Company rule, But still no self-rule for Indians.

BCW Bits:

- Dalhousie also introduced: Railways, Telegraph, Postal system.
- These improved connectivity but also helped military control.
- He applied the Doctrine of Lapse to annex states. His policies indirectly contributed to the Revolt of 1857.

- British reforms often had a dual purpose: development + control.
- ICS exams were held in London, limiting Indian participation

78. Arrange the following in the correct sequence. ***[BPSC WMO]***

i. Charles Wood's dispatch
ii. Macaulay's Minute
iii. Wardha Scheme of Basic Education
iv. Sadler University Commission

Choose the correct answer from the code given below:
Code:

A. (iv), (iii), (ii), (i)
B. (ii), (i), (iv), (iii)
C. (i), (ii), (iii), (iv)
D. (i), (ii), (iv), (iii)

Ans. B

Exp:

1. **Macaulay's Minute (1835):** Thomas Macaulay's minute advocated for the introduction of Western education and literature in India through the medium of English.
2. **Charles Wood's Dispatch (1854):** Often referred to as the "Magna Carta of English Education in India," this document proposed a comprehensive plan for education from the primary to the university level.
3. **Sadler University Commission (1917–1919):** Also known as the Calcutta University Commission, it was appointed to examine the problems of Calcutta University and made significant recommendations for secondary and higher education.
4. **Wardha Scheme of Basic Education (1937):** Proposed by Mahatma Gandhi, this scheme focused on "learning through activity" and emphasized free, compulsory education in the mother tongue for children aged 7–14.

79. Loknath Bal, Ambika Chakraborty, Kalpana Dutt and Ganesh Ghosh were associated with which of the following? ***[BPSC WMO]***

A. Ghadar Movement
B. Indian Republican Army
C. Indian National Army
D. Abhinav Bharat

Ans. B

Exp: Loknath Bal, Ambika Chakraborty, Kalpana Dutt, and Ganesh Ghosh were prominent revolutionaries associated with the **Chittagong Armoury Raid** of 1930.They operated under the leadership of **Surya Sen**, who founded the **Indian Republican Army** (Chittagong Branch) with the objective of overthrowing British rule through armed rebellion.

80. Which of the following leaders laid the foundation of a Federation Hall on 16 October 1905 to mark the indestructible unity of Bengal? ***[BPSC WMO]***

A. Rabindranath Tagore
B. Krishna Kumar Mitra
C. Ananda Mohan Bose
D. Surendranath Banerjee

Ans. C

Exp: On 16 October 1905, the day the partition of Bengal officially came into effect, **Ananda Mohan Bose** laid the foundation stone of the **Federation Hall** in Kolkata. This event was organized to symbolize the permanent unity of the Bengali people against the British policy of "divide and rule" used to partition the province. The ceremony was significant as it served as a powerful protest and a testament to the nationalist resolve of that period.

81. What was one main feature of the Rowlatt Act of 1919? ***[BPSC WMO]***

A. It allowed the government to arrest and jail people without trial in court
B. It ended all press censorship in India
C. It gave all Indians the right to vote
D. It made English and Hindi the official languages of India

Ans. A

Exp: The Rowlatt Act of 1919, officially titled the **Anarchical and Revolutionary Crimes Act**, was introduced under **Viceroy Lord Chelmsford** following recommendations from the **Sedition Committee chaired by Sir Sidney Rowlatt.** It was designed to indefinitely extend the **repressive powers of the Defence of India Act 1915,** an emergency measure enacted during World War I (1914–1918) to suppress militant nationalism, particularly in Punjab and Bengal.

- The Act granted the colonial administration extraordinary powers: police could search premises without warrants, and individuals could be detained for up to two years based solely on suspicion without trial. Special courts, consisting of a three-judge bench, were established where guilty verdicts were final, with no right to appeal. Furthermore, the **Act severely restricted civil liberties by banning public meetings, fairs, and peaceful protests.**
- The legislation sparked widespread Indian outrage. Leading Indian members of the **Viceroy's Legislative Council—including Muhammad Ali Jinnah, Mazhar-ul-Haq, Madan Mohan Malviya, and B.N. Sharma—resigned in protest.Mahatma Gandhi** subsequently launched the **Rowlatt Satyagraha from Bombay on 24 February 1919**, escalating into a nationwide movement that culminated in a **mass Satyagraha on 6 April 1919**, marking a pivotal moment in the struggle against British colonial rule.

82. He started his career as an Assistant Professor of Mathematics. He was the President of the Indian National

Congress thrice, and was the first Indian to become a Member of the House of Commons. Whom do the above sentences refer to? ***[BPSC WMO]***

A. Bhimrao Ambedkar
B. Dadabhai Naoroji
C. G. K. Gokhale
D. Mahatma Gandhi

Ans. B

Exp: Dadabhai Naoroji, widely revered as the "Grand Old Man of India," was a preeminent intellectual and political figure. Before dedicating his life to the nationalist cause, he pursued a career in academics, serving as a professor of mathematics and natural philosophy at Elphinstone College, Bombay.

- His contribution to Indian politics was monumental; he served as the President of the Indian National Congress on three separate occasions—in 1886, 1893, and 1906. Furthermore, he achieved a historic milestone in 1892 by becoming the first Indian to be elected as a Member of Parliament in the British House of Commons, representing Finsbury Central.
- Naoroji is also celebrated for pioneering the "Drain of Wealth" theory. In his seminal work, *Poverty and Un-British Rule in India*, he argued that British colonial rule was systematically draining India's wealth by transferring its resources to Britain without any adequate economic return.

83. The Charter Act of 1833 is mainly associated with which important change in the administration of British India? ***[BPSC WMO]***

A. It ended the rule of the British Crown and restored full power to the East India Company.
B. It transferred all civil and military powers to the British Parliament in London.
C. It abolished the Company's trade monopoly with China and made the Governor-General of Bengal the Governor-General of India.
D. It divided India into two presidencies with separate Governors-General.

Ans. C

Exp: The Charter Act of 1833, also known as the St. Helena Act due to the Crown taking control of the island from the East India Company. It was necessitated by the Company's vast territorial expansion between 1813 and 1833.

- This landmark legislation stripped the East India Company of all remaining trade monopolies, including those for tea and trade with China, effectively transforming it into a purely administrative body.
- The act redesignated the Governor-General of Bengal as the Governor-General of India, with Lord William Bentinck as the first to hold this office.
- It further expanded the Executive Council by adding a fourth member—a Law Member, first held by Lord Macaulay—and established a Law Commission to codify Indian laws.
- Additionally, the act centralized legislative authority by abolishing the lawmaking powers of provincial legislatures and mandated that all laws passed in India be reviewed by the British Parliament, thereafter formally designated as "Acts" rather than "Regulations."

84. Which of the following pair/pairs is/are not matched correctly? ***[BPSC WMO]***

Tribal Movements		Leaders
i.	Santhal Rebellion	Sidhu and Kanhu
ii.	Kol Rebellion	Alluri Sitaramaraju
iii.	Koya Rebellion	Buddho Bhagat
iv.	Khasi Rebellion	Tirut Singh

Choose the correct answer from the code given below:
Code:

A. (ii) and (iv)   B. (i) and (ii)
C. Only (ii)   D. (ii) and (iii)

Ans. D

Exp: The **Santhal Rebellion (1855–1856) in the Rajmahal Hills was led by Sidhu and Kanhu Murmu** against exploitation by colonial authorities and landlords.

- The **Kol Rebellion (1831–1832) in Chota Nagpur, organized by Buddho Bhagat, Joa Bhagat, and Jhindrai Manki,** was a violent reaction against the displacement of tribes by outsiders and harsh revenue demands.
- The **Koya Rebellion (1879–1880) in the Godavari region, spearheaded by Tomma Dora,** sought to resist oppressive taxation and infringement on forest rights.
- The **Khasi Rebellion (1829–1833) in Meghalaya was led by Tirot Singh** to defend indigenous political autonomy against British road-building projects.

These movements collectively reflected tribal resistance to colonial encroachment, though all were eventually suppressed by British military power.

85. Which of the following pair is not correctly matched? ***[BPSC WMO]***

A. Raja Ram Mohan Roy - Ramakrishna Mission
B. Lala Lajpat Rai - Arya Samaj
C. Annie Besant - Central Hindu College
D. Madam Blavatsky - Theosophical Society

Ans. A

Exp: The **Ramakrishna Mission** was founded by **Swami Vivekananda** in 1897, not by Raja Ram Mohan Roy.

- **Ram Mohan Roy** is instead associated with the **Brahmo Samaj,** which he established in 1828 to promote monotheism and social reform.

- **Lala Lajpat Rai** was a prominent leader and active supporter of the **Arya Samaj,** which was founded by Dayananda Saraswati;
- **Annie Besant** played a pivotal role in establishing the **Central Hindu College** in Varanasi in 1898;
- **Madam Blavatsky,** alongside Henry Steel Olcott, co-founded the **Theosophical Society** in 1875 to foster the study of comparative religion, philosophy, and science.

86. Who among the following was not member of the 1946 Cabinet Mission to India? ***[BPSC WMO]***

A. Sir Stafford Cripps
B. Lord Mountbatten
C. Lord Pethick-Lawrence
D. A.V.Alexander

Ans. B

Exp: The Cabinet Mission of 1946 was sent to India by the British government to discuss the transfer of power and devise a framework for a new constitution.

- The mission consisted of three members: **Sir Stafford Cripps** (President of the Board of Trade), **Lord Pethick-Lawrence** (Secretary of State for India), and **A.V. Alexander** (First Lord of the Admiralty).
- Lord Mountbatten was not a member of this mission; he arrived in India later, in 1947, as the last Viceroy to oversee the final process of partition and independence.

87. Which of the following was the first Hindi newspaper printed at Calcutta in 1826? ***[BPSC WMO]***

A. Udant Martanda
B. Praja Hitaishi
C. Benaras Akhbar
D. Buddhi Prakash

Ans. A

Exp: *Udant Martanda* (The Rising Sun) was the first Hindi language newspaper published in India. It was started on May 30, 1826, in Calcutta by Jugal Kishore Shukla. This pioneering publication played a crucial role in the development of Hindi journalism and the promotion of the language during the colonial period. May 30 is still celebrated as "Hindi Journalism Day" (*Hindi Patrakarita Diwas*) in India to commemorate this historic milestone.

88. The Permanent Settlement of 1793, introduced in Bengal, was mainly intended to ***[BPSC WMO]***

A. Abolish land revenue completely and make all peasants the owners of their holdings.
B. Nationalise all agricultural land and make the company the only landlord.
C. Fix land revenue permanently so that the Company could get a regular surplus and landlords would be encouraged to improve their lands.
D. Give British officials the power to change land revenue every year according to harvest.

Ans. C

Exp: Introduced by **Lord Cornwallis** in 1793 in Bengal, Bihar, and Orissa, the Permanent Settlement aimed to create a loyal landlord class that would invest in agricultural development. Under this system, landlords became hereditary owners with permanently fixed revenue obligations, theoretically incentivizing them to improve their holdings. However, it relied on the **Sunset Law**, which required strict payment deadlines or risk land auction. The system failed due to **absentee landlordism** and the lack of protections for peasants, who suffered under unlimited rent demands and exploitation, while the East India Company secured a stable, fixed income.

Bihar Special

8

History of Bihar

1. Who became Chief Minister of Bihar in March 1939? ***[71st BPSC]***

A. Anugraha Narain Singh
B. Rajendra Prasad
C. Shri Krishna Singh
D. None of the above

Ans. **D**

Exp: Shri Krishna Sinha was premier Since 1937. The dictionary meaning of 'Premier' is 'chief' also. There was no post of chief Minister created in 1939.

BCW Bits:

- **Bihar Kesari** — Shri Krishna Singh was the leader of the first congress ministry in the 1937 election of Bihar, that took place under the Government of India Act 1935.
- Indian national congress was an all-India party so easily won the election.
- **Kisan sabha** also contested the election on the issue of "**Bakasht Land**" but **couldn't win** a **single seat**.
- **On 15 February 1938** - the Congress Ministry at Bihar resigned under the leadership of Shri. Krishna Singh on the issue of release of **political prisoners**, i.e. **Yogendra Shukla** was given punishment and sent to Cellular jail at Andaman and Nicobar Island (Kalapani).
- Bihar Ministry again tendered its resignation on **31 October 1939**, in protest of the Linlithgow declaration of engulfment of India in World War-II, without taking consideration of Indian view points.

  Note: If in question, it is asked about 1st Congress ministry to be formed in Bihar, then the answer will be Shri Krishna Singh. If it is **1st Indian Ministry formation in Bihar**, then it will be **Md. Yunus.**

2. Which Congress leader visited Bihar in 1918 in order to strengthen the activities of the Home Rule League? ***[71st BPSC]***

A. B.G Tilak
B. Annie Besant
C. Mahatma Gandhi
D. M. A. Ansari

Ans. **B**

Exp: Mrs. **Annie Besant** visited **Bihar in 1918** in connection with the **Home Rule Movement**. She was a leader of the Congress and presided over its 1917 session.

3. Champaran Agrarian Committee was appointed in agreement between: ***[71st BPSC]***

A. Indian Government and Gandhiji
B. Indian Government and Raj Kumar Shukla
C. Indian Government and Rajendra Prasad
D. Indian Government and Anugraha Narain Prasad

Ans. **A**

Exp: The **Champaran Agrarian Committee** was appointed in **1917** during the Champaran Satyagraha. It was set up by the British Indian Government in response to complaints from indigo farmers in Champaran (Bihar) about oppressive plantation practices by European landlords.

- Gandhi negotiated with the British authorities, and the government appointed a committee under the agreement with Gandhi to investigate the conditions of the peasants and suggest reforms.
- The Champaran Movement (1917), led by Mahatma Gandhi, was a protest against the forced indigo cultivation imposed on farmers in the Champaran district of Bihar. The British government, facing widespread unrest, responded by establishing the Champaran Agrarian Committee to investigate the farmers' grievances and find a solution. This committee ultimately led to the **passage of the Champaran Agrarian Act in 1918.**

4. Who was the prominent leader of the Kisan Sabha movement in Bihar? ***[71st BPSC]***

A. Raj Kumar Shukla
B. Sahajanand Saraswati
C. Acharya Narendra Dev
D. Baba Ram Chandra

Ans. **B**

Exp: The Bihar Provincial Kisan Sabha was formed in 1929 at Sonepur with Sahjanand Saraswati as its President. In 1922-23, the local Kisan Sabha was formed by Sri Krishna Singh and Shah Mohd Zubair in the Munger region.

5. Who was murdered while hoisting the national flag under the area of Bikram Police Station? *[71st BPSC]*

A. Raghunath Singh of Gorakhari
B. Ram Narain of Bikrampur
C. Satyendra of Gorakhari
D. None of the above

Ans. **D**

Exp: The name given in options are different from the true name, i.e. Rang Nath Sharma alias Raghunath Sharma.

6. Who among the following was not an Acharya in The Nalanda Vishvavidyalaya? ***[71st BPSC]***

A. Nagarjuna
B. Kautilya
C. Sheelbhadra
D. Shantirakshit

Ans. **B**

Exp: There were several Acharyas Nagarjuna, sheelbhadra and Shantirakshit at the Nalanda Mahabihar. Kautilya was not an Acharya in Nalanda Vishwavidyalaya.

7. Among the following newspapers of Bihar, which was the first to get published? *[71st BPSC]*

A. Bihar Bandhu B. Patna Harkara
C. Akhbar-E-Bihar D. Bihar Herald

Ans. B

Exp: Among the following, Patna Harkara was the first newspaper published in Bihar, with its inaugural issue released on April 21, 1855, by owner-editor Sah Abu Torab. It was published from Khaja Kalan in Patna City, on the 1st, 11th, and 21st day of every month, featuring essays, government orders, and advertisements.

- It is recognized as the first newspaper to get published from Bihar.
- It supported English East India Company policies while opposing Christian missionary activities.
- It consisted of 12 pages and utilized correspondents known as Mukhbirs.
- The paper ceased publication in 1856, lasting for only a year.

8. Match List I with List II- *[71st BPSC]*

List-I (Event/Organisation)		List-II (Related Person)	
a.	Champaran Satyagrah	1.	Jayaprakash Narayan
b.	Azad Dasta	2.	Shyam Narayan Singh
c.	Individual Satyagrah in Bihar 1940	3.	Sahajanand Saraswati
d.	Kisan Sabha Bihar	4.	Dr. Rajendra Prasad

	a	b	c	d
A.	4	3	2	1
B.	2	3	1	4
C.	1	4	3	2
D.	4	1	2	3

Ans. D

Exp: Champaran Satyagraha → Dr. Rajendra Prasad (4)

- Champaran Satyagraha (1917) was Gandhi's first satyagraha in India. Dr. Rajendra Prasad, a native of Bihar, became one of Gandhi's closest associates during this movement.
- **Azad Dasta → Jayaprakash Narayan (1)**
- A secret revolutionary / guerrilla group formed during the Quit India Movement (1942). Formed at Bakri ka Tapu in Nepal.
- **Individual Satyagraha in Bihar (1940) → Shyam Narayan Singh (2)**
- Launched by Gandhiji in 1940, before Quit India. Shyam Narayan Singh, born in Bihar's Nalanda district, was a prominent freedom fighter and active participant in the 1940 Individual Satyagraha. A student leader at Patna College, he was recognized as a key leader of the 1940 movement in Bihar.
- Nationally→Vinoba Bhave (first satyagrahi)
- Bihar→Sri Krishna Singh (First Satyagrahi)
- **Kisan Sabha, Bihar → Swami Sahajanand Saraswati (3)**
- The Bihar Provincial Kisan Sabha (BPKS) is directly associated with Swami Sahajanand Saraswati because he was its founder, chief ideologue, and mass leader. Later played a key role in founding the All India Kisan Sabha (1936).

9. In which year, Nalanda University was re-established in Bihar? *[71st BPSC]*

A. 2005 B. 2007 C. 2008 D. 2014

Ans. D

Exp: The first residential university of the world founded at Nalanda by the magnanimity of Emperor Kumaragupta in 427CE and sustained by the conscientiousness of the learned monks and teachers, flourished for over 800 years till the end in 12th century CE.

- In March 2006, while addressing a joint session of Bihar State Legislative Assembly, the former President of India, Dr. A.P.J. Abdul Kalam proposed the revival of the ancient Nalanda.
- The Parliament of India passed the Nalanda University Act, 2010 and in September, 2014, the first batch of students were enrolled.

10. Consider the following statements about the Mauryan period in Bihar. *[BPSC ASO, Pre]*

1. Chandragupta Maurya established the Maurya Empire by overthrowing the Nanda dynasty.
2. Ashoka's edicts, found extensively in Bihar, promoted a policy of 'Dhamma' based on non-violence and tolerance.
3. Megasthenes, a Greek ambaésador, visited Pataliputra during the time of Bindusara.

How many of the statements given above are correct?

A. All three **B.** Only one and two
C. Only one **D.** None of the above

Ans. B

Exp: Statements 1 and 2 are correct, Megasthenes visited Patliputra at the time of Chandragupta.

- Chandragupta Maurya, with the guidance of Chanakya (Kautilya), overthrew Dhana Nanda, the last ruler of the Nanda dynasty. This event marks the foundation of the Mauryan Empire. Hence statement 1 is correct
- Ashoka issued **rock and pillar edicts** across his empire. In **Bihar**, important Ashokan sites include:
  - Lauriya Nandangarh
  - Lauriya Areraj
  - **Rampurva**

- These edicts emphasize:
  - Ahimsa (non-violence)
  - Religious tolerance
  - Moral conduct
  - Welfare of people
- Ashoka's Dhamma was a socio-ethical policy, not a religious doctrine. Hence Statement 2 is correct.
- Megasthenes was the ambassador of Seleucus I Nicator. He visited Pataliputra during the reign of Chandragupta Maurya, not Bindusara. His famous work Indica describes Mauryan administration, society, and the city of Pataliputra.
- During Bindusara's reign, another Greek ambassador, Deimachus, visited the Mauryan court. Hence statement 3 is incorrect.

11. Kunwar Singh fought his last-war against which English captain? ***[BPSC ASO, Pre]***

A. Le Grand B. Douglas C. Eyre D. Lugard

Ans. A

Exp: On 23 April 1858, Kunwar Singh had a victory near Jagdishpur over the force led by **Captain Le Grand.** This was his last war, as he died in his village on 26 April 1858.

- Babu Kunwar Singh, son of Raja Sahibzada Singh (zamindar of Jagdishpur, which is in the present-day Bhojpur (Ara) district of Bihar) acted as a prominent leader during the revolution of 1857. He led the Indian Rebellion of 1857 in Bihar. He was nearly 80 years old and in failing health when he was called upon to take up arms. He was assisted by his brother, Babu Amar Singh and his commander-in-chief, Hare Krishna Singh. After being pursued by Douglas, he retreated towards his home in Jagdishpur, Bihar.
- The mantle of the old chief now fell on his brother Amar Singh who, despite heavy odds, continued the struggle and for a considerable time, ran a parallel government in the district of Shahabad region.

BCW Bits:

- **23rd April** is celebrated as Veer Kunwar Diwas as he recaptured Jagdishpur.
- On **23 April 2022**, 78, 220 flags were hoisted at Duler (Dulaur) ground, Jagdishpur during "Veer Kunwar Singh Vijayotsav" and this has been listed in "The Guinness Book of World Records".
- In his name, **Veer Kunwar Singh University** was established in 1992.
- On **23 April 2025**, Shaurya Diwas (also known as Valour Day) was celebrated by the Indian Air Force's Surya Kiran Aerobatic Team (SKAT) in Bihar to commemorate the victory of Babu Veer Kunwar Singh over British forces during the 1857 Rebellion.

12. The Bihar Provincial Conference passed a resolution supporting Gandhi's Non-Cooperation on ***[BPSC ASO, Mains]***

A. 1 August, 1920 B. 2 August, 1920
C. 28 August, 1920 D. 1 September, 1920

Ans. C

Exp: The Bihar Provincial Conference passed a resolution supporting Mahatma Gandhi's **Non-Cooperation Movement** on **August 28, 1920.**

- The **Bihar Provincial Conference** served as the annual deliberative assembly of the **Bihar Provincial Congress Committee (BPCC)**. BPCC was a landmark political organisation established in **Patna** in **1908** by **Nawab Sarfaraz Hussain Khan**.
- It served as the primary regional catalyst for the **Indian National Congress** in Bihar, initially spearheaded by leaders such as **Ali Imam** and **Mazharul Haque** to advocate for Bihar's administrative separation from the Bengal Presidency. The organisation held its first session in **April 1908**, presided over by **Sir Syed Ali Imam**.
- The **12th session** of the conference, held on **August 28, 1920**, in **Bhagalpur**, marked a historic transformation under the presidency of **Dr Rajendra Prasad**. During this assembly, the conference formally adopted the resolution supporting **Mahatma Gandhi's Non-Cooperation Movement**, making Bihar one of the first provinces to institutionalise the Gandhian mandate. This shift moved the regional struggle from constitutional agitation to active mass resistance.
- Subsequent milestones, such as the **1921 Arrah conference**, further solidified this momentum by focusing on boycotting foreign goods and promoting indigenous Khadi, thereby firmly embedding Bihar in the national freedom struggle.

13. The Prince of Wales's visit to Patna was boycotted by the people on ***[BPSC ASO, Mains]***

A. 28 August, 1921 B. 22 December, 1921
C. 1 March, 1922 D. 12 March, 1922

Ans. B

Exp: The boycott of the Prince of Wales's visit to Patna on **December 22, 1921**, was a definitive moment for the **Non-Cooperation Movement** in Bihar. Organised by the **Bihar Provincial Congress Committee (BPCC)** under the leadership of **Dr Rajendra Prasad**, the city observed a complete **hartal**.

- Despite colonial efforts to organise a grand reception, the public's refusal to participate left the streets deserted, dealing a significant blow to British prestige. This "silent protest" demonstrated the deep-rooted nationalist sentiment and the success of Gandhian non-violence in mobilising the masses against imperial authority.

14. Who was the leader of the Indian Rebellion of 1857 from Jagdishpur, Bihar? ***[BPSC DSO, Pre]***

A. Kunwar Singh B. Devi Singh
C. Kadam Singh D. Chittur Singh

Ans. A

Exp: Babu Kunwar Singh, son of Raja Sahibzada Singh (zamindar of Jagdishpur, which is in the present-day Bhojpur (Ara) district of Bihar) acted as a prominent leader during the revolution of 1857. He led the Indian Rebellion of 1857 in Bihar.

15. Which of the following leaders was prominently associated with the Champaran Satyagraha (1917) in Bihar marking Mahatma Gandhi's first active involvement in Indian mass movements? ***[BPSC DSO, Pre]***

A. Rajendra Prasad
B. Jayaprakash Narayan
C. Shri Krishna Sinha
D. Anugrah Narayan Sinha

Ans. A

Exp: The Champaran Satyagraha of 1917 was the first Satyagraha movement led by Gandhiji in India. It was Raj Kumar Shukla, who persuaded Gandhi to take up the Champaran's peasant cause demanding changes in the Tinkathia system of Indigo cultivation.

- Gandhiji arrived in Patna on 10 April 1917. The people with him were **Brajkishore Prasad, Rajendra Prasad, A.N Singh, Narayan Singh, Ramnavmi Prasad, Acharya Kriplani, Mazharul Haque, Mahadev Desai, Narhari Parikh.**
- During his visit to Champaran**,** Gandhiji stayed at the house of **Sant Raut,** in **Amolwa village,** while on his first visit to Bettiah village on 22 April 1917, he took residence at **Hazarimal Dharmashala.**
- The Satyagraha was successful as the Tinkathia system was abolished and the peasants were left free to grow whatever crop they wanted.

16. When Bhakhtiyar Khalji attacked Tirhut in 1202-03 during his invasion of Bengal, what was the outcome for its ruler Raja Narshimhadeva? ***[BPSC DSO, Pre]***

A. He was killed in battle
B. He was imprisoned and exiled
C. He agreed to pay tribute and his territory was restored to him
D. His kingdom was permanently annexed by Khalji

Ans. C

Exp: When Bakhtiyar Khalji attacked Tirhut (in the Mithila region of modern-day northern Bihar) around 1202–1203 CE during his expansion from Bengal:

- He confronted Raja Narshimhadeva, the ruler of the Karnat dynasty in Tirhut.
- Instead of being killed or permanently losing his kingdom, Raja Narshimhadeva submitted to Bakhtiyar Khalji's forces.
- He agreed to pay tribute (a regular payment acknowledging Khalji's suzerainty) to avoid further destruction.
- As part of this agreement, his territory was returned to him and he continued to rule Tirhut, albeit as a tributary (vassal) under Khalji's over lordship.

17. The Bihar Youth League was established in 1928 during a session of the Bihari Youth Conference held at which of the following places? ***[BPSC DSO, Pre]***

A. Arrah B. Muzaffarpur
C. Patna D. Motihari

Ans. D

Exp: The Bihar Youth League was founded in November 1928 at Motihari (East Champaran, Bihar) during the session of the Bihari Students' Conference, under the guidance of revolutionary leader Professor Gyan Saha.

18. Which of the following pairs of leaders were responsible for organising the Kisan Sabba in Munger in 1922? ***[BPSC DSO, Pre]***

A. Swami Sahajanand Saraswati and Mohammad Zubair
B. Sri Krishna Singh and Mohammad Zubair
C. Swami Sahajanand Saraswati and Sri Krishna Singh
D. Mohammad Zubair and Rajendra Prasad

Ans. B

Exp: The Bihar Provincial Kisan Sabha was formed in 1929 at Sonepur with Sahjanand Saraswati as its President.

- In 1922-23, the local Kisan Sabha was formed by
- Sri Krishna Singh and Shah Mohd Zubair in the Munger region.

Bihar Kisan Sabha:

- 1st Conference Sonepur - President: Sahajanand Saraswati in 1929
- 2nd Conference Gaya-President: Purushottam Das Tandon in 1934
- 3rd Conference Hajipur-President: Sahajanand Saraswati in 1935
- All India Kisan Sabha was formed in 1936 at Lucknow with Swami Sahajanand as President and N.G. Ranga as its Secretary.

19. Who held the position of President of the Bihar Provincial Congress Committee during the year 1928-29? ***[BPSC DSO, Pre]***

A. Anugrah Narayan Singh
B. SriKrishna Singh
C. Braj Kishore Prasad
D. Rajendra Prasad

Ans. A

Exp: Anugrah Narayan Singh served as the President of the Bihar Provincial Congress Committee (BPCC) during the year 1928–29. He was a prominent leader of the Indian National Congress and actively involved in the freedom movement in Bihar.

20. When was Bihar Scientific Society established? ***[BPSC DSO, Pre]***

A. 1868 B. 1785 C. 1659 D. 1793

Ans. **A**

Exp: The **Bihar Scientific Society** was established in **1868** in Muzaffarpur by Syed Imdad Ali during British colonial rule. It was initially called the British Indian Association, and later (by 1872) came to be known as the Bihar Scientific Society.

21. Match the following institutions with their establishment years. ***[BPSC AEDO]***

	Institutions		Establishment year
1.	Patna College	i.	1921
2.	Prince of Wales Medical College	ii.	1863
3.	Bihar Vidyapeeth	iii.	1900
4.	Bihar School of Engineering	iv.	1940
5.	Patna Women's College	v.	1925

Code:

	1	2	3	4	5
A.	ii	v	i	iii	iv
B.	v	iii	iv	i	ii
C.	iii	iv	i	v	ii
D.	i	iv	ii	iii	v

Ans. **A**

Exp:

Patna College (1863)

- One of the oldest colleges in Bihar
- Established during British period
- Initially part of Calcutta University, later under Patna University
- Became a major centre of higher education in eastern India
- Recently in the news **for completing over 150 years of legacy**.

Prince of Wales Medical College (1925)

- Now known as Patna Medical College
- Established to improve medical education in Bihar
- Named after British royal title at that time
- Later became one of the largest hospitals in Bihar

Bihar Vidyapeeth (1921)

- Founded during **Non-Cooperation Movement**
- Associated with **Mahatma Gandhi and Dr. Rajendra Prasad.**
- Purpose to promote **national education (swadeshi system),** alternative to British-controlled education.

Bihar School of Engineering (1900)

- Later became Bihar College of Engineering (BCE), and BCE later became NIT Patna.
- One of the oldest engineering institutions in India
- Focus on technical education during the colonial period

Patna Women's College (1940)

- Established for women's higher education
- Played a key role in empowerment of women in Bihar.

BCW Bits:

- Patna University (1917) is among the **oldest universities in India**
- Bihar Vidyapeeth symbolized **educational nationalism**
- PMCH is being upgraded into a **mega hospital project**
- NIT Patna is now an **Institute of National Importance**
- Women's education expanded significantly after 1940s
- Many Bihar institutions were part of **freedom movement ecosystem**
- Colonial education aimed to create **administrative workforce**

22. The Bihar Provincial Kisan Sabha was founded by: ***[BPSC AEDO]***

A. Jaiprakash Narayan
B. Swami Sahajanand Saraswati
C. Rajendra Prasad
D. Shri Krishna Sinha

Ans. **B**

Exp: The Bihar Provincial Kisan Sabha was founded by Swami Sahajanand Saraswati in 1929.

- Swami Sahajanand Saraswati is regarded as one of the most important peasant leaders of India. He organized farmers against zamindari exploitation, high rents, illegal taxes, and oppression by landlords. The Bihar Provincial Kisan Sabha became an important platform for mobilizing peasants and raising agrarian issues in Bihar.
- The movement later influenced the formation of the All India Kisan Sabha in 1936, which became a major peasant organization at the national level. Swami Sahajanand believed that political freedom alone was not enough unless farmers and rural poor also received economic justice.
- **Jayaprakash Narayan** is an extremely important personality in modern Indian history. Initially associated with the Congress Socialist Party, he later became famous for leading the "Total Revolution" movement against corruption and authoritarianism during the 1970s. He also played a major role against the Emergency imposed in 1975.
- **Rajendra Prasad** was a major leader from Bihar and closely associated with Mahatma Gandhi. He participated

actively in the Champaran Satyagraha and later became the President of the Constituent Assembly as well as the first President of independent India.

- **Shri Krishna Sinha** also known as Sri Babu, was another major political leader of Bihar. He became the first Chief Minister of Bihar after independence and played an important role in the abolition of the zamindari system in the state.

BCW Bits:

- The All India Kisan Sabha was formed in Lucknow in 1936.
- N.G. Ranga was another important peasant leader of India.
- Champaran Satyagraha was related to indigo cultivation.
- The Permanent Settlement strengthened zamindari in eastern India.
- The Tebhaga Movement took place in Bengal.
- Bardoli Satyagraha was led by Sardar Vallabhbhai Patel.
- Bihar became an important center of agrarian politics during British rule.
- Socialist ideas became popular among many young nationalist leaders in the 1930s.
- Zamindari abolition became a major issue after independence.
- Peasant movements often merged economic demands with nationalist politics.

23. The first Chief Minister of Bihar after independence was: ***[BPSC AEDO]***

A. Jaiprakash Narayan
B. Dr. Anugrah Narayan Sinha
C. Shri Krishna Sinha
D. Dr. Rajendra Prasad

Ans. C

Exp: Shri Krishna Sinha, also known as "Sri Babu, " became the first Chief Minister of Bihar after independence. Shri Krishna Sinha was one of the most prominent political leaders of Bihar and played an active role in the Indian freedom struggle. He worked closely with national leaders like Mahatma Gandhi and Dr. Rajendra Prasad. After independence, he became the first Chief Minister of Bihar and remained an influential figure in the state's politics for many years.

- His tenure is especially remembered for land reform efforts and the abolition of the zamindari system in Bihar. These reforms were important because Bihar had a large agrarian population suffering under exploitative landlord practices since colonial times.
- **Jayaprakash Narayan** popularly known as JP, was a major socialist leader and later led the famous "Total Revolution" movement against corruption and authoritarian rule during the 1970s. He played a key role in mobilizing opposition against the Emergency imposed in 1975.
- **Anugrah Narayan Sinha** was one of the most respected leaders of Bihar and served as the first Deputy Chief Minister and Finance Minister of Bihar. He worked closely with Shri Krishna Sinha and contributed significantly to Bihar's administration and development.
- **Rajendra Prasad** immense contribution to both Bihar and national politics. He was closely associated with the Champaran Satyagraha, became President of the Constituent Assembly, and later became the first President of independent India.

BCW Bits:

- Bihar became a separate province in 1912.
- Patna is the capital of Bihar.
- The Bihar Legislative Assembly is unicameral along with a Legislative Council.
- Dr. Anugrah Narayan Sinha is known as "Bihar Vibhuti."
- Dr. Rajendra Prasad was born in Zeradei, Bihar.
- Jayaprakash Narayan was associated with the Congress Socialist Party.
- The Constituent Assembly drafted the Constitution of India.
- Zamindari abolition was an important post-independence reform.
- Bihar played a major role in India's freedom movement.

24. At which place the Bihar Provincial Congress Conference was held in 1920? ***[BPSC AEDO]***

A. Patna
B. Bhagalpur
C. Muzaffarpur
D. Gaya

Ans. B

Exp: On the 28th July, 1920, Mahatma Gandhi announced that he would start Non-co-operation movement against the British Government from the 1st of August 1920.
A significant decision was taken at the 12th session of Bihar Provincial conference which met at Bhagalpur on 28 and 29 August under the presidentship of Rajendra Prasad, who in his address in Hindi, "Put in a strong plea for Non co-operation".One noteworthy feature of this conference was that among its delegates a large number came from the villages.

- The resolution on Non co-operation was moved by Dharnidhar of Darbhanga and strongly supported by other members including Gorakh Prasad of motihari.

25. With which Kisan leader was Sahajanand Saraswati not associated? ***[BPSC AEDO]***

A. Bihar Provincial Kisan Sabha, 1929
B. Hunkar Hindi Weekly
C. Bakasht Movement
D. Krishak Samachar

Ans. **D**

Exp: Swami Sahajanand Saraswati was one of the most important peasant leaders of India, especially in Bihar. Originally a religious monk (swami), he later became a mass leader of farmers (kisan movement).

- Fought against: Zamindari system, Exploitation of peasants. Founder of organised peasant politics in Bihar. He is often called the "Father of the Kisan Movement in India"
- His major associations: Bihar Provincial Kisan Sabha (1929)- Founded in 1929 in Bihar, led by Sahajanand Saraswati. Aim: Protect farmers from zamindar exploitation, raise voice for land rights. He was directly associated with Hunkar (Hindi Weekly)- A Hindi newspaper/weekly linked to peasant issues, used to spread awareness among farmers. Associated with Sahajanand's movement, Bakasht Movement- Major peasant movement in Bihar, Focus-Land taken away by zamindars (bakasht land issue), Sahajanand Saraswati was a key leader behind it.
- Krishak Samachar: was a Hindi publication associated with the peasant movement in Bihar during the early 20th century.

26. In the nineteenth century, which city in Bihar became the main centre of the Wahabi Movement against British rule? ***[BPSC WMO]***

A. Patna B. Muzaffarpur
C. Gaya D. Bhagalpur

Ans. **A**

Exp: Patna served as the primary headquarters and nerve center of the Wahabi Movement in India, where leaders recruited Muslim youth for campaigns, initially against the Sikh ruler Maharaja Ranjit Singh and later against the British. Although the movement was originally founded by Abdul Wahab of Nejd, it was later spearheaded in India by Syed Ahmed Barelvi, whose death in the 1831 Battle of Balakot eventually led to the movement's decline. Ultimately, the movement failed to gain broad support, and the British exploited the resulting communal divisions to foster separatist tendencies in Indian society.

27. Who carried on prolonged guerrilla warfare against the Company's forces in the Kaimur Hills during 1857? [BPSC WMO]

A. Amar Singh B. Lt. Stanton
C. Kunwar Singh D. Eyre

Ans. **A**

Exp: During the 1857 revolt, after the passing of his brother, Veer Kunwar Singh, **Amar Singh** took command of the revolutionary forces. He retreated to the Kaimur Hills, where he organized and led a prolonged, effective guerrilla campaign against the British East India Company.

28. Consider the following statements about the Birsa Movement in Chotanagpur. ***[BPSC WMO]***

1. It arose among sections of the Adivasis in the hilly tract of Chotanagpur.
2. It was led by Shri Birsa, a Munda of the village Chalkad in Tamar Thana.
3. It began as a movement for internal religious reform and social purification.

Which of the above statements is/are correct?

A. Only 2 and 3 B. 1, 2 and 3
C. Only 1 and 2 D. Only 1 and 3

Ans. **B**

Exp: The Munda Rebellion, or Ulgulan, was a late 19th-century tribal uprising in the forested Chotanagpur plateau (modern-day Jharkhand). Led by **Birsa Munda**, a Munda tribal member from **Chalkad village in the Tamar region**, the movement began as **Birsait—a reformist campaign for social purification** that urged Adivasis to abandon practices like animal sacrifice, alcohol consumption, and witchcraft. It soon **evolved into a militant struggle against** exploitative **diku (outsider)** landlords and British colonial land laws. Although the rebellion was brutally suppressed in 1900 and resulted in Birsa's death in custody, its legacy compelled the British to enact the **Chotanagpur Tenancy Act of 1908**, securing vital land rights for the tribal community.

29. Syed Hasan Imam, President of the special session of the Indian National Congress held at Bombay in 1918, belonged to which region? ***[BPSC WMO]***

A. Bengal B. Uttar Pradesh
C. Punjab D. Bihar

Ans. **D**

Exp: Syed Hasan Imam was a prominent lawyer and nationalist leader from Bihar. He served as the President of the special session of the Indian National Congress held in Bombay in 1918. His deep association with Bihar's political landscape is further underscored by the fact that he presided over the first session of the Bihar Provincial Congress Committee (BPCC) at Patna in 1908, which had been formed earlier that same year by Nawab Sarfraz Hussain Khan.

30. Which of the following statements is/are correct? ***[BPSC WMO]***

1. Most rivers of North Bihar are antecedent rivers.
2. These rivers have a high tendency for lateral erosion.

Choose the correct option.

A. Only 2 B. Neither 1 nor 2
C. Only 1 D. Both 1 and 2

Ans. **A**

Exp: Most rivers of North Bihar, such as the **Kosi, Gandak, and Burhi Gandak**, are Himalayan **consequent rivers**

that follow the regional slope as they emerge onto the plains, rather than being antecedent rivers - like the Indus, Sutlej, or Brahmaputra that existed before the uplift of the Himalayas and maintained their course by cutting through them. These rivers flow across soft, unconsolidated alluvial deposits with a very low gradient, which gives them **a high tendency for lateral erosion.** This physical environment leads to frequent channel shifting and makes the region highly susceptible to severe, recurrent flooding.

31. The Bihari Students' Conference, an influential students' organisation in Bihar's Nationalist Movement, was first held at ***[BPSC WMO]***

A. Patna College, Patna
B. Presidency College, Madras
C. Calcutta University
D. Allahabad University

Ans. A

Exp: The Bihar Students' Conference was **established in 1906**, with **Dr. Rajendra Prasad** playing a pivotal role in its founding. The organization's very **first conference was held at Patna College, Patna**. This influential body was instrumental in engaging the youth of Bihar with the Indian freedom movement. Over the years, the conference held several notable sessions, including the **15th session at Daltonganj, presided over by C.F. Andrews**, and the **16th session at Hazaribagh, presided over by Sarla Devi.**

Bihar Geography

1. Which of the following parks/sanctuaries come under the project tiger in Bihar? ***[71st BPSC]***

A. Valmiki National Park
B. Udaipur Wildlife Sanctuary
C. Bhimbandh Wildlife Sanctuary
D. None of these

Ans. A

Exp: Project Tiger is a centrally sponsored scheme launched in 1973 to conserve the Bengal tiger and its habitat. In Bihar, the only tiger reserve under Project Tiger is Valmiki Tiger Reserve.

- **Valmiki Tiger Reserve** is located in **West Champaran** district of Bihar. It is the only tiger reserve in Bihar.
- **Udaipur Wildlife Sanctuary** – Located in **Gaya** district, not a tiger reserve, mainly for deer and birds.
- **Bhimbandh Wildlife Sanctuary** – Located in **Munger** district, known for various species, but not under Project Tiger.

2. Which of the following is the state animal of Bihar? ***[71st BPSC]***

A. Cow B. Bull C. Buffalo D. Horse

Ans. B

Exp: Bull is the wild ox or male Gaur. The official name is Gaur.

3. Karmanasa river originates in: ***[71st BPSC]***

A. Kaimur District B. Vaishali District
C. Patna District D. Bikram District

Ans. A

Exp:

River	**Origin**	**Related Information**
Ganga	Gangotri (Uttarakhand)	Enters Bihar from Chausa (Buxar); Total length- 2525 km., Bihar-445 km; Borders 12 districts of Bihar.
North Bihar Rivers		
Ghagra	Tibet	Drains into Ganga (Saran); 83 km in Bihar
Gandak	Nepal (Annapurna Range)	Drains into Ganga (near Hajipur); 260 km in Bihar.
Budhi Gandak	Someshwar Hills	Drains into Ganga (Khagaria); 320 km in Bihar.
Bagmati	Bagdwar region, Shivpuri Hills (Nepal)	Enters Bihar from Sitamarhi; Meets Koshi; 394 km.
Kamla	Mahabharat Hills (Nepal)	Meet Koshi.
Koshi	Gushai sthan (Nepal)	Drains into Ganga (Kursela, Katihar); 260km.
Mahananda	Darjeeling, West Bengal	Meets Ganga outside Bihar.
South Bihar Rivers		
Karmnasha	Vindhyachal (Kaimur)	Meets Ganga at Chausa; Forms the boundary with UP and Bihar.
Son	Amarkantak hills	Meets Ganga near Patna. Two important tributaries of Son are Rihand and North Koel. It separates the Kaimur plateau from the Chota Nagpur plateau.
Punpun	Palamu (Jharkhand)	Meets Ganga at Fatuha.
Phalgu/ Niranjana	Chotanagpur Pathar	Meets Punpun in Patna district.
Ajay	Jamui	Meets Ganga outside Bihar.
Kiyul	Hazaribagh (JH)	Meet Ganga at Lakhisarai.

Note:
- In 1873-74, one of the oldest irrigation systems in the country and the first irrigation project of Bihar, before independence was developed across the river Son, at Dehri-on-sone.
- In 1960, the construction of 1, 407 metres long Sone Barrage (Indrapuri Barrage) at Indrapuri was taken up. This fourth longest barrage in the world, was commissioned in 1968.

4. Gold is found in which district of Bihar? *[71st BPSC]*
   A. Gaya District B. Jamui District
   C. Nalanda District D. Bhagalpur District

Ans. B

Exp: As per **Media reports,** in India, the largest reserves of gold ore (primary) are located in **Bihar** (44%) followed by **Rajasthan** (25%), **Karnataka** (21%), **West Bengal** (3%), **Andhra Pradesh** (3%), **Jharkhand** (2%). The remaining 2% resources of ore are located in Chhattisgarh, Madhya Pradesh, Kerala, Maharashtra, and Tamil Nadu. The presence of gold in areas such as Karmatia, Jhajha, and Sono in the Jamui district, was confirmed by the Geological Survey of India.

5. Most of Bihar comes under which of the following climate types? *[71st BPSC]*
   A. Subtropical Monsoon B. Tropical Savannah
   C. Hot and dry D. Hot and Semi-Arid

Ans. A

Exp: Most of Bihar experiences a Subtropical Monsoon climate, which is characterized by:
- **Hot summers**: March to June, with temperatures often exceeding 40°C in the plains.
- **Cold winters**: December to February, with temperatures sometimes dropping below 5°C in northern Bihar.
- **Monsoon rains:** June to September, brought by the southwest monsoon, providing the majority of the annual rainfall (~1, 000–1, 200 mm).
- **High humidity** during monsoon and post-monsoon months.
- This climate supports intensive agriculture, especially rice, wheat, maize, and sugarcane.
- **Tropical Savannah B.** Found in central India and some southern regions, not Bihar.
- **Hot and dry C.** Typical of desert areas like Rajasthan.
- **Hot and Semi-Arid D.** Found in rain-shadow regions, not in Bihar.

6. Which districts of Bihar will benefit from the Kosi Mechi Link project *[71st BPSC]*
   A. Patna Nalanda Gaya Siwan
   B. Darbhanga Muzaffarpur Sitamarhi Begusarai
   C. Araria Purnia Kishanganj Katihar
   D. More than one of the above

Ans. C

Exp: Kosi Mechi Intra-State Link Project envisages diversion of part of surplus water of Kosi river for extending irrigation to Mahananda basin lying in Bihar by way of remodelling of existing Eastern Kosi Main Canal (EKMC) and extending the EKMC beyond its tail end at RD 41.30 km up to river Mechi at RD 117.50 km so that rivers Kosi and Mechi which flow through Bihar could be linked together within Bihar.

The Link Project will provide 2, 10, 516 hectares additional annual irrigation in Kharif season in Araria, Purnea, Kishanganj and Katihar districts of Bihar.

7. According to Koppen Climate Classification Scheme, Bihar mainly lies in which type of climate? *[BPSC ASO, Pre]*
   A. As B. Cwg C. Aw D. Bshw

Ans. B

Exp: The Koppen climatic division divided the world into five climate zones using colours and shades, based on criteria like temperature, and vegetation.
- The climate of North Bihar may be explained as **Cwg** as per Koppen's climatic division. It is warm and temperate. The summers have an ample amount of rainfall while there is very little rain in winter.
- The climate of South Bihar was explained as **Aw** as per Koppen's climatic division.

8. Gogabeel bird sanctuary is not located in which district of Bihar *[BPSC ASO, Pre]*
   i. Samastipur ii. Katihar
   iii. Kaimur iv. Nalanda
   A. i, iii and iv B. B. ii, iii and iv
   C. i, ii and iv D. None of the above

Ans. A

Exp: "Gogabeel" present in Katihar has the most significant number of aquatic birds in Bihar as per the latest Bird census conducted by the Department of Environment, Forest and Climate Change of Bihar in collaboration with the Wetland International Institute. It is the 1st community reserve in the state.

9. Consider the following statements: *[BPSC ASO, Pre]*
   1. Except for some hills of Champaran, the remaining plain area of North Bihar is completely flat.
   2. The plains of North Bihar are spread over approximately 90, 000 sq. km.
   3. The plains of South Bihar are spread over approximately 10, 568 sq. km.

   Which of the above statements is/are correct?
   A. Only 3 B. Only 1 C. 2and 3 D. Only 2

Ans. B

Exp: North Bihar is part of the Indo-Gangetic plain and is generally flat alluvial land; only small hills like those in

Champaran break this flat terrain. Hence This statement is correct

- The total Bihar plain (both North and South together) is around 90, 650 sq km, but **North Bihar alone** does not cover that much; some reliable sources give figures like about 56, 980 sq km for the northern plain. Hence the second statement is incorrect.
- The area figure cited here is not supported by standard geographical data — South Bihar plains are much larger (tens of thousands of square kilometres) and often include both plains and plateau/floodplain regions. Hence the third statement is incorrect.

10. Which of the following is correct statement about the sex ratio in Bihar between Census 2001 and Census 2011? ***[BPSC ASO, Pre]***

A. Overall sex ratio remained the same, but child sex ratio decreased
B. Overall sex ratio increased, but child sex ratio decreased
C. Both overall sex ratio and child sex ratio decreased
D. Both overall sex ratio and child sex ratio increased

Ans. C

Exp: According to official census figures:

- **Overall Sex Ratio** (number of females per 1000 males) in Bihar **slightly decreased** from **919 in 2001 to 918 in 2011.**
- **Child Sex Ratio** (0-6 years) **also decreased** from **942 in 2001 to 935 in 2011.**

So:

- Overall, the sex ratio fell **by 1 point.**
- Child sex ratio fell **by 7 points.**

This means **both ratios decreased** over the decade.

11. Which of the following statements regarding the State of Bihar during the period 2001—2011 is correct? ***[BPSC ASO, Pre]***

A. The percent growth of population decreased, but Bihar’s share in India’s total population increased
B. Both the percent growth of population and Bihar’s share in India’s total population increased
C. Both the percent growth of population and Bihar’s share in India’s total population decreased
D. The percent growth of population increased, but Bihar’s share in India’s total population decreased

Ans. A

Exp: Per cent Growth of Population:

- In 2001–2011, Bihar’s decadal population growth was 25.42%.
- In the previous decade, 1991–2001, it was 28.43%.
- This shows that the population growth rate decreased in 2011 compared to 2001.
- Bihar’s Share in India’s Total Population

Bihar’s share of India’s population in 2001 was 8.07%.

- In 2011, it increased to 8.60%. This means Bihar’s proportion of India’s total population increased, even though its growth rate slowed.

12. Bhimbandh Wildlife Sanctuary is located in ***[BPSC ASO, Mains]***

A. Munger B. West Champaran
C. East Champaran D. Saran

Ans. A

Exp: The **Bhimbandh Wildlife Sanctuary** was established under the **Wildlife (Protection) Act, 1972**, and is situated within the **Kharagpur Hills** of the **Munger** district. Geographically, it covers approximately 681 km^2 and is distinguished by its unique **natural hot springs**, such as **Rishi Kund and Sita Kund,** and a dense cover of **dry deciduous forests**. This protected area serves as a vital ecological corridor for diverse fauna, including **leopards, sloth bears**, and numerous migratory bird species.

- In terms of broader conservation, Bihar possesses **one National Park** and **12 Wildlife Sanctuaries**. The **Valmiki National Park** in West Champaran is the state's sole national park and primary tiger reserve.
- Other significant sanctuaries include the **Vikramshila Gangetic Dolphin Sanctuary** in Bhagalpur, the **Kanwar Lake Bird Sanctuary** (a Ramsar site) in Begusarai, and the **Kaimur Wildlife Sanctuary**, which is currently the largest in the state by area.

13. Kesaria Chaur wetland is located in which district of Bihar? ***[BPSC ASO, Mains]***

A. Muzaffarpur B. East Champaran
C. Saran D. Vaishali

Ans. B

Exp: The **Kesaria Chaur** is a vital perennial wetland located in the **East Champaran** district of Bihar. Geographically, **"Chaur"** refers to a low-lying waterlogged area common in the North Bihar plains, formed by shifting river courses.

- **Ramsar sites** are wetlands of international importance protected under the 1971 Ramsar Convention; India currently has **99** such sites, while Bihar has **6 (as of April 2026)**.
- Nationally, wetlands are governed by the **Wetlands (Conservation and Management) Rules, 2017**, and the **Environment (Protection) Act, 1986**.

Ramsar Sites in Bihar

1. **Kanwar Taal (Begusarai):** Bihar's first site (2020).
2. **Nagi Bird Sanctuary (Jamui):** Added in 2024.
3. **Nakti Bird Sanctuary (Jamui):** Added in 2024.
4. **Gogabil Lake (Katihar):** Added in October 2025.
5. **Gokul Jalashay (Buxar):** Added in September 2025.
6. **Udaipur Jheel (West Champaran):** Added in September 2025.

14. Ptilophyllum fossil was obtained from *[BPSC ASO, Mains]*

A. Vindhya Mountain B. Rajmahal Hills
C. Himalayan Range D. Jayanti and Khasi Hills

Ans. B

Exp: *Ptilophyllum* is a genus of extinct Bennettitalean plants (resembling modern cycads) that flourished during the Mesozoic era. Fossils of this genus are characteristic of the **Upper Gondwana** flora in India and are abundant in the **Rajmahal Hills** of Jharkhand. These "Rajmahal Flora" fossils are world-renowned among paleobotanists for their well-preserved structural details.

15. Bhagalpur (Bihar) city is specially known as *[BPSC DSO, Pre]*

A. Zardalu Mango B. Lentils Barley
C. Tussar Silk D. Oilseeds

Ans. C

Exp: Bhagalpur is known as the "silk city". Bhagalpur has been associated with the silk industry for hundreds of years and is famous all over India for its **Tussar Silk & Bhagalpuri Saree.** Silkworms are employed to produce the renowned Tussar Silk from which the Tussar Saree is manufactured. The Silk Institute and Agricultural University is located in the city. Bhagalpuri silk is made from cocoons of **Antheraea Paphia** silkworms. This species, also known as the Vanya silkworm, is native to India.

16. Jamui, Bihar, is known for its: *[BPSC DSO, Pre]*

A. Copper Reserve **B.** Gold Reserve
C. Zinc Reserve **D.** Iron Reserve

Ans. B

Exp: As per **Media reports,** in India, the largest reserves of gold ore (primary) are located in **Bihar** (44%), followed by **Rajasthan** (25%), **Karnataka** (21%), **West Bengal** (3%), **Andhra Pradesh** (3%), and **Jharkhand** (2%). The remaining 2% resources of ore are located in Chhattisgarh, Madhya Pradesh, Kerala, Maharashtra, and Tamil Nadu. The **presence of gold** in areas such as Karmatia, Jhajha, and **Sono in the Jamui** district was confirmed by the Geological Survey of India.

17. Which River is called the sorrow of Bihar? *[BPSC DSO, Pre]*

A. Ganga B. Kaveri C. Kosi D. Damodar

Ans. C

Exp: Kosi is known as the sorrow of Bihar.

BCW Bits:

- Kosi: Total length of Kosi is 729 km. The distance it travels in Bihar is 260 km. Also known as Sorrow of Bihar.
- Canals on the Kosi are:
  1. East Kosi,
  2. West Kosi
  3. Rajpur Canal
- The **Kosi-Mechi project** is designed to divert part of the Kosi River surplus water through the existing Hanuman Nagar Barrage.

18. Which of the following district of Bihar does not shares its borders with Nepal *[BPSC DSO, Pre]*

A. Madhubani B. Purnia
C. Araria D. Sitamarhi

Ans. B

Exp: Districts of Bihar bordering **Nepal** are West Champaran, East Champaran, Sitamarhi, Madhubani, Supaul, Araria, and Kishanganj.

BCW Bits:

- Kishanganj, Purnea, and Katihar are the 3 districts that border **West Bengal.**
- Eight Districts that share a border with **Uttar Pradesh** are West Champaran, Gopalganj, Siwan, Saran, Bhojpur, Buxar, Rohtas and Kaimur.
- Eight districts share a border with **Jharkhand** as well. These are: Rohtas, Aurangabad, Gaya, Nawada, Jamui, Banka, Bhagalpur and Katihar.

Note:

- West Champaran shares the border with both Nepal and Uttar Pradesh.
- Katihar shares a border with both West Bengal and Jharkhand.
- Rohtas shares a border with both Uttar Pradesh and Jharkhand.
- Kishanganj shares a border with both Nepal and West Bengal.

19. Which Sanctuary is the largest in Bihar? *[BPSC DSO, Pre]*

A. Kaimur
B. Bhimbandh
C. Valmiki
D. Sanjay Gandhi Udyan

Ans. A

Exp: In terms of area, and sequence of the above Wildlife, Bird sanctuaries in descending order are:

- Kaimur Sanctuary > Valmiki Wildlife Sanctuary > Bhimband Sanctuary > Gautam Buddha Sanctuary.
- A list of various wildlife/ bird sanctuaries of Bihar is given below.

Name	Location
Kaimur Wildlife Sanctuary	Kaimur
Valmiki National Park (VTR)	West Champaran

Barela Jheel Salim Ali-Jubba Sahni (Baraila) Bird Sanctuary	Vaishali
Bhimbandh Wildlife Sanctuary	Munger
Gautam Budha (Buddha) Wildlife Sanctuary– BR	Gaya
Kanwar Jheel Bird Sanctuary	Begusarai
Kusheshwar Asthan Bird Sanctuary	Darbhanga
Nagi Dam Bird Sanctuary	Jamui
Nakti Dam Bird Sanctuary	Jamui
Pant (Rajgir) Wildlife Sanctuary	Nalanda
Udaipur (Udaypur) Wildlife Sanctuary	West Champaran
Vikramshila Gangetic Dolphin Wildlife Sanctuary	Bhagalpur

20. Bhimbandh Wildlife Sanctuary is located in *[BPSC DSO, Pre]*

A. Munger B. West Champaran
C. East Champaran D. Saran

Ans. A

Exp: Bhimbandh wildlife sanctuary is situated in the Munger district. Bhimbandh wildlife sanctuary and Gautam Buddha wildlife sanctuary was established in 1976 and hence they are the oldest wildlife sanctuary in Bihar.

21. Which is the main river system of Bihar? *[BPSC DSO, Pre]*

A. Kosi B. Ganga
C. Gandak D. Damodar

Ans. B

Exp: The state of Bihar is primarily drained by rivers that are part of the Ganga River system. The Ghagra River is one of the major tributaries of the Ganga and flows through parts of Bihar, contributing to its drainage system.

22. In which of the following districts river Ganga enters into Bihar? *[BPSC DSO, Pre]*

A. Gopalganj B. Siwan
C. Rohtas D. Buxar

Ans. D

Exp: Ganga enters Bihar at Chausa (Buxar) and exits by making a boundary between Katihar and Bhagalpur. Patna district has the longest length of the Ganga River in Bihar.

BCW Bits:

- The number of districts situated on the bank of the Ganga River in Bihar state is 12.
- These 12 districts are Buxar, Bhojpur, Saran, Patna, Vaishali, Samastipur, Begusrai, Munger, Khagaria, Katihar, Bhagalpur, and Lakhisarai.
- In India, Ganga flows through mainly 5 states namely Uttarakhand, Uttar Pradesh, Bihar, Jharkhand & West Bengal.
- The river basin of Ganga extends to 11 states of India.
- Length of Ganga River in Bihar- 445 km.
- Total length of river Ganga- 2525 km.

23. DP Dhar prepared and launched: *[BPSC DSO, Pre]*

A. Bihar Fifth-Five-year plan
B. Bihar Sixth-Five-year plan
C. Bihar Seventh-Five-year plan
D. Bihar Fourth-Five-year plan

Ans. A

Exp: D.P. Dhar prepared and launched the Fifth Five-Year Plan of India. The final draft of this plan was prepared under his guidance and launched in 1974. The Fifth Plan's major objectives included removal of poverty ("Garibi Hatao") and attainment of self-reliance, with a focus on growth, employment, and better income distribution.

- The Fifth Five-Year Plan covered 1974–1979, although it was terminated early in 1978.
- D.P. Dhar was the Planning Commission's Deputy Chairman and played a central role in preparing and launching this plan.

24. What is the approximate percentage of geographical area covered under forest in Bihar? *[BPSC AEDO]*

A. 5% B. 7% C. 13% D. 11%

Ans. B

Exp: **Forest cover in India vs. Bihar:** India overall has around 21–24% forest cover (varies slightly by report). Bihar has relatively low forest cover compared to the national average due to high population density, extensive agricultural land use, and limited forested hilly regions.

- Bihar's forest cover: Bihar has roughly 7% forest cover of its total geographical area. Forests are mainly concentrated in West Champaran (the Valmiki Tiger Reserve area) and in some parts of the Kaimur hills. Compared to other states, Bihar is forest-deficient.

BCW Bits:

- India has around 21-24% forest cover, while Bihar has around 7%
- **Valmiki Tiger Reserve** is the only tiger reserve in Bihar.
- Bihar mainly lies in the Gangetic plains, which are more suitable for agriculture than for dense forest growth.

25. Which of the following towns is located in the westernmost part of Bihar? ***[BPSC AEDO]***

A. Bhagalpur B. Patna
C. Kaimur D. Katihar

Ans. C

Exp: **Bihar lies in the eastern part of India, but within the state itself:** West - Kaimur / Buxar region (border with Uttar Pradesh), **Center -Patna, East - Bhagalpur, Katihar (towards West Bengal).**

- Kaimur District lies in the south-western part of Bihar, along the Kaimur (Rohtas) plateau (an extension of Vindhyan Range) . It sits near the tri-junction of Bihar–Uttar Pradesh–Jharkhand, District HQ: Bhabua
- **Bihar shares borders with:** Uttar Pradesh (West), Jharkhand (South), West Bengal (East), Nepal (North).

BCW Bits:

- **Famous places of Kaimur:** Mundeshwari Temple, one of the oldest Hindu temples in India, and is dedicated to Lord Shiva and Goddess Shakti; **Kaimur Wildlife Sanctuary:** Known for waterfalls, forests, and wildlife; **Telhar Kund:** Scenic waterfall in the hills
- **The region is known for waterfalls and forest areas**
- **Bihar is divided by the Ganga River into North and South Bihar**
- **Most of Bihar lies in the fertile Indo-Gangetic plain**

26. Which of the following lakes is situated in Bihar? ***[BPSC AEDO]***

A. Anupam lake B. Sambar lake
C. Kama lake D. Sukhna lake

Ans. A

Exp: Bihar is mainly a **river-based state (Ganga basin)**, so it doesn't have many large, famous lakes like Rajasthan or Kashmir. Most lakes here are **Oxbow lakes (formed by rivers changing course) and wetlands used for fishing and agriculture.**

- Sambar (India's largest inland **saltwater lake**)- Rajasthan, Sukhna (man-made lake) - Chandigarh, Kama-MP

BCW Bits:

- Anupam lake is located in Kaimur District.
- Kanwar (Kabar) Lake is the largest freshwater oxbow lake in Asia.
- Baraila Lake (Vaishali) is another important wetland.
- These lakes are crucial for: Migratory birds, Fishing economy.
- Bihar wetlands are often formed due to flood-prone rivers like Kosi and Gandak.

27. In April 2022, a Geographical Indication (GI) tag was awarded to which of the following by the Government of India? ***[BPSC AEDO]***

A. Darbhanga Makhana B. Nalanda Makhana
C. Vaishali Makhana D. Mithila Makhana

Ans. D

Exp: Makhana is obtained from the plant Euryale ferox and is widely cultivated in the Mithila region of Bihar, especially in districts such as Darbhanga, Madhubani, Sitamarhi, and Supaul.

- Health benefits: High in protein and fibre, Rich in calcium and antioxidants, low-fat → good for heart and weight control. Bihar produces ~85–90% of India's makhana.
- GI Tag protects regional identity, prevents misuse of the name, and helps in branding and exports.
- The government has increased focus on the makhana sector, Proposal for Makhana Board in Bihar. Budget allocations for Processing, Marketing, Export promotion. Shows the rising importance of makhana in the economy.

BCW Bits:

- Makhana cultivation requires standing water for months. Harvesting is labour-intensive and traditional.
- Bihar exports makhana to countries like the USA and the Middle East.
- It is also called "Phool Makhana" and is used in Ayurvedic medicines.
- Considered a high-value cash crop.
- Government promotes it under agro-processing schemes
- Mithila Makhana received the GI tag in April 2022.

28. Food crops in order of importance of area cultivated in Bihar is: ***[BPSC AEDO]***

A. Maize, Pulses, Paddy, Wheat
B. Maize, Paddy, Wheat, Pulses
C. Paddy, Wheat, Maize, Pulses
D. Wheat, Paddy, Maize, Pulses

Ans. C

Exp: **Paddy** occupies the largest cultivated area in Bihar because rice is the main staple food of the state and large parts of Bihar receive sufficient monsoon rainfall suitable for paddy cultivation. Paddy cultivation is especially dominant in north Bihar where floodplain soils and water availability support rice farming.

- **Wheat** comes second and is mainly grown during the Rabi season. It is widely cultivated in districts having fertile alluvial soil and better irrigation facilities. After the Green Revolution and expansion of tube-well irrigation, wheat production increased significantly in Bihar.
- **Maize** ranks third in terms of cultivated area. Bihar is one of the leading maize-producing states in India, particularly in hybrid maize cultivation. In recent years, maize has become economically important because of its use in poultry feed, starch industries, and food processing.
- Pulses occupy comparatively less area than the above three crops. Even though pulses are important for protein

supply and soil fertility, their cultivated area is smaller in Bihar.

BCW Bits:

- Bihar mainly has alluvial soil brought by Himalayan rivers.
- Agriculture in Bihar is heavily dependent on monsoon rainfall.
- Paddy is mainly a Kharif crop.
- Wheat is mainly a Rabi crop.
- Maize can be grown in Kharif, Rabi, and Zaid seasons in Bihar.
- Major rivers of Bihar include Ganga, Kosi, Gandak, and Bagmati.
- North Bihar is more flood-prone than South Bihar.
- Pulses help in nitrogen fixation in soil.
- The Green Revolution had a greater impact in Punjab and Haryana than Bihar.
- Bihar is an important producer of litchi, makhana, and maize.

29. Which of the following statements about floods in Bihar are correct? ***[BPSC AEDO]***

1. Bihar is basically a plain area
2. Major rivers in Bihar are long with catchment in mountains
3. Bihar is economically backward

Out of these:

A. None of the above is correct
B. Only 1 and 2 are correct
C. All the three are correct
D. Only 1 is correct

Ans. **C**

Exp: **Statement 1** is correct because Bihar is predominantly a vast alluvial plain. Most parts of the state lie in the fertile Gangetic plains with very little variation in relief. Since the land is flat and low-lying, floodwater spreads easily over large areas and remains stagnant for longer.

- **Statement 2** is also correct. Most major rivers of Bihar, such as Kosi, Gandak, Bagmati, and Kamla, originate in the Himalayas or have their catchments in the mountainous regions of Nepal and Tibet. During the monsoon season, heavy rainfall in the Himalayan catchment areas causes these rivers to carry huge volumes of water and sediments into Bihar. Since Bihar lies downstream in the plains, flooding becomes frequent and severe. The Kosi River is often called the "Sorrow of Bihar" because of its frequent floods and shifting river channels. Himalayan rivers also carry large amounts of silt, which raises riverbeds over time and increases flood risk.
- **Statement 3** is also considered correct in the context of this question. Bihar has historically faced economic challenges, including limited industrial development, high population pressure on agriculture, and weaker infrastructure compared to many other Indian states. Economic backwardness increases flood vulnerability because disaster management infrastructure, such as embankments and drainage systems, and rehabilitation capacity are comparatively weaker.

BCW Bits:

- Kosi River is known as the "Sorrow of Bihar."
- Most rivers of North Bihar flow from Nepal into India.
- Floods are more common in North Bihar than South Bihar.
- Alluvial soil is highly fertile but flood-prone.
- Embankments are constructed to control river flooding.
- The Ganga divides Bihar into North Bihar and South Bihar.
- Sedimentation raises river beds and increases flood risk.
- Disaster management includes mitigation, preparedness, response, and rehabilitation.
- Bihar is one of the most densely populated states of India.
- Climate change is increasing the frequency of extreme rainfall events.

30. Which of the following states are almost identical in their shapes? ***[BPSC AEDO]***

A. Uttarakhand & Himachal Pradesh
B. Bihar and Jharkhand
C. Chhattisgarh and Odisha
D. None of the above

Ans. **A**

Exp: **Uttarakhand & Himachal Pradesh:** On the political map of India, both of these Western Himalayan states feature a roughly quadrilateral or parallelogram-like outline. Their geometric appearances, longitudinal/latitudinal alignments, and mountainous topographies create a strong visual symmetry.

- Bihar & Jharkhand: Bihar is distinctly rectangular or quadrilateral in shape, whereas Jharkhand forms an irregular polygon that does not mimic Bihar's outline.
- Chhattisgarh & Odisha: Chhattisgarh is famous for its unique shape resembling a seahorse (*Hippocampus*), while Odisha has a broad, coastal, and semi-triangular appearance

BCW Bits:

- Jharkhand was formed on 15 November 2000.
- Capital of Bihar = Patna.
- Capital of Jharkhand = Ranchi.
- Chotanagpur Plateau lies mainly in Jharkhand.
- River Ganga divides Bihar into North Bihar and South Bihar plains.
- Bihar shares a border with Nepal in the north.
- Odisha has a long coastline along the Bay of Bengal.
- Chhattisgarh is rich in coal and iron ore resources.

- Himachal Pradesh and Uttarakhand are part of the Himalayan mountain system.

31. Match List-I with List-II ***[BPSC AEDO]***

List-I (River)		List-II (Originates)	
I.	Mahananda	a.	Shivpuri Range (Nepal)
II.	Bagmati	b.	Mahabharat range (Nepal)
III.	Kamala	c.	Nampa (Nepal)
IV.	Ghagra	d.	Sikkim

Select the correct answer using the codes given below:

	I	II	III	IV
(A)	d	a	b	c
(B)	a	b	c	d
(C)	a	d	c	b
(D)	d	c	b	a

Ans. A

Exp: Mahananda River originates from the Mahaldiram Hills (Darjeeling). It later flows through North Bengal and Bihar before joining the Ganga. Bagmati River originates from the **Bagdwar region, Shivpuri Hills (Nepal)**. Bagmati is a very important river culturally as well as geographically. The famous Pashupatinath Temple is situated on its banks. After flowing through Nepal, the river enters Bihar and becomes flood-prone during heavy monsoon rainfall.

- Kamala River rises from the **Mahabharat Range of Nepal**. The Kamala River system is important in north Bihar and often causes flooding in the Mithila region. Many Himalayan rivers descending into the plains carry large amounts of silt because of steep slopes and erosion in the young fold mountains.
- Ghaghara River, which originates from **Tibet**. Ghaghara is one of the major tributaries of the Ganga and is known as Karnali in its upper course in Nepal. It is a perennial Himalayan River fed by glaciers and monsoon rainfall.
- Rivers originating in the Himalayas are perennial because they receive water from glaciers as well as rainfall. During monsoon months, these rivers carry excessive discharge and often create devastating floods in Bihar. That is why north Bihar is known as a flood-prone region. Himalayan rivers are generally youthful rivers in their upper course. They form deep gorges, carry huge sediment loads, and frequently change courses in the plains. Rivers like Kosi are famous for channel shifting and are called the "Sorrow of Bihar."

BCW Bits:

- Kosi River is called the "Sorrow of Bihar."
- Ghaghara is known as Karnali in Nepal.
- Bagmati is considered a holy river in Nepal.
- Most north Bihar rivers are Himalayan in origin.
- Himalayan rivers are perennial due to glacier + rainfall feeding.
- Gandak originates near the Nepal-Tibet border.
- Burhi Gandak is different from Gandak River.
- Mahananda joins the Ganga near Godagari region of Bangladesh area.
- North Bihar plains are highly flood-prone because of heavy silt deposition.
- Meandering and shifting river channels are common in the Gangetic plains.

32. How many districts of Bihar have similar name of their headquarter as the name of district? ***[BPSC AEDO]***

A. 30 B. 31 C. 29 D. 35

Ans. C

Exp: In Bihar, many districts have headquarters with the same name as the district itself. For example, Patna district has Patna as its headquarters, Gaya district has Gaya as headquarters, and Muzaffarpur district has Muzaffarpur as headquarters. The correct number of districts in Bihar where the district name and headquarters name are the same is **29.**

- The name of headquarters differs from the following districts:
  - Rohtas (Sasaram), Kaimur (Bhabua), Bhojpur (Ara), Nalanda (Bihar Sharif), Saran (Chhapra), West Champaran (Bettiah), East Champaran (Motihari), Sitamarhi (Dumra), Vaishali (Hajipur)

Note: At some places headquarter of Sitamarhi is considered as the same name as the district, in that case number of districts with same name and headquarters will become 30

BCW Bits:

- Bihar currently has 38 districts.
- Bihar is divided into 9 administrative divisions.
- Largest district of Bihar by area = West Champaran.
- Smallest district of Bihar by area = Sheohar.
- Most populous district of Bihar = Patna.
- Bihar was separated from Bengal Presidency in 1912.
- Jharkhand was carved out of Bihar in 2000.
- Bihar shares border with Nepal, Uttar Pradesh, Jharkhand, and West Bengal.

33. What is the position of Bihar after the separation of Jharkhand? Consider the following statements: ***[BPSC AEDO]***

1. Longitudinal extension remains the same
2. Latitudinal extension reduced by 2° 30′
3. It comprises 54.15% area of undivided Bihar

How many of the statements given above are correct?

A. Only one and three B. All three
C. Only one D. None of the above

Ans. A

Exp: Jharkhand was carved out from the southern part of Bihar on 15 November 2000. After this separation, Bihar lost most of its mineral-rich plateau region, forests, and industrial belt, while the remaining Bihar became largely a plain-dominated agricultural state. This political division also affected the geographical dimensions of Bihar.

- Bihar mainly lost its southern territory during bifurcation, while its east-west spread did not change significantly. Therefore, the longitudinal extent remained almost unchanged. Bihar did lose its southern extension after Jharkhand was separated, but the reduction was not as large as 2°30′. Bihar now comprises **54.15% area of undivided Bihar**. After bifurcation, the remaining Bihar retained a little more than half of the area of old Bihar, while Jharkhand occupied the remaining portion.

BCW Bits:

- Present Bihar has 38 districts.
- Bihar is mainly an alluvial plain state after bifurcation.
- Most mineral-rich areas of old Bihar went to Jharkhand.
- Chotanagpur Plateau is now largely in Jharkhand.
- Ganga River divides Bihar into North and South Bihar.
- Bihar is one of the most densely populated states of India.
- Major industries lost by Bihar after bifurcation included steel and mining sectors.
- Present Bihar economy is agriculture-dominated.
- Jharkhand became India's 28th state.

34. What are the major crops in Bihar? ***[BPSC AEDO]***

A. Rice, Wheat and Maize
B. Arhar, Urad and Moong
C. Sugarcane and Oil seeds
D. Cotton and Jute

Ans. A

Exp: **Rice, Wheat and Maize**. These are the principal food crops of Bihar and dominate the state's agricultural landscape in terms of cultivated area, production, and economic importance. Bihar is an agriculture-based state where a large share of the population depends on farming for livelihood. The fertile alluvial plains formed by rivers like the Ganga River, Kosi, Gandak, and Bagmati create ideal conditions for cultivation of food grains.

- Rice is the most important crop of Bihar and is mainly cultivated during the **Kharif season**. It requires high temperature, abundant rainfall, and fertile soil. Bihar receives monsoon rainfall from the southwest monsoon, which supports paddy cultivation across most districts. North Bihar, with its floodplain regions and moisture-rich soils, is especially suitable for rice farming. Since rice is the staple food of the state, it occupies a major share of agricultural land.
- Wheat is the second major crop and is grown mainly during the **Rabi season**. It requires cooler temperatures during growth and moderate climate during harvesting. After the monsoon season, Bihar's winter conditions become favorable for wheat cultivation. Irrigation from canals, tube wells, and rivers supports wheat farming in many districts. The spread of high-yielding variety seeds after the Green Revolution also improved wheat productivity in Bihar.
- Maize has emerged as another highly significant crop in Bihar. In fact, Bihar is among the leading maize-producing states in India. Maize is cultivated in both Kharif and Rabi seasons in several districts, particularly in north Bihar. It is used not only as food grain but also as animal feed and industrial raw material. Hybrid maize cultivation has expanded rapidly because of demand from poultry and starch industries.
- Sugarcane is an important commercial crop of Bihar, especially in districts like West Champaran and Gopalganj, where sugar mills historically developed. Oilseeds are also cultivated in some regions, but together they do not represent the primary agricultural identity of Bihar.
- Cotton requires black soil and comparatively dry climatic conditions, which are more suitable in states like Maharashtra and Gujarat. Jute is cultivated in limited areas of northeastern Bihar, especially near the border regions with West Bengal and Assam-type climatic zones.
- **Kharif crops** – sown with monsoon (rice, maize, jowar); **Rabi crops** – sown in winter (wheat, barley, mustard); **Zaid crops** – grown between Kharif and Rabi (watermelon, cucumber, vegetables).
- Agriculture in Bihar is heavily dependent on monsoon rainfall, although irrigation facilities have expanded over time. Floods in north Bihar and drought-like situations in south Bihar often affect agricultural productivity. Despite fertile land, challenges such as fragmented landholdings, low mechanization, migration of labour, and inadequate storage facilities continue to affect agricultural development in the state.
- Bihar is also known for horticulture and fruit production. Crops like litchi, banana, mango, and makhana have gained national recognition. The principal food grains dominating agricultural production are rice, wheat, and maize.

BCW Bits:

- Bihar lies mainly in the fertile Indo-Gangetic alluvial plain region.
- Hybrid maize cultivation has increased rapidly in north Bihar.
- Muzaffarpur is famous for Shahi Litchi production.

- Bihar is one of the leading producers of makhana in India.
- Floods caused by rivers like Kosi are a major agricultural challenge in north Bihar.
- South Bihar receives comparatively lower rainfall than north Bihar.
- Major irrigation sources in Bihar include canals, tube wells, ponds, and rivers.
- Jute cultivation in Bihar is limited mainly to northeastern districts.
- Agriculture contributes significantly to Bihar's employment despite gradual growth of the service sector.

Bihar Special Miscellaneous

1. Which ancient site is the UNESCO World Heritage Site present in the Bihar state of India? ***[BPSC ASO, Mains]***

A. Tomb of Sher Shah Suri
B. Ancient site of Vikramshila Monastery
C. Mahabodhi Vihar
D. Kumhrar, Patna

Ans. C

Exp: Bihar is home to **two** UNESCO World Heritage Sites: the **Mahabodhi Temple Complex** in Bodh Gaya (inscribed in **2002**) and the **Archaeological Site of Nalanda Mahavihara** (inscribed in **2016**). Mahabodhi Vihar is the holiest Buddhist site, marking the location where Siddhartha Gautama attained enlightenment.

- **Tomb of Sher Shah Suri:** A magnificent Indo-Islamic mausoleum in Sasaram. While a National Monument, it is not a UNESCO World Heritage site.
- **Ancient site of Vikramshila Monastery:** An 8th-century Buddhist university founded by King Dharmapala. It is an important archaeological site, but not UNESCO-designated.
- **Kumhrar, Patna:** Contains the archaeological remains of the ancient Mauryan capital, Pataliputra, including an 80-pillared hall. It is a major historical site but lacks UNESCO status.

2. Under the One District One Product (ODOP) initiative, which of the following districts of Bihar is incorrectly matched with its designated product? ***[BPSC ASO, Mains]***

A. Gaya – Jaggery
B. Muzaffarpur – Litchi
C. Nalanda – Makhana
D. More than one of the above

Ans. D

Exp: The **One District One Product (ODOP)** initiative is a central-sector vision aligned with the **Atmanirbhar Bharat** mission. It focuses on selecting, branding, and promoting one unique product from each district to transform them into export hubs.

Core Institutional Framework:

- **Launch Year:** The nationwide ODOP initiative was formally integrated into the **PM Formalisation of Micro Food Processing Enterprises (PMFME)** scheme in **2020**.
- **Nodal Ministries:**
  - **Ministry of Food Processing Industries (MoFPI):** Manages the food-based ODOPs (which cover most of Bihar's districts).
  - **Department for Promotion of Industry and Internal Trade (DPIIT):** Handles non-food items, including handicrafts, handlooms, and industrial products.
- State **Nodal Agency (Bihar):** Department of Industries, Government of Bihar.

Key Provisions and Features:

- **Financial Support:** Under the **PMFME scheme**, micro-enterprises receive a **35% credit-linked capital subsidy** (maximum ₹10 lakh) for upgrading technology and infrastructure.
- **Infrastructure Support:** Focus on creating **Common Infrastructure** like cold storage, testing labs, and processing centers within the district.
- **Market Integration:** The initiative provides support for branding and packaging to help local products meet international standards. It also facilitates integration with e-commerce platforms like **GeM (Government e-Marketplace)**.
- **Selection Criteria:** Products are chosen based on their existing ecosystem, historical presence, and potential for export and employment generation.

One District One Product List of Bihar:

BIHAR		
Sl. No.	**District**	**ODOP**
1	Araria	Makhana (Foxnut)
2	Arwal	Mango
3	Aurangabad	Strawberry
4	Banka	Katarni Rice
5	Begusarai	Chilly
6	Bhagalpur	Jardalu Mango
7	Bhojpur	Pea
8	Buxar	Mentha
9	Darbhanga	Makhana (Foxnut)
10	East Champaran	Litchi
11	Gaya	Mushroom
12	Gopalganj	Papaya

13	Jamui	Jackfruit
14	Jehanabad	Mushroom
15	Kaimur	Guava
16	Katihar	Makhana (Foxnut)
17	Khagaria	Banana
18	Kishanganj	Pineapple
19	Lakhisarai	Tomato
20	Madhepura	Mango
21	Madhubani	Makhana (Foxnut)
22	Munger	Lemon Grass (Aromatic Plant)
23	Muzaffarpur	Litchi
24	Nalanda	Potato
25	Nawada	Betel Vine
26	Patna	Onion
27	Purnia	Banana
28	Rohtas	Tomato
29	Saharsa	Makhana (Foxnut)
30	Samastipur	Turmeric
31	Saran	Tomato
32	Sheikhpura	Onion
33	Sheohar	Moringa
34	Sitamarhi	Litchi
35	Siwan	Mentha
36	Supaul	Makhana (Foxnut)
37	Vaishali	Honey
38	West Champaran	Sugarcane Products
TOTAL	**38**	

Based on the official **One District One Product List of Bihar** provided in the document; we can evaluate the matches as follows:

A. Gaya – Jaggery: This is **incorrectly matched**. According to the official list (Sl. No. 11), the designated product for **Gaya** is **Mushroom**.

B. Muzaffarpur – Litchi: This is **correctly matched**. The list (Sl. No. 23) confirms **Muzaffarpur** is designated for **Litchi**.

C. Nalanda – Makhana: This is **incorrectly matched**. The list (Sl. No. 24) designates **Potato** for **Nalanda**. **Makhana (Foxnut)** is assigned to other districts such as Araria, Darbhanga, and Katihar, Madhubani, Saharsa and Supaul.

Since both options A and C are incorrect based on the provided list, the answer is **D More than one of the above**.

3. Which Committee was formed to establish Veterinary College in Patna in 1927? ***[BPSC AEDO]***

A. Nathan Committee
B. Irwin Committee
C. Ganesh Dutt Committee
D. Syed Fakruddin Committee

Ans. C

Exp: Following the recommendations of the Ganesh Dutt Committee, Bihar veterinary college was established on April 2, 1927. Sir Syed Mohammad Fakhruddin (1868–1933), an important education reformer and Education Minister of Bihar and Orissa (1921–1933), played an important role in establishing this veterinary college at Patna. He also actively recruited qualified teachers from outside to improve educational standards.

4. As per 2011 Census, density of population per sq. km. in Bihar state was: ***[BPSC AEDO]***

A. 1025 B. 1076 C. 1081 D. 1106

Ans. D

Exp: According to the Census of India 2011, Bihar had a population density of **1106 persons per sq. km.**, making it one of the most densely populated states in India. Population density refers to the number of people living in one square kilometre area. It is an important demographic indicator because it helps us understand population pressure on land and resources.

- Population density is calculated using the formula: Population Density=Total Population/Total Area In Bihar's case, a very large population is concentrated within a relatively small geographical area. This creates high pressure on agriculture, landholding, employment, housing, education, healthcare, and infrastructure. Since Bihar is primarily an agrarian state with fertile alluvial plains, people historically settled densely along river valleys and agricultural regions.
- The 2011 Census recorded Bihar's population at more than 10 crore people. Despite being smaller in area compared to many other Indian states, Bihar had an extremely high concentration of population. This is why its density crossed 1100 persons per sq. km.
- Several geographical and historical factors contribute to Bihar's high density. The state lies in the fertile Indo-Gangetic plain. Rivers like the Ganga River, Kosi, Gandak, and Bagmati provide fertile alluvial soil suitable for agriculture. Fertile plains historically attract settlement because agriculture supports large populations. Additionally, Bihar has traditionally experienced high birth rates compared to many southern Indian states.
- High population density has both advantages and disadvantages. A dense population can provide an abundant labour force and a large consumer market, which may support industrial and economic growth. However, excessive density can also lead to unemployment,

poverty, pressure on natural resources, urban congestion, migration, and environmental stress. Bihar's large-scale migration to other states for employment is partly linked to high population pressure and limited industrialization.

- Demographic transition and development indicators: States with better literacy, healthcare, and urbanization generally experience lower population growth rates over time. Kerala and Tamil Nadu, for example, have lower fertility rates compared to Bihar. Therefore, population density questions are often indirectly connected with literacy, urbanization, migration, sex ratio, and economic development.
- In India, population density varies greatly across states. Bihar and West Bengal are among the most densely populated large states, while states like Arunachal Pradesh have very low density due to mountainous terrain, forest cover, and sparse settlement.

BCW Bits:

- Bihar became India's third most populous state after Maharashtra according to Census 2011.
- The least densely populated Indian state is Arunachal Pradesh.
- Delhi has one of the highest population densities among Union Territories.
- Census in India is conducted every 10 years under the Census Act of 1948.
- The first complete synchronous Census in India was conducted in 1881.
- Population density is expressed as persons per square kilometre.
- Bihar's high density is strongly linked with fertile alluvial plains and dependence on agriculture.
- Urbanization level in Bihar has historically remained lower than the national average.
- District-wise density in Bihar is highest in heavily urbanized and river plain regions.

Bihar Polity

1. The Elementary Education Act was implemented in Bihar and Orissa in 1919 in which compulsory education for children of what age was passed? ***[BPSC AEDO]***

   A. 6 – 10 years B. 5 – 7 years
   C. 4 – 8 years D. 3 – 5 years

Ans. A

Exp: The Elementary Education Act of 1919 in Bihar and Orissa aimed to promote compulsory primary education for children. Children between 6 to 10 years were brought under compulsory education.

- During the early 20th century, the British administration began expanding primary education policies **Literacy** levels were very low, social reformers and nationalist leaders were demanding wider education access, Education for common people.
- At that time Bihar and Orissa were part of a **single province** under British rule, Separate Bihar province came later.
- The Act tried to Increase school enrolment, reduce illiteracy, Make elementary education more systematic. But implementation remained limited due to poor infrastructure, a lack of schools, poverty, and social backwardness.
- It showed historical importance by the Early movement toward **compulsory education in India.** Growing realization that education was necessary for Administration, Social reform, Political awareness.
- **Before Independence the British** introduced several education-related measures, **Wood's Despatch (1854)**-"Magna Carta of English Education", **Hunter Commission (1882)** -focus on primary education, **Sadler Commission (1917)**-university reforms. Indian leaders believed Education should not only create clerks for British administration. It should build National consciousness, Scientific thinking, Social reform.

BCW Bits:

- Literacy in colonial India remained extremely low till independence
- Girls' education was especially neglected in many regions
- Mahatma Gandhi later proposed **Basic Education (Wardha Scheme)**
- Compulsory education became a Constitutional goal after independence
- Today, education is a **Fundamental Right under Article 21A**
- The Right to Education Act (2009) covers children aged **6–14 years**
- Bihar later saw expansion of mass education after independence
- Colonial education policies often focused more on urban areas.

Bihar Economy

1. Consider the following statements: ***[BPSC DSO, Pre]***

   Assertion [A]: Bihar's economy has seen significant growth in recent years.

   Reason [R]: Infrastructure development, agriculture and human development are playing positive roles

   Select the correct answer from the following codes:

A. Both [A] and [R] are true and [R] is the correct explanation of [A].
B. Both [A] and [R] are true but [R] is not the correct explanation of [A].
C. [A] is false but [R] is true.
D. [A] is true but [R] is false.

Ans. A

Exp: Assertion [A]: Bihar's economy has seen significant growth in recent years. This statement is true. Bihar has indeed experienced notable economic growth, often recording high growth rates among Indian states in recent years.

- **Reason [R]:** Infrastructure development, agriculture and human development are playing positive roles. This statement is also true. These sectors are key drivers of economic growth, and their development in Bihar has contributed to the state's economic progress.
- **Relationship:** The reason (R) directly explains the assertion (A). The positive roles played by infrastructure, agriculture, and human development are precisely why Bihar's economy has grown significantly. Therefore, R is the correct explanation for A.

2. Consider the following statements: ***[BPSC AEDO]***

1. In 2019–20, the percentage growth rate of Bihar Gross State Domestic Product was more than the percentage growth rate of GDP of India.
2. In 2018–19, the percentage growth rate of Bihar Gross State Domestic Product was less than the percentage growth rate of GDP of India.
3. In 2023–24, as per quick estimate, the percentage growth rate of GSDP of Bihar was 9.2.

How many statements given above are correct?

A. Only 1 and 2 B. Only 3
C. All three D. Only 1 and 3

Ans. D

Exp: Gross Domestic Product: - Total value of all goods and services produced in the whole country in one year. Gross State Domestic Product: - GSDP is the same concept but for a **state**. Bihar's GSDP = Total production and services within Bihar.

- High growth ≠ rich state, even though Bihar sometimes records High GSDP growth, It still faces Low industrialization, High poverty, Low per capita income because growth started from a smaller economic base.
- Bihar recorded a higher GSDP growth rate than India's GDP growth rate in 2019–20.
- As per quick estimates, Bihar's GSDP growth rate in 2023–24 was about 9.2%.
- Bihar's GSDP growth rate in 2018–19 was about 10.5%, whereas India's GDP growth rate during the same period was around 6.8%. (For recent data, refer to our Current Affairs Magazine)
- Bihar's economy is highly dependent on the **agriculture sector.**
- Bihar's per capita income remains among the lowest in India.
- Migration plays a major role in Bihar's economy through remittances.
- Infrastructure spending has increased in recent years.
- The service sector contributes significantly to urban growth in Bihar.
- GSDP estimates are released in: Advance estimates, Quick estimates, and Final estimates.
- Economic growth can fluctuate due to: Floods, Monsoon dependence, and Agricultural output.

Indian and World Geography

1. According to the Indus water treaty, three eastern rivers are allocated to India. Which of the following is not one of them? *[71st BPSC]*
   A. Jhelum
   B. Ravi
   C. Sutlej
   D. More than one of the above

Ans. A

Exp: Under the Indus Waters Treaty (1960), the waters of the six main rivers of the Indus Basin were divided between India and Pakistan based on their geographical location (Eastern vs. Western).

- **The Division of Rivers:** The treaty categorizes the rivers into two distinct groups:
- **Eastern Rivers (Allocated to India):** India has unrestricted rights to the waters of the Ravi, Beas, and Sutlej.
- **Western Rivers (Allocated to Pakistan):** These include the Indus, Jhelum, and Chenab. India is allowed limited use of these rivers for domestic consumption, agriculture, and 'run-of-the-river' hydroelectric projects, but cannot store the water.

2. Which of the following is the place of confluence of the Alaknanda and Bhagirathi? *[71st BPSC]*
   A. Vishnu Prayag
   B. Karan Prayag
   C. Rudraprayag
   D. Devprayag

Ans. D

Exp:

- Vishnuprayag: Alaknanda meets the Dhauliganga.
- Nandprayag: Alaknanda meets the Nandakini.
- Karnaprayag: Alaknanda meets the Pindar River.
- Rudraprayag: Alaknanda meets the Mandakini.
- Devprayag: Alaknanda meets the Bhagirathi (Formation of Ganga).

3. The two volcanic Islands in the Indian territory are- *[71st BPSC]*
   A. Kavaratti and new Moore
   B. Bitra and Kavaratti
   C. Pamban and barren
   D. Narcondam and Barren

Ans. D

Exp:

Barren Island:

- It is located about 135 km northeast of Port Blair.
- It is the only active volcano in South Asia.
- Its first recorded eruption was in 1787, and it has erupted several times since then most recently in 2017.

Narcondam Island:

- It is located further north of Barren Island.
- It is classified as a dormant volcano, meaning it hasn't erupted in recent history but could potentially become active again.
  - **Kavaratti:** This is a coral island and the capital of Lakshadweep.
  - **New Moore:** This was a small uninhabited sandbar island in the Bay of Bengal that has since been submerged due to sea-level rise.
  - **Pamban:** Also known as Rameswaram Island, it is a limestone island located between India and Sri Lanka.
  - **Bitra:** The smallest inhabited island in Lakshadweep, which is of coral origin.

4. Loktak is a- *[71st BPSC]*
   A. Valley
   B. Lake
   C. River
   D. Mountain series

Ans. B

Exp: **The Loktak Lake** is situated in Manipur. It is the largest freshwater lake in the North–East India known for its floating swamps called **Phumdi's**which look like an island and create the world's only floating National Park called **'KeibulLamjao National Park'** which is home to 'Sangai Deer' or Dancing Deer (IUCN status: Endangered).

5. Karmanasa river originate in: *[71st BPSC]*
   A. Kaimur district
   B. Vaishali district
   C. Patna district
   D. Bikram district

Ans. A

Exp: The Karmanasa River is a significant tributary of the Ganges in Bihar and Uttar Pradesh.

- Origin: It originates on the northern face of the Kaimur Range (near Sarodag) in the Kaimur district of Bihar.
- Course: It flows through the states of Bihar and Uttar Pradesh, acting as a natural boundary between the two for a considerable distance.
- Confluence: It joins the Ganges near a place called Chausa (Buxar district).
- Religious Significance: The name 'Karmanasa' literally means 'destroyer of religious merit' (Karma = deed, Nasa = destruction). In ancient times, it was considered an inauspicious river, and people believed that touching its water would destroy all their good deeds.

6. Which of the following statement/s is/are correct regarding the Indian cement industry? ***[71st BPSC]***

1. Around 87% of India's cement industry is concentrated in a few major states like Rajasthan, Andhra Pradesh, and Madhya Pradesh.
2. India is the largest producer of cement in the world.
3. Most cement plants in India are located far from raw material sources near to the port to support export.

Select the correct answer using the codes given below:

A. 1
B. 2
C. 1 and 3
D. More than one of the above

Ans. A

Exp:

Statement 1: Correct

- The cement industry in India is highly resource-localized. Because limestone (the primary raw material) is heavy and weight-losing during production, plants are clustered where limestone is abundant. Rajasthan, Andhra Pradesh, and Madhya Pradesh are among the top producers, collectively holding a massive share of the country's total capacity.

Statement 2:Incorrect

- India is actually the second-largest producer of cement in the world, accounting for over 7% of global installed capacity. China remains the undisputed largest producer by a significant margin.

Statement 3: Incorrect

- Cement is a 'weight-losing' industry. It takes about 1.6 tons of limestone and coal to produce 1 ton of cement. Therefore, most plants are located near the raw material sources (limestone quarries) to minimize transportation costs. While some plants exist near to ports to support export (like in Gujarat), the vast majority are inland near mineral belts.
  - **Global Rank** - 2nd (after China)
  - **First Plant** - Established in 1904 at Chennai (Madras)
  - **Raw Materials** - Limestone (primary), Silica, Alumina, and Gypsum
  - **Top States** - Rajasthan (Highest capacity), Andhra Pradesh, Gujarat, Madhya Pradesh.

7. The second largest river basin of India is- ***[71st BPSC]***

A. Mahanadi River basin
B. Narmada river basin
C. Godavari River basin
D. Kaveri River basin

Ans. C

Exp: The **Godavari** is India's second-longest **river** after the Ganges **River** it is known as the Dakshin Ganga.

- 'Dakshin Ganga' because it is the largest river of Peninsular India. The river rises in the Sahyadris, near Trimbakeshwar in the Nashik district of Maharashtra and flows across the Deccan Plateau from the Western to the Eastern Ghats and finally drains into the Bay of Bengal in Andhra Pradesh. Pranhita River is its largest tributary which is formed by the confluence of three river, Penganga, Wainganga and Wardha.
- The **Krishna River** is the third-longest in India, after the Ganges and Godavari. It is also the fourth-largest in terms of river basin area in India, after the Ganges, Indus and Godavari.

8. Which organisation recognized the Nagi bird sanctuary Site as an Important Bird and Biodiversity Area (IBA)? ***[71st BPSC]***

A. UNESCO
B. World Wildlife Fund
C. Bird Life International
D. More than one of the above

Ans. C

Exp: The Nagi Bird Sanctuary, located in the Jamui district of Bihar, India, is a critical site for avian conservation.

BirdLife International & IBAs

- The Important Bird and Biodiversity Area (IBA) designation is a global initiative led by BirdLife International. This organization uses a set of internationally agreed-upon criteria to identify sites that are critical for the long-term viability of bird populations.
- **Nagi & Nakti:** Both the Nagi and Nakti bird sanctuaries were recognized as IBAs due to their importance as wintering grounds for migratory birds, particularly those traveling along the Central Asian Flyway.
- **While BirdLife International designated it an IBA, it is worth noting a more recent milestone:** In 2024, the Nagi and Nakti Bird Sanctuaries were officially designated as Ramsar Sites (Wetlands of International Importance). This designation falls under the Ramsar Convention, which is often associated with UNESCO, but the IBA tag itself remains a BirdLife International program.

Biodiversity Profile

- **Migratory Species:** The sanctuary hosts species like the Bar-headed Goose, Northern Pintail, and Red-crested Pochard.
- **Dry-Land Habitat:** Unique among many bird sanctuaries, Nagi is surrounded by rocky hillocks and dry deciduous forests, providing a diverse ecosystem for both aquatic and terrestrial birds.

9. The UN climate conference held in Baku in 2024 concluded with an agreement that: ***[71st BPSC]***

A. Aims to ban fossil fuels globally by 2030.
B. Calls on developed countries to provide at least $500 billion annually to developing nations by 2030.
C. Calls on developed countries to provide at least $300 billion annually to developing nations by 2035.
D. More than one of the above

Ans. C

Exp: COP29 (29th Conference of the Parties to the UNFCCC) was held in Baku, Azerbaijan, in November 2024. Often dubbed the 'Finance COP, ' its primary goal was to establish a new financial target to replace the previous $100 billion annual commitment.

The center-piece of the Baku agreement was the New Collective Quantified Goal (NCQG).

- **The Core Target:** Developed countries committed to taking the lead in mobilizing at least $300 billion per year for climate action in developing nations by 2035.
- **The Larger Ambition:** The agreement also set a broader, 'aspirational' goal to scale up total climate finance (including private and public sources) to $1.3 trillion per year by 2035.
- **Tripling the Previous Goal:** The $300 billion target effectively triples the previous $100 billion goal set in 2009.
- **Carbon Markets:** Baku also saw the finalization of rules for Article 6 of the Paris Agreement, which governs international carbon markets, allowing for more structured trading of carbon credits.

10. Kanha national park is established in: ***[71st BPSC]***

A. Uttrakhand B. Assam
C. Madhya Pradesh D. Maharashtra

Ans. C

Exp: Kanha National Park, also known as Kanha Tiger Reserve, is the largest national park in Central India.

- **Location:** It is situated in the Maikal range of the Satpuras in the Mandla and Balaghat districts of Madhya Pradesh.
- **Species Recovery:** It is world-renowned for saving the Barasingha (Swamp Deer) from near extinction; it is the first tiger reserve in India to officially introduce a mascot, 'Bhoorsingh the Barasingha.'
- **Literary Connection:** The lush sal and bamboo forests of Kanha provided the inspiration for Rudyard Kipling's famous novel, The Jungle Book.
- **Conservation Status:** It was created as a wildlife sanctuary in 1933, declared a National Park in 1955, and became a Tiger Reserve in 1973 under Project Tiger.
- **Landscape:** The park is characterized by vast 'chaurs' (grassy meadows) and dense forests, making it one of the best places in the world to spot the Royal Bengal Tiger in the wild.

11. Which of the following parks/sanctuaries come under the project tiger in Bihar? ***[71st BPSC]***

A. Valmiki National Park
B. Udaipur Wildlife Sanctuary
C. Bhimbandh Wildlife Sanctuary
D. None of these

Ans. A

Exp: Project Tiger is a tiger conservation programme launched in **April 1973** by the Government of India during Prime Minister Indira Gandhi's tenure.

- The objective of **'Project Tiger'** was to save Royal Bengal tigers from becoming extinct.
- The theme of National Biodiversity Day in 2022 was **'Building a shared future for all life'.** It is celebrated on 22nd May.
- There are a total of **58 Tiger Reserves** in the country. In Bihar, there is only 1 Tiger Reserve at present **(Valmiki Tiger Reserve, West Champaran).**
- International Big Cat Alliance **(IBCA)** is a mega global alliance launched by India in April 2023 during the 50th anniversary of Project Tiger. The alliance aims at conservation of world's seven principal big cats, which include the tiger, lion, snow leopard, leopard, jaguar, puma, and cheetah.

12. The species which are in danger of extinction are known as? ***[71st BPSC]***

A. Vulnerable B. Rare
C. Endangered D. Extinct

Ans. C

Exp: Species are classified based on their risk of disappearance from the wild by the International Union for Conservation of Nature (IUCN) in its 'Red List of Threatened Species.'

- **Endangered Species:** These are species that are at a very high risk of extinction in the near future.
- **Key Criteria:** A significant decline in population (often 50–70% over 10 years), a very small geographic range, or a population size of fewer than 2,500 mature individuals.
- **Vulnerable:** Likely to become endangered unless the circumstances threatening its survival improve.
- **Rare:** Species with small total populations, often confined to specific geographical areas. Not always 'endangered, ' but easily become so.
- Endangered Species currently facing a very high risk of extinction in the world.
- **Extinct:** There is no reasonable doubt that the last individual has died.
- In technical conservation terms, the three categories of Vulnerable, Endangered, and Critically Endangered are collectively referred to as 'Threatened' species.

13. Ozone day is celebrated on: ***[71st BPSC]***

A. 16 September B. 5 June
C. 21 April D. 5 December

Ans. A

Exp: The World Ozone Day, officially known as the International Day for the Preservation of the Ozone Layer, is observed annually on 16th September to spread awareness about the depletion of the Ozone Layer and its impact on the environment.

- This date commemorates the signing of the Montreal Protocol in 1987. The Montreal Protocol is an international treaty designed to phase out the production and consumption of ozone-depleting substances (ODS) like Chlorofluorocarbons (CFCs).
  - **June 5:** World Environment Day (Led by UNEP)
  - **April 21:** National Civil Services Day (In India)
  - **December 5:** World Soil Day

14. In India, the Temperate Forest Research Centre is in which city? [71st BPSC]
A. Shimla B. Ranchi
C. Srinagar D. Dehradun

Ans. A

Exp: The Himalayan Forest Research Institute (HFRI), which serves as the primary temperate forest research center in India, is located in Shimla, Himachal Pradesh.
- Himalayan Forest Research Institute (Temperate) : Shimla
- Forest Research Institute (National HQ) : Dehradun
- Arid Forest Research Institute (AFRI) : Jodhpur
- Rain Forest Research Institute (RFRI) : Jorhat
- Tropical Forest Research Institute (TFRI) : Jabalpur

15. Ladang refers to: *[71st BPSC]*
A. Shifting agriculture
B. Plantation agriculture
C. Subsistence type of agriculture
D. Dry forming

Ans. A

Exp: Ladang is the specific name used for shifting cultivation (also known as slash-and-burn agriculture) in Indonesia and Malaysia.

Shifting Cultivation
- In this system, a piece of forest land is cleared by cutting down trees and burning them. The ashes act as a natural fertilizer. Crops are grown for a few years until the soil fertility declines, at which point the farmers move to a new patch of forest, allowing the old land to fallow and regrow.

Local Name	Region/Country
Ladang	Indonesia and Malaysia
Jhumming	North-East India
Milpa	Central America and Mexico
Roca	Brazil
Conuco	Venezuela
Masole	Congo (Central Africa)
Ray	Vietnam
Chengin	Philippines

Important Distinction
- **Plantation Agriculture:** Large-scale farming of a single cash crop (e.g., Tea, Coffee, Rubber) for commercial purposes.
- **Subsistence Agriculture:** Farming intended mainly to feed the farmer's family, with little or no surplus for trade.
- **Dry Farming:** A method of farming in semi-arid areas without the aid of irrigation, using drought-resistant crops.

16. Which of the following is a Kharif crop? *[71st BPSC]*
A. Sugar cane B. Barley
C. Gram D. Wheat

Ans. A

Exp: The crops which are sown in the **rainy season** are called Kharif crops. The rainy season in India is generally from June to September. **Paddy, maize, soybean, groundnut, sugarcane and cotton** are Kharif crops. Wheat is an example of the Rabi crop.

17. According to the UNDP human development index report 2025, life expectancy in India increased from 58.6 years in 1990 to How many years in 2023? *[71st BPSC]*
A. 72 years
B. 73 years
C. 74 years
D. More than one of the above

Ans. A

Exp: According to the UNDP Human Development Report 2025 (which is based on data collected through 2023), India's life expectancy at birth saw a significant rise from 58.6 years in 1990 to 72 years in 2023.

18. Atmosphere always has: *[BPSC ASO, Pre]*
A. Dust B. Air
C. Oxygen D. All of the above

Ans. D

Exp: The Earth's atmosphere is a complex mixture of gases and suspended particles that are held in place by gravity. While the proportions of these components can vary based on altitude and location, they are fundamental constituents of the atmospheric layers.

Composition of the Atmosphere
- **Air:** This is the primary substance of the atmosphere. It is a mechanical mixture of various gases, mainly Nitrogen (78%) and Oxygen (21%).
- **Oxygen:** As the second most abundant gas, it is essential for the survival of aerobic organisms and for the process of combustion.
- **Dust:** These are solid particles (aerosols) such as sea salt, fine soil, smoke, soot, and pollen. Dust particles are crucial because they act as hygroscopic nuclei around which water vapor condenses to form clouds.

19. Which of the following climatic zone lies around 40° to 60° latitude? *[BPSC ASO, Pre]*

A. Temperate B. Tropical
C. Arctic D. Subtropical

Ans. A

Exp: The Earth is divided into distinct latitudinal zones based on the intensity of solar radiation received. The region between 40° and 60° latitude in both the Northern and Southern Hemispheres is characterized by moderate temperatures and distinct seasonal changes, placing it firmly in the Temperate Zone.

Zone	Latitudinal Range	Characteristics
Tropical (Torrid)	0° to 23.5° N/S	High temperatures year-round; lies between the. Tropics of Cancer and Capricorn.
Subtropical	23.5° to 35°/40° N/S	Transitions between tropical and temperate; includes many of the world's major deserts.
Temperate	40° to 60°/66.5° N/S	Four distinct seasons; dominated by Westerly winds and cyclonic activity.
Arctic / Antarctic	66.5° to 90° N/S	Extremely cold; characterized by ice caps and tundra.

20. Which of the following is a medicinal plant used to treat certain type of cancer? [BPSC ASO, Pre]

A. Himalayan Birch B. Himalayan yew
C. Chir-pine D. Himalayan Oak

Ans. B

Exp: The Himalayan Yew (Taxus wallichiana) is a coniferous tree found in the temperate forests of the Himalayas. It is highly valued in modern medicine for its anti-cancer properties. The bark and needles of the Himalayan Yew contain a chemical compound called Taxol (or Paclitaxel). This compound is a powerful cytotoxic agent that inhibits cell division, making it highly effective in treating:

- Breast cancer
- Ovarian cancer
- Lung cancer

It is currently listed as Endangered on the IUCN Red List. Large-scale drying of its bark and needles for pharmaceutical companies has led to the death of many trees in regions like Himachal Pradesh and Arunachal Pradesh.

- **Himalayan Birch (Bhojpatra):** Historically used as paper for writing ancient scriptures; its bark has antiseptic properties but is not a primary cancer treatment.
- **Chir-pine:** Mostly used for timber and extracting turpentine oil and resin.
- **Himalayan Oak:** Vital for soil moisture and local fodder, but not used for specialized cancer drugs.

21. Which of the following is a primary treatment for water pollution? *[BPSC ASO, Pre]*

A. Sedimentation B. Sludge method
C. Removal of nitrates D. Trickling filter method

Ans. A

Exp: Water treatment is generally divided into three main stages: Primary, Secondary, and Tertiary. Primary treatment is a mechanical process designed to remove large, solid matter from sewage.

1. Primary Treatment (Physical Process): This is the first step in wastewater treatment. It focuses on the removal of physical particles through:
   - Screening: Removing large floating objects like rags, sticks, and plastics.
   - Sedimentation: Allowing the water to sit in large tanks so that heavier solids (grit and sand)
   - sink to the bottom as primary sludge, while grease and oil float to the top to be skimmed off.
2. Secondary Treatment (Biological Process):
   - The Sludge method (specifically Activated Sludge) and the Trickling filter method are parts of Secondary treatment.
   - These processes use microorganisms (bacteria) to break down organic matter that is dissolved in the water after the sedimentation process is complete.
3. Tertiary Treatment (Chemical/Advanced Process):
   - The Removal of nitrates and phosphates is part of Tertiary treatment.
   - This stage involves chemical or advanced biological processes to remove specific nutrients and contaminants that can cause 'Eutrophication' (excessive algae growth) in water bodies.

22. Which one of the following is a wrong statement? *[BPSC ASO, Pre]*

A. Most of the forest have been lost in tropical areas.
B. Greenhouse effect is a natural phenomena
C. Eutrophication is a natural phenomena in freshwater bodies
D. Ozone in upper part of atmosphere is harmful to animals

Ans. D

Exp: Ozone in the upper part of the atmosphere is harmful to animals (WRONG):

- Ozone in the stratosphere (upper part of the atmosphere) is actually 'Good Ozone.' It acts as a protective shield that absorbs harmful ultraviolet (UV) radiation from the sun. Without it, life on Earth would suffer from skin cancer, cataracts, and immune system damage. Ozone is only considered 'harmful' or 'pollutant' when it is in the troposphere (lower atmosphere), where it can damage respiratory systems.

- Most of the forest have been lost in tropical areas (CORRECT):
- Deforestation is significantly higher in tropical regions (like the Amazon or Southeast Asia) due to agricultural expansion, logging, and cattle ranching compared to temperate regions.
- Greenhouse effect is a natural phenomenon (CORRECT):
- The greenhouse effect is essential for life. Without natural greenhouse gases (like CO_2 and water vapor), the Earth's average temperature would be roughly -18°C instead of the comfortable 15°C it is today. It only becomes a problem ('Global Warming') when human activities increase these gas concentrations excessively.
- Eutrophication is a natural phenomenon in freshwater bodies (CORRECT):
- While we often hear about 'cultural eutrophication' caused by fertilizer runoff, eutrophication itself is a natural aging process of a lake. Over centuries, a lake naturally gathers nutrients and silt, eventually turning into a marsh and then dry land.

23. Which of the following rivers is related to the Polavaram project? ***[BPSC ASO, Mains]***

A. Ganga B. Yamuna
C. Godavari D. Kaveri

Ans. C

Exp: The **Polavaram Project** (officially the Indira Sagar Polavaram Project) is a multi-purpose national irrigation project located on the **Godavari River** in the Eluru and East Godavari districts of **Andhra Pradesh**.

Key Facts:

- **National Status:** It was accorded **"National Project Status"** under the Andhra Pradesh Reorganisation Act, 2014.
- **Purpose:** It is designed to provide irrigation benefits to nearly 3 lakh hectares, generate **960 MW of hydropower**, and supply drinking water to Visakhapatnam and surrounding villages.
- **Inter-basin Transfer:** A unique feature of this project is the diversion of surplus Godavari water to the **Krishna River** basin via the Right Canal.

Other River Projects:

- ○ **Ganga:** Features the **Farakka Barrage** (West Bengal) and **the Tehri Dam** (on the Bhagirathi, Uttarakhand).
- ○ **Yamuna:** Features the **Lakhwar-Vyasi Dam** (Uttarakhand) and **Hathni Kund Barrage** (Haryana).
- ○ **Kaveri:** Features the **Mettur Dam** (Tamil Nadu) and **Krishna Raja Sagara Dam** (Karnataka).

24. Which of the following rivers is related to Jog Fall? ***[BPSC ASO, Mains]***

A. Godavari B. Kaveri
C. Sharavati D. Son

Ans. C

Exp: **Jog Falls**, also known as Gerosoppa Falls, is a spectacular segmented waterfall located in the **Shivamogga** district of Karnataka, where the **Sharavati River** plunges approximately **253 meters (830 feet)** in four distinct cascades named Raja, Rani, Roarer, and Rocket.

- While Jog Falls is one of India's most famous plunge falls, **Kunchikal Falls** (455 meters) in Karnataka is the tallest in India, and **Angel Falls** in Venezuela (979 meters) holds the title of the world's tallest waterfall.
- In **Bihar**, the most prominent waterfalls include **Kakolat** in the Nawada district, **Karkat** and **Telhar Kund** in Kaimur, **Kashish,** Tutla **Bhawani**, **Manjhar Kund**, and **Dhua Kund** in the Rohtas district. **Kashish Waterfall** is the **highest waterfall in Bihar**.

25. What are the bases of agro-climatic regionalization proposed by the Planning Commission of India? ***[BPSC ASO, Mains]***

A. Relief, soil and drainage
B. Rainfall, temperature and cropped area
C. Vegetation, soil and land use
D. Relief, land use and Net Sown Area

Ans. B

Exp: In 1989, the Planning Commission divided India into **15 agro-climatic regions**. The primary objective was to maximise agricultural productivity through regional planning. The classification is based on physical factors like **rainfall** and **temperature**, alongside socio-economic variables like **cropped area**, land use, and water resources, to ensure sustainable development tailored to each region’s specific ecology.

- According to the Planning Commission’s classification, Bihar (along with eastern Uttar Pradesh) falls under **Zone 4: Middle Gangetic Plains Region**. This region is characterised by fertile alluvial soil and a sub-tropical climate.

26. The Bundelkhand Plateau covers which two parts of the following states? ***[BPSC ASO, Mains]***

A. Chhattisgarh and Jharkhand
B. Uttar Pradesh and Madhya Pradesh
C. Chhattisgarh and Uttar Pradesh
D. Madhya Pradesh and Chhattisgarh

Ans. B

Exp: India's physiography is broadly classified into the Northern Mountains, the Northern Plains, the Peninsular Plateau, the Coastal Plains, and the Islands. Within the **Peninsular Plateau**, a massive tableland composed of old crystalline rocks, the **Bundelkhand Plateau** spans parts of **Uttar Pradesh and Madhya Pradesh**.

- Furthermore, while the plateau dominates the southern landscape, the state of **Bihar lies in the Middle Gangetic Plain**, a fertile region of the Northern Plains characterised by its deep alluvial deposits and significant agricultural importance.

27. Given below are two statements: one labelled as Assertion (A) and the other labelled as Reason(R):

Assertion (A): The Eastern Coast of India is affected by tropical cyclones more than the Western Coast.

Reason (R): Tropical cyclone originates only in the Bay of Bengal.

In the context of the above two statements, which one of the following is correct? ***[BPSC ASO, Mains]***

A. Both A.and (R) are correct, but (R) does not explain (A)

B. Both A.and (R) are correct, and (R) explains (A)

C. (A) is correct, but (R) is false

D. (A) is false, but (R) is correct

Ans. C

Exp: India's long coastline faces a severe threat from tropical cyclones, which are massive, spinning storms that form over warm ocean waters, bringing destructive winds and heavy rainfall. However, this danger is highly unequal across the subcontinent. The Eastern Coast, bordering the Bay of Bengal, is a major hotspot that gets hit much harder and more frequently by intense storms. In contrast, the Western Coast, along the Arabian Sea, faces a significantly lower risk, experiencing rarer and generally weaker cyclones.

Types and Classification of Cyclones:

- The storms affecting the region are classified as tropical cyclones, which are categorized by the India Meteorological Department (IMD) into distinct levels based on their wind speeds. These range from moderate **Cyclonic Storms** to highly destructive **Severe and Very Severe Cyclonic Storms**. At the most extreme end are **Super Cyclones**, which possess immense energy and cause catastrophic coastal damage, massive storm surges, and widespread flooding.

The Bay of Bengal: Heat and Rivers

- The main reason for the high frequency of severe cyclones on the East Coast is the unique environment of the Bay of Bengal. Its waters are naturally warmer, providing the heat energy needed to fuel and grow these storms. Additionally, massive rivers like the Ganges and Brahmaputra dump huge amounts of fresh water into the Bay. This lighter fresh water sits on top of the salty ocean water like a blanket, preventing deeper, colder water from rising to the surface. This keeps the surface hot, constantly feeding passing storms with the energy they need to rapidly intensify.

The Arabian Sea: Winds and Cool Water

- Conversely, the environment of the Arabian Sea acts as a natural deterrent to storm growth. It features higher salinity and is exposed to stronger monsoonal winds that constantly stir the ocean. This stirring brings cold water up from the deep to the surface, cooling the sea down and depriving potential cyclones of thermal energy. Furthermore, the East Coast is perfectly positioned to catch the remnants of typhoons that travel all the way from the Pacific Ocean. These storms cross over Southeast Asia into the Bay of Bengal, where they quickly regain strength over the warm, quiet waters—a phenomenon that rarely reaches far enough west to affect the Arabian Sea.
- In summary, India's East Coast faces a much higher risk of severe cyclones than the West Coast because the Bay of Bengal is warmer, less stirred by winds, and layered with a freshwater blanket that traps heat; however, tropical cyclones do not originate *only* in the Bay of Bengal, as they also form in the Arabian Sea, even though those storms are generally rarer and weaker. Hence, the assertion is correct and the reason is false.

Therefore, the correct option is **(C) (A) is correct, but (R) is false**.

28. Kufri is a variety of ***[BPSC ASO, Mains]***

A. Potato B. Tomato

C. Brinjal D. Radish

Ans. A

Exp: "Kufri" refers to a series of potato varieties developed by the **Central Potato Research Institute (CPRI)**, which is headquartered in Kufri, Himachal Pradesh. Popular varieties include *Kufri Jyoti*, *Kufri Chandramukhi*, and *Kufri Bahar*. These varieties are specifically bred to be disease-resistant and suitable for different agro-climatic zones in India.

29. Which National Park is the new home of the Indian one-horned rhinoceros? ***[BPSC ASO, Mains]***

A. Kanha B. Corbett

C. Dudhwa D. Bandhavgarh

Ans. C

Exp: The Indian one-horned rhinoceros (**Rhinoceros unicornis**) is protected under **Schedule I of the Wildlife Protection Act (1972)**, ensuring the highest legal safeguards against poaching and habitat loss. To mitigate the risks of localized extinction in Assam, the **Rhino Reintroduction Programme** was initiated in 1984, translocating rhinos from Pobitara and Nepal to the **Sonaripur and Belrayan ranges** of **Dudhwa National Park**.

Rhino Reintroduction Programme:

- This movement established a critical "satellite population" within the **Terai-Arc Landscape**, where the rhino serves as an **umbrella species**, indirectly protecting sympatric wildlife such as the Swamp Deer and Bengal Florican through habitat preservation.
- Conservation efforts are further structured by the **Indian Rhino Vision 2020 (IRV 2020)**, a strategic scheme

launched by the Assam Forest Department in partnership with the WWF and International Rhino Foundation.

- The objective of this vision was to increase the total rhino population to at least **3, 000** across seven protected areas, ensuring genetic diversity and long-term viability.
- By expanding the species' range into Dudhwa, the program fulfills the **National Wildlife Action Plan's** mandate for meta-population management, shielding the species from catastrophic stochastic events like floods or disease outbreaks common in more concentrated habitats.

30. Match List-I with List-II and select the correct answer using the codes given below the lists: ***[BPSC ASO, Mains]***

List-I (Feature)		List-II (State)	
(a)	Most urbanised state	1.	Bihar
(b)	State having largest urban population	2.	Arunachal Pradesh
(c)	Most densely populated state	3.	Maharashtra
(d)	Least densely populated state	4.	Goa

Codes:
A. a-1, b-2, c-3, d-4 B. a-4, b-3, c-2, d-1
C. a-3, b-2, c-1, d-4 D. a-4, b-3, c-1, d-2

Ans. D

Exp: According to the official Census of India 2011, **Goa** is the most urbanised state by percentage (62.17%), while **Maharashtra** holds the largest total urban population. **Bihar** is the most densely populated state with 1, 106 persons per square kilometre, whereas **Arunachal Pradesh** is the least densely populated, with only 17 persons per square kilometre.

31. In which year will the 16th Indian Census be conducted? ***[BPSC ASO, Mains]***

A. 2026 B. 2027 C. 2028 D. 2029

Ans: B

Exp: The **16th Indian Census** (the 8th since Independence) is officially scheduled to be conducted in **2027** at an approved cost of **₹11, 718.24 crore**.

- According to the **Press Information Bureau (PIB)**, this will be India's first-ever **fully digital census**, replacing traditional paper-based methods with mobile applications (using the 'Bring Your Own Device' model) and a central monitoring portal.
- The exercise will follow a two-phase structure: the **Houselisting and Housing Census** will take place between **April and September 2026**, followed by the **Population Enumeration** in **February 2027** (with a reference date of March 1, 2027).
- A major citizen-centric innovation is the introduction of a **Self-Enumeration** facility via a dedicated portal in 16 languages, allowing households to securely enter their own details.
- Furthermore, the 2027 Census will include **caste enumeration** for the first time since 1931 and will utilize the mascots "**Pragati**" and "**Vikas**" to promote inclusive participation.

32. Which among the following are true about Indus water treaty, 1960? ***[BPSC DSO, Pre]***

1. The treaty was signed between Jawaharlal Nehru and Field Marshal Ayub Khan at Lahore
2. The Treaty allowed India to generate hydroelectric power from the chenab, Indus and Jhelum rivers.
3. The permanent Indus commission formed shall have one Commissioner each from India and Pakistan

Select the correct answer using the codes given below:
A. 2 and 3 B. 1 and 3
C. All of the above D. None of the above

Ans. A

Exp: The Indus Water Treaty was signed on September 19, 1960, by Prime Minister Jawaharlal Nehru and President Ayub Khan in Karachi, not Lahore.

- Under the treaty, the Western Rivers (the Indus, the Jhelum, and the Chenab) were allocated to Pakistan. However, India was granted specific rights to use these waters for:
- Domestic use (non-consumptive).
- Agriculture.
- Run-of-the-river hydroelectric projects (projects that do not require large storage/dams that significantly alter the flow).
- The treaty established a Permanent Indus Commission (PIC) to serve as a channel of communication and to resolve disputes. It consists of:
- One Commissioner from India.
- One Commissioner from Pakistan.
- The commissioners are required to meet at least once a year, alternating between the two countries.

33. Lakshadweep is made up of how many islands? ***[BPSC DSO, Pre]***

A. 30 B. 36 C. 37 D. 38

Ans. B

Exp: Lakshadweep, India's smallest Union Territory, is an archipelago consisting of 36 islands in the Arabian Sea. It is famous for its coral reefs and crystal-clear lagoons.

Key Facts About Lakshadweep

- Total Islands: 36 islands (covering an area of approximately 32 sq. km).
- Inhabited Islands: Only 10 of these islands are inhabited (Kavaratti, Agatti, Amini, Kadmat, Kiltan, Chetlat, Bitra, Andrott, Kalpeni, and Minicoy).

- **Origin:** These are atolls, meaning they are coral islands formed on submerged volcanic peaks.
- **Capital:** Kavaratti.
- **Main Channels:** *The **9 Degree Channel** separates the island of Minicoy from the main Lakshadweep archipelago.
- The **11 Degree Channel** divides the group into the Amindivi Islands (north) and Cannanore Islands (south).

34. Maitri Setu, The bridge connecting the Indian State of Tripura with Bangladesh has been built over which river? ***[BPSC DSO, Pre]***

A. Brahmaputra B. Gomti
C. Feni D. Sumli

Ans. C

Exp: The Maitri Setu (Friendship Bridge) is a significant infrastructure project aimed at boosting trade and connectivity between Northeast India and Bangladesh.

- **The River:** It is built over the Feni River, which flows between South Tripura and Bangladesh.
- **Location:** The bridge connects Sabroom in Tripura (India) with Ramgarh in Bangladesh.
- **Length:** The bridge is approximately 1.9 kilometers long.
- **Significance:** It provides Tripura with access to the Chittagong Port in Bangladesh, which is only about 80 km from Sabroom. This has led to Tripura being called the 'Gateway to Southeast Asia.'

35. Which of the following rivers originate from north of greater Himalayas? ***[BPSC DSO, Pre]***

A. Ravi B. Jhelum C. Chenab D. Sutlej

Ans. D

Exp: The Sutlej is one of the few rivers that are antecedent, meaning they existed before the Himalayas were fully formed and cut deep gorges through the mountains as they rose.

- The Sutlej originates from the Rakas Lake (near Lake Mansarovar) in the Tibetan Plateau. This region lies to the north of the Greater Himalayan range (also known as the Trans-Himalaya region).
- **The Jhelum:** Originates from the Verinag Spring at the foot of the Pir Panjal range in the Kashmir Valley.
- **The Chenab:** Formed by the confluence of two streams, the Chandra and the Bhaga, which rise in the Upper Himalayas near the Bara-lacha La pass in Lahaul and Spiti.
- **The Ravi:** Originates in the Kullu Hills of Himachal Pradesh, near the Rohtang Pass.
- **The Trans-Himalayan Rivers:** There are three major rivers that originate north of the Greater Himalayas and cut through them:
  - The Indus
  - The Sutlej
  - The Brahmaputra (Tsangpo)

36. Which Himalayan range formed later? ***[BPSC DSO, Pre]***

A. Greater Himalayas B. Middle Himalaya
C. Shivalik D. Zaskar

Ans. C

Exp: In the geological timeline of the Himalayas, the ranges were formed in successive phases as the Indian Plate collided with the Eurasian Plate. The formation moved from north to south, meaning the northernmost ranges are the oldest and the southernmost are the youngest.

Chronological Order of Formation

- The Himalayas didn't pop up all at once; they rose in 'waves' over millions of years:
- **Greater Himalayas (Himadri):** Formed first, roughly 50–40 million years ago during the Eocene epoch. These are the highest and most continuous peaks.
- **Middle/Lesser Himalayas (Himachal):** Formed next, about 25–30 million years ago during the Miocene epoch.
- **Shivalik (Outer Himalayas):** Formed last, approximately 2–20 million years ago (Plio-Pleistocene period). They are composed of unconsolidated sediments brought down by rivers from the higher ranges.

37. Match the following and choose the correct answer by using codes given below: ***[BPSC DSO, Pre]***

River		Tributary	
A.	Ganga	1.	Chambal
B.	Yamuna	2.	Bhima
C.	Krishna	3.	Kosi
D.	Mahanadi	4.	Sheonath

Code:
A. A-3, B-1, C-2, D-4 B. A-4, B-2, C-1, D-3
C. A-3, B-2, C-1, D-4 D. A-2, B-1, C-4, D-3

Ans. A

Exp: **Ganga - Kosi,** The river Kosi is known as the 'Sorrow of Bihar' due to its frequent course changes and flooding.

- **Yamuna-** Chambal, The Chambal is the most significant tributary of the Yamuna, famous for the 'Chambal Ravines.'
- **Krishna -** Bhima, The Bhima River is a major tributary that flows through Maharashtra and Karnataka before joining the Krishna.
- Mahanadi - Sheonath, The Sheonath is the first major tributary of the Mahanadi, joining it in Chhattisgarh.
- **Ganga Left Bank Tributaries:** Ramganga, Gomti, Ghaghara, Gandak, Kosi.
- **Ganga Right Bank Tributaries:** Yamuna (largest), Son, Punpun.
- **Yamuna Tributaries:** Chambal, Sind, Betwa, Ken.
- **Peninsular Rivers:** Krishna (Bhima, Tungabhadra, Koyna) and Mahanadi (Sheonath, Hasdeo, Mand, Tel).

38. Which river drains into Arabian Sea? *[BPSC DSO, Pre]*
A. Mahanadi B. Godavari C. Narmada D. Yamuna

Ans. C

Exp: **The Narmada:** Originating from Madhya Pradesh's Amarkantak Plateau, it is a west-flowing river emptying into the Arabian Sea. It forms the Narmada Valley and is culturally significant in Hindu traditions.

Info Bits:

- **The Mahanadi:** Originating in Chhattisgarh, it flows eastward into the Bay of Bengal. Known for the Hirakud Dam, it is vital for irrigation, hydroelectric power, and agriculture in Odisha and Chhattisgarh.
- **Godavari:** The longest peninsular river, originating in Maharashtra and flowing southeast into the Bay of Bengal.
- **Yamuna:** A major tributary of the Ganga River; it flows eastward/southeastward to merge with the Ganga at Prayagraj, which eventually drains into the Bay of Bengal.

39. Ranchi Plateau is an example of: *[BPSC DSO, Pre]*
A. True peneplain
B. Uplifted peneplain
C. Downwarped peneplain
D. Incipient peneplain

Ans. B

Exp: The Ranchi Plateau, located in Jharkhand is a classic example of an uplifted peneplain. A peneplain is a low-relief plain that has been worn down by erosion over a vast period of time until it is almost level. In the case of Ranchi, this flat surface was later pushed upward by tectonic forces.

- **Peneplain** - A near-level surface produced by long-term erosion.
- **Uplifted Peneplain** - A peneplain raised by tectonic forces (e.g., Ranchi Plateau).
- **Piedmont Plain** - A plain at the foot of a mountain

40. Match list I with list II and Select the correct answer from the code given below the lists: *[BPSC DSO, Pre]*

List I (Waterfall)		List II (Location)	
(a)	Dudhsagar	1.	Karnataka
(b)	Barkana	2.	Odisha
(c)	Khandadhar	3.	Himachal Pradesh
(d)	Palani	4.	Goa

Code :
A. a-4, b-1, c-2, d-3 B. a-1, b-2, c-3, d-4
C. a-4, b-3, c-2, d-1 D. a-3, b-2, c-4, d-1

Ans. A

Exp:

A. Dudhsagar - 4. Goa - Located on the Mandovi River, it's one of India's tallest waterfalls.
B. Barkana - 1. Karnataka - Formed by the Seetha River in the Shimoga district.
C. Khandadhar - 2.Odisha - Located in the Sundargarh district; it is a famous horse-tail type waterfall.
D. Palani - 3.Himachal Pradesh - Located near Kullu, often visited for its scenic trek.

41. According to koppen's climatic classification which of the following type of climate found in India? *[BPSC DSO, Pre]*
A. Af B. Am C. Ds D. Cf

Ans. B

Exp: Under Koppen's climate classification system, India features several climate types, but Am (Tropical Monsoon Climate) is one of the most prominent, especially along the West Coast and parts of the North East.

Koppen's Major Climate Zones in India

- Koppen divided India into several distinct regions based on temperature and precipitation:
- **Amw:** Tropical Monsoon (Western Coast).
- **As:** Tropical Savanna with a dry summer (Coromandel Coast/Tamil Nadu).
- **Aw:** Tropical Savanna (Most of the Peninsular plateau).
- **BWhw:** Hot Desert (Rajasthan).
- **Cwg:** Monsoon type with dry winters (Ganga Plain).
- **E:** Polar type (Himalayan region).

42. Which of the following in India is not associated with the production of coal? *[BPSC DSO, Pre]*
A. Girdih B. Bokaro
C. Buladila D. Karanpura

Ans. C

Exp:

River valley coalfield	State	Major Coalfields
The Son River Valley Coalfield	Madhya Pradesh & Uttar Pradesh	The Singrauli mining area (Sidhi District) in Madhya Pradesh is famous for coal production. Sohagpur, Umaria, Tatapani, and Ramkola are other major coalfields in Madhya Pradesh.
The Mahanadi River Valley Coalfield	Chhatisgarh & Odisha.	Korba district, Vishrampur, Jhilmil and Chirmir (Ambikapur district) in Chhattisgarh; Talcher (Dhenkanal district) and Rampur-Hingir (Sambalpur) in Odisha are major mining fields.

The Damodar River Valley Coalfield	Jharkhand & West Bengal.	Jharia (Jharkhand) is the largest coal mining field. Other major coalfields in Jharkhand are Chandrapura, Bokaro, Giridih, Karanpura and Ramgarh. Raniganj (West Bengal) is also a major coalfield located in the Damodar Valley coalfield.

43. Erosion of top soil and Run-off fertilizers and pesticides passing into water bodies are the examples of ***[BPSC DSO, Pre]***

A. Negative and positive Pollution
B. Positive and negative pollution
C. Positive and positive pollution
D. Negative and negative pollution

Ans. A

Exp: Positive Pollution: This occurs when the productivity of the soil decreases due to the addition of undesirable substances. When you apply excess fertilizers and pesticides, you are adding chemicals that alter the soil chemistry and eventually 'run off' into water bodies.

- **Negative Pollution:** This occurs when the productivity of the soil decreases due to the removal of its vital components. Erosion of topsoil is the classic example because the most fertile, nutrient-rich layer of the earth is being stripped away.

44. Among the following one is not a greenhouse gas, That is: ***[BPSC DSO, Pre]***

A. Nitrogen oxide B. CFCs
C. SO_2 D. Methane

Ans. C

Exp: The main greenhouse gases rising are carbon dioxide, methane, nitrous oxide, hydrochlorofluorocarbons (HCFCs), hydro-fluorocarbons (HFCs) and ozone in the lower atmosphere.

- Hydrogen is **not** a greenhouse gas.
- The greenhouse effect is how **heat is trapped close to Earth's surface** by 'greenhouse gases.' These heat-trapping gases can be thought of as a **blanket wrapped around Earth,** keeping the planet toastier than it would be without them.

Note: Nitrous Oxide (N_2O) is also called **'Laughing gas'** and is a greenhouse gas and ozone-depleting substance too.

45. Which of the following has maximum contribution in greenhouse effect in the atmosphere? ***[BPSC DSO, Pre]***

A. Carbon dioxide B. Carbon mono oxide
C. Sulphur dioxide D. Nitrous oxide

Ans. A

Exp: The Greenhouse Effect is the process by which certain gases in the Earth's atmosphere trap heat, preventing it from escaping into space and thereby keeping the planet warm. While several gases contribute to this, their impact varies based on their concentration and 'Global Warming Potential' (GWP).

- CO_2 is considered the most significant greenhouse gas not because it is the most potent, but because of its sheer abundance and long atmospheric lifetime.
- Sources: Burning of fossil fuels (coal, oil, and gas), deforestation, and industrial processes.
- **Contribution:** It is responsible for approximately 60% to 75% of the human-caused greenhouse effect.
- **Carbon Monoxide (CO):** It is not a direct greenhouse gas. However, it influences the lifetime of other greenhouse gases like methane.
- **Sulphur Dioxide:** This actually has a cooling effect on the atmosphere. It forms aerosols that reflect sunlight back into space, though it is a major cause of acid rain.
- **Nitrous Oxide:** It is a very potent greenhouse gas (nearly 300 times more effective at trapping heat than carbon dioxide), but its concentration in the atmosphere is much lower, so its total contribution is smaller.

46. How many biodiversity hotspots are found in India? ***[BPSC DSO, Pre]***

A. 4 B. 8 C. 12 D. 16

Ans. A

Exp: India is home to four of the world's 36 recognized biodiversity hotspots. A 'hotspot' is a region that contains high levels of species endemism (species found nowhere else) but is also under significant threat from human activities.

The Four Hotspots in India

- **The Himalayas:** It includes the entire Indian Himalayan region (and falls into Pakistan, Tibet, Nepal, and Bhutan). It is famous for species like the Snow Leopard and Red Panda.
- **The Western Ghats:** A mountain range running parallel to India's western coast. It is one of the 'hottest hotspots' and is home to the Lion-tailed Macaque and many endemic amphibians.
- **Indo-Burma Region:** This region includes the entire North-Eastern India (excluding the Himalayas), Myanmar, Thailand, Vietnam, Laos, Cambodia, and southern China.
- **Sundaland:** This hotspot primarily covers South-East Asia but includes the Nicobar Islands of India.

Criteria for a Hotspot

- To qualify as a biodiversity hotspot according to Conservation International, a region must meet two strict conditions:

1. Species Endemism: It must contain at least 1, 500 species of vascular plants as endemics.
2. Degree of Threat: It must have lost at least 70% of its original habitat.

Key Fact:

- It is important not to confuse Biodiversity Hotspots (4) with Biogeographic Zones (10) or Mega-diverse countries (India is one of 17).

47. As per 2011 census, what is the state of India's child sex ratio? ***[BPSC DSO, Pre]***

A. 933 B. 943 C. 972 D. 919

Ans. D

Exp: According to the 2011 Census, the Child Sex Ratio (CSR) in India—defined as the number of females per 1, 000 males in the 0–6 year age group—showed a declining trend, dropping from 927 in 2001 to 919 in 2011.

48. In which state Toda tribe is found? ***[BPSC DSO, Pre]***

A. Jharkhand B. Madhya Pradesh
C. Rajasthan D. Tamil Nadu

Ans. D

Exp: **Todas Tribe** is from Tamil Nadu (Nilgiri Hills) and belongs to the **Dravidian ethnic group.**

Info Bits:

- **Bhil/Bheel: This** Tribe is an ethnic group in western India. As of 2013, Bhils were the largest tribal group.
- **Gaddi Tribe:** Himachal Pradesh and Jammu and Kashmir.
- **Kotas Tribe,** also **Kothar** or **Kov** are found in the Nilgiris mountain range in Tamil Nadu.

49. Vansadhara Water Disputes Tribunal was set up in 2010 to resolve the water dispute between which of the following States? ***[BPSC AEDO]***

A. Odisha and Andhra Pradesh
B. Karnataka and Tamil Nadu
C. Odisha and Chhattisgarh
D. Goa, Karnataka and Maharashtra

Ans. A

Exp:

- **In India, rivers often flow across states. When two states disagree over water sharing or the construction of projects, the Central Government may establish a tribunal under the Inter-State River Water Disputes Act, 1956.**
- Vansadhara (Vamsadhara) is an **east-flowing river originating in the Eastern Ghats near Lanjigarh in Kalahandi district of Odisha**. It flows through **Odisha, Andhra Pradesh, and the Bay of Bengal.** Both states use its water for **irrigation and agriculture.**
- **Why the tribunal?** Disputes arose mainly over the use of river water and the construction of irrigation projects (like the Neradi Barrage). Odisha raised concerns regarding reduced water availability, while Andhra Pradesh sought greater utilisation of river water for irrigation projects. To resolve the dispute, the Vansadhara Water Disputes Tribunal was constituted in 2010.

BCW Bits:

- **Dispute tribunals- Article 262 of the Indian Constitution empowers Parliament to make laws regarding inter-state river water disputes and may bar courts, including the Supreme Court, from interfering.**
- **Other inter-state water disputes**
- **Cauvery-** Karnataka + Tamil Nadu
- **Mahanadi -** Chhattisgarh + Odisha
- **Mahadayi -** Goa + Karnataka + Maharashtra

50. Which of the following waterfalls of India is located in Goa? ***[BPSC AEDO]***

A. Landshing waterfalls B. Dhuandhar waterfalls
C. Nohkalikai waterfalls D. Dudhsagar waterfalls

Ans. D

Exp: A waterfall is formed when a river or stream flows over a steep drop or cliff, causing water to fall vertically. These are usually found in hilly or plateau regions with a sudden change in elevation.**Dhuandhar Waterfalls-** Located near Jabalpur in Madhya Pradesh, Formed by the Narmada River, Name means "smoke cascade" because of mist; **Nohkalikai Waterfalls :** Located near Cherrapunji in **Meghalaya,** One of the **tallest waterfalls in India,** Very famous in Northeast India; **Dudhsagar Waterfalls :** Located in Goa near the Goa–Karnataka border , Formed by the Mandovi River, Name means "Sea of Milk" because of its white foamy appearance, One of the tallest waterfalls in India.

BCW Bits:

- Dudhsagar becomes most powerful during monsoon season, attracting tourists.
- It is part of the Bhagwan Mahavir Wildlife Sanctuary.
- It lies in the Western Ghats, a biodiversity hotspot.
- A railway track passing nearby gives a very iconic scenic view.
- Waterfalls in India are often linked with plateau regions and monsoon rivers.

51. Bharatmala Pariyojana was launched in October 2017. The total length of National Highways approved in it was: ***[BPSC AEDO]***

A. 34, 800 km B. 40, 500 km
C. 41, 800 km D. 27, 900 km

Ans. A

Exp:

- **Bharatmala Pariyojana is a major road development**

project launched by the Government of India in 2017. Its main goal is to improve road connectivity across the country. Improve logistics efficiency and economic connectivity; develop border roads, coastal roads, and highways.

- **Key focus areas of the project:** Economic corridors (for faster transport of goods), Border & international connectivity roads, Coastal and port connectivity, Greenfield expressways (new highways). Basically, it's about creating a strong highway network across India.

BCW Bits:

- The project is implemented by the National Highways Authority of India (NHAI)
- It is part of India's effort to reduce logistics costs and improve supply chains.
- Bharatmala complements the Sagarmala Project (focused on ports and coastal development)
- It also helps in strategic connectivity, especially near border areas.

52. Zojila Pass lies in between: ***[BPSC AEDO]***

A. Kashmir Valley and Ladakh
B. Kashmir and Pakistan
C. Kashmir and Tibet
D. Kashmir and Himachal

Ans. A

Exp:

- A mountain pass is a natural route through a mountain range that allows people to travel from one side to the other. In India, passes are very important for connectivity, Trade, and military movement.
- **Zojila Pass-** lies in the Zaskar (Zanskar) range of the Himalayas; connects: Kashmir Valley (Srinagar side), Ladakh (Leh side). It lies on the strategic Srinagar–Leh Highway.
- **Important Mountain Passes of India-** In Himalayas (Jammu & Kashmir / Ladakh):- Zojila Pass - Kashmir ↔ Ladakh (Zanskar range), Khardung La - near Leh, one of highest motorable roads, Banihal Pass - connects Jammu ↔ Kashmir (Pir Panjal range); Himachal Pradesh:- Rohtang Pass → Manali ↔ Lahaul-Spiti (Pir Panjal range), Baralacha La → connects Himachal ↔ Ladakh; Uttarakhand:- Mana Pass → India ↔ Tibet (China), Niti Pass → trade route with Tibet; Sikkim:- Nathu La → India ↔ China (important trade route), Jelep La → historical trade route; Arunachal Pradesh:- Bomdi La → strategic military route, Diphu Pass → tri-junction (India–China–Myanmar).

BCW Bits:

- Zojila often remains **closed in winter due to heavy snowfall.**
- The government is building **Zojila Tunnel** to ensure all-weather connectivity.
- Zojila Pass is strategically important for Ladakh connectivity.
- During wars (like 1948, Kargil 1999), such passes were **strategically critical.**
- Himalayan passes are generally located at **high altitudes (3000–5000 m)**
- Many ancient passes were part of **Silk Route trade networks.**

53. The oldest mountain range in India is: ***[BPSC AEDO]***

A. Shivalik B. Aravali
C. Himachal Himalaya D. Greater Himalaya

Ans. B

Exp:

- In India, mountains are broadly of two types: Old fold mountains → highly eroded due to great age (like Aravalli); young fold mountains → recently formed, still rising (like the Himalayas). The Himalayas (Shivalik, Himachal, Greater Himalaya) are young; the Aravalli is very old.
- Shivalik, Outer Himalayas, Contains Duns (valleys) like Dehradun; lowest range of the Himalayas, very young and recently formed, known for loose rocks; fragile structure; Himachal Himalaya, Middle Himalayas, includes hill stations like Shimla, Mussoorie; Still part of the young Himalayas. Greater Himalaya: Highest and most rugged part; includes peaks like Everest and Kanchenjunga; young fold mountains. Aravali - One of the oldest fold mountain systems in India , formed during the Precambrian era (very ancient), extends from Gujarat → Rajasthan → Haryana → Delhi. Highest peak: Guru Shikhar (Mount Abu), highly eroded due to age.
- Mount Abu (Aravalli) - only hill station of Rajasthan; Jim Corbett National Park - Shivalik region, Great Himalayan National Park- Himachal, Kaziranga, Nanda Devi Biosphere - Himalayan biodiversity.

BCW Bits:

- Aravalli is older than the Himalayas by millions of years.
- It is heavily eroded and broken.
- Himalayas are still rising due to tectonic activity.
- Aravalli helps prevent the Thar Desert from spreading eastward.
- Many ancient civilisations developed near Aravalli mineral zones.
- Himalayas act as a climate barrier, blocking cold winds from Central Asia.
- Shivalik hills are prone to landslides due to loose soil .

54. Match List – I with List – II and select the correct answer using the code given below the lists. ***[BPSC AEDO]***

List – I (National Park)		List – II (Location)	
a.	Kaziranga	1.	Chikkamagaluru
b.	Kudremukh	2.	Palghat
c.	Silent Valley	3.	Nagpur
d.	Pench Valley	4.	Golaghat–Nagaon

Code:

	a	b	c	d
A.	3	4	2	1
B.	4	1	2	3
C.	1	3	4	2
D.	2	4	1	3

Ans. B

Exp:

- Kaziranga National Park is located in Assam Districts: Golaghat and Nagaon. Famous for: One-horned rhinoceros; Kudremukh National Park- located in Karnataka, in Chikkamagaluru district; Silent Valley National Park- located in Kerala, in Palakkad (Palghat) district, known for Dense evergreen forest, Rare species like lion-tailed macaque. Pench National Park is located in Madhya Pradesh & Maharashtra, Near Nagpur, and inspired the Jungle Book setting.

BCW Bits:

- Kaziranga is a UNESCO World Heritage Site.
- The Silent Valley movement was a major environmental conservation success in India.
- Kudremukh region is rich in iron ore deposits.
- Pench lies along the Satpura range ecosystem.
- Many national parks are part of Project Tiger reserves.
- Western Ghats are a global biodiversity hotspot.
- Floods in Kaziranga help maintain its grassland ecosystem.

55. Match List – I with List – II and select the correct answer from the code given below. [BPSC AEDO]

List – I (Industry)		List – II (Centre)	
a.	Sugar	1.	Raurkela
b.	Paper	2.	Maholi
c.	Cement	3.	Chandrahat
d.	Iron and steel	4.	Sawai Madhopur

Code:

	a	b	c	d
A.	3	1	2	4
B.	4	3	1	2
C.	4	2	3	1
D.	2	3	4	1

Ans. D

Exp:

- Raurkela is one of India's most famous iron and steel centers. Rourkela is known for the Rourkela Steel Plant, which was established with German collaboration during the Second Five-Year Plan. It is located near the mineral-rich Chotanagpur Plateau region where iron ore, coal, limestone, and manganese are available. Maholi, which is associated with the sugar industry. Sugar mills generally develop in areas where sugarcane cultivation is widespread. Sugarcane is a bulky and weight-losing crop because a large quantity of cane is required to produce a smaller quantity of sugar. Therefore, sugar factories are usually located near sugarcane-producing regions to reduce transportation costs. Chandrahat is known as a paper industry center. Paper industries require raw materials such as bamboo, wood pulp, grasses, recycled paper, and abundant water supply. Historically, paper mills in India developed near forest regions and water-rich locations. Sawai Madhopur, which is associated with the cement industry. Cement industries generally develop near limestone-producing areas because limestone is the most important raw material for cement manufacturing. Rajasthan is one of India's leading producers of limestone and cement. Several cement plants are located in the state because of the easy availability of raw materials.
- Industries such as sugar, cement, iron and steel, jute, and paper often develop near their raw material sources because transportation costs are significant. This principle is called material-oriented industrial location.

BCW Bits:

- Jamshedpur is called the Steel City of India and is home to TISCO (now Tata Steel).
- Bhilai Steel Plant was established with Soviet assistance.
- Durgapur Steel Plant was established with British collaboration.
- Limestone, gypsum, and clay are the major raw materials of the cement industry.
- The sugar industry is classified as an agro-based industry.
- The iron and steel industry is known as a basic or key industry because many other industries depend upon it.
- Newsprint paper is generally produced from softwood pulp and recycled paper.
- Rajasthan is one of India's leading cement-producing states because of abundant limestone reserves.
- The Second Five-Year Plan emphasized the development of heavy industries, especially steel plants.
- In industrial geography questions, always remember the trio: Raw Material + Power + Transport = Major Determinants of Industrial Location.

56. Which of the following lakes of India is located in Assam? ***[BPSC AEDO]***

A. Chapanala Lake B. Hamirsar Lake
C. Sala Lake D. Kolleru Lake

Ans. A

Exp:

- **Chapanala Lake, Assam** : Natural freshwater lake / wetland-type formation, Formed due to **riverine and floodplain processes** (common in the Brahmaputra valley), Supports **local biodiversity,** Linked with **wetland ecosystems,** Habitat to Aquatic plants, fish, migratory birds.
- **Hamirsar Lake, Bhuj (Kutch), Gujarat:** Artificial lake (man-made reservoir); **Kolleru Lake, Andhra Pradesh:** Natural freshwater lake, Formed between **Krishna and Godavari river deltas,** One of the **largest freshwater lakes in India,** Major **Ramsar site (wetland of international importance)**

BCW Bits:

- Assam's **Deepor Beel** is a famous Ramsar wetland
- Floodplains create highly fertile soil but also cause **frequent flooding**
- Kolleru Lake plays a role in **flood control between two rivers**
- Hamirsar Lake is central to **Bhuj city planning**
- Wetlands act as **natural water filters**
- Migratory birds use these lakes as **stopover points**
- Climate change is rapidly affecting **wetland ecosystems.**
- **Krishna and Godavari river water dispute:-**
  - **Krishna River, States involved:-**Maharashtra, Karnataka, Andhra Pradesh, Telangana; **Tribunals:- Krishna Water Disputes Tribunal I** (Justice Bachawat), **Krishna Water Disputes Tribunal II.**

  Core Issues:
  1. Allocation of river water among states
  2. Irrigation projects & dams (e.g., Nagarjuna Sagar)
  3. Post-2014 Andhra–Telangana bifurcation disputes
- **Godavari River Disputes:**
  - **Godavari River-**States involved- Maharashtra, Telangana, Andhra Pradesh, Chhattisgarh, Odisha, Karnataka.

  Tribunal:
- **Godavari Water Disputes Tribunal** (Justice Bachawat)

  Core Issues:
  1. Sharing of water among multiple states
  2. Construction of major projects (e.g., Polavaram Project)
  3. Upstream vs downstream usage conflicts
- **Dispute Tribunals in the Indian Constitution:-** Constitutional Provision the main provision is **Article 262.**
  - Parliament can make laws for:
  - **Adjudication of disputes** related to **inter-state rivers and river valleys**
  - It can also:
    - **Bar the jurisdiction of courts (including Supreme Court)** in such disputes. That's why water disputes go to **tribunals, not regular courts.**
- General tribunals (like administrative tribunals) → **Article 323A & Article 323B.**

57. Match the following: ***[BPSC AEDO]***

List – I (Festivals)		List – II (States)	
I.	Myoko Festival	a.	Arunachal Pradesh
II.	Chapchar Kut	b.	Manipur
III.	Sekrenyi	c.	Mizoram
IV.	Sangai	d.	Nagaland

Code:

	I	II	III	IV
A.	b	a	d	c
B.	a	c	d	b
C.	a	b	c	d
D.	b	d	a	c

Ans. B

Exp:

- Myoko Festival -Arunachal Pradesh, celebrated by the **Apatani tribe,** State- **Arunachal Pradesh,** Related to Friendship, Prosperity, Community bonding.
- Chapchar Kut - Mizoram, One of the biggest festivals of **Mizoram,** celebrated after Jungle clearing for shifting cultivation (jhum cultivation). Includes Dance, Music, Traditional dress.
- **Sekrenyi** is celebrated mainly by the **Angami tribe in Nagaland, a Festival of Purification, Youth,** and community rituals.
- **Sangai Festival,** Famous cultural festival of **Manipur,** Named after **Sangai deer** (state animal of Manipur) Promotes Tourism, Culture, Handicrafts.

BCW Bits:

- Sangai deer is found in **KeibulLamjao National Park**, the world's only floating national park
- Apatani tribe of Arunachal is known for unique wet rice cultivation methods
- Nagaland is called the **"Land of Festivals"**
- Hornbill Festival is another major festival of Nagaland
- Many Northeast festivals involve: Folk dances, Bamboo instruments.
- Tribal festivals often reflect strong connection with forests and ecology

58. Match List-I with List-II **[BPSC AEDO]**

List-I (State)		List-II (State Flower)	
(a)	Jharkhand	1.	Lotus
(b)	Karnataka	2.	Gladiolus
(c)	Odisha	3.	Palash
(d)	Punjab	4.	Ashoka

Select the correct answer using the codes given below:

	a	b	c	d
A.	3	1	4	2
B.	2	3	4	1
C.	1	2	3	4
D.	3	4	1	2

Ans. A

Exp:

- Jharkhand is correctly matched with Palash. Jharkhand has Palash as its state flower. Palash is also known as the "Flame of the Forest" because of its bright orange-red flowers. It is commonly found in forest regions of eastern and central India. Karnataka is matched with Lotus. Lotus is the state flower of Karnataka. It is also the national flower of India and holds religious and cultural importance. Odisha is matched with Ashoka flower. The Ashoka tree is culturally important in Indian traditions and is associated with ancient Indian literature and Buddhism. Punjab is matched with Gladiolus. Gladiolus is widely cultivated and known for its colorful ornamental flowers.

BCW Bits:

- Lotus is the national flower of India.
- Palash is also called "Flame of the Forest."
- The Ashoka tree is associated with Buddhist traditions.
- Gladiolus is mainly known as an ornamental flowering plant.
- The peacock is the national bird of India.
- Banyan tree is the national tree of India.
- The Royal Bengal Tiger is the national animal of India.
- Karnataka's state animal is the Indian Elephant.
- Odisha's state animal is the Sambar deer.
- Jharkhand was formed in the year 2000.

59. Which one is the highest Himalayan peak in India? **[BPSC AEDO]**

A. Kamet B. K2
C. Nanda Devi D. Mt. Everest

Ans. C

Exp:

- Kangchenjunga standing at an elevation of 8,586 meters (28,169 feet), it is located in the Himalayas on the border between India (Sikkim) and Nepal. It is the third-highest mountain peak in the world. However, as per the given options Nanda Devi will be the right answer.
- **K2** is the highest peak in India, at about 8, 611 meters. It is located in the Karakoram Range (Trans-Himalayas) and is also known as "Godwin Austen." K2 is the second-highest mountain peak in the world after Mount Everest. **Mount Everest** is the highest mountain peak in the world, but it is located on the border of Nepal and China, not in India. **Nanda Devi**, is the highest mountain peak located entirely within India. This is where confusion usually happens. Since K2 lies in the greater Kashmir region claimed by India, Nanda Devi is the highest peak fully inside Indian territory. **Kamet**, is also an important Himalayan peak located in Uttarakhand, but it is lower than both K2 and Nanda Devi.

BCW Bits:

- K2 is also called Godwin Austen.
- Kanchenjunga is the third-highest mountain peak in the world.
- Kanchenjunga lies on the India–Nepal border.
- The Karakoram Range is separate from the Greater Himalayas.
- Siachen Glacier is located in the Karakoram region.
- Mount Everest is called Sagarmatha in Nepal.
- Nanda Devi National Park is a UNESCO World Heritage Site.
- Himalayan ranges are divided into Himadri, Himachal, and Shiwalik.
- Fold mountains are formed due to tectonic plate collision.
- The Himalayas are geologically young mountains.

60. Which among the following pair/s is/are NOT correctly matched? **[BPSC AEDO]**

	Biosphere Reserve	State
A.	Dehang-Debang	Arunachal Pradesh
B.	Manas	Assam
C.	Similipal	Odisha
D.	Seshachalam	Tamil Nadu

Ans. D

Exp:

- **Dibang Biosphere Reserve**, also referred to in some older sources as Dehang-Debang, is located in Arunachal Pradesh. It is known for rich biodiversity, dense forests, and Himalayan wildlife. Arunachal Pradesh is one of the most biodiverse states in India because of its location in the Eastern Himalayas. **Manas Biosphere Reserve** is located in Assam. It is famous not only as a Biosphere Reserve but also as a National Park, Tiger Reserve, Elephant Reserve, and UNESCO World Heritage Site.

Similipal Biosphere Reserve is located in Odisha. It is famous for forests, waterfalls, tigers, elephants, and rich tribal biodiversity. Similipal is also part of the UNESCO World Network of Biosphere Reserves. **Seshachalam Biosphere Reserve** is located in Andhra Pradesh, not Tamil Nadu. Seshachalam Hills lie close to the Tamil Nadu border and the Tirupati region is nearby.

BCW Bits:

- Nilgiri Biosphere Reserve was the first Biosphere Reserve of India.
- Gulf of Mannar Biosphere Reserve is located in Tamil Nadu.
- Sundarbans Biosphere Reserve is famous for mangrove forests and Royal Bengal Tigers.
- Pachmarhi Biosphere Reserve is located in Madhya Pradesh.
- Khangchendzonga Biosphere Reserve is located in Sikkim.
- Nanda Devi Biosphere Reserve is located in Uttarakhand.
- Great Nicobar Biosphere Reserve is located in Andaman and Nicobar Islands.
- UNESCO launched the "Man and Biosphere Programme" in 1971.
- Biosphere Reserves aim to conserve biodiversity along with sustainable development.
- India has several UNESCO-recognized Biosphere Reserves.

61. Considering the topic of cancer, which of the following Iron & Steel plants is different (odd one)?

[BPSC AEDO]

A. Rourkela B. Bhilai
C. Bokaro D. Jamshedpur

Ans. D

Exp:

- The Tropic of Cancer passes through the middle part of India. It cuts across 8 Indian states — Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura, and Mizoram. In many geography questions, industrial towns located near this latitude are grouped together. Here, three steel plants are situated either close to or north of this belt, while one is comparatively farther south.
- Bokaro is located in Jharkhand, through which the Tropic of Cancer passes. Because of this, Bokaro is commonly associated with the Tropic of Cancer region. Similarly, Bhilai lies in Chhattisgarh, another state crossed by the Tropic of Cancer. Rourkela is also relatively close to this central belt of India.
- But Jamshedpur is different. Although it is also in eastern India and is an important iron and steel center, it lies comparatively farther south from the Tropic of Cancer.

BCW Bits:

- Tropic of Cancer = 23½° North Latitude.
- Rajasthan is the largest state through which Tropic of Cancer passes.
- Tripura is the smallest state through which it passes.
- Tropic of Cancer does NOT pass through Odisha.
- Jamshedpur is called the "Steel City of India."
- Jamshedpur steel plant was established by Tata Steel.
- Bhilai steel plant was set up with Soviet assistance.
- Rourkela steel plant was established with German collaboration.
- Bokaro steel plant was developed with Soviet cooperation.

62. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

[BPSC AEDO]

Assertion (A): Monsoon forests are called *par excellence.*
Reason (R): As they form the natural cover almost all over India.

In light of the above statements, choose the most appropriate answer from the options given below:

A. Both A.& (R) are correct and (R) is correct explanation of (A)
B. (A) is correct but (R) is not correct
C. (A) is not correct but (R) is correct
D. None of the above

Ans. A

Exp:

- The **Assertion** says that monsoon forests are called *par excellence*. The phrase *par excellence* means "the best example" or "the most typical representation." In simple words, it means these forests are considered the most characteristic forests of India. This statement is correct because India's climate is mainly controlled by the monsoon system, and these forests grow naturally under monsoon climatic conditions. **Reason statement-** forests form the natural cover almost all over India. Tropical deciduous forests are spread across a very large part of the country — from central India to eastern India and parts of southern India. Since they occupy such a vast geographical area, they are regarded as the dominant natural vegetation type of India.
- Monsoon forests mainly develop in regions receiving rainfall between about 70 cm and 200 cm. Since a large portion of India falls within this rainfall range, these forests become naturally widespread. They are called **deciduous forests** because the trees shed their leaves during the dry summer season to conserve water. This is an adaptation to seasonal rainfall conditions. These forests are broadly divided into two types — **Moist Deciduous Forests** and **Dry Deciduous Forests**. Moist deciduous forests are found in areas with comparatively higher rainfall, while dry deciduous forests occur in regions with

less rainfall. This distinction is very important for prelims because many questions are framed around rainfall ranges and tree species.

- Some major trees of monsoon forests are teak, sal, sandalwood, bamboo, mahua, palash, shisham, and tendu. Teak is more common in southern and central India, while Sal dominates eastern and northern India.

BCW Bits:

- Tropical Deciduous Forests are also called Monsoon Forests.
- They are the most widespread forests in India.
- Trees shed leaves mainly in summer to reduce water loss.
- Moist deciduous forests receive more rainfall than dry deciduous forests.
- Teak is the most important commercial hardwood tree of India.
- Sal forests are common in Jharkhand, Odisha, Chhattisgarh, and West Bengal.
- Thorn forests develop in areas receiving less than 70 cm rainfall.
- Mangrove forests are also called tidal forests.
- Tropical evergreen forests are found mainly in Western Ghats, Northeast India, and Andaman-Nicobar Islands.
- Tendu leaves from deciduous forests are used for making bidis.

63. Match List-I with List-II ***[BPSC AEDO]***

List-I		List-II	
(a)	Typical Hogback Topography	(I)	Himadri
(b)	Flat floored structural valleys	(II)	Lesser Himalayas
(c)	Broad synclinal valleys	(III)	Between Himadri and Himachal
(d)	Hispar glacier	(IV)	Between Siwalik and Himachal

Select the correct answer using the codes given below:

	a	b	c	d
A.	II	IV	III	I
B.	I	II	III	IV
C.	III	I	IV	II
D.	IV	III	I	II

Ans. A

Exp:

- The Himalayas are generally divided into three major parallel ranges — **Himadri (Greater Himalaya)**, **Himachal (Lesser Himalaya)**, and **Siwalik (Outer Himalaya)**. Each range has its own height, rock structure, relief features, and landforms.
- A hogback is a sharp ridge formed due to steeply tilted sedimentary rock layers. This kind of rugged folded topography is mainly associated with the **Lesser Himalayas (Himachal region)**. The rocks here are highly folded and compressed, creating steep ridges and valleys.
- **Flat floored structural valleys**. These valleys are mainly found **between the Siwalik and Himachal ranges**. Such valleys are called **Duns**. Examples include Dehradun and Kotli Dun. These valleys have flat floors because sediments accumulate in structural depressions formed between mountain ranges.
- **Broad synclinal valleys**. A syncline is a downward fold in rock layers. Broad synclinal valleys are generally found **between Himadri and Himachal**. One famous example is the Kashmir Valley, which is associated with synclinal structure and intermontane basin formation.
- **Hispar Glacier** belongs to the **Himadri or Greater Himalaya region**. The Himadri contains the highest peaks and major glaciers of the Himalayas because this zone remains permanently snow-covered. The Hispar Glacier lies in the Karakoram-Himalayan high mountain region.
- Himalayan geography is strongly linked with tectonic activity. The Himalayas are young fold mountains formed due to collision between the Indian Plate and Eurasian Plate. Because the mountains are geologically young, they are still rising and remain highly unstable, leading to landslides and earthquakes.

BCW Bits:

- Himadri is also called Greater Himalaya.
- Average height of Himadri is above 6000 meters.
- Siwalik is the youngest Himalayan range.
- Duns are longitudinal valleys found mainly in Uttarakhand.
- Dehradun is a famous Dun valley.
- Pir Panjal range lies in the Lesser Himalayas.
- Karewas of Kashmir are famous for saffron cultivation.
- Bhabar lies south of Siwalik and contains porous pebbles.
- Terai region lies south of Bhabar and is marshy in nature.

64. Which of the following statements is not correct? ***[BPSC AEDO]***

I. Satpura runs west–east
II. Satpura is known as Mahadeo hills in the central part of India
III. Satpura is northern boundary of Peninsular plateau
IV. Satpura is longer than Vindhya ranges

Code:

A. I B. III
C. II D. IV

Ans. No Answer

Exp:

- Unlike the folded Himalayas, the Satpura is a Horst (an uplifted fault-block mountain). It was formed due to crustal tension and tectonic block-faulting during the late Cretaceous and early Tertiary periods. The range is bounded by two parallel fault-guided depressions (rifts): the Narmada Graben immediately to its north and the Tapi (Tapti) Graben to its south. The western and central sectors are heavily capped by Deccan Trap basalts, while its eastern reaches expose ancient Archaean gneisses, schists, Gondwana coal-bearing sediments, and Vindhyan quartzites. The range stretches roughly 900 kilometers in an east–west alignment, running south of and parallel to the Vindhya Range. It traverses four states: Gujarat, Maharashtra, Madhya Pradesh, and Chhattisgarh. Mahadeo Hills in the central Madhya Pradesh is the central and highest section which is home to Pachmarhi and Dhupgarh. Dhupgarh (1,350 m): Located near Pachmarhi in the Mahadeo Hills, it is the highest point of the entire Satpura system and the highest peak in Madhya Pradesh.

  Note: Both statements III and IV are wrong as Vindhyas forms the northern boundary of Peninsular Plateau and length of Vindhya range is 1050km while Satpura is 900km only.
- Narmada River flows between the Vindhya range in the north and the Satpura range in the south. This is one of the most important map-based facts in Indian geography. The Narmada and Tapi rivers are special because unlike most Peninsular rivers, they flow westward into the Arabian Sea.

BCW Bits:

- Narmada River flows through a rift valley.
- Pachmarhi is called the "Queen of Satpura."
- Vindhya range forms part of the Central Highlands.
- Satpura ranges are older than the Himalayas.
- Black soil regions are found south of Satpura in Deccan Trap areas.
- Western Ghats are higher and more continuous than Eastern Ghats.
- Aravalli is one of the oldest fold mountain systems of the world.
- Satpura region contains rich teak forests and mineral deposits.

65. What is 'Marusthali' in Thar desert? ***[BPSC AEDO]***

A. Fertile land developed by rivers
B. Region of moving sands and low rainfall
C. Both sand dunes and dune
D. Transverse dune

Ans. B

Exp:

- The word "Maru" means desert and "Sthali" means region or land. So, Marusthali literally means "desert land." In geography, it refers to the hot dry region of the Thar Desert characterized by **very low rainfall, sparse vegetation, and shifting sand dunes**. One of the most important features of Marusthali is the presence of **moving sands**. Because vegetation cover is very little, strong desert winds keep shifting the sand from one place to another. This creates different types of sand dunes. Rainfall in this region is extremely low, often less than 25 cm annually in many places. Due to high temperature and dry winds, evaporation is also very high.
- The Thar Desert itself is climatically very important for India. During summer, this region gets intensely heated and develops a strong low-pressure area. This low pressure helps attract moisture-laden southwest monsoon winds toward the Indian subcontinent. So, even though deserts are dry regions, they still play an important role in India's monsoon system.

BCW Bits:

- Thar Desert is also called the Great Indian Desert.
- Most of the Thar Desert lies in Rajasthan.
- Luni River is the largest river of the Thar Desert region.
- Indira Gandhi Canal is also known as Rajasthan Canal.
- Barchans are crescent-shaped sand dunes.
- Longitudinal dunes are formed parallel to prevailing wind direction.
- Desert soils contain high soluble salts but low humus content.
- Thorny vegetation like cactus and khejri is common in desert areas.
- Hot dry summer winds of northwestern India are called "Loo."
- Sambhar Lake of Rajasthan is India's largest inland salt lake.

66. Which of the following Dun's is in the west of Dun valley? ***[BPSC AEDO]***

A. Patli Dun B. Kotli Dun
C. Pijori Dun D. None of the above

Ans. A

Exp:

- A **Dun** is a longitudinal valley formed between the **Siwalik Range** and the **Lesser Himalayas (Himachal Range)**. These valleys are flat-floored and are formed due to structural and tectonic processes. Since sediments accumulate in these depressions, the land becomes comparatively level and fertile. The most famous example is Dehradun, which lies between the Ganga and Yamuna rivers. But there are several other dun valleys in the Himalayan foothill region.
- Among the given options, **Patli Dun** lies in the western

part of the Dun valley region. It is located in the western Himalayan foothill area and is considered one of the western dun valleys.

- Kotli Dun, is also a dun valley but it is not situated in the extreme western part compared to Patli Dun. The Himalayas are geologically young fold mountains formed due to the collision between the Indian Plate and Eurasian Plate. Because of intense folding, faulting, and compression, several structural valleys developed between mountain ranges. Dun valleys are one such example. These valleys are important not only geographically but also agriculturally and economically because they contain fertile soil and relatively easier terrain compared to surrounding hills.
- Duns are different from intermontane plateaus like Kashmir Valley or Kathmandu Valley. Duns are specifically associated with the region between Siwalik and Lesser Himalayas, whereas larger intermontane valleys may occur between other Himalayan ranges as well.
- Himalayan structure in sequence Himadri (Greater Himalaya), Himachal (Lesser Himalaya), Siwalik (Outer Himalaya), Bhabar, Terai. Dun valleys are mainly located between Siwalik and Himachal.

BCW Bits:

- Dehradun is the most famous Dun valley.
- Duns are mainly found between Siwalik and Lesser Himalayas.
- Bhabar region lies south of Siwalik and contains porous pebbles.
- Terai region lies south of Bhabar and is marshy in nature.
- Siwalik is the youngest Himalayan range.
- Lesser Himalayas contain many hill stations like Shimla and Mussoorie.
- Kashmir Valley is an intermontane valley, not a Dun valley.
- Himalayas are examples of young fold mountains.
- Frequent landslides occur in Himalayas because mountains are geologically unstable.

67. Given below are two statements: One is labeled as Assertion (A) and the other is labeled as Reason (R).

[BPSC AEDO]

Assertion (A): Recently introduced, 'Urea Gold' combines urea with sulphur, minimizing wastage and enhancing plant nutrient uptake.

Reason (R): Many soils in India are deficient in organic carbon macronutrients and essential micronutrients.

In the light of the above statements, choose the most appropriate answer from the options given below:

A. (R) is correct but (A) is not correct
B. Both A.and (R) are correct but (R) is not the correct explanation of (A)
C. (A) is correct but (R) is not correct
D. Both A.and (R) are correct and (R) is the correct explanation of (A)

Ans. D

Exp:

- **Assertion (A)** is correct. Urea Gold is a recently introduced sulphur-coated urea fertilizer. It combines nitrogen from urea with sulphur, which helps improve nutrient efficiency, reduces nitrogen loss, and increases nutrient absorption by plants. Normally, a significant portion of ordinary urea gets wasted through volatilization, leaching, or runoff. By coating urea with sulphur, nutrient release becomes more gradual and efficient. Sulphur is itself an important plant nutrient, so the fertilizer provides dual benefits. **Reason (R)** is also correct. Many Indian soils suffer from deficiencies of organic carbon, nitrogen, sulphur, zinc, and other micronutrients.

  Continuous intensive farming, overuse of chemical fertilizers, and low organic matter have reduced soil fertility in many regions. Because of these deficiencies, fertilizers like Urea Gold are being promoted to improve nutrient use efficiency and soil health.
- The Reason also correctly explains the Assertion because the development of sulphur-coated urea is directly linked with nutrient deficiency and poor fertilizer efficiency in Indian agriculture.
- Indian soils are nutrient-deficient therefore improved fertilizers like sulphur-coated urea are introduced, to reduce wastage and improve nutrient uptake. Another important thing to understand is the difference between macronutrients, secondary nutrients, and micronutrients. For example Nitrogen, phosphorus, potassium → primary macronutrients; Sulphur, calcium, magnesium → secondary nutrients; Zinc, iron, boron → micronutrients.

BCW Bits:

- Urea mainly supplies nitrogen to plants.
- Nitrogen promotes vegetative growth in crops.
- Sulphur is important for protein synthesis in plants.
- Neem-coated urea is used to reduce misuse and improve efficiency.
- Soil Health Card Scheme was launched in 2015.
- Organic carbon improves soil fertility and water retention.
- Micronutrients are needed in small quantities but are essential for plant growth.
- Excessive fertilizer use can degrade soil quality over time.
- Balanced fertilization is important for sustainable agriculture.
- India is one of the largest consumers of fertilizers in the world.

68. Which of the following statements is/are correct?

[BPSC WMO]

1. Black soil is rich in lime, iron and magnesium.
2. Black soil is acidic in nature.

Choose the correct option.

A. Only 2 B. Neither 1 nor 2
C. Only 1 D. Both 1 and 2

Ans: C

Exp: Black soil (Regur) is rich in essential nutrients like lime, iron, magnesium, and potash, making it highly fertile and ideal for crops like cotton. It is typically alkaline or neutral in nature rather than acidic, as it is derived from the chemical weathering of basaltic volcanic rocks, which releases mineral bases that accumulate in the soil profile. It is found extensively in the Deccan Plateau region of India, covering states like Maharashtra, Gujarat, and Madhya Pradesh and is ideal for cotton cultivation.

69. Arrange the following crops in increasing order of water requirement. ***[BPSC WMO]***

1. Wheat 2. Rice
3. Bajra 4. Sugarcane

Choose the correct option.

A. 1 - 3 - 2 – 4 B. 3 - 1 - 2 - 4
C. 3 - 4 - 1 – 2 D. 3 - 2 - 1 - 4

Ans. B

Exp: The increasing order of water requirement for these crops is **Bajra < Wheat < Rice < Sugarcane**. Bajra (pearl millet) is a drought-resistant crop that requires the least amount of water, followed by wheat, which is a rabi crop requiring moderate irrigation. Rice is a water-intensive kharif crop that grows in flooded conditions, while sugarcane requires the highest amount of water throughout its long growth cycle, making it the most water-demanding among the four.

70. In which of the following places one cannot see the mid-day sun exactly overhead twice a year? ***[BPSC WMO]***

A. Ranchi B. Sagar (Madhya Pradesh)
C. Gandhi Nagar D. Jabalpur

Ans. B

Exp: The sun is seen exactly overhead at noon twice a year only at locations situated within the Tropics (between the Tropic of Cancer at **23.5°N** and the Tropic of Capricorn at **23.5°S**). Among the given options, Ranchi, Sagar, and Jabalpur all lie slightly south of the Tropic of Cancer, allowing the sun to be directly overhead twice annually as it moves north and south between the tropics. Conversely, Gandhinagar is situated slightly north of the Tropic of Cancer, meaning the sun never reaches the zenith (directly overhead) at that latitude.

71. In Thar desert 'Rohi' is related to which of the following? ***[BPSC WMO]***

A. Fertile land
B. Salt Lake
C. Region of moving sands
D. Low rainfall

Ans. A

Exp: 'Rohi' refers to the fertile, low-lying tracts of land situated between sand dunes in the Thar Desert of Rajasthan; these areas retain significant moisture and nutrients, making them the most agriculturally productive zones within the otherwise arid desert landscape.

72. Match the rivers with their major tributaries. ***[BPSC WMO]***

List - I (River)		List - II (Tributary)	
a.	Narmada	1.	Wainganga
b.	Godavari	2.	Son
c.	Yamuna	3.	Bhavani
d.	Kaveri	4.	Tawa

Choose the correct option.

A. A-1, B-4, C-2, D-3
B. A-2, B-1, C-4, D-3
C. A-4, B-1, C-3, D-2
D. A-4, B-1, C-2, D-3

Ans. D

Exp: The **Tawa** River is the largest tributary of the **Narmada**, joining it in the Hoshangabad district of Madhya Pradesh.

- The **Wainganga** is a major tributary that flows south to join the Godavari river system.
- The **Son** River is a significant tributary that flows north to join the Ganges; however, in the context of many standard geography resources, it is often associated with the broader Gangetic drainage basin, which includes the Yamuna-Ganges system.
- **(Note:** The official answer key was not released till the printing of this book, it is the best possible answer till then).
- The **Bhavani** River is a major tributary of the Kaveri, originating in the Western Ghats and joining it in Tamil Nadu.

73. Depsang and Rupsa are ***[BPSC WMO]***

A. High plateau B. High lakes
C. High mountains D. High plains

Ans: D

Exp: Depsang and Rupsa are both examples of high-altitude plains located in the Himalayan region of Ladakh, India. Characterized by vast, relatively flat, or gently undulating terrain at elevations exceeding 15, 000 feet, they form part of a high-altitude cold desert environment.

- **Depsang Plains:** Located in the strategic northern part of Ladakh near the Line of Actual Control (LAC), this is a massive, high-altitude gravelly plain situated at an elevation of approximately 17, 000 feet.

- **Rupshu/Rupsa:** This area, located in eastern Ladakh, consists of a high-altitude plateau and plain region, home to unique landscapes and famous high-altitude saltwater lakes like Tso Moriri.

World Geography

1. Maginot line exists between which countries? ***[71st BPSC]***

A. USA and Canada
B. France and Germany
C. Namibia and Angola
D. South Korea and North Korea

Ans. B

Exp: The **Maginot Line** separates **France** and **Germany,** while the Oder-Neisse Line separates Poland and Germany. The **49th Parallel line** separates the **USA and Canada**. The **38th Parallel line** separates **North Korea and South Korea**.

2. Where is Ground Zero? ***[71st BPSC]***

A. Greenwich B. New York
C. Indira point D. Sriharikota

Ans. B

Exp: Ground Zero is the term commonly used to describe the site in Lower Manhattan, New York City, where the 'Twin Towers' of the World Trade Centre once stood.

- **Context:** Following the terrorist attacks on September 11, 2001, the term became synonymous with the disaster site and the subsequent recovery and rebuilding efforts.
- **The Site Today:** It is now home to the 9/11 Memorial & Museum and the One World Trade Center (Freedom Tower).
- **General Definition:** In military and scientific terms, "ground zero" refers to the point on the Earth's surface directly below or above the explosion of a nuclear bomb or a major impact.

3. Which of the following isthmuses have been known as devil's neck? ***[71st Bpsc]***

A. Isthmuses of Panama B. Kra isthmuses
C. Karelian isthmuses D. Isthmuses of corinth

Ans. B

Exp: The Kra Isthmus is the narrowest part of the Malay Peninsula in Thailand. It has historically been referred to as the "Devil's Neck" (or sometimes the "Gorge of the Devil") due to several factors:

- **Difficult Terrain:** The area was notoriously difficult to navigate and traverse in the past due to dense tropical jungles, rugged terrain, and the prevalence of diseases.
- **Strategic Narrowness:** Much like a "bottleneck," this narrow strip of land separates the Gulf of Thailand (Pacific Ocean) from the Andaman Sea (Indian Ocean).
- **The Kra Canal Project:** For centuries, engineers have proposed cutting a canal through this "neck" to bypass the long journey through the Strait of Malacca, a project often met with immense logistical and political "hellish" challenges.

Quick Facts on Other Isthmuses:

- **Isthmus of Panama:** Connects North and South America; famous for the Panama Canal.
- **Isthmus of Corinth:** A narrow strip of land connecting the Peloponnese peninsula with the rest of the mainland of Greece.
- **Karelian Isthmus:** Located between the Gulf of Finland and Lake Ladoga in northwestern Russia, to the north of River Neva

4. Which is the largest river island in the world? ***[71st BPSC]***

A. Srirangam Island B. Majuli Island
C. Bhavani Island D. Agatti Island

Ans. B

Exp: Majuli is a lush, green environment located in the Brahmaputra River in the state of Assam, India.

- **World Record:** It was officially declared the largest river island in the world by the Guinness World Records in 2016 (surpassing Marajó in Brazil, though Marajó is technically larger, it is a river-marine island).
- **Cultural Hub:** It is the nerve center of neo-Vaishnavite culture in Assam, famous for its Sattras (monasteries).
- **District Status:** Majuli became the first island to be made a district in India.
- **Environmental Threat:** The island is currently shrinking due to severe soil erosion caused by the Brahmaputra river's flooding, losing a significant portion of its land area over the last century.
- Srirangam Island is a prominent river island in Tiruchirappalli, Tamil Nadu, formed by the Kaveri River and its northern distributary, the Kollidam (Coleroon) River. It is most famous for housing the Sri Ranganathaswamy Temple.
- Bhavani Island It is a river island situated near Vijayawada on the Bank of Krishna River in Andhra Pradesh.
- Agatti is a stunning 3.84 sq km coral island in the Lakshadweep archipelago, India. Famous for its crystal-clear turquoise lagoons, vibrant coral reefs, and pristine white sand beaches

5. The Pyrenees mountain is located in the middle portion of: ***[BPSC ASO, Pre]***

A. France and the Mediterranean Sea
B. Spain and France

C. Spain and the Mediterranean Sea
D. Spain and Portugal

Ans. B

Exp: The **Pyrenees Mountains** are a chain of mountains in Europe that create the border between France and Spain.

- The highest peak in the Pyrenees Mountains is **Pico de Aneto** in the Spanish Pyrenees, with an elevation of 3,404 m (11,168 feet).
- These mountain ranges are famous for **granite, iron ore** and **limestone.**
- **Treaty of Tordesillas was signed between Spain and Portugal in order to solve the dispute over territories that were discovered newly or explored by Christopher Columbus**

6. Which of the following option is the correct order of mountains from North to South in Asia? ***[BPSC ASO, Pre]***

A. Cherskiy, kolyma, stanovoy and yablonovy
B. Kolyma, cherskiy, stanovoy and yablonovy
C. Kolyma, stanovoy, cherskiy and yablonovy
D. Kolyma, cherskiy, yablonovy and stanovoy

Ans. A

Exp: These mountain ranges are located in eastern Siberia (Asia) and are arranged roughly in a north–south direction: Cherskiy Range lies the farthest north, close to the Arctic Circle.

- Kolyma Mountains are located south of the Cherskiy Range.
- Stanovoy Range lies further south, forming a watershed between Arctic and Pacific drainage systems.
- The Yablonovy Range is the southernmost, near Lake Baikal.
- Hence, from North to South, the correct sequence is:
- Cherskiy → Kolyma → Stanovoy → Yablonovy

7. Which among the following countries share their border with Ukraine? ***[BPSC DSO, Pre]***

1. Moldova 2. Belarus
3. Romania 4. Slovakia

Select the correct answer using the codes given below

A. 1, 2 and 3 B. 3 and 4
C. 2 and 4 D. All of the above

Ans. D

Exp: Ukraine shares land borders with all four listed countries:

- Moldova – along Ukraine's south-western side (including the Transnistria region)
- Belarus – to the north of Ukraine
- Romania – along the south-west (Carpathian region and Danube delta area)
- Slovakia – a short but important border on the western side
- Russia to the east and northeast Poland to the west Hungary to the southwest

8. The strait of hormuz is situated between which among the following gulfs? ***[BPSC DSO, Pre]***

1. Gulf of Persia 2. Gulf of Aden
3. Gulf of Khambhat 4. Gulf of Oman

Select the correct answer using the codes given below:

A. One and three B. One and four
C. Two and three D. Two and four

Ans. B

Exp: The Strait of Hormuz is a strategically important narrow passage that connects:

- The Gulf of Persia (Persian Gulf) on the west, and
- The Gulf of Oman on the east, which further opens into the Arabian Sea.

9. In which area of the world, the leaching process is very intensive and quick? ***[BPSC DSO, Pre]***

A. Moist tropical region B. Cold temperate region
C. Dry tropical region D. Mid latitude region

Ans. A

Exp: In moist tropical regions, high temperatures and heavy rainfall work together to accelerate chemical weathering and the breakdown of organic matter.

- The constant, intense flow of water through the soil (percolation) quickly flushes out soluble minerals and nutrients like silica and calcium.
- This leaves behind a nutrient-poor soil rich in iron and aluminum oxides, a process commonly known as laterization

10. In the world, viticulture, horticulture sericulture are well developed in the: ***[BPSC DSO, Pre]***

A. Monsoon region
B. Tropical Highlands
C. Mediterranean region
D. Marine wet Coast areas

Ans. C

Exp: Why these industries thrive there:

- **Viticulture (Grape cultivation):** The Mediterranean climate—characterized by warm, dry summers and mild, wet winters—is the gold standard for vineyards, as the long sunlight hours maximize sugar content in grapes.
- **Horticulture (Fruits and Vegetables):** Often called the "Orchard Lands of the World," this region specializes in citrus fruits, olives, and figs, utilizing intensive farming techniques and specialized irrigation.
- **Sericulture (Silk production):** Historically, the mild climate supported the growth of mulberry trees, which are essential for rearing silkworms, particularly in Mediterranean areas like Southern Europe and parts of the Levant

11. World's maximum volcanoes are found in the belt of: ***[BPSC DSO, Pre]***

A. Mid continental belt
B. Atlantic belt
C. Indian Ocean belt
D. Circum pacific belt

Ans. **D**

Exp: The Circum-Pacific belt, famously known as the "Ring of Fire," is a horseshoe-shaped zone that circles the Pacific Ocean. It is the most active volcanic and seismic region in the world for several reasons:

- **Tectonic Activity:** It is the meeting point of several tectonic plates. The heavy oceanic plates slide under the lighter continental plates (a process called subduction), creating intense heat and pressure that melts rock into magma.
- **Scale:** Roughly 75% of the world's active volcanoes and about 90% of the world's earthquakes occur along this belt.
- **Geography:** It stretches from the tip of South America, up the coast of North America, across the Bering Strait, down through Japan, and into New Zealand.

12. Biodiversity richest area in the world is: ***[BPSC DSO, Pre]***

A. Tropical region B. Polar region
C. Temperate region D. Ocean

Ans. **A**

Exp: Why the Tropics are Biodiversity Hotspots

- The tropical regions (specifically the area between the Tropic of Cancer and the Tropic of Capricorn) house more than half of the world's plant and animal species despite covering a relatively small portion of Earth's surface.
- **Constant Climate:** Unlike polar or temperate regions, the tropics do not experience harsh winters. This stable, warm climate allows for year-round growth and less "seasonal stress" on species.
- **Energy and Moisture:** High solar energy and abundant rainfall create high primary productivity. This massive amount of plant life supports complex food webs and diverse niches.
- **Evolutionary Time:** Many tropical areas have remained relatively undisturbed by ice ages for millions of years, allowing species more time to evolve and diversify.

13. The top two leading tin producing countries in the world are: ***[BPSC DSO, Pre]***

A. Peru and Bolivia B. Bolivia, Brazil
C. China and Indonesia D. Malaysia and Russia

Ans. **C**

Exp: China and Indonesia consistently rank as the top two producers of tin globally, accounting for a massive share of the world's total output.

- **China:** By far the largest producer, China's tin industry is concentrated primarily in the Yunnan and Guangxi provinces. It not only leads in mining but is also the world's largest consumer of tin due to its massive electronics manufacturing sector.
- **Indonesia:** Ranking second, Indonesia's production is centered on the islands of Bangka and Belitung. It is also one of the world's largest exporters of refined tin.
- **Other Noteworthy Producers:** While countries like Peru, Bolivia, and Myanmar are significant players, they typically fluctuate between the third and fifth positions.

14. The correct order of location of Great Lakes of America from the coast to the interior is: ***[BPSC AEDO]***

A. Michigan, Ontario, Erie, Huron
B. Ontario, Huron, Erie, Michigan
C. Ontario, Erie, Huron, Michigan
D. Erie, Ontario, Huron, Michigan

Ans. **C**

Exp:

- The **Great Lakes of North America** are a group of five large freshwater lakes located along the border of the **United States and Canada**. These lakes are Superior, Michigan, Huron, Erie, Ontario. The lake closest to the Atlantic side is **Lake Ontario.** Water from the Great Lakes flows outward through St. Lawrence River, into the Atlantic Ocean.
- **Great Lakes of North America:-** These lakes together form the largest freshwater lake system in the world by surface area. **Five Great Lakes-** A common mnemonic is,
  - **HOMES:-** H → Huron; O → Ontario; M → Michigan; E → Erie; S → Superior.
- **Lake Superior** is the largest freshwater lake by surface area in the world. **Lake Michigan** is the only Great Lake completely inside the USA. Great Lakes are important for Inland navigation, Trade, Industry, Fishing, Hydroelectric power.

BCW Bits:

- The Great Lakes were formed mainly by glacial action.
- St. Lawrence Seaway connects the Great Lakes to the Atlantic Ocean.
- Niagara Falls lies between Lake Erie and Lake Ontario.
- Chicago is located near Lake Michigan.
- Detroit lies between Lake Erie and Lake Huron.
- The Great Lakes region is highly industrialized.
- Freshwater lakes contain only a small fraction of Earth's total water.
- Lake Baikal in Russia is the deepest freshwater lake in the world.
- The Caspian Sea is the world's largest inland water body.
- Lake Superior is colder and deeper than Lake Erie.

15. Which among the following pair/s is/are NOT correctly matched? ***[BPSC AEDO]***

	Country	**Capital**
1.	Norway	Oslo
2.	Canada	Montreal
3.	Australia	Canberra
4.	South Africa	Cape Town

Select the correct answer using the codes given below:

A. 2, 3 and 4 B. 2 and 4
C. 1, 2 and 4 D. Only 2

Ans. D

Exp:

- **Pair 1** is correctly matched because Norway's capital is Oslo. Oslo is the political, cultural, and economic center of Norway.
- **Pair 2** is incorrect because Canada does not have Montreal as its capital. The capital of Canada is Ottawa. Montreal is an important French-speaking city, while Toronto is the largest city of Canada.
- **Pair 3** is correctly matched because Australia's capital is Canberra. Canberra was developed as the capital to avoid rivalry between Sydney and Melbourne.
- **Pair 4** is also correctly matched. South Africa has a unique system with three capitals: Pretoria – Executive capital, Cape Town – Legislative capital, Bloemfontein – Judicial capital. Since Cape Town is the legislative capital, the pair is considered correct.

BCW Bits:

- Ottawa is the capital of Canada.
- Toronto is the largest city in Canada.
- Montreal is famous for its French-speaking population.
- Canberra is the capital of Australia, not Sydney.
- South Africa has three capitals.
- Brasilia is the capital of Brazil, not Rio de Janeiro.
- Ankara is the capital of Turkey, not Istanbul.
- Naypyidaw is the capital of Myanmar.
- Astana is the present capital of Kazakhstan.
- Sri Jayawardenepura Kotte is the official capital of Sri Lanka.

16. Match List-I with List-II ***[BPSC AEDO]***

List-I (Country)		**List-II (Capital)**	
(a)	Lebanon	1.	Cairo
(b)	Egypt	2.	Beirut
(c)	Uganda	3.	Bratislava
(d)	Slovakia	4.	Kampala

Select the correct answer using the codes given below:

	a	b	c	d
A.	2	1	4	3
B.	3	4	1	2
C.	4	2	3	1
D.	1	3	2	4

Ans. A

Exp:

- Lebanon is a country located on the eastern coast of the Mediterranean Sea in West Asia, and its capital is Beirut. Beirut is an important port city and financial center of the Middle East.
- Egypt has Cairo as its capital. Cairo is located near the Nile River and is one of the largest cities in Africa and the Arab world.
- Uganda is located in East Africa, and its capital is Kampala.
- Slovakia has Bratislava as its capital. Slovakia became an independent country after the peaceful breakup of Czechoslovakia in 1993.

BCW Bits:

- Nile is the longest river associated with Egypt.
- Egypt connects Africa and Asia through the Sinai Peninsula.
- Uganda lies near Lake Victoria.
- Slovakia is a landlocked country in Europe.
- Czechoslovakia split into Czech Republic and Slovakia in 1993.
- Beirut is an important Mediterranean port city.
- Cairo is famous for the nearby Pyramids of Giza.
- Kampala is one of the major cities of East Africa.
- The official currency of Egypt is the Egyptian Pound.
- Slovakia is a member of the European Union.

Indian Economy

10

1. Which of the following statement/s is/are correct about Capability Poverty Measure (CPM) developed by the UNDP? ***[71st BPSC]***

1. CPM includes child malnutrition as a proxy for lack of health and nourishment.
2. Female illiteracy is included to reflect deprivation in education and knowledge.
3. It considers male illiteracy as a key indicator of capability poverty.
4. It includes lack of access to clean drinking water as an important indicator of capability poverty.

Select the correct answer using the codes given below:

A. 1 and 2
B. 1, 2 and 3
C. Only 1
D. More than one of the above

Ans. A

Exp: The CPM (introduced in Human Development Report 1996) focuses on human capabilities rather than income. It uses three indicators:

- **Lack of nourishment:** Proportion of children under 5 who are underweight. (Statement 1)
- **Lack of healthy reproduction:** Proportion of births unattended by trained personnel.
- **Female illiteracy:** As a proxy for deprivation in education/knowledge. (Statement 2)

2. Which organization jointly developed the National Urban Innovation Stack (NUIS) Digital Blueprint with the Ministry of Housing and Urban Affairs (MoHUA)? ***[71st BPSC]***

A. Ministry of Electronics and IT
B. National Institute of Urban Affairs (NIUA)
C. NITI Aayog
D. More than one of the above

Ans. B

Exp: The NUIS strategy and blueprint were released by the Ministry of Housing and Urban Affairs (MoHUA) in collaboration with the National Institute of Urban Affairs (NIUA) to strengthen the urban ecosystem.

3. Before the launch of Kisan Rin Portal (KRP), farmers used to submit manual claims under the Interest Subvention and Prompt Repayment Incentive schemes to which of the following institution/s? ***[71st BPSC]***

A. Ministry of Agriculture and Farmers Welfare and NABARD
B. Reserve Bank of India and NABARD
C. National Payments Corporation of India and Reserve Bank of India
D. More than one of the above

Ans. B

Exp: The Kisan Rin Portal, launched in September, 2023, digitized the Interest Subvention Scheme (ISS). Previously, banks had to manually submit claims to RBI (for commercial banks) and NABARD (for cooperative/rural banks), leading to delays and lack of transparency.

4. As per NFHS-5 data, which of the following state(s) recorded less than 6% of the population as multidimensionally poor in 2019-20? ***[71st BPSC]***

A. Jammu and Kashmir
B. Himachal Pradesh
C. Uttar Pradesh
D. More than one of the above

Ans. B

Exp: According to the NITI Aayog **Multidimensional Poverty Index** (based on NFHS-5), the states with very low poverty rates (<1-5%) include Kerala (<1%), Goa, Sikkim, Tamil Nadu, and **Himachal Pradesh** (approx 4.9%). States like UP and Bihar have much higher rates.

5. Match List-I with List-II. [71st BPSC]

List-I		List-II	
(a)	Perform, Achieve and Trade (PAT) scheme	1.	Introduced in the Union Budget for 2023-24
(b)	Mangrove Initiative for Shoreline Habitats & Tangible Incomes (MISHTI)	2.	Aims to reduce emissions in energy-intensive industries
(c)	The Lifestyle for Environment (LiFE) initiative	3.	Launched in 2019 as a national-level strategy
(d)	National Clean Air Programme (NCAP)	4.	Launched at COP26 in Glasgow in 2021

	(a)	(b)	(c)	(d)
A.	1	3	2	4
B.	3	4	1	2
C.	2	1	4	3
D.	More than one of the above			

Ans. C

Exp:

(a) **PAT** (Perform, Achieve, Trade): Related to Energy

Efficiency (reducing emissions in energy-intensive industries). (Matches 2).

(b) **MISHTI:** Mangrove Initiative (Shoreline Habitats), introduced in Union Budget 2023-24. (Matches 1)

(c) **LiFE** (Lifestyle for Environment): Concept introduced by PM Modi at COP26 in Glasgow (2021). (Matches 4).

(d) **NCAP** (National Clean Air Programme): Launched in 2019 as a national strategy to tackle air pollution. (Matches 3).

6. Share of Micro, Small and Medium Enterprises in the country's Gross Value Added (GVA) has increased from — in 2020-21 to — in 2022-23. ***[71st BPSC]***

A. 30.3%, 35.6%
B. 27.3%, 30.1%
C. 35.1%, 38.2%
D. More than one of the above

Ans. **B**

Exp: The share of MSME Gross Value Added (GVA) in India's total GVA dipped during the pandemic but recovered:
- 2020-21: 27.3%
- 2022-23: Recovered to roughly 30.1%.

7. The base year for GDP, CPI and IIP has been revised to the following: ***[BPSC ASO, Pre]***

A. 2022-23, 2022-23, 2023-24 respectively
B. 2023-24, 2023-24, 2022-23 respectively
C. 2023-24, 2022-23, 2022-23 respectively
D. 2022-23, 2023-24, 2022-23 respectively

Ans. **D**

Exp: According to the Union Budget 2025, the government (Ministry of Statistics) is revising the base years to better capture current economic structures:
- GDP & IIP: Revised to 2022-23.
- CPI (Consumer Price Index): Revised to 2024 (or fiscal 2023-24).

8. Below is a new MSMEs classification criterion as per Union budget 2025. Which is correct? [BPSC ASO, Pre]

	Enterprise category	Investment Limit	Turnover Limit
(a)	Micro	₹2.5 crore	₹`10 crore
(b)	Small	₹25 crore	₹`100 crore
(c)	Medium	₹250 crore	₹500 crore

A. a and c
B. a and b
C. a, b and c
D. b and c

Ans. **B**

Exp: The Union Budget 2025 introduced updated limits to redefine MSMEs, increasing the thresholds to encourage growth:
- Micro: Investment ₹ 2.5 Cr (prev 1 Cr), Turnover ₹ 10 Cr (prev 5 Cr). [Matches row a]
- Small: Investment ₹ 25 Cr (prev 10 Cr), Turnover ₹100 Cr (prev 50 Cr). [Matches row b]
- Medium: Investment ₹125 Cr (prev 50 Cr), Turnover ₹ 500 Cr (prev 250 Cr). [Row c in question says ₹250 Cr, which is incorrect; it is 125 Cr].

9. Which currency is recently added to IMF's Special Drawing Rights (SDRs) basket? ***[BPSC ASO, Pre]***

A. Chinese Renminbi
B. Euro
C. British Pound Sterling
D. Japanese Yen

Ans. **A**

Exp: The Chinese Renminbi (Yuan) was the most recent addition to the Special Drawing Rights (SDR) basket (effective October 2016), joining the US Dollar, Euro, Japanese Yen, and British Pound.

10. As per the National Family Health Survey (NFHS-5), 2019 - 2021, what proportion of households in Bihar live in a pucca house? ***[BPSC ASO, Pre]***

A. One-half
B. One-fourth
C. Two-thirds
D. One-third

Ans. **D**

Exp: As per NFHS-5 (2019-21), only about 34-35% (approx. one-third) of households in Bihar live in pucca houses, which is significantly lower than the national average (approx. 60%).

11. In Bihar, _______ of the households are in rural areas as per National Family Health Survey (NFHS-5), 2019 - 2021. ***[BPSC ASO, Pre]***

A. 84%
B. 78%
C. 89%
D. 80%

Ans. **A**

Exp: Bihar is one of the least urbanized states in India. Census 2011 recorded 88.7% rural population. NFHS-5 data reflects a similar trend, with approximately 84-85% of the sample households residing in rural areas.

12. The first Indian Commercial Bank which was wholly owned and managed by Indians was ***[BPSC ASO, Pre]***

A. State Bank of India
B. Central Bank of India
C. Punjab National Bank
D. Oudh Commercial Bank

Ans. **B**

Exp: Established in 1911, the Central Bank of India claims the distinction of being the first commercial bank "wholly owned and managed" by Indians.
- Punjab National Bank (1894) was the first bank started with Indian capital, but Central Bank stresses the "wholly managed" aspect from inception. Oudh Commercial Bank (1881) had Indian liability but limited management.

13. Which of the following cities is not related to BHEL? ***[BPSC ASO, Mains]***

A. Bhopal
B. Haridwar
C. Hyderabad
D. Lucknow

Ans. D

Exp: BHEL is India's largest engineering and manufacturing enterprise in the energy and infrastructure sectors. Established in 1964, it is a prestigious **Maharatna** central public sector undertaking. BHEL designs and manufactures equipment for power generation (thermal, hydro, gas, nuclear), transmission, transportation, and defence.

- While BHEL has a **Regional Marketing Centre** in **Lucknow**, its major manufacturing units are located in **Bhopal** (Heavy Electrical Plant), **Haridwar** (Heavy Electrical Equipment Plant), and **Hyderabad** (Heavy Power Equipment Plant). A significant project is the **1, 320 MW Maitree Super Thermal Power Project** in Bangladesh, India’s largest-ever turnkey power project export.

14. What is the focus of Central Government sponsored 'Atal Pension Yojana'? ***[BPSC ASO, Mains]***

A. Pension scheme for all unemployed youth
B. Pension scheme for aged widows
C. Pension scheme for workers engaged in the private unorganized sector
D. Pension scheme for women labourers

Ans. C

Exp: The **Atal Pension Yojana (APY)**, launched in **2015**, is a flagship social security scheme administered by the **PFRDA** under the Ministry of Finance. It targets Indian citizens aged **18–40** years, specifically those in the **unorganized sector**, requiring a minimum contribution period of 20 years to provide long-term financial stability.

- APY was introduced to replace the *Swavalamban Yojana* and address the lack of retirement provisions for informal workers:
  - **Target Group:** It focuses on workers in the **unorganized sector** (e.g., drivers, gardeners, home-based workers) who are not covered by any statutory social security scheme.
  - **Pension Tiers:** Based on contributions, it guarantees a fixed monthly pension of **₹1, 000, ₹2, 000, ₹3, 000, ₹4, 000, or ₹5, 000** upon reaching **age 60.**
  - **Triple Benefit:** The scheme ensures a pension for the **subscriber**, followed by a pension for the **spouse** after the subscriber's death, and finally the return of the full accumulated corpus to the **nominee**.
  - **Eligibility Update:** Since October 2022, any citizen who is or has been an **income-tax payer** is no longer eligible to join the scheme, further sharpening its focus on the underprivileged.
  - As of **November 30, 2025**, the scheme has crossed **8.45 crore (8, 45, 17, 419) total subscriber enrolments**.

15. Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason (R). ***[BPSC ASO, Mains]***

Assertion (A): India's National Monetisation Pipeline (NMP) is expected to strengthen fiscal consolidation by reducing the fiscal deficit without cutting capital expenditure.

Reason (R): Proceeds from asset monetisation are treated as capital receipts allowing them to be used for financing infrastructure creation rather than revenue expenditure.

Options:

A. Both A and (R) are true and (R) is the correct explanation of A.
B. Both A and (R) are true, but (R) is not the correct explanation of A.
C. A is true, but (R) is false
D. A is false, but (R) is true

Ans. A

Exp: The **National Monetisation Pipeline (NMP)**, developed in 2021 by **NITI Aayog**, is a four-year roadmap (FY 2022–2025) with an estimated potential of **₹6 lakh crore** aimed at unlocking value from operational **brownfield public assets**.

- By leasing rights to assets such as **roads (27%)**, **railways (25%)**, and **power (14%)** to private operators while retaining government ownership, the program creates a cycle of **"Asset Recycling."**
- This generates non-debt **capital receipts** that are reinvested into the **National Infrastructure Pipeline (NIP)**, allowing the government to maintain high **capital expenditure** and fund new greenfield projects without increasing the fiscal deficit or relying on further borrowing.

Assertion A isTrue:

- The NMP is a critical tool for **fiscal consolidation**. Fiscal consolidation refers to policies aimed at reducing government deficits and debt. By generating revenue from existing assets, the government can fund the **National Infrastructure Pipeline (NIP)** without relying solely on increased borrowing or taxation.
- This allows the government to maintain high **Capital Expenditure (CapEx)**—which has a high multiplier effect on the economy—while keeping the fiscal deficit under control.

Reason (R) is True:

- In government accounting, proceeds from asset monetisation are classified as **Capital Receipts**. Unlike revenue receipts (like taxes), capital receipts are typically used to create long-term assets or reduce liabilities.
- By channelling these funds specifically into greenfield infrastructure, the government ensures that "locked" capital is recycled back into the economy as productive infrastructure rather than being consumed by **Revenue Expenditure** (like salaries, subsidies, or interest payments).

- The reason directly explains how the fiscal goal is achieved: because the money is treated as a capital receipt (R), it provides a non-debt source of funding that replaces the need for deficit-increasing loans, thereby strengthening the fiscal position (A).

16. Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason (R). ***[BPSC ASO, Mains]***

Assertion (A): India's renewable energy sector is attracting large FDI inflows which are sufficient to meet the 2030 target of 500 GW non-fossil capacity.

Reason (R): Favourable policy frameworks like Production-Linked Incentive (PLI) for solar modules and Green Hydrogen Missions have removed all financing and regulatory hurdles for private investors.

Options:

A. Both A and (R) are true and (R) is the correct explanation of (A)
B. Both A and (R) are true, but (R) is not the correct explanation of (A)
C. A is true, but (R) is false
D. A is false, but (R) is true

Ans. C

Exp: India has emerged as one of the world's most attractive markets for renewable energy, driven by ambitious national targets and a robust policy framework. To achieve the **Panchamrit** goals announced at COP26, specifically reaching **500 GW of non-fossil fuel capacity by 2030**, the government has opened the sector to **100% Foreign Direct Investment (FDI)** under the automatic route. While strategic initiatives like the **National Green Hydrogen Mission** and **Production-Linked Incentive (PLI)** schemes have significantly accelerated private participation, the transition continues to navigate complex structural and regulatory landscapes.

Assertion A isTrue:

- India is currently a global leader in energy transition. According to the **Ministry of New and Renewable Energy (MNRE)**, the country has already surpassed the **200 GW** mark in non-fossil installed capacity.
- The sector is attracting record FDI—exceeding **US$ 15-20 billion** cumulatively—which experts suggest is sufficient in volume to keep India on track for its 2030 targets.
- India's track record of meeting its Paris Agreement commitments years ahead of schedule further validates that the current investment trajectory is aligned with the **500 GW**goal.

Reason (R) is False:

- While the **PLI Scheme** for high-efficiency solar modules and the **Green Hydrogen Mission** are transformative, the statement that they have removed **"all"** financing and regulatory hurdles is false. Significant challenges remain, including:
  - **Inter-state Transmission (ISTS) bottlenecks:** Difficulties in evacuating power from resource-rich states.
  - **Discom Health:** The financial stress of State Distribution Companies leads to payment delays for private developers.
  - **Land Acquisition:** Securing large, contiguous tracts of land for solar and wind parks remains a major regulatory and social hurdle.

17. Arrange the following in correct chronology from the codes given below: ***[BPSC ASO, Mains]***

(1) National Hydrogen Mission announced
(2) National Logistics Policy launched
(3) Atmanirbhar Bharat Abhiyan stimulus packages announced
(4) PLI scheme expanded to new sectors (textiles, drones, semiconductors)

Codes:

A. (3), (2), (4), (1) B. (3), (1), (4), (2)
C. (4), (1), (3), (2) D. (3), (1), (2), (4)

Ans. B

Exp: The Government of India launched a series of strategic initiatives between 2020 and 2022 to strengthen industrial output, enhance supply chain efficiency, and promote sustainable energy. These policies, spanning from the **Atmanirbhar Bharat Abhiyan** to the **National Logistics Policy**, form the core of India's multi-sectoral approach to achieving a self-reliant and competitive global economy.

Schemes:

1. **Atmanirbhar Bharat Abhiyan (Self-Reliant India Mission):**
   - **Announcement: May 12, 2020.**
   - A comprehensive stimulus package worth **₹20 lakh crore** (equivalent to 10% of India's GDP) was announced to support MSMEs, farmers, and the migrant workforce during the COVID-19 pandemic.
2. **National Hydrogen Mission (NHM):**
   - **Announcement: February 1, 2021.**
   - Formally announced in the Union Budget 2021-22, the mission aims to scale up green hydrogen production to **5 million metric tonnes (MMT)** per annum by 2030, facilitating the transition to clean energy.
3. **PLI Scheme Expansion (Textiles, Drones, Semiconductors):**
   - **Announcement/Expansion: Late 2021.**
   - **The Production Linked Incentive (PLI) Scheme** is a strategic reform initiative launched in 2020 to strengthen India's manufacturing ecosystem, reduce import dependence, and boost global competitiveness. Moving away from traditional input-based incentives,

it uses an outcome-linked framework that directly ties financial support to incremental sales over a base year.

- As of **December 31, 2025,** the scheme with **₹1.91 Lakh Crore** Outlay has achieved significant milestones across **14 strategic sectors** →namely Large-Scale Electronics Manufacturing (LSEM), IT Hardware, Pharmaceuticals, Bulk Drugs, Medical Devices, Automobiles and Auto Components, Advanced Chemistry Cell Batteries, Solar PV modules, Telecom & Networking Products, Food Processing, Textiles, Specialty Steel, White Goods, Drones & Drone Components by incentivizing incremental production and sales.

4. **National Logistics Policy (NLP):**
   - **Launch: September 17, 2022.**
   - Launched to reduce the cost of logistics in India from the current **13-14% of GDP** to single digits by 2030, improving the global competitiveness of Indian exports.

- By arranging these initiatives according to their official announcement and launch dates:
  - **Statement (3)** Atmanirbhar Bharat Abhiyan: **May 2020**
  - **Statement (1)** National Hydrogen Mission: **February 2021**
  - **Statement (4)** PLI Scheme Expansion: **Late 2021 (Sept–Dec)**
  - **Statement (2)** National Logistics Policy: **September 2022**

  The correct chronological sequence is **(3), (1), (4), (2)**.

18. Arrange the following in correct chronology from the codes given below: ***[BPSC ASO, Mains]***

(1) Pradhan Mantri Mudra Yojana (PMMY)
(2) Pradhan Mantri Jan Dhan Yojana
(3) PM-KISAN
(4) Atal Pension Yojana

Codes:

A. (2), (1), (4), (3)
B. (4), (1), (3), (2)
C. (4), (3), (1), (2)
D. (1), (2), (3), (4)

Ans. A

Exp: The Government of India promotes financial inclusion and social security for the unorganised sector and farmers through flagship schemes managed by the **Ministry of Finance** and the **Ministry of Agriculture & Farmers Welfare**. These initiatives utilise the **"JAM" trinity** (Jan Dhan, Aadhaar, and Mobile) to facilitate transparent **Direct Benefit Transfers (DBT)**, integrating banking, credit, and pension services to provide economic stability for vulnerable populations.

Schemes:

1. **Pradhan Mantri Jan Dhan Yojana (PMJDY):**
   - **Launch Date:** August 28, 2014.
   - **Objective:** To ensure access to financial services, namely, basic savings & deposit accounts, remittance, credit, insurance, and pension in an affordable manner.
   - As of **February 2026**, more than **57.78 crore** Jan-Dhan accounts have been opened under PMJDY, with total deposits of **₹2.94 lakh crore**. **56% Jan Dhan accounts belong to women.**

2. **Pradhan Mantri Mudra Yojana (PMMY):**
   - **Launch Date:** April 8, 2015.
   - **Objective:** To provide collateral-free loans up to ₹20 lakh to non-corporate, non-farm small/micro-enterprises.
   - **Categories:** Shishu (up to ₹50, 000), Kishor (₹50, 000 to ₹5 lakh), Tarun (₹5 lakh to ₹10 lakh), TarunPlus (₹10 lakh to ₹20 lakh).
   - Since inception, over **57.79 crore** loans (as of April 2026) have been sanctioned, with nearly 70% of the beneficiaries being women entrepreneurs.
   - **Women Borrowers**: A total of ₹9.02 lakh crore was disbursed under the Shishu category, ₹ 6.22 lakh crore under Kishor, and ₹ 1.09 lakh crore under the Tarun category.
   - **Minority Borrowers**: The disbursements amounted to ₹1.33 lakh crore under Shishu, ₹1.54 lakh crore under Kishor, and ₹ 0.62 lakh crore under Tarun.

3. **Atal Pension Yojana (APY):**
   - **Launch Date:** May 9, 2015 and **operationalised** from **1st June 2015**.
   - **Objective:** A pension scheme focused on the unorganised sector, providing a guaranteed minimum pension of ₹1, 000 to ₹5, 000 per month starting at age 60.
   - As of **May 2026**, APY has been implemented comprehensively across the country covering all States and Union Territories with the total gross enrolments having crossed **9.10 crore** and Assets Under Management **(AUM) exceeding ₹54, 000 crore**, with a record 1.35 crore subscribers added during FY 2025-26.
   - **Women** participation **under APY** reached a **record 55.14% during FY 2025-26**.
   - At the state level, **SLBCs such as Jharkhand (170%), Bihar (167%), and West Bengal (156%) were top achievers**, with Assam (130%) and Madhya Pradesh (133%) also performing strongly in **FY 2025-26.**

4. **PM-KISAN (Pradhan Mantri Kisan Samman Nidhi):**
   - **Launch Date:** February 24, 2019 (Effective from December 1, 2018).
   - **Objective:** To supplement the financial needs of all land-holding farmers by providing an income support of **₹6, 000 per year** in three equal instalments.

- More than **₹ 4.27 lakh crore** has been disbursed since its launch, making PM-KISAN one of the **world's largest DBT initiatives**.
- In its **22nd instalment** (disbursed on **March 13, 2026**), around ₹18, 640 crore was released.
- Under this 22nd instalment, nearly **9.32 crore eligible farmers** received financial assistance, which prominently includes a significant share of more than **2.15 crore women beneficiaries** .
- **₹60, 000 crore** has been allocated to the scheme in the **Union Budget 2026–27**.

Correct chronology as per official launch dates:

2. **Pradhan Mantri Jan Dhan Yojana:** August 2014
1. **Pradhan Mantri Mudra Yojana:** April 2015
4. **Atal Pension Yojana:** May 2015
3. **PM-KISAN:** February 2019

The correct chronological order is **(2), (1), (4), (3)**.

19. Consider the following statements:

(1) Agreement on Agriculture (AoA) under WTO classifies subsidies into Green Box, Blue Box and Amber Box.

(2) India's Minimum Support Price (MSP) operations are categorized under the Green Box subsidies.

(3) Special and Differential Treatment (S & DT) provisions under WTO provide developing countries like India with greater flexibility in implementing agreements.

Which of the statements given above is/are correct? ***[BPSC ASO, Mains]***

A. Only (1) and (2) B. Only (2)
C. Only (1) and (3) D. (1), (2) and (3)

Ans. C

Exp: The **Agreement on Agriculture (AoA)**, which came into effect in 1995, is a WTO treaty aimed at establishing a fair and market-oriented agricultural trading system. It operates on three main pillars: **Market Access**, **Export Subsidies**, and **Domestic Support**.

- Under the "Domestic Support" pillar, the WTO regulates the level of subsidies a government can provide to its farmers.
- These subsidies are categorized into "Boxes" based on their perceived impact on international trade:
  - **Green Box** (non-distorting),
  - **Blue Box** (distorting but with production limits), and
  - **Amber Box** (highly distorting).
- Developing countries like India are granted additional flexibility under **Special and Differential Treatment (S&DT)** to ensure food security and rural development.
- **Statement (1) is correct:** The AoA explicitly uses the **Box system** to classify domestic support. The Green Box includes "public services" programs (like R&D), the Blue Box covers direct payments under production-limiting programs, and the Amber Box includes all domestic support measures that support prices or are coupled with production.
- **Statement (2) is incorrect:** India's **Minimum Support Price (MSP)** is a price support mechanism. According to WTO rules, any support that maintains prices higher than the international "External Reference Price" is considered trade-distorting and falls under the **Amber Box**. It is not a Green Box subsidy because it directly influences production decisions and market prices.
- **Statement (3) is correct: Special and Differential Treatment (S&DT)** is a core WTO principle. It allows developing countries longer timeframes to meet reduction commitments, higher subsidy limits (*de minimis* levels of 10% compared to 5% for developed nations), and exemptions for certain investment and input subsidies to low-income producers.

Note: India frequently defends its MSP operations at the WTO using the **"Peace Clause"** (Bali Ministerial, 2013), which prevents other countries from challenging India's food procurement programs even if the Amber Box subsidy limits are exceeded, provided certain transparency conditions are met.

20. Consider the following statements regarding Fiscal Responsibility and Budget Management (FRBM) Act, 2003: ***[BPSC ASO, Mains]***

(1) The Act mandates elimination of revenue deficit and reduction of fiscal deficit to 3% of GDP.

(2) It prohibits the government from borrowing from the Reserve Bank of India, except under exceptional circumstances.

(3) The Act requires the Central Government to present a Medium-Term Fiscal Policy Statement along with the annual budget.

Which of the statements given above are correct?

A. Only (1) and (2) B. Only (2) and (3)
C. Only (1) and (3) D. All (1), (2) and (3)

Ans. D

Exp: Fiscal Responsibility and Budget Management (FRBM) Act, 2003:

- The **FRBM Act** was enacted to institutionalize financial discipline and ensure long-term macroeconomic stability by setting a legislative glide path for deficit reduction. It aims to eliminate fiscal profligacy and ensure that the government's borrowing remains within sustainable limits.

Core Provisions & Framework:

- **Deficit Targets:** The Act originally mandated eliminating the **Revenue Deficit** and reducing the **Fiscal Deficit** to **3% of GDP**.
- **RBI Borrowing:** It prohibits the Central Government from borrowing directly from the **Reserve Bank of**

India (RBI), ensuring monetary policy independence. Exceptions are allowed only for temporary "Ways and Means Advances" or under extraordinary circumstances (e.g., national security or natural calamity).

- **Mandatory Disclosures:** Along with the Annual Budget, the government must present three key policy statements:
  1. **Medium-Term Fiscal Policy Statement**
  2. **Fiscal Policy Strategy Statement**
  3. **Macro-economic Framework Statement**

Key Amendments (2018):

- Following the **N.K. Singh Committee** recommendations, the focus shifted toward debt sustainability:
  - **Debt-to-GDP Targets:** Aimed for a **40% limit for the Centre** and **60% for General Government** (Centre + States combined) debt.
  - **Escape Clause:** Formalised a provision allowing a **0.5% deviation** from fiscal deficit targets during structural reforms, war, or agricultural collapse.

Centre-State Dynamics & Regional Facts:

- **State Compliance:** States are required to pass their own FRBM Acts to limit their fiscal deficit to **3% of GSDP**.
- **Bihar Context:** Bihar enacted its **FRBM Act in 2006. Fiscal deficit** for 2026-27 is targeted at 3% of GSDP (Rs 39, 112 crore). In 2025-26, as per revised estimates, the fiscal deficit is expected to be 11.8% of GSDP, significantly higher than the budget estimate (3% of GSDP).

Verification of the options:

- **Statement (1):** The Act was specifically designed to provide a target-oriented transition toward fiscal health. It originally mandated **eliminating the revenue deficit** and set a ceiling on the **fiscal deficit at 3% of GDP**. While deadlines have been extended via amendments, these remain the core statutory benchmarks.
- **Statement (2):** To ensure that the government does not fuel inflation through "deficit financing" (printing money), the Act **prohibits the RBI from subscribing to primary issues of government securities**. Direct borrowing is permitted only under "exceptional circumstances, " such as national security or a collapse in agriculture.
- **Statement (3):** To increase transparency, the Act makes it legally binding for the government to present a **Medium-Term Fiscal Policy Statement** alongside the budget. This document outlines three-year rolling targets for fiscal indicators, ensuring the government is accountable for future financial planning.

Since all three statements of the question are fundamental components of the 2003 Act, **D.All (1), (2) and (3)** is the correct choice.

21. Match the following Five-Year Plans with their primary objectives: ***[BPSC ASO, Mains]***

List-I (Five-Year Plans)		List-II (Major Objectives)	
a.	First Plan (1951-56)	1.	Removal of poverty and attainment of self-reliance
b.	Second Plan (1956-61)	2.	Agricultural development with focus on irrigation and energy
c.	Third Plan (1961-66)	3.	Rapid industrialization based on Mahalanobis model
d.	Fifth Plan (1974-79)	4.	Self-sufficiency in food grains and expansion of basic industries

Codes:
A. a-2, b-3, c-4, d-1
B. a-1, b-4, c-2, d-3
C. a-4, b-3, c-2, d-1
D. a-2, b-1, c-3, d-4

Ans. A

Exp: The Five-Year Plans were a series of national economic programs designed to facilitate centralized and integrated development in post-independence India. Spearheaded by the **Planning Commission**, these blueprints evolved from addressing immediate post-partition crises, such as food shortages, to fostering long-term industrial growth and social equity through targeted sectoral investments.

The correct matching for the provided list is as follows:

(a) First Plan (1951–56) — 2. Agricultural development with focus on irrigation: Aimed at resolving the food crisis and inflation, this plan prioritised the primary sector and saw the initiation of major projects like the Bhakra-Nangal Dam.

(b) Second Plan (1956–61) — 3. Rapid industrialization based on Mahalanobis model: Developed by statistician **Prasanta Chandra Mahalanobis**, it shifted focus toward heavy industries and capital goods to build a self-reliant industrial base.

(c) Third Plan (1961–66) — 4. Self-sufficiency in food grains and expansion of basic industries: Known as the "Gadgil Yojana, " it sought to make the economy "take off" by balancing industrial expansion with food security.

(d) Fifth Plan (1974–79) — 1. Removal of poverty and attainment of self-reliance: This plan is historically significant for the "Garibi Hatao" (Abolish Poverty) slogan and emphasised the "Minimum Needs Programme."

Plan	Model / Key Figure	Primary Focus	Key Outcomes & Facts
First (1951-56)	**Harrod-Domar**	Agriculture, Irrigation, & Energy	**Success Rate:** Achieved 3.6% growth (Target: 2.1%). Focused on post-partition food security; initiated Bhakra, Hirakud, and Mettur dams.
Second (1956-61)	**P.C. Mahalanobis**	Rapid Industrialization (Heavy Industry)	Shifted toward a "Socialistic Pattern of Society." Established steel plants at **Bhilai, Durgapur, and Rourkela** with foreign aid.
Third (1961-66)	**John Sandy & S. Chakravarty**	Self-sufficiency in Food & Basic Industry	Known as **Gadgil Yojana**.Targeted a "self-reliant and self-generating economy." Failed due to the 1962/1965 wars and severe drought.
Fifth (1974-78)	**D.P. Dhar**	Poverty Alleviation & Self-Reliance	Introduced the **"Garibi Hatao"** slogan. Launched the Minimum Needs Programme (MNP). The plan was terminated one year early by the Janata Government.

- **Plan Holidays:** There were three annual plans between the 3rd and 4th Five-Year Plans (1966–69) due to economic instability.
- **Rolling Plan:** Introduced by the Morarji Desai-led government (1978–80) after terminating the Fifth Plan.
- **Final Plan:** The **12th Five-Year Plan (2012–17)** was the last one before the Planning Commission was replaced by **NITI Aayog** in 2015.

22. What does Blue Economy refer to: ***[BPSC DSO, Pre]***

A. Economic activities related to oceans, seas and coasts
B. Economy based on manufacturing of Electronic goods
C. Economic activities related to sky
D. Economic activities related to production of hydrogen driven vehicles

Ans. **A**

Exp: The "Blue Economy" is a concept defined by the World Bank as the sustainable use of ocean resources for economic growth, improved livelihoods, and jobs while preserving the health of the ocean ecosystem. It encompasses fisheries, maritime transport, renewable energy (offshore wind), and tourism.

Info Bits:

- **India's Vision:** India's blue economy policy is anchored in the SAGAR (Security and Growth for All in the Region) vision, articulated by PM Narendra Modi in 2015 it envisions a safe , secure and stable maritime environment by positioning India as a "net security provider" and promote sustainable Blue economy growth.
- **Deep Ocean Mission:** Launched by the Ministry of Earth Sciences in year 2021 to explore deep-sea resources (like polymetallic nodules).

23. The sector primarily responsible for infrastructure development in India is: ***[BPSC DSO, Pre]***

A. Public Works B. Education
C. Health D. Water Transport

Ans. **A**

Exp: The "Public Works" sector (managed by the Central and State Public Works Departments - CPWD/PWD) is the primary arm for executing infrastructure projects like roads, bridges, and government buildings.

- While Health and Education are social infrastructure, Public Works deals with physical infrastructure.

Note: PM Gati Shakti, A national master plan to provide multimodal connectivity infrastructure to various economic zones.

24. The main focus of Smart Cities Mission in India is: ***[BPSC DSO, Pre]***

A. Urban Poverty Alleviation
B. Development of smart infrastructure and technology
C. Information Technology
D. Telecommunication

Ans. **B**

Exp: The Smart Cities Mission (launched in 2015) aims to drive economic growth and improve the quality of life, robust infrastructure, sustainable solutions by enabling local area development and harnessing technology (Smart Solutions) to create smart outcomes for citizens.

- **Ministry:** Ministry of Housing and Urban Affairs (MoHUA).
- **Number of Cities:** 100 cities were selected to be developed as Smart Cities.

25. The concept of 'Green Infrastructure' is related with: ***[BPSC DSO, Pre]***

A. Rural Connectivity
B. Renewal Energy
C. Sustainable Urban Development
D. Water Supply and Sewerage

Ans. **C**

Exp: Green Infrastructure refers to a network of natural and semi-natural areas (parks, green roofs, vertical gardens) designed to deliver ecosystem services. It is a key component of **sustainable urban planning** to manage water, reduce heat islands, and improve air quality.

26. Which of the following is essential for the development of a competitive manufacturing sector in India? ***[BPSC DSO, Pre]***

A. Financial Services
B. Digital Infrastructure
C. Water Tourism
D. Transportation Infrastructure

Ans. **D**

Exp: Efficient transportation infrastructure (roads, railways, ports) is critical for reducing logistics costs, Lower logistics costs make manufacturing competitive globally. As per current assessment prepared by NCAER for DPIIT, logistics costs in India are estimated at about 7.97% of total GDP.

27. The infrastructure play role in a country's development is: ***[BPSC DSO, Pre]***

A. Reducing inequality
B. Creating sustainable growth by sparking innovation
C. Strengthening global competitiveness
D. All the above

Ans. **D**

Exp: Infrastructure has a multiplier effect on the economy:

- **Reducing Inequality:** By providing better access to essential services like transport, education, and healthcare in underserved areas, infrastructure helps level the playing field.
- **Sustainable Growth:** Modern infrastructure (e.g., high-speed internet, efficient energy grids) enables new industries, improves productivity, and fosters innovation, which are key drivers of sustainable economic growth.
- **Competitiveness:** Efficient ports, roads, and communication systems reduce business costs and improve logistics, making a country's goods and services more competitive in the international market.

28. Consider the following statements about NITI Aayog: ***[BPSC DSO, Pre]***

1. NITI Aayog was established on 1 January 2018
2. It was a replacement of the Planning Commission of India
3. Its full-form is National Institution for Transforming India
4. The Chairman of the NITI Aayog is the Prime Minister of India

Which of the above statements are True?

A. Statements (1), (2), (3) and (4)
B. Statements (2), (3) and (4)
C. Statements (1), (3) and (4)
D. Statements (1), (2) and (3)

Ans. **B**

Exp: NITI **(National Institution for Transforming India)** Aayog is a premiere policy **Think Tank** which was established in **2015** via an executive resolution by replacing the Planning Commission of India.

- It works on a **Bottom to Top approach** in contrast to Planning Commission (Top to bottom approach). NITI Aayog ensures more participation of states and hence strengthens the Federal Architecture of India.
- Ex-officio Chairman: Prime Minister
- **Trivia:**
  - 1st Vice Chairman: Arvind Pangariya
  - 1st CEO: Sindhushree Khullar
  - At present Vice Chairman: Mr. Suman Bery (Since
  - 1 May 2022)
  - CEO: Shri. B.V.R. Subrahmanyam (Since 25 February 2023)

Note: The 5-year plan is now replaced by **15-year Vision Document.**

29. Which of the following are functions of NITI Aayog? ***[BPSC DSO, Pre]***

1. To formulate credible plans at the village level
2. To focus on technology upgradation and capacity building for implementation of programmes and initiatives
3. Partnerships with National and International think tanks

Choose the correct answer with reference to above:

A. (1) and (2) only
B. (2) and (3) only
C. (1) and (3) only
D. All of the above

Ans. **D**

Exp: NITI Aayog functions as the government's premier "Think Tank" providing directional and policy inputs.

- **Village Level Plans:** It fosters Cooperative Federalism by evolving shared vision with states and developing mechanisms for village-level planning.
- **Technology:** Focuses on capacity building and tech upgradation.
- **Partnerships:** Collaborates with national and international experts.

30. Consider the following point-targets during plan periods: ***[BPSC DSO, Pre]***

1. Rapid Industrialisation on the development of basic and heavy industries.
2. Quality of Life and Growth with Social Justice.
3. Rapid and inclusive growth with Reduction of gender inequality.
4. Direct attack on the problem of poverty by creating conditions of an expanding economy

Which of the following is the correct chronological order of the events given above?

A. (1)-(4)-(2)-(3) B. (2)-(3)-(4)-(1)
C. (3)-(1)-(2)-(4) D. (4)-(2)-(1)-(3)

Ans. A

Exp: Rapid Industrialisation (Heavy Industries): 2nd Plan (1956–61) – Mahalanobis Model.

- Direct attack on poverty (Garibi Hatao): 5th Plan (1974–78).
- Growth with Social Justice: 9th Plan (1997–2002).
- Rapid and Inclusive Growth: 11th Plan (2007–12).

31. Which of the following is not an objective of the Twelfth five-year plan? ***[BPSC DSO, Pre]***

A. Zero Fiscal Deficit by 2015
B. Sustainable Growth
C. Faster Growth
D. Inclusive Growth

Ans. A

Exp: The theme of the 12th Plan (2012–2017) was "**Faster, Sustainable and More Inclusive Growth**". While it aimed to reduce the fiscal deficit, "Zero Fiscal Deficit by 2015" was not an objective.

Note: zero fiscal deficit by 2015 is clearly not the objective as India being developing economy and being a very extreme statement of attaining "Zero" with a date

32. What was the target growth rate of the tenth five year plan? *[BPSC DSO, Pre]*

A. 6% B. 7% C. 8% D. 9%

Ans. C

Exp: Target growth rate for 10th Five year plan was 8%. The actual achievement was approximately 7.6%.

33. Consider the following statements about NITI Aayog. ***[BPSC DSO, Pre]***

1. NITI Aayog has seven pillar objectives
2. It has three specialised wings

Which of the above statements is/are correct?

A. Only (1) B. Only (2)
C. Both (1) and (2) D. None of the above

Ans. C

Exp: **7 Pillars**: Pro-people, Pro-activity, Participation, Empowering, Inclusion of all, Equality, and Transparency.

- **3 Specialized Wings**: Research Wing, Consultancy Wing, and Team India Wing.

34. Consider the following statement: ***[BPSC DSO, Pre]***

Assertion [A]: NITI Aayog replaced the planning commission.

Reason [R]: This shift reflects a move towards a more dynamic and adaptable approach to economic planning in a globalized world.

Select the correct answer:

A. Both [A] and [R] are true and [R] is the correct explanation of [A]
B. Both [A] and [R] are true bu [R] is not the correct explanation of [A]
C. [A] is false but [R] is true
D. [A] is true but [R] is false

Ans. A

Exp: NITI Aayog replaced Planning commission in 2015. The shift from the Planning Commission (top-down approach) to NITI Aayog (bottom-up approach) was driven by the need for a more dynamic institution that could adapt to the changing global economic landscape and foster cooperative federalism.

35. Multi-Dimensional Poverty Index has been included in which of the following? ***[BPSC AEDO]***

A. World Development Report
B. Human Development Report
C. World Trade Report
D. World Hunger Report

Ans. B

Exp:

- Earlier, poverty was measured mainly through income levels. However, income alone does not fully reflect living conditions because a person may still:
  - lack proper education,
  - suffer from poor healthcare,
  - or live in inadequate housing conditions.
- Therefore, a broader method called the Multi-Dimensional Poverty Index (MPI) was developed.
- The MPI is included in the **Human Development Report (HDR)**, published by the **UNDP (United Nations Development Programme).** Since the HDR focuses on overall human well-being, the MPI forms an important part of the report.

BCW Bits:

- **World Development Report: Published** by the World Bank, focuses on development themes (jobs, climate, etc.).
- **World Trade Report: Published by WTO, Deals with international trade issues.**
- **World Hunger Report: Focuses only on hunger and food security.**
- MPI = Human Development Report (UNDP). HDI (Human Development Index) is also part of the same report (HDI + MPI = HDR)
- MPI was developed with help from the **Oxford Poverty and Human Development Initiative (OPHI).** It identifies **"how many people are poor" and "the intensity of poverty ".**
- Can be removed unless current data reference is intended.
- MPI is useful for governments to design **targeted welfare policies**.

36. Consider the following stages of demographic transition associated with economic development: ***[BPSC AEDO]***

I. High birth rate with low death rate
II. Low birth rate with low death rate
III. High birth rate with high death rate

Arrange these stages in correct chronological order from the code given below:

A. I, III, II
B. III, II, I
C. II, I, III
D. III, I, II

Ans. D

Exp:

- **The demographic transition** is a model that explains how a population changes as a country develops economically. As a country moves from poor to developing to developed, its birth and death rates change in a predictable pattern.
- **Birth rate**- number of people being born; **Death rate**- number of people dying; **Economic development**- improvement in: income, healthcare, education, living standards. These directly affect population growth.
- **High birth rate + High death rate,** poor living conditions, high disease and mortality and slow population growth. **High birth rate + Low death rate; improvement in healthcare, sanitation, and food supply; death rate declines while birth rate remains high, leading to** rapid population growth. **Low birth rate + Low death rate, Developed stage, Better education and family planning, Stable population growth**

BCW Bits:

- Europe went through this transition during the Industrial Revolution
- India is currently in the late Stage 2 / early Stage 3
- Some developed countries (like Japan) are now facing very low birth rates and an ageing population
- Government policies like family planning and the education of women play a big role in moving to later stages

37. Match List – I with the List – II. ***[BPSC AEDO]***

List – I		List – II	
a.	Black revolution	1.	Milk
b.	Blue revolution	2.	Food crop
c.	White revolution	3.	Petroleum
d.	Green revolution	4.	Fish

Select the correct answer using the code given below:

	a	b	c	d
A.	3	4	1	2
B.	3	1	4	2
C.	4	1	3	2
D.	1	3	4	2

Ans. A

Exp:

- These are major development drives in different sectors of the Indian economy. Each "colour revolution" is linked with a specific product or sector. **a. Black Revolution - Petroleum,** "Black" refers to **crude oil (black gold),** Focus: **increase in petroleum production; b. Blue Revolution-** Fish, Blue relates to water bodies (oceans, rivers), Focus: fisheries and aquaculture development; **c. White Revolution - Milk,** White = milk, Also known as **Operation Flood,** made India one of the largest milk producers; **Green Revolution- Food crops,** Green = agriculture, Focus: **increase in food grain production (wheat, rice).**

BCW Bits:

- White Revolution was led by Dr Verghese Kurien (Father of White Revolution)
- Operation Flood was launched in 1970.
- The Green Revolution in India was led by M.S. Swaminathan
- Blue Revolution gained importance with the growth of fish exports
- The petroleum sector is crucial for the energy security of India

38. Which of the following is *not* one of the weakest currencies in the world? ***[BPSC AEDO]***

A. Lebanese Pound
B. Laos Laotian Kip
C. Omani Rial
D. Iranian Rial

Ans. C

Exp:

- A currency is usually called: Weak when its value is very low compared to major currencies (like USD); Strong when its value is high and stable. Example: If 1 USD = 80 units → weaker currency, if 1 USD = 0.3 units → stronger currency."Weak" doesn't always mean a country is poor—it's about exchange value + stability.
- Currency strength depends on factors such as inflation, foreign exchange reserves, economic stability, trade balance, and monetary policy. Strongest currencies (generally): - Kuwaiti Dinar, Bahraini Dinar, Omani Rial, Jordanian Dinar, British Pound; Weakest currencies (generally): - Iranian Rial, Lebanese Pound, Vietnamese Dong, Indonesian Rupiah, Laotian Kip.
- India's currency (Rupee): - Moderately stable, Not very strong, not extremely weak, Influenced by- Oil imports, Global markets, India maintains balance through Forex reserves, RBI interventions.

BCW Bits:

- Some countries keep currency intentionally weaker to boost exports.

- Strong currency makes imports cheaper but exports costlier.
- Currency value can change daily due to forex market trading.
- Pegged currencies (like some Gulf countries) remain stable.
- Inflation is the biggest enemy of currency strength.
- Digital currencies and CBDCs are changing future currency dynamics.
- The exchange rate doesn't always reflect real economic strength fully.

39. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason (R): ***[BPSC AEDO]***

Assertion (A): 'Make in India' is the initiative launched in 2014.

Reason (R): Its main aim is to promote India's manufacturing domain to the world.

In the light of the above statements, choose the most appropriate answer from the options given below:

A. Both A and (R) are not correct
B. A is correct but (R) is not correct
C. A is not correct but (R) is correct
D. Both A and (R) are true and (R) is the correct explanation of (A)

Ans. D

Exp:

- The Assertion is correct because the "Make in India" initiative was officially launched in September 2014 by the Government of India. The programme was launched to encourage companies to manufacture in India and to make the country a global manufacturing hub.
- The **Reason statement** is also correct. One of the main objectives of Make in India was to promote India's manufacturing sector globally and attract both domestic and foreign investment. The initiative aimed to improve ease of doing business, increase industrial growth, create employment opportunities, and reduce excessive dependence on imports.
- The Reason correctly explains the assertion because the initiative was launched specifically to strengthen and promote India's manufacturing domain at the international level. The Make in India programme mainly focused on sectors like automobiles, electronics, defence manufacturing, textiles, railways, renewable energy, and pharmaceuticals. It also aimed to increase the share of manufacturing in India's GDP.

BCW Bits:

- Make in India was launched on 25 September 2014.
- The lion logo of Make in India is made of cogs/gears symbolizing manufacturing.
- "Startup India" was launched in 2016.
- "Digital India" was launched in 2015.
- The Ease of Doing Business ranking was an important area for reform under Make in India.
- FDI stands for Foreign Direct Investment.
- The manufacturing sector plays a major role in employment generation.
- Atmanirbhar Bharat also focuses on domestic production capacity.
- MSME stands for Micro, Small and Medium Enterprises.
- Industrial corridors are important for manufacturing and logistics development.

40. With reference to LiFE, consider the following statements: ***[BPSC AEDO]***

1. It is a Mission Lifestyle for the Environment.
2. Its aim is to promote sustainable tourism.
3. It aims to bring tourism to Indian villages.

Out of these:

A. None of them is correct
B. Only 1 and 2 are correct
C. All the three are correct
D. Only 1 is correct

Ans. D

Exp:

- LiFE stands for "Lifestyle for Environment." It is a global initiative introduced by India to encourage environmentally responsible lifestyles and sustainable consumption patterns. The idea is that environmental protection should not depend solely on governments and industries; individuals should also adopt eco-friendly habits in their daily lives. The initiative was launched by Narendra Modi during COP26 in Glasgow in 2021.
  It focuses on moving people from a "use-and-dispose" economy toward mindful and sustainable living. LiFE is not specifically aimed at promoting sustainable tourism. Although sustainable living may indirectly support ecotourism and environmentally responsible travel, tourism is not the mission's core objective. LiFE is not a programme for taking tourism to Indian villages. This statement is more related to rural tourism or village tourism initiatives run under other schemes. LiFE mainly focuses on environmental awareness, climate-conscious lifestyle changes, waste reduction, energy conservation, and sustainable consumption.

BCW Bits:

- LiFE stands for Lifestyle for Environment.
- LiFE was highlighted during COP26 held in Glasgow.
- COP stands for Conference of Parties under the UNFCCC.
- UNFCCC stands for United Nations Framework Convention on Climate Change.

- Sustainable development means meeting present needs without harming future generations.
- India has committed to achieving Net Zero emissions by 2070.
- The International Solar Alliance was launched by India and France.
- Mission LiFE encourages reduction of wasteful consumption.
- Climate change is mainly linked to greenhouse gas emissions.
- Renewable energy sources include solar, wind, hydro, and biomass.

41. Consider the following statements about the textile industries in India: ***[BPSC AEDO]***

1. The textile industry is a major employment generator and it accounts for about 11 percent of India's manufacturing GVA.
2. India is the fourth largest exporter of textiles and apparel and has a share of about 10 percent of the global trade in this segment.
3. India's technical textile industry is rapidly growing, ranking fifth globally.

Which of the above statement/s is/are correct?

A. 1 and 3 are correct B. 2 and 3 are correct
C. Only 1 is correct D. 1 and 2 are correct

Ans. A

Exp:

- The textile industry is one of India's largest employment-generating sectors after agriculture. It contributes significantly to manufacturing Gross Value Added (GVA), exports, and employment generation, especially for women and rural workers. The sector includes cotton textiles, handlooms, silk, jute, wool, garments, and technical textiles. It also has strong backward and forward linkages with agriculture, trade, and manufacturing. India is indeed one of the largest textile and apparel exporters in the world, but the claim of around 10% share in global trade is exaggerated. India's share is lower than this figure.
- India's technical textile industry has been growing rapidly and India is among the leading countries in this sector, ranking around fifth globally. Technical textiles differ from traditional textiles, they are designed primarily for functional and industrial purposes rather than for fashion or decoration. Examples include medical textiles, geotextiles, bulletproof jackets, fire-resistant fabrics, and automobile textiles.
  - The government is promoting this sector through schemes such as the National Technical Textiles Mission, given its importance to manufacturing, exports, infrastructure, and defence.

BCW Bits:

- Cotton textile industry is one of the oldest industries in India.
- Mumbai is known as the "Cottonopolis of India."
- Ahmedabad is called the "Manchester of India."
- India is one of the largest producers of cotton in the world.
- Textile sector contributes significantly to export earnings.
- Handloom sector provides large rural employment.
- Jute industry is concentrated mainly in West Bengal.
- Technical textiles are used in defence, healthcare, and infrastructure.
- The textile sector is labor-intensive in nature.
- PM MITRA Parks are being developed to strengthen textile manufacturing.

42. Which one of the following indicates that India is a fast growing economy? ***[BPSC AEDO]***

A. Policy Reforms B. Digital Transformation
C. Increasing FDI D. All the above

Ans. D

Exp:

- Economic growth generally means an increase in the production of goods and services in an economy over time, commonly measured through: GDP growth, Industrial expansion, Investment growth, Infrastructure development, Rising income levels, Employment generation. A fast-growing economy usually attracts investment, adopts technological innovation, improves governance, and creates a favorable business environment. All the options given in the question are connected with these processes.
- India has implemented several important reforms since the economic liberalization of 1991, such as: Liberalization of industries, GST reforms, Insolvency and Bankruptcy Code (IBC), Financial inclusion measures, Ease of Doing Business reforms. These reforms improve economic efficiency, encourage private investment, reduce bureaucratic barriers, and strengthen market confidence. For example, GST attempted to create a unified national market, while IBC improved mechanisms for resolving corporate insolvency. Such reforms are considered signs of a modernising and expanding economy.
- India has witnessed rapid digital expansion in areas such as: Digital payments, E-governance, Online banking, Digital identity systems, Startup ecosystem. Programs like Digital India, UPI payments, Aadhaar-linked services, Direct Benefit Transfer (DBT) have transformed service delivery and financial transactions. Digital transformation improves: Productivity, Financial inclusion, Transparency, Efficiency, Innovation. The rapid growth of India's digital economy is widely considered a major indicator of economic modernization and growth potential.

- Foreign Direct Investment (FDI) refers to investment made by foreign companies or individuals in businesses and productive assets of another country.

FDI → Foreign capital + technology + employment + economic growth

- Rising FDI inflows generally indicate: Investor confidence, Stable economic environment, Growth opportunities, Better business climate. FDI contributes to: Capital formation, Employment generation, Technology transfer, Export growth, and Infrastructure development. India has attracted increasing FDI in sectors such as: Services, Telecommunications, Manufacturing, Technology, and Renewable energy. Therefore, increasing FDI is definitely an indicator of a growing economy.
- Distinction between: **Economic growth** and **Economic development.** Economic growth mainly refers to an increase in output or income, while economic development includes broader improvements such as: Education, Healthcare, Living standards, Human development, and Social inclusion. India's recent policy framework often attempts to combine both growth and inclusion.

BCW Bits:

- GDP stands for **Gross Domestic Product**.
- India initiated major economic liberalization reforms in **1991**.
- FDI differs from FPI (Foreign Portfolio Investment) because FDI involves long-term productive investment.
- UPI stands for **Unified Payments Interface**.
- Reserve Bank of India regulates monetary policy in India.
- GST was introduced through the **101st Constitutional Amendment Act**.
- Digital payments expanded rapidly after UPI and Aadhaar integration.
- Ease of Doing Business reforms aims to improve investment climate.
- Economic growth without social inclusion can increase inequality.
- India is often described as one of the world's fastest-growing major economies due to high growth potential and demographic advantages.

43. Which one of the following is not a driving factor for increase in the contribution of service sector in India?

[BPSC AEDO]

A. Globalization and Liberalization
B. Technological advancement
C. Urbanization
D. Decrease in core industries

Ans. D

Exp:

- Three major sectors of an economy: **Primary Sector** → Agriculture and allied activities, **Secondary Sector** → Manufacturing and industries, **Tertiary Sector (Service Sector)** → Banking, insurance, IT, transport, communication, tourism, education, healthcare, trade, etc. India's economy has witnessed rapid growth of the tertiary sector, especially after economic reforms of 1991. Today, services contribute the largest share to India's GDP.
- After the 1991 economic reforms, India opened its economy to global markets through: Liberalization, Privatization, Globalization (LPG reforms). These reforms increased foreign investment, international trade, outsourcing, and integration with the global economy. As a result, sectors like: Information Technology (IT), Business Process Outsourcing (BPO), Financial services, Telecommunications, Aviation, Tourism expanded rapidly. Global companies started investing in India, and India became a major service-exporting nation, especially in software and IT-enabled services. Therefore, globalization and liberalization directly contributed to the expansion of the service sector.
- Technological development revolutionized communication, banking, education, healthcare, e-commerce, and digital services.

Technology → IT expansion + digital services + productivity growth

- India's rise as a global IT hub became possible because of: Internet expansion, Software development, Digital infrastructure, Telecommunications revolution. Technological advancement also enabled: Online banking, E-governance, Digital payments, Startup ecosystem, Remote service delivery. Thus, technological advancement is clearly a driving force behind service sector expansion.
- Urbanization increases demand for services such as: Transport, Education, Healthcare, Banking, Retail trade, Real estate, and Hospitality. As cities expand, service activities naturally grow because urban populations require more organized economic and social services. India's growing urban middle class has also boosted demand for: Entertainment, Tourism, Restaurants, Insurance, Financial services, Digital commerce. Therefore, urbanization strongly supports service sector growth.
- Core industries include sectors such as: Steel, Cement, Coal, Electricity, Refinery products, Natural gas. While economies may gradually shift toward services over time. The service sector growth in India mainly occurred because of: Reforms, Technology, Urbanization, Rising incomes, Global integration. Core Industries decline has no relation with the growth of service sectors in India. In fact, strong industrial growth often supports services because industries require: Banking, Transport, Logistics, Insurance, Communication, Marketing. Thus, industry and services are often complementary rather than

oppositional.
- **Growth of one sector does not necessarily require decline of another sector.**

BCW Bits:
- The service sector is also called the **tertiary sector**.
- IT and IT-enabled services are major contributors to India's service exports.
- Economic reforms in India began in **1991**.
- Reserve Bank of India regulates banking and monetary systems.
- Core industries are considered the backbone of industrial growth.
- India is among the world's leading exporters of software services.
- Structural transformation refers to shifting contribution among agriculture, industry, and services.
- Service sector growth often creates high-skilled employment opportunities.
- Digital India and startup growth accelerated service-sector expansion.
- Manufacturing and services are economically interconnected, not mutually exclusive.

44. Select the scheme which is not related to financial inclusion in India? ***[BPSC AEDO]***

A. PM Jeevan Jyoti Bima Yojana
B. Mudra Yojana
C. PM Suraksha Yojana
D. Mantri Suraksha Bima Yojana

Ans. C

Exp:
- Financial inclusion means ensuring that all sections of society — especially poor, rural, marginalized, and low-income people — have access to: Banking services, Savings accounts, Credit facilities, Insurance, Pension services, Digital payments. The objective is to bring people into the formal financial system and reduce dependence on informal moneylenders. India's financial inclusion drive accelerated through schemes such as: Pradhan Mantri Jan Dhan Yojana, Insurance schemes, Pension schemes, Mudra loans, Digital banking initiatives.
- Pradhan Mantri Jeevan Jyoti Bima Yojana (PMJJBY), which is an official government scheme related to financial inclusion. Launched in 2015, this scheme provides low-cost life insurance coverage to people in the age group of 18–50 years. It aims to provide affordable social security protection to economically weaker sections. Since insurance access is an important part of financial inclusion
- Pradhan Mantri Mudra Yojana was launched in 2015 to provide collateral-free loans to: Small entrepreneurs, Shopkeepers, Artisans, Street vendors, Micro enterprises. The objective is to bring small businesses into the formal credit system and promote self-employment. Thus, Mudra Yojana is strongly connected with financial inclusion.
- Pradhan Mantri Suraksha Bima Yojana provides accidental insurance coverage at a very low premium and is part of the broader social security and financial inclusion framework.

BCW Bits:
- Pradhan Mantri Jan Dhan Yojana was launched in **2014**.
- PMJJBY provides **life insurance** coverage.
- PMSBY provides **accident insurance** coverage.
- Mudra loans are categorized into: Shishu, Kishor, Tarun.
- Financial inclusion aims to reduce dependency on informal credit sources.
- Jan Dhan accounts are linked with: RuPay cards, Direct Benefit Transfer (DBT).
- Atal Pension Yojana targets workers in the unorganized sector.
- Digital payments became a major part of inclusion after UPI expansion.
- Reserve Bank of India play a major role in financial inclusion policies.
- Social security schemes are a major governance focus after 2014.

45. India's National unemployment rate in July, 2005, was: ***[BPSC AEDO]***

A. 3.6 percent B. 4.8 percent
C. 5.2 percent D. 6.4 percent

Ans. B

Exp:
- Unemployment is one of the most important indicators used to measure the health of an economy. In India, unemployment data is generally collected through large-scale surveys conducted by organizations like the National Sample Survey Office (NSSO) and later by the Periodic Labour Force Survey (PLFS). Around 2005, India was witnessing relatively high economic growth, but the problem of "jobless growth" was becoming an important concern. Despite GDP growth, employment generation was not increasing at the same pace. The national unemployment rate in July 2005 was estimated at around **4.8%.**
- Unemployment rate , It is calculated using the formula: Unemployment Rate=Number of Unemployed Persons/ Labour Force×100, Here, the labour force includes people who are either employed or actively seeking work. People who are not willing to work or are outside the working-age population are not counted.

BCW Bits:
- India's unemployment data is currently mainly released through the Periodic Labour Force Survey (PLFS).

- PLFS is conducted by the National Statistical Office (NSO).
- Disguised unemployment is most common in the agricultural sector.
- "Labour Force Participation Rate (LFPR)" and "Worker Population Ratio (WPR)" are different from unemployment rate.
- Seasonal unemployment is common in agriculture, sugar mills, tourism, and construction.
- "Jobless growth" means GDP rises without proportional employment growth.
- Educated unemployment is more visible in urban India.
- MGNREGA guarantees 100 days of wage employment to rural households.
- Female labour force participation in India has historically remained lower than many developing countries.

46. Which one of the following is not an example of infrastructure in a country? ***[BPSC AEDO]***

A. Private houses B. Transportation
C. Telecommunication D. Public Services

Ans. A

Exp:

- Infrastructure refers to the basic physical and organizational facilities that are necessary for the functioning and development of an economy and society. These facilities support production, trade, communication, public welfare, and overall economic growth. Transportation systems, telecommunication networks, electricity, water supply, banking, schools, hospitals, and public services are all considered infrastructure because they help society and the economy function smoothly.
- Private houses, however, are generally considered personal or private assets rather than infrastructure. A private house mainly serves individual or family needs and does not directly function as a public support system for economic activities on a national scale. This is why option A is correct. Houses are physical structures, but infrastructure is not simply about construction; it is about facilities that support broader economic and social functioning.
- Infrastructure is broadly divided into two categories: **Economic Infrastructure, Social Infrastructure.** Economic infrastructure includes transportation, communication, banking, irrigation, power supply, roads, railways, ports, and airports. These directly support economic activities such as production and trade. Social infrastructure includes education, healthcare, sanitation, housing schemes, and public welfare services that improve human capital and quality of life.
- Transportation is an important example of infrastructure. Transport systems like roads, railways, waterways, and airways connect markets, industries, and people. Efficient transportation reduces costs, improves mobility, increases trade, and promotes regional development. In India, projects like the National Highways Authority of India (NHAI) and Bharatmala Project are examples of infrastructure development aimed at boosting connectivity.
- Telecommunication is also a part of infrastructure. Modern economies depend heavily on communication networks such as mobile towers, internet systems, fiber optics, satellites, and digital services. Telecommunication has become even more important in the digital economy because banking, education, governance, and business activities increasingly rely on internet connectivity. In India, initiatives like Digital India aim to strengthen telecom and digital infrastructure.
- Public Servicesis also an example of infrastructure. Public services include facilities provided by the government for public welfare, such as police services, sanitation, public health systems, water supply, and education services. These may not always generate direct profit, but they are essential for social stability and economic productivity.
- Distinction between public goods and private goods. Infrastructure is often linked with public goods because many infrastructure services are non-exclusive and benefit large sections of society. For example, a highway benefits businesses, farmers, passengers, and industries simultaneously. A private house, on the other hand, mainly benefits its owner or residents and is privately consumed.
- Infrastructure plays a major role in economic development. Countries with strong infrastructure usually experience faster industrialization, higher investment, better employment opportunities, and improved living standards. Poor infrastructure creates bottlenecks in development. For example, weak electricity supply can reduce industrial output, and poor transport can increase logistics costs. This is why infrastructure development receives major attention in Five-Year Plans, budgets, and government policies.
- In developing countries like India, infrastructure gaps remain a major challenge. Rural roads, power distribution, internet access, railway modernization, and urban transport are still areas needing improvement. Public-private partnership (PPP) models are often used to finance large infrastructure projects because infrastructure development requires huge investment and long-term planning.

BCW Bits:

- Infrastructure is often called the "backbone of an economy."
- Economic infrastructure directly supports production and trade.

- Social infrastructure improves human capital and quality of life.
- The PPP model stands for Public-Private Partnership.
- Infrastructure investment generally has a multiplier effect on the economy.
- Roads, railways, ports, airports, and power grids are examples of core infrastructure.
- Digital infrastructure has become increasingly important in the 21st-century economy.
- India launched the National Infrastructure Pipeline (NIP) for large-scale infrastructure investment.
- Logistics costs in India has historically been higher than many developed countries due to infrastructure gaps.
- Smart Cities Mission focuses on improving urban infrastructure and governance.

47. Which one of the following committees is not related to tax-reforms in India? ***[BPSC AEDO]***

A. Chelliah Committee B. Tarapore Committee
C. Kelkar Committee D. Reforms Commission

Ans. B

Exp:

- The famous committee in Indian economic reforms is actually the **Tarapore Committee**, not "Taraporewala Committee." It was headed by S. S. Tarapore and dealt with **Capital Account Convertibility (CAC).** Capital Account Convertibility refers to the freedom to convert domestic currency into foreign currency and vice versa for capital transactions such as investments and asset purchases. The Tarapore Committee was therefore connected with external sector reforms and financial liberalization.
- The Raja J. Chelliah Committee was one of the most important tax reform committees formed during the economic liberalization period of the 1990s. It is officially known as the **Tax Reforms Committee**. The committee recommended Reduction in tax rates, Simplification of tax structure, Broadening of tax base, Rationalization of customs and excise duties. Its recommendations played a major role in modernizing India's taxation system after the 1991 economic reforms.
- The Vijay Kelkar Committee gave important recommendations on Direct taxes, Indirect taxes, Fiscal reforms, Tax administration modernization. The Kelkar Task Force strongly advocated Simplification of tax laws, Reduction in exemptions, Improvement in tax compliance, Expansion of service tax base. Many later GST-related and tax rationalization ideas were influenced by broader reform thinking associated with Kelkar recommendations.
- India's economic reforms after 1991. Following the Balance of Payments crisis, India initiated Liberalization, Privatization, Globalization (LPG reforms), Tax reforms became essential because India earlier had Very high tax rates, Complex tax structures, Narrow tax base, High evasion levels. Committees like Chelliah and Kelkar aimed to create a simpler, growth-oriented, and compliance-friendly tax system.

BCW Bits:

- The Chelliah Committee was formed in the early 1990s during economic liberalization.
- Manmohan Singh was Finance Minister during the 1991 reforms.
- The Tarapore Committee dealt with **Capital Account Convertibility (CAC)**.
- Capital Account Convertibility refers to free movement of capital across borders.
- The Kelkar Committee recommended simplification of both direct and indirect taxes.
- GST was introduced in India through the **101st Constitutional Amendment Act, 2016**.
- Narasimham Committee is associated with banking sector reforms.
- The LPG reforms stand for: Liberalization, Privatization, Globalization.
- India's 1991 reforms were triggered by a severe **Balance of Payments crisis**.
- Tax reforms aim to improve: Revenue collection, Compliance, Simplicity, Economic efficiency.

48. With reference to 'One Nation One Ration Card' scheme, consider the following statements. ***[BPSC WMO]***

1. Its objective is to introduce nationwide portability of ration card holders under National Food Security Act, 2013 (NFSA).
2. It has been made available across the country from July 1, 2020.
3. It has been launched by Ministry of Agriculture and Farmers' Welfare.

Which of the above statements is/are correct?

A. Only 1 and 2 B. 1, 2 and 3
C. Only 1 D. Only 1 and 3

Ans. A

Exp: The 'One Nation One Ration Card' (ONORC) scheme, launched by the Department of Food & Supplies and Consumer Affairs under the **Ministry of Consumer Affairs, Food & Public Distribution,** is a national portability initiative designed to ensure food security for all, including internal migrants, by enabling beneficiaries to access their National Food Security Act **(NFSA), 2013** entitlements from any Fair Price Shop across the country. Having successfully implemented the plan across all 36 States (Assam - 36th) and Union Territories, the scheme has institutionalized nationwide portability, facilitating over 2.07 billion transactions, and is further supported by the 'MERA RATION' mobile app—available in 13 languages—to provide users with real-time information. The One Nation One Ration Card scheme was available across the country from **July 1, 2020.**

Therefore, Statements 1 and 2 are correct, and Statement 3 is incorrect.

49. India's national income is estimated primarily by which agency? ***[BPSC WMO]***

A. National Statistical Office (NSO) (Earlier CSO)
B. Ministry of Finance
C. Reserve Bank of India (RBI)
D. NITI Aayog (Planning Commission)

Ans. **A**

Exp: The national income of India is primarily estimated by the **National Statistical Office (NSO)**, which operates under the Ministry of Statistics and Programme Implementation. The NSO was formed following the merger of the Central Statistical Office (CSO) and the National Sample Survey Office (NSSO), and it is responsible for compiling and releasing key economic data, including Gross Domestic Product (GDP) and national income statistics.

- The correct option is **A.National Statistical Office (NSO) (Earlier CSO)**.

50. Given below are the two statements marked as Assertion A and Reason (R). ***[BPSC WMO]***

Assertion (A): The National Infrastructure Pipeline (NIP) was designed as a forward-looking initiative to increase investment between FY20 and FY25.

Reason (R): The NIP includes over 9, 766 projects across 37 sub-sectors, which are tracked through the India Investment Grid portal.

Choose the correct answer from the options given below.

A. Both A and (R) are true, but (R) does not correctly explain (A)
B. A is false, but (R) is true
C. Both A and (R) are true, and (R) correctly explains (A)
D. A is true, but (R) is false

Ans. **C**

Exp: As per the Economic Survey 2024-25, The National Infrastructure Pipeline (NIP) is a forward-looking initiative designed to boost economic growth by targeting approximately ₹111 lakh crore in infrastructure investment between FY20 and FY25.As a centralized platform, it monitors over 9, 766 projects across 37 sub-sectors through the integrated India Investment Grid, providing the structural oversight necessary to realize its ambitious investment targets.

Both A and (R) are true, and (R) correctly explains A.because the NIP's massive, tracked portfolio provides the oversight needed to meet its investment goals.

The correct option is **(C)**.

51. Consider the following statements about inflation in India. ***[BPSC WMO]***

1. Retail headline inflation in India is measured using the Consumer Price Index (CPI).
2. Wholesale Price Index (WPI) inflation includes services.
3. The inflation target is 4% +/- 2% under the monetary policy framework of India.

Which of the above statements are correct?

A. Statements 1 and 3 are correct
B. Statements 1, 2 and 3 are correct
C. Statements 1 and 2 are correct
D. Statements 2 and 3 are correct

Ans. **A**

Exp: The Consumer Price Index (CPI) tracks the cost of a representative basket of everyday goods and services to measure retail inflation, whereas the Wholesale Price Index (WPI) primarily monitors price changes for goods at the wholesale stage. According to the Economic Survey 2025-26, India recorded its lowest headline inflation rate since the inception of the current CPI series, with an average of **1.7%** between April and December 2025.

This moderation was driven by a disinflationary trend in food and fuel—which constitute **52.7%** of the CPI basket—and India recorded one of the sharpest declines in headline inflation among major economies, falling by approximately **1.8 percentage points** in 2025.

- Regarding WPI, the Office of the Economic Adviser, DPIIT, is transitioning to a revised series with a base year of 2022-23, replacing the previous 2011-12 base year. This transition aligns with global best practices and IMF recommendations, incorporating new series for the Output Producer Price Index (OPPI), Input Producer Price Index (IPPI), and Service Producer Price Index (Service PPI) to provide a more comprehensive understanding of price movements.
- Therefore, Statement 1 is correct: CPI is the standard measure for retail headline inflation. **Statement 2 is incorrect:** WPI traditionally tracks goods; services are now monitored through the newly introduced Service PPI. **Statement 3 is correct:** India's monetary policy framework mandates an inflation target of **4% +/- 2%**. The correct option is **A. Statements 1 and 3 are correct**.